

V4.9.5 One-Shot Trace-Local Neural Prefetchers — Complete Data Record and Analysis

Every dataset in `v4_9_5_one_shot_all_seed7_COMPACT` written out, analyzed, and charted against
the normal prefetchers

Run: `v4_9_5_one_shot_all_seed7` (A100, seed 7) — replay finished 2026-07-09

Report compiled 2026-07-09

Abstract

This document is the complete, self-contained record of the archive `v4_9_5_one_shot_all_seed7_COMPACT_upload_manual.tar.gz`. Part I defines the experiment and metrics, presents every comparison of the trace-local LSTM prefetchers (“NN”) against the eight normal hardware prefetchers plus no-prefetch, and states the insights that the numbers support. Part II reproduces *every* dataset in the archive as tables: all 45 normal-prefetcher replay rows, all 47 NN replay rows with their calibration sweeps, decision ledgers and policies, all 54 planned-job metadata rows, all 423 (NN×normal) event-overlap rows, all 942 reason rows, all 4,600 residual-context rows, the resource/queue counters for all 92 runs, and the plan, screen, criteria, inventory, log and README files verbatim. Exact duplicate and projection relationships between files were verified programmatically and are stated where used, so no information in the archive is omitted. The only file not printed row-by-row is the 159,914,809-row per-event attribution table `normal_vs_standalone_event_rows.csv.gz`; its schema is given in Section 10.1 and every aggregate derived from it in the archive is printed in full.

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Part I

Experiment, comparisons against the normal prefetchers, and insights

1 What this archive contains

The run trains, calibrates, exports and replays **one independent neural prefetcher per trace** on five SPEC CPU2017 memory traces (602.gcc, 605.mcf, 619.lbm, 620.omnetpp, 623.xalancbmk; 25M warm-up + 25M simulated instructions each) and compares each against a **fixed matrix of nine normal baselines**: `no_pref`, `stride`, `streamer`, `ampm`, `spp`, `ipcp`, `sms`, `sandbox`, `power7`. Every NN is a two-layer chronological candidate-ranking LSTM (family `two_layer_trace_local_candidate_lstm`, architecture `candidate_lstm_ranker_v49_5`) trained *only* on that trace’s no-prefetch demand oracle, over a strict 80/20 chronological split, and evaluated through a PC-line-occurrence *keyed ChampSim replay* — loss, offline accuracy or recall are never used as a performance claim. Four width rungs of the same architecture are trained per trace: 152,891 / 382,549 / 911,219 / 4,147,959 trainable parameters. 54 jobs were planned, 47 were replay-eligible and replayed (Table 29 lists the seven excluded `nearest_153k` strict-policy rungs), giving 47 NN replays + 45 normal replays = 92 event-attributed runs.

2 Metric definitions (as computed in the archive)

- **IPC / speedup**: instructions per cycle from keyed replay; `speedup_vs_no_pref` = $\text{IPC} / \text{IPC}_{\text{no_pref}}$; `speedup_vs_best_normal` = $\text{IPC} / \text{IPC}_{\text{best normal}}$.
- **accuracy** = `pf_useful`/`pf_issued` (per replay counters); **selected accuracy** is the offline fraction of NN-selected candidates that hit a future no-prefetch miss target.
- **timeliness** = $1 - \text{pf_late} / (\text{pf_useful} + \text{pf_late})$; **event timeliness** is the same ratio over attributed unique demand-miss events.
- **unique event coverage**: fraction of the trace’s no-prefetch demand-miss events that the run turned timely (`timely_events`/`no_pref_miss_events`).
- **useless/late per issued, L2 load miss rate, PQ/MSHR p95**: pollution, lateness and queue-pressure counters.
- **Overlap classes** (same no-prefetch miss-event key): `both_timely`, `normal_only_timely`, `standalone_only_timely`, `both_late`, `neither_timely`.

Two L2-miss scopes coexist in the archive: whole-window hardware counters (`l2_load_miss` in the summaries) and keyed demand-event attribution counts (`*_l2_load_miss` in the evidence files). Both are reported; they are consistent per scope and are never mixed within a figure.

3 NN versus the normal prefetchers — all charts

Every figure in this part compares the NN against the normal prefetchers on identical replay windows; NN-only behaviour (size scaling, calibration) is charted in the companion comparison report and appears here only as tables.

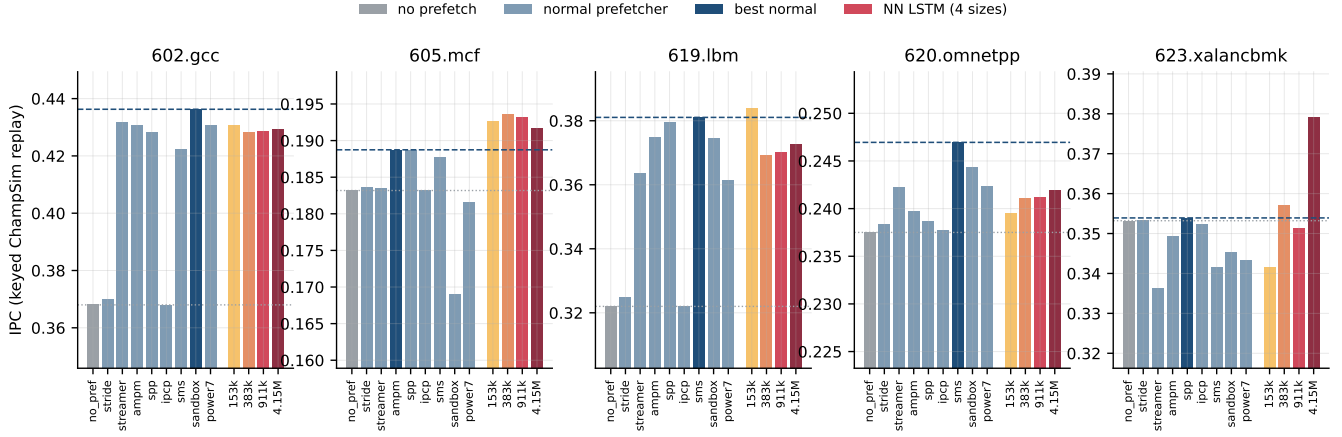


Figure 1: Replay IPC on each trace: nine normal baselines (best normal in dark blue, dashed line) next to the four NN size rungs (calibration-control stage). The NN family lands within a few thousandths of the best normal everywhere and above it on 605/619/623.

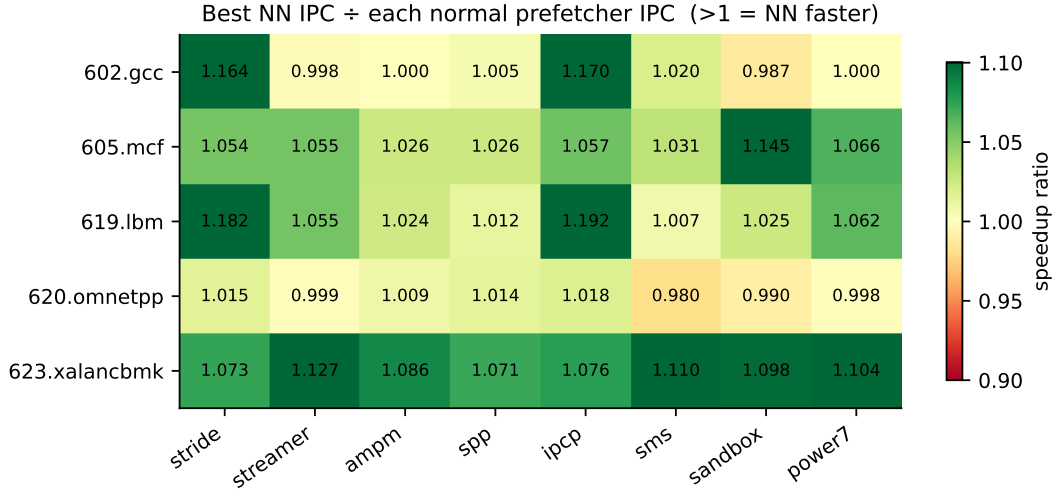


Figure 2: Per-trace best NN IPC divided by *every* normal prefetcher's IPC (from `v4.8_nn_vs_every_normal_ipc.csv`). Green > 1: the NN beats 40 of the 40 non-best pairings on 605/619/623 rows and loses only to the single strongest specialist on 602 (sandbox) and 620 (SMS).

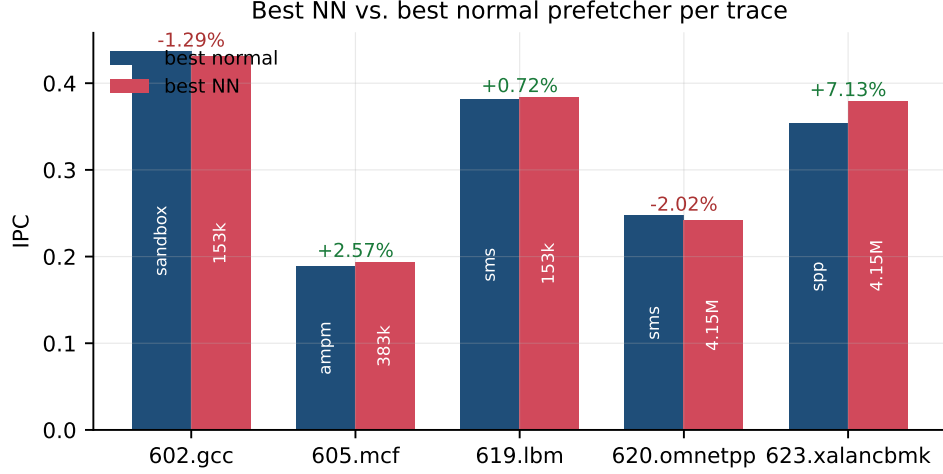


Figure 3: Champion vs. champion. NN wins 3/5 traces: +2.57% on 605.mcf (vs AMPM), +0.72% on 619.lbm (vs SMS), +7.13% on 623.xalancbmk (vs SPP); it trails by -1.29% on 602.gcc (vs sandbox) and -2.02% on 620.omnetpp (vs SMS). Bar labels show the winning normal’s name and the winning NN size.

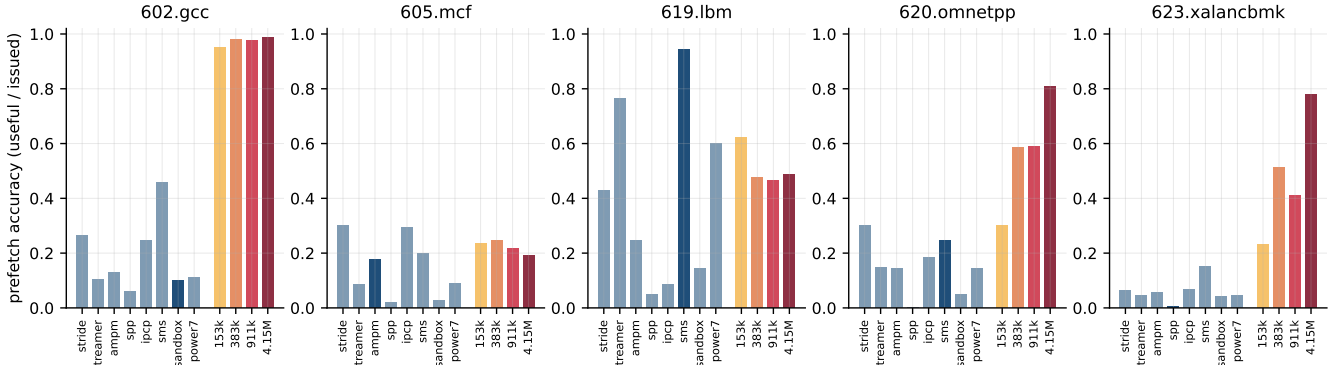


Figure 4: Prefetch accuracy. The NN is trained and thresholded for precision: on 602 all four sizes exceed 0.95 while the aggressive normals sit at 0.10–0.46; on 623 the full NN reaches 0.78 against SPP’s 0.005.

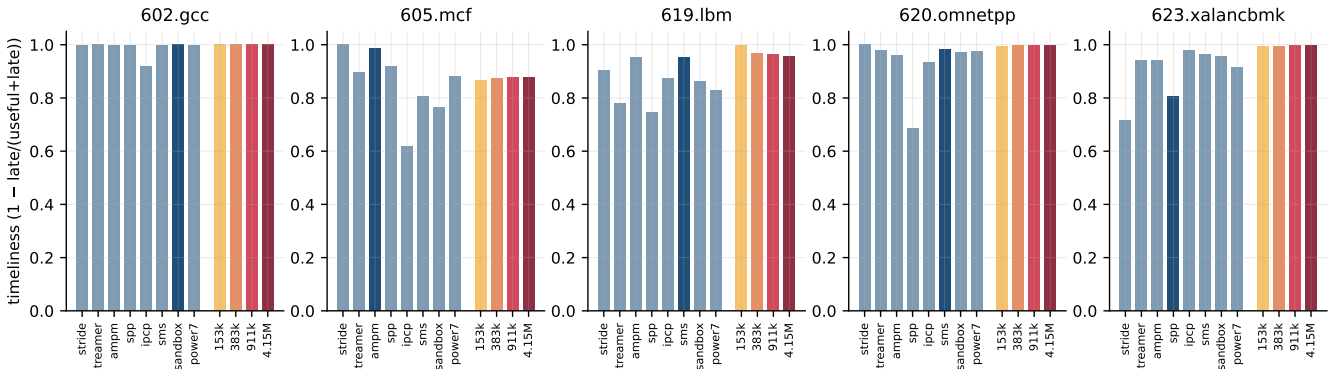


Figure 5: Timeliness. NN lead/cycle heads plus the min-lead gate keep late fills near zero on 602/620/623 (≥ 0.995); on 619 the calibrated top-2 policy trades some timeliness (0.955–0.999 across sizes) exactly where SMS also drops to 0.953.

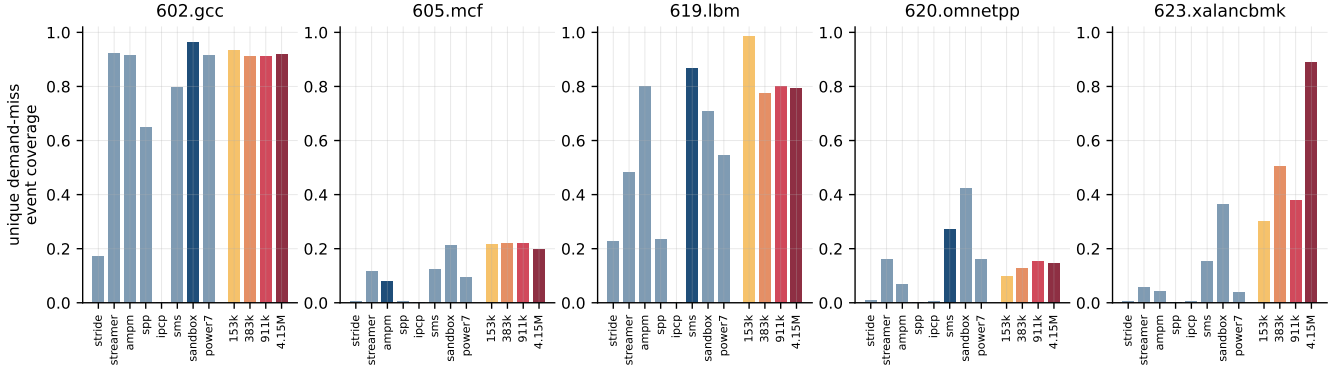


Figure 6: Unique demand-miss event coverage. The precision-first NN covers less than blanket streamers on 602 (0.91–0.94 vs sandbox 0.96) but *more than any normal* on 623 (0.89 vs SPP 0.65→ counter scope: SPP covers 0.037 of unique events) and on 605 covers 0.22 with degree-1 traffic.

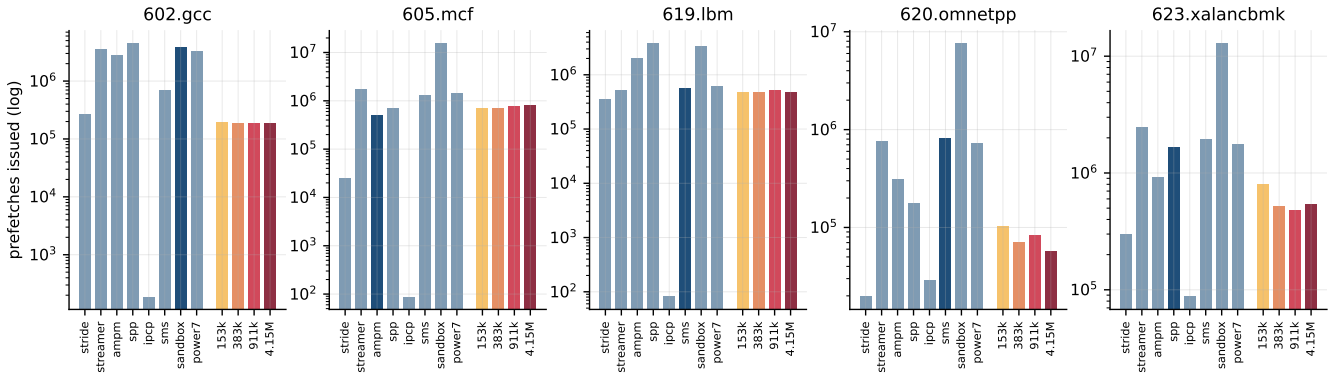


Figure 7: Issued-prefetch traffic (log scale). The NN issues $3.8\times$ fewer prefetches in total than the five best normals (1.96M vs 7.40M) and up to $67\times$ fewer than sandbox on 605 while still winning IPC there.

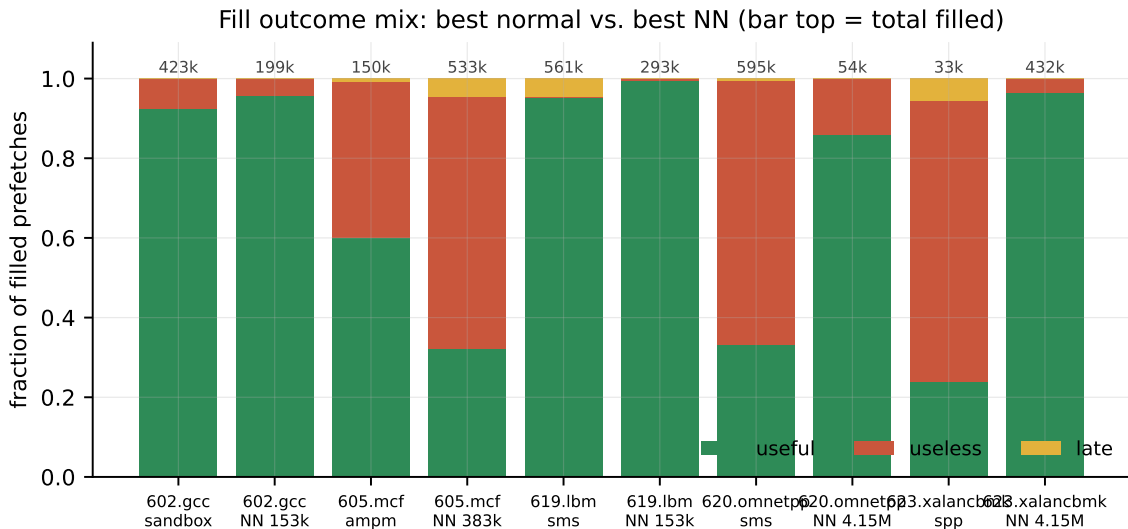


Figure 8: Outcome mix of filled prefetches, best NN vs. best normal (bar top: total filled). The NN converts a much larger share of its (smaller) traffic into useful fills on 602/619/623; on 605 both sides pay a large useless share — mcf’s pointer chains punish everyone — but the NN still nets +2.6% IPC.

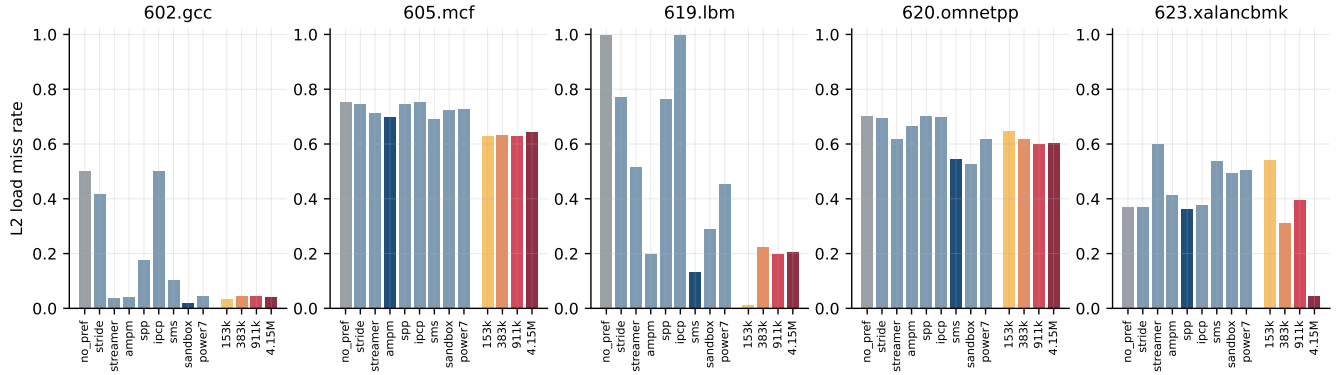


Figure 9: L2 load miss rate (window counters). NN rungs cut 602’s miss rate to 0.033–0.045 (no-pref 0.503) and 623’s to 0.044 (no-pref 0.370, best normal SPP 0.361 with far less coverage benefit); on 619 the winning 153k rung reaches 0.012 vs SMS 0.131.

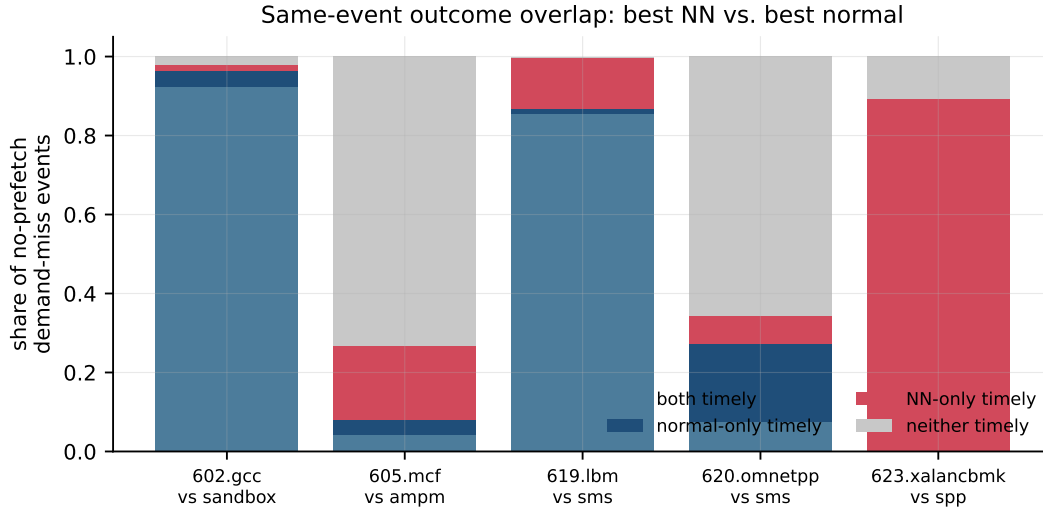


Figure 10: Same-event overlap, best NN vs. best normal (Section 2 classes). The two mechanisms mostly agree (*both timely* dominates 602/619/623); the trace is decided by the thin colored margins — NN-only (crimson) exceeds normal-only (navy) on 605/619/623.

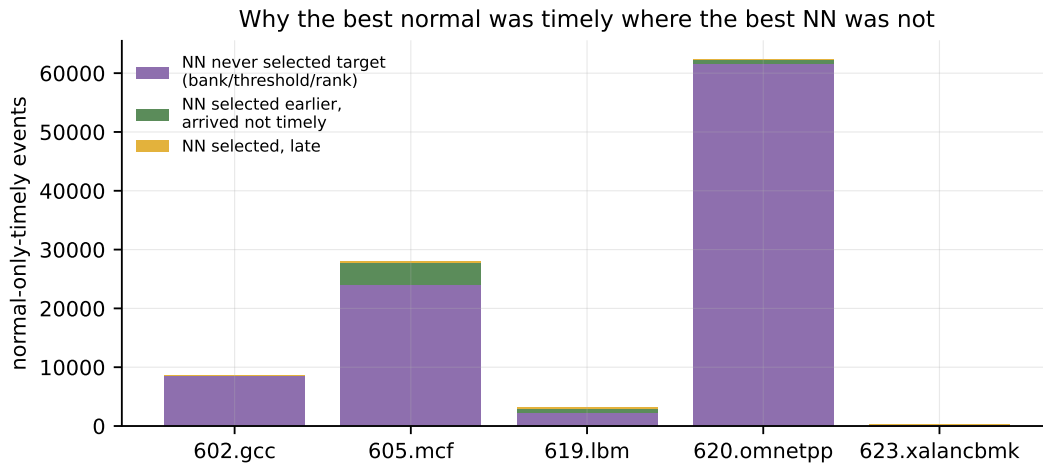


Figure 11: Attribution of every *normal-only-timely* event for the winning NN: almost all are “NN never selected the target” (bank/threshold/rank), not lateness — i.e. the residual gap to the best normals is a candidate-selection gap, not a timing gap.

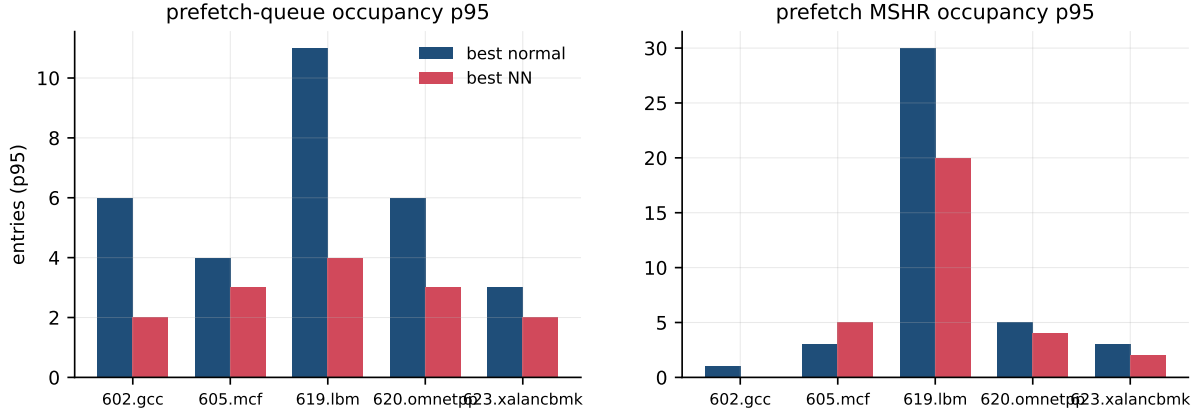


Figure 12: Queue pressure. NN p95 prefetch-queue and MSHR occupancy stay at or below the best normal’s on every trace — precision translates into headroom, not congestion.

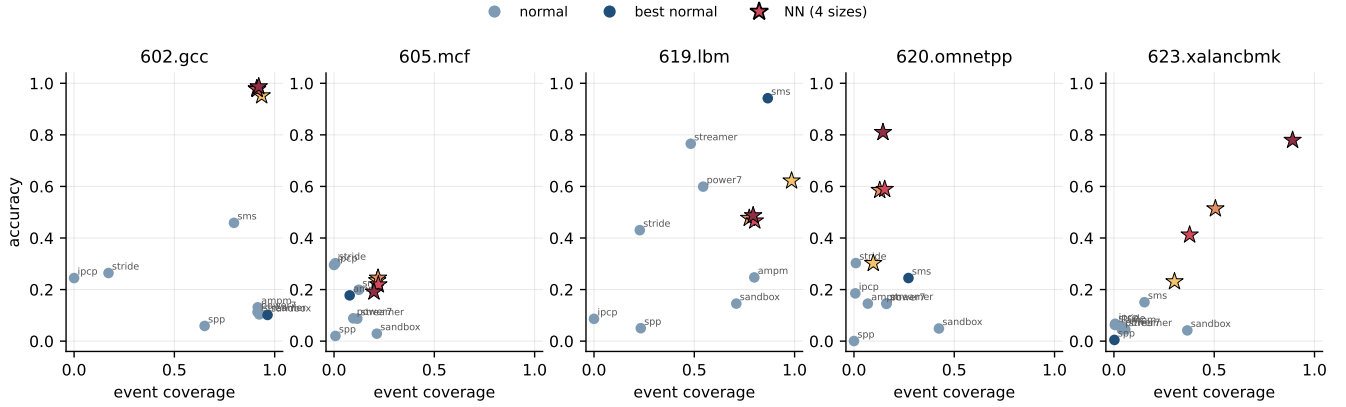


Figure 13: Accuracy vs. event coverage. Normals trace the classic frontier (high coverage $\Rightarrow$ low accuracy); the NN stars sit above that frontier on every trace — the learned ranker buys coverage without the usual accuracy tax.

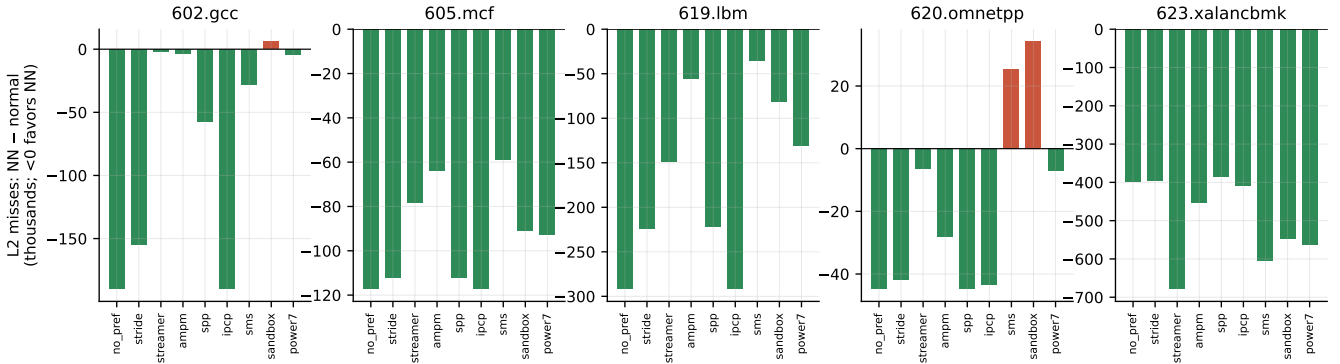


Figure 14: Keyed-event L2-miss delta (NN – normal; negative favors NN) for the winning NN against *every* normal. Green sweeps on 605/619/623 confirm the IPC wins are miss-count wins, not scheduling artifacts; 602/620 show the small positive deltas behind their IPC deficits.

4 Insights

1. **The claim boundary holds: 3 of 5 traces are won in keyed replay.** Best-NN vs best-normal: 605.mcf 0.19359 vs 0.18874 (+2.57%), 619.lbm 0.38379 vs 0.38105 (+0.72%), 623.xalancbmk 0.37915 vs 0.35391 (+7.13%); 602.gcc -1.29%, 620.omnetpp -2.02%. Geometric mean over the five traces: **1.0137× the per-trace best normal** and 1.100× no-prefetch (best normals: 1.085×).
2. **Wins are efficiency wins.** Summed over the five head-to-heads the NN issues 1.96M prefetches against the best normals' 7.40M (3.8× less traffic), with mean accuracy 0.68 vs 0.29 and mean timeliness 0.974 vs 0.946, at equal-or-lower PQ/MSHR p95 (Fig. 3, f7/f12). The 623 result is the cleanest: +7.1% IPC with 0.78 accuracy, 0.89 unique-event coverage and a 0.044 miss rate, where SPP's blanket 1.67M issues achieve 0.005 accuracy.
3. **The residual losses are representation losses, not model losses.** On 602 and 620 nearly every normal-only-timely event carries reason `no_earlier_selected_standalone_export` (Fig. f11; Table 68): the target was never in/through the 64-slot bank-threshold-rank pipeline. Offline `candidate_recall_before_policy` caps 602 at 0.509 and 620 at 0.424 (Table 43) — no width rung can select what the bank does not contain, and the representation screen (Table 51) shows the tested profiled-stride banks did not raise recall.
4. **Capacity is not the bottleneck — three winners are small.** The 602 and 619 winners are the *smallest* rung (152,891 params; 611KB checkpoint) and the 605 winner is 383k; only 620/623 favor the 4.15M reference. On 619 the 153k rung beats every larger sibling (0.38379 vs 0.372–0.377) — with the V31 timing bank the task is timing gating, which tiny hidden state handles, while larger models over-issue late.
5. **Calibration, not weights, is the fragile part.** The seven planned-but-excluded jobs are all `nearest_153k` strict-recipe rungs (Table 29), and the one replayed strict 153k on 602 (J07) collapses to 0.3803 IPC while the *same checkpoint* with a calibrated threshold scores 0.43066 (J03): a fixed 0.9 threshold starves the tiny model (41k issues vs 200k). The per-job sweeps (Table 54) document exactly which threshold each policy needed.
6. **High accuracy on both sides does not mean equal misses.** The 423-pair evidence table shows large `both_timely` cores with decisive one-sided margins (e.g. 623 winner vs SPP: 355,349 both, 46,349 NN-only, 8,869 normal-only), and the keyed miss deltas (Fig. f14) track IPC sign on every trace — validating the archive's own caution that outcome-level attribution, not accuracy, is the deciding evidence.
7. **Seed noise is an order of magnitude below the margins.** The only multi-seed cell (619 top-2 full policy, seeds 7/19/43) has IPC stdev 0.00107 (Table 77); the smallest reported head-to-head margin (619, +0.00274 IPC) is $\approx 2.6\sigma$, and 605/623 margins are $4.5\sigma/23\sigma$ equivalents.
8. **Where the NN loses, it loses to a specialist at its own game.** 602's sandbox wins by covering 0.965 of unique events at 1.92×-miss coverage with 3.8M issues; the NN matches its timeliness (1.000) and beats its accuracy 9.6× but leaves 8.6k events unselected. 620's SMS wins on raw coverage (0.31 vs NN 0.14) under the NN's deliberate 0.25 issue budget; the bank-recall ceiling above, not the budget, is the binding constraint (raising recall is the archive's own `next_focus` for both traces, Table 78).

Part II

Complete data record — every dataset in the archive

All 47 NN jobs are indexed by the IDs of Table 1; wide files are printed in column chunks with the ID repeated, and columns constant across all rows of a file are shown once above the chunks. Verified byte-identities and projections between files are stated in place so that nothing is double-printed or silently dropped.

5 Job index

Table 1: Job ID map: every NN training/replay job in the archive. All later tables use these IDs.

ID	full tag
J01	v4.9.5_605.mcf.s-994B__support16.fixed.nextline__strict__selected.full__seed7__one_shot.full.reference
J02	v4.9.5_605.mcf.s-994B__support16.fixed.nextline__strict__about.911k__seed7__one_shot.strict.size
J03	v4.9.5_605.mcf.s-994B__support16.fixed.nextline__strict__about.383k__seed7__one_shot.strict.size
J04	v4.9.5_605.mcf.s-994B__support16.fixed.nextline__strict__nearest.153k__seed7__one_shot.strict.size
J05	v4.9.5_605.mcf.s-994B__support16.fixed.nextline__calibration.control__selected.full__seed7__one_shot.calibration.control
J06	v4.9.5_605.mcf.s-994B__support16.fixed.nextline__calibration.control__about.911k__seed7__one_shot.calibration.control
J07	v4.9.5_605.mcf.s-994B__support16.fixed.nextline__calibration.control__about.383k__seed7__one_shot.calibration.control
J08	v4.9.5_605.mcf.s-994B__support16.fixed.nextline__calibration.control__nearest.153k__seed7__one_shot.calibration.control
J09	v4.9.5_623.xalancbm.s-700B__context.lru128__strict__selected.full__seed7__one_shot.full.reference
J10	v4.9.5_623.xalancbm.s-700B__context.lru128__strict__about.383k__seed7__one_shot.strict.size
J11	v4.9.5_623.xalancbm.s-700B__context.lru128__calibration.control__selected.full__seed7__one_shot.calibration.control
J12	v4.9.5_623.xalancbm.s-700B__context.lru128__calibration.control__about.911k__seed7__one_shot.calibration.control
J13	v4.9.5_623.xalancbm.s-700B__context.lru128__calibration.control__about.383k__seed7__one_shot.calibration.control
J14	v4.9.5_623.xalancbm.s-700B__context.lru128__calibration.control__nearest.153k__seed7__one_shot.calibration.control
J15	v4.9.5_602.gcc.s-734B__assoc.control__strict__selected.full__seed7__one_shot.full.reference
J16	v4.9.5_602.gcc.s-734B__assoc.control__strict__about.911k__seed7__one_shot.strict.size
J17	v4.9.5_602.gcc.s-734B__assoc.control__strict__about.383k__seed7__one_shot.strict.size
J18	v4.9.5_602.gcc.s-734B__assoc.control__strict__nearest.153k__seed7__one_shot.strict.size
J19	v4.9.5_602.gcc.s-734B__assoc.control__calibration.control__selected.full__seed7__one_shot.calibration.control
J20	v4.9.5_602.gcc.s-734B__assoc.control__calibration.control__about.911k__seed7__one_shot.calibration.control
J21	v4.9.5_602.gcc.s-734B__assoc.control__calibration.control__about.383k__seed7__one_shot.calibration.control
J22	v4.9.5_602.gcc.s-734B__assoc.control__calibration.control__nearest.153k__seed7__one_shot.calibration.control
J23	v4.9.5_620.omnetpp.s-874B__region.control__strict__selected.full__seed7__one_shot.full.reference
J24	v4.9.5_620.omnetpp.s-874B__region.control__strict__about.911k__seed7__one_shot.strict.size
J25	v4.9.5_620.omnetpp.s-874B__region.control__strict__about.383k__seed7__one_shot.strict.size
J26	v4.9.5_620.omnetpp.s-874B__region.control__calibration.control__selected.full__seed7__one_shot.calibration.control
J27	v4.9.5_620.omnetpp.s-874B__region.control__calibration.control__about.911k__seed7__one_shot.calibration.control
J28	v4.9.5_620.omnetpp.s-874B__region.control__calibration.control__about.383k__seed7__one_shot.calibration.control
J29	v4.9.5_620.omnetpp.s-874B__region.control__calibration.control__nearest.153k__seed7__one_shot.calibration.control
J30	v4.9.5_619.lbm.s-4268B__top2__strict__selected.full__seed7__one_shot.619.full.policy
J31	v4.9.5_619.lbm.s-4268B__top2__calibration.control__selected.full__seed7__one_shot.619.calibration.control
J32	v4.9.5_619.lbm.s-4268B__top3__strict__selected.full__seed7__one_shot.619.full.policy
J33	v4.9.5_619.lbm.s-4268B__top4__strict__selected.full__seed7__one_shot.619.full.policy
J34	v4.9.5_619.lbm.s-4268B__adaptive.score.gap__strict__selected.full__seed7__one_shot.619.full.policy
J35	v4.9.5_619.lbm.s-4268B__top2__strict__selected.full__seed19__one_shot.619.full.policy
J36	v4.9.5_619.lbm.s-4268B__top2__strict__selected.full__seed43__one_shot.619.full.policy
J37	v4.9.5_619.lbm.s-4268B__top2__strict__about.911k__seed7__one_shot.619.strict.size.policy
J38	v4.9.5_619.lbm.s-4268B__top2__calibration.control__about.911k__seed7__one_shot.619.calibration.control
J39	v4.9.5_619.lbm.s-4268B__top3__strict__about.911k__seed7__one_shot.619.strict.size.policy
J40	v4.9.5_619.lbm.s-4268B__top4__strict__about.911k__seed7__one_shot.619.strict.size.policy
J41	v4.9.5_619.lbm.s-4268B__adaptive.score.gap__strict__about.911k__seed7__one_shot.619.strict.size.policy
J42	v4.9.5_619.lbm.s-4268B__top2__strict__about.383k__seed7__one_shot.619.strict.size.policy
J43	v4.9.5_619.lbm.s-4268B__top2__calibration.control__about.383k__seed7__one_shot.619.calibration.control
J44	v4.9.5_619.lbm.s-4268B__top3__strict__about.383k__seed7__one_shot.619.strict.size.policy
J45	v4.9.5_619.lbm.s-4268B__top4__strict__about.383k__seed7__one_shot.619.strict.size.policy
J46	v4.9.5_619.lbm.s-4268B__adaptive.score.gap__strict__about.383k__seed7__one_shot.619.strict.size.policy
J47	v4.9.5_619.lbm.s-4268B__top2__calibration.control__nearest.153k__seed7__one_shot.619.calibration.control
J48	v4.9.5_623.xalancbm.s-700B__context.lru128__strict__about.911k__seed7__one_shot.strict.size
J49	v4.9.5_623.xalancbm.s-700B__context.lru128__strict__nearest.153k__seed7__one_shot.strict.size
J50	v4.9.5_620.omnetpp.s-874B__region.control__strict__nearest.153k__seed7__one_shot.strict.size
J51	v4.9.5_619.lbm.s-4268B__top2__strict__nearest.153k__seed7__one_shot.619.strict.size.policy
J52	v4.9.5_619.lbm.s-4268B__top3__strict__nearest.153k__seed7__one_shot.619.strict.size.policy
J53	v4.9.5_619.lbm.s-4268B__top4__strict__nearest.153k__seed7__one_shot.619.strict.size.policy
J54	v4.9.5_619.lbm.s-4268B__adaptive.score.gap__strict__nearest.153k__seed7__one_shot.619.strict.size.policy

6 Replay summaries

Table 2: Normal-prefetcher replay summary: columns constant across all rows (shown once)

column	constant value
nodup_mode	1
run_failed	0
fail_reason	-
log_missing	0

Table 3: replay_results/normal_summary_used.csv — all 45 normal-prefetcher replay rows (log paths in Appendix) — part 1/3

trace	<i>prefetcher</i>	<i>ipc</i>	<i>speedup-vs-no-pref</i>	<i>l2_loads</i>	<i>l2_load_miss</i>	<i>l2_load_miss_rate</i>	<i>miss_reduction-vs-no-pref</i>	<i>pf_requested</i>	<i>pf_dropped</i>
602.gcc.s-734B	no_pref	0.368	1	404,219	203,131	0.50253	0	0	0
602.gcc.s-734B	stride	0.37006	1.0056	404,219	168,277	0.4163	0.17158	267,553	0
602.gcc.s-734B	streamer	0.43161	1.17285	404,219	15,475	0.03828	0.92382	3,623,513	0
602.gcc.s-734B	ampm	0.43069	1.17035	404,219	17,032	0.04214	0.91615	2,846,050	0
602.gcc.s-734B	spp	0.42836	1.16402	404,219	70,873	0.17533	0.6511	4,491,857	0
602.gcc.s-734B	ipcp	0.36796	0.99989	404,219	203,131	0.50253	0	188	0
602.gcc.s-734B	sms	0.42213	1.14709	404,219	41,169	0.10185	0.79733	696,706	0
602.gcc.s-734B	sandbox	0.43628	1.18554	404,219	7,141	0.01767	0.96485	3,839,474	0
602.gcc.s-734B	power7	0.43075	1.17052	404,219	17,306	0.04281	0.9148	3,304,558	0
619.lbm.s-4268B	no_pref	0.322	1	295,037	294,639	0.99865	0	0	0
619.lbm.s-4268B	stride	0.32465	1.00823	295,026	227,228	0.7702	0.22879	350,381	0
619.lbm.s-4268B	streamer	0.36368	1.12944	295,032	152,322	0.51629	0.48302	524,800	0
619.lbm.s-4268B	ampm	0.3748	1.16398	295,027	58,689	0.19893	0.80081	2,008,610	0
619.lbm.s-4268B	spp	0.37933	1.17804	295,024	225,832	0.76547	0.23353	3,804,639	0
619.lbm.s-4268B	ipcp	0.322	1	295,039	294,643	0.99866	-1.358e-05	81	0
619.lbm.s-4268B	sms	0.38105	1.18339	295,028	38,705	0.13119	0.86864	568,017	0
619.lbm.s-4268B	sandbox	0.3744	1.16273	295,024	85,283	0.28907	0.71055	3,352,602	0
619.lbm.s-4268B	power7	0.36128	1.12199	295,027	134,096	0.45452	0.54488	615,642	0
605.mcf.s-994B	no_pref	0.18318	1	985,276	740,223	0.75128	0	0	0
605.mcf.s-994B	stride	0.18359	1.00224	985,274	735,395	0.74639	0.00652	25,125	0
605.mcf.s-994B	streamer	0.1835	1.00175	985,268	701,493	0.71198	0.05232	1,775,122	0
605.mcf.s-994B	ampm	0.18874	1.03035	985,273	687,090	0.69736	0.07178	509,840	0
605.mcf.s-994B	spp	0.18874	1.03035	985,273	735,406	0.7464	0.00651	710,459	0
605.mcf.s-994B	ipcp	0.18318	1	985,276	740,223	0.75128	0	88	0
605.mcf.s-994B	sms	0.18777	1.02506	985,269	682,187	0.69239	0.0784	1,287,553	60
605.mcf.s-994B	sandbox	0.16905	0.92286	985,257	714,230	0.72492	0.03512	15,930,213	428,375
605.mcf.s-994B	power7	0.18163	0.99154	985,267	715,851	0.72656	0.03293	1,466,157	0
620.omnetpp.s-874B	no_pref	0.2375	1	451,398	316,554	0.70127	0	0	0
620.omnetpp.s-874B	stride	0.23833	1.00349	451,396	313,694	0.69494	0.00903	19,579	0
620.omnetpp.s-874B	streamer	0.24228	1.02013	451,360	278,278	0.61653	0.12091	757,318	0
620.omnetpp.s-874B	ampm	0.23976	1.00952	451,359	300,054	0.66478	0.05212	310,419	0
620.omnetpp.s-874B	spp	0.23869	1.00501	451,389	316,347	0.70083	6.539e-04	174,696	0
620.omnetpp.s-874B	ipcp	0.23771	1.00088	451,406	315,134	0.69812	0.00449	29,016	0
620.omnetpp.s-874B	sms	0.24695	1.03979	451,284	246,242	0.54565	0.22212	808,086	0
620.omnetpp.s-874B	sandbox	0.24433	1.02876	451,253	237,594	0.52652	0.24944	7,809,601	186,257
620.omnetpp.s-874B	power7	0.24232	1.02029	451,341	278,931	0.61801	0.11885	730,714	0
623.xalancbmk.s-700B	no_pref	0.35321	1	1,218,494	450,350	0.3696	0	0	0
623.xalancbmk.s-700B	stride	0.3534	1.00054	1,218,495	448,287	0.3679	0.00458	299,845	0
623.xalancbmk.s-700B	streamer	0.33644	0.95252	1,218,352	730,241	0.59937	-0.6215	2,459,972	0
623.xalancbmk.s-700B	ampm	0.34924	0.98876	1,218,334	505,771	0.41513	-0.12306	924,845	0
623.xalancbmk.s-700B	spp	0.35391	1.00198	1,218,483	439,632	0.3608	0.0238	1,670,201	0
623.xalancbmk.s-700B	ipcp	0.35241	0.99774	1,218,487	461,190	0.37849	-0.02407	87,172	0
623.xalancbmk.s-700B	sms	0.34167	0.96733	1,218,827	656,622	0.53873	-0.45803	1,932,085	0
623.xalancbmk.s-700B	sandbox	0.34524	0.97744	1,218,507	600,003	0.49241	-0.3323	17,013,266	4,122,042
623.xalancbmk.s-700B	power7	0.3433	0.97194	1,218,439	615,468	0.50513	-0.36664	1,764,449	0

Table 4: replay_results/normal_summary_used.csv — all 45 normal-prefetcher replay rows (log paths in Appendix) — part 2/3

trace	<i>pf_issued</i>	<i>nodup_issued</i>	<i>pf_filled</i>	<i>pf_useful</i>	<i>pf_useless</i>	<i>pf_late</i>	<i>pq_merged_duplicate_proxy</i>	<i>accuracy</i>	<i>nodup_accuracy</i>
602.gcc.s-734B	0	0	0	0	0	0	0	0	0
602.gcc.s-734B	267,553	267,553	71,723	70,694	1,023	168	0	0.26422	0.26422
602.gcc.s-734B	3,623,513	3,620,275	392,083	375,582	16,394	49	3,238	0.10365	0.10374
602.gcc.s-734B	2,846,050	2,844,277	385,815	373,090	12,600	604	1,773	0.13109	0.13117
602.gcc.s-734B	4,491,857	4,485,769	266,257	265,086	1,162	387	6,088	0.05901	0.05909
602.gcc.s-734B	188	188	120	46	74	4	0	0.24468	0.24468
602.gcc.s-734B	696,706	696,706	343,138	319,593	23,409	200	0	0.45872	0.45872
602.gcc.s-734B	3,839,474	3,830,425	422,799	390,102	32,524	59	9,049	0.1016	0.10184
602.gcc.s-734B	3,304,558	3,162,983	386,269	372,751	13,449	128	141,575	0.1128	0.11785
619.lbm.s-4268B	0	0	0	0	0	0	0	0	0
619.lbm.s-4268B	350,381	350,375	150,816	150,814	0	15,702	6	0.43043	0.43044
619.lbm.s-4268B	524,800	524,667	402,791	401,794	990	112,603	133	0.76561	0.76581
619.lbm.s-4268B	2,008,610	1,517,402	496,725	496,363	348	25,272	491,208	0.24712	0.32711
619.lbm.s-4268B	3,804,639	2,314,384	190,672	190,662	10	65,229	1,490,255	0.05011	0.08238
619.lbm.s-4268B	81	81	10	7	3	1	0	0.08642	0.08642
619.lbm.s-4268B	568,017	568,017	535,220	535,130	84	26,099	0	0.9421	0.9421
619.lbm.s-4268B	3,352,602	2,335,419	489,356	488,051	1,296	77,378	1,017,183	0.14557	0.20898
619.lbm.s-4268B	615,642	599,811	369,761	368,891	862	76,926	15,831	0.5992	0.61501
605.mcf.s-994B	0	0	0	0	0	0	0	0	0
605.mcf.s-994B	25,125	25,121	7,762	7,595	167	1	4	0.30229	0.30234
605.mcf.s-994B	1,775,122	1,687,533	842,607	154,695	686,999	18,210	87,589	0.08715	0.09167
605.mcf.s-994B	509,840	508,759	149,168	90,495	58,586	1,305	1,081	0.1775	0.17787
605.mcf.s-994B	710,459	705,508	19,585	14,384	5,178	1,257	4,951	0.02025	0.02039
605.mcf.s-994B	88	88	42	26	16	16	0	0.29545	0.29545
605.mcf.s-994B	1,287,493	1,287,493	893,091	256,319	635,113	61,126	0	0.19908	0.19908
605.mcf.s-994B	15,501,838	13,238,296	7,641,216	449,411	7,188,334	139,527	2,263,542	0.02899	0.03395
605.mcf.s-994B	1,466,157	1,386,191	687,520	129,697	556,672	17,755	79,966	0.08846	0.09356
620.omnetpp.s-874B	0	0	0	0	0	0	0	0	0
620.omnetpp.s-874B	19,579	19,578	7,781	5,928	1,849	0	1	0.30277	0.30279
620.omnetpp.s-874B	757,318	659,178	397,943	110,831	286,074	2,538	98,140	0.14635	0.16814
620.omnetpp.s-874B	310,419	289,753	143,933	45,147	98,338	1,868	20,666	0.14544	0.15581
620.omnetpp.s-874B	174,696	171,086	127	74	53	34	3,610	4.236e-04	4.325e-04
620.omnetpp.s-874B	29,016	29,016	14,872	5,374	9,433	374	0	0.18521	0.18521
620.omnetpp.s-874B	808,086	808,086	593,171	197,737	394,038	3,655	0	0.2447	0.2447
620.omnetpp.s-874B	7,623,344	6,663,110	3,656,950	373,815	3,279,950	10,848	960,234	0.04904	0.0561
620.omnetpp.s-874B	730,714	633,770	378,404	105,531	271,622	2,751	96,944	0.14442	0.16651
623.xalancbmk.s-700B	0	0	0	0	0	0	0	0	0
623.xalancbmk.s-700B	299,845	268,834	19,224	18,849	373	7,498	31,011	0.06286	0.07011
623.xalancbmk.s-700B	2,459,972	2,136,580	828,293	111,140	715,938	6,770	323,392	0.04518	0.05202
623.xalancbmk.s-700B	924,845	839,364	139,608	52,884	86,456	3,276	85,481	0.05718	0.063
623.xalancbmk.s-700B	1,670,201	1,612,741	30,949	7,816	23,108	1,870	57,460	0.00468	0.00485
623.xalancbmk.s-700B	87,172	87,172	16,769	5,863	10,856	118	0	0.06726	0.06726
623.xalancbmk.s-700B	1,932,085	1,932,085	1,057,158	291,589	764,626	11,070	0	0.15092	0.15092
623.xalancbmk.s-700B	12,891,224	10,755,534	5,030,567	536,632	4,491,270	23,596	2,135,690	0.04163	0.04989
623.xalancbmk.s-700B	1,764,449	1,503,898	560,701	82,274	477,415	7,460	260,551	0.04663	0.05471

Table 5: replay_results/normal_summary_used.csv — all 45 normal-prefetcher replay rows (log paths in Appendix) — part 3/3

trace	<i>selected_accuracy</i>	<i>coverage_vs_no_pref_l2_miss</i>	<i>useful_per_l2_miss_self</i>	<i>timeliness</i>	<i>late_per_issued</i>	<i>drop_rate</i>	<i>useless_per_issued</i>
602.gcc.s-734B	0	0	0	0	0	0	0
602.gcc.s-734B	0.26422	0.34802	0.4201	0.99763	6.279e-04	0	0.00382
602.gcc.s-734B	0.10374	1.84896	24.27024	0.99987	1.352e-05	0	0.00452
602.gcc.s-734B	0.13117	1.8367	21.90524	0.99838	2.122e-04	0	0.00443

Table 5: replay_results/normal_summary_used.csv — all 45 normal-prefetcher replay rows (log paths in Appendix) — part 3/3 (cont.)

trace	<i>selected-accuracy</i>	<i>coverage-vs-no-pref-l2-miss</i>	<i>useful-per-l2-miss-self</i>	<i>timeliness</i>	<i>late-per-issued</i>	<i>drop-rate</i>	<i>useless-per-issued</i>
602.gcc.s-734B	0.05909	1.305	3.7403	0.99854	8.616e-05	0	2.587e-04
602.gcc.s-734B	0.24468	2.265e-04	2.265e-04	0.92	0.02128	0	0.39362
602.gcc.s-734B	0.45872	1.57333	7.76295	0.99937	2.871e-04	0	0.0336
602.gcc.s-734B	0.10184	1.92045	54.62848	0.99985	1.537e-05	0	0.00847
602.gcc.s-734B	0.11785	1.83503	21.53883	0.99966	3.873e-05	0	0.00407
619.lbm.s-4268B	0	0	0	0	0	0	0
619.lbm.s-4268B	0.43044	0.51186	0.66371	0.9057	0.04481	0	0
619.lbm.s-4268B	0.76581	1.36368	2.63779	0.7811	0.21456	0	0.00189
619.lbm.s-4268B	0.32711	1.68465	8.45751	0.95155	0.01258	0	1.733e-04
619.lbm.s-4268B	0.08238	0.6471	0.84426	0.74509	0.01714	0	2.628e-06
619.lbm.s-4268B	0.08642	2.376e-05	2.376e-05	0.875	0.01235	0	0.03704
619.lbm.s-4268B	0.9421	1.81622	13.82586	0.9535	0.04595	0	1.479e-04
619.lbm.s-4268B	0.20898	1.65644	5.72272	0.86315	0.02308	0	3.866e-04
619.lbm.s-4268B	0.61501	1.25201	2.75095	0.82745	0.12495	0	0.0014
605.mcf.s-994B	0	0	0	0	0	0	0
605.mcf.s-994B	0.30234	0.01026	0.01033	0.99987	3.980e-05	0	0.00665
605.mcf.s-994B	0.09167	0.20898	0.22052	0.89468	0.01026	0	0.38702
605.mcf.s-994B	0.17787	0.12225	0.13171	0.98578	0.00256	0	0.11491
605.mcf.s-994B	0.02039	0.01943	0.01956	0.91963	0.00177	0	0.00729
605.mcf.s-994B	0.29545	3.512e-05	3.512e-05	0.61905	0.18182	0	0.18182
605.mcf.s-994B	0.19908	0.34627	0.37573	0.80744	0.04748	4.660e-05	0.49329
605.mcf.s-994B	0.03395	0.60713	0.62922	0.76309	0.009	0.02689	0.46371
605.mcf.s-994B	0.09356	0.17521	0.18118	0.87959	0.01211	0	0.37968
620.omnetpp.s-874B	0	0	0	0	0	0	0
620.omnetpp.s-874B	0.30279	0.01873	0.0189	1	0	0	0.09444
620.omnetpp.s-874B	0.16814	0.35012	0.39827	0.97761	0.00335	0	0.37775
620.omnetpp.s-874B	0.15581	0.14262	0.15046	0.96027	0.00602	0	0.31679
620.omnetpp.s-874B	4.325e-04	2.338e-04	2.339e-04	0.68519	1.946e-04	0	3.034e-04
620.omnetpp.s-874B	0.18521	0.01698	0.01705	0.93493	0.01289	0	0.3251
620.omnetpp.s-874B	0.2447	0.62465	0.80302	0.98185	0.00452	0	0.48762
620.omnetpp.s-874B	0.0561	1.18089	1.57334	0.9718	0.00142	0.02385	0.43025
620.omnetpp.s-874B	0.16651	0.33337	0.37834	0.97459	0.00376	0	0.37172
623.xalancbmk.s-700B	0	0	0	0	0	0	0
623.xalancbmk.s-700B	0.07011	0.04185	0.04205	0.71541	0.02501	0	0.00124
623.xalancbmk.s-700B	0.05202	0.24679	0.1522	0.94258	0.00275	0	0.29104
623.xalancbmk.s-700B	0.063	0.11743	0.10456	0.94167	0.00354	0	0.09348
623.xalancbmk.s-700B	0.00485	0.01736	0.01778	0.80694	0.00112	0	0.01384
623.xalancbmk.s-700B	0.06726	0.01302	0.01271	0.98027	0.00135	0	0.12454
623.xalancbmk.s-700B	0.15092	0.64747	0.44407	0.96342	0.00573	0	0.39575
623.xalancbmk.s-700B	0.04989	1.19159	0.89438	0.95788	0.00183	0.24228	0.3484
623.xalancbmk.s-700B	0.05471	0.18269	0.13368	0.91687	0.00423	0	0.27057

Table 6: NN (LSTM) replay summary: columns constant across all rows (shown once)

column	constant value
architecture_family	two_layer_trace_local_candidate_lstm
architecture_id	candidate_lstm_ranker_v49_5
bank_slots	64
candidate_bank_storage_bytes	0
candidate_index_bytes_estimate	0
candidate_index_entries	0
candidate_rate_limit_filtered_count	0
candidate_table_array_bytes	0
drop_rate	0
fail_reason	—
fallback_emitted_rows	0
fallback_invalid_rows	0
fallback_profile_table_bytes	0
fallback_suppressed_dedup_rows	0
fallback_suppressed_rate_limit_rows	0
implementation_revision	v4.9.5_one_shot_train_export_replay_matrix
job_status	ok
keyed_dropped_invalid_address	0

column	value
keyed_mapped.cycle_pc.line.rows	0
keyed_unmatched.rows	0
layer_count	2
ledger_error	–
list_replayer_instantiated	1
log_missing	0
nodup_mode	1
one_shot_phase	1
performance_gate.status	pending_keyed_replay
pf_dropped	0
pre_replay_matrix	1
prefetcher	standalone_lstm
replay_eligible	1
replay_layout	event_attribution_layout
replay_transport.ok	1
run_failed	0
selected_target.late.proxy.count	0
sequence_len	1,024
train_fraction	0.8
zero_action.export	0

Table 7: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 1/9

ID	accuracy	bank_id	best_normal	best_normal_ipc	best_val_loss	calibration_anchor_issue_per_event	calibration_anchor_source	calibration_candidate_count	calibration_events	calibration_measured_issue_per_event
J21	0.97942	assoc.v40	sandbox	0.43628	0.15911	0.46871	fresh_selected_full_strict_metadata	14	40,422	0.46784
J20	0.97648	assoc.v40	sandbox	0.43628	0.14666	0.46871	fresh_selected_full_strict_metadata	14	40,422	0.47291
J22	0.95211	assoc.v40	sandbox	0.43628	0.45777	0.46871	fresh_selected_full_strict_metadata	6	40,422	0.49468
J19	0.98708	assoc.v40	sandbox	0.43628	0.10358	0.46871	fresh_selected_full_strict_metadata	14	40,422	0.47393
J15	0.98708	assoc.v40	sandbox	0.43628	0.10358	–	strict_frozen_policy	–	40,422	–
J17	0.97942	assoc.v40	sandbox	0.43628	0.15911	–	strict_frozen_policy	–	40,422	–
J16	0.97648	assoc.v40	sandbox	0.43628	0.14666	–	strict_frozen_policy	–	40,422	–
J18	0.97816	assoc.v40	sandbox	0.43628	0.45777	–	strict_frozen_policy	–	40,422	–
J07	0.24505	support16_fixed_nextline	ampm	0.18874	1.13896	0.58491	fresh_selected_full_strict_metadata	14	98,528	0.57688
J06	0.21887	support16_fixed_nextline	ampm	0.18874	1.12211	0.58491	fresh_selected_full_strict_metadata	14	98,528	0.67382
J08	0.23496	support16_fixed_nextline	ampm	0.18874	1.47089	0.58491	fresh_selected_full_strict_metadata	14	98,528	0.72308
J05	0.19112	support16_fixed_nextline	ampm	0.18874	1.35867	0.58491	fresh_selected_full_strict_metadata	14	98,528	0.67166
J01	0.2338	support16_fixed_nextline	ampm	0.18874	1.35867	–	strict_frozen_policy	–	98,528	–
J03	0.3214	support16_fixed_nextline	ampm	0.18874	1.13896	–	strict_frozen_policy	–	98,528	–
J02	0.31676	support16_fixed_nextline	ampm	0.18874	1.12211	–	strict_frozen_policy	–	98,528	–
J04	0.37373	support16_fixed_nextline	ampm	0.18874	1.47089	–	strict_frozen_policy	–	98,528	–
J43	0.47696	v31_timing	sms	0.38105	0.19444	1.63299	fresh_selected_full_strict_metadata	14	29,504	1.56101
J38	0.46669	v31_timing	sms	0.38105	0.18678	1.63299	fresh_selected_full_strict_metadata	14	29,504	1.65615
J47	0.62177	v31_timing	sms	0.38105	0.50708	1.63299	fresh_selected_full_strict_metadata	14	29,504	1.59151
J31	0.48703	v31_timing	sms	0.38105	0.17532	1.63299	fresh_selected_full_strict_metadata	14	29,504	1.56718
J34	0.28693	v31_timing	sms	0.38105	0.17532	–	strict_frozen_policy	–	29,504	–
J30	0.48703	v31_timing	sms	0.38105	0.17532	–	strict_frozen_policy	–	29,504	–
J35	0.48573	v31_timing	sms	0.38105	0.17681	–	strict_frozen_policy	–	29,504	–
J36	0.47856	v31_timing	sms	0.38105	0.17511	–	strict_frozen_policy	–	29,504	–
J32	0.36266	v31_timing	sms	0.38105	0.17532	–	strict_frozen_policy	–	29,504	–
J33	0.28693	v31_timing	sms	0.38105	0.17532	–	strict_frozen_policy	–	29,504	–
J46	0.30247	v31_timing	sms	0.38105	0.19444	–	strict_frozen_policy	–	29,504	–
J41	0.28777	v31_timing	sms	0.38105	0.18678	–	strict_frozen_policy	–	29,504	–
J42	0.47696	v31_timing	sms	0.38105	0.19444	–	strict_frozen_policy	–	29,504	–
J37	0.46669	v31_timing	sms	0.38105	0.18678	–	strict_frozen_policy	–	29,504	–
J44	0.36734	v31_timing	sms	0.38105	0.19444	–	strict_frozen_policy	–	29,504	–
J39	0.35451	v31_timing	sms	0.38105	0.18678	–	strict_frozen_policy	–	29,504	–
J45	0.30247	v31_timing	sms	0.38105	0.19444	–	strict_frozen_policy	–	29,504	–
J40	0.28777	v31_timing	sms	0.38105	0.18678	–	strict_frozen_policy	–	29,504	–
J28	0.58513	region_pair_residual_pair	sms	0.24695	0.81741	0.12661	fresh_selected_full_strict_metadata	14	45,140	0.14541
J27	0.5891	region_pair_residual_pair	sms	0.24695	0.78336	0.12661	fresh_selected_full_strict_metadata	14	45,140	0.1749
J29	0.30192	region_pair_residual_pair	sms	0.24695	1.26069	0.12661	fresh_selected_full_strict_metadata	14	45,140	0.21777

Table 7: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 1/9 (cont.)

ID	accuracy	bank_id	best_normal	best_normal_ipc	best_val_loss	calibration_anchor_issue-per-event		calibration_anchor_source	calibration_candidate-count	calibration-events	calibration_measured-issue-per-event	cali
J26	0.80988	region_pair_residual_pair	sms	0.24695	0.6926	0.12661	fresh_selected_full_strict_metadata	14	45,140	0.11845		
J23	0.80988	region_pair_residual_pair	sms	0.24695	0.6926	—	strict_frozen_policy	—	45,140	—		
J25	0.84886	region_pair_residual_pair	sms	0.24695	0.81741	—	strict_frozen_policy	—	45,140	—		
J24	0.8511	region_pair_residual_pair	sms	0.24695	0.78336	—	strict_frozen_policy	—	45,140	—		
J13	0.51368	context_v40	spp	0.35391	0.69583	0.39596	fresh_selected_full_strict_metadata	14	121,849	0.37662		
J12	0.4123	context_v40	spp	0.35391	0.63947	0.39596	fresh_selected_full_strict_metadata	14	121,849	0.34722		
J14	0.231	context_v40	spp	0.35391	1.31471	0.39596	fresh_selected_full_strict_metadata	14	121,849	0.58622		
J11	0.77952	context_v40	spp	0.35391	0.31902	0.39596	fresh_selected_full_strict_metadata	14	121,849	0.39832		
J09	0.82283	context_v40	spp	0.35391	0.31902	—	strict_frozen_policy	—	121,849	—		
J10	0.73031	context_v40	spp	0.35391	0.69583	—	strict_frozen_policy	—	121,849	—		

Table 8: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, paths in Appendix) — part 2/9

ID	calibration_policy_state-scope	calibration-split	calibration-stage	calibration-target_issue-per-event	calibration-total-suffix-events	candidate-absent-count	candidate-absent-pct	candidate-dedup-filtered-count
J21	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.46871	40,422	198,666	0.49149	65,679
J20	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.46871	40,422	198,666	0.49149	59,599
J22	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.46871	40,422	198,666	0.49149	64,003
J19	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.46871	40,422	198,666	0.49149	39,083
J15	—	none	not_used	—	0	198,666	0.49149	39,083
J17	—	none	not_used	—	0	198,666	0.49149	65,679
J16	—	none	not_used	—	0	198,666	0.49149	59,599
J18	—	none	not_used	—	0	198,666	0.49149	0
J07	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.58491	98,528	386,535	0.43883	223,091
J06	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.58491	98,528	386,535	0.43883	221,191
J08	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.58491	98,528	386,535	0.43883	184,488
J05	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.58491	98,528	386,535	0.43883	224,409
J01	—	none	not_used	—	0	386,535	0.43883	221,568
J03	—	none	not_used	—	0	386,535	0.43883	216,729
J02	—	none	not_used	—	0	386,535	0.43883	206,825
J04	—	none	not_used	—	0	386,535	0.43883	102,668
J43	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	1.63299	29,504	950	0.00322	0
J38	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	1.63299	29,504	950	0.00322	0
J47	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	1.63299	29,504	950	0.00322	0
J31	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	1.63299	29,504	950	0.00322	0
J34	—	none	not_used	—	0	950	0.00322	0
J30	—	none	not_used	—	0	950	0.00322	0
J35	—	none	not_used	—	0	950	0.00322	0
J36	—	none	not_used	—	0	950	0.00322	0
J32	—	none	not_used	—	0	950	0.00322	0
J33	—	none	not_used	—	0	950	0.00322	0
J46	—	none	not_used	—	0	950	0.00322	0
J41	—	none	not_used	—	0	950	0.00322	0
J42	—	none	not_used	—	0	950	0.00322	0
J37	—	none	not_used	—	0	950	0.00322	0
J44	—	none	not_used	—	0	950	0.00322	0

Table 8: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, paths in Appendix) — part 2/9 (cont.)

ID	calibration_policy_state_scope	calibration_split	calibration_stage	calibration_target_issue_per_event	calibration_total_suffix_events	candidate_absent_count	candidate_absent_pct	candidate_dedup_filtered_count
J39	—	none	not_used	—	0	950	0.00322	0
J45	—	none	not_used	—	0	950	0.00322	0
J40	—	none	not_used	—	0	950	0.00322	0
J28	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.12661	45,140	259,864	0.57569	16,770
J27	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.12661	45,140	259,864	0.57569	19,053
J29	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.12661	45,140	259,864	0.57569	18,632
J26	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.12661	45,140	259,864	0.57569	13,484
J23	—	none	not_used	—	0	259,864	0.57569	13,484
J25	—	none	not_used	—	0	259,864	0.57569	5,421
J24	—	none	not_used	—	0	259,864	0.57569	5,913
J13	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.39596	121,849	85,499	0.07245	271,836
J12	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.39596	121,849	85,499	0.07245	179,387
J14	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.39596	121,849	85,499	0.07245	93,419
J11	reset_at_calibration_boundary	validation_calibration_prefix	calibrated_scalar_threshold	0.39596	121,849	85,499	0.07245	599,953
J09	—	none	not_used	—	0	85,499	0.07245	591,250
J10	—	none	not_used	—	0	85,499	0.07245	30,581

Table 9: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 3/9

ID	candidate_threshold_filtered_count	checkpoint_bytes	continuous_projection_dim	correct_rank_recall_top1	correct_rank_recall_top2	correct_rank_recall_top4	cycles	embedding_dim	eval_issue_per_event	eval_selected_accuracy	eval_selected_timely_proxy
J21	24,316	1,544,132	4	0.47091	0.48049	0.48772	58,363,793	4	0.4694	0.95573	0.95573
J20	22,159	3,658,820	8	0.47337	0.48161	0.49048	58,336,856	10	0.47472	0.9519	0.9519
J22	18,260	625,604	1	0.45802	0.4682	0.47544	58,049,894	1	0.49354	0.81348	0.81348
J19	18,847	16,605,316	24	0.47902	0.48467	0.49182	58,242,690	40	0.47939	0.95789	0.95789
J15	18,847	16,605,316	24	0.47902	0.48467	0.49182	58,242,690	40	0.47635	0.96175	0.96175
J17	24,316	1,544,132	4	0.47091	0.48049	0.48772	58,363,793	4	0.46848	0.95847	0.95847
J16	22,159	3,658,820	8	0.47337	0.48161	0.49048	58,336,856	10	0.47362	0.95398	0.95398
J18	166,384	625,604	1	0.45802	0.4682	0.47544	65,737,676	1	0.09393	0.96115	0.96115
J07	64,098	1,544,132	4	0.15086	0.24367	0.36318	129,136,700	4	0.53178	0.03323	0.03323
J06	46,858	3,658,820	8	0.15072	0.24306	0.36264	129,364,635	10	0.64448	0.02591	0.02591
J08	86,873	625,604	1	0.15516	0.24426	0.35841	129,811,199	1	0.71113	0.02229	0.02229
J05	63,014	16,605,316	24	0.13896	0.22808	0.34574	130,393,869	40	0.64349	0.0233	0.0233
J01	79,474	16,605,316	24	0.13896	0.22808	0.34574	132,030,776	40	0.00221	0	0
J03	86,305	1,544,132	4	0.15086	0.24367	0.36318	129,816,388	4	0.06986	0.08332	0.08332
J02	89,878	3,658,820	8	0.15072	0.24306	0.36264	129,976,864	10	0.16752	0.0867	0.0867
J04	210,123	625,604	1	0.15516	0.24426	0.35841	131,807,191	1	2.030e-05	0	0
J43	44,423	1,544,132	4	0.99339	0.99519	0.99595	67,708,850	4	1.65818	0.99775	0.99775
J38	32,369	3,658,820	8	0.99352	0.99556	0.99597	67,565,075	10	1.73953	0.99809	0.99809
J47	6,155	625,604	1	0.96542	0.98665	0.99146	65,139,558	1	1.59375	0.9705	0.9705
J31	44,182	16,605,316	24	0.9952	0.99601	0.99636	67,124,103	40	1.66469	0.99914	0.99914
J34	44,182	16,605,316	24	0.9952	0.99601	0.99636	66,568,079	40	2.91569	0.99918	0.99918
J30	44,182	16,605,316	24	0.9952	0.99601	0.99636	67,124,103	40	1.61593	0.9993	0.9993
J35	47,284	16,605,316	24	0.99539	0.99609	0.99639	67,053,705	40	1.59016	0.99924	0.99924
J36	48,989	16,605,316	24	0.99539	0.99612	0.99635	67,496,477	40	1.58245	0.99919	0.99919
J32	44,182	16,605,316	24	0.9952	0.99601	0.99636	66,737,813	40	2.28152	0.99935	0.99935
J33	44,182	16,605,316	24	0.9952	0.99601	0.99636	66,568,079	40	2.91569	0.99918	0.99918

Table 9: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 3/9 (cont.)

ID	candidate_threshold_filtered_count	checkpoint_bytes	continuous_projection_dim	correct_rank_recall_top1	correct_rank_recall_top2	correct_rank_recall_top4	cycles	embedding_dim	eval_issue_per_event	eval_selected_accuracy	eval_selected_timely_proxy
J46	44,423	1,544,132	4	0.99339	0.99519	0.99595	66,579,137	4	2.80281	0.99805	0.99805
J41	32,369	3,658,820	8	0.99352	0.99556	0.99597	66,347,037	10	3.02039	0.99729	0.99729
J42	44,423	1,544,132	4	0.99339	0.99519	0.99595	67,708,850	4	1.6096	0.99797	0.99797
J37	32,369	3,658,820	8	0.99352	0.99556	0.99597	67,565,075	10	1.69784	0.9974	0.9974
J44	44,423	1,544,132	4	0.99339	0.99519	0.99595	66,776,567	4	2.26207	0.99788	0.99788
J39	32,369	3,658,820	8	0.99352	0.99556	0.99597	66,394,056	10	2.41688	0.99716	0.99716
J45	44,423	1,544,132	4	0.99339	0.99519	0.99595	66,579,137	4	2.80281	0.99805	0.99805
J40	32,369	3,658,820	8	0.99352	0.99556	0.99597	66,347,037	10	3.02039	0.99729	0.99729
J28	137,595	1,544,132	4	0.11133	0.16775	0.23775	103,712,108	4	0.14229	0.31356	0.31356
J27	130,486	3,658,820	8	0.12112	0.17422	0.24312	103,642,650	10	0.1751	0.30516	0.30516
J29	136,848	625,604	1	0.07313	0.12421	0.20403	104,375,640	1	0.21887	0.14211	0.14211
J26	139,006	16,605,316	24	0.15473	0.20531	0.26964	103,325,709	40	0.1218	0.54674	0.54674
J23	139,006	16,605,316	24	0.15473	0.20531	0.26964	103,325,709	40	0.12013	0.5538	0.5538
J25	166,825	1,544,132	4	0.11133	0.16775	0.23775	104,326,737	4	0.04901	0.60588	0.60588
J24	166,721	3,658,820	8	0.12112	0.17422	0.24312	104,344,151	10	0.04999	0.62087	0.62087
J13	596,813	1,544,132	4	0.36947	0.53083	0.69269	70,011,496	4	0.35494	0.38881	0.38881
J12	705,723	3,658,820	8	0.36709	0.5325	0.70156	71,170,857	10	0.32757	0.32197	0.32197
J14	750,357	625,604	1	0.42435	0.51368	0.63446	73,209,519	1	0.55963	0.14168	0.14168
J11	138,708	16,605,316	24	0.81544	0.83443	0.8523	65,937,378	40	0.37995	0.64168	0.64168
J09	147,667	16,605,316	24	0.81544	0.83443	0.8523	65,994,870	40	0.34311	0.72585	0.72585
J10	982,569	1,544,132	4	0.36947	0.53083	0.69269	69,974,084	4	0.0772	0.64973	0.64973

Table 10: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 4/9

ID	evaluation_events	experiment_lane	exported_list_bytes	exported_rows	hidden_dim	instructions	ipc	ipc_delta_vs_best_normal	issue_per_event	keyed_direct_index_rows	keyed_direct_index_rows_verified_pc_line
J21	40,422	B_representation_coverage_repair	34,873,796	188,865	40	25,000,003	0.42835	-0.00793	0.46723	188,865	188,865
J20	40,422	B_representation_coverage_repair	35,075,947	189,926	40	25,000,003	0.42855	-0.00773	0.46986	189,926	189,926
J22	40,422	B_representation_coverage_repair	37,438,601	199,519	4	25,000,003	0.43066	-0.00562	0.49359	199,519	199,519
J19	40,422	B_representation_coverage_repair	34,643,413	189,461	160	25,000,003	0.42924	-0.00704	0.46871	189,461	189,461
J15	80,844	B_representation_coverage_repair	34,643,413	189,461	160	25,000,003	0.42924	-0.00704	0.46871	189,461	189,461
J17	80,844	B_representation_coverage_repair	34,873,796	188,865	40	25,000,003	0.42835	-0.00793	0.46723	188,865	188,865
J16	80,844	B_representation_coverage_repair	35,075,947	189,926	40	25,000,003	0.42855	-0.00773	0.46986	189,926	189,926
J18	80,844	B_representation_coverage_repair	7,630,295	41,212	4	25,000,003	0.3803	-0.05598	0.10195	41,212	41,212
J07	98,528	A_reproduce_then_strict_compress	151,698,974	703,571	40	25,000,000	0.19359	0.00485	0.71409	703,571	703,571
J06	98,528	A_reproduce_then_strict_compress	170,946,726	793,436	40	25,000,000	0.19325	0.00451	0.80529	793,436	793,436
J08	98,528	A_reproduce_then_strict_compress	153,470,356	712,142	4	25,000,000	0.19259	0.00385	0.72278	712,142	712,142
J05	98,528	A_reproduce_then_strict_compress	179,758,308	824,198	160	25,000,000	0.19173	0.00299	0.83651	824,198	824,198
J01	197,056	A_reproduce_then_strict_compress	127,612,219	576,298	160	25,000,000	0.18935	6.100e-04	0.58491	576,298	576,298
J03	197,056	A_reproduce_then_strict_compress	108,273,649	496,321	40	25,000,000	0.19258	0.00384	0.50374	496,321	496,321
J02	197,056	A_reproduce_then_strict_compress	108,949,608	500,058	40	25,000,000	0.19234	0.0036	0.50753	500,058	500,058
J04	197,056	A_reproduce_then_strict_compress	70,912,527	325,300	4	25,000,000	0.18967	9.300e-04	0.33016	325,300	325,300
J43	29,504	C_fixed_score_timing_policy	91,046,602	479,746	40	25,000,003	0.36923	-0.01182	1.62605	479,746	479,746
J38	29,504	C_fixed_score_timing_policy	96,203,648	507,607	40	25,000,003	0.37001	-0.01104	1.72049	507,607	507,607
J47	29,504	C_fixed_score_timing_policy	87,572,266	470,007	4	25,000,003	0.38379	0.00274	1.59304	470,007	470,007

Table 10: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 4/9 (cont.)

ID	evaluation_events	experiment_lane	exported_list_bytes	exported_rows	hidden_dim	instructions	ipc	ipc_delta-vs_best_normal	issue_per_event	keyed_direct_index_rows	keyed_direct_index_rows_verified_pc_line
J31	29,504	C.fixed_score.timing_policy	89,657,152	481,792	160	25,000,003	0.37244	-0.00861	1.63299	481,792	481,792
J34	59,008	C.fixed_score.timing_policy	167,603,438	871,355	160	25,000,003	0.37556	-0.00549	2.95338	871,355	871,355
J30	59,008	C.fixed_score.timing_policy	89,657,152	481,792	160	25,000,003	0.37244	-0.00861	1.63299	481,792	481,792
J35	59,008	C.fixed_score.timing_policy	89,565,832	474,738	160	25,000,003	0.37284	-0.00821	1.60908	474,738	474,738
J36	59,008	C.fixed_score.timing_policy	89,252,399	473,634	160	25,000,003	0.37039	-0.01066	1.60534	473,634	473,634
J32	59,008	C.fixed_score.timing_policy	126,876,970	680,908	160	25,000,003	0.3746	-0.00645	2.30787	680,908	680,908
J33	59,008	C.fixed_score.timing_policy	162,375,308	871,355	160	25,000,003	0.37556	-0.00549	2.95338	871,355	871,355
J46	59,008	C.fixed_score.timing_policy	162,640,415	835,216	40	25,000,003	0.37549	-0.00556	2.83089	835,216	835,216
J41	59,008	C.fixed_score.timing_policy	176,522,485	906,048	40	25,000,003	0.37681	-0.00424	3.07096	906,048	906,048
J42	59,008	C.fixed_score.timing_policy	91,046,602	479,746	40	25,000,003	0.36923	-0.01182	1.62605	479,746	479,746
J37	59,008	C.fixed_score.timing_policy	96,203,648	507,607	40	25,000,003	0.37001	-0.01104	1.72049	507,607	507,607
J44	59,008	C.fixed_score.timing_policy	127,565,258	674,417	40	25,000,003	0.37438	-0.00667	2.28587	674,417	674,417
J39	59,008	C.fixed_score.timing_policy	136,952,219	723,587	40	25,000,003	0.37654	-0.00451	2.45253	723,587	723,587
J45	59,008	C.fixed_score.timing_policy	157,629,119	835,216	40	25,000,003	0.37549	-0.00556	2.83089	835,216	835,216
J40	59,008	C.fixed_score.timing_policy	171,086,197	906,048	40	25,000,003	0.37681	-0.00424	3.07096	906,048	906,048
J28	45,140	B.representation.coverage.repair	14,385,979	70,187	40	25,000,001	0.24105	-0.0059	0.15549	70,187	70,187
J27	45,140	B.representation.coverage.repair	17,208,736	83,788	40	25,000,001	0.24121	-0.00574	0.18562	83,788	83,788
J29	45,140	B.representation.coverage.repair	21,018,122	102,458	4	25,000,001	0.23952	-0.00743	0.22698	102,458	102,458
J26	45,140	B.representation.coverage.repair	11,570,139	57,152	160	25,000,001	0.24195	-0.005	0.12661	57,152	57,152
J23	90,280	B.representation.coverage.repair	11,570,139	57,152	160	25,000,001	0.24195	-0.005	0.12661	57,152	57,152
J25	90,280	B.representation.coverage.repair	4,969,201	24,752	40	25,000,001	0.23963	-0.00732	0.05483	24,752	24,752
J24	90,280	B.representation.coverage.repair	4,976,185	24,823	40	25,000,001	0.23959	-0.00736	0.05499	24,823	24,823
J13	121,850	A.reproduce.then_strict_compress	110,952,042	517,067	40	25,000,000	0.35708	0.00317	0.42435	517,067	517,067
J12	121,850	A.reproduce.then_strict_compress	103,811,658	480,655	40	25,000,000	0.35127	-0.00264	0.39447	480,655	480,655
J14	121,850	A.reproduce.then_strict_compress	171,184,755	796,561	4	25,000,000	0.34149	-0.01242	0.65373	796,561	796,561
J11	121,850	A.reproduce.then_strict_compress	113,045,533	535,537	160	25,000,000	0.37915	0.02524	0.43951	535,537	535,537
J09	243,699	A.reproduce.then_strict_compress	101,475,908	482,477	160	25,000,000	0.37882	0.02491	0.39596	482,477	482,477
J10	243,699	A.reproduce.then_strict_compress	23,753,070	112,484	40	25,000,000	0.35728	0.00337	0.09231	112,484	112,484

Table 11: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 5/9

ID	keyed_entries	keyed_trigger_coverage	keyed_unique_trigger_keys	l2_load_hit	l2_load_miss	l2_load_miss_rate	l2_loads	late_per_issued	list_loaded_entries	list_loaded_trigger_keys	model_class
J21	188,865	1	120,319	386,065	18,154	0.04491	404,219	5.295e-06	188,865	120,319	GCC602Prefetcher
J20	189,926	1	128,864	386,546	17,673	0.04372	404,219	5.265e-06	189,926	128,864	GCC602Prefetcher
J22	199,519	1	146,705	391,051	13,168	0.03258	404,219	1.002e-05	199,519	146,705	GCC602Prefetcher
J19	189,461	1	151,478	388,101	16,118	0.03987	404,219	0	189,461	151,478	GCC602Prefetcher
J15	189,461	1	151,478	388,101	16,118	0.03987	404,219	0	189,461	151,478	GCC602Prefetcher
J17	188,865	1	120,319	386,065	18,154	0.04491	404,219	5.295e-06	188,865	120,319	GCC602Prefetcher
J16	189,926	1	128,864	386,546	17,673	0.04372	404,219	5.265e-06	189,926	128,864	GCC602Prefetcher
J18	41,212	1	41,210	241,399	162,820	0.4028	404,219	2.426e-05	41,212	41,210	GCC602Prefetcher
J07	703,571	0.99427	703,571	362,043	623,232	0.63255	985,275	0.03513	703,571	703,571	MCF605DependencyPrefetcher
J06	793,436	0.99393	793,436	365,790	619,482	0.62874	985,272	0.03071	793,436	793,436	MCF605DependencyPrefetcher
J08	712,142	0.9954	712,142	364,639	620,636	0.62991	985,275	0.03575	712,142	712,142	MCF605DependencyPrefetcher
J05	824,198	0.99489	824,198	350,794	634,483	0.64396	985,277	0.02644	824,198	824,198	MCF605DependencyPrefetcher
J01	576,298	0.99468	576,298	329,535	655,741	0.66554	985,276	0.00303	576,298	576,298	MCF605DependencyPrefetcher
J03	496,321	0.99493	496,321	351,671	633,604	0.64307	985,275	0.00483	496,321	496,321	MCF605DependencyPrefetcher
J02	500,058	0.99395	500,058	355,854	629,420	0.63883	985,274	0.02839	500,058	500,058	MCF605DependencyPrefetcher

Table 11: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 5/9 (cont.)

ID	keyed_entries	keyed_trigger_coverage	keyed_unique_trigger_keys	l2_load_hit	l2_load_miss	l2_load_miss_rate	l2_loads	late_per_issued	list_loaded_entries	list_loaded_trigger_keys	model_class
J04	325,300	0.99471	325,300	341,038	644,237	0.65387	985,275	0.00802	325,300	325,300	MCF605DependencyPrefetcher
J43	479,746	0.99828	250,033	228,847	66,193	0.22435	295,040	0.01581	479,746	250,033	LBM619TimingPrefetcher
J38	507,607	0.99854	262,085	236,989	58,058	0.19678	295,047	0.0174	507,607	262,085	LBM619TimingPrefetcher
J47	470,007	0.99692	292,884	291,517	3,516	0.01192	295,033	8.949e-04	470,007	292,884	LBM619TimingPrefetcher
J31	481,792	0.99819	250,011	234,646	60,410	0.20474	295,056	0.02271	481,792	250,011	LBM619TimingPrefetcher
J34	871,355	0.99785	250,011	249,940	45,132	0.15295	295,072	0.00282	871,355	250,011	LBM619TimingPrefetcher
J30	481,792	0.99819	250,011	234,646	60,410	0.20474	295,056	0.02271	481,792	250,011	LBM619TimingPrefetcher
J35	474,738	0.99854	246,907	230,693	64,362	0.21814	295,055	0.02288	474,738	246,907	LBM619TimingPrefetcher
J36	473,634	0.99844	245,182	226,759	68,307	0.2315	295,066	0.02269	473,634	245,182	LBM619TimingPrefetcher
J32	680,908	0.99785	250,011	246,855	48,208	0.16338	295,063	0.00665	680,908	250,011	LBM619TimingPrefetcher
J33	871,355	0.99785	250,011	249,940	45,132	0.15295	295,072	0.00282	871,355	250,011	LBM619TimingPrefetcher
J46	835,216	0.9978	250,033	252,513	42,534	0.14416	295,047	0.00199	835,216	250,033	LBM619TimingPrefetcher
J41	906,048	0.99788	262,085	260,697	34,372	0.11649	295,069	0.00171	906,048	262,085	LBM619TimingPrefetcher
J42	479,746	0.99828	250,033	228,847	66,193	0.22435	295,040	0.01581	479,746	250,033	LBM619TimingPrefetcher
J37	507,607	0.99854	262,085	236,989	58,058	0.19678	295,047	0.0174	507,607	262,085	LBM619TimingPrefetcher
J44	674,417	0.99791	250,033	247,657	47,395	0.16063	295,052	0.00497	674,417	250,033	LBM619TimingPrefetcher
J39	723,587	0.99811	262,085	256,514	38,561	0.13068	295,075	0.00499	723,587	262,085	LBM619TimingPrefetcher
J45	835,216	0.9978	250,033	252,513	42,534	0.14416	295,047	0.00199	835,216	250,033	LBM619TimingPrefetcher
J40	906,048	0.99788	262,085	260,697	34,372	0.11649	295,069	0.00171	906,048	262,085	LBM619TimingPrefetcher
J28	70,187	0.9996	70,187	172,780	278,633	0.61725	451,413	0.00124	70,187	70,187	OMNET620RegionPrefetcher
J27	83,788	0.99947	83,788	180,637	270,804	0.59987	451,441	0.00147	83,788	83,788	OMNET620RegionPrefetcher
J29	102,458	0.9995	102,458	158,913	292,473	0.64794	451,386	0.00157	102,458	102,458	OMNET620RegionPrefetcher
J26	57,152	0.99967	57,152	179,700	271,764	0.60196	451,464	9.277e-04	57,152	57,152	OMNET620RegionPrefetcher
J23	57,152	0.99967	57,152	179,700	271,764	0.60196	451,464	9.277e-04	57,152	57,152	OMNET620RegionPrefetcher
J25	24,752	1	24,752	155,358	296,075	0.65586	451,433	8.484e-04	24,752	24,752	OMNET620RegionPrefetcher
J24	24,823	0.99968	24,823	155,392	296,018	0.65576	451,410	7.254e-04	24,823	24,823	OMNET620RegionPrefetcher
J13	517,067	0.99827	517,067	838,232	381,025	0.31251	1,219,257	0.00229	517,067	517,067	XALAN623ContextPrefetcher
J12	480,655	0.99784	480,655	736,598	481,518	0.3953	1,218,116	0.00143	480,655	480,655	XALAN623ContextPrefetcher
J14	796,561	0.99698	796,561	559,303	658,974	0.54091	1,218,277	0.00153	796,561	796,561	XALAN623ContextPrefetcher
J11	535,537	0.99864	535,537	1,166,065	53,216	0.04365	1,219,281	0.00155	535,537	535,537	XALAN623ContextPrefetcher
J09	482,477	0.99884	482,477	1,156,692	62,526	0.05128	1,219,218	0.00148	482,477	482,477	XALAN623ContextPrefetcher
J10	112,484	0.99999	112,484	834,695	383,674	0.31491	1,218,369	0.00235	112,484	112,484	XALAN623ContextPrefetcher

Table 12: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 6/9

ID	model_key	model_size	no_pref_ipc	nodup_accuracy	nodup_issued	online_global_issue_budget	online_non_neural_bytes	parameters	pf_filled	pf_issued	pf_late
J21	assoc_control	about_383k	0.368	0.97942	188,865	404,219	4,104	382,549	188,628	188,865	1
J20	assoc_control	about_911k	0.368	0.97648	189,926	404,219	4,104	911,219	189,644	189,926	1
J22	assoc_control	nearest_153k	0.368	0.95211	199,519	404,219	4,104	152,891	198,612	199,519	2
J19	assoc_control	selected_full	0.368	0.98708	189,461	404,219	4,104	4,147,959	189,322	189,461	0
J15	assoc_control	selected_full	0.368	0.98708	189,461	404,219	4,104	4,147,959	189,322	189,461	0
J17	assoc_control	about_383k	0.368	0.97942	188,865	404,219	4,104	382,549	188,628	188,865	1
J16	assoc_control	about_911k	0.368	0.97648	189,926	404,219	4,104	911,219	189,644	189,926	1
J18	assoc_control	nearest_153k	0.368	0.97816	41,212	404,219	4,104	152,891	41,130	41,212	1
J07	605_support16.fixed_nextline	about_383k	0.18318	0.24506	699,529	985,276	4,104	382,549	508,735	699,538	24,576
J06	605_support16.fixed_nextline	about_911k	0.18318	0.21888	788,611	985,276	4,104	911,219	589,310	788,620	24,216
J08	605_support16.fixed_nextline	nearest_153k	0.18318	0.23496	708,859	985,276	4,104	152,891	539,748	708,864	25,342
J05	605_support16.fixed_nextline	selected_full	0.18318	0.19112	819,974	985,276	4,104	4,147,959	523,945	819,983	21,679
J01	605_support16.fixed_nextline	selected_full	0.18318	0.23381	573,224	985,276	4,104	4,147,959	329,301	573,234	1,738
J03	605_support16.fixed_nextline	about_383k	0.18318	0.32141	493,799	985,276	4,104	382,549	355,459	493,807	2,387
J02	605_support16.fixed_nextline	about_911k	0.18318	0.31677	497,025	985,276	4,104	911,219	340,781	497,031	14,110
J04	605_support16.fixed_nextline	nearest_153k	0.18318	0.37373	323,577	985,276	4,104	152,891	256,949	323,579	2,596

Table 12: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 6/9 (cont.)

ID		<i>model_key</i>	<i>model_size</i>	<i>no_pref_ipc</i>	<i>nodup-accuracy</i>	<i>nodup-issued</i>	<i>online-global-issue-budget</i>	<i>online-non-neural-bytes</i>	<i>parameters</i>	<i>pf_filled</i>	<i>pf_issued</i>	<i>pf_late</i>
J43		619_timing_score_model	about_383k	0.322	0.54406	419,880	590,074	8	382,549	228,958	478,946	7,573
J38		619_timing_score_model	about_911k	0.322	0.52234	452,919	590,074	8	911,219	237,118	506,926	8,820
J47		619_timing_score_model	nearest_153k	0.322	0.70323	413,984	590,074	8	152,891	292,497	468,215	419
J31		619_timing_score_model	selected_full	0.322	0.56368	415,530	590,074	8	4,147,959	234,345	480,929	10,921
J34		619_timing_score_model	selected_full	0.322	0.35186	709,125	1,180,148	8	4,147,959	249,633	869,593	2,451
J30		619_timing_score_model	selected_full	0.322	0.56368	415,530	590,074	8	4,147,959	234,345	480,929	10,921
J35		619_timing_score_model	selected_full	0.322	0.57036	403,734	590,074	8	4,147,959	230,393	474,081	10,845
J36		619_timing_score_model	selected_full	0.322	0.5598	404,308	590,074	8	4,147,959	226,420	472,941	10,729
J32		619_timing_score_model	selected_full	0.322	0.42701	577,100	885,111	8	4,147,959	246,551	679,499	4,516
J33		619_timing_score_model	selected_full	0.322	0.35186	709,125	1,180,148	8	4,147,959	249,633	869,593	2,451
J46		619_timing_score_model	about_383k	0.322	0.36943	682,406	1,180,148	8	382,549	252,861	833,464	1,656
J41		619_timing_score_model	about_911k	0.322	0.35582	731,455	1,180,148	8	911,219	260,980	904,426	1,544
J42		619_timing_score_model	about_383k	0.322	0.54406	419,880	590,074	8	382,549	228,958	478,946	7,573
J37		619_timing_score_model	about_911k	0.322	0.52234	452,919	590,074	8	911,219	237,118	506,926	8,820
J44		619_timing_score_model	about_383k	0.322	0.42893	576,412	885,111	8	382,549	247,958	673,058	3,348
J39		619_timing_score_model	about_911k	0.322	0.41413	618,354	885,111	8	911,219	256,764	722,342	3,607
J45		619_timing_score_model	about_383k	0.322	0.36943	682,406	1,180,148	8	382,549	252,861	833,464	1,656
J40		619_timing_score_model	about_911k	0.322	0.35582	731,455	1,180,148	8	911,219	260,980	904,426	1,544
J28		region_control	about_383k	0.2375	0.58513	70,159	112,849	4,104	382,549	64,351	70,159	87
J27		region_control	about_911k	0.2375	0.5891	83,744	112,849	4,104	911,219	74,693	83,744	123
J29		region_control	nearest_153k	0.2375	0.30193	102,406	112,849	4,104	152,891	84,957	102,407	161
J26		region_control	selected_full	0.2375	0.80988	57,133	112,849	4,104	4,147,959	53,811	57,133	53
J23		region_control	selected_full	0.2375	0.80988	57,133	112,849	4,104	4,147,959	53,811	57,133	53
J25		region_control	about_383k	0.2375	0.84886	24,752	112,849	4,104	382,549	24,551	24,752	21
J24		region_control	about_911k	0.2375	0.8511	24,815	112,849	4,104	911,219	24,513	24,815	18
J13		623_context_lru128	about_383k	0.35321	0.5137	516,160	1,218,494	2,056	382,549	437,838	516,174	1,180
J12		623_context_lru128	about_911k	0.35321	0.41233	479,582	1,218,494	2,056	911,219	415,164	479,615	684
J14		623_context_lru128	nearest_153k	0.35321	0.231	794,147	1,218,494	2,056	152,891	754,243	794,157	1,212
J11		623_context_lru128	selected_full	0.35321	0.77971	534,685	1,218,494	2,056	4,147,959	431,296	534,810	828
J09		623_context_lru128	selected_full	0.35321	0.82299	481,820	1,218,494	2,056	4,147,959	404,134	481,918	712
J10		623_context_lru128	about_383k	0.35321	0.73031	112,483	1,218,494	2,056	382,549	92,744	112,483	264

Table 13: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 7/9

ID	<i>pf_requested</i>	<i>pf_useful</i>	<i>pf_useless</i>	<i>policy-mode</i>	<i>policy-tag</i>	<i>pq-merged-duplicate-proxy</i>	<i>replay_ipc</i>	<i>replayer-emitted-candidates</i>	<i>replayer-final-loaded-trigger-keys</i>	<i>replayer-matched-trigger-keys</i>	<i>replayer-runtime_l2_loads</i>
J21	188,865	184,979	3,614	calibration_control	assoc_control	0	0.42835	188,865	120,319	120,319	404,219
J20	189,926	185,459	4,152	calibration_control	assoc_control	0	0.42855	189,926	128,864	128,864	404,219
J22	199,519	189,965	8,575	calibration_control	assoc_control	0	0.43066	199,519	146,705	146,705	404,219
J19	189,461	187,013	2,284	calibration_control	assoc_control	0	0.42924	189,461	151,478	151,478	404,219
J15	189,461	187,013	2,284	strict	assoc_control	0	0.42924	189,461	151,478	151,478	404,219
J17	188,865	184,979	3,614	strict	assoc_control	0	0.42835	188,865	120,319	120,319	404,219
J16	189,926	185,459	4,152	strict	assoc_control	0	0.42855	189,926	128,864	128,864	404,219
J18	41,212	40,312	813	strict	assoc_control	0	0.3803	41,212	41,210	41,210	404,219
J07	699,538	171,424	336,523	calibration_control	support16_fixed_nextline	9	0.19359	699,538	703,571	699,538	985,275
J06	788,620	172,608	415,179	calibration_control	support16_fixed_nextline	9	0.19325	788,620	793,436	788,620	985,272
J08	708,864	166,555	371,979	calibration_control	support16_fixed_nextline	5	0.19259	708,864	712,142	708,864	985,275
J05	819,983	156,713	366,271	calibration_control	support16_fixed_nextline	9	0.19173	819,983	824,198	819,983	985,277
J01	573,234	134,024	195,270	strict	support16_fixed_nextline	10	0.18935	573,234	576,298	573,234	985,276
J03	493,807	158,710	196,722	strict	support16_fixed_nextline	8	0.19258	493,807	496,321	493,807	985,275

Table 13: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 7/9 (cont.)

ID	<i>pf_requested</i>	<i>pf_useful</i>	<i>pf_useless</i>	<i>policy_mode</i>	<i>policy_tag</i>	<i>pq_merged_duplicate_proxy</i>	<i>replay_ipc</i>	<i>replayer-emitted_candidates</i>	<i>replayer-final_loaded_trigger_keys</i>	<i>replayer-matched_trigger_keys</i>	<i>replayer-runtime_l2_loads</i>
J02	497,031	157,442	183,325	strict	support16_fixed_nextline	6	0.19234	497,031	500,058	497,031	985,274
J04	323,579	120,930	136,019	strict	support16_fixed_nextline	2	0.18967	323,579	325,300	323,579	985,275
J43	478,946	228,439	512	calibration_control	top2	59,066	0.36923	478,946	250,033	249,603	295,040
J38	506,926	236,578	533	calibration_control	top2	54,007	0.37001	506,926	262,085	261,702	295,047
J47	468,215	291,124	1,347	calibration_control	top2	54,231	0.38379	468,215	292,884	291,982	295,033
J31	480,929	234,227	117	calibration_control	top2	65,399	0.37244	480,929	250,011	249,559	295,056
J34	869,593	249,511	121	strict	adaptive_score_gap	160,468	0.37556	869,593	250,011	249,474	295,072
J30	480,929	234,227	117	strict	top2	65,399	0.37244	480,929	250,011	249,559	295,056
J35	474,081	230,274	110	strict	top2	70,347	0.37284	474,081	246,907	246,546	295,055
J36	472,941	226,333	86	strict	top2	68,633	0.37039	472,941	245,182	244,799	295,066
J32	679,499	246,430	120	strict	top3	102,399	0.3746	679,499	250,011	249,473	295,063
J33	869,593	249,511	121	strict	top4	160,468	0.37556	869,593	250,011	249,474	295,072
J46	833,464	252,101	751	strict	adaptive_score_gap	151,058	0.37549	833,464	250,033	249,483	295,047
J41	904,426	260,265	705	strict	adaptive_score_gap	172,971	0.37681	904,426	262,085	261,530	295,069
J42	478,946	228,439	512	strict	top2	59,066	0.36923	478,946	250,033	249,603	295,040
J37	506,926	236,578	533	strict	top2	54,007	0.37001	506,926	262,085	261,702	295,047
J44	673,058	247,243	706	strict	top3	96,646	0.37438	673,058	250,033	249,510	295,052
J39	722,342	256,078	676	strict	top3	103,988	0.37654	722,342	262,085	261,589	295,075
J45	833,464	252,101	751	strict	top4	151,058	0.37549	833,464	250,033	249,483	295,047
J40	904,426	260,265	705	strict	top4	172,971	0.37681	904,426	262,085	261,530	295,069
J28	70,159	41,052	23,084	calibration_control	region_control	0	0.24105	70,159	70,187	70,159	451,413
J27	83,744	49,334	25,124	calibration_control	region_control	0	0.24121	83,744	83,788	83,744	451,441
J29	102,407	30,919	53,581	calibration_control	region_control	1	0.23952	102,407	102,458	102,407	451,386
J26	57,133	46,271	7,465	calibration_control	region_control	0	0.24195	57,133	57,152	57,133	451,464
J23	57,133	46,271	7,465	strict	region_control	0	0.24195	57,133	57,152	57,133	451,464
J25	24,752	21,011	3,503	strict	region_control	0	0.23963	24,752	24,752	24,752	451,433
J24	24,815	21,120	3,358	strict	region_control	0	0.23959	24,815	24,823	24,815	451,410
J13	516,174	265,149	172,096	calibration_control	context_lru128	14	0.35708	516,174	517,067	516,174	1,219,257
J12	479,615	197,747	216,704	calibration_control	context_lru128	33	0.35127	479,615	480,655	479,615	1,218,116
J14	794,157	183,448	569,458	calibration_control	context_lru128	10	0.34149	794,157	796,561	794,157	1,218,277
J11	534,810	416,897	14,274	calibration_control	context_lru128	125	0.37915	534,810	535,537	534,810	1,219,281
J09	481,918	396,535	7,533	strict	context_lru128	98	0.37882	481,918	482,477	481,918	1,219,218
J10	112,483	82,147	10,540	strict	context_lru128	0	0.35728	112,483	112,484	112,483	1,218,369

Table 14: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 8/9

ID	<i>route_id</i>	<i>seed</i>	<i>selected_accuracy</i>	<i>selected_timely_proxy</i>	<i>speedup-vs-best_normal</i>	<i>speedup-vs-no-pref</i>	<i>stage</i>	<i>stage-reference_mode</i>	<i>threshold-effective</i>
J21	assoc_control	7	0.95218	0.95218	0.98182	1.16399	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.9
J20	assoc_control	7	0.95088	0.95088	0.98228	1.16454	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.9
J22	assoc_control	7	0.79469	0.79469	0.98712	1.17027	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.75758
J19	assoc_control	7	0.9711	0.9711	0.98386	1.16641	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.9
J15	assoc_control	7	0.9711	0.9711	0.98386	1.16641	one_shot_full_reference	strict_frozen_policy	0.9
J17	assoc_control	7	0.95218	0.95218	0.98182	1.16399	one_shot_strict_size	strict_frozen_policy	0.9
J16	assoc_control	7	0.95088	0.95088	0.98228	1.16454	one_shot_strict_size	strict_frozen_policy	0.9
J18	assoc_control	7	0.95031	0.95031	0.87169	1.03342	one_shot_strict_size	strict_frozen_policy	0.9
J07	support16_fixed_nextline	7	0.15124	0.15124	1.0257	1.05683	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.43349
J06	support16_fixed_nextline	7	0.1377	0.1377	1.0239	1.05497	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.39924
J08	support16_fixed_nextline	7	0.15486	0.15486	1.0204	1.05137	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.43634
J05	support16_fixed_nextline	7	0.12049	0.12049	1.01584	1.04668	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.41942
J01	support16_fixed_nextline	7	0.15917	0.15917	1.00323	1.03368	one_shot_full_reference	strict_frozen_policy	0.5

Table 14: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 8/9 (cont.)

ID	route_id	seed	selected_accuracy	selected_timely-proxy	speedup-vs-best-normal	speedup-vs-no-pref	stage	stage-reference_mode	threshold_effective
J03	support16_fixed_nextline	7	0.20259	0.20259	1.02035	1.05132	one_shot_strict_size	strict_frozen_policy	0.5
J02	support16_fixed_nextline	7	0.20823	0.20823	1.01907	1.05001	one_shot_strict_size	strict_frozen_policy	0.5
J04	support16_fixed_nextline	7	0.27242	0.27242	1.00493	1.03543	one_shot_strict_size	strict_frozen_policy	0.5
J43	top2	7	0.99855	0.99855	0.96898	1.14668	one_shot_619_calibration_control	calibration_control_not_strict_stage_b	0.98557
J38	top2	7	0.99854	0.99854	0.97103	1.1491	one_shot_619_calibration_control	calibration_control_not_strict_stage_b	0.98557
J47	top2	7	0.96866	0.96866	1.00719	1.19189	one_shot_619_calibration_control	calibration_control_not_strict_stage_b	0.79846
J31	top2	7	0.99965	0.99965	0.9774	1.15665	one_shot_619_calibration_control	calibration_control_not_strict_stage_b	0.98557
J34	adaptive_score_gap	7	0.99971	0.99971	0.98559	1.16634	one_shot_619_full_policy	strict_frozen_policy	0.98557
J30	top2	7	0.99965	0.99965	0.9774	1.15665	one_shot_619_full_policy	strict_frozen_policy	0.98557
J35	top2	19	0.99966	0.99966	0.97845	1.15789	one_shot_619_full_policy	strict_frozen_policy	0.98557
J36	top2	43	0.99969	0.99969	0.97202	1.15028	one_shot_619_full_policy	strict_frozen_policy	0.98557
J32	top3	7	0.99971	0.99971	0.98307	1.16335	one_shot_619_full_policy	strict_frozen_policy	0.98557
J33	top4	7	0.99971	0.99971	0.98559	1.16634	one_shot_619_full_policy	strict_frozen_policy	0.98557
J46	adaptive_score_gap	7	0.99867	0.99867	0.98541	1.16612	one_shot_619_strict_size_policy	strict_frozen_policy	0.98557
J41	adaptive_score_gap	7	0.99871	0.99871	0.98887	1.17022	one_shot_619_strict_size_policy	strict_frozen_policy	0.98557
J42	top2	7	0.99855	0.99855	0.96898	1.14668	one_shot_619_strict_size_policy	strict_frozen_policy	0.98557
J37	top2	7	0.99854	0.99854	0.97103	1.1491	one_shot_619_strict_size_policy	strict_frozen_policy	0.98557
J44	top3	7	0.99853	0.99853	0.9825	1.16267	one_shot_619_strict_size_policy	strict_frozen_policy	0.98557
J39	top3	7	0.99856	0.99856	0.98816	1.16938	one_shot_619_strict_size_policy	strict_frozen_policy	0.98557
J45	top4	7	0.99867	0.99867	0.98541	1.16612	one_shot_619_strict_size_policy	strict_frozen_policy	0.98557
J40	top4	7	0.99871	0.99871	0.98887	1.17022	one_shot_619_strict_size_policy	strict_frozen_policy	0.98557
J28	region_control	7	0.33811	0.33811	0.97611	1.01495	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.55722
J27	region_control	7	0.32748	0.32748	0.97676	1.01562	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.56726
J29	region_control	7	0.146	0.146	0.96991	1.00851	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.53857
J26	region_control	7	0.59272	0.59272	0.97975	1.01874	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.65
J23	region_control	7	0.59272	0.59272	0.97975	1.01874	one_shot_full_reference	strict_frozen_policy	0.65
J25	region_control	7	0.62023	0.62023	0.97036	1.00897	one_shot_strict_size	strict_frozen_policy	0.65
J24	region_control	7	0.63385	0.63385	0.9702	1.0088	one_shot_strict_size	strict_frozen_policy	0.65
J13	context_lru128	7	0.3872	0.3872	1.00896	1.01096	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.57529
J12	context_lru128	7	0.31852	0.31852	0.99254	0.99451	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.58546
J14	context_lru128	7	0.14131	0.14131	0.96491	0.96682	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.50697
J11	context_lru128	7	0.66306	0.66306	1.07132	1.07344	one_shot_calibration_control	calibration_control_not_strict_stage_b	0.57753
J09	context_lru128	7	0.73654	0.73654	1.07039	1.07251	one_shot_full_reference	strict_frozen_policy	0.65
J10	context_lru128	7	0.65465	0.65465	1.00952	1.01152	one_shot_strict_size	strict_frozen_policy	0.65

Table 15: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 9/9

ID	trace	train_end	useful_per_12_miss_self	useless_per_12_missed
J21	602.gcc_s-734B	323,247	10.18943	0.01914
J20	602.gcc_s-734B	323,247	10.49392	0.02186
J22	602.gcc_s-734B	323,247	14.42626	0.04298
J19	602.gcc_s-734B	323,247	11.60274	0.01206
J15	602.gcc_s-734B	323,247	11.60274	0.01206
J17	602.gcc_s-734B	323,247	10.18943	0.01914
J16	602.gcc_s-734B	323,247	10.49392	0.02186
J18	602.gcc_s-734B	323,247	0.24759	0.01973
J07	605.mcf_s-994B	788,092	0.27506	0.48106
J06	605.mcf_s-994B	788,092	0.27863	0.52646
J08	605.mcf_s-994B	788,092	0.26836	0.52475
J05	605.mcf_s-994B	788,092	0.24699	0.44668
J01	605.mcf_s-994B	788,092	0.20439	0.34065
J03	605.mcf_s-994B	788,092	0.25049	0.39838
J02	605.mcf_s-994B	788,092	0.25014	0.36884
J04	605.mcf_s-994B	788,092	0.18771	0.42036

Table 15: replay_results/lstm_summary.csv — all 47 NN replay rows (long-text columns are expanded in Tables 19, 20, 54, 16; paths in Appendix) — part 9/9 (cont.)

ID		<i>trace</i>	<i>train_end</i>	<i>useful_per_12-miss-self</i>	<i>useless_per_issued</i>
J43	619.lbm_s-4268B	235,901	3.45111	0.00107	
J38	619.lbm_s-4268B	235,901	4.07486	0.00103	
J47	619.lbm_s-4268B	235,901	82.79977	0.00288	
J31	619.lbm_s-4268B	235,901	3.87729	2.433e-04	
J34	619.lbm_s-4268B	235,901	5.52847	1.391e-04	
J30	619.lbm_s-4268B	235,901	3.87729	2.433e-04	
J35	619.lbm_s-4268B	235,901	3.57779	2.320e-04	
J36	619.lbm_s-4268B	235,901	3.31347	1.818e-04	
J32	619.lbm_s-4268B	235,901	5.11181	1.766e-04	
J33	619.lbm_s-4268B	235,901	5.52847	1.391e-04	
J46	619.lbm_s-4268B	235,901	5.92705	9.011e-04	
J41	619.lbm_s-4268B	235,901	7.57201	7.795e-04	
J42	619.lbm_s-4268B	235,901	3.45111	0.00107	
J37	619.lbm_s-4268B	235,901	4.07486	0.00103	
J44	619.lbm_s-4268B	235,901	5.21665	0.00103	
J39	619.lbm_s-4268B	235,901	6.64085	9.358e-04	
J45	619.lbm_s-4268B	235,901	5.92705	9.011e-04	
J40	619.lbm_s-4268B	235,901	7.57201	7.795e-04	
J28	620.omnetpp_s-874B	360,990	0.14733	0.32902	
J27	620.omnetpp_s-874B	360,990	0.18218	0.30001	
J29	620.omnetpp_s-874B	360,990	0.10572	0.52322	
J26	620.omnetpp_s-874B	360,990	0.17026	0.13066	
J23	620.omnetpp_s-874B	360,990	0.17026	0.13066	
J25	620.omnetpp_s-874B	360,990	0.07097	0.14152	
J24	620.omnetpp_s-874B	360,990	0.07135	0.13532	
J13	623.xalancbmk_s-700B	974,667	0.69588	0.33341	
J12	623.xalancbmk_s-700B	974,667	0.41067	0.45183	
J14	623.xalancbmk_s-700B	974,667	0.27838	0.71706	
J11	623.xalancbmk_s-700B	974,667	7.83405	0.02669	
J09	623.xalancbmk_s-700B	974,667	6.34192	0.01563	
J10	623.xalancbmk_s-700B	974,667	0.21411	0.0937	

Table 16: Per-trace candidate sources and feature contract (from lstm_summary columns `candidate_sources`, `feature_contract`)

ID	trace	model key	candidate sources	feature contract
J21	602.gcc_s-734B	assoc_control	v31_pc.delta;v31_pc.pair;v31_global.delta;v31_global.pair	["causal PC/page/line history", "association/residual candidates", "optional train-prefix signed stride sources"]
J07	605.mcf_s-994B	605_support16_fixed_ctx3_phase	ctx3;phase;predecessor;pc;temporal;assoc.ab5	["causal PC/page/line history", "committed dependency sidecar", "support16 dependency-edge + fixed next-line bank"]
J43	619.lbm_s-4268B	619_timing_score_model	v31_pc.delta;v31_pc.pair;v31_global.delta;v31_global.pair	["causal PC/page/line history", "V31 timing bank", "lead/cycle timing heads"]
J28	620.omnetpp_s-874B	region_control	ctx3;phase;offset;predecessor;pc;region.pair	["causal PC/page/line history", "region/residual pair candidates", "optional train-prefix signed stride sources"]
J13	623.xalancbmk_s-700B	623_context_lru128	ctx3;phase;offset	["causal PC/page/line/context history", "context hierarchy candidates", "LRU128 policy"]

Table 17: replay_results/lstm_winners.csv — per-trace NN winner (selection rule: max replay IPC among current-run NN candidates); non-key columns duplicate Table 7 — part 1/2

ID		<i>trace</i>	<i>model_size</i>	<i>winner_rank_within_trace</i>	<i>beats_best_normal</i>	<i>best_normal</i>	<i>best_normal_ipc</i>	<i>replay_ipc</i>	<i>speedup_vs_best_normal</i>	<i>speedup_vs_no_pref</i>
J22	602.gcc.s-734B	nearest_153k		1	0	sandbox	0.43628	0.43066	0.98712	1.17027
J07	605.mcf.s-994B	about_383k		1	1	ampm	0.18874	0.19359	1.0257	1.05683
J47	619.lbm.s-4268B	nearest_153k		1	1	sms	0.38105	0.38379	1.00719	1.19189
J26	620.omnetpp.s-874B	selected_full		1	0	sms	0.24695	0.24195	0.97975	1.01874
J11	623.xalancbmk.s-700B	selected_full		1	1	spp	0.35391	0.37915	1.07132	1.07344

Table 18: replay_results/lstm_winners.csv — per-trace NN winner (selection rule: max replay IPC among current-run NN candidates); non-key columns duplicate Table 7 — part 2/2

ID	<i>accuracy</i>	<i>selected_accuracy</i>	<i>timeliness</i>	<i>pf-issued</i>	<i>pf-useful</i>	<i>pf-useless</i>	<i>pf_late</i>	<i>winner_selection</i>
J22	0.95211	0.79469	0.99999	199,519	189,965	8,575	2	max_replay_ipc.among_current_run_nn.candidates
J07	0.24505	0.15124	0.87461	699,538	171,424	336,523	24,576	max_replay_ipc.among_current_run_nn.candidates
J47	0.62177	0.96866	0.99856	468,215	291,124	1,347	419	max_replay_ipc.among_current_run_nn.candidates
J26	0.80988	0.59272	0.99886	57,133	46,271	7,465	53	max_replay_ipc.among_current_run_nn.candidates
J11	0.77952	0.66306	0.99802	534,810	416,897	14,274	828	max_replay_ipc.among_current_run_nn.candidates

Table 19: Effective replay policy per job (parsed from lstm_summary policy / policy_recipe JSON)

ID	<i>trace</i>	<i>model_size</i>	<i>stage</i>	<i>policy_tag</i>	<i>threshold</i>	<i>degree</i>	<i>dedup</i>	<i>max_issue</i>	<i>min_lead</i>	<i>min_cycle</i>	<i>recipe_threshold</i>
J21	602.gcc.s-734B	about_383k	one_shot_calibration_control	assoc_control	0.9	4	256	1	4	0	0.9
J20	602.gcc.s-734B	about_911k	one_shot_calibration_control	assoc_control	0.9	4	256	1	4	0	0.9
J22	602.gcc.s-734B	nearest_153k	one_shot_calibration_control	assoc_control	0.75758	4	256	1	4	0	0.9
J19	602.gcc.s-734B	selected_full	one_shot_calibration_control	assoc_control	0.9	4	256	1	4	0	0.9
J15	602.gcc.s-734B	selected_full	one_shot_full_reference	assoc_control	0.9	4	256	1	4	0	0.9
J17	602.gcc.s-734B	about_383k	one_shot_strict_size	assoc_control	0.9	4	256	1	4	0	0.9
J16	602.gcc.s-734B	about_911k	one_shot_strict_size	assoc_control	0.9	4	256	1	4	0	0.9
J18	602.gcc.s-734B	nearest_153k	one_shot_strict_size	assoc_control	0.9	4	256	1	4	0	0.9
J07	605.mcf.s-994B	about_383k	one_shot_calibration_control	support16_fixed_nextline	0.43349	1	256	1	4	0	0.5
J06	605.mcf.s-994B	about_911k	one_shot_calibration_control	support16_fixed_nextline	0.39924	1	256	1	4	0	0.5
J08	605.mcf.s-994B	nearest_153k	one_shot_calibration_control	support16_fixed_nextline	0.43634	1	256	1	4	0	0.5
J05	605.mcf.s-994B	selected_full	one_shot_calibration_control	support16_fixed_nextline	0.41942	1	256	1	4	0	0.5
J01	605.mcf.s-994B	selected_full	one_shot_full_reference	support16_fixed_nextline	0.5	1	256	1	4	0	0.5
J03	605.mcf.s-994B	about_383k	one_shot_strict_size	support16_fixed_nextline	0.5	1	256	1	4	0	0.5
J02	605.mcf.s-994B	about_911k	one_shot_strict_size	support16_fixed_nextline	0.5	1	256	1	4	0	0.5
J04	605.mcf.s-994B	nearest_153k	one_shot_strict_size	support16_fixed_nextline	0.5	1	256	1	4	0	0.5
J43	619.lbm.s-4268B	about_383k	one_shot_619_calibration_control	top2	0.98557	2	0	2	8	0	0.98557
J38	619.lbm.s-4268B	about_911k	one_shot_619_calibration_control	top2	0.98557	2	0	2	8	0	0.98557
J47	619.lbm.s-4268B	nearest_153k	one_shot_619_calibration_control	top2	0.79846	2	0	2	8	0	0.98557
J31	619.lbm.s-4268B	selected_full	one_shot_619_calibration_control	top2	0.98557	2	0	2	8	0	0.98557
J34	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	adaptive_score_gap	0.98557	4	0	4	8	0	0.98557
J30	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	top2	0.98557	2	0	2	8	0	0.98557
J35	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	top2	0.98557	2	0	2	8	0	0.98557
J36	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	top2	0.98557	2	0	2	8	0	0.98557
J32	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	top3	0.98557	3	0	3	8	0	0.98557
J33	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	top4	0.98557	4	0	4	8	0	0.98557
J46	619.lbm.s-4268B	about_383k	one_shot_619_strict_size_policy	adaptive_score_gap	0.98557	4	0	4	8	0	0.98557
J41	619.lbm.s-4268B	about_911k	one_shot_619_strict_size_policy	adaptive_score_gap	0.98557	4	0	4	8	0	0.98557
J42	619.lbm.s-4268B	about_383k	one_shot_619_strict_size_policy	top2	0.98557	2	0	2	8	0	0.98557

Table 19: Effective replay policy per job (parsed from lstm_summary policy / policy_recipe JSON) (cont.)

ID	trace	model_size	stage	policy_tag	threshold	degree	dedup	max_issue	min_lead	min_cycle	recipe_threshold
J37	619.lbm.s-4268B	about_911k	one_shot_619_strict_size.policy	top2	0.98557	2	0	2	8	0	0.98557
J44	619.lbm.s-4268B	about_383k	one_shot_619_strict_size.policy	top3	0.98557	3	0	3	8	0	0.98557
J39	619.lbm.s-4268B	about_911k	one_shot_619_strict_size.policy	top3	0.98557	3	0	3	8	0	0.98557
J45	619.lbm.s-4268B	about_383k	one_shot_619_strict_size.policy	top4	0.98557	4	0	4	8	0	0.98557
J40	619.lbm.s-4268B	about_911k	one_shot_619_strict_size.policy	top4	0.98557	4	0	4	8	0	0.98557
J28	620.omnetpp.s-874B	about_383k	one_shot_calibration_control	region_control	0.55722	1	256	0.25	4	0	0.65
J27	620.omnetpp.s-874B	about_911k	one_shot_calibration_control	region_control	0.56726	1	256	0.25	4	0	0.65
J29	620.omnetpp.s-874B	nearest_153k	one_shot_calibration_control	region_control	0.53857	1	256	0.25	4	0	0.65
J26	620.omnetpp.s-874B	selected_full	one_shot_calibration_control	region_control	0.65	1	256	0.25	4	0	0.65
J23	620.omnetpp.s-874B	selected_full	one_shot_full_reference	region_control	0.65	1	256	0.25	4	0	0.65
J25	620.omnetpp.s-874B	about_383k	one_shot_strict_size	region_control	0.65	1	256	0.25	4	0	0.65
J24	620.omnetpp.s-874B	about_911k	one_shot_strict_size	region_control	0.65	1	256	0.25	4	0	0.65
J13	623.xalancbm.k.s-700B	about_383k	one_shot_calibration_control	context_lru128	0.57529	1	128	1	4	0	0.65
J12	623.xalancbm.k.s-700B	about_911k	one_shot_calibration_control	context_lru128	0.58546	1	128	1	4	0	0.65
J14	623.xalancbm.k.s-700B	nearest_153k	one_shot_calibration_control	context_lru128	0.50697	1	128	1	4	0	0.65
J11	623.xalancbm.k.s-700B	selected_full	one_shot_calibration_control	context_lru128	0.57753	1	128	1	4	0	0.65
J09	623.xalancbm.k.s-700B	selected_full	one_shot_full_reference	context_lru128	0.65	1	128	1	4	0	0.65
J10	623.xalancbm.k.s-700B	about_383k	one_shot_strict_size	context_lru128	0.65	1	128	1	4	0	0.65

Table 20: Offline decision-ledger summary per job (parsed from ledger_summary JSON; counts over validation future-target events) — part 1/2

ID	trace	model_size	stage	events	future_target_events	candidate_absent	target_selected	below_threshold	dedup_lru	rank_cut	unseen
J21	602.gcc.s-734B	about_383k	one_shot_calibration_control	80,844	80,839	42,184	23,269	2,102	13,284	0	5
J20	602.gcc.s-734B	about_911k	one_shot_calibration_control	80,844	80,839	42,184	25,324	1,568	11,763	0	5
J22	602.gcc.s-734B	nearest_153k	one_shot_calibration_control	80,844	80,839	42,184	25,303	779	12,573	0	5
J19	602.gcc.s-734B	selected_full	one_shot_calibration_control	80,844	80,839	42,184	29,845	923	7,887	0	5
J15	602.gcc.s-734B	selected_full	one_shot_full_reference	80,844	80,839	42,184	29,845	923	7,887	0	5
J17	602.gcc.s-734B	about_383k	one_shot_strict_size	80,844	80,839	42,184	23,269	2,102	13,284	0	5
J16	602.gcc.s-734B	about_911k	one_shot_strict_size	80,844	80,839	42,184	25,324	1,568	11,763	0	5
J18	602.gcc.s-734B	nearest_153k	one_shot_strict_size	80,844	80,839	42,184	7,299	31,356	0	0	5
J07	605.mcf.s-994B	about_383k	one_shot_calibration_control	197,056	197,052	180,810	3,895	10,798	14	1,535	4
J06	605.mcf.s-994B	about_911k	one_shot_calibration_control	197,056	197,052	180,810	3,732	9,441	644	2,425	4
J08	605.mcf.s-994B	nearest_153k	one_shot_calibration_control	197,056	197,052	180,810	3,569	12,055	13	605	4
J05	605.mcf.s-994B	selected_full	one_shot_calibration_control	197,056	197,052	180,810	3,296	12,212	14	720	4
J01	605.mcf.s-994B	selected_full	one_shot_full_reference	197,056	197,052	180,810	0	16,242	0	0	4
J03	605.mcf.s-994B	about_383k	one_shot_strict_size	197,056	197,052	180,810	1,147	15,093	0	2	4
J02	605.mcf.s-994B	about_911k	one_shot_strict_size	197,056	197,052	180,810	2,862	13,309	17	54	4
J04	605.mcf.s-994B	nearest_153k	one_shot_strict_size	197,056	197,052	180,810	0	16,242	0	0	4
J43	619.lbm.s-4268B	about_383k	one_shot_619_calibration_control	59,008	59,003	201	33,900	9,388	15,513	1	5
J38	619.lbm.s-4268B	about_911k	one_shot_619_calibration_control	59,008	59,003	201	34,017	7,094	17,691	0	5
J47	619.lbm.s-4268B	nearest_153k	one_shot_619_calibration_control	59,008	59,003	201	45,456	1,268	12,077	1	5
J31	619.lbm.s-4268B	selected_full	one_shot_619_calibration_control	59,008	59,003	201	29,217	9,321	20,264	0	5
J34	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	59,008	59,003	201	27,340	9,321	22,141	0	5
J30	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	59,008	59,003	201	29,217	9,321	20,264	0	5
J35	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	59,008	59,003	201	29,048	9,940	19,814	0	5
J36	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	59,008	59,003	201	29,020	10,443	19,339	0	5
J32	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	59,008	59,003	201	27,559	9,321	21,922	0	5
J33	619.lbm.s-4268B	selected_full	one_shot_619_full_policy	59,008	59,003	201	27,340	9,321	22,141	0	5
J46	619.lbm.s-4268B	about_383k	one_shot_619_strict_size.policy	59,008	59,003	201	31,278	9,388	18,136	0	5
J41	619.lbm.s-4268B	about_911k	one_shot_619_strict_size.policy	59,008	59,003	201	31,676	7,094	20,032	0	5
J42	619.lbm.s-4268B	about_383k	one_shot_619_strict_size.policy	59,008	59,003	201	33,900	9,388	15,513	1	5
J37	619.lbm.s-4268B	about_911k	one_shot_619_strict_size.policy	59,008	59,003	201	34,017	7,094	17,691	0	5
J44	619.lbm.s-4268B	about_383k	one_shot_619_strict_size.policy	59,008	59,003	201	31,517	9,388	17,897	0	5
J39	619.lbm.s-4268B	about_911k	one_shot_619_strict_size.policy	59,008	59,003	201	31,931	7,094	19,777	0	5
J45	619.lbm.s-4268B	about_383k	one_shot_619_strict_size.policy	59,008	59,003	201	31,278	9,388	18,136	0	5
J40	619.lbm.s-4268B	about_911k	one_shot_619_strict_size.policy	59,008	59,003	201	31,676	7,094	20,032	0	5
J28	620.omnetpp.s-874B	about_383k	one_shot_calibration_control	90,280	90,274	67,798	4,182	12,953	2,938	2,403	6
J27	620.omnetpp.s-874B	about_911k	one_shot_calibration_control	90,280	90,274	67,798	4,930	11,524	3,314	2,708	6
J29	620.omnetpp.s-874B	nearest_153k	one_shot_calibration_control	90,280	90,274	67,798	2,815	12,328	3,489	3,844	6

Table 20: Offline decision-ledger summary per job (parsed from `ledger_summary` JSON; counts over validation future-target events) — part 1/2 (cont.)

ID	<i>trace</i>	<i>model_size</i>	<i>stage</i>	<i>events</i>	<i>future-target-events</i>	<i>candidate-absent</i>	<i>target-selected</i>	<i>below-threshold</i>	<i>dedup-lru</i>	<i>rank-cut</i>	<i>unseen</i>
J26	620.omnetpp.s-874B	selected_full	one_shot_calibration_control	90,280	90,274	67,798	6,006	13,058	2,352	1,060	6
J23	620.omnetpp.s-874B	selected_full	one_shot_full_reference	90,280	90,274	67,798	6,006	13,058	2,352	1,060	6
J25	620.omnetpp.s-874B	about_383k	one_shot_strict_size	90,280	90,274	67,798	2,681	18,133	942	720	6
J24	620.omnetpp.s-874B	about_911k	one_shot_strict_size	90,280	90,274	67,798	2,802	18,063	1,035	576	6
J13	623.xalancbmk.s-700B	about_383k	one_shot_calibration_control	243,699	218,408	34,430	34,543	98,048	46,805	4,582	25,291
J12	623.xalancbmk.s-700B	about_911k	one_shot_calibration_control	243,699	218,408	34,430	26,341	117,173	31,012	9,452	25,291
J14	623.xalancbmk.s-700B	nearest_153k	one_shot_calibration_control	243,699	218,408	34,430	19,611	125,272	16,025	23,070	25,291
J11	623.xalancbmk.s-700B	selected_full	one_shot_calibration_control	243,699	218,408	34,430	61,198	17,779	104,824	177	25,291
J09	623.xalancbmk.s-700B	selected_full	one_shot_full_reference	243,699	218,408	34,430	60,669	20,645	102,591	73	25,291
J10	623.xalancbmk.s-700B	about_383k	one_shot_strict_size	243,699	218,408	34,430	12,203	165,221	5,227	1,327	25,291

Table 21: Offline decision-ledger summary per job (parsed from `ledger_summary` JSON; counts over validation future-target events) — part 2/2

ID	<i>present-not-selected</i>
J21	15,386
J20	13,331
J22	13,352
J19	8,810
J15	8,810
J17	15,386
J16	13,331
J18	31,356
J07	12,347
J06	12,510
J08	12,673
J05	12,946
J01	16,242
J03	15,095
J02	13,380
J04	16,242
J43	24,902
J38	24,785
J47	13,346
J31	29,585
J34	31,462
J30	29,585
J35	29,754
J36	29,782
J32	31,243
J33	31,462
J46	27,524
J41	27,126
J42	24,902
J37	24,785
J44	27,285
J39	26,871
J45	27,524
J40	27,126
J28	18,294
J27	17,546
J29	19,661
J26	16,470
J23	16,470
J25	19,795
J24	19,674
J13	149,435

Table 21: Offline decision-ledger summary per job (parsed from `ledger_summary` JSON; counts over validation future-target events) — part 2/2 (cont.)

ID	<i>present_not_selected</i>
J12	157,637
J14	164,367
J11	122,780
J09	123,309
J10	171,775

7 Plan and offline artifacts (`run_artifacts/`)

File-identity notes. `run_artifacts/v4.9_all_job_metadata.csv` is byte-identical to `plan/v4.9_offline_export_acceptance`; `plan/v4.9_combined_replay_plan.csv` is byte-identical to `replay_results/replay_plan.csv` and is exactly this table restricted to the 47 replay-eligible IDs (J01–J47 excluding the seven listed in Table 29); `plan/v4.9_all_job_metadata.csv`, `plan/v4.9_offline_export_acceptance.csv.orig` and `plan/v4.9_one_shot_train_export_manifest.csv` are byte-identical to each other (pre-replay snapshot, pipe-format tags) and are printed as Table 31; `plan/v4.9_combined_replay_plan.csv` is Table 38.

Table 22: `v4.9_all_job_metadata` (final): columns constant across all rows (shown once)

column	constant value
<code>architecture_id</code>	<code>candidate_lstm_ranker_v49.5</code>
<code>architecture_family</code>	<code>two_layer_trace_local_candidate_lstm</code>
<code>implementation_revision</code>	<code>v4.9.5_one_shot_train_export_replay_matrix</code>
<code>layer_count</code>	2
<code>sequence_len</code>	1,024
<code>bank_slots</code>	64
<code>train_fraction</code>	0.8
<code>candidate_rate_limit_filtered_count</code>	0
<code>selected_target_late_proxy_count</code>	0
<code>candidate_table_array_bytes</code>	0
<code>candidate_index_entries</code>	0
<code>candidate_index_bytes_estimate</code>	0
<code>candidate_bank_storage_bytes</code>	0
<code>fallback_profile_table_bytes</code>	0
<code>fallback_emitted_rows</code>	0
<code>fallback_suppressed_dedup_rows</code>	0
<code>fallback_suppressed_rate_limit_rows</code>	0
<code>fallback_invalid_rows</code>	0
<code>ledger_error</code>	–
<code>pre_replay_matrix</code>	1
<code>one_shot_phase</code>	1

Table 23: `run_artifacts/v4.9_all_job_metadata.csv` — final metadata for all 54 planned jobs —

ID	<i>trace</i>	<i>route_id</i>	<i>model_key</i>	<i>stage</i>	<i>policy_mode</i>	<i>model_size</i>	<i>seed</i>
J01	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_full_reference	strict	selected_full	7 4,1
J02	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	about_911k	7 9
J03	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	about_383k	7 38
J04	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	nearest_153k	7 15
J05	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	selected_full	7 4,1
J06	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	about_911k	7 9
J07	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	about_383k	7 38
J08	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	nearest_153k	7 15

Table 23: run_artifacts/v4.9_all_job_metadata.csv — final metadata for all 54 planned jobs — part 1/6

ID	trace	route_id	model_key	stage	policy_mode	model_size	seed
J09	623.xalancbmk.s-700B	context_lru128	623.context_lru128	one_shot_full_reference	strict	selected_full	7 4,14
J10	623.xalancbmk.s-700B	context_lru128	623.context_lru128	one_shot_strict_size	strict	about_383k	7 38
J11	623.xalancbmk.s-700B	context_lru128	623.context_lru128	one_shot_calibration_control	calibration_control	selected_full	7 4,14
J12	623.xalancbmk.s-700B	context_lru128	623.context_lru128	one_shot_calibration_control	calibration_control	about_911k	7 91
J13	623.xalancbmk.s-700B	context_lru128	623.context_lru128	one_shot_calibration_control	calibration_control	about_383k	7 38
J14	623.xalancbmk.s-700B	context_lru128	623.context_lru128	one_shot_calibration_control	calibration_control	nearest_153k	7 15
J15	602.gcc.s-734B	assoc_control	assoc_control	one_shot_full_reference	strict	selected_full	7 4,14
J16	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	about_911k	7 91
J17	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	about_383k	7 38
J18	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	nearest_153k	7 15
J19	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	selected_full	7 4,14
J20	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	about_911k	7 91
J21	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	about_383k	7 38
J22	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	nearest_153k	7 15
J23	620.omnetpp.s-874B	region_control	region_control	one_shot_full_reference	strict	selected_full	7 4,14
J24	620.omnetpp.s-874B	region_control	region_control	one_shot_strict_size	strict	about_911k	7 91
J25	620.omnetpp.s-874B	region_control	region_control	one_shot_strict_size	strict	about_383k	7 38
J26	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	selected_full	7 4,14
J27	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	about_911k	7 91
J28	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	about_383k	7 38
J29	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	nearest_153k	7 15
J30	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,14
J31	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_calibration_control	calibration_control	selected_full	7 4,14
J32	619.lbm.s-4268B	top3	619.timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,14
J33	619.lbm.s-4268B	top4	619.timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,14
J34	619.lbm.s-4268B	adaptive_score_gap	619.timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,14
J35	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_full_policy	strict	selected_full	19 4,14
J36	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_full_policy	strict	selected_full	43 4,14
J37	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 91
J38	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_calibration_control	calibration_control	about_911k	7 91
J39	619.lbm.s-4268B	top3	619.timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 91
J40	619.lbm.s-4268B	top4	619.timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 91
J41	619.lbm.s-4268B	adaptive_score_gap	619.timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 91
J42	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 38
J43	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_calibration_control	calibration_control	about_383k	7 38
J44	619.lbm.s-4268B	top3	619.timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 38
J45	619.lbm.s-4268B	top4	619.timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 38
J46	619.lbm.s-4268B	adaptive_score_gap	619.timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 38
J47	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_calibration_control	calibration_control	nearest_153k	7 15
J48	623.xalancbmk.s-700B	context_lru128	623.context_lru128	one_shot_strict_size	strict	about_911k	7 91
J49	623.xalancbmk.s-700B	context_lru128	623.context_lru128	one_shot_strict_size	strict	nearest_153k	7 15
J50	620.omnetpp.s-874B	region_control	region_control	one_shot_strict_size	strict	nearest_153k	7 15
J51	619.lbm.s-4268B	top2	619.timing_score_model	one_shot_619_strict_size_policy	strict	nearest_153k	7 15
J52	619.lbm.s-4268B	top3	619.timing_score_model	one_shot_619_strict_size_policy	strict	nearest_153k	7 15
J53	619.lbm.s-4268B	top4	619.timing_score_model	one_shot_619_strict_size_policy	strict	nearest_153k	7 15
J54	619.lbm.s-4268B	adaptive_score_gap	619.timing_score_model	one_shot_619_strict_size_policy	strict	nearest_153k	7 15

Table 24: run_artifacts/v4.9_all_job_metadata.csv — final metadata for all 54 planned jobs — part 2/6

ID	hidden_dim	continuous_projection_dim	bank_id	policy_tag	stage_reference_mode	train_end	calibration_events	evaluation_events	selected_accuracy	selected_timely_precision
J01	160	24	support16.fixed_nextline	support16.fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.15917	0.15917
J02	40	8	support16.fixed_nextline	support16.fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.20823	0.20823
J03	40	4	support16.fixed_nextline	support16.fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.20259	0.20259
J04	4	1	support16.fixed_nextline	support16.fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.27242	0.27242

Table 24: run_artifacts/v4.9_all_job_metadata.csv — final metadata for all 54 planned jobs — part 2/6 (cont.)

ID	hidden_dim	continuous_projection_dim	bank_id	policy_tag	stage_reference_mode	train_end	calibration_events	evaluation_events	selected_accuracy	selected_timely_pro
J05	160	24	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage.b	788,092	98,528	98,528	0.12049	0.12049
J06	40	8	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage.b	788,092	98,528	98,528	0.1377	0.1377
J07	40	4	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage.b	788,092	98,528	98,528	0.15124	0.15124
J08	4	1	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage.b	788,092	98,528	98,528	0.15486	0.15486
J09	160	24	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0.73654	0.73654
J10	40	4	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0.65465	0.65465
J11	160	24	context_v40	context_lru128	calibration_control_not_strict_stage.b	974,667	121,849	121,850	0.66306	0.66306
J12	40	8	context_v40	context_lru128	calibration_control_not_strict_stage.b	974,667	121,849	121,850	0.31852	0.31852
J13	40	4	context_v40	context_lru128	calibration_control_not_strict_stage.b	974,667	121,849	121,850	0.3872	0.3872
J14	4	1	context_v40	context_lru128	calibration_control_not_strict_stage.b	974,667	121,849	121,850	0.14131	0.14131
J15	160	24	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.9711	0.9711
J16	40	8	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95088	0.95088
J17	40	4	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95218	0.95218
J18	4	1	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95031	0.95031
J19	160	24	assoc_v40	assoc_control	calibration_control_not_strict_stage.b	323,247	40,422	40,422	0.9711	0.9711
J20	40	8	assoc_v40	assoc_control	calibration_control_not_strict_stage.b	323,247	40,422	40,422	0.95088	0.95088
J21	40	4	assoc_v40	assoc_control	calibration_control_not_strict_stage.b	323,247	40,422	40,422	0.95218	0.95218
J22	4	1	assoc_v40	assoc_control	calibration_control_not_strict_stage.b	323,247	40,422	40,422	0.79469	0.79469
J23	160	24	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.59272	0.59272
J24	40	8	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.63385	0.63385
J25	40	4	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.62023	0.62023
J26	160	24	region_pair_residual_pair	region_control	calibration_control_not_strict_stage.b	360,990	45,140	45,140	0.59272	0.59272
J27	40	8	region_pair_residual_pair	region_control	calibration_control_not_strict_stage.b	360,990	45,140	45,140	0.32748	0.32748
J28	40	4	region_pair_residual_pair	region_control	calibration_control_not_strict_stage.b	360,990	45,140	45,140	0.33811	0.33811
J29	4	1	region_pair_residual_pair	region_control	calibration_control_not_strict_stage.b	360,990	45,140	45,140	0.146	0.146
J30	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99965	0.99965
J31	160	24	v31_timing	top2	calibration_control_not_strict_stage.b	235,901	29,504	29,504	0.99965	0.99965
J32	160	24	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.99971
J33	160	24	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.99971
J34	160	24	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.99971
J35	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99966	0.99966
J36	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99969	0.99969
J37	40	8	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99854	0.99854
J38	40	8	v31_timing	top2	calibration_control_not_strict_stage.b	235,901	29,504	29,504	0.99854	0.99854
J39	40	8	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99856	0.99856
J40	40	8	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99871	0.99871
J41	40	8	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99871	0.99871
J42	40	4	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99855	0.99855
J43	40	4	v31_timing	top2	calibration_control_not_strict_stage.b	235,901	29,504	29,504	0.99855	0.99855
J44	40	4	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99853	0.99853
J45	40	4	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99867	0.99867
J46	40	4	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99867	0.99867
J47	4	1	v31_timing	top2	calibration_control_not_strict_stage.b	235,901	29,504	29,504	0.96866	0.96866
J48	40	8	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0	0
J49	4	1	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0	0
J50	4	1	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0	0
J51	4	1	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0	0
J52	4	1	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0	0
J53	4	1	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0	0
J54	4	1	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0	0

Table 25: run_artifacts/v4.9_all_job_metadata.csv — final metadata for all 54 planned jobs — part 3/6

ID	eval_selected_accuracy	eval_selected_timely_proxy	eval_issue_per_event	candidate_recall_before_policy	correct_rank_recall_top1	correct_rank_recall_top2	correct_rank_recall_top4	candidate_absent_count	candidate_absent_pct	candidate_rank_filtered_count	candidate_threshold_filtered_count
J01	0	0	0.00221	0.56117	0.13896	0.22808	0.34574	386,535	0.43883	101,518	79,474
J02	0.0867	0.0867	0.16752	0.56117	0.15072	0.24306	0.36264	386,535	0.43883	93,458	89,878
J03	0.08332	0.08332	0.06986	0.56117	0.15086	0.24367	0.36318	386,535	0.43883	90,708	86,305
J04	0	0	2.030e-05	0.56117	0.15516	0.24426	0.35841	386,535	0.43883	92,881	210,123
J05	0.0233	0.0233	0.64349	0.56117	0.13896	0.22808	0.34574	386,535	0.43883	107,559	63,014
J06	0.02591	0.02591	0.64448	0.56117	0.15072	0.24306	0.36264	386,535	0.43883	116,981	46,858
J07	0.03323	0.03323	0.53178	0.56117	0.15086	0.24367	0.36318	386,535	0.43883	100,691	64,098
J08	0.02229	0.02229	0.71113	0.56117	0.15516	0.24426	0.35841	386,535	0.43883	112,650	86,873
J09	0.72585	0.72585	0.34311	0.92755	0.81544	0.83443	0.8523	85,499	0.07245	337	147,667
J10	0.64973	0.64973	0.0772	0.92755	0.36947	0.53083	0.69269	85,499	0.07245	7,828	982,569
J11	0.64168	0.64168	0.37995	0.92755	0.81544	0.83443	0.8523	85,499	0.07245	864	138,708
J12	0.32197	0.32197	0.32757	0.92755	0.36709	0.5325	0.70156	85,499	0.07245	56,408	705,723
J13	0.38881	0.38881	0.35494	0.92755	0.36947	0.53083	0.69269	85,499	0.07245	25,757	596,813
J14	0.14168	0.14168	0.55963	0.92755	0.42435	0.51368	0.63446	85,499	0.07245	138,279	750,357
J15	0.96175	0.96175	0.47635	0.50851	0.47902	0.48467	0.49182	198,666	0.49149	0	18,847
J16	0.95398	0.95398	0.47362	0.50851	0.47337	0.48161	0.49048	198,666	0.49149	0	22,159
J17	0.95847	0.95847	0.46848	0.50851	0.47091	0.48049	0.48772	198,666	0.49149	0	24,316
J18	0.96115	0.96115	0.09393	0.50851	0.45802	0.4682	0.47544	198,666	0.49149	0	166,384
J19	0.95789	0.95789	0.47939	0.50851	0.47902	0.48467	0.49182	198,666	0.49149	0	18,847
J20	0.9519	0.9519	0.47472	0.50851	0.47337	0.48161	0.49048	198,666	0.49149	0	22,159
J21	0.95573	0.95573	0.4694	0.50851	0.47091	0.48049	0.48772	198,666	0.49149	0	24,316
J22	0.81348	0.81348	0.49354	0.50851	0.45802	0.4682	0.47544	198,666	0.49149	2	18,260
J23	0.5538	0.5538	0.12013	0.42431	0.15473	0.20531	0.26964	259,864	0.57569	5,163	139,006
J24	0.62087	0.62087	0.04999	0.42431	0.12112	0.17422	0.24312	259,864	0.57569	3,160	166,721
J25	0.60588	0.60588	0.04901	0.42431	0.11133	0.16775	0.23775	259,864	0.57569	3,930	166,825
J26	0.54674	0.54674	0.1218	0.42431	0.15473	0.20531	0.26964	259,864	0.57569	5,163	139,006
J27	0.30516	0.30516	0.1751	0.42431	0.12112	0.17422	0.24312	259,864	0.57569	14,550	130,486
J28	0.31356	0.31356	0.14229	0.42431	0.11133	0.16775	0.23775	259,864	0.57569	13,432	137,595
J29	0.14211	0.14211	0.21887	0.42431	0.07313	0.12421	0.20403	259,864	0.57569	21,089	136,848
J30	0.9993	0.9993	1.61593	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
J31	0.99914	0.99914	1.66469	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
J32	0.99935	0.99935	2.28152	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
J33	0.99918	0.99918	2.91569	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
J34	0.99918	0.99918	2.91569	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
J35	0.99924	0.99924	1.59016	0.99678	0.99539	0.99609	0.99639	950	0.00322	0	47,284
J36	0.99919	0.99919	1.58245	0.99678	0.99539	0.99612	0.99635	950	0.00322	0	48,989
J37	0.9974	0.9974	1.69784	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
J38	0.99809	0.99809	1.73953	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
J39	0.99716	0.99716	2.41688	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
J40	0.99729	0.99729	3.02039	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
J41	0.99729	0.99729	3.02039	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
J42	0.99797	0.99797	1.6096	0.99678	0.99339	0.99519	0.99595	950	0.00322	12	44,423
J43	0.99775	0.99775	1.65818	0.99678	0.99339	0.99519	0.99595	950	0.00322	12	44,423
J44	0.99788	0.99788	2.26207	0.99678	0.99339	0.99519	0.99595	950	0.00322	1	44,423
J45	0.99805	0.99805	2.80281	0.99678	0.99339	0.99519	0.99595	950	0.00322	0	44,423
J46	0.99805	0.99805	2.80281	0.99678	0.99339	0.99519	0.99595	950	0.00322	0	44,423
J47	0.9705	0.9705	1.59375	0.99678	0.96542	0.98665	0.99146	950	0.00322	800	6,155
J48	0	0	0	0.92755	0.36709	0.5325	0.70156	85,499	0.07245	0	1,094,616
J49	0	0	0	0.92755	0.42435	0.51368	0.63446	85,499	0.07245	0	1,094,616
J50	0	0	0	0.42431	0.07313	0.12421	0.20403	259,864	0.57569	0	191,528
J51	0	0	0	0.99678	0.96542	0.98665	0.99146	950	0.00322	0	294,082
J52	0	0	0	0.99678	0.96542	0.98665	0.99146	950	0.00322	0	294,082
J53	0	0	0	0.99678	0.96542	0.98665	0.99146	950	0.00322	0	294,082
J54	0	0	0	0.99678	0.96542	0.98665	0.99146	950	0.00322	0	294,082

Table 26: run_artifacts/v4.9_all_job_metadata.csv — final metadata for all 54 planned jobs — part 4/6

ID	candidate_dedup_filtered_count	exported_rows	exported_list_bytes	checkpoint_bytes	online_non_neural_bytes	online_global_issue_budget	best_val_loss	job_status	zero_action_export	replay_eligible	performance_gate_status
J01	221,568	576,298	127,612,219	16,605,316	4,104	985,276	1.35867	ok	0	1	pending_keyed_replay
J02	206,825	500,058	108,949,608	3,658,820	4,104	985,276	1.12211	ok	0	1	pending_keyed_replay
J03	216,729	496,321	108,273,649	1,544,132	4,104	985,276	1.13896	ok	0	1	pending_keyed_replay
J04	102,668	325,300	70,912,527	625,604	4,104	985,276	1.47089	ok	0	1	pending_keyed_replay
J05	224,409	824,198	179,758,308	16,605,316	4,104	985,276	1.35867	ok	0	1	pending_keyed_replay
J06	221,191	793,436	170,946,726	3,658,820	4,104	985,276	1.12211	ok	0	1	pending_keyed_replay
J07	223,091	703,571	151,698,974	1,544,132	4,104	985,276	1.13896	ok	0	1	pending_keyed_replay
J08	184,488	712,142	153,470,356	625,604	4,104	985,276	1.47089	ok	0	1	pending_keyed_replay
J09	591,250	482,477	101,475,908	16,605,316	2,056	1,218,494	0.31902	ok	0	1	pending_keyed_replay
J10	30,581	112,484	23,753,070	1,544,132	2,056	1,218,494	0.69583	ok	0	1	pending_keyed_replay
J11	599,953	535,537	113,045,533	16,605,316	2,056	1,218,494	0.31902	ok	0	1	pending_keyed_replay
J12	179,387	480,655	103,811,658	3,658,820	2,056	1,218,494	0.63947	ok	0	1	pending_keyed_replay
J13	271,836	517,067	110,952,042	1,544,132	2,056	1,218,494	0.69583	ok	0	1	pending_keyed_replay
J14	93,419	796,561	171,184,755	625,604	2,056	1,218,494	1.31471	ok	0	1	pending_keyed_replay
J15	39,083	189,461	34,643,413	16,605,316	4,104	404,219	0.10358	ok	0	1	pending_keyed_replay
J16	59,599	189,926	35,075,947	3,658,820	4,104	404,219	0.14666	ok	0	1	pending_keyed_replay
J17	65,679	188,865	34,873,796	1,544,132	4,104	404,219	0.15911	ok	0	1	pending_keyed_replay
J18	0	41,212	7,630,295	625,604	4,104	404,219	0.45777	ok	0	1	pending_keyed_replay
J19	39,083	189,461	34,643,413	16,605,316	4,104	404,219	0.10358	ok	0	1	pending_keyed_replay
J20	59,599	189,926	35,075,947	3,658,820	4,104	404,219	0.14666	ok	0	1	pending_keyed_replay
J21	65,679	188,865	34,873,796	1,544,132	4,104	404,219	0.15911	ok	0	1	pending_keyed_replay
J22	64,003	199,519	37,438,601	625,604	4,104	404,219	0.45777	ok	0	1	pending_keyed_replay
J23	13,484	57,152	11,570,139	16,605,316	4,104	112,849	0.6926	ok	0	1	pending_keyed_replay
J24	5,913	24,823	4,976,185	3,658,820	4,104	112,849	0.78336	ok	0	1	pending_keyed_replay
J25	5,421	24,752	4,969,201	1,544,132	4,104	112,849	0.81741	ok	0	1	pending_keyed_replay
J26	13,484	57,152	11,570,139	16,605,316	4,104	112,849	0.6926	ok	0	1	pending_keyed_replay
J27	19,053	83,788	17,208,736	3,658,820	4,104	112,849	0.78336	ok	0	1	pending_keyed_replay
J28	16,770	70,187	14,385,979	1,544,132	4,104	112,849	0.81741	ok	0	1	pending_keyed_replay
J29	18,632	102,458	21,018,122	625,604	4,104	112,849	1.26069	ok	0	1	pending_keyed_replay
J30	0	481,792	89,657,152	16,605,316	8	590,074	0.17532	ok	0	1	pending_keyed_replay
J31	0	481,792	89,657,152	16,605,316	8	590,074	0.17532	ok	0	1	pending_keyed_replay
J32	0	680,908	126,876,970	16,605,316	8	885,111	0.17532	ok	0	1	pending_keyed_replay
J33	0	871,355	162,375,308	16,605,316	8	1,180,148	0.17532	ok	0	1	pending_keyed_replay
J34	0	871,355	167,603,438	16,605,316	8	1,180,148	0.17532	ok	0	1	pending_keyed_replay
J35	0	474,738	89,565,832	16,605,316	8	590,074	0.17681	ok	0	1	pending_keyed_replay
J36	0	473,634	89,252,399	16,605,316	8	590,074	0.17511	ok	0	1	pending_keyed_replay
J37	0	507,607	96,203,648	3,658,820	8	590,074	0.18678	ok	0	1	pending_keyed_replay
J38	0	507,607	96,203,648	3,658,820	8	590,074	0.18678	ok	0	1	pending_keyed_replay
J39	0	723,587	136,952,219	3,658,820	8	885,111	0.18678	ok	0	1	pending_keyed_replay
J40	0	906,048	171,086,197	3,658,820	8	1,180,148	0.18678	ok	0	1	pending_keyed_replay
J41	0	906,048	176,522,485	3,658,820	8	1,180,148	0.18678	ok	0	1	pending_keyed_replay
J42	0	479,746	91,046,602	1,544,132	8	590,074	0.19444	ok	0	1	pending_keyed_replay
J43	0	479,746	91,046,602	1,544,132	8	590,074	0.19444	ok	0	1	pending_keyed_replay
J44	0	674,417	127,565,258	1,544,132	8	885,111	0.19444	ok	0	1	pending_keyed_replay
J45	0	835,216	157,629,119	1,544,132	8	1,180,148	0.19444	ok	0	1	pending_keyed_replay
J46	0	835,216	162,640,415	1,544,132	8	1,180,148	0.19444	ok	0	1	pending_keyed_replay
J47	0	470,007	87,572,266	625,604	8	590,074	0.50708	ok	0	1	pending_keyed_replay
J48	0	0	191	3,658,820	2,056	1,218,494	0.63947	abstain	1	0	abstain_zero_action
J49	0	0	191	625,604	2,056	1,218,494	1.31471	abstain	1	0	abstain_zero_action
J50	0	0	191	625,604	4,104	112,849	1.26069	abstain	1	0	abstain_zero_action
J51	0	0	191	625,604	8	590,074	0.50708	abstain	1	0	abstain_zero_action
J52	0	0	191	625,604	8	885,111	0.50708	abstain	1	0	abstain_zero_action
J53	0	0	191	625,604	8	1,180,148	0.50708	abstain	1	0	abstain_zero_action
J54	0	0	191	625,604	8	1,180,148	0.50708	abstain	1	0	abstain_zero_action

Table 27: run_artifacts/v4.9_all_job_metadata.csv — final metadata for all 54 planned

ID	fail_reason	threshold_recipe	threshold_effective	calibration_stage	calibration_split	calibration_anchor_issue_per_event	calibration
J01	—	0.5	0.5	not_used	none	—	strict_frozen
J02	—	0.5	0.5	not_used	none	—	strict_frozen
J03	—	0.5	0.5	not_used	none	—	strict_frozen
J04	—	0.5	0.5	not_used	none	—	strict_frozen
J05	—	0.5	0.41942	calibrated_scalar_threshold	validation_calibration_prefix	0.58491	fresh_selected_full_strict_me
J06	—	0.5	0.39924	calibrated_scalar_threshold	validation_calibration_prefix	0.58491	fresh_selected_full_strict_me
J07	—	0.5	0.43349	calibrated_scalar_threshold	validation_calibration_prefix	0.58491	fresh_selected_full_strict_me
J08	—	0.5	0.43634	calibrated_scalar_threshold	validation_calibration_prefix	0.58491	fresh_selected_full_strict_me
J09	—	0.65	0.65	not_used	none	—	strict_frozen
J10	—	0.65	0.65	not_used	none	—	strict_frozen
J11	—	0.65	0.57753	calibrated_scalar_threshold	validation_calibration_prefix	0.39596	fresh_selected_full_strict_me
J12	—	0.65	0.58546	calibrated_scalar_threshold	validation_calibration_prefix	0.39596	fresh_selected_full_strict_me
J13	—	0.65	0.57529	calibrated_scalar_threshold	validation_calibration_prefix	0.39596	fresh_selected_full_strict_me
J14	—	0.65	0.50697	calibrated_scalar_threshold	validation_calibration_prefix	0.39596	fresh_selected_full_strict_me
J15	—	0.9	0.9	not_used	none	—	strict_frozen
J16	—	0.9	0.9	not_used	none	—	strict_frozen
J17	—	0.9	0.9	not_used	none	—	strict_frozen
J18	—	0.9	0.9	not_used	none	—	strict_frozen
J19	—	0.9	0.9	calibrated_scalar_threshold	validation_calibration_prefix	0.46871	fresh_selected_full_strict_me
J20	—	0.9	0.9	calibrated_scalar_threshold	validation_calibration_prefix	0.46871	fresh_selected_full_strict_me
J21	—	0.9	0.9	calibrated_scalar_threshold	validation_calibration_prefix	0.46871	fresh_selected_full_strict_me
J22	—	0.9	0.75758	calibrated_scalar_threshold	validation_calibration_prefix	0.46871	fresh_selected_full_strict_me
J23	—	0.65	0.65	not_used	none	—	strict_frozen
J24	—	0.65	0.65	not_used	none	—	strict_frozen
J25	—	0.65	0.65	not_used	none	—	strict_frozen
J26	—	0.65	0.65	calibrated_scalar_threshold	validation_calibration_prefix	0.12661	fresh_selected_full_strict_me
J27	—	0.65	0.56726	calibrated_scalar_threshold	validation_calibration_prefix	0.12661	fresh_selected_full_strict_me
J28	—	0.65	0.55722	calibrated_scalar_threshold	validation_calibration_prefix	0.12661	fresh_selected_full_strict_me
J29	—	0.65	0.53857	calibrated_scalar_threshold	validation_calibration_prefix	0.12661	fresh_selected_full_strict_me
J30	—	0.98557	0.98557	not_used	none	—	strict_frozen
J31	—	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_prefix	1.63299	fresh_selected_full_strict_me
J32	—	0.98557	0.98557	not_used	none	—	strict_frozen
J33	—	0.98557	0.98557	not_used	none	—	strict_frozen
J34	—	0.98557	0.98557	not_used	none	—	strict_frozen
J35	—	0.98557	0.98557	not_used	none	—	strict_frozen
J36	—	0.98557	0.98557	not_used	none	—	strict_frozen
J37	—	0.98557	0.98557	not_used	none	—	strict_frozen
J38	—	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_prefix	1.63299	fresh_selected_full_strict_me
J39	—	0.98557	0.98557	not_used	none	—	strict_frozen
J40	—	0.98557	0.98557	not_used	none	—	strict_frozen
J41	—	0.98557	0.98557	not_used	none	—	strict_frozen
J42	—	0.98557	0.98557	not_used	none	—	strict_frozen
J43	—	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_prefix	1.63299	fresh_selected_full_strict_me
J44	—	0.98557	0.98557	not_used	none	—	strict_frozen
J45	—	0.98557	0.98557	not_used	none	—	strict_frozen
J46	—	0.98557	0.98557	not_used	none	—	strict_frozen
J47	—	0.98557	0.79846	calibrated_scalar_threshold	validation_calibration_prefix	1.63299	fresh_selected_full_strict_me
J48	zero_exported_rows_no_forced_floor	0.65	0.65	not_used	none	—	strict_frozen
J49	zero_exported_rows_no_forced_floor	0.65	0.65	not_used	none	—	strict_frozen
J50	zero_exported_rows_no_forced_floor	0.65	0.65	not_used	none	—	strict_frozen
J51	zero_exported_rows_no_forced_floor	0.98557	0.98557	not_used	none	—	strict_frozen
J52	zero_exported_rows_no_forced_floor	0.98557	0.98557	not_used	none	—	strict_frozen
J53	zero_exported_rows_no_forced_floor	0.98557	0.98557	not_used	none	—	strict_frozen
J54	zero_exported_rows_no_forced_floor	0.98557	0.98557	not_used	none	—	strict_frozen

Table 28: run_artifacts/v4.9.all.job.metadata.csv — final metadata for all 54 planned jobs — part 6/6

ID	<i>calibration_measured_issue_per_event</i>	<i>calibration_target_issue_per_event</i>	<i>calibration_policy_state_scope</i>
J01	—	—	—
J02	—	—	—
J03	—	—	—
J04	—	—	—
J05	0.67166	0.58491	reset_at_calibration_boundary
J06	0.67382	0.58491	reset_at_calibration_boundary
J07	0.57688	0.58491	reset_at_calibration_boundary
J08	0.72308	0.58491	reset_at_calibration_boundary
J09	—	—	—
J10	—	—	—
J11	0.39832	0.39596	reset_at_calibration_boundary
J12	0.34722	0.39596	reset_at_calibration_boundary
J13	0.37662	0.39596	reset_at_calibration_boundary
J14	0.58622	0.39596	reset_at_calibration_boundary
J15	—	—	—
J16	—	—	—
J17	—	—	—
J18	—	—	—
J19	0.47393	0.46871	reset_at_calibration_boundary
J20	0.47291	0.46871	reset_at_calibration_boundary
J21	0.46784	0.46871	reset_at_calibration_boundary
J22	0.49468	0.46871	reset_at_calibration_boundary
J23	—	—	—
J24	—	—	—
J25	—	—	—
J26	0.11845	0.12661	reset_at_calibration_boundary
J27	0.1749	0.12661	reset_at_calibration_boundary
J28	0.14541	0.12661	reset_at_calibration_boundary
J29	0.21777	0.12661	reset_at_calibration_boundary
J30	—	—	—
J31	1.56718	1.63299	reset_at_calibration_boundary
J32	—	—	—
J33	—	—	—
J34	—	—	—
J35	—	—	—
J36	—	—	—
J37	—	—	—
J38	1.65615	1.63299	reset_at_calibration_boundary
J39	—	—	—
J40	—	—	—
J41	—	—	—
J42	—	—	—
J43	1.56101	1.63299	reset_at_calibration_boundary
J44	—	—	—
J45	—	—	—
J46	—	—	—
J47	1.59151	1.63299	reset_at_calibration_boundary
J48	—	—	—
J49	—	—	—
J50	—	—	—
J51	—	—	—
J52	—	—	—
J53	—	—	—
J54	—	—	—

Table 29: The 7 planned jobs excluded from the final replay plan (present in metadata, absent from `combined_replay`)

ID	trace	model_size	stage	policy_tag	job_status	replay_eligible	performance_gate_status
J48	623.xalancbmks-s-700B	about_911k	one_shot_strict_size	context_lru128	abstain	0	abstain_zero_action zero_exported

Table 29: The 7 planned jobs excluded from the final replay plan (present in metadata, absent from combined_replay (cont.)

ID	trace	model_size	stage	policy_tag	job_status	replay_eligible	performance_gate_status	
J49	623.xalancbmk.s-700B	nearest_153k	one_shot_strict_size	context_lru128	abstain	0	abstain_zero_action	zero_exported
J50	620.omnetpp.s-874B	nearest_153k	one_shot_strict_size	region_control	abstain	0	abstain_zero_action	zero_exported
J51	619.lbm.s-4268B	nearest_153k	one_shot_619_strict_size_policy	top2	abstain	0	abstain_zero_action	zero_exported
J52	619.lbm.s-4268B	nearest_153k	one_shot_619_strict_size_policy	top3	abstain	0	abstain_zero_action	zero_exported
J53	619.lbm.s-4268B	nearest_153k	one_shot_619_strict_size_policy	top4	abstain	0	abstain_zero_action	zero_exported
J54	619.lbm.s-4268B	nearest_153k	one_shot_619_strict_size_policy	adaptive_score_gap	abstain	0	abstain_zero_action	zero_exported

Table 30: pre-replay manifest: columns constant across all rows (shown once)

column	constant value
architecture_id	candidate_lstm_ranker_v49.5
architecture_family	two_layer_trace_local_candidate_lstm
implementation_revision	v4.9.5_one_shot_train_export_replay_matrix
layer_count	2
sequence_len	1,024
bank_slots	64
train_fraction	0.8
candidate_rate_limit_filtered_count	0
selected_target_late_proxy_count	0
candidate_table_array_bytes	0
candidate_index_entries	0
candidate_index_bytes_estimate	0
candidate_bank_storage_bytes	0
fallback_profile_table_bytes	0
fallback_emitted_rows	0
fallback_suppressed_dedup_rows	0
fallback_suppressed_rate_limit_rows	0
fallback_invalid_rows	0
ledger_error	—
pre_replay_matrix	1
one_shot_phase	1

Table 31: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) — pre-replay — part 1/6

row	trace	route_id	model_key	stage	policy_mode	model_size	seed
1	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_full_reference	strict	selected_full	7 4,1
2	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	about_911k	7 9
3	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	about_383k	7 3
4	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	nearest_153k	7 1
5	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	selected_full	7 4,1
6	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	about_911k	7 9
7	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	about_383k	7 3
8	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	nearest_153k	7 1
9	623.xalancbmk.s-700B	context_lru128	623_context_lru128	one_shot_full_reference	strict	selected_full	7 4,1
10	623.xalancbmk.s-700B	context_lru128	623_context_lru128	one_shot_strict_size	strict	about_383k	7 3
11	623.xalancbmk.s-700B	context_lru128	623_context_lru128	one_shot_calibration_control	calibration_control	selected_full	7 4,1
12	623.xalancbmk.s-700B	context_lru128	623_context_lru128	one_shot_calibration_control	calibration_control	about_911k	7 9
13	623.xalancbmk.s-700B	context_lru128	623_context_lru128	one_shot_calibration_control	calibration_control	about_383k	7 3
14	623.xalancbmk.s-700B	context_lru128	623_context_lru128	one_shot_calibration_control	calibration_control	nearest_153k	7 1
15	602.gcc.s-734B	assoc_control	assoc_control	one_shot_full_reference	strict	selected_full	7 4,1
16	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	about_911k	7 9
17	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	about_383k	7 3
18	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	nearest_153k	7 1
19	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	selected_full	7 4,1
20	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	about_911k	7 9
21	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	about_383k	7 3
22	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	nearest_153k	7 1
23	620.omnetpp.s-874B	region_control	region_control	one_shot_full_reference	strict	selected_full	7 4,1
24	620.omnetpp.s-874B	region_control	region_control	one_shot_strict_size	strict	about_911k	7 9
25	620.omnetpp.s-874B	region_control	region_control	one_shot_strict_size	strict	about_383k	7 3

Table 31: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) — pre-replay
— part 1/6 (cont.)

row	trace	route_id	model_key	stage	policy_mode	model_size	seed
26	620.omnetpp_s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	selected_full	7 4,1
27	620.omnetpp_s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	about_911k	7 9
28	620.omnetpp_s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	about_383k	7 3
29	620.omnetpp_s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	nearest_153k	7 1
30	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,1
31	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_calibration_control	calibration_control	selected_full	7 4,1
32	619.lbm_s-4268B	top3	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,1
33	619.lbm_s-4268B	top4	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,1
34	619.lbm_s-4268B	adaptive_score_gap	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,1
35	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	19 4,1
36	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	43 4,1
37	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 9
38	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_calibration_control	calibration_control	about_911k	7 9
39	619.lbm_s-4268B	top3	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 9
40	619.lbm_s-4268B	top4	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 9
41	619.lbm_s-4268B	adaptive_score_gap	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 9
42	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 3
43	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_calibration_control	calibration_control	about_383k	7 3
44	619.lbm_s-4268B	top3	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 3
45	619.lbm_s-4268B	top4	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 3
46	619.lbm_s-4268B	adaptive_score_gap	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 3
47	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_calibration_control	calibration_control	nearest_153k	7 1
48	623.xalancbmk_s-700B	context_lru128	623_context_lru128	one_shot_strict_size	strict	about_911k	7 9
49	623.xalancbmk_s-700B	context_lru128	623_context_lru128	one_shot_strict_size	strict	nearest_153k	7 1
50	620.omnetpp_s-874B	region_control	region_control	one_shot_strict_size	strict	nearest_153k	7 1
51	619.lbm_s-4268B	top2	619_timing_score_model	one_shot_619_strict_size_policy	strict	nearest_153k	7 1
52	619.lbm_s-4268B	top3	619_timing_score_model	one_shot_619_strict_size_policy	strict	nearest_153k	7 1
53	619.lbm_s-4268B	top4	619_timing_score_model	one_shot_619_strict_size_policy	strict	nearest_153k	7 1
54	619.lbm_s-4268B	adaptive_score_gap	619_timing_score_model	one_shot_619_strict_size_policy	strict	nearest_153k	7 1

Table 32: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) — pre-replay snapshot, 54 rows
— part 2/6

row	hidden_dim	continuous_projection_dim	bank_id	policy_tag	stage_reference_mode	train_end	calibration_events	evaluation_events	selected_accuracy	selected_timely-pr
1	160	24	support16_fixed_nextline	support16_fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.15917	0.159
2	40	8	support16_fixed_nextline	support16_fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.20823	0.208
3	40	4	support16_fixed_nextline	support16_fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.20259	0.202
4	4	1	support16_fixed_nextline	support16_fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.27242	0.272
5	160	24	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage_b	788,092	98,528	98,528	0.12049	0.120
6	40	8	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage_b	788,092	98,528	98,528	0.1377	0.13
7	40	4	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage_b	788,092	98,528	98,528	0.15124	0.151
8	4	1	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage_b	788,092	98,528	98,528	0.15486	0.154
9	160	24	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0.73654	0.736
10	40	4	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0.65465	0.654
11	160	24	context_v40	context_lru128	calibration_control_not_strict_stage_b	974,667	121,849	121,850	0.66306	0.663
12	40	8	context_v40	context_lru128	calibration_control_not_strict_stage_b	974,667	121,849	121,850	0.31852	0.318
13	40	4	context_v40	context_lru128	calibration_control_not_strict_stage_b	974,667	121,849	121,850	0.3872	0.38
14	4	1	context_v40	context_lru128	calibration_control_not_strict_stage_b	974,667	121,849	121,850	0.14131	0.141
15	160	24	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.9711	0.97
16	40	8	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95088	0.950
17	40	4	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95218	0.952
18	4	1	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95031	0.950
19	160	24	assoc_v40	assoc_control	calibration_control_not_strict_stage_b	323,247	40,422	40,422	0.9711	0.97

Table 32: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) — pre-replay snapshot, 54 rows — part 2/6 (cont.)

row	hidden-dim	continuous-projection-dim	bank-id	policy-tag	stage-reference-mode	train-end	calibration-events	evaluation-events	selected-accuracy	selected-timely-pr
20	40	8	assoc_v40	assoc_control	calibration_control_not_strict_stage_b	323,247	40,422	40,422	0.95088	0.950
21	40	4	assoc_v40	assoc_control	calibration_control_not_strict_stage_b	323,247	40,422	40,422	0.95218	0.952
22	4	1	assoc_v40	assoc_control	calibration_control_not_strict_stage_b	323,247	40,422	40,422	0.79469	0.794
23	160	24	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.59272	0.592
24	40	8	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.63385	0.633
25	40	4	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.62023	0.620
26	160	24	region_pair_residual_pair	region_control	calibration_control_not_strict_stage_b	360,990	45,140	45,140	0.59272	0.592
27	40	8	region_pair_residual_pair	region_control	calibration_control_not_strict_stage_b	360,990	45,140	45,140	0.32748	0.327
28	40	4	region_pair_residual_pair	region_control	calibration_control_not_strict_stage_b	360,990	45,140	45,140	0.33811	0.338
29	4	1	region_pair_residual_pair	region_control	calibration_control_not_strict_stage_b	360,990	45,140	45,140	0.146	0.1
30	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99965	0.999
31	160	24	v31_timing	top2	calibration_control_not_strict_stage_b	235,901	29,504	29,504	0.99965	0.999
32	160	24	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.999
33	160	24	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.999
34	160	24	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.999
35	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99966	0.999
36	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99969	0.999
37	40	8	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99854	0.998
38	40	8	v31_timing	top2	calibration_control_not_strict_stage_b	235,901	29,504	29,504	0.99854	0.998
39	40	8	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99856	0.998
40	40	8	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99871	0.998
41	40	8	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99871	0.998
42	40	4	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99855	0.998
43	40	4	v31_timing	top2	calibration_control_not_strict_stage_b	235,901	29,504	29,504	0.99855	0.998
44	40	4	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99853	0.998
45	40	4	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99867	0.998
46	40	4	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99867	0.998
47	4	1	v31_timing	top2	calibration_control_not_strict_stage_b	235,901	29,504	29,504	0.96866	0.968
48	40	8	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0	
49	4	1	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0	
50	4	1	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0	
51	4	1	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0	
52	4	1	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0	
53	4	1	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0	
54	4	1	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0	

Table 33: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) — pre-replay snapshot, 54 rows — part 3/6

row	eval-selected-accuracy	eval-selected-timely-proxy	eval-issue-per-event	candidate-recall-before-policy	correct-rank-recall-top1	correct-rank-recall-top2	correct-rank-recall-top4	candidate-absent-count	candidate-absent-pct	candidate-rank-filtered-count	candidate-threshold-filtered-count
1	0	0	0.00221	0.56117	0.13896	0.22808	0.34574	386,535	0.43883	101,518	79,474
2	0.0867	0.0867	0.16752	0.56117	0.15072	0.24306	0.36264	386,535	0.43883	93,458	89,878
3	0.08332	0.08332	0.06986	0.56117	0.15086	0.24367	0.36318	386,535	0.43883	90,708	86,305
4	0	0	2.030e-05	0.56117	0.15516	0.24426	0.35841	386,535	0.43883	92,881	210,123
5	0.0233	0.0233	0.64349	0.56117	0.13896	0.22808	0.34574	386,535	0.43883	107,559	63,014
6	0.02591	0.02591	0.64448	0.56117	0.15072	0.24306	0.36264	386,535	0.43883	116,981	46,858
7	0.03323	0.03323	0.53178	0.56117	0.15086	0.24367	0.36318	386,535	0.43883	100,691	64,098
8	0.02229	0.02229	0.71113	0.56117	0.15516	0.24426	0.35841	386,535	0.43883	112,650	86,873
9	0.72585	0.72585	0.34311	0.92755	0.81544	0.83443	0.8523	85,499	0.07245	337	147,667
10	0.64973	0.64973	0.0772	0.92755	0.36947	0.53083	0.69269	85,499	0.07245	7,828	982,569

Table 33: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) — pre-replay snapshot, 54 rows
— part 3/6 (cont.)

row	<i>eval_selected_accuracy</i>	<i>eval_selected_timely_proxy</i>	<i>eval_issue_per_event</i>	<i>candidate_recall_before_policy</i>	<i>correct_rank_recall_top1</i>	<i>correct_rank_recall_top2</i>	<i>correct_rank_recall_top4</i>	<i>candidate_absent_count</i>	<i>candidate_absent_pct</i>	<i>candidate_rank_filtered_count</i>	<i>candidate_threshold_filtered_count</i>
11	0.64168	0.64168	0.37995	0.92755	0.81544	0.83443	0.8523	85,499	0.07245	864	138,708
12	0.32197	0.32197	0.32757	0.92755	0.36709	0.5325	0.70156	85,499	0.07245	56,408	705,723
13	0.38881	0.38881	0.35494	0.92755	0.36947	0.53083	0.69269	85,499	0.07245	25,757	596,813
14	0.14168	0.14168	0.55963	0.92755	0.42435	0.51368	0.63446	85,499	0.07245	138,279	750,357
15	0.96175	0.96175	0.47635	0.50851	0.47902	0.48467	0.49182	198,666	0.49149	0	18,847
16	0.95398	0.95398	0.47362	0.50851	0.47337	0.48161	0.49048	198,666	0.49149	0	22,159
17	0.95847	0.95847	0.46848	0.50851	0.47091	0.48049	0.48772	198,666	0.49149	0	24,316
18	0.96115	0.96115	0.09393	0.50851	0.45802	0.4682	0.47544	198,666	0.49149	0	166,384
19	0.95789	0.95789	0.47939	0.50851	0.47902	0.48467	0.49182	198,666	0.49149	0	18,847
20	0.9519	0.9519	0.47472	0.50851	0.47337	0.48161	0.49048	198,666	0.49149	0	22,159
21	0.95573	0.95573	0.4694	0.50851	0.47091	0.48049	0.48772	198,666	0.49149	0	24,316
22	0.81348	0.81348	0.49354	0.50851	0.45802	0.4682	0.47544	198,666	0.49149	2	18,260
23	0.5538	0.5538	0.12013	0.42431	0.15473	0.20531	0.26964	259,864	0.57569	5,163	139,006
24	0.62087	0.62087	0.04999	0.42431	0.12112	0.17422	0.24312	259,864	0.57569	3,160	166,721
25	0.60588	0.60588	0.04901	0.42431	0.11133	0.16775	0.23775	259,864	0.57569	3,930	166,825
26	0.54674	0.54674	0.1218	0.42431	0.15473	0.20531	0.26964	259,864	0.57569	5,163	139,006
27	0.30516	0.30516	0.1751	0.42431	0.12112	0.17422	0.24312	259,864	0.57569	14,550	130,486
28	0.31356	0.31356	0.14229	0.42431	0.11133	0.16775	0.23775	259,864	0.57569	13,432	137,595
29	0.14211	0.14211	0.21887	0.42431	0.07313	0.12421	0.20403	259,864	0.57569	21,089	136,848
30	0.9993	0.9993	1.61593	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
31	0.99914	0.99914	1.66469	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
32	0.99935	0.99935	2.28152	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
33	0.99918	0.99918	2.91569	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
34	0.99918	0.99918	2.91569	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
35	0.99924	0.99924	1.59016	0.99678	0.99539	0.99609	0.99639	950	0.00322	0	47,284
36	0.99919	0.99919	1.58245	0.99678	0.99539	0.99612	0.99635	950	0.00322	0	48,989
37	0.9974	0.9974	1.69784	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
38	0.99809	0.99809	1.73953	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
39	0.99716	0.99716	2.41688	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
40	0.99729	0.99729	3.02039	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
41	0.99729	0.99729	3.02039	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
42	0.99797	0.99797	1.6096	0.99678	0.99339	0.99519	0.99595	950	0.00322	12	44,423
43	0.99775	0.99775	1.65818	0.99678	0.99339	0.99519	0.99595	950	0.00322	12	44,423
44	0.99788	0.99788	2.26207	0.99678	0.99339	0.99519	0.99595	950	0.00322	1	44,423
45	0.99805	0.99805	2.80281	0.99678	0.99339	0.99519	0.99595	950	0.00322	0	44,423
46	0.99805	0.99805	2.80281	0.99678	0.99339	0.99519	0.99595	950	0.00322	0	44,423
47	0.9705	0.9705	1.59375	0.99678	0.96542	0.98665	0.99146	950	0.00322	800	6,155
48	0	0	0	0.92755	0.36709	0.5325	0.70156	85,499	0.07245	0	1,094,616
49	0	0	0	0.92755	0.42435	0.51368	0.63446	85,499	0.07245	0	1,094,616
50	0	0	0	0.42431	0.07313	0.12421	0.20403	259,864	0.57569	0	191,528
51	0	0	0	0.99678	0.96542	0.98665	0.99146	950	0.00322	0	294,082
52	0	0	0	0.99678	0.96542	0.98665	0.99146	950	0.00322	0	294,082
53	0	0	0	0.99678	0.96542	0.98665	0.99146	950	0.00322	0	294,082
54	0	0	0	0.99678	0.96542	0.98665	0.99146	950	0.00322	0	294,082

Table 34: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) — pre-replay snapshot, 54 rows
— part 4/6

row	candidate-dedup-filtered-count	exported-rows	exported-list-bytes	checkpoint-bytes	online-non-neural-bytes	online-global-issue-budget	best-val-loss	job-status	zero-action-export	replay-eligible	performance-gate-status
1	221,568	576,298	127,612,219	16,605,316	4,104	985,276	1.35867	ok	0	1	pending_keyed_replay
2	206,825	500,058	108,949,608	3,658,820	4,104	985,276	1.12211	ok	0	1	pending_keyed_replay
3	216,729	496,321	108,273,649	1,544,132	4,104	985,276	1.13896	ok	0	1	pending_keyed_replay
4	102,668	325,300	70,912,527	625,604	4,104	985,276	1.47089	ok	0	1	pending_keyed_replay
5	224,409	824,198	179,758,308	16,605,316	4,104	985,276	1.35867	ok	0	1	pending_keyed_replay
6	221,191	793,436	170,946,726	3,658,820	4,104	985,276	1.12211	ok	0	1	pending_keyed_replay
7	223,091	703,571	151,698,974	1,544,132	4,104	985,276	1.13896	ok	0	1	pending_keyed_replay
8	184,488	712,142	153,470,356	625,604	4,104	985,276	1.47089	ok	0	1	pending_keyed_replay
9	591,250	482,477	101,475,908	16,605,316	2,056	1,218,494	0.31902	ok	0	1	pending_keyed_replay
10	30,581	112,484	23,753,070	1,544,132	2,056	1,218,494	0.69583	ok	0	1	pending_keyed_replay
11	599,953	535,537	113,045,533	16,605,316	2,056	1,218,494	0.31902	ok	0	1	pending_keyed_replay
12	179,387	480,655	103,811,658	3,658,820	2,056	1,218,494	0.63947	ok	0	1	pending_keyed_replay
13	271,836	517,067	110,952,042	1,544,132	2,056	1,218,494	0.69583	ok	0	1	pending_keyed_replay
14	93,419	796,561	171,184,755	625,604	2,056	1,218,494	1.31471	ok	0	1	pending_keyed_replay
15	39,083	189,461	34,643,413	16,605,316	4,104	404,219	0.10358	ok	0	1	pending_keyed_replay
16	59,599	189,926	35,075,947	3,658,820	4,104	404,219	0.14666	ok	0	1	pending_keyed_replay
17	65,679	188,865	34,873,796	1,544,132	4,104	404,219	0.15911	ok	0	1	pending_keyed_replay
18	0	41,212	7,630,295	625,604	4,104	404,219	0.45777	ok	0	1	pending_keyed_replay
19	39,083	189,461	34,643,413	16,605,316	4,104	404,219	0.10358	ok	0	1	pending_keyed_replay
20	59,599	189,926	35,075,947	3,658,820	4,104	404,219	0.14666	ok	0	1	pending_keyed_replay
21	65,679	188,865	34,873,796	1,544,132	4,104	404,219	0.15911	ok	0	1	pending_keyed_replay
22	64,003	199,519	37,438,601	625,604	4,104	404,219	0.45777	ok	0	1	pending_keyed_replay
23	13,484	57,152	11,570,139	16,605,316	4,104	112,849	0.6926	ok	0	1	pending_keyed_replay
24	5,913	24,823	4,976,185	3,658,820	4,104	112,849	0.78336	ok	0	1	pending_keyed_replay
25	5,421	24,752	4,969,201	1,544,132	4,104	112,849	0.81741	ok	0	1	pending_keyed_replay
26	13,484	57,152	11,570,139	16,605,316	4,104	112,849	0.6926	ok	0	1	pending_keyed_replay
27	19,053	83,788	17,208,736	3,658,820	4,104	112,849	0.78336	ok	0	1	pending_keyed_replay
28	16,770	70,187	14,385,979	1,544,132	4,104	112,849	0.81741	ok	0	1	pending_keyed_replay
29	18,632	102,458	21,018,122	625,604	4,104	112,849	1.26069	ok	0	1	pending_keyed_replay
30	0	481,792	89,657,152	16,605,316	8	590,074	0.17532	ok	0	1	pending_keyed_replay
31	0	481,792	89,657,152	16,605,316	8	590,074	0.17532	ok	0	1	pending_keyed_replay
32	0	680,908	126,876,970	16,605,316	8	885,111	0.17532	ok	0	1	pending_keyed_replay
33	0	871,355	162,375,308	16,605,316	8	1,180,148	0.17532	ok	0	1	pending_keyed_replay
34	0	871,355	167,603,438	16,605,316	8	1,180,148	0.17532	ok	0	1	pending_keyed_replay
35	0	474,738	89,565,832	16,605,316	8	590,074	0.17681	ok	0	1	pending_keyed_replay
36	0	473,634	89,252,399	16,605,316	8	590,074	0.17511	ok	0	1	pending_keyed_replay
37	0	507,607	96,203,648	3,658,820	8	590,074	0.18678	ok	0	1	pending_keyed_replay
38	0	507,607	96,203,648	3,658,820	8	590,074	0.18678	ok	0	1	pending_keyed_replay
39	0	723,587	136,952,219	3,658,820	8	885,111	0.18678	ok	0	1	pending_keyed_replay
40	0	906,048	171,086,197	3,658,820	8	1,180,148	0.18678	ok	0	1	pending_keyed_replay
41	0	906,048	176,522,485	3,658,820	8	1,180,148	0.18678	ok	0	1	pending_keyed_replay
42	0	479,746	91,046,602	1,544,132	8	590,074	0.19444	ok	0	1	pending_keyed_replay
43	0	479,746	91,046,602	1,544,132	8	590,074	0.19444	ok	0	1	pending_keyed_replay
44	0	674,417	127,565,258	1,544,132	8	885,111	0.19444	ok	0	1	pending_keyed_replay
45	0	835,216	157,629,119	1,544,132	8	1,180,148	0.19444	ok	0	1	pending_keyed_replay
46	0	835,216	162,640,415	1,544,132	8	1,180,148	0.19444	ok	0	1	pending_keyed_replay
47	0	470,007	87,572,266	625,604	8	590,074	0.50708	ok	0	1	pending_keyed_replay
48	0	0	191	3,658,820	2,056	1,218,494	0.63947	abstain	1	0	abstain_zero_action
49	0	0	191	625,604	2,056	1,218,494	1.31471	abstain	1	0	abstain_zero_action
50	0	0	191	625,604	4,104	112,849	1.26069	abstain	1	0	abstain_zero_action
51	0	0	191	625,604	8	590,074	0.50708	abstain	1	0	abstain_zero_action
52	0	0	191	625,604	8	885,111	0.50708	abstain	1	0	abstain_zero_action
53	0	0	191	625,604	8	1,180,148	0.50708	abstain	1	0	abstain_zero_action
54	0	0	191	625,604	8	1,180,148	0.50708	abstain	1	0	abstain_zero_action

Table 35: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) —
— part 5/6

row	fail_reason	threshold_recipe	threshold_effective	calibration_stage	calibration_split	calibration_anchor_issue_per_event	calibrat:
1	—	0.5	0.5	not_used	none	—	strict_frozen
2	—	0.5	0.5	not_used	none	—	strict_frozen
3	—	0.5	0.5	not_used	none	—	strict_frozen
4	—	0.5	0.5	not_used	none	—	strict_frozen
5	—	0.5	0.41942	calibrated_scalar_threshold	validation_calibration_prefix	0.58491	fresh_selected_full_strict_m
6	—	0.5	0.39924	calibrated_scalar_threshold	validation_calibration_prefix	0.58491	fresh_selected_full_strict_m
7	—	0.5	0.43349	calibrated_scalar_threshold	validation_calibration_prefix	0.58491	fresh_selected_full_strict_m
8	—	0.5	0.43634	calibrated_scalar_threshold	validation_calibration_prefix	0.58491	fresh_selected_full_strict_m
9	—	0.65	0.65	not_used	none	—	strict_frozen
10	—	0.65	0.65	not_used	none	—	strict_frozen
11	—	0.65	0.57753	calibrated_scalar_threshold	validation_calibration_prefix	0.39596	fresh_selected_full_strict_m
12	—	0.65	0.58546	calibrated_scalar_threshold	validation_calibration_prefix	0.39596	fresh_selected_full_strict_m
13	—	0.65	0.57529	calibrated_scalar_threshold	validation_calibration_prefix	0.39596	fresh_selected_full_strict_m
14	—	0.65	0.50697	calibrated_scalar_threshold	validation_calibration_prefix	0.39596	fresh_selected_full_strict_m
15	—	0.9	0.9	not_used	none	—	strict_frozen
16	—	0.9	0.9	not_used	none	—	strict_frozen
17	—	0.9	0.9	not_used	none	—	strict_frozen
18	—	0.9	0.9	not_used	none	—	strict_frozen
19	—	0.9	0.9	calibrated_scalar_threshold	validation_calibration_prefix	0.46871	fresh_selected_full_strict_m
20	—	0.9	0.9	calibrated_scalar_threshold	validation_calibration_prefix	0.46871	fresh_selected_full_strict_m
21	—	0.9	0.9	calibrated_scalar_threshold	validation_calibration_prefix	0.46871	fresh_selected_full_strict_m
22	—	0.9	0.75758	calibrated_scalar_threshold	validation_calibration_prefix	0.46871	fresh_selected_full_strict_m
23	—	0.65	0.65	not_used	none	—	strict_frozen
24	—	0.65	0.65	not_used	none	—	strict_frozen
25	—	0.65	0.65	not_used	none	—	strict_frozen
26	—	0.65	0.65	calibrated_scalar_threshold	validation_calibration_prefix	0.12661	fresh_selected_full_strict_m
27	—	0.65	0.56726	calibrated_scalar_threshold	validation_calibration_prefix	0.12661	fresh_selected_full_strict_m
28	—	0.65	0.55722	calibrated_scalar_threshold	validation_calibration_prefix	0.12661	fresh_selected_full_strict_m
29	—	0.65	0.53857	calibrated_scalar_threshold	validation_calibration_prefix	0.12661	fresh_selected_full_strict_m
30	—	0.98557	0.98557	not_used	none	—	strict_frozen
31	—	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_prefix	1.63299	fresh_selected_full_strict_m
32	—	0.98557	0.98557	not_used	none	—	strict_frozen
33	—	0.98557	0.98557	not_used	none	—	strict_frozen
34	—	0.98557	0.98557	not_used	none	—	strict_frozen
35	—	0.98557	0.98557	not_used	none	—	strict_frozen
36	—	0.98557	0.98557	not_used	none	—	strict_frozen
37	—	0.98557	0.98557	not_used	none	—	strict_frozen
38	—	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_prefix	1.63299	fresh_selected_full_strict_m
39	—	0.98557	0.98557	not_used	none	—	strict_frozen
40	—	0.98557	0.98557	not_used	none	—	strict_frozen
41	—	0.98557	0.98557	not_used	none	—	strict_frozen
42	—	0.98557	0.98557	not_used	none	—	strict_frozen
43	—	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_prefix	1.63299	fresh_selected_full_strict_m
44	—	0.98557	0.98557	not_used	none	—	strict_frozen
45	—	0.98557	0.98557	not_used	none	—	strict_frozen
46	—	0.98557	0.98557	not_used	none	—	strict_frozen
47	—	0.98557	0.79846	calibrated_scalar_threshold	validation_calibration_prefix	1.63299	fresh_selected_full_strict_m
48	zero_exported_rows_no_forced_floor	0.65	0.65	not_used	none	—	strict_frozen
49	zero_exported_rows_no_forced_floor	0.65	0.65	not_used	none	—	strict_frozen
50	zero_exported_rows_no_forced_floor	0.65	0.65	not_used	none	—	strict_frozen
51	zero_exported_rows_no_forced_floor	0.98557	0.98557	not_used	none	—	strict_frozen
52	zero_exported_rows_no_forced_floor	0.98557	0.98557	not_used	none	—	strict_frozen
53	zero_exported_rows_no_forced_floor	0.98557	0.98557	not_used	none	—	strict_frozen
54	zero_exported_rows_no_forced_floor	0.98557	0.98557	not_used	none	—	strict_frozen

Table 36: plan/v4.9_all_job_metadata.csv ($\equiv$ export-acceptance.orig $\equiv$ train-export manifest) — pre-replay snapshot, 54 rows
— part 6/6

row	calibration_measured_issue_per_event	calibration_target_issue_per_event	calibration_policy_state_scope
1	—	—	—
2	—	—	—
3	—	—	—
4	—	—	—
5	0.67166	0.58491	reset_at_calibration_boundary
6	0.67382	0.58491	reset_at_calibration_boundary
7	0.57688	0.58491	reset_at_calibration_boundary
8	0.72308	0.58491	reset_at_calibration_boundary
9	—	—	—
10	—	—	—
11	0.39832	0.39596	reset_at_calibration_boundary
12	0.34722	0.39596	reset_at_calibration_boundary
13	0.37662	0.39596	reset_at_calibration_boundary
14	0.58622	0.39596	reset_at_calibration_boundary
15	—	—	—
16	—	—	—
17	—	—	—
18	—	—	—
19	0.47393	0.46871	reset_at_calibration_boundary
20	0.47291	0.46871	reset_at_calibration_boundary
21	0.46784	0.46871	reset_at_calibration_boundary
22	0.49468	0.46871	reset_at_calibration_boundary
23	—	—	—
24	—	—	—
25	—	—	—
26	0.11845	0.12661	reset_at_calibration_boundary
27	0.1749	0.12661	reset_at_calibration_boundary
28	0.14541	0.12661	reset_at_calibration_boundary
29	0.21777	0.12661	reset_at_calibration_boundary
30	—	—	—
31	1.56718	1.63299	reset_at_calibration_boundary
32	—	—	—
33	—	—	—
34	—	—	—
35	—	—	—
36	—	—	—
37	—	—	—
38	1.65615	1.63299	reset_at_calibration_boundary
39	—	—	—
40	—	—	—
41	—	—	—
42	—	—	—
43	1.56101	1.63299	reset_at_calibration_boundary
44	—	—	—
45	—	—	—
46	—	—	—
47	1.59151	1.63299	reset_at_calibration_boundary
48	—	—	—
49	—	—	—
50	—	—	—
51	—	—	—
52	—	—	—
53	—	—	—
54	—	—	—

Table 37: original replay plan: columns constant across all rows (shown once)

column	constant value
architecture_id	candidate_lstm_ranker_v49_5
architecture_family	two_layer_trace_local_candidate_lstm
implementation_revision	v4_9_5_one_shot_train_export_replay_matrix
layer_count	2
sequence_len	1,024
bank_slots	64
train_fraction	0.8
candidate_rate_limit_filtered_count	0
selected_target_late_proxy_count	0
candidate_table_array_bytes	0
candidate_index_entries	0
candidate_index_bytes_estimate	0
candidate_bank_storage_bytes	0
fallback_profile_table_bytes	0
fallback_emitted_rows	0
fallback_suppressed_dedup_rows	0
fallback_suppressed_rate_limit_rows	0
fallback_invalid_rows	0
ledger_error	–
job_status	ok
zero_action_export	0
replay_eligible	1
performance_gate_status	pending_keyed_replay
fail_reason	–
pre_replay_matrix	1
one_shot_phase	1

Table 38: plan/v4_9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows –

row	trace	route_id	model_key	stage	policy_mode	model_size	seed
1	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	about_383k	7 3
2	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	about_911k	7 9
3	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	nearest_153k	7 1
4	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	selected_full	7 4,1
5	602.gcc.s-734B	assoc_control	assoc_control	one_shot_full_reference	strict	selected_full	7 4,1
6	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	about_383k	7 3
7	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	about_911k	7 9
8	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	nearest_153k	7 1
9	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	about_383k	7 3
10	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	about_911k	7 9
11	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	nearest_153k	7 1
12	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_calibration_control	calibration_control	selected_full	7 4,1
13	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_full_reference	strict	selected_full	7 4,1
14	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	about_383k	7 3
15	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	about_911k	7 9
16	605.mcf.s-994B	support16_fixed_nextline	605_support16_fixed_nextline	one_shot_strict_size	strict	nearest_153k	7 1
17	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_calibration_control	calibration_control	about_383k	7 3
18	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_calibration_control	calibration_control	about_911k	7 9
19	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_calibration_control	calibration_control	nearest_153k	7 1
20	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_calibration_control	calibration_control	selected_full	7 4,1
21	619.lbm.s-4268B	adaptive_score_gap	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,1
22	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,1
23	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	19 4,1
24	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	43 4,1
25	619.lbm.s-4268B	top3	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,1
26	619.lbm.s-4268B	top4	619_timing_score_model	one_shot_619_full_policy	strict	selected_full	7 4,1
27	619.lbm.s-4268B	adaptive_score_gap	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 3
28	619.lbm.s-4268B	adaptive_score_gap	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 9
29	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 3
30	619.lbm.s-4268B	top2	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 9
31	619.lbm.s-4268B	top3	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 3
32	619.lbm.s-4268B	top3	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 9
33	619.lbm.s-4268B	top4	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_383k	7 3
34	619.lbm.s-4268B	top4	619_timing_score_model	one_shot_619_strict_size_policy	strict	about_911k	7 9
35	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	about_383k	7 3

Table 38: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — p

row	trace	route_id	model_key	stage	policy_mode	model_size	seed
36	620.omnetpp_s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	about.911k	7 9
37	620.omnetpp_s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	nearest.153k	7 1
38	620.omnetpp_s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	selected_full	7 4,1
39	620.omnetpp_s-874B	region_control	region_control	one_shot_full_reference	strict	selected_full	7 4,1
40	620.omnetpp_s-874B	region_control	region_control	one_shot_strict_size	strict	about.383k	7 3
41	620.omnetpp_s-874B	region_control	region_control	one_shot_strict_size	strict	about.911k	7 9
42	623.xalancbmk_s-700B	context_lru128	623_context_lru128	one_shot_calibration_control	calibration_control	about.383k	7 3
43	623.xalancbmk_s-700B	context_lru128	623_context_lru128	one_shot_calibration_control	calibration_control	about.911k	7 9
44	623.xalancbmk_s-700B	context_lru128	623_context_lru128	one_shot_calibration_control	calibration_control	nearest.153k	7 1
45	623.xalancbmk_s-700B	context_lru128	623_context_lru128	one_shot_calibration_control	calibration_control	selected_full	7 4,1
46	623.xalancbmk_s-700B	context_lru128	623_context_lru128	one_shot_full_reference	strict	selected_full	7 4,1
47	623.xalancbmk_s-700B	context_lru128	623_context_lru128	one_shot_strict_size	strict	about.383k	7 3

Table 39: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — part 2/5

row	hidden_dim	continuous_projection_dim	bank_id	policy_tag	stage-reference_mode	train_end	calibration_events	evaluation_events	selected_accuracy	selected_timely-pr
1	40	4	assoc_v40	assoc_control	calibration_control_not_strict_stage_b	323,247	40,422	40,422	0.95218	0.952
2	40	8	assoc_v40	assoc_control	calibration_control_not_strict_stage_b	323,247	40,422	40,422	0.95088	0.950
3	4	1	assoc_v40	assoc_control	calibration_control_not_strict_stage_b	323,247	40,422	40,422	0.79469	0.794
4	160	24	assoc_v40	assoc_control	calibration_control_not_strict_stage_b	323,247	40,422	40,422	0.9711	0.97
5	160	24	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.9711	0.97
6	40	4	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95218	0.952
7	40	8	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95088	0.950
8	4	1	assoc_v40	assoc_control	strict_frozen_policy	323,247	40,422	80,844	0.95031	0.950
9	40	4	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage_b	788,092	98,528	98,528	0.15124	0.151
10	40	8	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage_b	788,092	98,528	98,528	0.1377	0.13
11	4	1	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage_b	788,092	98,528	98,528	0.15486	0.154
12	160	24	support16_fixed_nextline	support16_fixed_nextline	calibration_control_not_strict_stage_b	788,092	98,528	98,528	0.12049	0.120
13	160	24	support16_fixed_nextline	support16_fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.15917	0.159
14	40	4	support16_fixed_nextline	support16_fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.20259	0.202
15	40	8	support16_fixed_nextline	support16_fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.20823	0.208
16	4	1	support16_fixed_nextline	support16_fixed_nextline	strict_frozen_policy	788,092	98,528	197,056	0.27242	0.272
17	40	4	v31_timing	top2	calibration_control_not_strict_stage_b	235,901	29,504	29,504	0.99855	0.998
18	40	8	v31_timing	top2	calibration_control_not_strict_stage_b	235,901	29,504	29,504	0.99854	0.998
19	4	1	v31_timing	top2	calibration_control_not_strict_stage_b	235,901	29,504	29,504	0.96866	0.968
20	160	24	v31_timing	top2	calibration_control_not_strict_stage_b	235,901	29,504	29,504	0.99965	0.999
21	160	24	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.999
22	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99965	0.999
23	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99966	0.999
24	160	24	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99969	0.999
25	160	24	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.999
26	160	24	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99971	0.999
27	40	4	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99867	0.998
28	40	8	v31_timing	adaptive_score_gap	strict_frozen_policy	235,901	29,504	59,008	0.99871	0.998
29	40	4	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99855	0.998
30	40	8	v31_timing	top2	strict_frozen_policy	235,901	29,504	59,008	0.99854	0.998
31	40	4	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99853	0.998
32	40	8	v31_timing	top3	strict_frozen_policy	235,901	29,504	59,008	0.99856	0.998
33	40	4	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99867	0.998
34	40	8	v31_timing	top4	strict_frozen_policy	235,901	29,504	59,008	0.99871	0.998
35	40	4	region_pair_residual_pair	region_control	calibration_control_not_strict_stage_b	360,990	45,140	45,140	0.33811	0.338
36	40	8	region_pair_residual_pair	region_control	calibration_control_not_strict_stage_b	360,990	45,140	45,140	0.32748	0.327
37	4	1	region_pair_residual_pair	region_control	calibration_control_not_strict_stage_b	360,990	45,140	45,140	0.146	0.1
38	160	24	region_pair_residual_pair	region_control	calibration_control_not_strict_stage_b	360,990	45,140	45,140	0.59272	0.592

Table 39: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — part 2/5 (cont.)

row	hidden_dim	continuous_projection_dim	bank_id	policy_tag	stage_reference_mode	train_end	calibration_events	evaluation_events	selected_accuracy	selected_timely-pr
39	160	24	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.59272	0.592
40	40	4	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.62023	0.620
41	40	8	region_pair_residual_pair	region_control	strict_frozen_policy	360,990	45,140	90,280	0.63385	0.633
42	40	4	context_v40	context_lru128	calibration_control_not_strict_stage_b	974,667	121,849	121,850	0.3872	0.38
43	40	8	context_v40	context_lru128	calibration_control_not_strict_stage_b	974,667	121,849	121,850	0.31852	0.318
44	4	1	context_v40	context_lru128	calibration_control_not_strict_stage_b	974,667	121,849	121,850	0.14131	0.141
45	160	24	context_v40	context_lru128	calibration_control_not_strict_stage_b	974,667	121,849	121,850	0.66306	0.663
46	160	24	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0.73654	0.736
47	40	4	context_v40	context_lru128	strict_frozen_policy	974,667	121,849	243,699	0.65465	0.654

Table 40: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — part 3/5

row	eval_selected_accuracy	eval_selected_timely_proxy	eval_issue_per_event	candidate_recall_before_policy	correct_rank_recall_top1	correct_rank_recall_top2	correct_rank_recall_top4	candidate_absent_count	candidate_absent_pct	candidate_rank_filtered_count	candidate_threshold_filtered_count
1	0.95573	0.95573	0.4694	0.50851	0.47091	0.48049	0.48772	198,666	0.49149	0	24,316
2	0.9519	0.9519	0.47472	0.50851	0.47337	0.48161	0.49048	198,666	0.49149	0	22,159
3	0.81348	0.81348	0.49354	0.50851	0.45802	0.4682	0.47544	198,666	0.49149	2	18,260
4	0.95789	0.95789	0.47939	0.50851	0.47902	0.48467	0.49182	198,666	0.49149	0	18,847
5	0.96175	0.96175	0.47635	0.50851	0.47902	0.48467	0.49182	198,666	0.49149	0	18,847
6	0.95847	0.95847	0.46848	0.50851	0.47091	0.48049	0.48772	198,666	0.49149	0	24,316
7	0.95398	0.95398	0.47362	0.50851	0.47337	0.48161	0.49048	198,666	0.49149	0	22,159
8	0.96115	0.96115	0.09393	0.50851	0.45802	0.4682	0.47544	198,666	0.49149	0	166,384
9	0.03323	0.03323	0.53178	0.56117	0.15086	0.24367	0.36318	386,535	0.43883	100,691	64,098
10	0.02591	0.02591	0.64448	0.56117	0.15072	0.24306	0.36264	386,535	0.43883	116,981	46,858
11	0.02229	0.02229	0.71113	0.56117	0.15516	0.24426	0.35841	386,535	0.43883	112,650	86,873
12	0.0233	0.0233	0.64349	0.56117	0.13896	0.22808	0.34574	386,535	0.43883	107,559	63,014
13	0	0	0.00221	0.56117	0.13896	0.22808	0.34574	386,535	0.43883	101,518	79,474
14	0.08332	0.08332	0.06986	0.56117	0.15086	0.24367	0.36318	386,535	0.43883	90,708	86,305
15	0.0867	0.0867	0.16752	0.56117	0.15072	0.24306	0.36264	386,535	0.43883	93,458	89,878
16	0	0	2.030e-05	0.56117	0.15516	0.24426	0.35841	386,535	0.43883	92,881	210,123
17	0.99775	0.99775	1.65818	0.99678	0.99339	0.99519	0.99595	950	0.00322	12	44,423
18	0.99809	0.99809	1.73953	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
19	0.9705	0.9705	1.59375	0.99678	0.96542	0.98665	0.99146	950	0.00322	800	6,155
20	0.99914	0.99914	1.66469	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
21	0.99918	0.99918	2.91569	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
22	0.9993	0.9993	1.61593	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
23	0.99924	0.99924	1.59016	0.99678	0.99539	0.99609	0.99639	950	0.00322	0	47,284
24	0.99919	0.99919	1.58245	0.99678	0.99539	0.99612	0.99635	950	0.00322	0	48,989
25	0.99935	0.99935	2.28152	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
26	0.99918	0.99918	2.91569	0.99678	0.9952	0.99601	0.99636	950	0.00322	0	44,182
27	0.99805	0.99805	2.80281	0.99678	0.99339	0.99519	0.99595	950	0.00322	0	44,423
28	0.99729	0.99729	3.02039	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
29	0.99797	0.99797	1.6096	0.99678	0.99339	0.99519	0.99595	950	0.00322	12	44,423
30	0.9974	0.9974	1.69784	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
31	0.99788	0.99788	2.26207	0.99678	0.99339	0.99519	0.99595	950	0.00322	1	44,423
32	0.99716	0.99716	2.41688	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
33	0.99805	0.99805	2.80281	0.99678	0.99339	0.99519	0.99595	950	0.00322	0	44,423
34	0.99729	0.99729	3.02039	0.99678	0.99352	0.99556	0.99597	950	0.00322	0	32,369
35	0.31356	0.31356	0.14229	0.42431	0.11133	0.16775	0.23775	259,864	0.57569	13,432	137,595
36	0.30516	0.30516	0.1751	0.42431	0.12112	0.17422	0.24312	259,864	0.57569	14,550	130,486
37	0.14211	0.14211	0.21887	0.42431	0.07313	0.12421	0.20403	259,864	0.57569	21,089	136,848
38	0.54674	0.54674	0.1218	0.42431	0.15473	0.20531	0.26964	259,864	0.57569	5,163	139,006

Table 40: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — part 3/5 (cont.)

row	eval_selected_accuracy	eval_selected_timely_proxy	eval_issue_per_event	candidate_recall_before_policy	correct_rank_recall_top1	correct_rank_recall_top2	correct_rank_recall_top4	candidate_absent_count	candidate_absent_pct	candidate_rank_filtered_count	candidate_threshold_filtered_count
39	0.5538	0.5538	0.12013	0.42431	0.15473	0.20531	0.26964	259,864	0.57569	5,163	139,006
40	0.60588	0.60588	0.04901	0.42431	0.11133	0.16775	0.23775	259,864	0.57569	3,930	166,825
41	0.62087	0.62087	0.04999	0.42431	0.12112	0.17422	0.24312	259,864	0.57569	3,160	166,721
42	0.38881	0.38881	0.35494	0.92755	0.36947	0.53083	0.69269	85,499	0.07245	25,757	596,813
43	0.32197	0.32197	0.32757	0.92755	0.36709	0.5325	0.70156	85,499	0.07245	56,408	705,723
44	0.14168	0.14168	0.55963	0.92755	0.42435	0.51368	0.63446	85,499	0.07245	138,279	750,357
45	0.64168	0.64168	0.37995	0.92755	0.81544	0.83443	0.8523	85,499	0.07245	864	138,708
46	0.72585	0.72585	0.34311	0.92755	0.81544	0.83443	0.8523	85,499	0.07245	337	147,667
47	0.64973	0.64973	0.0772	0.92755	0.36947	0.53083	0.69269	85,499	0.07245	7,828	982,569

Table 41: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — part 4/5

row	candidate_dedup_filtered_count	exported_rows	exported_list_bytes	checkpoint_bytes	online_non_neural_bytes	online_global_issue_budget	best_val_loss	threshold_recipe	threshold_effective	calibration_stage	calibration
1	65,679	188,865	34,873,796	1,544,132	4,104	404,219	0.15911	0.9	0.9	calibrated_scalar_threshold	validation_calibration_pre
2	59,599	189,926	35,075,947	3,658,820	4,104	404,219	0.14666	0.9	0.9	calibrated_scalar_threshold	validation_calibration_pre
3	64,003	199,519	37,438,601	625,604	4,104	404,219	0.45777	0.9	0.75758	calibrated_scalar_threshold	validation_calibration_pre
4	39,083	189,461	34,643,413	16,605,316	4,104	404,219	0.10358	0.9	0.9	calibrated_scalar_threshold	validation_calibration_pre
5	39,083	189,461	34,643,413	16,605,316	4,104	404,219	0.10358	0.9	0.9	not_used	no
6	65,679	188,865	34,873,796	1,544,132	4,104	404,219	0.15911	0.9	0.9	not_used	no
7	59,599	189,926	35,075,947	3,658,820	4,104	404,219	0.14666	0.9	0.9	not_used	no
8	0	41,212	7,630,295	625,604	4,104	404,219	0.45777	0.9	0.9	not_used	no
9	223,091	703,571	151,698,974	1,544,132	4,104	985,276	1.13896	0.5	0.43349	calibrated_scalar_threshold	validation_calibration_pre
10	221,191	793,436	170,946,726	3,658,820	4,104	985,276	1.12211	0.5	0.39924	calibrated_scalar_threshold	validation_calibration_pre
11	184,488	712,142	153,470,356	625,604	4,104	985,276	1.47089	0.5	0.43634	calibrated_scalar_threshold	validation_calibration_pre
12	224,409	824,198	179,758,308	16,605,316	4,104	985,276	1.35867	0.5	0.41942	calibrated_scalar_threshold	validation_calibration_pre
13	221,568	576,298	127,612,219	16,605,316	4,104	985,276	1.35867	0.5	0.5	not_used	no
14	216,729	496,321	108,273,649	1,544,132	4,104	985,276	1.13896	0.5	0.5	not_used	no
15	206,825	500,058	108,949,608	3,658,820	4,104	985,276	1.12211	0.5	0.5	not_used	no
16	102,668	325,300	70,912,527	625,604	4,104	985,276	1.47089	0.5	0.5	not_used	no
17	0	479,746	91,046,602	1,544,132	8	590,074	0.19444	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_pre
18	0	507,607	96,203,648	3,658,820	8	590,074	0.18678	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_pre
19	0	470,007	87,572,266	625,604	8	590,074	0.50708	0.98557	0.79846	calibrated_scalar_threshold	validation_calibration_pre
20	0	481,792	89,657,152	16,605,316	8	590,074	0.17532	0.98557	0.98557	calibrated_scalar_threshold	validation_calibration_pre
21	0	871,355	167,603,438	16,605,316	8	1,180,148	0.17532	0.98557	0.98557	not_used	no
22	0	481,792	89,657,152	16,605,316	8	590,074	0.17532	0.98557	0.98557	not_used	no
23	0	474,738	89,565,832	16,605,316	8	590,074	0.17681	0.98557	0.98557	not_used	no
24	0	473,634	89,252,399	16,605,316	8	590,074	0.17511	0.98557	0.98557	not_used	no
25	0	680,908	126,876,970	16,605,316	8	885,111	0.17532	0.98557	0.98557	not_used	no
26	0	871,355	162,375,308	16,605,316	8	1,180,148	0.17532	0.98557	0.98557	not_used	no
27	0	835,216	162,640,415	1,544,132	8	1,180,148	0.19444	0.98557	0.98557	not_used	no
28	0	906,048	176,522,485	3,658,820	8	1,180,148	0.18678	0.98557	0.98557	not_used	no
29	0	479,746	91,046,602	1,544,132	8	590,074	0.19444	0.98557	0.98557	not_used	no
30	0	507,607	96,203,648	3,658,820	8	590,074	0.18678	0.98557	0.98557	not_used	no
31	0	674,417	127,565,258	1,544,132	8	885,111	0.19444	0.98557	0.98557	not_used	no
32	0	723,587	136,952,219	3,658,820	8	885,111	0.18678	0.98557	0.98557	not_used	no
33	0	835,216	157,629,119	1,544,132	8	1,180,148	0.19444	0.98557	0.98557	not_used	no
34	0	906,048	171,086,197	3,658,820	8	1,180,148	0.18678	0.98557	0.98557	not_used	no
35	16,770	70,187	14,385,979	1,544,132	4,104	112,849	0.81741	0.65	0.55722	calibrated_scalar_threshold	validation_calibration_pre
36	19,053	83,788	17,208,736	3,658,820	4,104	112,849	0.78336	0.65	0.56726	calibrated_scalar_threshold	validation_calibration_pre

Table 41: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — part 4/5 (cont.)

row	candidate_dedup_filtered_count	exported_rows	exported_list_bytes	checkpoint_bytes	online_non_neural_bytes	online_global_issue_budget	best_val_loss	threshold_recipe	threshold_effective	calibration_stage	calibration_recipe
37	18,632	102,458	21,018,122	625,604	4,104	112,849	1.26069	0.65	0.53857	calibrated_scalar_threshold	validation_calibration_recipe
38	13,484	57,152	11,570,139	16,605,316	4,104	112,849	0.6926	0.65	0.65	calibrated_scalar_threshold	validation_calibration_recipe
39	13,484	57,152	11,570,139	16,605,316	4,104	112,849	0.6926	0.65	0.65	not_used	not_used
40	5,421	24,752	4,969,201	1,544,132	4,104	112,849	0.81741	0.65	0.65	not_used	not_used
41	5,913	24,823	4,976,185	3,658,820	4,104	112,849	0.78336	0.65	0.65	not_used	not_used
42	271,836	517,067	110,952,042	1,544,132	2,056	1,218,494	0.69583	0.65	0.57529	calibrated_scalar_threshold	validation_calibration_recipe
43	179,387	480,655	103,811,658	3,658,820	2,056	1,218,494	0.63947	0.65	0.58546	calibrated_scalar_threshold	validation_calibration_recipe
44	93,419	796,561	171,184,755	625,604	2,056	1,218,494	1.31471	0.65	0.50697	calibrated_scalar_threshold	validation_calibration_recipe
45	599,953	535,537	113,045,533	16,605,316	2,056	1,218,494	0.31902	0.65	0.57753	calibrated_scalar_threshold	validation_calibration_recipe
46	591,250	482,477	101,475,908	16,605,316	2,056	1,218,494	0.31902	0.65	0.65	not_used	not_used
47	30,581	112,484	23,753,070	1,544,132	2,056	1,218,494	0.69583	0.65	0.65	not_used	not_used

Table 42: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — part 5/5

row	calibration_anchor_issue_per_event	calibration_anchor_source	calibration_policy_events	calibration_total_suffix_events	experiment_lane	calibration_candidate_count	calibration_measured_issue_per_event	calibration_target_issue_per_event
1	0.46871	fresh_selected_full_strict_metadata	40,422	40,422	B_representation_coverage_repair	14	0.46784	0.46871
2	0.46871	fresh_selected_full_strict_metadata	40,422	40,422	B_representation_coverage_repair	14	0.47291	0.46871
3	0.46871	fresh_selected_full_strict_metadata	40,422	40,422	B_representation_coverage_repair	6	0.49468	0.46871
4	0.46871	fresh_selected_full_strict_metadata	40,422	40,422	B_representation_coverage_repair	14	0.47393	0.46871
5	—	strict_frozen_policy	0	0	B_representation_coverage_repair	—	—	—
6	—	strict_frozen_policy	0	0	B_representation_coverage_repair	—	—	—
7	—	strict_frozen_policy	0	0	B_representation_coverage_repair	—	—	—
8	—	strict_frozen_policy	0	0	B_representation_coverage_repair	—	—	—
9	0.58491	fresh_selected_full_strict_metadata	50,000	98,528	A_reproduce_then_strict_compress	14	0.57688	0.58491
10	0.58491	fresh_selected_full_strict_metadata	50,000	98,528	A_reproduce_then_strict_compress	14	0.67382	0.58491
11	0.58491	fresh_selected_full_strict_metadata	50,000	98,528	A_reproduce_then_strict_compress	14	0.72308	0.58491
12	0.58491	fresh_selected_full_strict_metadata	50,000	98,528	A_reproduce_then_strict_compress	14	0.67166	0.58491
13	—	strict_frozen_policy	0	0	A_reproduce_then_strict_compress	—	—	—
14	—	strict_frozen_policy	0	0	A_reproduce_then_strict_compress	—	—	—
15	—	strict_frozen_policy	0	0	A_reproduce_then_strict_compress	—	—	—
16	—	strict_frozen_policy	0	0	A_reproduce_then_strict_compress	—	—	—
17	1.63299	fresh_selected_full_strict_metadata	29,504	29,504	C_fixed_score_timing_policy	14	1.56101	1.63299
18	1.63299	fresh_selected_full_strict_metadata	29,504	29,504	C_fixed_score_timing_policy	14	1.65615	1.63299
19	1.63299	fresh_selected_full_strict_metadata	29,504	29,504	C_fixed_score_timing_policy	14	1.59151	1.63299
20	1.63299	fresh_selected_full_strict_metadata	29,504	29,504	C_fixed_score_timing_policy	14	1.56718	1.63299
21	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
22	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
23	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
24	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
25	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
26	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
27	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
28	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
29	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
30	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
31	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
32	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—
33	—	strict_frozen_policy	0	0	C_fixed_score_timing_policy	—	—	—

Table 42: plan/v4.9_combined_replay_plan.csv.orig — original (pre-final) replay plan, 47 rows — part 5/5 (con

row	calibration_anchor_issue_per_event		calibration_anchor_source	calibration_policy_events	calibration_total_suffix_events		experiment_lane	calibration_candidate_count	calibration_measured_issue_per_event	calibration_target_issue_per_event
34	—		strict_frozen_policy	0	0	C.fixed_score_timing_policy	—	—		
35	0.12661	fresh_selected_full_strict_metadata		45,140	45,140	B_representation_coverage_repair	14	0.14541	0.12661	0.12661
36	0.12661	fresh_selected_full_strict_metadata		45,140	45,140	B_representation_coverage_repair	14	0.1749	0.12661	0.12661
37	0.12661	fresh_selected_full_strict_metadata		45,140	45,140	B_representation_coverage_repair	14	0.21777	0.12661	0.12661
38	0.12661	fresh_selected_full_strict_metadata		45,140	45,140	B_representation_coverage_repair	14	0.11845	0.12661	0.12661
39	—		strict_frozen_policy	0	0	B_representation_coverage_repair	—	—		
40	—		strict_frozen_policy	0	0	B_representation_coverage_repair	—	—		
41	—		strict_frozen_policy	0	0	B_representation_coverage_repair	—	—		
42	0.39596	fresh_selected_full_strict_metadata		50,000	121,849	A_reproduce_then_strict_compress	14	0.37662	0.39596	0.39596
43	0.39596	fresh_selected_full_strict_metadata		50,000	121,849	A_reproduce_then_strict_compress	14	0.34722	0.39596	0.39596
44	0.39596	fresh_selected_full_strict_metadata		50,000	121,849	A_reproduce_then_strict_compress	14	0.58622	0.39596	0.39596
45	0.39596	fresh_selected_full_strict_metadata		50,000	121,849	A_reproduce_then_strict_compress	14	0.39832	0.39596	0.39596
46	—		strict_frozen_policy	0	0	A_reproduce_then_strict_compress	—	—		
47	—		strict_frozen_policy	0	0	A_reproduce_then_strict_compress	—	—		

Table 43: plan/v4.9_offline_causal_diagnostics.csv — offline causal diagnostics for all jobs — part 1/5

ID	trace	route_id	architecture_id	model_size	seed	policy_emits	issue_per_event	selected_accuracy	selected_timely_proxy	candidate_recall_before_policy
—	605.mcf.s-994B	support16_fixed_nextline	candidate_lstm_ranker_v49.5	selected_full	7	576,298	0.58491	0.15917	0.15917	0.56117
—	605.mcf.s-994B	support16_fixed_nextline	candidate_lstm_ranker_v49.5	about_911k	7	500,058	0.50753	0.20823	0.20823	0.56117
—	605.mcf.s-994B	support16_fixed_nextline	candidate_lstm_ranker_v49.5	about_383k	7	496,321	0.50374	0.20259	0.20259	0.56117
—	605.mcf.s-994B	support16_fixed_nextline	candidate_lstm_ranker_v49.5	nearest_153k	7	325,300	0.33016	0.27242	0.27242	0.56117
—	605.mcf.s-994B	support16_fixed_nextline	candidate_lstm_ranker_v49.5	selected_full	7	824,198	0.83651	0.12049	0.12049	0.56117
—	605.mcf.s-994B	support16_fixed_nextline	candidate_lstm_ranker_v49.5	about_911k	7	793,436	0.80529	0.1377	0.1377	0.56117
—	605.mcf.s-994B	support16_fixed_nextline	candidate_lstm_ranker_v49.5	about_383k	7	703,571	0.71409	0.15124	0.15124	0.56117
—	605.mcf.s-994B	support16_fixed_nextline	candidate_lstm_ranker_v49.5	nearest_153k	7	712,142	0.72278	0.15486	0.15486	0.56117
—	623.xalancbm.s-700B	context_lru128	candidate_lstm_ranker_v49.5	selected_full	7	482,477	0.39596	0.73654	0.73654	0.92755
—	623.xalancbm.s-700B	context_lru128	candidate_lstm_ranker_v49.5	about_383k	7	112,484	0.09231	0.65465	0.65465	0.92755
—	623.xalancbm.s-700B	context_lru128	candidate_lstm_ranker_v49.5	selected_full	7	535,537	0.43951	0.66306	0.66306	0.92755
—	623.xalancbm.s-700B	context_lru128	candidate_lstm_ranker_v49.5	about_911k	7	480,655	0.39447	0.31852	0.31852	0.92755
—	623.xalancbm.s-700B	context_lru128	candidate_lstm_ranker_v49.5	about_383k	7	517,067	0.42435	0.3872	0.3872	0.92755
—	623.xalancbm.s-700B	context_lru128	candidate_lstm_ranker_v49.5	nearest_153k	7	796,561	0.65373	0.14131	0.14131	0.92755
—	602.gcc.s-734B	assoc_control	candidate_lstm_ranker_v49.5	selected_full	7	189,461	0.46871	0.9711	0.9711	0.50851
—	602.gcc.s-734B	assoc_control	candidate_lstm_ranker_v49.5	about_911k	7	189,926	0.46986	0.95088	0.95088	0.50851
—	602.gcc.s-734B	assoc_control	candidate_lstm_ranker_v49.5	about_383k	7	188,865	0.46723	0.95218	0.95218	0.50851
—	602.gcc.s-734B	assoc_control	candidate_lstm_ranker_v49.5	nearest_153k	7	41,212	0.10195	0.95031	0.95031	0.50851
—	602.gcc.s-734B	assoc_control	candidate_lstm_ranker_v49.5	selected_full	7	189,461	0.46871	0.9711	0.9711	0.50851
—	602.gcc.s-734B	assoc_control	candidate_lstm_ranker_v49.5	about_911k	7	189,926	0.46986	0.95088	0.95088	0.50851
—	602.gcc.s-734B	assoc_control	candidate_lstm_ranker_v49.5	about_383k	7	188,865	0.46723	0.95218	0.95218	0.50851
—	602.gcc.s-734B	assoc_control	candidate_lstm_ranker_v49.5	nearest_153k	7	199,519	0.49359	0.79469	0.79469	0.50851
—	620.omnetpp.s-874B	region_control	candidate_lstm_ranker_v49.5	selected_full	7	57,152	0.12661	0.59272	0.59272	0.42431
—	620.omnetpp.s-874B	region_control	candidate_lstm_ranker_v49.5	about_911k	7	24,823	0.05499	0.63385	0.63385	0.42431
—	620.omnetpp.s-874B	region_control	candidate_lstm_ranker_v49.5	about_383k	7	24,752	0.05483	0.62023	0.62023	0.42431
—	620.omnetpp.s-874B	region_control	candidate_lstm_ranker_v49.5	selected_full	7	57,152	0.12661	0.59272	0.59272	0.42431
—	620.omnetpp.s-874B	region_control	candidate_lstm_ranker_v49.5	about_911k	7	83,788	0.18562	0.32748	0.32748	0.42431
—	620.omnetpp.s-874B	region_control	candidate_lstm_ranker_v49.5	about_383k	7	70,187	0.15549	0.33811	0.33811	0.42431
—	620.omnetpp.s-874B	region_control	candidate_lstm_ranker_v49.5	nearest_153k	7	102,458	0.22698	0.146	0.146	0.42431
—	619.lbm.s-4268B	top2	candidate_lstm_ranker_v49.5	selected_full	7	481,792	1.63299	0.99965	0.99965	0.99678
—	619.lbm.s-4268B	top2	candidate_lstm_ranker_v49.5	selected_full	7	481,792	1.63299	0.99965	0.99965	0.99678

Table 43: plan/v4.9_offline_causal_diagnostics.csv — offline causal diagnostics for all jobs — part 1/5 (cont.)

ID	trace	route_id	architecture_id	model_size	seed	policy_emits	issue_per_event	selected_accuracy	selected_timely_proxy	candidate_recall_before_policy
—	619.lbm_s-4268B	top3	candidate_lstm_ranker_v49.5	selected_full	7	680,908	2.30787	0.99971	0.99971	0.99678
—	619.lbm_s-4268B	top4	candidate_lstm_ranker_v49.5	selected_full	7	871,355	2.95338	0.99971	0.99971	0.99678
—	619.lbm_s-4268B	adaptive_score_gap	candidate_lstm_ranker_v49.5	selected_full	7	871,355	2.95338	0.99971	0.99971	0.99678
—	619.lbm_s-4268B	top2	candidate_lstm_ranker_v49.5	selected_full	19	474,738	1.60908	0.99966	0.99966	0.99678
—	619.lbm_s-4268B	top2	candidate_lstm_ranker_v49.5	selected_full	43	473,634	1.60534	0.99969	0.99969	0.99678
—	619.lbm_s-4268B	top2	candidate_lstm_ranker_v49.5	about_911k	7	507,607	1.72049	0.99854	0.99854	0.99678
—	619.lbm_s-4268B	top2	candidate_lstm_ranker_v49.5	about_911k	7	507,607	1.72049	0.99854	0.99854	0.99678
—	619.lbm_s-4268B	top3	candidate_lstm_ranker_v49.5	about_911k	7	723,587	2.45253	0.99856	0.99856	0.99678
—	619.lbm_s-4268B	top4	candidate_lstm_ranker_v49.5	about_911k	7	906,048	3.07096	0.99871	0.99871	0.99678
—	619.lbm_s-4268B	adaptive_score_gap	candidate_lstm_ranker_v49.5	about_911k	7	906,048	3.07096	0.99871	0.99871	0.99678
—	619.lbm_s-4268B	top2	candidate_lstm_ranker_v49.5	about_383k	7	479,746	1.62605	0.99855	0.99855	0.99678
—	619.lbm_s-4268B	top2	candidate_lstm_ranker_v49.5	about_383k	7	479,746	1.62605	0.99855	0.99855	0.99678
—	619.lbm_s-4268B	top3	candidate_lstm_ranker_v49.5	about_383k	7	674,417	2.28587	0.99853	0.99853	0.99678
—	619.lbm_s-4268B	top4	candidate_lstm_ranker_v49.5	about_383k	7	835,216	2.83089	0.99867	0.99867	0.99678
—	619.lbm_s-4268B	adaptive_score_gap	candidate_lstm_ranker_v49.5	about_383k	7	835,216	2.83089	0.99867	0.99867	0.99678
—	619.lbm_s-4268B	top2	candidate_lstm_ranker_v49.5	nearest_153k	7	470,007	1.59304	0.96866	0.96866	0.99678
—	623.xalancbm_k-s-700B	context_lru128	candidate_lstm_ranker_v49.5	about_911k	7	0	0	0	0	0.92755
—	623.xalancbm_k-s-700B	context_lru128	candidate_lstm_ranker_v49.5	nearest_153k	7	0	0	0	0	0.92755
—	620.omnetpp-s-874B	region_control	candidate_lstm_ranker_v49.5	nearest_153k	7	0	0	0	0	0.42431
—	619.lbm_s-4268B	top2	candidate_lstm_ranker_v49.5	nearest_153k	7	0	0	0	0	0.99678
—	619.lbm_s-4268B	top3	candidate_lstm_ranker_v49.5	nearest_153k	7	0	0	0	0	0.99678
—	619.lbm_s-4268B	top4	candidate_lstm_ranker_v49.5	nearest_153k	7	0	0	0	0	0.99678
—	619.lbm_s-4268B	adaptive_score_gap	candidate_lstm_ranker_v49.5	nearest_153k	7	0	0	0	0	0.99678

Table 44: plan/v4.9_offline_causal_diagnostics.csv — offline causal diagnostics for all jobs — part 2/5

ID	dedup_skips	invalid_skips	threshold_skips	degree_skips	rate_limit_skips	max_issue_budget	exported_useful_proxy	exported_timely_proxy	events	future_target_events
—	1,661,563	301	56,311,054	4,508,448	0	985,276	91,730	91,730	985,276	880,825
—	1,418,460	4	57,722,376	3,416,766	0	985,276	104,129	104,129	985,276	880,825
—	1,454,917	307	57,643,956	3,462,163	0	985,276	100,548	100,548	985,276	880,825
—	702,873	0	59,634,972	2,394,519	0	985,276	88,618	88,618	985,276	880,825
—	1,723,450	7,223	55,111,497	5,391,296	0	985,276	99,308	99,308	985,276	880,825
—	1,661,874	1,376	54,901,250	5,699,728	0	985,276	109,260	109,260	985,276	880,825
—	1,633,679	8,100	56,412,056	4,300,258	0	985,276	106,410	106,410	985,276	880,825
—	1,200,478	3,839	57,004,681	4,136,524	0	985,276	110,279	110,279	985,276	880,825
—	5,169,922	1	72,138,777	192,439	0	1,218,494	355,362	355,362	1,218,494	1,180,115
—	206,852	85	76,976,560	687,635	0	1,218,494	73,638	73,638	1,218,494	1,180,115
—	6,089,376	88	71,008,254	350,361	0	1,218,494	355,091	355,091	1,218,494	1,180,115
—	534,363	7,339	75,102,860	1,858,399	0	1,218,494	153,098	153,098	1,218,494	1,180,115
—	1,728,689	13,099	70,890,706	4,834,055	0	1,218,494	200,210	200,210	1,218,494	1,180,115
—	208,579	6,127	75,154,505	1,817,844	0	1,218,494	112,561	112,561	1,218,494	1,180,115
—	1,213,772	0	24,463,454	3,329	0	404,219	183,985	183,985	404,219	404,214
—	873,037	0	24,805,774	1,279	0	404,219	180,597	180,597	404,219	404,214
—	845,484	0	24,834,372	1,295	0	404,219	179,833	179,833	404,219	404,214
—	5	0	25,828,799	0	0	404,219	39,164	39,164	404,219	404,214
—	1,213,772	0	24,463,454	3,329	0	404,219	183,985	183,985	404,219	404,214
—	873,037	0	24,805,774	1,279	0	404,219	180,597	180,597	404,219	404,214
—	845,484	0	24,834,372	1,295	0	404,219	179,833	179,833	404,219	404,214
—	708,912	0	24,961,199	386	0	404,219	158,556	158,556	404,219	404,214
—	44,516	7	28,692,845	94,952	0	112,849	33,875	33,875	451,398	451,392
—	16,149	0	28,802,900	45,600	0	112,849	15,734	15,734	451,398	451,392
—	18,385	5	28,781,577	64,753	0	112,849	15,352	15,352	451,398	451,392
—	44,516	7	28,692,845	94,952	0	112,849	33,875	33,875	451,398	451,392
—	95,247	747	28,207,071	502,619	0	112,849	27,439	27,439	451,398	451,392

Table 44: plan/v4.9_offline_causal_diagnostics.csv — offline causal diagnostics for all jobs — part 2/5 (cont.)

ID	dedup_skips	invalid_skips	threshold_skips	degree_skips	rate_limit_skips	max_issue_budget	exported_useful_proxy	exported_timely_proxy	events	future_target_events
—	85,564	1,039	28,195,821	536,861	0	112,849	23,731	23,731	451,398	451,392
—	113,092	53	28,246,026	427,843	0	112,849	14,959	14,959	451,398	451,392
—	0	0	17,797,647	602,929	0	590,074	481,624	481,624	295,037	295,032
—	0	0	17,797,647	602,929	0	590,074	481,624	481,624	295,037	295,032
—	0	0	17,797,647	403,813	0	885,111	680,711	680,711	295,037	295,032
—	0	0	17,797,647	213,366	0	1,180,148	871,102	871,102	295,037	295,032
—	0	0	17,797,647	213,366	0	1,180,148	871,102	871,102	295,037	295,032
—	0	0	17,838,417	569,213	0	590,074	474,576	474,576	295,037	295,032
—	0	0	17,842,767	565,967	0	590,074	473,487	473,487	295,037	295,032
—	0	0	17,807,960	566,801	0	590,074	506,868	506,868	295,037	295,032
—	0	0	17,807,960	566,801	0	590,074	506,868	506,868	295,037	295,032
—	0	0	17,807,960	350,821	0	885,111	722,542	722,542	295,037	295,032
—	0	0	17,807,960	168,360	0	1,180,148	904,879	904,879	295,037	295,032
—	0	0	17,807,960	168,360	0	1,180,148	904,879	904,879	295,037	295,032
—	0	0	17,926,989	475,633	0	590,074	479,052	479,052	295,037	295,032
—	0	0	17,926,989	475,633	0	590,074	479,052	479,052	295,037	295,032
—	0	0	17,926,989	280,962	0	885,111	673,425	673,425	295,037	295,032
—	0	0	17,926,989	120,163	0	1,180,148	834,103	834,103	295,037	295,032
—	0	0	17,926,989	120,163	0	1,180,148	834,103	834,103	295,037	295,032
—	0	0	18,215,554	196,807	0	590,074	455,278	455,278	295,037	295,032
—	0	0	77,983,616	0	0	1,218,494	0	0	1,218,494	1,180,115
—	0	0	77,983,616	0	0	1,218,494	0	0	1,218,494	1,180,115
—	0	0	28,889,472	0	0	112,849	0	0	451,398	451,392
—	0	0	18,882,368	0	0	590,074	0	0	295,037	295,032
—	0	0	18,882,368	0	0	885,111	0	0	295,037	295,032
—	0	0	18,882,368	0	0	1,180,148	0	0	295,037	295,032
—	0	0	18,882,368	0	0	1,180,148	0	0	295,037	295,032

Table 45: plan/v4.9_offline_causal_diagnostics.csv — offline causal diagnostics for all jobs

ID	candidate_absent	candidate_rank_filtered	candidate_threshold_filtered	candidate_degree_filtered	candidate_dedup_filtered	candidate_rate_limit_filtered	candidate_invalid_filtered	target_selected	selected_target_proxy
—	386,535	101,518	79,474	101,518	221,568	0	0	91,730	0 {"below_threshold": 79474, "candidate_absent": 101518}
—	386,535	93,458	89,878	93,458	206,825	0	0	104,129	0 {"below_threshold": 89878, "candidate_absent": 93458}
—	386,535	90,708	86,305	90,708	216,729	0	0	100,548	0 {"below_threshold": 86305, "candidate_absent": 90708}
—	386,535	92,881	210,123	92,881	102,668	0	0	88,618	0 {"below_threshold": 210123, "candidate_absent": 92881}
—	386,535	107,559	63,014	107,559	224,409	0	0	99,308	0 {"below_threshold": 63014, "candidate_absent": 107559}
—	386,535	116,981	46,858	116,981	221,191	0	0	109,260	0 {"below_threshold": 46858, "candidate_absent": 116981}
—	386,535	100,691	64,098	100,691	223,091	0	0	106,410	0 {"below_threshold": 64098, "candidate_absent": 100691}
—	386,535	112,650	86,873	112,650	184,488	0	0	110,279	0 {"below_threshold": 86873, "candidate_absent": 112650}
—	85,499	337	147,667	337	591,250	0	0	355,362	0 {"below_threshold": 147667, "candidate_absent": 337}
—	85,499	7,828	982,569	7,828	30,581	0	0	73,638	0 {"below_threshold": 982569, "candidate_absent": 7828}
—	85,499	864	138,708	864	599,953	0	0	355,091	0 {"below_threshold": 138708, "candidate_absent": 864}
—	85,499	56,408	705,723	56,408	179,387	0	0	153,098	0 {"below_threshold": 705723, "candidate_absent": 56408}
—	85,499	25,757	596,813	25,757	271,836	0	0	200,210	0 {"below_threshold": 596813, "candidate_absent": 25757}
—	85,499	138,279	750,357	138,279	93,419	0	0	112,561	0 {"below_threshold": 750357, "candidate_absent": 138279}
—	198,666	0	18,847	0	39,083	0	0	147,618	0 {"below_threshold": 18847, "candidate_absent": 0}
—	198,666	0	22,159	0	59,599	0	0	123,790	0 {"below_threshold": 22159, "candidate_absent": 0}
—	198,666	0	24,316	0	65,679	0	0	115,553	0 {"below_threshold": 24316, "candidate_absent": 0}
—	198,666	0	166,384	0	0	0	0	39,164	0 {"below_threshold": 166384, "candidate_absent": 0}
—	198,666	0	18,847	0	39,083	0	0	147,618	0 {"below_threshold": 18847, "candidate_absent": 0}
—	198,666	0	22,159	0	59,599	0	0	123,790	0 {"below_threshold": 22159, "candidate_absent": 0}
—	198,666	0	24,316	0	65,679	0	0	115,553	0 {"below_threshold": 24316, "candidate_absent": 0}
—	198,666	2	18,260	2	64,003	0	0	123,283	0 {"below_threshold": 18260, "candidate_absent": 2}
—	259,864	5,163	139,006	5,163	13,484	0	0	33,875	0 {"below_threshold": 139006, "candidate_absent": 5163}
—	259,864	3,160	166,721	3,160	5,913	0	0	15,734	0 {"below_threshold": 166721, "candidate_absent": 3160}

Table 45: plan/v4.9_offline-causal-diagnostics.csv — offline causal diagnostics for all jobs —

ID	<i>candidate_absent</i>	<i>candidate_rank_filtered</i>	<i>candidate_threshold_filtered</i>	<i>candidate_degree_filtered</i>	<i>candidate_dedup_filtered</i>	<i>candidate_rate_limit_filtered</i>	<i>candidate_invalid_filtered</i>	<i>target_selected</i>	<i>selected_target_late_proxy</i>	
–	259,864	3,930	166,825	3,930	5,421	0	0	15,352	0	{"below_threshold": 166825, "candidate_absent"
–	259,864	5,163	139,006	5,163	13,484	0	0	33,875	0	{"below_threshold": 139006, "candidate_absent"
–	259,864	14,550	130,486	14,550	19,053	0	0	27,439	0	{"below_threshold": 130486, "candidate_absent"
–	259,864	13,432	137,595	13,432	16,770	0	0	23,731	0	{"below_threshold": 137595, "candidate_absent"
–	259,864	21,089	136,848	21,089	18,632	0	0	14,959	0	{"below_threshold": 136848, "candidate_absent"
–	950	0	44,182	0	0	0	0	249,900	0	
–	950	0	44,182	0	0	0	0	249,900	0	
–	950	0	44,182	0	0	0	0	249,900	0	
–	950	0	44,182	0	0	0	0	249,900	0	
–	950	0	44,182	0	0	0	0	249,900	0	
–	950	0	47,284	0	0	0	0	246,798	0	
–	950	0	48,989	0	0	0	0	245,093	0	
–	950	0	32,369	0	0	0	0	261,713	0	
–	950	0	32,369	0	0	0	0	261,713	0	
–	950	0	32,369	0	0	0	0	261,713	0	
–	950	0	32,369	0	0	0	0	261,713	0	
–	950	0	32,369	0	0	0	0	261,713	0	
–	950	12	44,423	12	0	0	0	249,647	0	{"below_threshold":
–	950	12	44,423	12	0	0	0	249,647	0	{"below_threshold":
–	950	1	44,423	1	0	0	0	249,658	0	{"below_threshold":
–	950	0	44,423	0	0	0	0	249,659	0	
–	950	0	44,423	0	0	0	0	249,659	0	
–	950	800	6,155	800	0	0	0	287,127	0	{"below_threshold":
–	85,499	0	1,094,616	0	0	0	0	0	0	
–	85,499	0	1,094,616	0	0	0	0	0	0	
–	259,864	0	191,528	0	0	0	0	0	0	
–	950	0	294,082	0	0	0	0	0	0	
–	950	0	294,082	0	0	0	0	0	0	
–	950	0	294,082	0	0	0	0	0	0	
–	950	0	294,082	0	0	0	0	0	0	

Table 46: plan/v4.9_offline-causal-diagnostics.csv — offline causal diagnostics for all jobs — part 4/5

ID	<i>candidate_absent_pct</i>	<i>candidate_rank_filtered_pct</i>	<i>candidate_threshold_filtered_pct</i>	<i>candidate_degree_filtered_pct</i>	<i>candidate_dedup_filtered_pct</i>	<i>candidate_rate_limit_filtered_pct</i>	<i>candidate_invalid_filtered_pct</i>	<i>target_selected_pct</i>	<i>selected_target_late_proxy_pct</i>	<i>target_events</i>
–	0.43883	0.11525	0.09023	0.11525	0.25155	0	0	0.10414	0	880,825
–	0.43883	0.1061	0.10204	0.1061	0.23481	0	0	0.11822	0	880,825
–	0.43883	0.10298	0.09798	0.10298	0.24605	0	0	0.11415	0	880,825
–	0.43883	0.10545	0.23855	0.10545	0.11656	0	0	0.10061	0	880,825
–	0.43883	0.12211	0.07154	0.12211	0.25477	0	0	0.11274	0	880,825
–	0.43883	0.13281	0.0532	0.13281	0.25112	0	0	0.12404	0	880,825
–	0.43883	0.11431	0.07277	0.11431	0.25328	0	0	0.12081	0	880,825
–	0.43883	0.12789	0.09863	0.12789	0.20945	0	0	0.1252	0	880,825
–	0.07245	2.856e-04	0.12513	2.856e-04	0.50101	0	0	0.30112	0	1,180,115
–	0.07245	0.00663	0.8326	0.00663	0.02591	0	0	0.0624	0	1,180,115
–	0.07245	7.321e-04	0.11754	7.321e-04	0.50839	0	0	0.3009	0	1,180,115
–	0.07245	0.0478	0.59801	0.0478	0.15201	0	0	0.12973	0	1,180,115
–	0.07245	0.02183	0.50572	0.02183	0.23035	0	0	0.16965	0	1,180,115
–	0.07245	0.11717	0.63583	0.11717	0.07916	0	0	0.09538	0	1,180,115
–	0.49149	0	0.04663	0	0.09669	0	0	0.3652	0	404,214
–	0.49149	0	0.05482	0	0.14744	0	0	0.30625	0	404,214
–	0.49149	0	0.06016	0	0.16249	0	0	0.28587	0	404,214

Table 46: plan/v4.9_offline_causal_diagnostics.csv — offline causal diagnostics for all jobs — part 4/5 (cont.)

ID	<i>candidate_absent_pct</i>	<i>candidate_rank_filtered_pct</i>	<i>candidate_threshold_filtered_pct</i>	<i>candidate_degree_filtered_pct</i>	<i>candidate_dedup_filtered_pct</i>	<i>candidate_rate_limit_filtered_pct</i>	<i>candidate_invalid_filtered_pct</i>	<i>target_selected_pct</i>	<i>selected_target_late_proxy_pct</i>	<i>target_events</i>
–	0.49149	0	0.41162	0	0	0	0	0.09689	0	404,214
–	0.49149	0	0.04663	0	0.09669	0	0	0.3652	0	404,214
–	0.49149	0	0.05482	0	0.14744	0	0	0.30625	0	404,214
–	0.49149	0	0.06016	0	0.16249	0	0	0.28587	0	404,214
–	0.49149	4.948e-06	0.04517	4.948e-06	0.15834	0	0	0.30499	0	404,214
–	0.57569	0.01144	0.30795	0.01144	0.02987	0	0	0.07505	0	451,392
–	0.57569	0.007	0.36935	0.007	0.0131	0	0	0.03486	0	451,392
–	0.57569	0.00871	0.36958	0.00871	0.01201	0	0	0.03401	0	451,392
–	0.57569	0.01144	0.30795	0.01144	0.02987	0	0	0.07505	0	451,392
–	0.57569	0.03223	0.28907	0.03223	0.04221	0	0	0.06079	0	451,392
–	0.57569	0.02976	0.30482	0.02976	0.03715	0	0	0.05257	0	451,392
–	0.57569	0.04672	0.30317	0.04672	0.04128	0	0	0.03314	0	451,392
–	0.00322	0	0.14975	0	0	0	0	0.84703	0	295,032
–	0.00322	0	0.14975	0	0	0	0	0.84703	0	295,032
–	0.00322	0	0.14975	0	0	0	0	0.84703	0	295,032
–	0.00322	0	0.14975	0	0	0	0	0.84703	0	295,032
–	0.00322	0	0.14975	0	0	0	0	0.84703	0	295,032
–	0.00322	0	0.16027	0	0	0	0	0.83651	0	295,032
–	0.00322	0	0.16605	0	0	0	0	0.83073	0	295,032
–	0.00322	0	0.10971	0	0	0	0	0.88707	0	295,032
–	0.00322	0	0.10971	0	0	0	0	0.88707	0	295,032
–	0.00322	0	0.10971	0	0	0	0	0.88707	0	295,032
–	0.00322	0	0.10971	0	0	0	0	0.88707	0	295,032
–	0.00322	0	0.10971	0	0	0	0	0.88707	0	295,032
–	0.00322	4.067e-05	0.15057	4.067e-05	0	0	0	0.84617	0	295,032
–	0.00322	4.067e-05	0.15057	4.067e-05	0	0	0	0.84617	0	295,032
–	0.00322	3.389e-06	0.15057	3.389e-06	0	0	0	0.84621	0	295,032
–	0.00322	0	0.15057	0	0	0	0	0.84621	0	295,032
–	0.00322	0	0.15057	0	0	0	0	0.84621	0	295,032
–	0.00322	0.00271	0.02086	0.00271	0	0	0	0.97321	0	295,032
–	0.07245	0	0.92755	0	0	0	0	0	0	1,180,115
–	0.07245	0	0.92755	0	0	0	0	0	0	1,180,115
–	0.57569	0	0.42431	0	0	0	0	0	0	451,392
–	0.00322	0	0.99678	0	0	0	0	0	0	295,032
–	0.00322	0	0.99678	0	0	0	0	0	0	295,032
–	0.00322	0	0.99678	0	0	0	0	0	0	295,032
–	0.00322	0	0.99678	0	0	0	0	0	0	295,032

Table 47: plan/v4.9_offline_causal_diagnostics.csv — offline causal diagnostics for all jobs — part 5/5

ID	<i>candidate_absent_count</i>	<i>correct_rank_recall_top1</i>	<i>correct_rank_recall_top2</i>	<i>correct_rank_recall_top4</i>	<i>rank_recall_before_policy</i>
–	386,535	0.13896	0.22808	0.34574	0.56117
–	386,535	0.15072	0.24306	0.36264	0.56117
–	386,535	0.15086	0.24367	0.36318	0.56117
–	386,535	0.15516	0.24426	0.35841	0.56117
–	386,535	0.13896	0.22808	0.34574	0.56117
–	386,535	0.15072	0.24306	0.36264	0.56117
–	386,535	0.15086	0.24367	0.36318	0.56117
–	386,535	0.15516	0.24426	0.35841	0.56117
–	85,499	0.81544	0.83443	0.8523	0.92755
–	85,499	0.36947	0.53083	0.69269	0.92755
–	85,499	0.81544	0.83443	0.8523	0.92755
–	85,499	0.36709	0.5325	0.70156	0.92755

Table 47: plan/v4.9_offline_causal_diagnostics.csv — offline causal diagnostics for all jobs — part 5/5 (cont.)

ID	<i>candidate_absent_count</i>	<i>correct_rank_recall_top1</i>	<i>correct_rank_recall_top2</i>	<i>correct_rank_recall_top4</i>	<i>rank_recall_before_policy</i>
—	85,499	0.36947	0.53083	0.69269	0.92755
—	85,499	0.42435	0.51368	0.63446	0.92755
—	198,666	0.47902	0.48467	0.49182	0.50851
—	198,666	0.47337	0.48161	0.49048	0.50851
—	198,666	0.47091	0.48049	0.48772	0.50851
—	198,666	0.45802	0.4682	0.47544	0.50851
—	198,666	0.47902	0.48467	0.49182	0.50851
—	198,666	0.47337	0.48161	0.49048	0.50851
—	198,666	0.47091	0.48049	0.48772	0.50851
—	198,666	0.45802	0.4682	0.47544	0.50851
—	259,864	0.15473	0.20531	0.26964	0.42431
—	259,864	0.12112	0.17422	0.24312	0.42431
—	259,864	0.11133	0.16775	0.23775	0.42431
—	259,864	0.15473	0.20531	0.26964	0.42431
—	259,864	0.12112	0.17422	0.24312	0.42431
—	259,864	0.11133	0.16775	0.23775	0.42431
—	259,864	0.07313	0.12421	0.20403	0.42431
—	950	0.9952	0.99601	0.99636	0.99678
—	950	0.9952	0.99601	0.99636	0.99678
—	950	0.9952	0.99601	0.99636	0.99678
—	950	0.9952	0.99601	0.99636	0.99678
—	950	0.9952	0.99601	0.99636	0.99678
—	950	0.99539	0.99609	0.99639	0.99678
—	950	0.99539	0.99612	0.99635	0.99678
—	950	0.99352	0.99556	0.99597	0.99678
—	950	0.99352	0.99556	0.99597	0.99678
—	950	0.99352	0.99556	0.99597	0.99678
—	950	0.99352	0.99556	0.99597	0.99678
—	950	0.99352	0.99556	0.99597	0.99678
—	950	0.99339	0.99519	0.99595	0.99678
—	950	0.99339	0.99519	0.99595	0.99678
—	950	0.99339	0.99519	0.99595	0.99678
—	950	0.99339	0.99519	0.99595	0.99678
—	950	0.99339	0.99519	0.99595	0.99678
—	950	0.96542	0.98665	0.99146	0.99678
—	85,499	0.36709	0.5325	0.70156	0.92755
—	85,499	0.42435	0.51368	0.63446	0.92755
—	259,864	0.07313	0.12421	0.20403	0.42431
—	950	0.96542	0.98665	0.99146	0.99678
—	950	0.96542	0.98665	0.99146	0.99678
—	950	0.96542	0.98665	0.99146	0.99678
—	950	0.96542	0.98665	0.99146	0.99678

Table 48: parameter/storage accounting: columns constant across all rows (shown once)

column	constant value
architecture_id	candidate_lstm_ranker_v49_5
layer_count	2
sequence_len	1,024
bank_slots	64
candidate_table_array_bytes	0
candidate_index_entries	0
candidate_index_bytes_estimate	0
candidate_bank_storage_bytes	0
fallback_profile_table_bytes	0
fallback_emitted_rows	0
fallback_suppressed_dedup_rows	0
fallback_suppressed_rate_limit_rows	0
fallback_invalid_rows	0

Table 49: plan/v4.9_parameter_storage_table.csv — neural parameters and non-neural storage accounting per job —

ID	trace	route_id	model_key	stage	policy_mode	model_size	seed
—	605.mcf.s-994B	support16.fixed_nextline	605.support16.fixed_nextline	one_shot_full_reference	strict	selected_full	7 4,14
—	605.mcf.s-994B	support16.fixed_nextline	605.support16.fixed_nextline	one_shot_strict_size	strict	about.911k	7 91
—	605.mcf.s-994B	support16.fixed_nextline	605.support16.fixed_nextline	one_shot_strict_size	strict	about.383k	7 38
—	605.mcf.s-994B	support16.fixed_nextline	605.support16.fixed_nextline	one_shot_strict_size	strict	nearest.153k	7 15
—	605.mcf.s-994B	support16.fixed_nextline	605.support16.fixed_nextline	one_shot_calibration_control	calibration_control	selected_full	7 4,14
—	605.mcf.s-994B	support16.fixed_nextline	605.support16.fixed_nextline	one_shot_calibration_control	calibration_control	about.911k	7 91
—	605.mcf.s-994B	support16.fixed_nextline	605.support16.fixed_nextline	one_shot_calibration_control	calibration_control	about.383k	7 38
—	605.mcf.s-994B	support16.fixed_nextline	605.support16.fixed_nextline	one_shot_calibration_control	calibration_control	nearest.153k	7 15
—	623.xalancbmk.s-700B	context.lru128	623.context.lru128	one_shot_full_reference	strict	selected_full	7 4,14
—	623.xalancbmk.s-700B	context.lru128	623.context.lru128	one_shot_strict_size	strict	about.383k	7 38
—	623.xalancbmk.s-700B	context.lru128	623.context.lru128	one_shot_calibration_control	calibration_control	selected_full	7 4,14
—	623.xalancbmk.s-700B	context.lru128	623.context.lru128	one_shot_calibration_control	calibration_control	about.911k	7 91
—	623.xalancbmk.s-700B	context.lru128	623.context.lru128	one_shot_calibration_control	calibration_control	about.383k	7 38
—	623.xalancbmk.s-700B	context.lru128	623.context.lru128	one_shot_calibration_control	calibration_control	nearest.153k	7 15
—	602.gcc.s-734B	assoc_control	assoc_control	one_shot_full_reference	strict	selected_full	7 4,14
—	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	about.911k	7 91
—	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	about.383k	7 38
—	602.gcc.s-734B	assoc_control	assoc_control	one_shot_strict_size	strict	nearest.153k	7 15
—	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	selected_full	7 4,14
—	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	about.911k	7 91
—	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	about.383k	7 38
—	602.gcc.s-734B	assoc_control	assoc_control	one_shot_calibration_control	calibration_control	nearest.153k	7 15
—	620.omnetpp.s-874B	region_control	region_control	one_shot_full_reference	strict	selected_full	7 4,14
—	620.omnetpp.s-874B	region_control	region_control	one_shot_strict_size	strict	about.911k	7 91
—	620.omnetpp.s-874B	region_control	region_control	one_shot_strict_size	strict	about.383k	7 38
—	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	selected_full	7 4,14
—	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	about.911k	7 91
—	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	about.383k	7 38
—	620.omnetpp.s-874B	region_control	region_control	one_shot_calibration_control	calibration_control	nearest.153k	7 15
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_full_policy	strict	selected_full	7 4,14
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_calibration_control	calibration_control	selected_full	7 4,14
—	619.lbm.s-4268B	top3	619.timing_score_model	one_shot.619_full_policy	strict	selected_full	7 4,14
—	619.lbm.s-4268B	top4	619.timing_score_model	one_shot.619_full_policy	strict	selected_full	7 4,14
—	619.lbm.s-4268B	adaptive_score_gap	619.timing_score_model	one_shot.619_full_policy	strict	selected_full	7 4,14
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_full_policy	strict	selected_full	19 4,14
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_full_policy	strict	selected_full	43 4,14
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_strict_size_policy	strict	about.911k	7 91
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_calibration_control	calibration_control	about.911k	7 91
—	619.lbm.s-4268B	top3	619.timing_score_model	one_shot.619_strict_size_policy	strict	about.911k	7 91
—	619.lbm.s-4268B	top4	619.timing_score_model	one_shot.619_strict_size_policy	strict	about.911k	7 91
—	619.lbm.s-4268B	adaptive_score_gap	619.timing_score_model	one_shot.619_strict_size_policy	strict	about.911k	7 91
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_strict_size_policy	strict	about.383k	7 38
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_calibration_control	calibration_control	about.383k	7 38
—	619.lbm.s-4268B	top3	619.timing_score_model	one_shot.619_strict_size_policy	strict	about.383k	7 38
—	619.lbm.s-4268B	top4	619.timing_score_model	one_shot.619_strict_size_policy	strict	about.383k	7 38
—	619.lbm.s-4268B	adaptive_score_gap	619.timing_score_model	one_shot.619_strict_size_policy	strict	about.383k	7 38
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_calibration_control	calibration_control	nearest.153k	7 15
—	623.xalancbmk.s-700B	context.lru128	623.context.lru128	one_shot_strict_size	strict	about.911k	7 91
—	623.xalancbmk.s-700B	context.lru128	623.context.lru128	one_shot_strict_size	strict	nearest.153k	7 15
—	620.omnetpp.s-874B	region_control	region_control	one_shot_strict_size	strict	nearest.153k	7 15
—	619.lbm.s-4268B	top2	619.timing_score_model	one_shot.619_strict_size_policy	strict	nearest.153k	7 15
—	619.lbm.s-4268B	top3	619.timing_score_model	one_shot.619_strict_size_policy	strict	nearest.153k	7 15
—	619.lbm.s-4268B	top4	619.timing_score_model	one_shot.619_strict_size_policy	strict	nearest.153k	7 15
—	619.lbm.s-4268B	adaptive_score_gap	619.timing_score_model	one_shot.619_strict_size_policy	strict	nearest.153k	7 15

Table 50: plan/v4.9_parameter_storage_table.csv — neural parameters and non-neural storage accounting per job — part 2/2

ID	<i>continuous-projection-dim</i>	<i>checkpoint-bytes</i>	<i>exported-rows</i>	<i>exported-list-bytes</i>	<i>online-non-neural-bytes</i>	<i>online-global-issue-budget</i>
—	24	16,605,316	576,298	127,612,219	4,104	985,276
—	8	3,658,820	500,058	108,949,608	4,104	985,276
—	4	1,544,132	496,321	108,273,649	4,104	985,276
—	1	625,604	325,300	70,912,527	4,104	985,276
—	24	16,605,316	824,198	179,758,308	4,104	985,276
—	8	3,658,820	793,436	170,946,726	4,104	985,276
—	4	1,544,132	703,571	151,698,974	4,104	985,276
—	1	625,604	712,142	153,470,356	4,104	985,276
—	24	16,605,316	482,477	101,475,908	2,056	1,218,494
—	4	1,544,132	112,484	23,753,070	2,056	1,218,494
—	24	16,605,316	535,537	113,045,533	2,056	1,218,494
—	8	3,658,820	480,655	103,811,658	2,056	1,218,494
—	4	1,544,132	517,067	110,952,042	2,056	1,218,494
—	1	625,604	796,561	171,184,755	2,056	1,218,494
—	24	16,605,316	189,461	34,643,413	4,104	404,219
—	8	3,658,820	189,926	35,075,947	4,104	404,219
—	4	1,544,132	188,865	34,873,796	4,104	404,219
—	1	625,604	41,212	7,630,295	4,104	404,219
—	24	16,605,316	189,461	34,643,413	4,104	404,219
—	8	3,658,820	189,926	35,075,947	4,104	404,219
—	4	1,544,132	188,865	34,873,796	4,104	404,219
—	1	625,604	199,519	37,438,601	4,104	404,219
—	24	16,605,316	57,152	11,570,139	4,104	112,849
—	8	3,658,820	24,823	4,976,185	4,104	112,849
—	4	1,544,132	24,752	4,969,201	4,104	112,849
—	24	16,605,316	57,152	11,570,139	4,104	112,849
—	8	3,658,820	83,788	17,208,736	4,104	112,849
—	4	1,544,132	70,187	14,385,979	4,104	112,849
—	1	625,604	102,458	21,018,122	4,104	112,849
—	24	16,605,316	481,792	89,657,152	8	590,074
—	24	16,605,316	481,792	89,657,152	8	590,074
—	24	16,605,316	680,908	126,876,970	8	885,111
—	24	16,605,316	871,355	162,375,308	8	1,180,148
—	24	16,605,316	871,355	167,603,438	8	1,180,148
—	24	16,605,316	474,738	89,565,832	8	590,074
—	24	16,605,316	473,634	89,252,399	8	590,074
—	8	3,658,820	507,607	96,203,648	8	590,074
—	8	3,658,820	507,607	96,203,648	8	590,074
—	8	3,658,820	723,587	136,952,219	8	885,111
—	8	3,658,820	906,048	171,086,197	8	1,180,148
—	8	3,658,820	906,048	176,522,485	8	1,180,148
—	4	1,544,132	479,746	91,046,602	8	590,074
—	4	1,544,132	479,746	91,046,602	8	590,074
—	4	1,544,132	674,417	127,565,258	8	885,111
—	4	1,544,132	835,216	157,629,119	8	1,180,148
—	4	1,544,132	835,216	162,640,415	8	1,180,148
—	1	625,604	470,007	87,572,266	8	590,074
—	8	3,658,820	0	191	2,056	1,218,494
—	1	625,604	0	191	2,056	1,218,494
—	1	625,604	0	191	4,104	112,849
—	1	625,604	0	191	8	590,074
—	1	625,604	0	191	8	885,111
—	1	625,604	0	191	8	1,180,148
—	1	625,604	0	191	8	1,180,148

Table 51: plan/v4.9_representation_screen.csv — candidate-bank representation screen (held-out recall gate before any training) — part 1/2

trace/route	base_bank	candidate_sources	candidate_bank_size
602.gcc.s-734B assoc_control	assoc.v40	v31_pc_delta;v31_pc_pair;v31_global_delta;v31_global_pair;fixed;assoc_abs	0
602.gcc.s-734B pc_stride_reach	assoc.v40	v31_pc_delta;v31_pc_pair;v31_global_delta;v31_global_pair;fixed;profiled_pc_stride	0
602.gcc.s-734B dual_stride_reach	assoc.v40	v31_pc_delta;v31_pc_pair;v31_global_delta;v31_global_pair;profiled_pc_stride;profiled_global_stride	0
620.omnetpp.s-874B region_control	region_pair_residual_pair	ctx3;phase;offset;predecessor;pc;region_pair;residual_pair	0
620.omnetpp.s-874B pc_stride_reach	region_pair_residual_pair	ctx3;phase;offset;predecessor;pc;region_pair;profiled_pc_stride	0
620.omnetpp.s-874B global_stride_reach	region_pair_residual_pair	ctx3;phase;offset;predecessor;pc;region_pair;profiled_global_stride	0
620.omnetpp.s-874B global_plus_pc_stride	region_pair_residual_pair	ctx3;phase;offset;predecessor;pc;region_pair;profiled_pc_stride;profiled_global_stride	0

Table 52: plan/v4.9_representation_screen.csv — candidate-bank representation screen (held-out recall gate before any training) — part 2/2

trace/route	candidate_absent_count	candidate_absent_pct	control_candidate_recall	recall_gain_vs_control	representation_gate_pass	decision
602.gcc.s-734B assoc_control	42,184	0.52183	0.47817	0	1	control
602.gcc.s-734B pc_stride_reach	42,276	0.52297	0.47817	-0.00114	0	do_not_train_no_representation_gain
602.gcc.s-734B dual_stride_reach	42,275	0.52295	0.47817	-0.00113	0	do_not_train_no_representation_gain
620.omnetpp.s-874B region_control	67,798	0.75102	0.24898	0	1	control
620.omnetpp.s-874B pc_stride_reach	67,936	0.75255	0.24898	-0.00153	0	do_not_train_no_representation_gain
620.omnetpp.s-874B global_stride_reach	67,311	0.74563	0.24898	0.00539	0	do_not_train_no_representation_gain
620.omnetpp.s-874B global_plus_pc_stride	67,716	0.75012	0.24898	9.083e-04	0	do_not_train_no_representation_gain

Table 53: plan/v4.9_stage_b_criteria.json — Stage-B acceptance gates

gate	requirement
accuracy_ratio_min	0.95
coverage_ratio_min	0.95
duplicate_fraction_delta_max	0.05
ipc_ratio_min	0.995
issue_rate_ratio_max	1.1
issue_ratio_max	1.1
mshr_p95_ratio_max	1.1
one_shot_matrix_rule	pre-replay matrix only; strict size rows are candidates, not accepted Stage-B results until keyed replay
pq.p95_ratio_max	1.1
reference_requirement	one_shot_all emits selected_full and all smaller strict rungs before replay; acceptance is computed only
rejected_fraction_delta_max	0.05
selected_accuracy_ratio_min	0.95

strict_stage_b_only	1
timeliness_ratio_min	0.95
zero_action_rule	ABSTAIN_ZERO_ACTIONS; replay_eligible=0; never force a floor export

8 Calibration sweeps (full)

Table 54: Full calibration threshold sweeps for every job (parsed from `calibration_tested_thresholds`; 272 rows)

ID	trace	model_size	threshold	measured_issue_per_event	issue_error	selected_accuracy	policy_emits
J21	602.gcc_s-734B	about_383k	0.01137	1	0.53129	0.06546	40,422
J21	602.gcc_s-734B	about_383k	0.02102	1	0.53129	0.06561	40,422
J21	602.gcc_s-734B	about_383k	0.03163	1	0.53129	0.06593	40,422
J21	602.gcc_s-734B	about_383k	0.04553	1	0.53129	0.06635	40,422
J21	602.gcc_s-734B	about_383k	0.06309	1	0.53129	0.06736	40,422
J21	602.gcc_s-734B	about_383k	0.08559	1	0.53129	0.06917	40,422
J21	602.gcc_s-734B	about_383k	0.11655	1	0.53129	0.07192	40,422
J21	602.gcc_s-734B	about_383k	0.15202	1	0.53129	0.07496	40,422
J21	602.gcc_s-734B	about_383k	0.17876	1	0.53129	0.07944	40,422
J21	602.gcc_s-734B	about_383k	0.20877	1	0.53129	0.10544	40,422
J21	602.gcc_s-734B	about_383k	0.27353	0.99077	0.52206	0.20785	40,049
J21	602.gcc_s-734B	about_383k	0.54339	0.51742	0.04871	0.52603	20,915
J21	602.gcc_s-734B	about_383k	0.9	0.46784	8.695e-04	0.96124	18,911
J21	602.gcc_s-734B	about_383k	0.91558	0.46396	0.00475	0.96603	18,754
J20	602.gcc_s-734B	about_911k	0.00189	1	0.53129	0.0659	40,422
J20	602.gcc_s-734B	about_911k	0.00726	1	0.53129	0.06687	40,422
J20	602.gcc_s-734B	about_911k	0.01572	1	0.53129	0.06816	40,422
J20	602.gcc_s-734B	about_911k	0.0286	1	0.53129	0.06927	40,422
J20	602.gcc_s-734B	about_911k	0.04726	1	0.53129	0.07018	40,422
J20	602.gcc_s-734B	about_911k	0.07148	1	0.53129	0.0712	40,422
J20	602.gcc_s-734B	about_911k	0.10069	1	0.53129	0.07179	40,422
J20	602.gcc_s-734B	about_911k	0.13803	1	0.53129	0.07209	40,422
J20	602.gcc_s-734B	about_911k	0.17833	1	0.53129	0.07897	40,422
J20	602.gcc_s-734B	about_911k	0.22127	1	0.53129	0.09967	40,422
J20	602.gcc_s-734B	about_911k	0.28503	1	0.53129	0.17107	40,422
J20	602.gcc_s-734B	about_911k	0.51177	0.51662	0.04792	0.54791	20,883
J20	602.gcc_s-734B	about_911k	0.9	0.47291	0.0042	0.95611	19,116
J20	602.gcc_s-734B	about_911k	0.93065	0.4642	0.00451	0.97005	18,764
J22	602.gcc_s-734B	nearest_153k	0.29342	1	0.53129	0.06435	40,422
J22	602.gcc_s-734B	nearest_153k	0.31004	1	0.53129	0.09317	40,422
J22	602.gcc_s-734B	nearest_153k	0.3313	1	0.53129	0.18052	40,422
J22	602.gcc_s-734B	nearest_153k	0.59225	0.51074	0.04203	0.37825	20,645
J22	602.gcc_s-734B	nearest_153k	0.75758	0.49468	0.02597	0.80311	19,996
J22	602.gcc_s-734B	nearest_153k	0.9	0.09925	0.36946	0.96311	4,012
J19	602.gcc_s-734B	selected_full	1.508e-05	1	0.53129	0.066	40,422
J19	602.gcc_s-734B	selected_full	1.205e-04	1	0.53129	0.06605	40,422
J19	602.gcc_s-734B	selected_full	5.296e-04	1	0.53129	0.06623	40,422
J19	602.gcc_s-734B	selected_full	0.00209	1	0.53129	0.06774	40,422
J19	602.gcc_s-734B	selected_full	0.0096	1	0.53129	0.07286	40,422
J19	602.gcc_s-734B	selected_full	0.03212	1	0.53129	0.07939	40,422
J19	602.gcc_s-734B	selected_full	0.06385	1	0.53129	0.0851	40,422
J19	602.gcc_s-734B	selected_full	0.10032	1	0.53129	0.09619	40,422
J19	602.gcc_s-734B	selected_full	0.14075	1	0.53129	0.11286	40,422
J19	602.gcc_s-734B	selected_full	0.18757	1	0.53129	0.14418	40,422
J19	602.gcc_s-734B	selected_full	0.24981	1	0.53129	0.21553	40,422
J19	602.gcc_s-734B	selected_full	0.39524	0.58636	0.11766	0.55371	23,702
J19	602.gcc_s-734B	selected_full	0.9	0.47393	0.00522	0.9657	19,157
J19	602.gcc_s-734B	selected_full	0.98939	0.45003	0.01868	0.99373	18,191
J07	605.mcf_s-994B	about_383k	0.03283	1	0.41509	0.01936	50,000
J07	605.mcf_s-994B	about_383k	0.09549	1	0.41509	0.01936	50,000
J07	605.mcf_s-994B	about_383k	0.17726	1	0.41509	0.01936	50,000
J07	605.mcf_s-994B	about_383k	0.20242	0.99996	0.41505	0.01936	49,998
J07	605.mcf_s-994B	about_383k	0.23631	0.99964	0.41473	0.01935	49,982
J07	605.mcf_s-994B	about_383k	0.24889	0.9993	0.41439	0.01937	49,965
J07	605.mcf_s-994B	about_383k	0.26581	0.99004	0.40513	0.01903	49,502

Table 54: Full calibration threshold sweeps for every job (parsed from `calibration_tested_thresholds`; 272 rows) (cont.)

ID		<i>trace</i>	<i>model_size</i>	<i>threshold</i>	<i>measured_issue_per_event</i>	<i>issue_error</i>	<i>selected_accuracy</i>	<i>policy_emits</i>
J07	605.mcf.s-994B	about_383k	0.32502		0.98066	0.39575	0.01901	49,033
J07	605.mcf.s-994B	about_383k	0.33671		0.97862	0.39371	0.01905	48,931
J07	605.mcf.s-994B	about_383k	0.35477		0.97062	0.38571	0.0192	48,531
J07	605.mcf.s-994B	about_383k	0.38231		0.91798	0.33307	0.02017	45,899
J07	605.mcf.s-994B	about_383k	0.40498		0.78982	0.20491	0.02299	39,491
J07	605.mcf.s-994B	about_383k	0.43349		0.57688	0.00803	0.03037	28,844
J07	605.mcf.s-994B	about_383k	0.5		0.07924	0.50567	0.07673	3,962
J06	605.mcf.s-994B	about_911k	0.04803		1	0.41509	0.0186	50,000
J06	605.mcf.s-994B	about_911k	0.08255		1	0.41509	0.0186	50,000
J06	605.mcf.s-994B	about_911k	0.18446		1	0.41509	0.0186	50,000
J06	605.mcf.s-994B	about_911k	0.22158		1	0.41509	0.0186	50,000
J06	605.mcf.s-994B	about_911k	0.25189		0.99998	0.41507	0.0186	49,999
J06	605.mcf.s-994B	about_911k	0.27082		0.99904	0.41413	0.01862	49,952
J06	605.mcf.s-994B	about_911k	0.2887		0.99552	0.41061	0.01864	49,776
J06	605.mcf.s-994B	about_911k	0.32891		0.99198	0.40707	0.01871	49,599
J06	605.mcf.s-994B	about_911k	0.35017		0.98606	0.40115	0.01878	49,303
J06	605.mcf.s-994B	about_911k	0.36527		0.91034	0.32543	0.01984	45,517
J06	605.mcf.s-994B	about_911k	0.38395		0.79082	0.20591	0.02233	39,541
J06	605.mcf.s-994B	about_911k	0.39924		0.67382	0.08891	0.02567	33,691
J06	605.mcf.s-994B	about_911k	0.42701		0.38062	0.20429	0.0433	19,031
J06	605.mcf.s-994B	about_911k	0.5		0.1621	0.42281	0.07761	8,105
J08	605.mcf.s-994B	nearest_153k	0.25421		1	0.41509	0.019	50,000
J08	605.mcf.s-994B	nearest_153k	0.26914		1	0.41509	0.019	50,000
J08	605.mcf.s-994B	nearest_153k	0.27978		1	0.41509	0.019	50,000
J08	605.mcf.s-994B	nearest_153k	0.31378		1	0.41509	0.019	50,000
J08	605.mcf.s-994B	nearest_153k	0.32878		1	0.41509	0.019	50,000
J08	605.mcf.s-994B	nearest_153k	0.33762		0.9999	0.41499	0.019	49,995
J08	605.mcf.s-994B	nearest_153k	0.34935		0.99782	0.41291	0.01902	49,891
J08	605.mcf.s-994B	nearest_153k	0.37744		0.9771	0.39219	0.0176	48,855
J08	605.mcf.s-994B	nearest_153k	0.39112		0.9751	0.39019	0.01762	48,755
J08	605.mcf.s-994B	nearest_153k	0.39639		0.9722	0.38729	0.01767	48,610
J08	605.mcf.s-994B	nearest_153k	0.40629		0.95946	0.37455	0.01789	47,973
J08	605.mcf.s-994B	nearest_153k	0.42094		0.91924	0.33433	0.01847	45,962
J08	605.mcf.s-994B	nearest_153k	0.43634		0.72308	0.13817	0.02191	36,154
J08	605.mcf.s-994B	nearest_153k	0.5		0	0.58491	0	0
J05	605.mcf.s-994B	selected_full	0.12213		1	0.41509	0.01708	50,000
J05	605.mcf.s-994B	selected_full	0.14874		1	0.41509	0.01708	50,000
J05	605.mcf.s-994B	selected_full	0.18309		1	0.41509	0.01708	50,000
J05	605.mcf.s-994B	selected_full	0.20817		1	0.41509	0.01708	50,000
J05	605.mcf.s-994B	selected_full	0.23512		1	0.41509	0.01708	50,000
J05	605.mcf.s-994B	selected_full	0.25028		0.99996	0.41505	0.01708	49,998
J05	605.mcf.s-994B	selected_full	0.27395		0.99962	0.41471	0.01703	49,981
J05	605.mcf.s-994B	selected_full	0.31298		0.9937	0.40879	0.01695	49,685
J05	605.mcf.s-994B	selected_full	0.32984		0.98068	0.39577	0.01585	49,034
J05	605.mcf.s-994B	selected_full	0.35865		0.9764	0.39149	0.01561	48,820
J05	605.mcf.s-994B	selected_full	0.38788		0.9574	0.37249	0.01588	47,870
J05	605.mcf.s-994B	selected_full	0.40157		0.88386	0.29895	0.01695	44,193
J05	605.mcf.s-994B	selected_full	0.41942		0.67166	0.08675	0.02117	33,583
J05	605.mcf.s-994B	selected_full	0.5		0.0022	0.58271	0	110
J43	619.lbm.s-4268B	about_383k	0		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	1.192e-07		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	5.006e-06		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	6.818e-05		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	3.905e-04		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	0.00136		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	0.00385		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	0.01116		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	0.02859		2	0.36701	0.98395	59,008
J43	619.lbm.s-4268B	about_383k	0.08269		1.9998	0.36681	0.98405	59,002
J43	619.lbm.s-4268B	about_383k	0.30684		1.99749	0.3645	0.98519	58,934
J43	619.lbm.s-4268B	about_383k	0.92556		1.8206	0.18761	0.99516	53,715
J43	619.lbm.s-4268B	about_383k	0.98557		1.56101	0.07198	0.9982	46,056
J43	619.lbm.s-4268B	about_383k	0.99358		1.33945	0.29354	0.99947	39,519
J38	619.lbm.s-4268B	about_911k	7.439e-07		2	0.36701	0.98448	59,008
J38	619.lbm.s-4268B	about_911k	1.849e-05		2	0.36701	0.98448	59,008
J38	619.lbm.s-4268B	about_911k	1.022e-04		2	0.36701	0.98448	59,008
J38	619.lbm.s-4268B	about_911k	4.672e-04		2	0.36701	0.98448	59,008

Table 54: Full calibration threshold sweeps for every job (parsed from `calibration_tested_thresholds`; 272 rows) (cont.)

ID	<i>trace</i>	<i>model_size</i>	<i>threshold</i>	<i>measured_issue_per_event</i>	<i>issue_error</i>	<i>selected_accuracy</i>	<i>policy_emits</i>
J38	619.lbm.s-4268B	about_911k	0.00161	2	0.36701	0.98448	59,008
J38	619.lbm.s-4268B	about_911k	0.00396	2	0.36701	0.98448	59,008
J38	619.lbm.s-4268B	about_911k	0.00768	2	0.36701	0.98448	59,008
J38	619.lbm.s-4268B	about_911k	0.015	2	0.36701	0.98448	59,008
J38	619.lbm.s-4268B	about_911k	0.02954	2	0.36701	0.98448	59,008
J38	619.lbm.s-4268B	about_911k	0.07924	1.99997	0.36698	0.98449	59,007
J38	619.lbm.s-4268B	about_911k	0.26543	1.99902	0.36603	0.98496	58,979
J38	619.lbm.s-4268B	about_911k	0.93634	1.86378	0.23079	0.9944	54,989
J38	619.lbm.s-4268B	about_911k	0.98557	1.65615	0.02316	0.99668	48,863
J38	619.lbm.s-4268B	about_911k	0.99644	1.33982	0.29317	0.99896	39,530
J47	619.lbm.s-4268B	nearest_153k	0.11149	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.17331	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.18867	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.19676	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.21184	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.22137	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.23665	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.25166	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.27362	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.34355	2	0.36701	0.94706	59,008
J47	619.lbm.s-4268B	nearest_153k	0.56603	1.99417	0.36118	0.94983	58,836
J47	619.lbm.s-4268B	nearest_153k	0.68197	1.98773	0.35474	0.95137	58,646
J47	619.lbm.s-4268B	nearest_153k	0.79846	1.59151	0.04148	0.96463	46,956
J47	619.lbm.s-4268B	nearest_153k	0.98557	0	1.63299	0	0
J31	619.lbm.s-4268B	selected_full	4.630e-08	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	2.200e-06	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	1.818e-05	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	8.886e-05	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	3.841e-04	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	9.887e-04	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	0.00218	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	0.00551	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	0.01596	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	0.051	2	0.36701	0.98617	59,008
J31	619.lbm.s-4268B	selected_full	0.27375	1.99559	0.36261	0.98835	58,878
J31	619.lbm.s-4268B	selected_full	0.90202	1.86429	0.2313	0.99698	55,004
J31	619.lbm.s-4268B	selected_full	0.98557	1.56718	0.06581	0.99946	46,238
J31	619.lbm.s-4268B	selected_full	0.99834	1.36849	0.2645	0.9999	40,376
J28	620.omnetpp.s-874B	about_383k	0.05626	0.25	0.12339	0.05379	11,285
J28	620.omnetpp.s-874B	about_383k	0.12485	0.25	0.12339	0.05379	11,285
J28	620.omnetpp.s-874B	about_383k	0.18689	0.25	0.12339	0.05379	11,285
J28	620.omnetpp.s-874B	about_383k	0.23978	0.25	0.12339	0.05494	11,285
J28	620.omnetpp.s-874B	about_383k	0.29177	0.25	0.12339	0.05654	11,285
J28	620.omnetpp.s-874B	about_383k	0.34667	0.25	0.12339	0.06052	11,285
J28	620.omnetpp.s-874B	about_383k	0.40471	0.25	0.12339	0.06575	11,285
J28	620.omnetpp.s-874B	about_383k	0.45386	0.25	0.12339	0.0763	11,285
J28	620.omnetpp.s-874B	about_383k	0.4919	0.25	0.12339	0.08658	11,285
J28	620.omnetpp.s-874B	about_383k	0.5035	0.25	0.12339	0.09544	11,285
J28	620.omnetpp.s-874B	about_383k	0.50824	0.25	0.12339	0.12025	11,285
J28	620.omnetpp.s-874B	about_383k	0.52536	0.25	0.12339	0.18077	11,285
J28	620.omnetpp.s-874B	about_383k	0.55722	0.14541	0.0188	0.33029	6,564
J28	620.omnetpp.s-874B	about_383k	0.65	0.04925	0.07736	0.62258	2,223
J27	620.omnetpp.s-874B	about_911k	0.0631	0.25	0.12339	0.06389	11,285
J27	620.omnetpp.s-874B	about_911k	0.12404	0.25	0.12339	0.06389	11,285
J27	620.omnetpp.s-874B	about_911k	0.18216	0.25	0.12339	0.06398	11,285
J27	620.omnetpp.s-874B	about_911k	0.23723	0.25	0.12339	0.06469	11,285
J27	620.omnetpp.s-874B	about_911k	0.28903	0.25	0.12339	0.06673	11,285
J27	620.omnetpp.s-874B	about_911k	0.33785	0.25	0.12339	0.07027	11,285
J27	620.omnetpp.s-874B	about_911k	0.38718	0.25	0.12339	0.07408	11,285
J27	620.omnetpp.s-874B	about_911k	0.43658	0.25	0.12339	0.08755	11,285
J27	620.omnetpp.s-874B	about_911k	0.48494	0.25	0.12339	0.1027	11,285
J27	620.omnetpp.s-874B	about_911k	0.50543	0.25	0.12339	0.11422	11,285
J27	620.omnetpp.s-874B	about_911k	0.51014	0.25	0.12339	0.13779	11,285
J27	620.omnetpp.s-874B	about_911k	0.53046	0.25	0.12339	0.20346	11,285
J27	620.omnetpp.s-874B	about_911k	0.56726	0.1749	0.04829	0.31894	7,895
J27	620.omnetpp.s-874B	about_911k	0.65	0.05058	0.07604	0.63163	2,283
J29	620.omnetpp.s-874B	nearest_153k	0.16026	0.25	0.12339	0.04617	11,285

Table 54: Full calibration threshold sweeps for every job (parsed from `calibration_tested_thresholds`; 272 rows) (cont.)

ID		<i>trace</i>	<i>model_size</i>	<i>threshold</i>	<i>measured_issue_per_event</i>	<i>issue_error</i>	<i>selected_accuracy</i>	<i>policy_emits</i>
J29	620.omnetpp.s-874B	nearest_153k	0.22822	0.25	0.12339	0.04617	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.28114	0.25	0.12339	0.04634	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.31738	0.25	0.12339	0.04643	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.353	0.25	0.12339	0.04705	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.38169	0.25	0.12339	0.04847	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.42281	0.25	0.12339	0.04909	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.46555	0.25	0.12339	0.05512	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.50869	0.25	0.12339	0.06043	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.51272	0.25	0.12339	0.06105	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.5171	0.25	0.12339	0.07266	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.51998	0.25	0.12339	0.09677	11,285	
J29	620.omnetpp.s-874B	nearest_153k	0.53857	0.21777	0.09116	0.14354	9,830	
J29	620.omnetpp.s-874B	nearest_153k	0.65	0	0.12661	0	0	
J26	620.omnetpp.s-874B	selected_full	0.0222	0.25	0.12339	0.07683	11,285	
J26	620.omnetpp.s-874B	selected_full	0.05593	0.25	0.12339	0.07692	11,285	
J26	620.omnetpp.s-874B	selected_full	0.10194	0.25	0.12339	0.07692	11,285	
J26	620.omnetpp.s-874B	selected_full	0.16091	0.25	0.12339	0.07718	11,285	
J26	620.omnetpp.s-874B	selected_full	0.23132	0.25	0.12339	0.07887	11,285	
J26	620.omnetpp.s-874B	selected_full	0.30513	0.25	0.12339	0.08259	11,285	
J26	620.omnetpp.s-874B	selected_full	0.37181	0.25	0.12339	0.08436	11,285	
J26	620.omnetpp.s-874B	selected_full	0.42244	0.25	0.12339	0.08861	11,285	
J26	620.omnetpp.s-874B	selected_full	0.46313	0.25	0.12339	0.10749	11,285	
J26	620.omnetpp.s-874B	selected_full	0.49951	0.25	0.12339	0.12379	11,285	
J26	620.omnetpp.s-874B	selected_full	0.52637	0.25	0.12339	0.16181	11,285	
J26	620.omnetpp.s-874B	selected_full	0.54989	0.25	0.12339	0.21249	11,285	
J26	620.omnetpp.s-874B	selected_full	0.5894	0.23254	0.10593	0.33591	10,497	
J26	620.omnetpp.s-874B	selected_full	0.65	0.11845	0.00816	0.56106	5,347	
J13	623.xalancbmk.s-700B	about_383k	0.0554	0.9909	0.59494	0.19197	49,545	
J13	623.xalancbmk.s-700B	about_383k	0.10358	0.99066	0.5947	0.19201	49,533	
J13	623.xalancbmk.s-700B	about_383k	0.16232	0.98844	0.59248	0.19244	49,422	
J13	623.xalancbmk.s-700B	about_383k	0.22337	0.98178	0.58582	0.19389	49,089	
J13	623.xalancbmk.s-700B	about_383k	0.27144	0.95386	0.5579	0.19959	47,693	
J13	623.xalancbmk.s-700B	about_383k	0.31678	0.85386	0.4579	0.22287	42,693	
J13	623.xalancbmk.s-700B	about_383k	0.39016	0.76204	0.36608	0.24938	38,102	
J13	623.xalancbmk.s-700B	about_383k	0.49279	0.71034	0.31438	0.26652	35,517	
J13	623.xalancbmk.s-700B	about_383k	0.53899	0.677	0.28104	0.27459	33,850	
J13	623.xalancbmk.s-700B	about_383k	0.55239	0.61838	0.22242	0.29157	30,919	
J13	623.xalancbmk.s-700B	about_383k	0.56296	0.52988	0.13392	0.32498	26,494	
J13	623.xalancbmk.s-700B	about_383k	0.57529	0.37662	0.01934	0.39254	18,831	
J13	623.xalancbmk.s-700B	about_383k	0.60658	0.178	0.21796	0.52899	8,900	
J13	623.xalancbmk.s-700B	about_383k	0.65	0.08016	0.3158	0.66218	4,008	
J12	623.xalancbmk.s-700B	about_911k	0.01786	0.99144	0.59548	0.18795	49,572	
J12	623.xalancbmk.s-700B	about_911k	0.06806	0.99106	0.5951	0.18798	49,553	
J12	623.xalancbmk.s-700B	about_911k	0.1364	0.99046	0.5945	0.18803	49,523	
J12	623.xalancbmk.s-700B	about_911k	0.19787	0.98812	0.59216	0.1885	49,406	
J12	623.xalancbmk.s-700B	about_911k	0.25289	0.94228	0.54632	0.19769	47,114	
J12	623.xalancbmk.s-700B	about_911k	0.30953	0.87296	0.477	0.21348	43,648	
J12	623.xalancbmk.s-700B	about_911k	0.3796	0.78598	0.39002	0.23708	39,299	
J12	623.xalancbmk.s-700B	about_911k	0.49065	0.7145	0.31854	0.25996	35,725	
J12	623.xalancbmk.s-700B	about_911k	0.57553	0.6804	0.28444	0.26728	34,020	
J12	623.xalancbmk.s-700B	about_911k	0.58234	0.63198	0.23602	0.27153	31,599	
J12	623.xalancbmk.s-700B	about_911k	0.58347	0.5922	0.19624	0.27818	29,610	
J12	623.xalancbmk.s-700B	about_911k	0.58437	0.51074	0.11478	0.29036	25,537	
J12	623.xalancbmk.s-700B	about_911k	0.58546	0.34722	0.04874	0.32273	17,361	
J12	623.xalancbmk.s-700B	about_911k	0.65	0	0.39596	0	0	
J14	623.xalancbmk.s-700B	nearest_153k	0.29376	0.99252	0.59656	0.11002	49,626	
J14	623.xalancbmk.s-700B	nearest_153k	0.35015	0.97634	0.58038	0.11193	48,817	
J14	623.xalancbmk.s-700B	nearest_153k	0.42506	0.84358	0.44762	0.1294	42,179	
J14	623.xalancbmk.s-700B	nearest_153k	0.47233	0.76418	0.36822	0.1423	38,209	
J14	623.xalancbmk.s-700B	nearest_153k	0.49236	0.71728	0.32132	0.15054	35,864	
J14	623.xalancbmk.s-700B	nearest_153k	0.50639	0.68978	0.29382	0.15127	34,489	
J14	623.xalancbmk.s-700B	nearest_153k	0.5067	0.68742	0.29146	0.15065	34,371	
J14	623.xalancbmk.s-700B	nearest_153k	0.50676	0.68416	0.2882	0.14982	34,208	
J14	623.xalancbmk.s-700B	nearest_153k	0.5068	0.68082	0.28486	0.14911	34,041	
J14	623.xalancbmk.s-700B	nearest_153k	0.50683	0.6772	0.28124	0.14814	33,860	
J14	623.xalancbmk.s-700B	nearest_153k	0.50687	0.67158	0.27562	0.147	33,579	
J14	623.xalancbmk.s-700B	nearest_153k	0.50691	0.65634	0.26038	0.14584	32,817	

Table 54: Full calibration threshold sweeps for every job (parsed from `calibration_tested_thresholds`; 272 rows) (cont.)

ID	<i>trace</i>	<i>model_size</i>	<i>threshold</i>	<i>measured_issue_per_event</i>	<i>issue_error</i>	<i>selected_accuracy</i>	<i>policy_emits</i>
J14	623.xalancbmk_s-700B	nearest_153k	0.50697	0.58622	0.19026	0.14196	29,311
J14	623.xalancbmk_s-700B	nearest_153k	0.65	0	0.39596	0	0
J11	623.xalancbmk_s-700B	selected_full	4.468e-06	0.9936	0.59764	0.25596	49,680
J11	623.xalancbmk_s-700B	selected_full	2.018e-04	0.9933	0.59734	0.25604	49,665
J11	623.xalancbmk_s-700B	selected_full	0.00152	0.99242	0.59646	0.25628	49,621
J11	623.xalancbmk_s-700B	selected_full	0.00631	0.98948	0.59352	0.25704	49,474
J11	623.xalancbmk_s-700B	selected_full	0.01927	0.98228	0.58632	0.25899	49,114
J11	623.xalancbmk_s-700B	selected_full	0.04715	0.96856	0.5726	0.26266	48,428
J11	623.xalancbmk_s-700B	selected_full	0.09657	0.94706	0.5511	0.26883	47,353
J11	623.xalancbmk_s-700B	selected_full	0.1671	0.90292	0.50696	0.28226	45,146
J11	623.xalancbmk_s-700B	selected_full	0.24656	0.8366	0.44064	0.30526	41,830
J11	623.xalancbmk_s-700B	selected_full	0.31284	0.72642	0.33046	0.35247	36,321
J11	623.xalancbmk_s-700B	selected_full	0.37557	0.55908	0.16312	0.45764	27,954
J11	623.xalancbmk_s-700B	selected_full	0.57753	0.39832	0.00236	0.66293	19,916
J11	623.xalancbmk_s-700B	selected_full	0.65	0.35348	0.04248	0.7433	17,674
J11	623.xalancbmk_s-700B	selected_full	0.93145	0.21468	0.18128	0.98127	10,734

9 Event evidence (replay_results/evidence/)

Table 55: normal evidence: columns constant across all rows (shown once)

column	constant value
fail_reason	–
log_missing	0
nodup_mode	1
run_failed	0

Table 56: evidence/normal_prefetcher_evidence.csv — normal-prefetcher event evidence (45 rows) — part 1/4

accuracy	<i>coverage_vs_no_pref_l2_miss</i>	<i>drop_rate</i>	<i>event_late</i>	<i>event_residual</i>	<i>event_timeliness_over_covered</i>	<i>event_timely</i>	<i>evidence_tags</i>
0	0	0	0	203,131	0	0	– 0
0.26422	0.34802	0	164	168,110	0.99532	34,854	low_unique_event_coverage 0.37
0.10365	1.84896	0	47	15,414	0.99975	187,580	– 0.43
0.13109	1.8367	0	600	16,396	0.99678	186,005	– 0.43
0.05901	1.305	0	384	70,471	0.9971	132,205	– 0.42
0.24468	2.265e-04	0	1	203,130	0	0	high_useless_fraction 0.36
0.45872	1.57333	0	173	40,963	0.99893	161,912	– 0.42
0.1016	1.92045	0	43	7,064	0.99978	195,894	– 0.43
0.1128	1.83503	0	125	17,169	0.99933	185,747	– 0.43
0	0	0	0	294,639	0	0	– 0
0.43043	0.51186	0	15,664	211,245	0.81103	67,229	low_unique_event_coverage 0.32
0.76561	1.36368	0	112,238	39,792	0.55882	142,164	– 0.36
0.24712	1.68465	0	24,838	33,578	0.90457	235,435	– 0.3
0.05011	0.6471	0	63,167	162,156	0.52099	68,702	low_unique_event_coverage 0.37
0.08642	2.376e-05	0	0	294,491	0	0	– 0
0.9421	1.81622	0	25,203	13,298	0.91014	255,270	– 0.38
0.14557	1.65644	0	76,845	8,199	0.73119	209,025	– 0.3
0.5992	1.25201	0	76,585	57,217	0.67674	160,327	– 0.36

Table 56: evidence/normal_prefetcher_evidence.csv — normal-prefetcher event evidence (45 rows) — part 1/4 (cont.)

accuracy	coverage-vs-no-pref.l2-miss	drop-rate	event_late	event_residual	event_timeliness-over-covered	event_timely	evidence-tags
0	0	0	0	740,223	0	0	— 0.18
0.30229	0.01026	0	0	734,820	1	4,858	low_unique_event_coverage;very_low_unique_event_coverage 0.18
0.08715	0.20898	0	15,919	626,627	0.84351	85,808	low_unique_event_coverage;high_useless_fraction 0.1
0.1775	0.12225	0	1,232	668,468	0.97908	57,672	low_unique_event_coverage;very_low_unique_event_coverage 0.18
0.02025	0.01943	0	1,089	722,964	0.82562	5,156	low_unique_event_coverage;very_low_unique_event_coverage 0.18
0.29545	3.512e-05	0	0	740,223	0	0	— 0.18
0.19908	0.34627	4.660e-05	54,251	582,137	0.62805	91,604	low_unique_event_coverage;high_useless_fraction 0.18
0.02899	0.60713	0.02689	109,635	460,390	0.59013	157,849	low_unique_event_coverage;high_useless_fraction 0.16
0.08846	0.17521	0	15,522	644,718	0.82062	71,010	low_unique_event_coverage;very_low_unique_event_coverage;high_useless_fraction 0.18
0	0	0	0	316,554	0	0	— 0.2
0.30277	0.01873	0	0	313,502	1	2,990	low_unique_event_coverage;very_low_unique_event_coverage 0.23
0.14635	0.35012	0	1,932	261,401	0.96377	51,399	low_unique_event_coverage;high_useless_fraction 0.24
0.14544	0.14262	0	878	292,579	0.96143	21,888	low_unique_event_coverage;very_low_unique_event_coverage;high_useless_fraction 0.23
4.236e-04	2.338e-04	0	13	315,658	0.79032	49	low_unique_event_coverage;very_low_unique_event_coverage 0.23
0.18521	0.01698	0	292	313,839	0.87988	2,139	low_unique_event_coverage;very_low_unique_event_coverage;high_useless_fraction 0.23
0.2447	0.62465	0	1,731	225,755	0.98027	86,007	high_useless_fraction 0.24
0.04904	1.18089	0.02385	8,004	171,771	0.94365	134,047	high_useless_fraction 0.24
0.14442	0.33337	0	1,997	261,512	0.96255	51,322	low_unique_event_coverage;high_useless_fraction 0.24
0	0	0	0	450,350	0	0	— 0.35
0.06286	0.04185	0	7,469	440,312	0.22068	2,115	low_unique_event_coverage;very_low_unique_event_coverage 0.3
0.04518	0.24679	0	3,711	415,300	0.87344	25,610	low_unique_event_coverage;very_low_unique_event_coverage;high_useless_fraction 0.33
0.05718	0.11743	0	2,673	423,874	0.87576	18,842	low_unique_event_coverage;very_low_unique_event_coverage 0.34
0.00468	0.01736	0	1,837	434,440	0.38127	1,132	low_unique_event_coverage;very_low_unique_event_coverage 0.35
0.06726	0.01302	0	79	446,603	0.97272	2,817	low_unique_event_coverage;very_low_unique_event_coverage 0.35
0.15092	0.64747	0	4,908	368,045	0.9332	68,566	low_unique_event_coverage;high_useless_fraction 0.34
0.04163	1.19159	0.24228	9,037	274,702	0.948	164,738	high_useless_fraction 0.34
0.04663	0.18269	0	4,522	422,903	0.79227	17,247	low_unique_event_coverage;very_low_unique_event_coverage;high_useless_fraction 0.3

Table 57: evidence/normal_prefetcher_evidence.csv — normal-prefetcher event evidence (45 rows) — part 2/4

accuracy	l2_loads	late-per-issued	miss-reduction-vs-no-pref	nodup-accuracy	nodup-issued	pf_dropped	pf_filled	pf_issued	pf_late	pf_requested
0	404,219	0	0	0	0	0	0	0	0	0
0.26422	404,219	6.279e-04	0.17158	0.26422	267,553	0	71,723	267,553	168	267,553
0.10365	404,219	1.352e-05	0.92382	0.10374	3,620,275	0	392,083	3,623,513	49	3,623,513
0.13109	404,219	2.122e-04	0.91615	0.13117	2,844,277	0	385,815	2,846,050	604	2,846,050
0.05901	404,219	8.616e-05	0.6511	0.05909	4,485,769	0	266,257	4,491,857	387	4,491,857
0.24468	404,219	0.02128	0	0.24468	188	0	120	188	4	188
0.45872	404,219	2.871e-04	0.79733	0.45872	696,706	0	343,138	696,706	200	696,706
0.1016	404,219	1.537e-05	0.96485	0.10184	3,830,425	0	422,799	3,839,474	59	3,839,474
0.1128	404,219	3.873e-05	0.9148	0.11785	3,162,983	0	386,269	3,304,558	128	3,304,558
0	295,037	0	0	0	0	0	0	0	0	0
0.43043	295,026	0.04481	0.22879	0.43044	350,375	0	150,816	350,381	15,702	350,381
0.76561	295,032	0.21456	0.48302	0.76581	524,667	0	402,791	524,800	112,603	524,800
0.24712	295,027	0.01258	0.80081	0.32711	1,517,402	0	496,725	2,008,610	25,272	2,008,610
0.05011	295,024	0.01714	0.23353	0.08238	2,314,384	0	190,672	3,804,639	65,229	3,804,639
0.08642	295,039	0.01235	-1.358e-05	0.08642	81	0	10	81	1	81
0.9421	295,028	0.04595	0.86864	0.9421	568,017	0	535,220	568,017	26,099	568,017
0.14557	295,024	0.02308	0.71055	0.20898	2,335,419	0	489,356	3,352,602	77,378	3,352,602
0.5992	295,027	0.12495	0.54488	0.61501	599,811	0	369,761	615,642	76,926	615,642
0	985,276	0	0	0	0	0	0	0	0	0
0.30229	985,274	3.980e-05	0.00652	0.30234	25,121	0	7,762	25,125	1	25,125
0.08715	985,268	0.01026	0.05232	0.09167	1,687,533	0	842,607	1,775,122	18,210	1,775,122
0.1775	985,273	0.00256	0.07178	0.17787	508,759	0	149,168	509,840	1,305	509,840

Table 57: evidence/normal_prefetcher_evidence.csv — normal-prefetcher event evidence (45 rows) — part 2/4 (cont.)

accuracy	<i>l2_loads</i>	<i>late_per_issued</i>	<i>miss_reduction_vs_no_pref</i>	<i>nodup_accuracy</i>	<i>nodup_issued</i>	<i>pf_dropped</i>	<i>pf_filled</i>	<i>pf_issued</i>	<i>pf_late</i>	<i>pf_requested</i>
0.02025	985,273	0.00177	0.00651	0.02039	705,508	0	19,585	710,459	1,257	710,459
0.29545	985,276	0.18182	0	0.29545	88	0	42	88	16	88
0.19908	985,269	0.04748	0.0784	0.19908	1,287,493	60	893,091	1,287,493	61,126	1,287,553
0.02899	985,257	0.009	0.03512	0.03395	13,238,296	428,375	7,641,216	15,501,838	139,527	15,930,213
0.08846	985,267	0.01211	0.03293	0.09356	1,386,191	0	687,520	1,466,157	17,755	1,466,157
0	451,398	0	0	0	0	0	0	0	0	0
0.30277	451,396	0	0.00903	0.30279	19,578	0	7,781	19,579	0	19,579
0.14635	451,360	0.00335	0.12091	0.16814	659,178	0	397,943	757,318	2,538	757,318
0.14544	451,359	0.00602	0.05212	0.15581	289,753	0	143,933	310,419	1,868	310,419
4.236e-04	451,389	1.946e-04	6.539e-04	4.325e-04	171,086	0	127	174,696	34	174,696
0.18521	451,406	0.01289	0.00449	0.18521	29,016	0	14,872	29,016	374	29,016
0.2447	451,284	0.00452	0.22212	0.2447	808,086	0	593,171	808,086	3,655	808,086
0.04904	451,253	0.00142	0.24944	0.0561	6,663,110	186,257	3,656,950	7,623,344	10,848	7,809,601
0.14442	451,341	0.00376	0.11885	0.16651	633,770	0	378,404	730,714	2,751	730,714
0	1,218,494	0	0	0	0	0	0	0	0	0
0.06286	1,218,495	0.02501	0.00458	0.07011	268,834	0	19,224	299,845	7,498	299,845
0.04518	1,218,352	0.00275	-0.6215	0.05202	2,136,580	0	828,293	2,459,972	6,770	2,459,972
0.05718	1,218,334	0.00354	-0.12306	0.063	839,364	0	139,608	924,845	3,276	924,845
0.00468	1,218,483	0.00112	0.0238	0.00485	1,612,741	0	30,949	1,670,201	1,870	1,670,201
0.06726	1,218,487	0.00135	-0.02407	0.06726	87,172	0	16,769	87,172	118	87,172
0.15092	1,218,827	0.00573	-0.45803	0.15092	1,932,085	0	1,057,158	1,932,085	11,070	1,932,085
0.04163	1,218,507	0.00183	-0.3323	0.04989	10,755,534	4,122,042	5,030,567	12,891,224	23,596	17,013,266
0.04663	1,218,439	0.00423	-0.36664	0.05471	1,503,898	0	560,701	1,764,449	7,460	1,764,449

Table 58: evidence/normal_prefetcher_evidence.csv — normal-prefetcher event evidence (45 rows) — part 3/4

accuracy	<i>pf_useful</i>	<i>pf_useless</i>	<i>pq_merged_duplicate_proxy</i>	<i>prefetcher</i>	<i>selected_accuracy</i>	<i>speedup_vs_no_pref</i>	<i>timeliness</i>	<i>trace</i>	<i>unique_event_coverage</i>	<i>useful_per_l2_miss_self</i>
0	0	0	0	no_pref	0	1	0	602.gcc.s-734B	0	0
0.26422	70,694	1,023	0	stride	0.26422	1.0056	0.99763	602.gcc.s-734B	0.17158	0.4201
0.10365	375,582	16,394	3,238	streamer	0.10374	1.17285	0.99987	602.gcc.s-734B	0.92344	24.27024
0.13109	373,090	12,600	1,773	ampm	0.13117	1.17035	0.99838	602.gcc.s-734B	0.91569	21.90524
0.05901	265,086	1,162	6,088	spp	0.05909	1.16402	0.99854	602.gcc.s-734B	0.65084	3.7403
0.24468	46	74	0	ipcp	0.24468	0.99989	0.92	602.gcc.s-734B	0	2.265e-04
0.45872	319,593	23,409	0	sms	0.45872	1.14709	0.99937	602.gcc.s-734B	0.79708	7.76295
0.1016	390,102	32,524	9,049	sandbox	0.10184	1.18554	0.99985	602.gcc.s-734B	0.96437	54.62848
0.1128	372,751	13,449	141,575	power7	0.11785	1.17052	0.99966	602.gcc.s-734B	0.91442	21.53883
0	0	0	0	no_pref	0	1	0	619.lbm.s-4268B	0	0
0.43043	150,814	0	6	stride	0.43044	1.00823	0.9057	619.lbm.s-4268B	0.22817	0.66371
0.76561	401,794	990	133	streamer	0.76581	1.12944	0.7811	619.lbm.s-4268B	0.4825	2.63779
0.24712	496,363	348	491,208	ampm	0.32711	1.16398	0.95155	619.lbm.s-4268B	0.79906	8.45751
0.05011	190,662	10	1,490,255	spp	0.08238	1.17804	0.74509	619.lbm.s-4268B	0.23317	0.84426
0.08642	7	3	0	ipcp	0.08642	1	0.875	619.lbm.s-4268B	0	2.376e-05
0.9421	535,130	84	0	sms	0.9421	1.18339	0.9535	619.lbm.s-4268B	0.86638	13.82586
0.14557	488,051	1,296	1,017,183	sandbox	0.20898	1.16273	0.86315	619.lbm.s-4268B	0.70943	5.72272
0.5992	368,891	862	15,831	power7	0.61501	1.12199	0.82745	619.lbm.s-4268B	0.54415	2.75095
0	0	0	0	no_pref	0	1	0	605.mcf.s-994B	0	0
0.30229	7,595	167	4	stride	0.30234	1.00224	0.99987	605.mcf.s-994B	0.00656	0.01033
0.08715	154,695	686,999	87,589	streamer	0.09167	1.00175	0.89468	605.mcf.s-994B	0.11592	0.22052
0.1775	90,495	58,586	1,081	ampm	0.17787	1.03035	0.98578	605.mcf.s-994B	0.07791	0.13171
0.02025	14,384	5,178	4,951	spp	0.02039	1.03035	0.91963	605.mcf.s-994B	0.00697	0.01956
0.29545	26	16	0	ipcp	0.29545	1	0.61905	605.mcf.s-994B	0	3.512e-05
0.19908	256,319	635,113	0	sms	0.19908	1.02506	0.80744	605.mcf.s-994B	0.12375	0.37573
0.02899	449,411	7,188,334	2,263,542	sandbox	0.03395	0.92286	0.76309	605.mcf.s-994B	0.21325	0.62922
0.08846	129,697	556,672	79,966	power7	0.09356	0.99154	0.87959	605.mcf.s-994B	0.09593	0.18118

Table 58: evidence/normal_prefetcher_evidence.csv — normal-prefetcher event evidence (45 rows) — part 3/4 (cont.)

accuracy	<i>pf_useful</i>	<i>pf_useless</i>	<i>pq_merged_duplicate_proxy</i>	<i>prefetcher</i>	<i>selected_accuracy</i>	<i>speedup_vs_no_pref</i>	<i>timeliness</i>		<i>trace</i>	<i>unique_event_coverage</i>	<i>useful_per_l2_miss_self</i>
0	0	0	0	no_pref	0	1	0	620.omnetpp_s-874B		0	0
0.30277	5,928	1,849	1	stride	0.30279	1.00349	1	620.omnetpp_s-874B		0.00945	0.0189
0.14635	110,831	286,074	98,140	streamer	0.16814	1.02013	0.97761	620.omnetpp_s-874B		0.16237	0.39827
0.14544	45,147	98,338	20,666	ampm	0.15581	1.00952	0.96027	620.omnetpp_s-874B		0.06914	0.15046
4.236e-04	74	53	3,610	spp	4.325e-04	1.00501	0.68519	620.omnetpp_s-874B	1.548e-04	2.339e-04	
0.18521	5,374	9,433	0	ipcp	0.18521	1.00088	0.93493	620.omnetpp_s-874B		0.00676	0.01705
0.2447	197,737	394,038	0	sms	0.2447	1.03979	0.98185	620.omnetpp_s-874B		0.2717	0.80302
0.04904	373,815	3,279,950	960,234	sandbox	0.0561	1.02876	0.9718	620.omnetpp_s-874B		0.42346	1.57334
0.14442	105,531	271,622	96,944	power7	0.16651	1.02029	0.97459	620.omnetpp_s-874B		0.16213	0.37834
0	0	0	0	no_pref	0	1	0	623.xalancbmk_s-700B		0	0
0.06286	18,849	373	31,011	stride	0.07011	1.00054	0.71541	623.xalancbmk_s-700B		0.0047	0.04205
0.04518	111,140	715,938	323,392	streamer	0.05202	0.95252	0.94258	623.xalancbmk_s-700B		0.05687	0.1522
0.05718	52,884	86,456	85,481	ampm	0.063	0.98876	0.94167	623.xalancbmk_s-700B		0.04184	0.10456
0.00468	7,816	23,108	57,460	spp	0.00485	1.00198	0.80694	623.xalancbmk_s-700B		0.00251	0.01778
0.06726	5,863	10,856	0	ipcp	0.06726	0.99774	0.98027	623.xalancbmk_s-700B		0.00626	0.01271
0.15092	291,589	764,626	0	sms	0.15092	0.96733	0.96342	623.xalancbmk_s-700B		0.15225	0.44407
0.04163	536,632	4,491,270	2,135,690	sandbox	0.04989	0.97744	0.95788	623.xalancbmk_s-700B		0.3658	0.89438
0.04663	82,274	477,415	260,551	power7	0.05471	0.97194	0.91687	623.xalancbmk_s-700B		0.0383	0.13368

Table 59: evidence/normal_prefetcher_evidence.csv — normal-prefetcher event evidence (45 rows) — part 4/4

accuracy	<i>useless_per_issued</i>
0	0
0.26422	0.00382
0.10365	0.00452
0.13109	0.00443
0.05901	2.587e-04
0.24468	0.39362
0.45872	0.0336
0.1016	0.00847
0.1128	0.00407
0	0
0.43043	0
0.76561	0.00189
0.24712	1.733e-04
0.05011	2.628e-06
0.08642	0.03704
0.9421	1.479e-04
0.14557	3.866e-04
0.5992	0.0014
0	0
0.30229	0.00665
0.08715	0.38702
0.1775	0.11491
0.02025	0.00729
0.29545	0.18182
0.19908	0.49329
0.02899	0.46371
0.08846	0.37968
0	0
0.30277	0.09444
0.14635	0.37775
0.14544	0.31679
4.236e-04	3.034e-04
0.18521	0.3251
0.2447	0.48762
0.04904	0.43025

Table 59: evidence/normal_prefetcher_evidence.csv — normal-prefetcher event evidence (45 rows) — part 4/4 (cont.)

accuracy	useless_per_issued
0.14442	0.37172
0	0
0.06286	0.00124
0.04518	0.29104
0.05718	0.09348
0.00468	0.01384
0.06726	0.12454
0.15092	0.39575
0.04163	0.3484
0.04663	0.27057

Table 60: evidence/standalone_prefetcher_evidence.csv — columns NOT already in Table 7 (all shared column values identical)

ID	trace	event_timely	event_late	event_residual	unique_event_coverage	event_timeliness_over_covered	
J21	602.gcc.s-734B	184,979	1	18,150	0.91064	0.99999	
J20	602.gcc.s-734B	185,459	1	17,671	0.913	0.99999	
J22	602.gcc.s-734B	189,962	2	13,083	0.93517	0.99999	
J19	602.gcc.s-734B	187,013	0	16,118	0.92065	1	
J15	602.gcc.s-734B	187,013	0	16,118	0.92065	1	
J17	602.gcc.s-734B	184,979	1	18,150	0.91064	0.99999	
J16	602.gcc.s-734B	185,459	1	17,671	0.913	0.99999	
J18	602.gcc.s-734B	40,312	1	162,773	0.19845	0.99998	
J07	605.mcf.s-994B	162,393	24,401	548,097	0.21938	0.86937	low_...
J06	605.mcf.s-994B	163,549	24,016	547,031	0.22095	0.87196	low_...
J08	605.mcf.s-994B	159,346	25,165	551,727	0.21527	0.86361	low_...
J05	605.mcf.s-994B	147,332	21,477	566,639	0.19904	0.87277	low_...
J01	605.mcf.s-994B	124,736	1,609	610,653	0.16851	0.98727	low_...
J03	605.mcf.s-994B	150,314	2,272	583,788	0.20307	0.98511	low_...
J02	605.mcf.s-994B	149,835	13,994	571,954	0.20242	0.91458	low_...
J04	605.mcf.s-994B	119,231	2,575	615,671	0.16107	0.97886	low_...
J43	619.lbm.s-4268B	228,038	7,540	58,387	0.77396	0.96799	
J38	619.lbm.s-4268B	236,144	8,770	49,023	0.80147	0.96419	
J47	619.lbm.s-4268B	290,231	416	2,972	0.98504	0.99857	
J31	619.lbm.s-4268B	233,746	10,890	49,264	0.79333	0.95548	
J34	619.lbm.s-4268B	248,889	2,439	42,442	0.84473	0.9903	
J30	619.lbm.s-4268B	233,746	10,890	49,264	0.79333	0.95548	
J35	619.lbm.s-4268B	229,904	10,801	53,252	0.78029	0.95513	
J36	619.lbm.s-4268B	225,923	10,703	57,285	0.76678	0.95477	
J32	619.lbm.s-4268B	245,839	4,505	43,446	0.83437	0.982	
J33	619.lbm.s-4268B	248,889	2,439	42,442	0.84473	0.9903	
J46	619.lbm.s-4268B	251,524	1,645	40,622	0.85367	0.9935	
J41	619.lbm.s-4268B	259,585	1,538	32,602	0.88103	0.99411	
J42	619.lbm.s-4268B	228,038	7,540	58,387	0.77396	0.96799	
J37	619.lbm.s-4268B	236,144	8,770	49,023	0.80147	0.96419	
J44	619.lbm.s-4268B	246,715	3,331	43,758	0.83735	0.98668	
J39	619.lbm.s-4268B	255,462	3,578	34,731	0.86703	0.98619	
J45	619.lbm.s-4268B	251,524	1,645	40,622	0.85367	0.9935	
J40	619.lbm.s-4268B	259,585	1,538	32,602	0.88103	0.99411	
J28	620.omnetpp.s-874B	40,630	85	275,413	0.12835	0.99791	low_...
J27	620.omnetpp.s-874B	48,487	120	267,360	0.15317	0.99753	low_...
J29	620.omnetpp.s-874B	30,449	161	285,460	0.09619	0.99474	low_unique_event_coverage;very_low_...
J26	620.omnetpp.s-874B	45,782	53	270,343	0.14463	0.99884	
J23	620.omnetpp.s-874B	45,782	53	270,343	0.14463	0.99884	
J25	620.omnetpp.s-874B	20,937	20	295,524	0.06614	0.99905	low_unique_event...
J24	620.omnetpp.s-874B	20,939	18	295,440	0.06615	0.99914	low_unique_event...
J13	623.xalancbmk.s-700B	227,823	1,028	218,108	0.50588	0.99551	
J12	623.xalancbmk.s-700B	169,978	577	275,422	0.37744	0.99662	
J14	623.xalancbmk.s-700B	135,984	613	310,237	0.30195	0.99551	
J11	623.xalancbmk.s-700B	401,150	749	42,845	0.89075	0.99814	
J09	623.xalancbmk.s-700B	388,893	652	55,607	0.86354	0.99833	
J10	623.xalancbmk.s-700B	80,808	263	367,905	0.17943	0.99676	

Projection notes. evidence/cache_miss_comparison.csv and analysis/normal_vs_standalone_target_attribution.csv were verified to be exact column subsets of this table with identical values; printing this table therefore reproduces

both files in full.

Table 61: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 1/4

ID	normal	both_late	both_timely	ipc-delta-standalone-minus-normal	l2-miss-delta-standalone-minus-normal	l2-miss-rate-delta-standalone-minus-normal	neither_timely	normal-event-coverage	normal-event-timeliness	normal_ipc	normal_l2-load-miss
J21	no_pref	—	—	0.06035	-184,977	-0.45762	18,151	0	0	0.368	203,131
J21	stride	—	33,929	0.05829	-150,123	-0.37139	17,082	0.17158	0.99532	0.37006	168,277
J21	streamer	—	176,943	-0.00326	2,679	0.00663	7,478	0.92344	0.99975	0.43161	15,475
J21	ampm	1	176,660	-0.00234	1,122	0.00278	8,602	0.91569	0.99678	0.43069	17,032
J21	spp	—	126,141	-1.000e-05	-52,719	-0.13042	12,059	0.65084	0.9971	0.42836	70,873
J21	ipcp	—	—	0.06039	-184,977	-0.45762	18,150	0	0	0.36796	203,131
J21	sms	—	152,952	0.00622	-23,015	-0.05694	9,057	0.79708	0.99893	0.42213	41,169
J21	sandbox	—	182,431	-0.00793	11,013	0.02725	4,648	0.96437	0.99978	0.43628	7,141
J21	power7	—	175,388	-0.0024	848	0.0021	7,683	0.91442	0.99933	0.43075	17,306
J20	no_pref	—	—	0.06055	-185,458	-0.45881	17,671	0	0	0.368	203,131
J20	stride	—	33,985	0.05849	-150,604	-0.37258	16,658	0.17158	0.99532	0.37006	168,277
J20	streamer	—	177,599	-0.00306	2,198	0.00544	7,654	0.92344	0.99975	0.43161	15,475
J20	ampm	1	177,291	-0.00214	641	0.00159	8,755	0.91569	0.99678	0.43069	17,032
J20	spp	—	126,610	1.900e-04	-53,200	-0.13161	12,049	0.65084	0.9971	0.42836	70,873
J20	ipcp	—	—	0.06059	-185,458	-0.45881	17,670	0	0	0.36796	203,131
J20	sms	—	153,325	0.00642	-23,496	-0.05813	8,952	0.79708	0.99893	0.42213	41,169
J20	sandbox	—	182,922	-0.00773	10,532	0.02606	4,659	0.96437	0.99978	0.43628	7,141
J20	power7	—	176,032	-0.0022	367	9.079e-04	7,847	0.91442	0.99933	0.43075	17,306
J22	no_pref	—	—	0.06266	-189,963	-0.46995	13,167	0	0	0.368	203,131
J22	stride	—	34,315	0.0606	-155,109	-0.38373	12,486	0.17158	0.99532	0.37006	168,277
J22	streamer	—	181,346	-9.500e-04	-2,307	-0.00571	6,899	0.92344	0.99975	0.43161	15,475
J22	ampm	1	181,041	-3.000e-05	-3,864	-0.00956	8,020	0.91569	0.99678	0.43069	17,032
J22	spp	—	129,244	0.0023	-57,705	-0.14276	10,187	0.65084	0.9971	0.42836	70,873
J22	ipcp	—	—	0.0627	-189,963	-0.46995	13,166	0	0	0.36796	203,131
J22	sms	—	156,725	0.00853	-28,001	-0.06927	7,851	0.79708	0.99893	0.42213	41,169
J22	sandbox	—	187,284	-0.00562	6,027	0.01491	4,518	0.96437	0.99978	0.43628	7,141
J22	power7	—	179,740	-9.000e-05	-4,138	-0.01024	7,053	0.91442	0.99933	0.43075	17,306
J19	no_pref	—	—	0.06124	-187,013	-0.46265	16,118	0	0	0.368	203,131
J19	stride	—	34,213	0.05918	-152,159	-0.37643	15,333	0.17158	0.99532	0.37006	168,277
J19	streamer	—	179,146	-0.00237	643	0.00159	7,648	0.92344	0.99975	0.43161	15,475
J19	ampm	—	178,826	-0.00145	-914	-0.00226	8,737	0.91569	0.99678	0.43069	17,032
J19	spp	—	127,711	8.800e-04	-54,755	-0.13546	11,599	0.65084	0.9971	0.42836	70,873
J19	ipcp	—	—	0.06128	-187,013	-0.46265	16,117	0	0	0.36796	203,131
J19	sms	—	154,707	0.00711	-25,051	-0.06197	8,779	0.79708	0.99893	0.42213	41,169
J19	sandbox	—	184,466	-0.00704	8,977	0.02221	4,649	0.96437	0.99978	0.43628	7,141
J19	power7	—	177,563	-0.00151	-1,188	-0.00294	7,825	0.91442	0.99933	0.43075	17,306
J15	no_pref	—	—	0.06124	-187,013	-0.46265	16,118	0	0	0.368	203,131
J15	stride	—	34,213	0.05918	-152,159	-0.37643	15,333	0.17158	0.99532	0.37006	168,277
J15	streamer	—	179,146	-0.00237	643	0.00159	7,648	0.92344	0.99975	0.43161	15,475
J15	ampm	—	178,826	-0.00145	-914	-0.00226	8,737	0.91569	0.99678	0.43069	17,032
J15	spp	—	127,711	8.800e-04	-54,755	-0.13546	11,599	0.65084	0.9971	0.42836	70,873
J15	ipcp	—	—	0.06128	-187,013	-0.46265	16,117	0	0	0.36796	203,131
J15	sms	—	154,707	0.00711	-25,051	-0.06197	8,779	0.79708	0.99893	0.42213	41,169
J15	sandbox	—	184,466	-0.00704	8,977	0.02221	4,649	0.96437	0.99978	0.43628	7,141
J15	power7	—	177,563	-0.00151	-1,188	-0.00294	7,825	0.91442	0.99933	0.43075	17,306
J17	no_pref	—	—	0.06035	-184,977	-0.45762	18,151	0	0	0.368	203,131
J17	stride	—	33,929	0.05829	-150,123	-0.37139	17,082	0.17158	0.99532	0.37006	168,277
J17	streamer	—	176,943	-0.00326	2,679	0.00663	7,478	0.92344	0.99975	0.43161	15,475
J17	ampm	1	176,660	-0.00234	1,122	0.00278	8,602	0.91569	0.99678	0.43069	17,032
J17	spp	—	126,141	-1.000e-05	-52,719	-0.13042	12,059	0.65084	0.9971	0.42836	70,873
J17	ipcp	—	—	0.06039	-184,977	-0.45762	18,150	0	0	0.36796	203,131
J17	sms	—	152,952	0.00622	-23,015	-0.05694	9,057	0.79708	0.99893	0.42213	41,169
J17	sandbox	—	182,431	-0.00793	11,013	0.02725	4,648	0.96437	0.99978	0.43628	7,141
J17	power7	—	175,388	-0.0024	848	0.0021	7,683	0.91442	0.99933	0.43075	17,306
J16	no_pref	—	—	0.06055	-185,458	-0.45881	17,671	0	0	0.368	203,131
J16	stride	—	33,985	0.05849	-150,604	-0.37258	16,658	0.17158	0.99532	0.37006	168,277
J16	streamer	—	177,599	-0.00306	2,198	0.00544	7,654	0.92344	0.99975	0.43161	15,475
J16	ampm	1	177,291	-0.00214	641	0.00159	8,755	0.91569	0.99678	0.43069	17,032
J16	spp	—	126,610	1.900e-04	-53,200	-0.13161	12,049	0.65084	0.9971	0.42836	70,873
J16	ipcp	—	—	0.06059	-185,458	-0.45881	17,670	0	0	0.36796	203,131
J16	sms	—	153,325	0.00642	-23,496	-0.05813	8,952	0.79708	0.99893	0.42213	41,169
J16	sandbox	—	182,922	-0.00773	10,532	0.02606	4,659	0.96437	0.99978	0.43628	7,141
J16	power7	—	176,032	-0.0022	367	9.079e-04	7,847	0.91442	0.99933	0.43075	17,306
J18	no_pref	—	—	0.0123	-40,311	-0.09973	162,818	0	0	0.368	203,131
J18	stride	—	3,718	0.01024	-5,457	-0.0135	131,526	0.17158	0.99532	0.37006	168,277
J18	streamer	—	38,567	-0.05131	147,345	0.36452	13,765	0.92344	0.99975	0.43161	15,475
J18	ampm	1	38,523	-0.05039	145,788	0.36067	14,853	0.91569	0.99678	0.43069	17,032
J18	spp	—	26,721	-0.04806	91,947	0.22747	57,046	0.65084	0.9971	0.42836	70,873
J18	ipcp	—	—	0.01234	-40,311	-0.09973	162,817	0	0	0.36796	203,131
J18	sms	—	33,186	-0.04183	121,651	0.30095	33,926	0.79708	0.99893	0.42213	41,169
J18	sandbox	—	39,822	-0.05598	155,679	0.38514	6,704	0.96437	0.99978	0.43628	7,141
J18	power7	—	38,257	-0.05045	145,514	0.35999	15,214	0.91442	0.99933	0.43075	17,306
J07	no_pref	—	—	0.01041	-116,991	-0.11874	553,429	0	0	0.18318	740,223
J07	stride	—	3,315	0.01	-112,163	-0.11384	551,903	0.00656	1	0.18359	735,395
J07	streamer	2,908	34,844	0.01009	-78,261	-0.07944	494,718	0.11592	0.84351	0.1835	701,493
J07	ampm	389	29,654	0.00485	-63,858	-0.06481	525,098	0.07791	0.97908	0.18874	687,090
J07	spp	143	3,521	0.00485	-112,174	-0.11385	551,192	0.00697	0.82562	0.18874	735,406
J07	ipcp	—	—	0.01041	-116,991	-0.11874	553,429	0	0	0.18318	740,223
J07	sms	14,988	39,799	0.00582	-58,955	-0.05984	475,781	0.12375	0.62805	0.18777	682,187

Table 61: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 1/4 (cont.)

ID	normal	both_late	both_timely	ipc_delta_standalone_minus_normal	l2_miss_delta_standalone_minus_normal	l2_miss_rate_delta_standalone_minus_normal	neither_timely	normal_event_coverage	normal_event_timeliness	normal_ipc	normal_l2_load_miss
J07	sandbox	17,800	49,409	0.02454	-90,998	-0.09237	383,008	0.21325	0.59013	0.16905	714,230
J07	power7	2,987	28,835	0.01196	-92,619	-0.09401	503,870	0.09593	0.82062	0.18163	715,851
J06	no-pref	-	-	0.01007	-120,741	-0.12254	552,658	0	0	0.18318	740,223
J06	stride	-	3,250	0.00966	-115,913	-0.11764	551,065	0.00656	1	0.18359	735,395
J06	streamer	3,082	34,404	0.00975	-82,011	-0.08324	494,071	0.11592	0.84351	0.1835	701,493
J06	ampm	401	29,008	0.00451	-67,608	-0.06862	523,738	0.07791	0.97908	0.18874	687,090
J06	spp	175	3,446	0.00451	-115,924	-0.11766	550,374	0.00697	0.82562	0.18874	735,406
J06	ipcp	-	-	0.01007	-120,741	-0.12254	552,658	0	0	0.18318	740,223
J06	sms	14,332	39,441	0.00548	-62,705	-0.06364	473,820	0.12375	0.62805	0.18777	682,187
J06	sandbox	17,355	48,657	0.0242	-94,748	-0.09618	381,561	0.21325	0.59013	0.16905	714,230
J06	power7	3,169	28,669	0.01162	-96,369	-0.09781	503,470	0.09593	0.82062	0.18163	715,851
J08	no-pref	-	-	0.00941	-119,587	-0.12137	555,712	0	0	0.18318	740,223
J08	stride	-	2,576	0.009	-114,759	-0.11647	553,443	0.00656	1	0.18359	735,395
J08	streamer	2,992	28,167	0.00909	-80,857	-0.08207	490,445	0.11592	0.84351	0.1835	701,493
J08	ampm	407	22,702	0.00385	-66,454	-0.06745	520,422	0.07791	0.97908	0.18874	687,090
J08	spp	170	2,776	0.00385	-114,770	-0.11649	552,760	0.00697	0.82562	0.18874	735,406
J08	ipcp	-	-	0.00941	-119,587	-0.12137	555,712	0	0	0.18318	740,223
J08	sms	15,063	32,756	0.00482	-61,551	-0.06248	471,962	0.12375	0.62805	0.18777	682,187
J08	sandbox	18,174	42,932	0.02354	-93,594	-0.09501	380,024	0.21325	0.59013	0.16905	714,230
J08	power7	3,075	23,997	0.01096	-95,215	-0.09664	501,422	0.09593	0.82062	0.18163	715,851
J05	no-pref	-	-	0.00855	-105,740	-0.10732	571,414	0	0	0.18318	740,223
J05	stride	-	2,773	0.00814	-100,912	-0.10242	569,340	0.00656	1	0.18359	735,395
J05	streamer	2,491	29,094	0.00823	-67,010	-0.06802	506,249	0.11592	0.84351	0.1835	701,493
J05	ampm	350	23,903	0.00299	-52,607	-0.0534	537,287	0.07791	0.97908	0.18874	687,090
J05	spp	123	2,911	0.00299	-100,923	-0.10243	568,514	0.00697	0.82562	0.18874	735,406
J05	ipcp	-	-	0.00855	-105,740	-0.10732	571,414	0	0	0.18318	740,223
J05	sms	13,239	33,049	0.00396	-47,704	-0.04842	483,949	0.12375	0.62805	0.18777	682,187
J05	sandbox	15,723	42,649	0.02268	-79,747	-0.08095	389,317	0.21325	0.59013	0.16905	714,230
J05	power7	2,561	25,035	0.0101	-81,368	-0.08259	517,348	0.09593	0.82062	0.18163	715,851
J01	no-pref	-	-	0.00617	-84,482	-0.08574	613,878	0	0	0.18318	740,223
J01	stride	-	1,050	0.00576	-79,654	-0.08085	610,081	0.00656	1	0.18359	735,395
J01	streamer	77	13,914	0.00585	-45,752	-0.04644	529,964	0.11592	0.84351	0.1835	701,493
J01	ampm	4	9,704	6.100e-04	-31,349	-0.03182	564,965	0.07791	0.97908	0.18874	687,090
J01	spp	7	876	6.100e-04	-79,665	-0.08086	608,821	0.00697	0.82562	0.18874	735,406
J01	ipcp	-	-	0.00617	-84,482	-0.08574	613,878	0	0	0.18318	740,223
J01	sms	289	17,361	0.00158	-26,446	-0.02685	496,818	0.12375	0.62805	0.18777	682,187
J01	sandbox	488	25,581	0.0203	-58,489	-0.05938	394,267	0.21325	0.59013	0.16905	714,230
J01	power7	74	13,426	0.00772	-60,110	-0.06101	544,693	0.09593	0.82062	0.18163	715,851
J03	no-pref	-	-	0.0094	-106,619	-0.10821	587,637	0	0	0.18318	740,223
J03	stride	-	3,034	0.00899	-101,791	-0.10331	585,827	0.00656	1	0.18359	735,395
J03	streamer	130	30,030	0.00908	-67,889	-0.06891	520,101	0.11592	0.84351	0.1835	701,493
J03	ampm	14	25,764	0.00384	-53,486	-0.05429	554,788	0.07791	0.97908	0.18874	687,090
J03	spp	9	3,117	0.00384	-101,802	-0.10332	584,858	0.00697	0.82562	0.18874	735,406
J03	ipcp	-	-	0.0094	-106,619	-0.10821	587,637	0	0	0.18318	740,223
J03	sms	504	34,146	0.00481	-48,583	-0.04931	488,891	0.12375	0.62805	0.18777	682,187
J03	sandbox	787	42,744	0.02353	-80,626	-0.08184	387,732	0.21325	0.59013	0.16905	714,230
J03	power7	131	25,501	0.01095	-82,247	-0.08348	530,777	0.09593	0.82062	0.18163	715,851
J02	no-pref	-	-	0.00916	-110,803	-0.11246	576,394	0	0	0.18318	740,223
J02	stride	-	2,628	0.00875	-105,975	-0.10756	574,177	0.00656	1	0.18359	735,395
J02	streamer	1,770	28,948	0.00884	-72,073	-0.07315	510,380	0.11592	0.84351	0.1835	701,493
J02	ampm	207	24,114	0.0036	-57,670	-0.05853	542,247	0.07791	0.97908	0.18874	687,090
J02	spp	98	2,689	0.0036	-105,986	-0.10757	573,275	0.00697	0.82562	0.18874	735,406
J02	ipcp	-	-	0.00916	-110,803	-0.11246	576,394	0	0	0.18318	740,223
J02	sms	7,849	33,595	0.00457	-52,767	-0.05356	484,719	0.12375	0.62805	0.18777	682,187
J02	sandbox	9,679	41,917	0.02329	-84,810	-0.08609	387,835	0.21325	0.59013	0.16905	714,230
J02	power7	1,812	24,431	0.01071	-86,431	-0.08773	521,032	0.09593	0.82062	0.18163	715,851
J04	no-pref	-	-	0.00649	-95,986	-0.09742	618,417	0	0	0.18318	740,223
J04	stride	-	697	0.00608	-91,158	-0.09252	614,267	0.00656	1	0.18359	735,395
J04	streamer	128	10,758	0.00617	-57,256	-0.05812	531,173	0.11592	0.84351	0.1835	701,493
J04	ampm	6	7,183	9.300e-04	-42,853	-0.04349	566,927	0.07791	0.97908	0.18874	687,090
J04	spp	14	657	9.300e-04	-91,169	-0.09253	613,155	0.00697	0.82562	0.18874	735,406
J04	ipcp	-	-	0.00649	-95,986	-0.09742	618,417	0	0	0.18318	740,223
J04	sms	561	14,407	0.0019	-37,950	-0.03852	499,787	0.12375	0.62805	0.18777	682,187
J04	sandbox	906	21,348	0.02062	-69,993	-0.07105	395,992	0.21325	0.59013	0.16905	714,230
J04	power7	127	10,381	0.00804	-71,614	-0.07269	545,988	0.09593	0.82062	0.18163	715,851
J43	no-pref	-	-	0.04723	-228,446	-0.7743	59,061	0	0	0.322	294,639
J43	stride	838	59,576	0.04458	-161,035	-0.54584	51,415	0.22817	0.81103	0.32465	227,228
J43	streamer	4,279	122,315	0.00555	-86,129	-0.29194	27,240	0.4825	0.55882	0.36368	152,322
J43	ampm	788	201,716	-0.00557	7,504	0.02543	27,593	0.79906	0.90457	0.3748	58,689
J43	spp	1,991	55,227	-0.0101	-159,639	-0.54112	35,866	0.23317	0.52099	0.37933	225,832
J43	ipcp	-	-	0.04723	-228,450	-0.77431	59,061	0	0	0.322	294,643
J43	sms	1,005	208,911	-0.01182	27,488	0.09316	8,292	0.86638	0.91014	0.38105	38,705
J43	sandbox	3,172	168,902	-0.00517	-19,090	-0.06472	8,315	0.70943	0.73119	0.3744	85,283
J43	power7	2,883	137,279	0.00795	-67,903	-0.23017	28,803	0.54415	0.67674	0.36128	134,096
J38	no-pref	-	-	0.04801	-236,581	-0.80188	49,725	0	0	0.322	294,639
J38	stride	1,238	57,342	0.04536	-169,170	-0.57342	40,563	0.22817	0.81103	0.32465	227,228
J38	streamer	4,004	119,679	0.00633	-94,264	-0.31951	19,550	0.4825	0.55882	0.36368	152,322
J38	ampm	594	202,611	-0.00479	-631	-0.00215	19,814	0.79906	0.90457	0.3748	58,689
J38	spp	2,630	57,062	-0.00932	-167,774	-0.56869	29,367	0.23317	0.52099	0.37933	225,832
J38	ipcp	-	-	0.04801	-236,585	-0.80188	49,725	0	0	0.322	294,643
J38	sms	1,228	212,365	-0.01104	19,353	0.06558	6,513	0.86638	0.91014	0.38105	38,705

Table 61: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 1/4 (cont.)

ID	normal	both_late	both_timely	ipc_delta_standalone_minus_normal	l2_miss_delta_standalone_minus_normal	l2_miss_rate_delta_standalone_minus_normal	neither_timely	normal_event_coverage	normal_event_timeliness	normal_ipc	normal_l2_load_miss
J38	sandbox	3,868	173,429	-0.00439	-27,225	-0.0923	6,455	0.70943	0.73119	0.3744	85,283
J38	power7	2,820	135,762	0.00873	-76,038	-0.25775	21,537	0.54415	0.67674	0.36128	134,096
J47	no_pref	-	-	0.06179	-291,123	-0.98673	3,992	0	0	0.322	294,639
J47	stride	82	66,678	0.05914	-223,712	-0.75828	3,473	0.22817	0.81103	0.32465	227,228
J47	streamer	185	140,336	0.02011	-148,806	-0.50437	1,336	0.4825	0.55882	0.36368	152,322
J47	ampm	44	232,994	0.00899	-55,173	-0.18701	1,266	0.79906	0.90457	0.3748	58,689
J47	spp	120	67,673	0.00446	-222,316	-0.75355	2,296	0.23317	0.52099	0.37933	225,832
J47	ipcp	-	-	0.06179	-291,127	-0.98674	3,992	0	0	0.322	294,643
J47	sms	70	252,027	0.00274	-35,189	-0.11927	739	0.86638	0.91014	0.38105	38,705
J47	sandbox	167	206,443	0.00939	-81,767	-0.27715	684	0.70943	0.73119	0.3744	85,283
J47	power7	135	158,487	0.02251	-130,580	-0.4426	1,620	0.54415	0.67674	0.36128	134,096
J31	no_pref	-	-	0.05044	-234,229	-0.79391	50,003	0	0	0.322	294,639
J31	stride	840	59,438	0.04779	-166,818	-0.56546	42,687	0.22817	0.81103	0.32465	227,228
J31	streamer	5,156	123,246	0.00876	-91,912	-0.31155	25,843	0.4825	0.55882	0.36368	152,322
J31	ampm	1,099	206,418	-0.00236	1,721	0.00581	25,668	0.79906	0.90457	0.3748	58,689
J31	spp	3,088	57,006	-0.00689	-165,422	-0.56073	30,367	0.23317	0.52099	0.37933	225,832
J31	ipcp	-	-	0.05044	-234,233	-0.79392	50,003	0	0	0.322	294,643
J31	sms	1,393	212,242	-0.00861	21,705	0.07355	7,004	0.86638	0.91014	0.38105	38,705
J31	sandbox	4,729	171,801	-0.00196	-24,873	-0.08433	6,499	0.70943	0.73119	0.3744	85,283
J31	power7	3,625	139,115	0.01116	-73,686	-0.24978	27,089	0.54415	0.67674	0.36128	134,096
J34	no_pref	-	-	0.05356	-249,507	-0.8457	43,311	0	0	0.322	294,639
J34	stride	86	62,743	0.05091	-182,096	-0.61724	37,238	0.22817	0.81103	0.32465	227,228
J34	streamer	763	130,539	0.01188	-107,190	-0.36334	25,309	0.4825	0.55882	0.36368	152,322
J34	ampm	493	219,759	7.600e-04	-13,557	-0.04598	25,438	0.79906	0.90457	0.3748	58,689
J34	spp	621	60,005	-0.00377	-180,700	-0.61252	26,565	0.23317	0.52099	0.37933	225,832
J34	ipcp	-	-	0.05356	-249,511	-0.84571	43,311	0	0	0.322	294,643
J34	sms	651	225,625	-0.00549	6,427	0.02176	6,785	0.86638	0.91014	0.38105	38,705
J34	sandbox	1,022	181,418	0.00116	-40,151	-0.13612	6,489	0.70943	0.73119	0.3744	85,283
J34	power7	629	147,452	0.01428	-88,964	-0.30157	26,103	0.54415	0.67674	0.36128	134,096
J30	no_pref	-	-	0.05044	-234,229	-0.79391	50,003	0	0	0.322	294,639
J30	stride	840	59,438	0.04779	-166,818	-0.56546	42,687	0.22817	0.81103	0.32465	227,228
J30	streamer	5,156	123,246	0.00876	-91,912	-0.31155	25,843	0.4825	0.55882	0.36368	152,322
J30	ampm	1,099	206,418	-0.00236	1,721	0.00581	25,668	0.79906	0.90457	0.3748	58,689
J30	spp	3,088	57,006	-0.00689	-165,422	-0.56073	30,367	0.23317	0.52099	0.37933	225,832
J30	ipcp	-	-	0.05044	-234,233	-0.79392	50,003	0	0	0.322	294,643
J30	sms	1,393	212,242	-0.00861	21,705	0.07355	7,004	0.86638	0.91014	0.38105	38,705
J30	sandbox	4,729	171,801	-0.00196	-24,873	-0.08433	6,499	0.70943	0.73119	0.3744	85,283
J30	power7	3,625	139,115	0.01116	-73,686	-0.24978	27,089	0.54415	0.67674	0.36128	134,096
J35	no_pref	-	-	0.05084	-230,277	-0.78052	53,934	0	0	0.322	294,639
J35	stride	1,028	58,529	0.04819	-162,866	-0.55206	45,917	0.22817	0.81103	0.32465	227,228
J35	streamer	5,441	122,626	0.00916	-87,960	-0.29815	27,808	0.4825	0.55882	0.36368	152,322
J35	ampm	973	204,141	-0.00196	5,673	0.01921	27,704	0.79906	0.90457	0.3748	58,689
J35	spp	3,096	56,979	-0.00649	-161,470	-0.54733	33,250	0.23317	0.52099	0.37933	225,832
J35	ipcp	-	-	0.05084	-230,281	-0.78052	53,934	0	0	0.322	294,643
J35	sms	1,410	210,097	-0.00821	25,657	0.08694	7,822	0.86638	0.91014	0.38105	38,705
J35	sandbox	4,626	170,088	-0.00156	-20,921	-0.07094	7,623	0.70943	0.73119	0.3744	85,283
J35	power7	3,495	137,234	0.01156	-69,734	-0.23639	28,857	0.54415	0.67674	0.36128	134,096
J36	no_pref	-	-	0.04839	-226,332	-0.76715	58,013	0	0	0.322	294,639
J36	stride	870	56,800	0.04574	-158,921	-0.5387	48,165	0.22817	0.81103	0.32465	227,228
J36	streamer	5,223	119,666	0.00671	-84,015	-0.28479	27,989	0.4825	0.55882	0.36368	152,322
J36	ampm	901	200,342	-0.00441	9,618	0.03257	27,333	0.79906	0.90457	0.3748	58,689
J36	spp	3,012	56,039	-0.00894	-157,525	-0.53397	35,371	0.23317	0.52099	0.37933	225,832
J36	ipcp	-	-	0.04839	-226,336	-0.76716	58,013	0	0	0.322	294,643
J36	sms	1,234	205,872	-0.01066	29,602	0.10031	7,012	0.86638	0.91014	0.38105	38,705
J36	sandbox	4,492	166,250	-0.00401	-16,976	-0.05757	7,011	0.70943	0.73119	0.3744	85,283
J36	power7	3,595	134,473	0.00911	-65,789	-0.22302	29,461	0.54415	0.67674	0.36128	134,096
J32	no_pref	-	-	0.0526	-246,431	-0.83527	44,295	0	0	0.322	294,639
J32	stride	204	62,242	0.04995	-179,020	-0.60681	37,959	0.22817	0.81103	0.32465	227,228
J32	streamer	1,875	129,193	0.01092	-104,114	-0.35291	25,426	0.4825	0.55882	0.36368	152,322
J32	ampm	690	217,094	-2.000e-04	-10,481	-0.03555	25,475	0.79906	0.90457	0.3748	58,689
J32	spp	1,201	59,430	-0.00473	-177,624	-0.60209	27,152	0.23317	0.52099	0.37933	225,832
J32	ipcp	-	-	0.0526	-246,435	-0.83528	44,295	0	0	0.322	294,643
J32	sms	795	222,920	-0.00645	9,503	0.03219	6,822	0.86638	0.91014	0.38105	38,705
J32	sandbox	1,875	179,477	2.000e-04	-37,075	-0.12569	6,501	0.70943	0.73119	0.3744	85,283
J32	power7	1,451	146,018	0.01332	-85,888	-0.29114	26,282	0.54415	0.67674	0.36128	134,096
J33	no_pref	-	-	0.05356	-249,507	-0.8457	43,311	0	0	0.322	294,639
J33	stride	86	62,743	0.05091	-182,096	-0.61724	37,238	0.22817	0.81103	0.32465	227,228
J33	streamer	763	130,539	0.01188	-107,190	-0.36334	25,309	0.4825	0.55882	0.36368	152,322
J33	ampm	493	219,759	7.600e-04	-13,557	-0.04598	25,438	0.79906	0.90457	0.3748	58,689
J33	spp	621	60,005	-0.00377	-180,700	-0.61252	26,565	0.23317	0.52099	0.37933	225,832
J33	ipcp	-	-	0.05356	-249,511	-0.84571	43,311	0	0	0.322	294,643
J33	sms	651	225,625	-0.00549	6,427	0.02176	6,785	0.86638	0.91014	0.38105	38,705
J33	sandbox	1,022	181,418	0.00116	-40,151	-0.13612	6,489	0.70943	0.73119	0.3744	85,283
J33	power7	629	147,452	0.01428	-88,964	-0.30157	26,103	0.54415	0.67674	0.36128	134,096
J46	no_pref	-	-	0.05349	-252,105	-0.85449	41,470	0	0	0.322	294,639
J46	stride	130	62,309	0.05084	-184,694	-0.62604	35,072	0.22817	0.81103	0.32465	227,228
J46	streamer	716	127,938	0.01181	-109,788	-0.37213	19,121	0.4825	0.55882	0.36368	152,322
J46	ampm	402	216,424	6.900e-04	-16,155	-0.05477	19,979	0.79906	0.90457	0.3748	58,689
J46	spp	461	60,518	-0.00384	-183,298	-0.62131	25,627	0.23317	0.52099	0.37933	225,832
J46	ipcp	-	-	0.05349	-252,109	-0.8545	41,470	0	0	0.322	294,643
J46	sms	279	227,102	-0.00556	3,829	0.01297	7,072	0.86638	0.91014	0.38105	38,705

Table 61: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 1/4 (cont.)

ID	normal	both_late	both_timely	ipc_delta_standalone_minus_normal	l2_miss_delta_standalone_minus_normal	l2_miss_rate_delta_standalone_minus_normal	neither_timely	normal_event_coverage	normal_event_timeliness	normal_ipc	normal_l2_load_miss
J46	sandbox	639	183,781	0.00109	-42,749	-0.14491	7,390	0.70943	0.73119	0.3744	85,283
J46	power7	585	144,783	0.01421	-91,562	-0.31036	20,136	0.54415	0.67674	0.36128	134,096
J41	no_pref	-	-	0.05481	-260,267	-0.88216	33,516	0	0	0.322	294,639
J41	stride	112	62,415	0.05216	-192,856	-0.65371	27,235	0.22817	0.81103	0.32465	227,228
J41	streamer	540	130,987	0.01313	-117,950	-0.3998	15,629	0.4825	0.55882	0.36368	152,322
J41	ampm	259	219,583	0.00201	-24,317	-0.08244	15,790	0.79906	0.90457	0.3748	58,689
J41	spp	419	62,375	-0.00252	-191,460	-0.64898	21,089	0.23317	0.52099	0.37933	225,832
J41	ipcp	-	-	0.05481	-260,271	-0.88217	33,516	0	0	0.322	294,643
J41	sms	335	232,090	-0.00424	-4,333	-0.0147	6,123	0.86638	0.91014	0.38105	38,705
J41	sandbox	645	188,525	0.00241	-50,911	-0.17258	6,167	0.70943	0.73119	0.3744	85,283
J41	power7	381	147,770	0.01553	-99,724	-0.33803	16,449	0.54415	0.67674	0.36128	134,096
J42	no_pref	-	-	0.04723	-228,446	-0.7743	59,061	0	0	0.322	294,639
J42	stride	838	59,576	0.04458	-161,035	-0.54584	51,415	0.22817	0.81103	0.32465	227,228
J42	streamer	4,279	122,315	0.00555	-86,129	-0.29194	27,240	0.4825	0.55882	0.36368	152,322
J42	ampm	788	201,716	-0.00557	7,504	0.02543	27,593	0.79906	0.90457	0.3748	58,689
J42	spp	1,991	55,227	-0.0101	-159,639	-0.54112	35,866	0.23317	0.52099	0.37933	225,832
J42	ipcp	-	-	0.04723	-228,450	-0.77431	59,061	0	0	0.322	294,643
J42	sms	1,005	208,911	-0.01182	27,488	0.09316	8,292	0.86638	0.91014	0.38105	38,705
J42	sandbox	3,172	168,902	-0.00517	-19,090	-0.06472	8,315	0.70943	0.73119	0.3744	85,283
J42	power7	2,883	137,279	0.00795	-67,903	-0.23017	28,803	0.54415	0.67674	0.36128	134,096
J37	no_pref	-	-	0.04801	-236,581	-0.80188	49,725	0	0	0.322	294,639
J37	stride	1,238	57,342	0.04536	-169,170	-0.57342	40,563	0.22817	0.81103	0.32465	227,228
J37	streamer	4,004	119,679	0.00633	-94,264	-0.31951	19,550	0.4825	0.55882	0.36368	152,322
J37	ampm	594	202,611	-0.00479	-631	-0.00215	19,814	0.79906	0.90457	0.3748	58,689
J37	spp	2,630	57,062	-0.00932	-167,774	-0.56869	29,367	0.23317	0.52099	0.37933	225,832
J37	ipcp	-	-	0.04801	-236,585	-0.80188	49,725	0	0	0.322	294,643
J37	sms	1,228	212,365	-0.01104	19,353	0.06558	6,513	0.86638	0.91014	0.38105	38,705
J37	sandbox	3,868	173,429	-0.00439	-27,225	-0.0923	6,455	0.70943	0.73119	0.3744	85,283
J37	power7	2,820	135,762	0.00873	-76,038	-0.25775	21,537	0.54415	0.67674	0.36128	134,096
J44	no_pref	-	-	0.05238	-247,244	-0.83802	44,593	0	0	0.322	294,639
J44	stride	374	61,582	0.04973	-179,833	-0.60956	37,910	0.22817	0.81103	0.32465	227,228
J44	streamer	1,808	126,761	0.0107	-104,927	-0.35566	20,639	0.4825	0.55882	0.36368	152,322
J44	ampm	555	213,371	-4.200e-04	-11,294	-0.03829	21,446	0.79906	0.90457	0.3748	58,689
J44	spp	928	59,474	-0.00495	-178,437	-0.60484	27,558	0.23317	0.52099	0.37933	225,832
J44	ipcp	-	-	0.05238	-247,248	-0.83803	44,593	0	0	0.322	294,643
J44	sms	491	223,428	-0.00667	8,690	0.02944	7,337	0.86638	0.91014	0.38105	38,705
J44	sandbox	1,322	180,840	-2.000e-05	-37,888	-0.12844	7,627	0.70943	0.73119	0.3744	85,283
J44	power7	1,224	143,038	0.0131	-86,701	-0.29389	21,743	0.54415	0.67674	0.36128	134,096
J39	no_pref	-	-	0.05454	-256,078	-0.86797	35,599	0	0	0.322	294,639
J39	stride	545	61,081	0.05189	-188,667	-0.63951	28,728	0.22817	0.81103	0.32465	227,228
J39	streamer	1,594	129,171	0.01286	-113,761	-0.38561	16,183	0.4825	0.55882	0.36368	152,322
J39	ampm	349	216,226	0.00174	-20,128	-0.06825	16,294	0.79906	0.90457	0.3748	58,689
J39	spp	1,034	61,555	-0.00279	-187,271	-0.63479	22,213	0.23317	0.52099	0.37933	225,832
J39	ipcp	-	-	0.05454	-256,082	-0.86798	35,599	0	0	0.322	294,643
J39	sms	502	228,491	-0.00451	-144	-5.089e-04	6,156	0.86638	0.91014	0.38105	38,705
J39	sandbox	1,407	185,726	0.00214	-46,722	-0.15839	6,212	0.70943	0.73119	0.3744	85,283
J39	power7	1,055	145,473	0.01526	-95,535	-0.32384	17,050	0.54415	0.67674	0.36128	134,096
J45	no_pref	-	-	0.05349	-252,105	-0.85449	41,470	0	0	0.322	294,639
J45	stride	130	62,309	0.05084	-184,694	-0.62604	35,072	0.22817	0.81103	0.32465	227,228
J45	streamer	716	127,938	0.01181	-109,788	-0.37213	19,121	0.4825	0.55882	0.36368	152,322
J45	ampm	402	216,424	6.900e-04	-16,155	-0.05477	19,979	0.79906	0.90457	0.3748	58,689
J45	spp	461	60,518	-0.00384	-183,298	-0.62131	25,627	0.23317	0.52099	0.37933	225,832
J45	ipcp	-	-	0.05349	-252,109	-0.8545	41,470	0	0	0.322	294,643
J45	sms	279	227,102	-0.00556	3,829	0.01297	7,072	0.86638	0.91014	0.38105	38,705
J45	sandbox	639	183,781	0.00109	-42,749	-0.14491	7,390	0.70943	0.73119	0.3744	85,283
J45	power7	585	144,783	0.01421	-91,562	-0.31036	20,136	0.54415	0.67674	0.36128	134,096
J40	no_pref	-	-	0.05481	-260,267	-0.88216	33,516	0	0	0.322	294,639
J40	stride	112	62,415	0.05216	-192,856	-0.65371	27,235	0.22817	0.81103	0.32465	227,228
J40	streamer	540	130,987	0.01313	-117,950	-0.3998	15,629	0.4825	0.55882	0.36368	152,322
J40	ampm	259	219,583	0.00201	-24,317	-0.08244	15,790	0.79906	0.90457	0.3748	58,689
J40	spp	419	62,375	-0.00252	-191,460	-0.64898	21,089	0.23317	0.52099	0.37933	225,832
J40	ipcp	-	-	0.05481	-260,271	-0.88217	33,516	0	0	0.322	294,643
J40	sms	335	232,090	-0.00424	-4,333	-0.0147	6,123	0.86638	0.91014	0.38105	38,705
J40	sandbox	645	188,525	0.00241	-50,911	-0.17258	6,167	0.70943	0.73119	0.3744	85,283
J40	power7	381	147,770	0.01553	-99,724	-0.33803	16,449	0.54415	0.67674	0.36128	134,096
J28	no_pref	-	-	0.00355	-37,921	-0.08403	275,839	0	0	0.2375	316,554
J28	stride	-	37	0.00272	-35,061	-0.0777	272,886	0.00945	1	0.23833	313,694
J28	streamer	3	25,032	-0.00123	355	7.140e-04	247,732	0.16237	0.96377	0.24228	278,278
J28	ampm	2	12,079	0.00129	-21,421	-0.04753	265,426	0.06914	0.96143	0.23976	300,054
J28	spp	-	28	0.00236	-37,714	-0.08358	275,817	1.548e-04	0.79032	0.23869	316,347
J28	ipcp	1	1,085	0.00334	-36,501	-0.08087	274,572	0.00676	0.87988	0.23771	315,134
J28	sms	2	21,128	-0.0059	32,391	0.0716	209,325	0.2717	0.98027	0.24695	246,242
J28	sandbox	1	37,158	-0.00328	41,039	0.09073	171,384	0.42346	0.94365	0.24433	237,594
J28	power7	3	24,794	-0.00127	-298	-7.587e-04	247,532	0.16213	0.96255	0.24232	278,931
J27	no_pref	-	-	0.00371	-45,750	-0.10141	267,947	0	0	0.2375	316,554
J27	stride	-	83	0.00288	-42,890	-0.09508	265,040	0.00945	1	0.23833	313,694
J27	streamer	12	27,279	-0.00107	-7,474	-0.01667	242,277	0.16237	0.96377	0.24228	278,278
J27	ampm	3	12,035	0.00145	-29,250	-0.06491	257,466	0.06914	0.96143	0.23976	300,054
J27	spp	-	29	0.00252	-45,543	-0.10096	267,925	1.548e-04	0.79032	0.23869	316,347
J27	ipcp	-	1,069	0.0035	-44,330	-0.09825	266,644	0.00676	0.87988	0.23771	315,134
J27	sms	3	24,791	-0.00574	24,562	0.05422	205,128	0.2717	0.98027	0.24695	246,242

Table 61: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 1/4 (cont.)

ID	normal	both_late	both_timely	ipc_delta_standalone_minus_normal	l2_miss_delta_standalone_minus_normal	l2_miss_rate_delta_standalone_minus_normal	neither_timely	normal_event_coverage	normal_event_timeliness	normal_ipc	normal_l2_load_miss
J27	sandbox	20	41,110	-0.00312	33,210	0.07335	167,835	0.42346	0.94365	0.24433	237,594
J27	power7	13	26,928	-0.00111	-8,127	-0.01814	241,987	0.16213	0.96255	0.24232	278,931
J29	no_pref	-	-	0.00202	-24,081	-0.05333	285,944	0	0	0.2375	316,554
J29	stride	-	100	0.00119	-21,221	-0.047	283,054	0.00945	1	0.23833	313,694
J29	streamer	9	19,619	-0.00276	14,195	0.03141	252,469	0.16237	0.96377	0.24228	278,278
J29	ampm	8	8,913	-2.400e-04	-7,581	-0.01683	272,461	0.06914	0.96143	0.23976	300,054
J29	spp	-	21	8.300e-04	-23,874	-0.05289	285,912	1.548e-04	0.79032	0.23869	316,347
J29	ipcp	5	871	0.00181	-22,661	-0.05017	284,585	0.00676	0.87988	0.23771	315,134
J29	sms	20	14,926	-0.00743	46,231	0.1023	213,274	0.2717	0.98027	0.24695	246,242
J29	sandbox	26	28,247	-0.00481	54,879	0.12142	172,434	0.42346	0.94365	0.24433	237,594
J29	power7	8	18,941	-0.0028	13,542	0.02994	251,837	0.16213	0.96255	0.24232	278,931
J26	no_pref	-	-	0.00445	-44,790	-0.09931	270,719	0	0	0.2375	316,554
J26	stride	-	1,161	0.00362	-41,930	-0.09298	268,890	0.00945	1	0.23833	313,694
J26	streamer	10	27,379	-3.300e-04	-6,514	-0.01457	244,974	0.16237	0.96377	0.24228	278,278
J26	ampm	3	13,216	0.00219	-28,290	-0.06282	261,439	0.06914	0.96143	0.23976	300,054
J26	spp	-	34	0.00326	-44,583	-0.09887	270,704	1.548e-04	0.79032	0.23869	316,347
J26	ipcp	-	1,084	0.00424	-43,370	-0.09615	269,437	0.00676	0.87988	0.23771	315,134
J26	sms	1	23,631	-0.005	25,522	0.05631	206,669	0.2717	0.98027	0.24695	246,242
J26	sandbox	10	40,795	-0.00238	34,170	0.07544	170,027	0.42346	0.94365	0.24433	237,594
J26	power7	8	26,815	-3.700e-04	-7,167	-0.01604	244,434	0.16213	0.96255	0.24232	278,931
J23	no_pref	-	-	0.00445	-44,790	-0.09931	270,719	0	0	0.2375	316,554
J23	stride	-	1,161	0.00362	-41,930	-0.09298	268,890	0.00945	1	0.23833	313,694
J23	streamer	10	27,379	-3.300e-04	-6,514	-0.01457	244,974	0.16237	0.96377	0.24228	278,278
J23	ampm	3	13,216	0.00219	-28,290	-0.06282	261,439	0.06914	0.96143	0.23976	300,054
J23	spp	-	34	0.00326	-44,583	-0.09887	270,704	1.548e-04	0.79032	0.23869	316,347
J23	ipcp	-	1,084	0.00424	-43,370	-0.09615	269,437	0.00676	0.87988	0.23771	315,134
J23	sms	1	23,631	-0.005	25,522	0.05631	206,669	0.2717	0.98027	0.24695	246,242
J23	sandbox	10	40,795	-0.00238	34,170	0.07544	170,027	0.42346	0.94365	0.24433	237,594
J23	power7	8	26,815	-3.700e-04	-7,167	-0.01604	244,434	0.16213	0.96255	0.24232	278,931
J25	no_pref	-	-	0.00213	-20,479	-0.04542	295,597	0	0	0.2375	316,554
J25	stride	-	5	0.0013	-17,619	-0.03909	292,612	0.00945	1	0.23833	313,694
J25	streamer	-	14,812	-0.00265	17,797	0.03932	257,171	0.16237	0.96377	0.24228	278,278
J25	ampm	-	7,417	-1.300e-04	-3,979	-0.00892	280,356	0.06914	0.96143	0.23976	300,054
J25	spp	-	20	9.400e-04	-20,272	-0.04497	295,566	1.548e-04	0.79032	0.23869	316,347
J25	ipcp	-	66	0.00192	-19,059	-0.04226	293,233	0.00676	0.87988	0.23771	315,134
J25	sms	-	10,401	-0.00732	49,833	0.11021	218,291	0.2717	0.98027	0.24695	246,242
J25	sandbox	-	19,086	-0.0047	58,481	0.12934	172,905	0.42346	0.94365	0.24433	237,594
J25	power7	1	14,778	-0.00269	17,144	0.03785	257,165	0.16213	0.96255	0.24232	278,931
J24	no_pref	-	-	0.00209	-20,536	-0.04551	295,597	0	0	0.2375	316,554
J24	stride	-	4	0.00126	-17,676	-0.03918	292,611	0.00945	1	0.23833	313,694
J24	streamer	-	15,551	-0.00269	17,740	0.03923	257,949	0.16237	0.96377	0.24228	278,278
J24	ampm	-	7,323	-1.700e-04	-4,036	-0.00902	280,258	0.06914	0.96143	0.23976	300,054
J24	spp	-	24	9.000e-04	-20,329	-0.04507	295,569	1.548e-04	0.79032	0.23869	316,347
J24	ipcp	1	624	0.00188	-19,116	-0.04235	293,810	0.00676	0.87988	0.23771	315,134
J24	sms	-	10,395	-0.00736	49,776	0.11012	218,297	0.2717	0.98027	0.24695	246,242
J24	sandbox	1	19,727	-0.00474	58,424	0.12924	173,657	0.42346	0.94365	0.24433	237,594
J24	power7	-	15,248	-0.00273	17,087	0.03776	257,720	0.16213	0.96255	0.24232	278,931
J13	no_pref	-	-	0.00387	-69,325	-0.05709	221,499	0	0	0.35321	450,350
J13	stride	35	1,157	0.00368	-67,262	-0.0554	218,250	0.0047	0.22068	0.3534	448,287
J13	streamer	35	13,992	0.02064	-349,216	-0.28686	207,955	0.05687	0.87344	0.33644	730,241
J13	ampm	2	10,618	0.00784	-124,746	-0.10263	211,522	0.04184	0.87576	0.34924	505,771
J13	spp	1	465	0.00317	-58,607	-0.0483	219,935	0.00251	0.38127	0.35391	439,632
J13	ipcp	-	1,259	0.00467	-80,165	-0.06599	219,886	0.00626	0.97272	0.35241	461,190
J13	sms	11	34,667	0.01541	-275,597	-0.22623	184,529	0.15225	0.9332	0.34167	656,622
J13	sandbox	35	82,247	0.01184	-218,978	-0.1799	134,958	0.3658	0.948	0.34524	600,003
J13	power7	43	9,204	0.01378	-234,443	-0.19262	211,268	0.0383	0.79227	0.3433	615,468
J12	no_pref	-	-	-0.00194	31,168	0.0257	279,795	0	0	0.35321	450,350
J12	stride	1	1,172	-0.00213	33,231	0.0274	276,812	0.0047	0.22068	0.3534	448,287
J12	streamer	3	13,053	0.01483	-248,723	-0.20407	265,047	0.05687	0.87344	0.33644	730,241
J12	ampm	4	11,421	0.00203	-24,253	-0.01984	271,000	0.04184	0.87576	0.34924	505,771
J12	spp	-	405	-0.00264	41,886	0.03449	278,310	0.00251	0.38127	0.35391	439,632
J12	ipcp	-	12	-0.00114	20,328	0.0168	276,912	0.00626	0.97272	0.35241	461,190
J12	sms	152	22,654	0.0096	-175,104	-0.14344	230,299	0.15225	0.9332	0.34167	656,622
J12	sandbox	3	65,364	0.00603	-118,485	-0.09711	175,229	0.3658	0.948	0.34524	600,003
J12	power7	2	8,284	0.00797	-133,950	-0.10983	268,580	0.0383	0.79227	0.3433	615,468
J14	no_pref	-	-	-0.01172	208,624	0.17131	313,753	0	0	0.35321	450,350
J14	stride	63	875	-0.01191	210,687	0.173	308,420	0.0047	0.22068	0.3534	448,287
J14	streamer	67	9,444	0.00505	-71,267	-0.05846	295,401	0.05687	0.87344	0.33644	730,241
J14	ampm	48	7,227	-0.00775	153,203	0.12577	300,401	0.04184	0.87576	0.34924	505,771
J14	spp	18	417	-0.01242	199,342	0.1801	311,866	0.00251	0.38127	0.35391	439,632
J14	ipcp	1	106	-0.01092	197,784	0.16241	310,975	0.00626	0.97272	0.35241	461,190
J14	sms	17	19,375	-1.800e-04	2,352	0.00217	260,662	0.15225	0.9332	0.34167	656,622
J14	sandbox	76	49,533	-0.00375	58,971	0.0485	192,667	0.3658	0.948	0.34524	600,003
J14	power7	100	6,202	-0.00181	43,506	0.03578	300,031	0.0383	0.79227	0.3433	615,468
J11	no_pref	-	-	0.02594	-397,134	-0.32595	48,451	0	0	0.35321	450,350
J11	stride	70	1,591	0.02575	-395,071	-0.32426	47,023	0.0047	0.22068	0.3534	448,287
J11	streamer	88	21,447	0.04271	-677,025	-0.55572	43,818	0.05687	0.87344	0.33644	730,241
J11	ampm	88	15,930	0.02991	-452,555	-0.37149	45,153	0.04184	0.87576	0.34924	505,771
J11	spp	30	931	0.02524	-386,416	-0.31716	47,899	0.00251	0.38127	0.35391	439,632
J11	ipcp	1	2,376	0.02674	-407,974	-0.33485	47,954	0.00626	0.97272	0.35241	461,190
J11	sms	31	60,215	0.03748	-603,406	-0.49509	39,831	0.15225	0.9332	0.34167	656,622

Table 61: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 1/4 (cont.)

ID	normal	<i>both_late</i>	<i>both_timely</i>	<i>ipc_delta_standalone_minus_normal</i>	<i>l2_miss_delta_standalone_minus_normal</i>	<i>l2_miss_rate_delta_standalone_minus_normal</i>	<i>neither_timely</i>	<i>normal_event_coverage</i>	<i>normal_event_timeliness</i>	<i>normal_ipc</i>	<i>normal_l2_load_miss</i>
J11	sandbox	85	146,828	0.03391	-546,787	-0.44876	29,760	0.3658	0.948	0.34524	600,003
J11	power7	80	14,489	0.03585	-562,252	-0.46148	45,007	0.0383	0.79227	0.3433	615,468
J09	no-pref	—	—	0.02561	-387,824	-0.31831	60,805	0	0	0.35321	450,350
J09	stride	41	1,273	0.02542	-385,761	-0.31662	58,050	0.0047	0.22068	0.3534	448,287
J09	streamer	51	19,292	0.04238	-667,715	-0.54808	53,590	0.05687	0.87344	0.33644	730,241
J09	ampm	56	14,045	0.02958	-443,245	-0.36385	55,239	0.04184	0.87576	0.34924	505,771
J09	spp	22	825	0.02491	-377,106	-0.30952	59,857	0.00251	0.38127	0.35391	439,632
J09	ipcp	1	2,283	0.02641	-398,664	-0.32721	60,212	0.00626	0.97272	0.35241	461,190
J09	sms	114	57,748	0.03715	-594,096	-0.48745	49,387	0.15225	0.9332	0.34167	656,622
J09	sandbox	53	141,488	0.03358	-537,477	-0.44112	36,321	0.3658	0.948	0.34524	600,003
J09	power7	40	13,035	0.03552	-552,942	-0.45384	55,328	0.0383	0.79227	0.3433	615,468
J10	no-pref	—	—	0.00407	-66,676	-0.05469	369,279	0	0	0.35321	450,350
J10	stride	—	—	0.00388	-64,613	-0.05299	359,696	0.0047	0.22068	0.3534	448,287
J10	streamer	—	1,318	0.02084	-346,567	-0.28446	341,280	0.05687	0.87344	0.33644	730,241
J10	ampm	—	547	0.00804	-122,097	-0.10023	348,344	0.04184	0.87576	0.34924	505,771
J10	spp	—	42	0.00337	-55,958	-0.04589	366,379	0.00251	0.38127	0.35391	439,632
J10	ipcp	—	12	0.00487	-77,516	-0.06359	366,395	0.00626	0.97272	0.35241	461,190
J10	sms	—	9,912	0.01561	-272,948	-0.22382	306,403	0.15225	0.9332	0.34167	656,622
J10	sandbox	1	26,661	0.01204	-216,329	-0.1775	223,360	0.3658	0.948	0.34524	600,003
J10	power7	—	990	0.01398	-231,794	-0.19022	348,513	0.0383	0.79227	0.3433	615,468

Table 62: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 2/4

ID	normal	<i>normal_l2_load_miss_rate</i>	<i>normal_late_standalone_not_timely</i>	<i>normal_only_timely</i>	<i>normal_pf_issued</i>	<i>normal_pf_late</i>	<i>normal_pf_useless</i>	<i>normal_selected_accuracy</i>	<i>reason_no_earlier_selected_standalone_export</i>	<i>reason_standalone_selected_but_late</i>	<i>reason_standalone_selected_earlier_but_not_timely</i>
J21	no-pref	0.50253	—	—	0	0	0	0	—	—	—
J21	stride	0.4163	144	925	267,553	168	1,023	0.26422	925	—	—
J21	streamer	0.03828	36	10,637	3,623,513	49	16,394	0.10374	10,586	—	51
J21	ampm	0.04214	204	9,345	2,846,050	604	12,600	0.13117	9,294	—	51
J21	spp	0.17533	29	6,064	4,491,857	387	1,162	0.05909	6,034	1	29
J21	ipcp	0.50253	1	—	188	4	74	0.24468	—	—	—
J21	sms	0.10185	135	8,960	696,706	200	23,409	0.45872	8,918	1	41
J21	sandbox	0.01767	41	13,463	3,839,474	59	32,524	0.10184	13,408	1	54
J21	power7	0.04281	109	10,359	3,304,558	128	13,449	0.11785	10,308	—	51
J20	no-pref	0.50253	—	—	0	0	0	0	—	—	—
J20	stride	0.4163	144	869	267,553	168	1,023	0.26422	869	—	—
J20	streamer	0.03828	36	9,981	3,623,513	49	16,394	0.10374	9,950	—	31
J20	ampm	0.04214	202	8,714	2,846,050	604	12,600	0.13117	8,684	—	30
J20	spp	0.17533	28	5,595	4,491,857	387	1,162	0.05909	5,579	1	15
J20	ipcp	0.50253	1	—	188	4	74	0.24468	—	—	—
J20	sms	0.10185	133	8,587	696,706	200	23,409	0.45872	8,561	1	25
J20	sandbox	0.01767	41	12,972	3,839,474	59	32,524	0.10184	12,937	1	34
J20	power7	0.04281	109	9,715	3,304,558	128	13,449	0.11785	9,684	—	31
J22	no-pref	0.50253	—	—	0	0	0	0	—	—	—
J22	stride	0.4163	142	539	267,553	168	1,023	0.26422	539	—	—
J22	streamer	0.03828	35	6,234	3,623,513	49	16,394	0.10374	6,212	1	21
J22	ampm	0.04214	183	4,964	2,846,050	604	12,600	0.13117	4,942	—	22
J22	spp	0.17533	20	2,961	4,491,857	387	1,162	0.05909	2,957	1	3
J22	ipcp	0.50253	1	—	188	4	74	0.24468	—	—	—
J22	sms	0.10185	130	5,187	696,706	200	23,409	0.45872	5,173	1	13
J22	sandbox	0.01767	41	8,610	3,839,474	59	32,524	0.10184	8,579	2	29
J22	power7	0.04281	108	6,007	3,304,558	128	13,449	0.11785	5,985	1	21

Table 62: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 2/4 (cont.)

ID	normal	normal_l2_load_miss_rate	normal_late_standalone_not_timely	normal_only_timely	normal_pf_issued	normal_pf_late	normal_pf_useless	normal_selected_accuracy	reason_no_earlier_selected_standalone-export	reason_standalone_selected_but_late	reason_standalone_selected_earlier_but_not_timely
J19	no_pref	0.50253	—	—	0	0	0	0	—	—	—
J19	stride	0.4163	144	641	267,553	168	1,023	0.26422	641	—	—
J19	streamer	0.03828	36	8,434	3,623,513	49	16,394	0.10374	8,412	—	22
J19	ampm	0.04214	202	7,179	2,846,050	604	12,600	0.13117	7,160	—	19
J19	spp	0.17533	25	4,494	4,491,857	387	1,162	0.05909	4,491	—	3
J19	ipcp	0.50253	1	—	188	4	74	0.24468	—	—	—
J19	sms	0.10185	134	7,205	696,706	200	23,409	0.45872	7,192	—	13
J19	sandbox	0.01767	41	11,428	3,839,474	59	32,524	0.10184	11,402	—	26
J19	power7	0.04281	109	8,184	3,304,558	128	13,449	0.11785	8,162	—	22
J15	no_pref	0.50253	—	—	0	0	0	0	—	—	—
J15	stride	0.4163	144	641	267,553	168	1,023	0.26422	641	—	—
J15	streamer	0.03828	36	8,434	3,623,513	49	16,394	0.10374	8,412	—	22
J15	ampm	0.04214	202	7,179	2,846,050	604	12,600	0.13117	7,160	—	19
J15	spp	0.17533	25	4,494	4,491,857	387	1,162	0.05909	4,491	—	3
J15	ipcp	0.50253	1	—	188	4	74	0.24468	—	—	—
J15	sms	0.10185	134	7,205	696,706	200	23,409	0.45872	7,192	—	13
J15	sandbox	0.01767	41	11,428	3,839,474	59	32,524	0.10184	11,402	—	26
J15	power7	0.04281	109	8,184	3,304,558	128	13,449	0.11785	8,162	—	22
J17	no_pref	0.50253	—	—	0	0	0	0	—	—	—
J17	stride	0.4163	144	925	267,553	168	1,023	0.26422	925	—	—
J17	streamer	0.03828	36	10,637	3,623,513	49	16,394	0.10374	10,586	—	51
J17	ampm	0.04214	204	9,345	2,846,050	604	12,600	0.13117	9,294	—	51
J17	spp	0.17533	29	6,064	4,491,857	387	1,162	0.05909	6,034	1	29
J17	ipcp	0.50253	1	—	188	4	74	0.24468	—	—	—
J17	sms	0.10185	135	8,960	696,706	200	23,409	0.45872	8,918	1	41
J17	sandbox	0.01767	41	13,463	3,839,474	59	32,524	0.10184	13,408	1	54
J17	power7	0.04281	109	10,359	3,304,558	128	13,449	0.11785	10,308	—	51
J16	no_pref	0.50253	—	—	0	0	0	0	—	—	—
J16	stride	0.4163	144	869	267,553	168	1,023	0.26422	869	—	—
J16	streamer	0.03828	36	9,981	3,623,513	49	16,394	0.10374	9,950	—	31
J16	ampm	0.04214	202	8,714	2,846,050	604	12,600	0.13117	8,684	—	30
J16	spp	0.17533	28	5,595	4,491,857	387	1,162	0.05909	5,579	1	15
J16	ipcp	0.50253	1	—	188	4	74	0.24468	—	—	—
J16	sms	0.10185	133	8,587	696,706	200	23,409	0.45872	8,561	1	25
J16	sandbox	0.01767	41	12,972	3,839,474	59	32,524	0.10184	12,937	1	34
J16	power7	0.04281	109	9,715	3,304,558	128	13,449	0.11785	9,684	—	31
J18	no_pref	0.50253	—	—	0	0	0	0	—	—	—
J18	stride	0.4163	156	31,136	267,553	168	1,023	0.26422	31,129	—	7
J18	streamer	0.03828	40	149,013	3,623,513	49	16,394	0.10374	148,544	—	469
J18	ampm	0.04214	483	147,482	2,846,050	604	12,600	0.13117	147,013	—	469
J18	spp	0.17533	289	105,484	4,491,857	387	1,162	0.05909	105,147	1	336
J18	ipcp	0.50253	1	—	188	4	74	0.24468	—	—	—
J18	sms	0.10185	167	128,726	696,706	200	23,409	0.45872	128,275	1	450
J18	sandbox	0.01767	43	156,072	3,839,474	59	32,524	0.10184	155,587	1	484
J18	power7	0.04281	114	147,490	3,304,558	128	13,449	0.11785	147,021	—	469
J07	no_pref	0.75128	—	—	0	0	0	0	—	—	—
J07	stride	0.74639	—	1,543	25,125	1	167	0.30234	1,183	17	343
J07	streamer	0.71198	9,079	50,964	1,775,122	18,210	686,999	0.09167	45,295	1,332	4,337
J07	ampm	0.69736	653	28,018	509,840	1,305	58,586	0.17787	24,091	340	3,587
J07	spp	0.7464	617	1,635	710,459	1,257	5,178	0.02039	1,079	15	541
J07	ipcp	0.75128	—	—	88	16	16	0.29545	—	—	—
J07	sms	0.69239	26,633	51,805	1,287,493	61,126	635,113	0.19908	46,465	790	4,550
J07	sandbox	0.72492	66,142	108,440	15,501,838	139,527	7,188,334	0.03395	97,935	4,161	6,344
J07	power7	0.72656	8,597	42,175	1,466,157	17,755	556,672	0.09356	36,657	1,213	4,305
J06	no_pref	0.75128	—	—	0	0	0	0	—	—	—
J06	stride	0.74639	—	1,608	25,125	1	167	0.30234	1,223	15	370
J06	streamer	0.71198	8,544	51,404	1,775,122	18,210	686,999	0.09167	45,823	1,361	4,220
J06	ampm	0.69736	594	28,664	509,840	1,305	58,586	0.17787	24,846	338	3,480
J06	spp	0.7464	587	1,710	710,459	1,257	5,178	0.02039	1,191	13	506
J06	ipcp	0.75128	—	—	88	16	16	0.29545	—	—	—
J06	sms	0.69239	27,517	52,163	1,287,493	61,126	635,113	0.19908	46,976	842	4,345
J06	sandbox	0.72492	66,096	109,192	15,501,838	139,527	7,188,334	0.03395	98,913	4,191	6,088
J06	power7	0.72656	8,088	42,341	1,466,157	17,755	556,672	0.09356	36,972	1,241	4,128
J08	no_pref	0.75128	—	—	0	0	0	0	—	—	—
J08	stride	0.74639	—	2,282	25,125	1	167	0.30234	1,807	13	462
J08	streamer	0.71198	9,030	57,641	1,775,122	18,210	686,999	0.09167	51,570	1,404	4,667
J08	ampm	0.69736	670	34,970	509,840	1,305	58,586	0.17787	30,759	350	3,861
J08	spp	0.7464	585	2,380	710,459	1,257	5,178	0.02039	1,755	13	612
J08	ipcp	0.75128	—	—	88	16	16	0.29545	—	—	—
J08	sms	0.69239	25,828	58,848	1,287,493	61,126	635,113	0.19908	53,229	926	4,693
J08	sandbox	0.72492	65,056	114,917	15,501,838	139,527	7,188,334	0.03395	104,141	4,285	6,491
J08	power7	0.72656	8,550	47,013	1,466,157	17,755	556,672	0.09356	41,226	1,273	4,514
J05	no_pref	0.75128	—	—	0	0	0	0	—	—	—
J05	stride	0.74639	—	2,085	25,125	1	167	0.30234	1,716	11	358
J05	streamer	0.71198	9,655	56,714	1,775,122	18,210	686,999	0.09167	51,532	1,204	3,978
J05	ampm	0.69736	677	33,769	509,840	1,305	58,586	0.17787	30,085	319	3,365
J05	spp	0.7464	668	2,245	710,459	1,257	5,178	0.02039	1,698	13	534
J05	ipcp	0.75128	—	—	88	16	16	0.29545	—	—	—

Table 62: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 2/4 (cont.)

ID	normal	normal_l2_load_miss_rate	normal_late_standalone_not_timely	normal_only_timely	normal_pf_issued	normal_pf_late	normal_pf_useless	normal_selected_accuracy	reason_no_earlier_selected_standalone-export	reason_standalone_selected_but_late	reason_standalone_selected_earlier_but_not_timely
J05 sms	0.69239	29,637	58,555	1,287,493	61,126	635,113	0.19908	53,825	727	4,003	
J05 sandbox	0.72492	70,573	115,200	15,501,838	139,527	7,188,334	0.03395	105,986	3,676	5,538	
J05 power7	0.72656	9,176	45,975	1,466,157	17,755	556,672	0.09356	40,995	1,085	3,895	
J01 no_pref	0.75128	-	-	0	0	0	0	-	-	-	
J01 stride	0.74639	-	3,808	25,125	1	167	0.30234	3,610	11	187	
J01 streamer	0.71198	12,173	71,894	1,775,122	18,210	686,999	0.09167	68,713	153	3,028	
J01 ampm	0.69736	1,033	47,968	509,840	1,305	58,586	0.17787	45,366	88	2,514	
J01 spp	0.7464	787	4,280	710,459	1,257	5,178	0.02039	3,927	10	343	
J01 icpc	0.75128	-	-	88	16	16	0.29545	-	-	-	
J01 sms	0.69239	43,017	74,243	1,287,493	61,126	635,113	0.19908	70,932	200	3,111	
J01 sandbox	0.72492	87,637	132,268	15,501,838	139,527	7,188,334	0.03395	127,782	294	4,192	
J01 power7	0.72656	11,753	57,584	1,466,157	17,755	556,672	0.09356	54,545	152	2,887	
J03 no_pref	0.75128	-	-	0	0	0	0	-	-	-	
J03 stride	0.74639	-	1,824	25,125	1	167	0.30234	1,428	14	382	
J03 streamer	0.71198	11,957	55,778	1,775,122	18,210	686,999	0.09167	50,974	199	4,605	
J03 ampm	0.69736	1,032	31,908	509,840	1,305	58,586	0.17787	27,863	91	3,954	
J03 spp	0.7464	753	2,039	710,459	1,257	5,178	0.02039	1,395	13	631	
J03 icpc	0.75128	-	-	88	16	16	0.29545	-	-	-	
J03 sms	0.69239	41,547	57,458	1,287,493	61,126	635,113	0.19908	52,434	259	4,765	
J03 sandbox	0.72492	85,206	115,105	15,501,838	139,527	7,188,334	0.03395	108,464	406	6,235	
J03 power7	0.72656	11,545	45,509	1,466,157	17,755	556,672	0.09356	40,830	194	4,485	
J02 no_pref	0.75128	-	-	0	0	0	0	-	-	-	
J02 stride	0.74639	-	2,230	25,125	1	167	0.30234	1,482	13	735	
J02 streamer	0.71198	9,943	56,860	1,775,122	18,210	686,999	0.09167	50,147	789	5,924	
J02 ampm	0.69736	801	33,558	509,840	1,305	58,586	0.17787	28,111	212	5,235	
J02 spp	0.7464	665	2,467	710,459	1,257	5,178	0.02039	1,490	13	964	
J02 icpc	0.75128	-	-	88	16	16	0.29545	-	-	-	
J02 sms	0.69239	34,248	58,009	1,287,493	61,126	635,113	0.19908	51,319	582	6,108	
J02 sandbox	0.72492	75,014	115,932	15,501,838	139,527	7,188,334	0.03395	105,970	2,387	7,575	
J02 power7	0.72656	9,514	46,579	1,466,157	17,755	556,672	0.09356	40,000	731	5,848	
J04 no_pref	0.75128	-	-	0	0	0	0	-	-	-	
J04 stride	0.74639	-	4,161	25,125	1	167	0.30234	3,811	11	339	
J04 streamer	0.71198	12,382	75,050	1,775,122	18,210	686,999	0.09167	70,779	188	4,083	
J04 ampm	0.69736	1,092	50,489	509,840	1,305	58,586	0.17787	47,112	91	3,286	
J04 spp	0.7464	779	4,499	710,459	1,257	5,178	0.02039	4,051	16	432	
J04 icpc	0.75128	-	-	88	16	16	0.29545	-	-	-	
J04 sms	0.69239	41,756	77,197	1,287,493	61,126	635,113	0.19908	72,651	323	4,223	
J04 sandbox	0.72492	86,337	136,501	15,501,838	139,527	7,188,334	0.03395	130,241	413	5,847	
J04 power7	0.72656	11,989	60,629	1,466,157	17,755	556,672	0.09356	56,515	189	3,925	
J43 no_pref	0.99865	-	-	0	0	0	0	-	-	-	
J43 stride	0.7702	2,151	7,653	350,381	15,702	0	0.43044	5,371	2,158	124	
J43 streamer	0.51629	14,897	19,849	524,800	112,603	990	0.76581	16,782	2,925	142	
J43 ampm	0.19893	4,443	33,719	2,008,610	25,272	348	0.32711	26,807	6,694	218	
J43 spp	0.76547	11,129	13,475	3,804,639	65,229	10	0.08238	12,022	1,409	44	
J43 icpc	0.99866	-	-	81	1	3	0.08642	-	-	-	
J43 sms	0.13119	10,596	46,359	568,017	26,099	84	0.9421	39,968	6,186	205	
J43 sandbox	0.28907	14,967	40,123	3,352,602	77,378	1,296	0.20898	35,602	4,344	177	
J43 power7	0.45452	11,218	23,048	615,642	76,926	862	0.61501	18,877	4,008	163	
J38 no_pref	0.99865	-	-	0	0	0	0	-	-	-	
J38 stride	0.7702	2,142	9,887	350,381	15,702	0	0.43044	6,890	2,867	130	
J38 streamer	0.51629	11,688	22,485	524,800	112,603	990	0.76581	18,354	3,998	133	
J38 ampm	0.19893	4,637	32,824	2,008,610	25,272	348	0.32711	25,073	7,550	201	
J38 spp	0.76547	10,245	11,640	3,804,639	65,229	10	0.08238	10,052	1,527	61	
J38 icpc	0.99866	-	-	81	1	3	0.08642	-	-	-	
J38 sms	0.13119	7,582	42,905	568,017	26,099	84	0.9421	35,418	7,275	212	
J38 sandbox	0.28907	12,430	35,596	3,352,602	77,378	1,296	0.20898	30,664	4,756	176	
J38 power7	0.45452	8,511	24,565	615,642	76,926	862	0.61501	19,520	4,888	157	
J47 no_pref	0.99865	-	-	0	0	0	0	-	-	-	
J47 stride	0.7702	73	551	350,381	15,702	0	0.43044	120	105	326	
J47 streamer	0.51629	1,012	1,828	524,800	112,603	990	0.76581	1,324	184	320	
J47 ampm	0.19893	617	2,441	2,008,610	25,272	348	0.32711	1,561	332	548	
J47 spp	0.76547	748	1,029	3,804,639	65,229	10	0.08238	815	81	133	
J47 icpc	0.99866	-	-	81	1	3	0.08642	-	-	-	
J47 sms	0.13119	350	3,243	568,017	26,099	84	0.9421	2,327	340	576	
J47 sandbox	0.28907	965	2,582	3,352,602	77,378	1,296	0.20898	1,855	239	488	
J47 power7	0.45452	757	1,840	615,642	76,926	862	0.61501	1,195	225	420	
J31 no_pref	0.99865	-	-	0	0	0	0	-	-	-	
J31 stride	0.7702	2,137	7,791	350,381	15,702	0	0.43044	5,050	2,612	129	
J31 streamer	0.51629	9,895	18,918	524,800	112,603	990	0.76581	14,125	4,653	140	
J31 ampm	0.19893	4,374	29,017	2,008,610	25,272	348	0.32711	19,732	9,056	229	
J31 spp	0.76547	9,860	11,696	3,804,639	65,229	10	0.08238	9,712	1,920	64	
J31 icpc	0.99866	-	-	81	1	3	0.08642	-	-	-	
J31 sms	0.13119	9,061	43,028	568,017	26,099	84	0.9421	33,707	9,090	231	
J31 sandbox	0.28907	12,217	37,224	3,352,602	77,378	1,296	0.20898	31,103	5,937	184	
J31 power7	0.45452	7,238	21,212	615,642	76,926	862	0.61501	15,499	5,536	177	
J34 no_pref	0.99865	-	-	0	0	0	0	-	-	-	
J34 stride	0.7702	1,782	4,486	350,381	15,702	0	0.43044	4,095	195	196	
J34 streamer	0.51629	7,242	11,625	524,800	112,603	990	0.76581	10,565	865	195	

Table 62: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 2/4 (cont.)

ID	normal	normal_l2_load_miss_rate	normal_late_standalone_not_timely	normal_only_timely	normal_pf_issued	normal_pf_late	normal_pf_useless	normal_selected_accuracy	reason_no_earlier_selected_standalone-export	reason_standalone_selected_but_late	reason_standalone_selected_earlier_but_not_timely
J34	ampm	0.19893	3,393	15,676	2,008,610	25,272	348	0.32711	14,196	1,196	284
J34	spp	0.76547	8,407	8,697	3,804,639	65,229	10	0.08238	8,263	358	76
J34	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J34	sms	0.13119	8,502	29,645	568,017	26,099	84	0.9421	27,739	1,621	285
J34	sandbox	0.28907	10,417	27,607	3,352,602	77,378	1,296	0.20898	26,175	1,202	230
J34	power7	0.45452	5,246	12,875	615,642	76,926	862	0.61501	11,724	913	238
J30	no-pref	0.99865	—	—	0	0	0	0	—	—	—
J30	stride	0.7702	2,137	7,791	350,381	15,702	0	0.43044	5,050	2,612	129
J30	streamer	0.51629	9,895	18,918	524,800	112,603	990	0.76581	14,125	4,653	140
J30	ampm	0.19893	4,374	29,017	2,008,610	25,272	348	0.32711	19,732	9,056	229
J30	spp	0.76547	9,860	11,696	3,804,639	65,229	10	0.08238	9,712	1,920	64
J30	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J30	sms	0.13119	9,061	43,028	568,017	26,099	84	0.9421	33,707	9,090	231
J30	sandbox	0.28907	12,217	37,224	3,352,602	77,378	1,296	0.20898	31,103	5,937	184
J30	power7	0.45452	7,238	21,212	615,642	76,926	862	0.61501	15,499	5,536	177
J35	no-pref	0.99865	—	—	0	0	0	0	—	—	—
J35	stride	0.7702	2,379	8,700	350,381	15,702	0	0.43044	5,544	3,062	94
J35	streamer	0.51629	11,210	19,538	524,800	112,603	990	0.76581	14,822	4,622	94
J35	ampm	0.19893	4,280	31,294	2,008,610	25,272	348	0.32711	21,788	9,344	162
J35	spp	0.76547	10,815	11,723	3,804,639	65,229	10	0.08238	9,823	1,854	46
J35	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J35	sms	0.13119	10,015	45,173	568,017	26,099	84	0.9421	35,936	9,076	161
J35	sandbox	0.28907	13,445	38,937	3,352,602	77,378	1,296	0.20898	32,744	6,071	122
J35	power7	0.45452	7,945	23,093	615,642	76,926	862	0.61501	17,013	5,961	119
J36	no-pref	0.99865	—	—	0	0	0	0	—	—	—
J36	stride	0.7702	2,506	10,429	350,381	15,702	0	0.43044	7,197	3,087	145
J36	streamer	0.51629	12,322	22,498	524,800	112,603	990	0.76581	17,565	4,796	137
J36	ampm	0.19893	4,938	35,093	2,008,610	25,272	348	0.32711	25,526	9,351	216
J36	spp	0.76547	11,907	12,663	3,804,639	65,229	10	0.08238	10,681	1,928	54
J36	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J36	sms	0.13119	10,664	49,398	568,017	26,099	84	0.9421	40,108	9,061	229
J36	sandbox	0.28907	14,267	42,775	3,352,602	77,378	1,296	0.20898	36,556	6,040	179
J36	power7	0.45452	8,643	25,854	615,642	76,926	862	0.61501	19,734	5,945	175
J32	no-pref	0.99865	—	—	0	0	0	0	—	—	—
J32	stride	0.7702	1,868	4,987	350,381	15,702	0	0.43044	4,291	519	177
J32	streamer	0.51629	7,615	12,971	524,800	112,603	990	0.76581	11,079	1,717	175
J32	ampm	0.19893	3,568	18,341	2,008,610	25,272	348	0.32711	14,997	3,089	255
J32	spp	0.76547	8,628	9,272	3,804,639	65,229	10	0.08238	8,445	757	70
J32	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J32	sms	0.13119	8,599	32,350	568,017	26,099	84	0.9421	28,623	3,476	251
J32	sandbox	0.28907	10,645	29,548	3,352,602	77,378	1,296	0.20898	26,931	2,399	218
J32	power7	0.45452	5,547	14,309	615,642	76,926	862	0.61501	12,270	1,843	196
J33	no-pref	0.99865	—	—	0	0	0	0	—	—	—
J33	stride	0.7702	1,782	4,486	350,381	15,702	0	0.43044	4,095	195	196
J33	streamer	0.51629	7,242	11,625	524,800	112,603	990	0.76581	10,565	865	195
J33	ampm	0.19893	3,393	15,676	2,008,610	25,272	348	0.32711	14,196	1,196	284
J33	spp	0.76547	8,407	8,697	3,804,639	65,229	10	0.08238	8,263	358	76
J33	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J33	sms	0.13119	8,502	29,645	568,017	26,099	84	0.9421	27,739	1,621	285
J33	sandbox	0.28907	10,417	27,607	3,352,602	77,378	1,296	0.20898	26,175	1,202	230
J33	power7	0.45452	5,246	12,875	615,642	76,926	862	0.61501	11,724	913	238
J46	no-pref	0.99865	—	—	0	0	0	0	—	—	—
J46	stride	0.7702	1,776	4,920	350,381	15,702	0	0.43044	4,415	298	207
J46	streamer	0.51629	8,901	14,226	524,800	112,603	990	0.76581	13,246	778	202
J46	ampm	0.19893	3,655	19,011	2,008,610	25,272	348	0.32711	17,543	1,175	293
J46	spp	0.76547	7,938	8,184	3,804,639	65,229	10	0.08238	7,824	279	81
J46	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J46	sms	0.13119	7,568	28,168	568,017	26,099	84	0.9421	26,547	1,338	283
J46	sandbox	0.28907	9,830	25,244	3,352,602	77,378	1,296	0.20898	23,994	994	256
J46	power7	0.45452	6,660	15,544	615,642	76,926	862	0.61501	14,424	870	250
J41	no-pref	0.99865	—	—	0	0	0	0	—	—	—
J41	stride	0.7702	1,737	4,814	350,381	15,702	0	0.43044	4,357	270	187
J41	streamer	0.51629	7,235	11,177	524,800	112,603	990	0.76581	10,443	525	209
J41	ampm	0.19893	2,734	15,852	2,008,610	25,272	348	0.32711	14,675	860	317
J41	spp	0.76547	6,321	6,327	3,804,639	65,229	10	0.08238	6,025	221	81
J41	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J41	sms	0.13119	5,311	23,180	568,017	26,099	84	0.9421	21,779	1,098	303
J41	sandbox	0.28907	7,617	20,500	3,352,602	77,378	1,296	0.20898	19,470	768	262
J41	power7	0.45452	5,153	12,557	615,642	76,926	862	0.61501	11,670	643	244
J42	no-pref	0.99865	—	—	0	0	0	0	—	—	—
J42	stride	0.7702	2,151	7,653	350,381	15,702	0	0.43044	5,371	2,158	124
J42	streamer	0.51629	14,897	19,849	524,800	112,603	990	0.76581	16,782	2,925	142
J42	ampm	0.19893	4,443	33,719	2,008,610	25,272	348	0.32711	26,807	6,694	218
J42	spp	0.76547	11,129	13,475	3,804,639	65,229	10	0.08238	12,022	1,409	44
J42	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J42	sms	0.13119	10,596	46,359	568,017	26,099	84	0.9421	39,968	6,186	205
J42	sandbox	0.28907	14,967	40,123	3,352,602	77,378	1,296	0.20898	35,602	4,344	177
J42	power7	0.45452	11,218	23,048	615,642	76,926	862	0.61501	18,877	4,008	163

Table 62: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 2/4 (cont.)

ID	normal	normal_l2_load-miss-rate	normal_late_standalone-not_timely	normal_only_timely	normal_pf_issued	normal_pf_late	normal_pf_useless	normal_selected-accuracy	reason_no-earlier_selected_standalone-export	reason_standalone-selected_but_late	reason_standalone-selected_earlier_but_not_timely
J37	no_pref	0.99865	—	—	0	0	0	0	—	—	—
J37	stride	0.7702	2,142	9,887	350,381	15,702	0	0.43044	6,890	2,867	130
J37	streamer	0.51629	11,688	22,485	524,800	112,603	990	0.76581	18,354	3,998	133
J37	ampm	0.19893	4,637	32,824	2,008,610	25,272	348	0.32711	25,073	7,550	201
J37	spp	0.76547	10,245	11,640	3,804,639	65,229	10	0.08238	10,052	1,527	61
J37	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J37	sms	0.13119	7,582	42,905	568,017	26,099	84	0.9421	35,418	7,275	212
J37	sandbox	0.28907	12,430	35,596	3,352,602	77,378	1,296	0.20898	30,664	4,756	176
J37	power7	0.45452	8,511	24,565	615,642	76,926	862	0.61501	19,520	4,888	157
J44	no_pref	0.99865	—	—	0	0	0	0	—	—	—
J44	stride	0.7702	1,862	5,647	350,381	15,702	0	0.43044	4,645	826	176
J44	streamer	0.51629	9,869	15,403	524,800	112,603	990	0.76581	13,906	1,318	179
J44	ampm	0.19893	3,770	22,064	2,008,610	25,272	348	0.32711	19,110	2,687	267
J44	spp	0.76547	8,430	9,228	3,804,639	65,229	10	0.08238	8,530	623	75
J44	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J44	sms	0.13119	8,141	31,842	568,017	26,099	84	0.9421	28,846	2,727	269
J44	sandbox	0.28907	10,779	28,185	3,352,602	77,378	1,296	0.20898	25,965	1,998	222
J44	power7	0.45452	7,361	17,289	615,642	76,926	862	0.61501	15,277	1,800	212
J39	no_pref	0.99865	—	—	0	0	0	0	—	—	—
J39	stride	0.7702	1,869	6,148	350,381	15,702	0	0.43044	4,832	1,146	170
J39	streamer	0.51629	7,882	12,993	524,800	112,603	990	0.76581	11,336	1,459	198
J39	ampm	0.19893	2,861	19,209	2,008,610	25,272	348	0.32711	16,164	2,765	280
J39	spp	0.76547	6,800	7,147	3,804,639	65,229	10	0.08238	6,519	561	67
J39	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J39	sms	0.13119	5,601	26,779	568,017	26,099	84	0.9421	23,569	2,937	273
J39	sandbox	0.28907	8,137	23,299	3,352,602	77,378	1,296	0.20898	21,013	2,049	237
J39	power7	0.45452	5,617	14,854	615,642	76,926	862	0.61501	12,707	1,922	225
J45	no_pref	0.99865	—	—	0	0	0	0	—	—	—
J45	stride	0.7702	1,776	4,920	350,381	15,702	0	0.43044	4,415	298	207
J45	streamer	0.51629	8,901	14,226	524,800	112,603	990	0.76581	13,246	778	202
J45	ampm	0.19893	3,655	19,011	2,008,610	25,272	348	0.32711	17,543	1,175	293
J45	spp	0.76547	7,938	8,184	3,804,639	65,229	10	0.08238	7,824	279	81
J45	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J45	sms	0.13119	7,568	28,168	568,017	26,099	84	0.9421	26,547	1,338	283
J45	sandbox	0.28907	9,830	25,244	3,352,602	77,378	1,296	0.20898	23,994	994	256
J45	power7	0.45452	6,660	15,544	615,642	76,926	862	0.61501	14,424	870	250
J40	no_pref	0.99865	—	—	0	0	0	0	—	—	—
J40	stride	0.7702	1,737	4,814	350,381	15,702	0	0.43044	4,357	270	187
J40	streamer	0.51629	7,235	11,177	524,800	112,603	990	0.76581	10,443	525	209
J40	ampm	0.19893	2,734	15,852	2,008,610	25,272	348	0.32711	14,675	860	317
J40	spp	0.76547	6,321	6,327	3,804,639	65,229	10	0.08238	6,025	221	81
J40	icpc	0.99866	—	—	81	1	3	0.08642	—	—	—
J40	sms	0.13119	5,311	23,180	568,017	26,099	84	0.9421	21,779	1,098	303
J40	sandbox	0.28907	7,617	20,500	3,352,602	77,378	1,296	0.20898	19,470	768	262
J40	power7	0.45452	5,153	12,557	615,642	76,926	862	0.61501	11,670	643	244
J28	no_pref	0.70127	—	—	0	0	0	0	—	—	—
J28	stride	0.69494	—	2,953	19,579	0	1,849	0.30279	2,952	—	1
J28	streamer	0.61653	1,797	26,367	757,318	2,538	286,074	0.16814	25,963	57	347
J28	ampm	0.66478	618	9,809	310,419	1,868	98,338	0.15581	9,630	14	165
J28	spp	0.70083	1	21	174,696	34	53	4.325e-04	21	—	—
J28	icpc	0.69812	213	1,054	29,016	374	9,433	0.18521	970	—	84
J28	sms	0.54565	1,682	64,879	808,086	3,655	394,038	0.2447	64,196	47	636
J28	sandbox	0.52652	7,642	96,889	7,623,344	10,848	3,279,950	0.0561	95,925	76	888
J28	power7	0.61801	1,834	26,528	730,714	2,751	271,622	0.16651	26,130	55	343
J27	no_pref	0.70127	—	—	0	0	0	0	—	—	—
J27	stride	0.69494	—	2,907	19,579	0	1,849	0.30279	2,894	—	13
J27	streamer	0.61653	1,589	24,120	757,318	2,538	286,074	0.16814	23,629	39	452
J27	ampm	0.66478	634	9,853	310,419	1,868	98,338	0.15581	9,548	6	299
J27	spp	0.70083	2	20	174,696	34	53	4.325e-04	20	—	—
J27	icpc	0.69812	233	1,070	29,016	374	9,433	0.18521	904	—	166
J27	sms	0.54565	1,644	61,216	808,086	3,655	394,038	0.2447	60,546	41	629
J27	sandbox	0.52652	7,241	92,937	7,623,344	10,848	3,279,950	0.0561	91,903	66	968
J27	power7	0.61801	1,604	24,394	730,714	2,751	271,622	0.16651	23,907	38	449
J29	no_pref	0.70127	—	—	0	0	0	0	—	—	—
J29	stride	0.69494	—	2,890	19,579	0	1,849	0.30279	2,852	—	38
J29	streamer	0.61653	1,725	31,780	757,318	2,538	286,074	0.16814	31,140	30	610
J29	ampm	0.66478	515	12,975	310,419	1,868	98,338	0.15581	12,564	7	404
J29	spp	0.70083	4	28	174,696	34	53	4.325e-04	28	—	—
J29	icpc	0.69812	103	1,268	29,016	374	9,433	0.18521	1,220	12	36
J29	sms	0.54565	1,670	71,081	808,086	3,655	394,038	0.2447	70,300	81	700
J29	sandbox	0.52652	7,821	105,800	7,623,344	10,848	3,279,950	0.0561	104,443	111	1,246
J29	power7	0.61801	1,756	32,381	730,714	2,751	271,622	0.16651	31,747	30	604
J26	no_pref	0.70127	—	—	0	0	0	0	—	—	—
J26	stride	0.69494	—	1,829	19,579	0	1,849	0.30279	1,826	—	3
J26	streamer	0.61653	1,734	24,020	757,318	2,538	286,074	0.16814	23,295	9	716
J26	ampm	0.66478	615	8,672	310,419	1,868	98,338	0.15581	8,127	7	538
J26	spp	0.70083	—	15	174,696	34	53	4.325e-04	14	—	1
J26	icpc	0.69812	227	1,055	29,016	374	9,433	0.18521	957	—	98

Table 62: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 2/4 (cont.)

ID	normal	normal_l2_load_miss_rate	normal_late_standalone_not_timely	normal_only_timely	normal_pf_issued	normal_pf_late	normal_pf_useless	normal_selected_accuracy	reason_no_earlier_selected_standalone-export	reason_standalone_selected_but_late	reason_standalone_selected_earlier_but_not_timely
J26	sms	0.54565	1,692	62,376	808,086	3,655	394,038	0.2447	61,592	18	766
J26	sandbox	0.52652	7,458	93,252	7,623,344	10,848	3,279,950	0.0561	92,013	18	1,221
J26	power7	0.61801	1,788	24,507	730,714	2,751	271,622	0.16651	23,786	10	711
J23	no-pref	0.70127	—	—	0	0	0	0	—	—	—
J23	stride	0.69494	—	1,829	19,579	0	1,849	0.30279	1,826	—	3
J23	streamer	0.61653	1,734	24,020	757,318	2,538	286,074	0.16814	23,295	9	716
J23	ampm	0.66478	615	8,672	310,419	1,868	98,338	0.15581	8,127	7	538
J23	spp	0.70083	—	15	174,696	34	53	4.325e-04	14	—	1
J23	icpc	0.69812	227	1,055	29,016	374	9,433	0.18521	957	—	98
J23	sms	0.54565	1,692	62,376	808,086	3,655	394,038	0.2447	61,592	18	766
J23	sandbox	0.52652	7,458	93,252	7,623,344	10,848	3,279,950	0.0561	92,013	18	1,221
J23	power7	0.61801	1,788	24,507	730,714	2,751	271,622	0.16651	23,786	10	711
J25	no-pref	0.70127	—	—	0	0	0	0	—	—	—
J25	stride	0.69494	—	2,985	19,579	0	1,849	0.30279	2,985	—	—
J25	streamer	0.61653	1,859	36,587	757,318	2,538	286,074	0.16814	36,136	20	431
J25	ampm	0.66478	770	14,471	310,419	1,868	98,338	0.15581	14,209	—	262
J25	spp	0.70083	2	29	174,696	34	53	4.325e-04	28	—	1
J25	icpc	0.69812	291	2,073	29,016	374	9,433	0.18521	2,071	—	2
J25	sms	0.54565	1,715	75,606	808,086	3,655	394,038	0.2447	75,341	15	250
J25	sandbox	0.52652	7,751	114,961	7,623,344	10,848	3,279,950	0.0561	114,415	20	526
J25	power7	0.61801	1,907	36,544	730,714	2,751	271,622	0.16651	36,096	19	429
J24	no-pref	0.70127	—	—	0	0	0	0	—	—	—
J24	stride	0.69494	—	2,986	19,579	0	1,849	0.30279	2,985	—	1
J24	streamer	0.61653	1,815	35,848	757,318	2,538	286,074	0.16814	35,257	15	576
J24	ampm	0.66478	774	14,565	310,419	1,868	98,338	0.15581	14,385	—	180
J24	spp	0.70083	3	25	174,696	34	53	4.325e-04	25	—	—
J24	icpc	0.69812	272	1,515	29,016	374	9,433	0.18521	1,432	—	83
J24	sms	0.54565	1,700	75,612	808,086	3,655	394,038	0.2447	75,204	12	396
J24	sandbox	0.52652	7,637	114,320	7,623,344	10,848	3,279,950	0.0561	113,527	17	776
J24	power7	0.61801	1,818	36,074	730,714	2,751	271,622	0.16651	35,479	15	580
J13	no-pref	0.3696	—	—	0	0	0	0	—	—	—
J13	stride	0.3679	2,291	958	299,845	7,498	373	0.07011	775	—	183
J13	streamer	0.59937	1,952	11,618	2,459,972	6,770	715,938	0.05202	9,988	26	1,604
J13	ampm	0.41513	1,799	8,224	924,845	3,276	86,456	0.063	6,832	46	1,346
J13	spp	0.3608	899	667	1,670,201	1,870	23,108	0.00485	620	2	45
J13	icpc	0.37849	62	1,558	87,172	118	10,856	0.06726	1,471	7	80
J13	sms	0.53873	3,156	33,899	1,932,085	11,070	764,626	0.15092	27,235	85	6,579
J13	sandbox	0.49241	4,168	82,491	12,891,224	23,596	4,491,270	0.04989	70,368	118	12,005
J13	power7	0.50513	2,192	8,043	1,764,449	7,460	477,415	0.05471	7,042	4	997
J12	no-pref	0.3696	—	—	0	0	0	0	—	—	—
J12	stride	0.3679	2,040	943	299,845	7,498	373	0.07011	742	—	201
J12	streamer	0.59937	2,197	12,557	2,459,972	6,770	715,938	0.05202	11,066	6	1,485
J12	ampm	0.41513	1,378	7,421	924,845	3,276	86,456	0.063	6,215	4	1,202
J12	spp	0.3608	758	727	1,670,201	1,870	23,108	0.00485	696	—	31
J12	icpc	0.37849	78	2,805	87,172	118	10,856	0.06726	2,797	—	8
J12	sms	0.53873	3,590	45,912	1,932,085	11,070	764,626	0.15092	41,152	6	4,754
J12	sandbox	0.49241	5,353	99,374	12,891,224	23,596	4,491,270	0.04989	88,909	161	10,304
J12	power7	0.50513	2,255	8,963	1,764,449	7,460	477,415	0.05471	7,930	3	1,030
J14	no-pref	0.3696	—	—	0	0	0	0	—	—	—
J14	stride	0.3679	4,099	1,240	299,845	7,498	373	0.07011	418	6	816
J14	streamer	0.59937	2,301	16,166	2,459,972	6,770	715,938	0.05202	7,894	115	8,157
J14	ampm	0.41513	1,858	11,615	924,845	3,276	86,456	0.063	4,340	121	7,154
J14	spp	0.3608	1,176	715	1,670,201	1,870	23,108	0.00485	400	4	311
J14	icpc	0.37849	67	2,711	87,172	118	10,856	0.06726	2,663	—	48
J14	sms	0.53873	4,034	49,191	1,932,085	11,070	764,626	0.15092	33,947	134	15,110
J14	sandbox	0.49241	6,081	115,205	12,891,224	23,596	4,491,270	0.04989	79,886	200	35,119
J14	power7	0.50513	2,704	11,045	1,764,449	7,460	477,415	0.05471	5,897	27	5,121
J11	no-pref	0.3696	—	—	0	0	0	0	—	—	—
J11	stride	0.3679	907	524	299,845	7,498	373	0.07011	384	3	137
J11	streamer	0.59937	510	4,163	2,459,972	6,770	715,938	0.05202	2,926	40	1,197
J11	ampm	0.41513	469	2,912	924,845	3,276	86,456	0.063	2,011	83	818
J11	spp	0.3608	354	201	1,670,201	1,870	23,108	0.00485	173	3	25
J11	icpc	0.37849	56	441	87,172	118	10,856	0.06726	420	—	21
J11	sms	0.53873	437	8,351	1,932,085	11,070	764,626	0.15092	7,057	168	1,126
J11	sandbox	0.49241	841	17,910	12,891,224	23,596	4,491,270	0.04989	15,848	60	2,002
J11	power7	0.50513	703	2,758	1,764,449	7,460	477,415	0.05471	1,973	17	768
J09	no-pref	0.3696	—	—	0	0	0	0	—	—	—
J09	stride	0.3679	1,913	842	299,845	7,498	373	0.07011	623	—	219
J09	streamer	0.59937	918	6,318	2,459,972	6,770	715,938	0.05202	4,450	21	1,847
J09	ampm	0.41513	807	4,797	924,845	3,276	86,456	0.063	3,348	38	1,411
J09	spp	0.3608	642	307	1,670,201	1,870	23,108	0.00485	258	1	48
J09	icpc	0.37849	59	534	87,172	118	10,856	0.06726	519	—	15
J09	sms	0.53873	695	10,818	1,932,085	11,070	764,626	0.15092	9,243	95	1,480
J09	sandbox	0.49241	1,307	23,250	12,891,224	23,596	4,491,270	0.04989	20,169	73	3,008
J09	power7	0.50513	1,269	4,212	1,764,449	7,460	477,415	0.05471	3,041	4	1,167
J10	no-pref	0.3696	—	—	0	0	0	0	—	—	—
J10	stride	0.3679	7,468	2,115	299,845	7,498	373	0.07011	2,102	—	13
J10	streamer	0.59937	3,707	24,292	2,459,972	6,770	715,938	0.05202	24,160	—	132

Table 62: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 2/4 (cont.)

ID	normal	<i>normal_l2_load_miss_rate</i>	<i>normal_late_standalone_not_timely</i>	<i>normal_only_timely</i>	<i>normal_pf_issued</i>	<i>normal_pf_late</i>	<i>normal_pf_useless</i>	<i>normal_selected_accuracy</i>	<i>reason_no_earlier_selected_standalone-export</i>	<i>reason_standalone-selected_but_late</i>	<i>reason_standalone-selected_earlier_but_not_timely</i>
J10	ampm	0.41513	2,640	18,295	924,845	3,276	86,456	0.063	18,187	—	108
J10	spp	0.3608	1,810	1,090	1,670,201	1,870	23,108	0.00485	1,065	—	25
J10	ipcp	0.37849	79	2,805	87,172	118	10,856	0.06726	2,803	—	2
J10	sms	0.53873	4,223	58,654	1,932,085	11,070	764,626	0.15092	54,703	1	3,950
J10	sandbox	0.49241	7,856	138,077	12,891,224	23,596	4,491,270	0.04989	130,271	14	7,792
J10	power7	0.50513	4,509	16,257	1,764,449	7,460	477,415	0.05471	16,162	—	95

Table 63: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 3/4

ID	normal	<i>standalone-event-coverage</i>	<i>standalone-event-timeliness</i>	<i>standalone-ipc</i>	<i>standalone_l2_load_miss</i>	<i>standalone_l2_load_miss_rate</i>	<i>standalone_late_normal_not_timely</i>	<i>standalone-only-timely</i>	<i>standalone-pf_issued</i>	<i>standalone-pf_late</i>	<i>standalone-pf-useless</i>
J21	no_pref	0.91064	0.99999	0.42835	18,154	0.04491	1	184,979	188,865	1	3,614
J21	stride	0.91064	0.99999	0.42835	18,154	0.04491	1	151,050	188,865	1	3,614
J21	streamer	0.91064	0.99999	0.42835	18,154	0.04491	1	8,036	188,865	1	3,614
J21	ampm	0.91064	0.99999	0.42835	18,154	0.04491	—	8,319	188,865	1	3,614
J21	spp	0.91064	0.99999	0.42835	18,154	0.04491	—	58,838	188,865	1	3,614
J21	ipcp	0.91064	0.99999	0.42835	18,154	0.04491	1	184,979	188,865	1	3,614
J21	sms	0.91064	0.99999	0.42835	18,154	0.04491	—	32,027	188,865	1	3,614
J21	sandbox	0.91064	0.99999	0.42835	18,154	0.04491	—	2,548	188,865	1	3,614
J21	power7	0.91064	0.99999	0.42835	18,154	0.04491	1	9,591	188,865	1	3,614
J20	no_pref	0.913	0.99999	0.42855	17,673	0.04372	1	185,459	189,926	1	4,152
J20	stride	0.913	0.99999	0.42855	17,673	0.04372	1	151,474	189,926	1	4,152
J20	streamer	0.913	0.99999	0.42855	17,673	0.04372	1	7,860	189,926	1	4,152
J20	ampm	0.913	0.99999	0.42855	17,673	0.04372	—	8,168	189,926	1	4,152
J20	spp	0.913	0.99999	0.42855	17,673	0.04372	—	58,849	189,926	1	4,152
J20	ipcp	0.913	0.99999	0.42855	17,673	0.04372	1	185,459	189,926	1	4,152
J20	sms	0.913	0.99999	0.42855	17,673	0.04372	—	32,134	189,926	1	4,152
J20	sandbox	0.913	0.99999	0.42855	17,673	0.04372	—	2,537	189,926	1	4,152
J20	power7	0.913	0.99999	0.42855	17,673	0.04372	1	9,427	189,926	1	4,152
J22	no_pref	0.93517	0.99999	0.43066	13,168	0.03258	2	189,962	199,519	2	8,575
J22	stride	0.93517	0.99999	0.43066	13,168	0.03258	2	155,647	199,519	2	8,575
J22	streamer	0.93517	0.99999	0.43066	13,168	0.03258	1	8,616	199,519	2	8,575
J22	ampm	0.93517	0.99999	0.43066	13,168	0.03258	1	8,921	199,519	2	8,575
J22	spp	0.93517	0.99999	0.43066	13,168	0.03258	1	60,718	199,519	2	8,575
J22	ipcp	0.93517	0.99999	0.43066	13,168	0.03258	2	189,962	199,519	2	8,575
J22	sms	0.93517	0.99999	0.43066	13,168	0.03258	1	33,237	199,519	2	8,575
J22	sandbox	0.93517	0.99999	0.43066	13,168	0.03258	—	2,678	199,519	2	8,575
J22	power7	0.93517	0.99999	0.43066	13,168	0.03258	1	10,222	199,519	2	8,575
J19	no_pref	0.92065	1	0.42924	16,118	0.03987	—	187,013	189,461	0	2,284
J19	stride	0.92065	1	0.42924	16,118	0.03987	—	152,800	189,461	0	2,284
J19	streamer	0.92065	1	0.42924	16,118	0.03987	—	7,867	189,461	0	2,284
J19	ampm	0.92065	1	0.42924	16,118	0.03987	—	8,187	189,461	0	2,284
J19	spp	0.92065	1	0.42924	16,118	0.03987	—	59,302	189,461	0	2,284
J19	ipcp	0.92065	1	0.42924	16,118	0.03987	—	187,013	189,461	0	2,284
J19	sms	0.92065	1	0.42924	16,118	0.03987	—	32,306	189,461	0	2,284
J19	sandbox	0.92065	1	0.42924	16,118	0.03987	—	2,547	189,461	0	2,284
J19	power7	0.92065	1	0.42924	16,118	0.03987	—	9,450	189,461	0	2,284
J15	no_pref	0.92065	1	0.42924	16,118	0.03987	—	187,013	189,461	0	2,284
J15	stride	0.92065	1	0.42924	16,118	0.03987	—	152,800	189,461	0	2,284
J15	streamer	0.92065	1	0.42924	16,118	0.03987	—	7,867	189,461	0	2,284
J15	ampm	0.92065	1	0.42924	16,118	0.03987	—	8,187	189,461	0	2,284
J15	spp	0.92065	1	0.42924	16,118	0.03987	—	59,302	189,461	0	2,284
J15	ipcp	0.92065	1	0.42924	16,118	0.03987	—	187,013	189,461	0	2,284
J15	sms	0.92065	1	0.42924	16,118	0.03987	—	32,306	189,461	0	2,284
J15	sandbox	0.92065	1	0.42924	16,118	0.03987	—	2,547	189,461	0	2,284
J15	power7	0.92065	1	0.42924	16,118	0.03987	—	9,450	189,461	0	2,284
J17	no_pref	0.91064	0.99999	0.42835	18,154	0.04491	1	184,979	188,865	1	3,614

Table 63: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 3/4 (cont.)

ID	normal	<i>standalone-event-coverage</i>	<i>standalone-event-timeliness</i>	<i>standalone-ipc</i>	<i>standalone-l2-load-miss</i>	<i>standalone-l2-load-miss-rate</i>	<i>standalone-late-normal-not-timely</i>	<i>standalone-only-timely</i>	<i>standalone-pf-issued</i>	<i>standalone-pf-late</i>	<i>standalone-pf-timelineless</i>
J17	stride	0.91064	0.99999	0.42835	18,154	0.04491	1	151,050	188,865	1	3,614
J17	streamer	0.91064	0.99999	0.42835	18,154	0.04491	1	8,036	188,865	1	3,614
J17	ampm	0.91064	0.99999	0.42835	18,154	0.04491	—	8,319	188,865	1	3,614
J17	spp	0.91064	0.99999	0.42835	18,154	0.04491	—	58,838	188,865	1	3,614
J17	ipcp	0.91064	0.99999	0.42835	18,154	0.04491	1	184,979	188,865	1	3,614
J17	sms	0.91064	0.99999	0.42835	18,154	0.04491	—	32,027	188,865	1	3,614
J17	sandbox	0.91064	0.99999	0.42835	18,154	0.04491	—	2,548	188,865	1	3,614
J17	power7	0.91064	0.99999	0.42835	18,154	0.04491	1	9,591	188,865	1	3,614
J16	no_pref	0.913	0.99999	0.42855	17,673	0.04372	1	185,459	189,926	1	4,152
J16	stride	0.913	0.99999	0.42855	17,673	0.04372	1	151,474	189,926	1	4,152
J16	streamer	0.913	0.99999	0.42855	17,673	0.04372	1	7,860	189,926	1	4,152
J16	ampm	0.913	0.99999	0.42855	17,673	0.04372	—	8,168	189,926	1	4,152
J16	spp	0.913	0.99999	0.42855	17,673	0.04372	—	58,849	189,926	1	4,152
J16	ipcp	0.913	0.99999	0.42855	17,673	0.04372	1	185,459	189,926	1	4,152
J16	sms	0.913	0.99999	0.42855	17,673	0.04372	—	32,134	189,926	1	4,152
J16	sandbox	0.913	0.99999	0.42855	17,673	0.04372	—	2,537	189,926	1	4,152
J16	power7	0.913	0.99999	0.42855	17,673	0.04372	1	9,427	189,926	1	4,152
J18	no_pref	0.19845	0.99998	0.3803	162,820	0.4028	1	40,312	41,212	1	813
J18	stride	0.19845	0.99998	0.3803	162,820	0.4028	1	36,594	41,212	1	813
J18	streamer	0.19845	0.99998	0.3803	162,820	0.4028	1	1,745	41,212	1	813
J18	ampm	0.19845	0.99998	0.3803	162,820	0.4028	—	1,789	41,212	1	813
J18	spp	0.19845	0.99998	0.3803	162,820	0.4028	—	13,591	41,212	1	813
J18	ipcp	0.19845	0.99998	0.3803	162,820	0.4028	1	40,312	41,212	1	813
J18	sms	0.19845	0.99998	0.3803	162,820	0.4028	—	7,126	41,212	1	813
J18	sandbox	0.19845	0.99998	0.3803	162,820	0.4028	—	490	41,212	1	813
J18	power7	0.19845	0.99998	0.3803	162,820	0.4028	1	2,055	41,212	1	813
J07	no_pref	0.21938	0.86937	0.19359	623,232	0.63255	24,401	162,393	699,538	24,576	336,523
J07	stride	0.21938	0.86937	0.19359	623,232	0.63255	24,384	159,078	699,538	24,576	336,523
J07	streamer	0.21938	0.86937	0.19359	623,232	0.63255	20,161	127,549	699,538	24,576	336,523
J07	ampm	0.21938	0.86937	0.19359	623,232	0.63255	23,672	132,739	699,538	24,576	336,523
J07	spp	0.21938	0.86937	0.19359	623,232	0.63255	24,243	158,872	699,538	24,576	336,523
J07	ipcp	0.21938	0.86937	0.19359	623,232	0.63255	24,401	162,393	699,538	24,576	336,523
J07	sms	0.21938	0.86937	0.19359	623,232	0.63255	8,623	122,594	699,538	24,576	336,523
J07	sandbox	0.21938	0.86937	0.19359	623,232	0.63255	2,440	112,984	699,538	24,576	336,523
J07	power7	0.21938	0.86937	0.19359	623,232	0.63255	20,201	133,558	699,538	24,576	336,523
J06	no_pref	0.22095	0.87196	0.19325	619,482	0.62874	24,016	163,549	788,620	24,216	415,179
J06	stride	0.22095	0.87196	0.19325	619,482	0.62874	24,001	160,299	788,620	24,216	415,179
J06	streamer	0.22095	0.87196	0.19325	619,482	0.62874	19,573	129,145	788,620	24,216	415,179
J06	ampm	0.22095	0.87196	0.19325	619,482	0.62874	23,277	134,541	788,620	24,216	415,179
J06	spp	0.22095	0.87196	0.19325	619,482	0.62874	23,828	160,103	788,620	24,216	415,179
J06	ipcp	0.22095	0.87196	0.19325	619,482	0.62874	24,016	163,549	788,620	24,216	415,179
J06	sms	0.22095	0.87196	0.19325	619,482	0.62874	8,842	124,108	788,620	24,216	415,179
J06	sandbox	0.22095	0.87196	0.19325	619,482	0.62874	2,470	114,892	788,620	24,216	415,179
J06	power7	0.22095	0.87196	0.19325	619,482	0.62874	19,606	134,880	788,620	24,216	415,179
J08	no_pref	0.21527	0.86361	0.19259	620,636	0.62991	25,165	159,346	708,864	25,342	371,979
J08	stride	0.21527	0.86361	0.19259	620,636	0.62991	25,152	156,770	708,864	25,342	371,979
J08	streamer	0.21527	0.86361	0.19259	620,636	0.62991	20,769	131,179	708,864	25,342	371,979
J08	ampm	0.21527	0.86361	0.19259	620,636	0.62991	24,408	136,644	708,864	25,342	371,979
J08	spp	0.21527	0.86361	0.19259	620,636	0.62991	24,982	156,570	708,864	25,342	371,979
J08	ipcp	0.21527	0.86361	0.19259	620,636	0.62991	25,165	159,346	708,864	25,342	371,979
J08	sms	0.21527	0.86361	0.19259	620,636	0.62991	9,176	126,590	708,864	25,342	371,979
J08	sandbox	0.21527	0.86361	0.19259	620,636	0.62991	2,706	116,414	708,864	25,342	371,979
J08	power7	0.21527	0.86361	0.19259	620,636	0.62991	20,817	135,349	708,864	25,342	371,979
J05	no_pref	0.19904	0.87277	0.19173	634,483	0.64396	21,477	147,332	819,983	21,679	366,271
J05	stride	0.19904	0.87277	0.19173	634,483	0.64396	21,466	144,559	819,983	21,679	366,271
J05	streamer	0.19904	0.87277	0.19173	634,483	0.64396	17,782	118,238	819,983	21,679	366,271
J05	ampm	0.19904	0.87277	0.19173	634,483	0.64396	20,808	123,429	819,983	21,679	366,271
J05	spp	0.19904	0.87277	0.19173	634,483	0.64396	21,341	144,421	819,983	21,679	366,271
J05	ipcp	0.19904	0.87277	0.19173	634,483	0.64396	21,477	147,332	819,983	21,679	366,271
J05	sms	0.19904	0.87277	0.19173	634,483	0.64396	7,511	114,283	819,983	21,679	366,271
J05	sandbox	0.19904	0.87277	0.19173	634,483	0.64396	2,078	104,683	819,983	21,679	366,271
J05	power7	0.19904	0.87277	0.19173	634,483	0.64396	17,831	122,297	819,983	21,679	366,271
J01	no_pref	0.16851	0.98727	0.18935	655,741	0.66554	1,609	124,736	573,234	1,738	195,270
J01	stride	0.16851	0.98727	0.18935	655,741	0.66554	1,598	123,686	573,234	1,738	195,270
J01	streamer	0.16851	0.98727	0.18935	655,741	0.66554	1,379	110,822	573,234	1,738	195,270
J01	ampm	0.16851	0.98727	0.18935	655,741	0.66554	1,517	115,032	573,234	1,738	195,270
J01	spp	0.16851	0.98727	0.18935	655,741	0.66554	1,592	123,860	573,234	1,738	195,270
J01	ipcp	0.16851	0.98727	0.18935	655,741	0.66554	1,609	124,736	573,234	1,738	195,270
J01	sms	0.16851	0.98727	0.18935	655,741	0.66554	1,120	107,375	573,234	1,738	195,270
J01	sandbox	0.16851	0.98727	0.18935	655,741	0.66554	827	99,155	573,234	1,738	195,270
J01	power7	0.16851	0.98727	0.18935	655,741	0.66554	1,383	111,310	573,234	1,738	195,270
J03	no_pref	0.20307	0.98511	0.19258	633,604	0.64307	2,272	150,314	493,807	2,387	196,722
J03	stride	0.20307	0.98511	0.19258	633,604	0.64307	2,258	147,280	493,807	2,387	196,722
J03	streamer	0.20307	0.98511	0.19258	633,604	0.64307	1,943	120,284	493,807	2,387	196,722
J03	ampm	0.20307	0.98511	0.19258	633,604	0.64307	2,167	124,550	493,807	2,387	196,722
J03	spp	0.20307	0.98511	0.19258	633,604	0.64307	2,250	147,197	493,807	2,387	196,722
J03	ipcp	0.20307	0.98511	0.19258	633,604	0.64307	2,272	150,314	493,807	2,387	196,722
J03	sms	0.20307	0.98511	0.19258	633,604	0.64307	1,509	116,168	493,807	2,387	196,722
J03	sandbox	0.20307	0.98511	0.19258	633,604	0.64307	1,079	107,570	493,807	2,387	196,722
J03	power7	0.20307	0.98511	0.19258	633,604	0.64307	1,947	124,813	493,807	2,387	196,722
J02	no_pref	0.20242	0.91458	0.19234	629,420	0.63883	13,994	149,835	497,031	14,110	183,325
J02	stride	0.20242	0.91458	0.19234	629,420	0.63883	13,981	147,207	497,031	14,110	183,325
J02	streamer	0.20242	0.91458	0.19234	629,420	0.63883	11,435	120,887	497,031	14,110	183,325
J02	ampm	0.20242	0.91458	0.19234	629,420	0.63883	13,575	125,721	497,031	14,110	183,325
J02	spp	0.20242	0.91458	0.19234	629,420	0.63883	13,883	147,146	497,031	14,110	183,325

Table 63: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 3/4 (cont.)

ID	normal	standalone-event-coverage	standalone-event-timeliness	standalone-ipc	standalone-l2-load-miss	standalone-l2-load-miss-rate	standalone-late-normal-not-timely	standalone-only-timely	standalone-pf-issued	standalone-pf-late	standalone-pf-useless
J02	ipcp	0.20242	0.91458	0.19234	629,420	0.63883	13,994	149,835	497,031	14,110	183,325
J02	sms	0.20242	0.91458	0.19234	629,420	0.63883	5,563	116,240	497,031	14,110	183,325
J02	sandbox	0.20242	0.91458	0.19234	629,420	0.63883	1,928	107,918	497,031	14,110	183,325
J02	power7	0.20242	0.91458	0.19234	629,420	0.63883	11,451	125,404	497,031	14,110	183,325
J04	no_pref	0.16107	0.97886	0.18967	644,237	0.65387	2,575	119,231	323,579	2,596	136,019
J04	stride	0.16107	0.97886	0.18967	644,237	0.65387	2,564	118,534	323,579	2,596	136,019
J04	streamer	0.16107	0.97886	0.18967	644,237	0.65387	2,259	108,473	323,579	2,596	136,019
J04	ampm	0.16107	0.97886	0.18967	644,237	0.65387	2,478	112,048	323,579	2,596	136,019
J04	spp	0.16107	0.97886	0.18967	644,237	0.65387	2,545	118,574	323,579	2,596	136,019
J04	ipcp	0.16107	0.97886	0.18967	644,237	0.65387	2,575	119,231	323,579	2,596	136,019
J04	sms	0.16107	0.97886	0.18967	644,237	0.65387	1,691	104,824	323,579	2,596	136,019
J04	sandbox	0.16107	0.97886	0.18967	644,237	0.65387	1,256	97,883	323,579	2,596	136,019
J04	power7	0.16107	0.97886	0.18967	644,237	0.65387	2,259	108,850	323,579	2,596	136,019
J43	no_pref	0.77396	0.96799	0.36923	66,193	0.22435	7,540	228,038	478,946	7,573	512
J43	stride	0.77396	0.96799	0.36923	66,193	0.22435	4,544	168,462	478,946	7,573	512
J43	streamer	0.77396	0.96799	0.36923	66,193	0.22435	336	105,723	478,946	7,573	512
J43	ampm	0.77396	0.96799	0.36923	66,193	0.22435	58	26,322	478,946	7,573	512
J43	spp	0.77396	0.96799	0.36923	66,193	0.22435	4,140	172,811	478,946	7,573	512
J43	ipcp	0.77396	0.96799	0.36923	66,193	0.22435	7,540	228,038	478,946	7,573	512
J43	sms	0.77396	0.96799	0.36923	66,193	0.22435	349	19,127	478,946	7,573	512
J43	sandbox	0.77396	0.96799	0.36923	66,193	0.22435	24	59,136	478,946	7,573	512
J43	power7	0.77396	0.96799	0.36923	66,193	0.22435	649	90,759	478,946	7,573	512
J38	no_pref	0.80147	0.96419	0.37001	58,058	0.19678	8,770	236,144	506,926	8,820	533
J38	stride	0.80147	0.96419	0.37001	58,058	0.19678	4,665	178,802	506,926	8,820	533
J38	streamer	0.80147	0.96419	0.37001	58,058	0.19678	768	116,465	506,926	8,820	533
J38	ampm	0.80147	0.96419	0.37001	58,058	0.19678	626	33,533	506,926	8,820	533
J38	spp	0.80147	0.96419	0.37001	58,058	0.19678	4,613	179,082	506,926	8,820	533
J38	ipcp	0.80147	0.96419	0.37001	58,058	0.19678	8,770	236,144	506,926	8,820	533
J38	sms	0.80147	0.96419	0.37001	58,058	0.19678	267	23,779	506,926	8,820	533
J38	sandbox	0.80147	0.96419	0.37001	58,058	0.19678	146	62,715	506,926	8,820	533
J38	power7	0.80147	0.96419	0.37001	58,058	0.19678	1,062	100,382	506,926	8,820	533
J47	no_pref	0.98504	0.99857	0.38379	3,516	0.01192	416	290,231	468,215	419	1,347
J47	stride	0.98504	0.99857	0.38379	3,516	0.01192	229	223,553	468,215	419	1,347
J47	streamer	0.98504	0.99857	0.38379	3,516	0.01192	47	149,895	468,215	419	1,347
J47	ampm	0.98504	0.99857	0.38379	3,516	0.01192	40	57,237	468,215	419	1,347
J47	spp	0.98504	0.99857	0.38379	3,516	0.01192	215	222,558	468,215	419	1,347
J47	ipcp	0.98504	0.99857	0.38379	3,516	0.01192	416	290,231	468,215	419	1,347
J47	sms	0.98504	0.99857	0.38379	3,516	0.01192	6	38,204	468,215	419	1,347
J47	sandbox	0.98504	0.99857	0.38379	3,516	0.01192	10	83,788	468,215	419	1,347
J47	power7	0.98504	0.99857	0.38379	3,516	0.01192	56	131,744	468,215	419	1,347
J31	no_pref	0.79333	0.95548	0.37244	60,410	0.20474	10,890	233,746	480,929	10,921	117
J31	stride	0.79333	0.95548	0.37244	60,410	0.20474	7,438	174,308	480,929	10,921	117
J31	streamer	0.79333	0.95548	0.37244	60,410	0.20474	1,081	110,500	480,929	10,921	117
J31	ampm	0.79333	0.95548	0.37244	60,410	0.20474	735	27,328	480,929	10,921	117
J31	spp	0.79333	0.95548	0.37244	60,410	0.20474	5,882	176,740	480,929	10,921	117
J31	ipcp	0.79333	0.95548	0.37244	60,410	0.20474	10,890	233,746	480,929	10,921	117
J31	sms	0.79333	0.95548	0.37244	60,410	0.20474	407	21,504	480,929	10,921	117
J31	sandbox	0.79333	0.95548	0.37244	60,410	0.20474	224	61,945	480,929	10,921	117
J31	power7	0.79333	0.95548	0.37244	60,410	0.20474	1,729	94,631	480,929	10,921	117
J34	no_pref	0.84473	0.9903	0.37556	45,132	0.15295	2,439	248,889	869,593	2,451	121
J34	stride	0.84473	0.9903	0.37556	45,132	0.15295	2,158	186,146	869,593	2,451	121
J34	streamer	0.84473	0.9903	0.37556	45,132	0.15295	811	118,350	869,593	2,451	121
J34	ampm	0.84473	0.9903	0.37556	45,132	0.15295	750	29,130	869,593	2,451	121
J34	spp	0.84473	0.9903	0.37556	45,132	0.15295	1,460	188,884	869,593	2,451	121
J34	ipcp	0.84473	0.9903	0.37556	45,132	0.15295	2,439	248,889	869,593	2,451	121
J34	sms	0.84473	0.9903	0.37556	45,132	0.15295	167	23,264	869,593	2,451	121
J34	sandbox	0.84473	0.9903	0.37556	45,132	0.15295	215	67,471	869,593	2,451	121
J34	power7	0.84473	0.9903	0.37556	45,132	0.15295	897	101,437	869,593	2,451	121
J30	no_pref	0.79333	0.95548	0.37244	60,410	0.20474	10,890	233,746	480,929	10,921	117
J30	stride	0.79333	0.95548	0.37244	60,410	0.20474	7,438	174,308	480,929	10,921	117
J30	streamer	0.79333	0.95548	0.37244	60,410	0.20474	1,081	110,500	480,929	10,921	117
J30	ampm	0.79333	0.95548	0.37244	60,410	0.20474	735	27,328	480,929	10,921	117
J30	spp	0.79333	0.95548	0.37244	60,410	0.20474	5,882	176,740	480,929	10,921	117
J30	ipcp	0.79333	0.95548	0.37244	60,410	0.20474	10,890	233,746	480,929	10,921	117
J30	sms	0.79333	0.95548	0.37244	60,410	0.20474	407	21,504	480,929	10,921	117
J30	sandbox	0.79333	0.95548	0.37244	60,410	0.20474	224	61,945	480,929	10,921	117
J30	power7	0.79333	0.95548	0.37244	60,410	0.20474	1,729	94,631	480,929	10,921	117
J35	no_pref	0.78029	0.95513	0.37284	64,362	0.21814	10,801	229,904	474,081	10,845	110
J35	stride	0.78029	0.95513	0.37284	64,362	0.21814	6,711	171,375	474,081	10,845	110
J35	streamer	0.78029	0.95513	0.37284	64,362	0.21814	738	107,278	474,081	10,845	110
J35	ampm	0.78029	0.95513	0.37284	64,362	0.21814	484	25,763	474,081	10,845	110
J35	spp	0.78029	0.95513	0.37284	64,362	0.21814	5,851	172,925	474,081	10,845	110
J35	ipcp	0.78029	0.95513	0.37284	64,362	0.21814	10,801	229,904	474,081	10,845	110
J35	sms	0.78029	0.95513	0.37284	64,362	0.21814	315	19,807	474,081	10,845	110
J35	sandbox	0.78029	0.95513	0.37284	64,362	0.21814	104	59,816	474,081	10,845	110
J35	power7	0.78029	0.95513	0.37284	64,362	0.21814	1,345	92,670	474,081	10,845	110
J36	no_pref	0.76678	0.95477	0.37039	68,307	0.2315	10,703	225,923	472,941	10,729	86
J36	stride	0.76678	0.95477	0.37039	68,307	0.2315	6,746	169,123	472,941	10,729	86
J36	streamer	0.76678	0.95477	0.37039	68,307	0.2315	684	106,257	472,941	10,729	86
J36	ampm	0.76678	0.95477	0.37039	68,307	0.2315	451	25,581	472,941	10,729	86
J36	spp	0.76678	0.95477	0.37039	68,307	0.2315	5,763	169,884	472,941	10,729	86
J36	ipcp	0.76678	0.95477	0.37039	68,307	0.2315	10,703	225,923	472,941	10,729	86
J36	sms	0.76678	0.95477	0.37039	68,307	0.2315	408	20,051	472,941	10,729	86
J36	sandbox	0.76678	0.95477	0.37039	68,307	0.2315	171	59,673	472,941	10,729	86
J36	power7	0.76678	0.95477	0.37039	68,307	0.2315	1,163	91,450	472,941	10,729	86

Table 63: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 3/4 (cont.)

ID	normal	standalone-event-coverage	standalone-event-timeliness	standalone-ipc	standalone-l2-load-miss	standalone-l2-load-miss-rate	standalone-late-normal_not_timely	standalone-only-timely	standalone-pf-issued	standalone-pf-late	standalone-pf-useless
J32	no_pref	0.83437	0.982	0.3746	48,208	0.16338	4,505	245,839	679,499	4,516	120
J32	stride	0.83437	0.982	0.3746	48,208	0.16338	3,782	183,597	679,499	4,516	120
J32	streamer	0.83437	0.982	0.3746	48,208	0.16338	913	116,646	679,499	4,516	120
J32	ampm	0.83437	0.982	0.3746	48,208	0.16338	726	28,745	679,499	4,516	120
J32	spp	0.83437	0.982	0.3746	48,208	0.16338	2,547	186,409	679,499	4,516	120
J32	ipcp	0.83437	0.982	0.3746	48,208	0.16338	4,505	245,839	679,499	4,516	120
J32	sms	0.83437	0.982	0.3746	48,208	0.16338	234	22,919	679,499	4,516	120
J32	sandbox	0.83437	0.982	0.3746	48,208	0.16338	231	66,362	679,499	4,516	120
J32	power7	0.83437	0.982	0.3746	48,208	0.16338	1,211	99,821	679,499	4,516	120
J33	no_pref	0.84473	0.9903	0.37556	45,132	0.15295	2,439	248,889	869,593	2,451	121
J33	stride	0.84473	0.9903	0.37556	45,132	0.15295	2,158	186,146	869,593	2,451	121
J33	streamer	0.84473	0.9903	0.37556	45,132	0.15295	811	118,350	869,593	2,451	121
J33	ampm	0.84473	0.9903	0.37556	45,132	0.15295	750	29,130	869,593	2,451	121
J33	spp	0.84473	0.9903	0.37556	45,132	0.15295	1,460	188,884	869,593	2,451	121
J33	ipcp	0.84473	0.9903	0.37556	45,132	0.15295	2,439	248,889	869,593	2,451	121
J33	sms	0.84473	0.9903	0.37556	45,132	0.15295	167	23,264	869,593	2,451	121
J33	sandbox	0.84473	0.9903	0.37556	45,132	0.15295	215	67,471	869,593	2,451	121
J33	power7	0.84473	0.9903	0.37556	45,132	0.15295	897	101,437	869,593	2,451	121
J46	no_pref	0.85367	0.9935	0.37549	42,534	0.14416	1,645	251,524	833,464	1,656	751
J46	stride	0.85367	0.9935	0.37549	42,534	0.14416	1,217	189,215	833,464	1,656	751
J46	streamer	0.85367	0.9935	0.37549	42,534	0.14416	151	123,586	833,464	1,656	751
J46	ampm	0.85367	0.9935	0.37549	42,534	0.14416	68	35,100	833,464	1,656	751
J46	spp	0.85367	0.9935	0.37549	42,534	0.14416	905	191,006	833,464	1,656	751
J46	ipcp	0.85367	0.9935	0.37549	42,534	0.14416	1,645	251,524	833,464	1,656	751
J46	sms	0.85367	0.9935	0.37549	42,534	0.14416	28	24,422	833,464	1,656	751
J46	sandbox	0.85367	0.9935	0.37549	42,534	0.14416	12	67,743	833,464	1,656	751
J46	power7	0.85367	0.9935	0.37549	42,534	0.14416	190	106,741	833,464	1,656	751
J41	no_pref	0.88103	0.99411	0.37681	34,372	0.11649	1,538	259,585	904,426	1,544	705
J41	stride	0.88103	0.99411	0.37681	34,372	0.11649	1,156	197,170	904,426	1,544	705
J41	streamer	0.88103	0.99411	0.37681	34,372	0.11649	473	128,598	904,426	1,544	705
J41	ampm	0.88103	0.99411	0.37681	34,372	0.11649	419	40,002	904,426	1,544	705
J41	spp	0.88103	0.99411	0.37681	34,372	0.11649	898	197,210	904,426	1,544	705
J41	ipcp	0.88103	0.99411	0.37681	34,372	0.11649	1,538	259,585	904,426	1,544	705
J41	sms	0.88103	0.99411	0.37681	34,372	0.11649	105	27,495	904,426	1,544	705
J41	sandbox	0.88103	0.99411	0.37681	34,372	0.11649	125	71,060	904,426	1,544	705
J41	power7	0.88103	0.99411	0.37681	34,372	0.11649	514	111,815	904,426	1,544	705
J42	no_pref	0.77396	0.96799	0.36923	66,193	0.22435	7,540	228,038	478,946	7,573	512
J42	stride	0.77396	0.96799	0.36923	66,193	0.22435	4,544	168,462	478,946	7,573	512
J42	streamer	0.77396	0.96799	0.36923	66,193	0.22435	336	105,723	478,946	7,573	512
J42	ampm	0.77396	0.96799	0.36923	66,193	0.22435	58	26,322	478,946	7,573	512
J42	spp	0.77396	0.96799	0.36923	66,193	0.22435	4,140	172,811	478,946	7,573	512
J42	ipcp	0.77396	0.96799	0.36923	66,193	0.22435	7,540	228,038	478,946	7,573	512
J42	sms	0.77396	0.96799	0.36923	66,193	0.22435	349	19,127	478,946	7,573	512
J42	sandbox	0.77396	0.96799	0.36923	66,193	0.22435	24	59,136	478,946	7,573	512
J42	power7	0.77396	0.96799	0.36923	66,193	0.22435	649	90,759	478,946	7,573	512
J37	no_pref	0.80147	0.96419	0.37001	58,058	0.19678	8,770	236,144	506,926	8,820	533
J37	stride	0.80147	0.96419	0.37001	58,058	0.19678	4,665	178,802	506,926	8,820	533
J37	streamer	0.80147	0.96419	0.37001	58,058	0.19678	768	116,465	506,926	8,820	533
J37	ampm	0.80147	0.96419	0.37001	58,058	0.19678	626	33,533	506,926	8,820	533
J37	spp	0.80147	0.96419	0.37001	58,058	0.19678	4,613	179,082	506,926	8,820	533
J37	ipcp	0.80147	0.96419	0.37001	58,058	0.19678	8,770	236,144	506,926	8,820	533
J37	sms	0.80147	0.96419	0.37001	58,058	0.19678	267	23,779	506,926	8,820	533
J37	sandbox	0.80147	0.96419	0.37001	58,058	0.19678	146	62,715	506,926	8,820	533
J37	power7	0.80147	0.96419	0.37001	58,058	0.19678	1,062	100,382	506,926	8,820	533
J44	no_pref	0.83735	0.98668	0.37438	47,395	0.16063	3,331	246,715	673,058	3,348	706
J44	stride	0.83735	0.98668	0.37438	47,395	0.16063	2,131	185,133	673,058	3,348	706
J44	streamer	0.83735	0.98668	0.37438	47,395	0.16063	205	119,954	673,058	3,348	706
J44	ampm	0.83735	0.98668	0.37438	47,395	0.16063	89	33,344	673,058	3,348	706
J44	spp	0.83735	0.98668	0.37438	47,395	0.16063	1,780	187,241	673,058	3,348	706
J44	ipcp	0.83735	0.98668	0.37438	47,395	0.16063	3,331	246,715	673,058	3,348	706
J44	sms	0.83735	0.98668	0.37438	47,395	0.16063	113	23,287	673,058	3,348	706
J44	sandbox	0.83735	0.98668	0.37438	47,395	0.16063	11	65,875	673,058	3,348	706
J44	power7	0.83735	0.98668	0.37438	47,395	0.16063	307	103,677	673,058	3,348	706
J39	no_pref	0.86703	0.98619	0.37654	38,561	0.13068	3,578	255,462	722,342	3,607	676
J39	stride	0.86703	0.98619	0.37654	38,561	0.13068	1,887	194,381	722,342	3,607	676
J39	streamer	0.86703	0.98619	0.37654	38,561	0.13068	525	126,291	722,342	3,607	676
J39	ampm	0.86703	0.98619	0.37654	38,561	0.13068	464	39,236	722,342	3,607	676
J39	spp	0.86703	0.98619	0.37654	38,561	0.13068	1,983	193,907	722,342	3,607	676
J39	ipcp	0.86703	0.98619	0.37654	38,561	0.13068	3,578	255,462	722,342	3,607	676
J39	sms	0.86703	0.98619	0.37654	38,561	0.13068	139	26,971	722,342	3,607	676
J39	sandbox	0.86703	0.98619	0.37654	38,561	0.13068	122	69,736	722,342	3,607	676
J39	power7	0.86703	0.98619	0.37654	38,561	0.13068	601	109,989	722,342	3,607	676
J45	no_pref	0.85367	0.9935	0.37549	42,534	0.14416	1,645	251,524	833,464	1,656	751
J45	stride	0.85367	0.9935	0.37549	42,534	0.14416	1,217	189,215	833,464	1,656	751
J45	streamer	0.85367	0.9935	0.37549	42,534	0.14416	151	123,586	833,464	1,656	751
J45	ampm	0.85367	0.9935	0.37549	42,534	0.14416	68	35,100	833,464	1,656	751
J45	spp	0.85367	0.9935	0.37549	42,534	0.14416	905	191,006	833,464	1,656	751
J45	ipcp	0.85367	0.9935	0.37549	42,534	0.14416	1,645	251,524	833,464	1,656	751
J45	sms	0.85367	0.9935	0.37549	42,534	0.14416	28	24,422	833,464	1,656	751
J45	sandbox	0.85367	0.9935	0.37549	42,534	0.14416	12	67,743	833,464	1,656	751
J45	power7	0.85367	0.9935	0.37549	42,534	0.14416	190	106,741	833,464	1,656	751
J40	no_pref	0.88103	0.99411	0.37681	34,372	0.11649	1,538	259,585	904,426	1,544	705
J40	stride	0.88103	0.99411	0.37681	34,372	0.11649	1,156	197,170	904,426	1,544	705
J40	streamer	0.88103	0.99411	0.37681	34,372	0.11649	473	128,598	904,426	1,544	705
J40	ampm	0.88103	0.99411	0.37681	34,372	0.11649	419	40,002	904,426	1,544	705

Table 63: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 3/4 (cont.)

ID	normal	<i>standalone-event-coverage</i>	<i>standalone-event-timeliness</i>	<i>standalone-ipc</i>	<i>standalone-l2-load-miss</i>	<i>standalone-l2-load-miss-rate</i>	<i>standalone-late-normal-not-timely</i>	<i>standalone-only-timely</i>	<i>standalone-pf-issued</i>	<i>standalone-pf-late</i>	<i>standalone-pf-useless</i>
J40	spp	0.88103	0.99411	0.37681	34,372	0.11649	898	197,210	904,426	1,544	705
J40	ipcp	0.88103	0.99411	0.37681	34,372	0.11649	1,538	259,585	904,426	1,544	705
J40	sms	0.88103	0.99411	0.37681	34,372	0.11649	105	27,495	904,426	1,544	705
J40	sandbox	0.88103	0.99411	0.37681	34,372	0.11649	125	71,060	904,426	1,544	705
J40	power7	0.88103	0.99411	0.37681	34,372	0.11649	514	111,815	904,426	1,544	705
J28	no_pref	0.12835	0.99791	0.24105	278,633	0.61725	85	40,630	70,159	87	23,084
J28	stride	0.12835	0.99791	0.24105	278,633	0.61725	85	40,593	70,159	87	23,084
J28	streamer	0.12835	0.99791	0.24105	278,633	0.61725	25	15,598	70,159	87	23,084
J28	ampm	0.12835	0.99791	0.24105	278,633	0.61725	69	28,551	70,159	87	23,084
J28	spp	0.12835	0.99791	0.24105	278,633	0.61725	85	40,602	70,159	87	23,084
J28	ipcp	0.12835	0.99791	0.24105	278,633	0.61725	84	39,545	70,159	87	23,084
J28	sms	0.12835	0.99791	0.24105	278,633	0.61725	36	19,502	70,159	87	23,084
J28	sandbox	0.12835	0.99791	0.24105	278,633	0.61725	8	3,472	70,159	87	23,084
J28	power7	0.12835	0.99791	0.24105	278,633	0.61725	27	15,836	70,159	87	23,084
J27	no_pref	0.15317	0.99753	0.24121	270,804	0.59987	120	48,487	83,744	123	25,124
J27	stride	0.15317	0.99753	0.24121	270,804	0.59987	120	48,404	83,744	123	25,124
J27	streamer	0.15317	0.99753	0.24121	270,804	0.59987	69	21,208	83,744	123	25,124
J27	ampm	0.15317	0.99753	0.24121	270,804	0.59987	111	36,452	83,744	123	25,124
J27	spp	0.15317	0.99753	0.24121	270,804	0.59987	120	48,458	83,744	123	25,124
J27	ipcp	0.15317	0.99753	0.24121	270,804	0.59987	120	47,418	83,744	123	25,124
J27	sms	0.15317	0.99753	0.24121	270,804	0.59987	76	23,696	83,744	123	25,124
J27	sandbox	0.15317	0.99753	0.24121	270,804	0.59987	34	7,377	83,744	123	25,124
J27	power7	0.15317	0.99753	0.24121	270,804	0.59987	69	21,559	83,744	123	25,124
J29	no_pref	0.09619	0.99474	0.23952	292,473	0.64794	161	30,449	102,407	161	53,581
J29	stride	0.09619	0.99474	0.23952	292,473	0.64794	161	30,349	102,407	161	53,581
J29	streamer	0.09619	0.99474	0.23952	292,473	0.64794	122	10,830	102,407	161	53,581
J29	ampm	0.09619	0.99474	0.23952	292,473	0.64794	146	21,536	102,407	161	53,581
J29	spp	0.09619	0.99474	0.23952	292,473	0.64794	161	30,428	102,407	161	53,581
J29	ipcp	0.09619	0.99474	0.23952	292,473	0.64794	144	29,578	102,407	161	53,581
J29	sms	0.09619	0.99474	0.23952	292,473	0.64794	60	15,523	102,407	161	53,581
J29	sandbox	0.09619	0.99474	0.23952	292,473	0.64794	24	2,202	102,407	161	53,581
J29	power7	0.09619	0.99474	0.23952	292,473	0.64794	123	11,508	102,407	161	53,581
J26	no_pref	0.14463	0.99884	0.24195	271,764	0.60196	53	45,782	57,133	53	7,465
J26	stride	0.14463	0.99884	0.24195	271,764	0.60196	53	44,621	57,133	53	7,465
J26	streamer	0.14463	0.99884	0.24195	271,764	0.60196	34	18,403	57,133	53	7,465
J26	ampm	0.14463	0.99884	0.24195	271,764	0.60196	43	32,566	57,133	53	7,465
J26	spp	0.14463	0.99884	0.24195	271,764	0.60196	53	45,748	57,133	53	7,465
J26	ipcp	0.14463	0.99884	0.24195	271,764	0.60196	53	44,698	57,133	53	7,465
J26	sms	0.14463	0.99884	0.24195	271,764	0.60196	34	22,151	57,133	53	7,465
J26	sandbox	0.14463	0.99884	0.24195	271,764	0.60196	25	4,987	57,133	53	7,465
J26	power7	0.14463	0.99884	0.24195	271,764	0.60196	35	18,967	57,133	53	7,465
J23	no_pref	0.14463	0.99884	0.24195	271,764	0.60196	53	45,782	57,133	53	7,465
J23	stride	0.14463	0.99884	0.24195	271,764	0.60196	53	44,621	57,133	53	7,465
J23	streamer	0.14463	0.99884	0.24195	271,764	0.60196	34	18,403	57,133	53	7,465
J23	ampm	0.14463	0.99884	0.24195	271,764	0.60196	43	32,566	57,133	53	7,465
J23	spp	0.14463	0.99884	0.24195	271,764	0.60196	53	45,748	57,133	53	7,465
J23	ipcp	0.14463	0.99884	0.24195	271,764	0.60196	53	44,698	57,133	53	7,465
J23	sms	0.14463	0.99884	0.24195	271,764	0.60196	34	22,151	57,133	53	7,465
J23	sandbox	0.14463	0.99884	0.24195	271,764	0.60196	25	4,987	57,133	53	7,465
J23	power7	0.14463	0.99884	0.24195	271,764	0.60196	35	18,967	57,133	53	7,465
J25	no_pref	0.06614	0.99905	0.23963	296,075	0.65586	20	20,937	24,752	21	3,503
J25	stride	0.06614	0.99905	0.23963	296,075	0.65586	20	20,932	24,752	21	3,503
J25	streamer	0.06614	0.99905	0.23963	296,075	0.65586	—	6,125	24,752	21	3,503
J25	ampm	0.06614	0.99905	0.23963	296,075	0.65586	20	13,520	24,752	21	3,503
J25	spp	0.06614	0.99905	0.23963	296,075	0.65586	20	20,917	24,752	21	3,503
J25	ipcp	0.06614	0.99905	0.23963	296,075	0.65586	20	20,871	24,752	21	3,503
J25	sms	0.06614	0.99905	0.23963	296,075	0.65586	5	10,536	24,752	21	3,503
J25	sandbox	0.06614	0.99905	0.23963	296,075	0.65586	—	1,851	24,752	21	3,503
J25	power7	0.06614	0.99905	0.23963	296,075	0.65586	—	6,159	24,752	21	3,503
J24	no_pref	0.06615	0.99914	0.23959	296,018	0.65576	18	20,939	24,815	18	3,358
J24	stride	0.06615	0.99914	0.23959	296,018	0.65576	18	20,935	24,815	18	3,358
J24	streamer	0.06615	0.99914	0.23959	296,018	0.65576	3	5,388	24,815	18	3,358
J24	ampm	0.06615	0.99914	0.23959	296,018	0.65576	18	13,616	24,815	18	3,358
J24	spp	0.06615	0.99914	0.23959	296,018	0.65576	18	20,915	24,815	18	3,358
J24	ipcp	0.06615	0.99914	0.23959	296,018	0.65576	17	20,315	24,815	18	3,358
J24	sms	0.06615	0.99914	0.23959	296,018	0.65576	6	10,544	24,815	18	3,358
J24	sandbox	0.06615	0.99914	0.23959	296,018	0.65576	—	1,212	24,815	18	3,358
J24	power7	0.06615	0.99914	0.23959	296,018	0.65576	3	5,691	24,815	18	3,358
J13	no_pref	0.50588	0.99551	0.35708	381,025	0.31251	1,028	227,823	516,174	1,180	172,096
J13	stride	0.50588	0.99551	0.35708	381,025	0.31251	993	226,666	516,174	1,180	172,096
J13	streamer	0.50588	0.99551	0.35708	381,025	0.31251	967	213,831	516,174	1,180	172,096
J13	ampm	0.50588	0.99551	0.35708	381,025	0.31251	980	217,205	516,174	1,180	172,096
J13	spp	0.50588	0.99551	0.35708	381,025	0.31251	1,025	227,358	516,174	1,180	172,096
J13	ipcp	0.50588	0.99551	0.35708	381,025	0.31251	1,021	226,564	516,174	1,180	172,096
J13	sms	0.50588	0.99551	0.35708	381,025	0.31251	932	193,156	516,174	1,180	172,096
J13	sandbox	0.50588	0.99551	0.35708	381,025	0.31251	875	145,576	516,174	1,180	172,096
J13	power7	0.50588	0.99551	0.35708	381,025	0.31251	981	218,619	516,174	1,180	172,096
J12	no_pref	0.37744	0.99662	0.35127	481,518	0.3953	577	169,978	479,615	684	216,704
J12	stride	0.37744	0.99662	0.35127	481,518	0.3953	576	168,806	479,615	684	216,704
J12	streamer	0.37744	0.99662	0.35127	481,518	0.3953	568	156,925	479,615	684	216,704
J12	ampm	0.37744	0.99662	0.35127	481,518	0.3953	569	158,557	479,615	684	216,704
J12	spp	0.37744	0.99662	0.35127	481,518	0.3953	577	169,573	479,615	684	216,704
J12	ipcp	0.37744	0.99662	0.35127	481,518	0.3953	577	169,966	479,615	684	216,704
J12	sms	0.37744	0.99662	0.35127	481,518	0.3953	419	147,324	479,615	684	216,704
J12	sandbox	0.37744	0.99662	0.35127	481,518	0.3953	413	104,614	479,615	684	216,704

Table 63: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 3/4 (cont.)

ID	normal	<i>standalone-event-coverage</i>	<i>standalone-event-timeliness</i>	<i>standalone-ipc</i>	<i>standalone_l2-load-miss</i>	<i>standalone_l2-load-miss-rate</i>	<i>standalone_late-normal_not-timely</i>	<i>standalone-only-timely</i>	<i>standalone-pf-issued</i>	<i>standalone-pf-late</i>	<i>standalone-pf-useless</i>
J12	power7	0.37744	0.99662	0.35127	481,518	0.3953	572	161,694	479,615	684	216,704
J14	no_pref	0.30195	0.99551	0.34149	658,974	0.54091	613	135,984	794,157	1,212	569,458
J14	stride	0.30195	0.99551	0.34149	658,974	0.54091	544	135,109	794,157	1,212	569,458
J14	streamer	0.30195	0.99551	0.34149	658,974	0.54091	431	126,540	794,157	1,212	569,458
J14	ampm	0.30195	0.99551	0.34149	658,974	0.54091	444	128,757	794,157	1,212	569,458
J14	spp	0.30195	0.99551	0.34149	658,974	0.54091	591	135,567	794,157	1,212	569,458
J14	ipcp	0.30195	0.99551	0.34149	658,974	0.54091	612	135,878	794,157	1,212	569,458
J14	sms	0.30195	0.99551	0.34149	658,974	0.54091	462	116,609	794,157	1,212	569,458
J14	sandbox	0.30195	0.99551	0.34149	658,974	0.54091	337	86,451	794,157	1,212	569,458
J14	power7	0.30195	0.99551	0.34149	658,974	0.54091	486	129,782	794,157	1,212	569,458
J11	no_pref	0.89075	0.99814	0.37915	53,216	0.04365	749	401,150	534,810	828	14,274
J11	stride	0.89075	0.99814	0.37915	53,216	0.04365	676	399,559	534,810	828	14,274
J11	streamer	0.89075	0.99814	0.37915	53,216	0.04365	621	379,703	534,810	828	14,274
J11	ampm	0.89075	0.99814	0.37915	53,216	0.04365	578	385,220	534,810	828	14,274
J11	spp	0.89075	0.99814	0.37915	53,216	0.04365	716	400,219	534,810	828	14,274
J11	ipcp	0.89075	0.99814	0.37915	53,216	0.04365	748	398,774	534,810	828	14,274
J11	sms	0.89075	0.99814	0.37915	53,216	0.04365	550	340,935	534,810	828	14,274
J11	sandbox	0.89075	0.99814	0.37915	53,216	0.04365	604	254,322	534,810	828	14,274
J11	power7	0.89075	0.99814	0.37915	53,216	0.04365	652	386,661	534,810	828	14,274
J09	no_pref	0.86354	0.99833	0.37882	62,526	0.05128	652	388,893	481,918	712	7,533
J09	stride	0.86354	0.99833	0.37882	62,526	0.05128	611	387,620	481,918	712	7,533
J09	streamer	0.86354	0.99833	0.37882	62,526	0.05128	580	369,601	481,918	712	7,533
J09	ampm	0.86354	0.99833	0.37882	62,526	0.05128	558	374,848	481,918	712	7,533
J09	spp	0.86354	0.99833	0.37882	62,526	0.05128	629	388,068	481,918	712	7,533
J09	ipcp	0.86354	0.99833	0.37882	62,526	0.05128	651	386,610	481,918	712	7,533
J09	sms	0.86354	0.99833	0.37882	62,526	0.05128	443	331,145	481,918	712	7,533
J09	sandbox	0.86354	0.99833	0.37882	62,526	0.05128	526	247,405	481,918	712	7,533
J09	power7	0.86354	0.99833	0.37882	62,526	0.05128	608	375,858	481,918	712	7,533
J10	no_pref	0.17943	0.99676	0.35728	383,674	0.31491	263	80,808	112,483	264	10,540
J10	stride	0.17943	0.99676	0.35728	383,674	0.31491	263	80,808	112,483	264	10,540
J10	streamer	0.17943	0.99676	0.35728	383,674	0.31491	263	79,490	112,483	264	10,540
J10	ampm	0.17943	0.99676	0.35728	383,674	0.31491	263	80,261	112,483	264	10,540
J10	spp	0.17943	0.99676	0.35728	383,674	0.31491	263	80,766	112,483	264	10,540
J10	ipcp	0.17943	0.99676	0.35728	383,674	0.31491	263	80,796	112,483	264	10,540
J10	sms	0.17943	0.99676	0.35728	383,674	0.31491	262	70,896	112,483	264	10,540
J10	sandbox	0.17943	0.99676	0.35728	383,674	0.31491	248	54,147	112,483	264	10,540
J10	power7	0.17943	0.99676	0.35728	383,674	0.31491	263	79,818	112,483	264	10,540

Table 64: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 4/4

ID	normal	<i>standalone_selected-accuracy</i>
J21	no_pref	0.95218
J21	stride	0.95218
J21	streamer	0.95218
J21	ampm	0.95218
J21	spp	0.95218
J21	ipcp	0.95218
J21	sms	0.95218
J21	sandbox	0.95218
J21	power7	0.95218
J20	no_pref	0.95088
J20	stride	0.95088
J20	streamer	0.95088
J20	ampm	0.95088
J20	spp	0.95088
J20	ipcp	0.95088
J20	sms	0.95088
J20	sandbox	0.95088
J20	power7	0.95088
J22	no_pref	0.79469
J22	stride	0.79469
J22	streamer	0.79469
J22	ampm	0.79469
J22	spp	0.79469
J22	ipcp	0.79469
J22	sms	0.79469

Table 64: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 4/4 (cont.)

ID	normal	<i>standalone_selected_accuracy</i>
J22	sandbox	0.79469
J22	power7	0.79469
J19	no_pref	0.9711
J19	stride	0.9711
J19	streamer	0.9711
J19	ampm	0.9711
J19	spp	0.9711
J19	ipcp	0.9711
J19	sms	0.9711
J19	sandbox	0.9711
J19	power7	0.9711
J15	no_pref	0.9711
J15	stride	0.9711
J15	streamer	0.9711
J15	ampm	0.9711
J15	spp	0.9711
J15	ipcp	0.9711
J15	sms	0.9711
J15	sandbox	0.9711
J15	power7	0.9711
J17	no_pref	0.95218
J17	stride	0.95218
J17	streamer	0.95218
J17	ampm	0.95218
J17	spp	0.95218
J17	ipcp	0.95218
J17	sms	0.95218
J17	sandbox	0.95218
J17	power7	0.95218
J16	no_pref	0.95088
J16	stride	0.95088
J16	streamer	0.95088
J16	ampm	0.95088
J16	spp	0.95088
J16	ipcp	0.95088
J16	sms	0.95088
J16	sandbox	0.95088
J16	power7	0.95088
J18	no_pref	0.95031
J18	stride	0.95031
J18	streamer	0.95031
J18	ampm	0.95031
J18	spp	0.95031
J18	ipcp	0.95031
J18	sms	0.95031
J18	sandbox	0.95031
J18	power7	0.95031
J07	no_pref	0.15124
J07	stride	0.15124
J07	streamer	0.15124
J07	ampm	0.15124
J07	spp	0.15124
J07	ipcp	0.15124
J07	sms	0.15124
J07	sandbox	0.15124
J07	power7	0.15124
J06	no_pref	0.1377
J06	stride	0.1377
J06	streamer	0.1377
J06	ampm	0.1377
J06	spp	0.1377
J06	ipcp	0.1377
J06	sms	0.1377
J06	sandbox	0.1377
J06	power7	0.1377
J08	no_pref	0.15486
J08	stride	0.15486
J08	streamer	0.15486
J08	ampm	0.15486
J08	spp	0.15486
J08	ipcp	0.15486
J08	sms	0.15486
J08	sandbox	0.15486
J08	power7	0.15486
J05	no_pref	0.12049
J05	stride	0.12049
J05	streamer	0.12049
J05	ampm	0.12049
J05	spp	0.12049
J05	ipcp	0.12049
J05	sms	0.12049
J05	sandbox	0.12049
J05	power7	0.12049
J01	no_pref	0.15917
J01	stride	0.15917
J01	streamer	0.15917
J01	ampm	0.15917
J01	spp	0.15917

Table 64: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 4/4 (cont.)

ID	normal	<i>standalone_selected_accuracy</i>
J01	ipcp	0.15917
J01	sms	0.15917
J01	sandbox	0.15917
J01	power7	0.15917
J03	no_pref	0.20259
J03	stride	0.20259
J03	streamer	0.20259
J03	ampm	0.20259
J03	spp	0.20259
J03	ipcp	0.20259
J03	sms	0.20259
J03	sandbox	0.20259
J03	power7	0.20259
J02	no_pref	0.20823
J02	stride	0.20823
J02	streamer	0.20823
J02	ampm	0.20823
J02	spp	0.20823
J02	ipcp	0.20823
J02	sms	0.20823
J02	sandbox	0.20823
J02	power7	0.20823
J04	no_pref	0.27242
J04	stride	0.27242
J04	streamer	0.27242
J04	ampm	0.27242
J04	spp	0.27242
J04	ipcp	0.27242
J04	sms	0.27242
J04	sandbox	0.27242
J04	power7	0.27242
J43	no_pref	0.99855
J43	stride	0.99855
J43	streamer	0.99855
J43	ampm	0.99855
J43	spp	0.99855
J43	ipcp	0.99855
J43	sms	0.99855
J43	sandbox	0.99855
J43	power7	0.99855
J38	no_pref	0.99854
J38	stride	0.99854
J38	streamer	0.99854
J38	ampm	0.99854
J38	spp	0.99854
J38	ipcp	0.99854
J38	sms	0.99854
J38	sandbox	0.99854
J38	power7	0.99854
J47	no_pref	0.96866
J47	stride	0.96866
J47	streamer	0.96866
J47	ampm	0.96866
J47	spp	0.96866
J47	ipcp	0.96866
J47	sms	0.96866
J47	sandbox	0.96866
J47	power7	0.96866
J31	no_pref	0.99965
J31	stride	0.99965
J31	streamer	0.99965
J31	ampm	0.99965
J31	spp	0.99965
J31	ipcp	0.99965
J31	sms	0.99965
J31	sandbox	0.99965
J31	power7	0.99965
J34	no_pref	0.99971
J34	stride	0.99971
J34	streamer	0.99971
J34	ampm	0.99971
J34	spp	0.99971
J34	ipcp	0.99971
J34	sms	0.99971
J34	sandbox	0.99971
J34	power7	0.99971
J30	no_pref	0.99965
J30	stride	0.99965
J30	streamer	0.99965
J30	ampm	0.99965
J30	spp	0.99965
J30	ipcp	0.99965
J30	sms	0.99965
J30	sandbox	0.99965
J30	power7	0.99965
J35	no_pref	0.99966
J35	stride	0.99966
J35	streamer	0.99966

Table 64: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 4/4 (cont.)

ID	normal	<i>standalone_selected_accuracy</i>
J35	ampm	0.99966
J35	spp	0.99966
J35	ipcp	0.99966
J35	sms	0.99966
J35	sandbox	0.99966
J35	power7	0.99966
J36	no_pref	0.99969
J36	stride	0.99969
J36	streamer	0.99969
J36	ampm	0.99969
J36	spp	0.99969
J36	ipcp	0.99969
J36	sms	0.99969
J36	sandbox	0.99969
J36	power7	0.99969
J32	no_pref	0.99971
J32	stride	0.99971
J32	streamer	0.99971
J32	ampm	0.99971
J32	spp	0.99971
J32	ipcp	0.99971
J32	sms	0.99971
J32	sandbox	0.99971
J32	power7	0.99971
J33	no_pref	0.99971
J33	stride	0.99971
J33	streamer	0.99971
J33	ampm	0.99971
J33	spp	0.99971
J33	ipcp	0.99971
J33	sms	0.99971
J33	sandbox	0.99971
J33	power7	0.99971
J46	no_pref	0.99867
J46	stride	0.99867
J46	streamer	0.99867
J46	ampm	0.99867
J46	spp	0.99867
J46	ipcp	0.99867
J46	sms	0.99867
J46	sandbox	0.99867
J46	power7	0.99867
J41	no_pref	0.99871
J41	stride	0.99871
J41	streamer	0.99871
J41	ampm	0.99871
J41	spp	0.99871
J41	ipcp	0.99871
J41	sms	0.99871
J41	sandbox	0.99871
J41	power7	0.99871
J42	no_pref	0.99855
J42	stride	0.99855
J42	streamer	0.99855
J42	ampm	0.99855
J42	spp	0.99855
J42	ipcp	0.99855
J42	sms	0.99855
J42	sandbox	0.99855
J42	power7	0.99855
J37	no_pref	0.99854
J37	stride	0.99854
J37	streamer	0.99854
J37	ampm	0.99854
J37	spp	0.99854
J37	ipcp	0.99854
J37	sms	0.99854
J37	sandbox	0.99854
J37	power7	0.99854
J44	no_pref	0.99853
J44	stride	0.99853
J44	streamer	0.99853
J44	ampm	0.99853
J44	spp	0.99853
J44	ipcp	0.99853
J44	sms	0.99853
J44	sandbox	0.99853
J44	power7	0.99853
J39	no_pref	0.99856
J39	stride	0.99856
J39	streamer	0.99856
J39	ampm	0.99856
J39	spp	0.99856
J39	ipcp	0.99856
J39	sms	0.99856
J39	sandbox	0.99856
J39	power7	0.99856
J45	no_pref	0.99867

Table 64: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job $\times$ normal prefetcher) event-overlap rows — part 4/4 (cont.)

ID	normal	<i>standalone_selected_accuracy</i>
J45	stride	0.99867
J45	streamer	0.99867
J45	ampm	0.99867
J45	spp	0.99867
J45	ipcp	0.99867
J45	sms	0.99867
J45	sandbox	0.99867
J45	power7	0.99867
J40	no_pref	0.99871
J40	stride	0.99871
J40	streamer	0.99871
J40	ampm	0.99871
J40	spp	0.99871
J40	ipcp	0.99871
J40	sms	0.99871
J40	sandbox	0.99871
J40	power7	0.99871
J28	no_pref	0.33811
J28	stride	0.33811
J28	streamer	0.33811
J28	ampm	0.33811
J28	spp	0.33811
J28	ipcp	0.33811
J28	sms	0.33811
J28	sandbox	0.33811
J28	power7	0.33811
J27	no_pref	0.32748
J27	stride	0.32748
J27	streamer	0.32748
J27	ampm	0.32748
J27	spp	0.32748
J27	ipcp	0.32748
J27	sms	0.32748
J27	sandbox	0.32748
J27	power7	0.32748
J29	no_pref	0.146
J29	stride	0.146
J29	streamer	0.146
J29	ampm	0.146
J29	spp	0.146
J29	ipcp	0.146
J29	sms	0.146
J29	sandbox	0.146
J29	power7	0.146
J26	no_pref	0.59272
J26	stride	0.59272
J26	streamer	0.59272
J26	ampm	0.59272
J26	spp	0.59272
J26	ipcp	0.59272
J26	sms	0.59272
J26	sandbox	0.59272
J26	power7	0.59272
J23	no_pref	0.59272
J23	stride	0.59272
J23	streamer	0.59272
J23	ampm	0.59272
J23	spp	0.59272
J23	ipcp	0.59272
J23	sms	0.59272
J23	sandbox	0.59272
J23	power7	0.59272
J25	no_pref	0.62023
J25	stride	0.62023
J25	streamer	0.62023
J25	ampm	0.62023
J25	spp	0.62023
J25	ipcp	0.62023
J25	sms	0.62023
J25	sandbox	0.62023
J25	power7	0.62023
J24	no_pref	0.63385
J24	stride	0.63385
J24	streamer	0.63385
J24	ampm	0.63385
J24	spp	0.63385
J24	ipcp	0.63385
J24	sms	0.63385
J24	sandbox	0.63385
J24	power7	0.63385
J13	no_pref	0.3872
J13	stride	0.3872
J13	streamer	0.3872
J13	ampm	0.3872
J13	spp	0.3872
J13	ipcp	0.3872
J13	sms	0.3872
J13	sandbox	0.3872

Table 64: evidence/normal_vs_standalone_evidence.csv — all 423 (NN job × normal prefetcher) event-overlap rows — part 4/4 (cont.)

ID	normal	standalone_selected_accuracy
J13	power7	0.3872
J12	no_pref	0.31852
J12	stride	0.31852
J12	streamer	0.31852
J12	ampm	0.31852
J12	spp	0.31852
J12	ipcp	0.31852
J12	sms	0.31852
J12	sandbox	0.31852
J12	power7	0.31852
J14	no_pref	0.14131
J14	stride	0.14131
J14	streamer	0.14131
J14	ampm	0.14131
J14	spp	0.14131
J14	ipcp	0.14131
J14	sms	0.14131
J14	sandbox	0.14131
J14	power7	0.14131
J11	no_pref	0.66306
J11	stride	0.66306
J11	streamer	0.66306
J11	ampm	0.66306
J11	spp	0.66306
J11	ipcp	0.66306
J11	sms	0.66306
J11	sandbox	0.66306
J11	power7	0.66306
J09	no_pref	0.73654
J09	stride	0.73654
J09	streamer	0.73654
J09	ampm	0.73654
J09	spp	0.73654
J09	ipcp	0.73654
J09	sms	0.73654
J09	sandbox	0.73654
J09	power7	0.73654
J10	no_pref	0.65465
J10	stride	0.65465
J10	streamer	0.65465
J10	ampm	0.65465
J10	spp	0.65465
J10	ipcp	0.65465
J10	sms	0.65465
J10	sandbox	0.65465
J10	power7	0.65465

Table 65: Column cache_miss_reading of the same file (one row per pair)

ID	normal	reading
J21	no_pref	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J21	stride	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J21	streamer	NN has more L2 misses and lower measured timely-event coverage.
J21	ampm	NN has more L2 misses and lower measured timely-event coverage.
J21	spp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J21	ipcp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J21	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J21	sandbox	NN has more L2 misses and lower measured timely-event coverage.
J21	power7	NN has more L2 misses and lower measured timely-event coverage.
J20	no_pref	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J20	stride	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J20	streamer	NN has more L2 misses and lower measured timely-event coverage.
J20	ampm	NN has more L2 misses and lower measured timely-event coverage.
J20	spp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J20	ipcp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J20	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J20	sandbox	NN has more L2 misses and lower measured timely-event coverage.
J20	power7	NN has more L2 misses and lower measured timely-event coverage.
J22	no_pref	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J22	stride	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J22	streamer	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J22	ampm	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J22	spp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J22	ipcp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J22	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J22	sandbox	NN has more L2 misses and lower measured timely-event coverage.
J22	power7	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J19	no_pref	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J19	stride	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J19	streamer	NN has more L2 misses and lower measured timely-event coverage.
J19	ampm	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J19	spp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J19	ipcp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J19	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J19	sandbox	NN has more L2 misses and lower measured timely-event coverage.

ID	normal	reading
J24	power7	NN has more L2 misses and lower measured timely-event coverage.
J13	no_pref	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J13	stride	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J13	streamer	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J13	ampm	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J13	spp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J13	ipcp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J13	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J13	sandbox	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J13	power7	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J12	no_pref	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J12	stride	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J12	streamer	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J12	ampm	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J12	spp	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J12	ipcp	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J12	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J12	sandbox	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J12	power7	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J14	no_pref	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J14	stride	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J14	streamer	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J14	ampm	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J14	spp	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J14	ipcp	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J14	sms	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J14	sandbox	NN has more L2 misses and lower measured timely-event coverage.
J14	power7	NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
J11	no_pref	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J11	stride	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J11	streamer	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J11	ampm	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J11	spp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J11	ipcp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J11	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J11	sandbox	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J11	power7	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	no_pref	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	stride	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	streamer	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	ampm	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	spp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	ipcp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	sandbox	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J09	power7	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J10	no_pref	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J10	stride	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J10	streamer	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J10	ampm	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J10	spp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J10	ipcp	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J10	sms	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.
J10	sandbox	NN has fewer L2 misses; inspect overlap and resource counters before assigning cause.
J10	power7	NN has fewer L2 misses and no lower measured timely-event coverage/timeliness.

Table 66: evidence/trace_profile.summary.csv — shipped with all metric fields empty in this archive (only

branch_fraction	compressed_trace_bytes	full_trace.counted	load_instruction_fraction	memory_instruction_fraction	records_read	store_instruction_fraction	take
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-

10 Event analysis (replay results/analysis/)

Table 67: analysis/run_unique_event_outcomes.csv — unique demand-miss event outcomes for all 92 runs (normals by name, NN jobs by ID)

trace	family	variant	event_rows_observed	unmatched_event_keys	no_pref_miss_events	timely_events	late_events	residual_events	unique_event_coverage	timeliness-over-covered
602.gcc.s-734B	normal	no_pref	404,219	0	203,131	0	0	203,131	0	0
602.gcc.s-734B	normal	stride	404,219	3,370	203,131	34,854	164	168,110	0.17158	0.99532
602.gcc.s-734B	normal	streamer	404,219	40,527	203,131	187,580	47	15,414	0.92344	0.99975
602.gcc.s-734B	normal	ampm	404,219	40,569	203,131	186,005	600	16,396	0.91569	0.99678
602.gcc.s-734B	normal	spp	404,219	30,824	203,131	132,205	384	70,471	0.65084	0.9971
602.gcc.s-734B	normal	ipcp	404,219	6	203,131	0	1	203,130	0	0
602.gcc.s-734B	normal	sms	404,219	35,180	203,131	161,912	173	40,963	0.79708	0.99893
602.gcc.s-734B	normal	sandbox	404,219	42,180	203,131	195,894	43	7,064	0.96437	0.99978

Table 67: analysis/run_unique_event_outcomes.csv — unique demand-miss event outcomes for all 92 runs (normals by name, NN jobs by ID) (cont.)

trace	family	variant	event-rows-observed	unmatched-event-keys	no-pref-miss-events	timely-events	late-events	residual-events	unique-event-coverage	timeliness-over-covered
602.gcc.s-734B	normal	power7	404,219	40,166	203,131	185,747	125	17,169	0.91442	0.99933
602.gcc.s-734B	standalone	J21	404,219	40,351	203,131	184,979	1	18,150	0.91064	0.99999
602.gcc.s-734B	standalone	J20	404,219	40,432	203,131	185,459	1	17,671	0.913	0.99999
602.gcc.s-734B	standalone	J22	404,219	41,343	203,131	189,962	2	13,083	0.93517	0.99999
602.gcc.s-734B	standalone	J19	404,219	40,775	203,131	187,013	0	16,118	0.92065	1
602.gcc.s-734B	standalone	J15	404,219	40,775	203,131	187,013	0	16,118	0.92065	1
602.gcc.s-734B	standalone	J17	404,219	40,351	203,131	184,979	1	18,150	0.91064	0.99999
602.gcc.s-734B	standalone	J16	404,219	40,432	203,131	185,459	1	17,671	0.913	0.99999
602.gcc.s-734B	standalone	J18	404,219	6,241	203,131	40,312	1	162,773	0.19845	0.99998
605.mcf.s-994B	normal	no-pref	985,276	0	740,223	0	0	740,223	0	0
605.mcf.s-994B	normal	stride	985,274	646	740,223	4,858	0	734,820	0.00656	1
605.mcf.s-994B	normal	streamer	985,268	15,434	740,223	85,808	15,919	626,627	0.11592	0.84351
605.mcf.s-994B	normal	amprn	985,273	15,289	740,223	57,672	1,232	668,468	0.07791	0.97908
605.mcf.s-994B	normal	spp	985,273	12,741	740,223	5,156	1,089	722,964	0.00697	0.82562
605.mcf.s-994B	normal	ipcp	985,276	0	740,223	0	0	740,223	0	0
605.mcf.s-994B	normal	sms	985,312	15,608	740,223	91,604	54,251	582,137	0.12375	0.62805
605.mcf.s-994B	normal	sandbox	1,130,709	161,820	740,223	157,849	109,635	460,390	0.21325	0.59013
605.mcf.s-994B	normal	power7	985,411	12,533	740,223	71,010	15,522	644,718	0.09593	0.82062
605.mcf.s-994B	standalone	J07	985,275	5,947	740,223	162,393	24,401	548,097	0.21938	0.86937
605.mcf.s-994B	standalone	J06	985,272	6,215	740,223	163,549	24,016	547,031	0.22095	0.87196
605.mcf.s-994B	standalone	J08	985,275	4,739	740,223	159,346	25,165	551,727	0.21527	0.86361
605.mcf.s-994B	standalone	J05	985,277	4,942	740,223	147,332	21,477	566,639	0.19904	0.87277
605.mcf.s-994B	standalone	J01	985,276	3,363	740,223	124,736	1,609	610,653	0.16851	0.98727
605.mcf.s-994B	standalone	J03	985,275	4,396	740,223	150,314	2,272	583,788	0.20307	0.98511
605.mcf.s-994B	standalone	J02	985,274	4,998	740,223	149,835	13,994	571,954	0.20242	0.91458
605.mcf.s-994B	standalone	J04	985,275	3,328	740,223	119,231	2,575	615,671	0.16107	0.97886
619.lbm.s-4268B	normal	no-pref	295,037	0	294,639	0	0	294,639	0	0
619.lbm.s-4268B	normal	stride	295,026	385	294,639	67,229	15,664	211,245	0.22817	0.81103
619.lbm.s-4268B	normal	streamer	295,032	345	294,639	142,164	112,238	39,792	0.4825	0.55882
619.lbm.s-4268B	normal	amprn	295,027	672	294,639	235,435	24,838	33,578	0.79906	0.90457
619.lbm.s-4268B	normal	spp	295,024	480	294,639	68,702	63,167	162,156	0.23317	0.52099
619.lbm.s-4268B	normal	ipcp	295,039	86	294,639	0	0	294,491	0	0
619.lbm.s-4268B	normal	sms	295,545	1,260	294,639	255,270	25,203	13,298	0.86638	0.91014
619.lbm.s-4268B	normal	sandbox	295,024	448	294,639	209,025	76,845	8,199	0.70943	0.73119
619.lbm.s-4268B	normal	power7	295,027	393	294,639	160,327	76,585	57,217	0.54415	0.67674
619.lbm.s-4268B	standalone	J43	295,040	579	294,639	228,038	7,540	58,387	0.77396	0.96799
619.lbm.s-4268B	standalone	J38	295,047	604	294,639	236,144	8,770	49,023	0.80147	0.96419
619.lbm.s-4268B	standalone	J47	295,033	912	294,639	290,231	416	2,972	0.98504	0.99857
619.lbm.s-4268B	standalone	J31	295,056	634	294,639	233,746	10,890	49,264	0.79333	0.95548
619.lbm.s-4268B	standalone	J34	300,760	6,462	294,639	248,889	2,439	42,442	0.84473	0.9903
619.lbm.s-4268B	standalone	J30	295,056	634	294,639	233,746	10,890	49,264	0.79333	0.95548
619.lbm.s-4268B	standalone	J35	295,055	581	294,639	229,904	10,801	53,252	0.78029	0.95513
619.lbm.s-4268B	standalone	J36	295,066	626	294,639	225,923	10,703	57,285	0.76678	0.95477
619.lbm.s-4268B	standalone	J32	295,594	1,272	294,639	245,839	4,505	43,446	0.83437	0.982
619.lbm.s-4268B	standalone	J33	300,760	6,462	294,639	248,889	2,439	42,442	0.84473	0.9903
619.lbm.s-4268B	standalone	J46	295,164	864	294,639	251,524	1,645	40,622	0.85367	0.9935
619.lbm.s-4268B	standalone	J41	295,187	928	294,639	259,585	1,538	32,602	0.88103	0.99411
619.lbm.s-4268B	standalone	J42	295,040	579	294,639	228,038	7,540	58,387	0.77396	0.96799
619.lbm.s-4268B	standalone	J37	295,047	604	294,639	236,144	8,770	49,023	0.80147	0.96419
619.lbm.s-4268B	standalone	J44	295,052	737	294,639	246,715	3,331	43,758	0.83735	0.98668
619.lbm.s-4268B	standalone	J39	295,075	765	294,639	255,462	3,578	34,731	0.86703	0.98619
619.lbm.s-4268B	standalone	J45	295,164	864	294,639	251,524	1,645	40,622	0.85367	0.9935
619.lbm.s-4268B	standalone	J40	295,187	928	294,639	259,585	1,538	32,602	0.88103	0.99411
620.omnetpp.s-874B	normal	no-pref	451,398	0	316,554	0	0	316,554	0	0
620.omnetpp.s-874B	normal	stride	451,396	37	316,554	2,990	0	313,502	0.00945	1
620.omnetpp.s-874B	normal	streamer	451,360	1,470	316,554	51,399	1,932	261,401	0.16237	0.96377
620.omnetpp.s-874B	normal	amprn	451,359	767	316,554	21,888	878	292,579	0.06914	0.96143
620.omnetpp.s-874B	normal	spp	451,389	762	316,554	49	13	315,658	1.548e-04	0.79032
620.omnetpp.s-874B	normal	ipcp	451,406	87	316,554	2,139	292	313,839	0.00676	0.87988
620.omnetpp.s-874B	normal	sms	451,284	3,317	316,554	86,007	1,731	225,755	0.2717	0.98027
620.omnetpp.s-874B	normal	sandbox	466,225	19,396	316,554	134,047	8,004	171,771	0.42346	0.94365
620.omnetpp.s-874B	normal	power7	451,341	1,294	316,554	51,322	1,997	261,512	0.16213	0.96255
620.omnetpp.s-874B	standalone	J28	451,413	221	316,554	40,630	85	275,413	0.12835	0.99791
620.omnetpp.s-874B	standalone	J27	451,441	280	316,554	48,487	120	267,360	0.15317	0.99753
620.omnetpp.s-874B	standalone	J29	451,386	233	316,554	30,449	161	285,460	0.09619	0.99474
620.omnetpp.s-874B	standalone	J26	451,464	243	316,554	45,782	53	270,343	0.14463	0.99884

Table 67: analysis/run_unique_event_outcomes.csv — unique demand-miss event outcomes for all 92 runs (normals by name, NN jobs by ID) (cont.)

trace	family	variant	event-rows-observed	unmatched-event-keys	no-pref-miss-events	timely-events	late-events	residual-events	unique-event-coverage	timeliness-over-covered
620.omnetpp.s-874B	standalone	J23	451,464	243	316,554	45,782	53	270,343	0.14463	0.99884
620.omnetpp.s-874B	standalone	J25	451,433	99	316,554	20,937	20	295,524	0.06614	0.99905
620.omnetpp.s-874B	standalone	J24	451,410	143	316,554	20,939	18	295,440	0.06615	0.99914
623.xalancbm.k.s-700B	normal	no-pref	1,218,494	0	450,350	0	0	450,350	0	0
623.xalancbm.k.s-700B	normal	stride	1,218,495	21	450,350	2,115	7,469	440,312	0.0047	0.22068
623.xalancbm.k.s-700B	normal	streamer	1,218,352	2,424	450,350	25,610	3,711	415,300	0.05687	0.87344
623.xalancbm.k.s-700B	normal	ampm	1,218,334	872	450,350	18,842	2,673	423,874	0.04184	0.87576
623.xalancbm.k.s-700B	normal	spp	1,218,483	222	450,350	1,132	1,837	434,440	0.00251	0.38127
623.xalancbm.k.s-700B	normal	ipcp	1,218,487	349	450,350	2,817	79	446,603	0.00626	0.97272
623.xalancbm.k.s-700B	normal	sms	1,218,827	4,800	450,350	68,566	4,908	368,045	0.15225	0.9332
623.xalancbm.k.s-700B	normal	sandbox	1,218,513	2,343	450,350	164,738	9,037	274,702	0.3658	0.948
623.xalancbm.k.s-700B	normal	power7	1,218,439	1,477	450,350	17,247	4,522	422,903	0.0383	0.79227
623.xalancbm.k.s-700B	standalone	J13	1,219,257	3,036	450,350	227,823	1,028	218,108	0.50588	0.99551
623.xalancbm.k.s-700B	standalone	J12	1,218,225	2,096	450,350	169,978	577	275,422	0.37744	0.99662
623.xalancbm.k.s-700B	standalone	J14	1,245,962	30,544	450,350	135,984	613	310,237	0.30195	0.99551
623.xalancbm.k.s-700B	standalone	J11	1,219,281	2,543	450,350	401,150	749	42,845	0.89075	0.99814
623.xalancbm.k.s-700B	standalone	J09	1,219,218	2,422	450,350	388,893	652	55,607	0.86354	0.99833
623.xalancbm.k.s-700B	standalone	J10	1,218,369	567	450,350	80,808	263	367,905	0.17943	0.99676

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J21	ampm	no_earlier_selected_standalone_export	9,294
J21	ampm	standalone_selected_earlier_but_not_timely	51
J21	power7	no_earlier_selected_standalone_export	10,308
J21	power7	standalone_selected_earlier_but_not_timely	51
J21	sandbox	no_earlier_selected_standalone_export	13,408
J21	sandbox	standalone_selected_but_late	1
J21	sandbox	standalone_selected_earlier_but_not_timely	54
J21	sms	no_earlier_selected_standalone_export	8,918
J21	sms	standalone_selected_but_late	1
J21	sms	standalone_selected_earlier_but_not_timely	41
J21	spp	no_earlier_selected_standalone_export	6,034
J21	spp	standalone_selected_but_late	1
J21	spp	standalone_selected_earlier_but_not_timely	29
J21	streamer	no_earlier_selected_standalone_export	10,586
J21	streamer	standalone_selected_earlier_but_not_timely	51
J21	stride	no_earlier_selected_standalone_export	925
J20	ampm	no_earlier_selected_standalone_export	8,684
J20	ampm	standalone_selected_earlier_but_not_timely	30
J20	power7	no_earlier_selected_standalone_export	9,684
J20	power7	standalone_selected_earlier_but_not_timely	31
J20	sandbox	no_earlier_selected_standalone_export	12,937
J20	sandbox	standalone_selected_but_late	1
J20	sandbox	standalone_selected_earlier_but_not_timely	34
J20	sms	no_earlier_selected_standalone_export	8,561
J20	sms	standalone_selected_but_late	1
J20	sms	standalone_selected_earlier_but_not_timely	25
J20	spp	no_earlier_selected_standalone_export	5,579
J20	spp	standalone_selected_but_late	1
J20	spp	standalone_selected_earlier_but_not_timely	15
J20	streamer	no_earlier_selected_standalone_export	9,950
J20	streamer	standalone_selected_earlier_but_not_timely	31
J20	stride	no_earlier_selected_standalone_export	869
J22	ampm	no_earlier_selected_standalone_export	4,942
J22	ampm	standalone_selected_earlier_but_not_timely	22
J22	power7	no_earlier_selected_standalone_export	5,985
J22	power7	standalone_selected_but_late	1
J22	power7	standalone_selected_earlier_but_not_timely	21
J22	sandbox	no_earlier_selected_standalone_export	8,579
J22	sandbox	standalone_selected_but_late	2
J22	sandbox	standalone_selected_earlier_but_not_timely	29
J22	sms	no_earlier_selected_standalone_export	5,173

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J22	sms	standalone_selected_but_late	1
J22	sms	standalone_selected_earlier_but_not_timely	13
J22	spp	no_earlier_selected_standalone_export	2,957
J22	spp	standalone_selected_but_late	1
J22	spp	standalone_selected_earlier_but_not_timely	3
J22	streamer	no_earlier_selected_standalone_export	6,212
J22	streamer	standalone_selected_but_late	1
J22	streamer	standalone_selected_earlier_but_not_timely	21
J22	stride	no_earlier_selected_standalone_export	539
J19	ampm	no_earlier_selected_standalone_export	7,160
J19	ampm	standalone_selected_earlier_but_not_timely	19
J19	power7	no_earlier_selected_standalone_export	8,162
J19	power7	standalone_selected_earlier_but_not_timely	22
J19	sandbox	no_earlier_selected_standalone_export	11,402
J19	sandbox	standalone_selected_earlier_but_not_timely	26
J19	sms	no_earlier_selected_standalone_export	7,192
J19	sms	standalone_selected_earlier_but_not_timely	13
J19	spp	no_earlier_selected_standalone_export	4,491
J19	spp	standalone_selected_earlier_but_not_timely	3
J19	streamer	no_earlier_selected_standalone_export	8,412
J19	streamer	standalone_selected_earlier_but_not_timely	22
J19	stride	no_earlier_selected_standalone_export	641
J17	ampm	no_earlier_selected_standalone_export	9,294
J17	ampm	standalone_selected_earlier_but_not_timely	51
J17	power7	no_earlier_selected_standalone_export	10,308
J17	power7	standalone_selected_earlier_but_not_timely	51
J17	sandbox	no_earlier_selected_standalone_export	13,408
J17	sandbox	standalone_selected_but_late	1
J17	sandbox	standalone_selected_earlier_but_not_timely	54
J17	sms	no_earlier_selected_standalone_export	8,918
J17	sms	standalone_selected_but_late	1
J17	sms	standalone_selected_earlier_but_not_timely	41
J17	spp	no_earlier_selected_standalone_export	6,034
J17	spp	standalone_selected_but_late	1
J17	spp	standalone_selected_earlier_but_not_timely	29
J17	streamer	no_earlier_selected_standalone_export	10,586
J17	streamer	standalone_selected_earlier_but_not_timely	51
J17	stride	no_earlier_selected_standalone_export	925
J16	ampm	no_earlier_selected_standalone_export	8,684
J16	ampm	standalone_selected_earlier_but_not_timely	30
J16	power7	no_earlier_selected_standalone_export	9,684
J16	power7	standalone_selected_earlier_but_not_timely	31
J16	sandbox	no_earlier_selected_standalone_export	12,937
J16	sandbox	standalone_selected_but_late	1
J16	sandbox	standalone_selected_earlier_but_not_timely	34
J16	sms	no_earlier_selected_standalone_export	8,561
J16	sms	standalone_selected_but_late	1
J16	sms	standalone_selected_earlier_but_not_timely	25
J16	spp	no_earlier_selected_standalone_export	5,579
J16	spp	standalone_selected_but_late	1
J16	spp	standalone_selected_earlier_but_not_timely	15
J16	streamer	no_earlier_selected_standalone_export	9,950
J16	streamer	standalone_selected_earlier_but_not_timely	31
J16	stride	no_earlier_selected_standalone_export	869
J18	ampm	no_earlier_selected_standalone_export	147,013
J18	ampm	standalone_selected_earlier_but_not_timely	469
J18	power7	no_earlier_selected_standalone_export	147,021
J18	power7	standalone_selected_earlier_but_not_timely	469
J18	sandbox	no_earlier_selected_standalone_export	155,587
J18	sandbox	standalone_selected_but_late	1
J18	sandbox	standalone_selected_earlier_but_not_timely	484
J18	sms	no_earlier_selected_standalone_export	128,275
J18	sms	standalone_selected_but_late	1
J18	sms	standalone_selected_earlier_but_not_timely	450
J18	spp	no_earlier_selected_standalone_export	105,147
J18	spp	standalone_selected_but_late	1
J18	spp	standalone_selected_earlier_but_not_timely	336
J18	streamer	no_earlier_selected_standalone_export	148,544
J18	streamer	standalone_selected_earlier_but_not_timely	469
J18	stride	no_earlier_selected_standalone_export	31,129
J18	stride	standalone_selected_earlier_but_not_timely	7
J15	ampm	no_earlier_selected_standalone_export	7,160
J15	ampm	standalone_selected_earlier_but_not_timely	19
J15	power7	no_earlier_selected_standalone_export	8,162
J15	power7	standalone_selected_earlier_but_not_timely	22
J15	sandbox	no_earlier_selected_standalone_export	11,402

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J15	sandbox	standalone_selected_earlier_but_not_timely	26
J15	sms	no_earlier_selected_standalone_export	7,192
J15	sms	standalone_selected_earlier_but_not_timely	13
J15	spp	no_earlier_selected_standalone_export	4,491
J15	spp	standalone_selected_earlier_but_not_timely	3
J15	streamer	no_earlier_selected_standalone_export	8,412
J15	streamer	standalone_selected_earlier_but_not_timely	22
J15	stride	no_earlier_selected_standalone_export	641
J07	ampm	no_earlier_selected_standalone_export	24,091
J07	ampm	standalone_selected_but_late	340
J07	ampm	standalone_selected_earlier_but_not_timely	3,587
J07	power7	no_earlier_selected_standalone_export	36,657
J07	power7	standalone_selected_but_late	1,213
J07	power7	standalone_selected_earlier_but_not_timely	4,305
J07	sandbox	no_earlier_selected_standalone_export	97,935
J07	sandbox	standalone_selected_but_late	4,161
J07	sandbox	standalone_selected_earlier_but_not_timely	6,344
J07	sms	no_earlier_selected_standalone_export	46,465
J07	sms	standalone_selected_but_late	790
J07	sms	standalone_selected_earlier_but_not_timely	4,550
J07	spp	no_earlier_selected_standalone_export	1,079
J07	spp	standalone_selected_but_late	15
J07	spp	standalone_selected_earlier_but_not_timely	541
J07	streamer	no_earlier_selected_standalone_export	45,295
J07	streamer	standalone_selected_but_late	1,332
J07	streamer	standalone_selected_earlier_but_not_timely	4,337
J07	stride	no_earlier_selected_standalone_export	1,183
J07	stride	standalone_selected_but_late	17
J07	stride	standalone_selected_earlier_but_not_timely	343
J06	ampm	no_earlier_selected_standalone_export	24,846
J06	ampm	standalone_selected_but_late	338
J06	ampm	standalone_selected_earlier_but_not_timely	3,480
J06	power7	no_earlier_selected_standalone_export	36,972
J06	power7	standalone_selected_but_late	1,241
J06	power7	standalone_selected_earlier_but_not_timely	4,128
J06	sandbox	no_earlier_selected_standalone_export	98,913
J06	sandbox	standalone_selected_but_late	4,191
J06	sandbox	standalone_selected_earlier_but_not_timely	6,088
J06	sms	no_earlier_selected_standalone_export	46,976
J06	sms	standalone_selected_but_late	842
J06	sms	standalone_selected_earlier_but_not_timely	4,345
J06	spp	no_earlier_selected_standalone_export	1,191
J06	spp	standalone_selected_but_late	13
J06	spp	standalone_selected_earlier_but_not_timely	506
J06	streamer	no_earlier_selected_standalone_export	45,823
J06	streamer	standalone_selected_but_late	1,361
J06	streamer	standalone_selected_earlier_but_not_timely	4,220
J06	stride	no_earlier_selected_standalone_export	1,223
J06	stride	standalone_selected_but_late	15
J06	stride	standalone_selected_earlier_but_not_timely	370
J08	ampm	no_earlier_selected_standalone_export	30,759
J08	ampm	standalone_selected_but_late	350
J08	ampm	standalone_selected_earlier_but_not_timely	3,861
J08	power7	no_earlier_selected_standalone_export	41,226
J08	power7	standalone_selected_but_late	1,273
J08	power7	standalone_selected_earlier_but_not_timely	4,514
J08	sandbox	no_earlier_selected_standalone_export	104,141
J08	sandbox	standalone_selected_but_late	4,285
J08	sandbox	standalone_selected_earlier_but_not_timely	6,491
J08	sms	no_earlier_selected_standalone_export	53,229
J08	sms	standalone_selected_but_late	926
J08	sms	standalone_selected_earlier_but_not_timely	4,693
J08	spp	no_earlier_selected_standalone_export	1,755
J08	spp	standalone_selected_but_late	13
J08	spp	standalone_selected_earlier_but_not_timely	612
J08	streamer	no_earlier_selected_standalone_export	51,570
J08	streamer	standalone_selected_but_late	1,404
J08	streamer	standalone_selected_earlier_but_not_timely	4,667
J08	stride	no_earlier_selected_standalone_export	1,807
J08	stride	standalone_selected_but_late	13
J08	stride	standalone_selected_earlier_but_not_timely	462
J05	ampm	no_earlier_selected_standalone_export	30,085
J05	ampm	standalone_selected_but_late	319
J05	ampm	standalone_selected_earlier_but_not_timely	3,365
J05	power7	no_earlier_selected_standalone_export	40,995
J05	power7	standalone_selected_but_late	1,085

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J05	power7	standalone_selected_earlier_but_not_timely	3,895
J05	sandbox	no_earlier_selected_standalone_export	105,986
J05	sandbox	standalone_selected_but_late	3,676
J05	sandbox	standalone_selected_earlier_but_not_timely	5,538
J05	sms	no_earlier_selected_standalone_export	53,825
J05	sms	standalone_selected_but_late	727
J05	sms	standalone_selected_earlier_but_not_timely	4,003
J05	spp	no_earlier_selected_standalone_export	1,698
J05	spp	standalone_selected_but_late	13
J05	spp	standalone_selected_earlier_but_not_timely	534
J05	streamer	no_earlier_selected_standalone_export	51,532
J05	streamer	standalone_selected_but_late	1,204
J05	streamer	standalone_selected_earlier_but_not_timely	3,978
J05	stride	no_earlier_selected_standalone_export	1,716
J05	stride	standalone_selected_but_late	11
J05	stride	standalone_selected_earlier_but_not_timely	358
J03	ampm	no_earlier_selected_standalone_export	27,863
J03	ampm	standalone_selected_but_late	91
J03	ampm	standalone_selected_earlier_but_not_timely	3,954
J03	power7	no_earlier_selected_standalone_export	40,830
J03	power7	standalone_selected_but_late	194
J03	power7	standalone_selected_earlier_but_not_timely	4,485
J03	sandbox	no_earlier_selected_standalone_export	108,464
J03	sandbox	standalone_selected_but_late	406
J03	sandbox	standalone_selected_earlier_but_not_timely	6,235
J03	sms	no_earlier_selected_standalone_export	52,434
J03	sms	standalone_selected_but_late	259
J03	sms	standalone_selected_earlier_but_not_timely	4,765
J03	spp	no_earlier_selected_standalone_export	1,395
J03	spp	standalone_selected_but_late	13
J03	spp	standalone_selected_earlier_but_not_timely	631
J03	streamer	no_earlier_selected_standalone_export	50,974
J03	streamer	standalone_selected_but_late	199
J03	streamer	standalone_selected_earlier_but_not_timely	4,605
J03	stride	no_earlier_selected_standalone_export	1,428
J03	stride	standalone_selected_but_late	14
J03	stride	standalone_selected_earlier_but_not_timely	382
J02	ampm	no_earlier_selected_standalone_export	28,111
J02	ampm	standalone_selected_but_late	212
J02	ampm	standalone_selected_earlier_but_not_timely	5,235
J02	power7	no_earlier_selected_standalone_export	40,000
J02	power7	standalone_selected_but_late	731
J02	power7	standalone_selected_earlier_but_not_timely	5,848
J02	sandbox	no_earlier_selected_standalone_export	105,970
J02	sandbox	standalone_selected_but_late	2,387
J02	sandbox	standalone_selected_earlier_but_not_timely	7,575
J02	sms	no_earlier_selected_standalone_export	51,319
J02	sms	standalone_selected_but_late	582
J02	sms	standalone_selected_earlier_but_not_timely	6,108
J02	spp	no_earlier_selected_standalone_export	1,490
J02	spp	standalone_selected_but_late	13
J02	spp	standalone_selected_earlier_but_not_timely	964
J02	streamer	no_earlier_selected_standalone_export	50,147
J02	streamer	standalone_selected_but_late	789
J02	streamer	standalone_selected_earlier_but_not_timely	5,924
J02	stride	no_earlier_selected_standalone_export	1,482
J02	stride	standalone_selected_but_late	13
J02	stride	standalone_selected_earlier_but_not_timely	735
J04	ampm	no_earlier_selected_standalone_export	47,112
J04	ampm	standalone_selected_but_late	91
J04	ampm	standalone_selected_earlier_but_not_timely	3,286
J04	power7	no_earlier_selected_standalone_export	56,515
J04	power7	standalone_selected_but_late	189
J04	power7	standalone_selected_earlier_but_not_timely	3,925
J04	sandbox	no_earlier_selected_standalone_export	130,241
J04	sandbox	standalone_selected_but_late	413
J04	sandbox	standalone_selected_earlier_but_not_timely	5,847
J04	sms	no_earlier_selected_standalone_export	72,651
J04	sms	standalone_selected_but_late	323
J04	sms	standalone_selected_earlier_but_not_timely	4,223
J04	spp	no_earlier_selected_standalone_export	4,051
J04	spp	standalone_selected_but_late	16
J04	spp	standalone_selected_earlier_but_not_timely	432
J04	streamer	no_earlier_selected_standalone_export	70,779
J04	streamer	standalone_selected_but_late	188
J04	streamer	standalone_selected_earlier_but_not_timely	4,083

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J04	stride	no_earlier_selected_standalone_export	3,811
J04	stride	standalone_selected_but_late	11
J04	stride	standalone_selected_earlier_but_not_timely	339
J01	ampm	no_earlier_selected_standalone_export	45,366
J01	ampm	standalone_selected_but_late	88
J01	ampm	standalone_selected_earlier_but_not_timely	2,514
J01	power7	no_earlier_selected_standalone_export	54,545
J01	power7	standalone_selected_but_late	152
J01	power7	standalone_selected_earlier_but_not_timely	2,887
J01	sandbox	no_earlier_selected_standalone_export	127,782
J01	sandbox	standalone_selected_but_late	294
J01	sandbox	standalone_selected_earlier_but_not_timely	4,192
J01	sms	no_earlier_selected_standalone_export	70,932
J01	sms	standalone_selected_but_late	200
J01	sms	standalone_selected_earlier_but_not_timely	3,111
J01	spp	no_earlier_selected_standalone_export	3,927
J01	spp	standalone_selected_but_late	10
J01	spp	standalone_selected_earlier_but_not_timely	343
J01	streamer	no_earlier_selected_standalone_export	68,713
J01	streamer	standalone_selected_but_late	153
J01	streamer	standalone_selected_earlier_but_not_timely	3,028
J01	stride	no_earlier_selected_standalone_export	3,610
J01	stride	standalone_selected_but_late	11
J01	stride	standalone_selected_earlier_but_not_timely	187
J46	ampm	no_earlier_selected_standalone_export	17,543
J46	ampm	standalone_selected_but_late	1,175
J46	ampm	standalone_selected_earlier_but_not_timely	293
J46	power7	no_earlier_selected_standalone_export	14,424
J46	power7	standalone_selected_but_late	870
J46	power7	standalone_selected_earlier_but_not_timely	250
J46	sandbox	no_earlier_selected_standalone_export	23,994
J46	sandbox	standalone_selected_but_late	994
J46	sandbox	standalone_selected_earlier_but_not_timely	256
J46	sms	no_earlier_selected_standalone_export	26,547
J46	sms	standalone_selected_but_late	1,338
J46	sms	standalone_selected_earlier_but_not_timely	283
J46	spp	no_earlier_selected_standalone_export	7,824
J46	spp	standalone_selected_but_late	279
J46	spp	standalone_selected_earlier_but_not_timely	81
J46	streamer	no_earlier_selected_standalone_export	13,246
J46	streamer	standalone_selected_but_late	778
J46	streamer	standalone_selected_earlier_but_not_timely	202
J46	stride	no_earlier_selected_standalone_export	4,415
J46	stride	standalone_selected_but_late	298
J46	stride	standalone_selected_earlier_but_not_timely	207
J41	ampm	no_earlier_selected_standalone_export	14,675
J41	ampm	standalone_selected_but_late	860
J41	ampm	standalone_selected_earlier_but_not_timely	317
J41	power7	no_earlier_selected_standalone_export	11,670
J41	power7	standalone_selected_but_late	643
J41	power7	standalone_selected_earlier_but_not_timely	244
J41	sandbox	no_earlier_selected_standalone_export	19,470
J41	sandbox	standalone_selected_but_late	768
J41	sandbox	standalone_selected_earlier_but_not_timely	262
J41	sms	no_earlier_selected_standalone_export	21,779
J41	sms	standalone_selected_but_late	1,098
J41	sms	standalone_selected_earlier_but_not_timely	303
J41	spp	no_earlier_selected_standalone_export	6,025
J41	spp	standalone_selected_but_late	221
J41	spp	standalone_selected_earlier_but_not_timely	81
J41	streamer	no_earlier_selected_standalone_export	10,443
J41	streamer	standalone_selected_but_late	525
J41	streamer	standalone_selected_earlier_but_not_timely	209
J41	stride	no_earlier_selected_standalone_export	4,357
J41	stride	standalone_selected_but_late	270
J41	stride	standalone_selected_earlier_but_not_timely	187
J34	ampm	no_earlier_selected_standalone_export	14,196
J34	ampm	standalone_selected_but_late	1,196
J34	ampm	standalone_selected_earlier_but_not_timely	284
J34	power7	no_earlier_selected_standalone_export	11,724
J34	power7	standalone_selected_but_late	913
J34	power7	standalone_selected_earlier_but_not_timely	238
J34	sandbox	no_earlier_selected_standalone_export	26,175
J34	sandbox	standalone_selected_but_late	1,202
J34	sandbox	standalone_selected_earlier_but_not_timely	230
J34	sms	no_earlier_selected_standalone_export	27,739

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J34	sms	standalone_selected_but_late	1,621
J34	sms	standalone_selected_earlier_but_not_timely	285
J34	spp	no_earlier_selected_standalone_export	8,263
J34	spp	standalone_selected_but_late	358
J34	spp	standalone_selected_earlier_but_not_timely	76
J34	streamer	no_earlier_selected_standalone_export	10,565
J34	streamer	standalone_selected_but_late	865
J34	streamer	standalone_selected_earlier_but_not_timely	195
J34	stride	no_earlier_selected_standalone_export	4,095
J34	stride	standalone_selected_but_late	195
J34	stride	standalone_selected_earlier_but_not_timely	196
J43	ampm	no_earlier_selected_standalone_export	26,807
J43	ampm	standalone_selected_but_late	6,694
J43	ampm	standalone_selected_earlier_but_not_timely	218
J43	power7	no_earlier_selected_standalone_export	18,877
J43	power7	standalone_selected_but_late	4,008
J43	power7	standalone_selected_earlier_but_not_timely	163
J43	sandbox	no_earlier_selected_standalone_export	35,602
J43	sandbox	standalone_selected_but_late	4,344
J43	sandbox	standalone_selected_earlier_but_not_timely	177
J43	sms	no_earlier_selected_standalone_export	39,968
J43	sms	standalone_selected_but_late	6,186
J43	sms	standalone_selected_earlier_but_not_timely	205
J43	spp	no_earlier_selected_standalone_export	12,022
J43	spp	standalone_selected_but_late	1,409
J43	spp	standalone_selected_earlier_but_not_timely	44
J43	streamer	no_earlier_selected_standalone_export	16,782
J43	streamer	standalone_selected_but_late	2,925
J43	streamer	standalone_selected_earlier_but_not_timely	142
J43	stride	no_earlier_selected_standalone_export	5,371
J43	stride	standalone_selected_but_late	2,158
J43	stride	standalone_selected_earlier_but_not_timely	124
J38	ampm	no_earlier_selected_standalone_export	25,073
J38	ampm	standalone_selected_but_late	7,550
J38	ampm	standalone_selected_earlier_but_not_timely	201
J38	power7	no_earlier_selected_standalone_export	19,520
J38	power7	standalone_selected_but_late	4,888
J38	power7	standalone_selected_earlier_but_not_timely	157
J38	sandbox	no_earlier_selected_standalone_export	30,664
J38	sandbox	standalone_selected_but_late	4,756
J38	sandbox	standalone_selected_earlier_but_not_timely	176
J38	sms	no_earlier_selected_standalone_export	35,418
J38	sms	standalone_selected_but_late	7,275
J38	sms	standalone_selected_earlier_but_not_timely	212
J38	spp	no_earlier_selected_standalone_export	10,052
J38	spp	standalone_selected_but_late	1,527
J38	spp	standalone_selected_earlier_but_not_timely	61
J38	streamer	no_earlier_selected_standalone_export	18,354
J38	streamer	standalone_selected_but_late	3,998
J38	streamer	standalone_selected_earlier_but_not_timely	133
J38	stride	no_earlier_selected_standalone_export	6,890
J38	stride	standalone_selected_but_late	2,867
J38	stride	standalone_selected_earlier_but_not_timely	130
J47	ampm	no_earlier_selected_standalone_export	1,561
J47	ampm	standalone_selected_but_late	332
J47	ampm	standalone_selected_earlier_but_not_timely	548
J47	power7	no_earlier_selected_standalone_export	1,195
J47	power7	standalone_selected_but_late	225
J47	power7	standalone_selected_earlier_but_not_timely	420
J47	sandbox	no_earlier_selected_standalone_export	1,855
J47	sandbox	standalone_selected_but_late	239
J47	sandbox	standalone_selected_earlier_but_not_timely	488
J47	sms	no_earlier_selected_standalone_export	2,327
J47	sms	standalone_selected_but_late	340
J47	sms	standalone_selected_earlier_but_not_timely	576
J47	spp	no_earlier_selected_standalone_export	815
J47	spp	standalone_selected_but_late	81
J47	spp	standalone_selected_earlier_but_not_timely	133
J47	streamer	no_earlier_selected_standalone_export	1,324
J47	streamer	standalone_selected_but_late	184
J47	streamer	standalone_selected_earlier_but_not_timely	320
J47	stride	no_earlier_selected_standalone_export	120
J47	stride	standalone_selected_but_late	105
J47	stride	standalone_selected_earlier_but_not_timely	326
J31	ampm	no_earlier_selected_standalone_export	19,732
J31	ampm	standalone_selected_but_late	9,056

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J31	ampm	standalone_selected_earlier_but_not_timely	229
J31	power7	no_earlier_selected_standalone_export	15,499
J31	power7	standalone_selected_but_late	5,536
J31	power7	standalone_selected_earlier_but_not_timely	177
J31	sandbox	no_earlier_selected_standalone_export	31,103
J31	sandbox	standalone_selected_but_late	5,937
J31	sandbox	standalone_selected_earlier_but_not_timely	184
J31	sms	no_earlier_selected_standalone_export	33,707
J31	sms	standalone_selected_but_late	9,090
J31	sms	standalone_selected_earlier_but_not_timely	231
J31	spp	no_earlier_selected_standalone_export	9,712
J31	spp	standalone_selected_but_late	1,920
J31	spp	standalone_selected_earlier_but_not_timely	64
J31	streamer	no_earlier_selected_standalone_export	14,125
J31	streamer	standalone_selected_but_late	4,653
J31	streamer	standalone_selected_earlier_but_not_timely	140
J31	stride	no_earlier_selected_standalone_export	5,050
J31	stride	standalone_selected_but_late	2,612
J31	stride	standalone_selected_earlier_but_not_timely	129
J42	ampm	no_earlier_selected_standalone_export	26,807
J42	ampm	standalone_selected_but_late	6,694
J42	ampm	standalone_selected_earlier_but_not_timely	218
J42	power7	no_earlier_selected_standalone_export	18,877
J42	power7	standalone_selected_but_late	4,008
J42	power7	standalone_selected_earlier_but_not_timely	163
J42	sandbox	no_earlier_selected_standalone_export	35,602
J42	sandbox	standalone_selected_but_late	4,344
J42	sandbox	standalone_selected_earlier_but_not_timely	177
J42	sms	no_earlier_selected_standalone_export	39,968
J42	sms	standalone_selected_but_late	6,186
J42	sms	standalone_selected_earlier_but_not_timely	205
J42	spp	no_earlier_selected_standalone_export	12,022
J42	spp	standalone_selected_but_late	1,409
J42	spp	standalone_selected_earlier_but_not_timely	44
J42	streamer	no_earlier_selected_standalone_export	16,782
J42	streamer	standalone_selected_but_late	2,925
J42	streamer	standalone_selected_earlier_but_not_timely	142
J42	stride	no_earlier_selected_standalone_export	5,371
J42	stride	standalone_selected_but_late	2,158
J42	stride	standalone_selected_earlier_but_not_timely	124
J37	ampm	no_earlier_selected_standalone_export	25,073
J37	ampm	standalone_selected_but_late	7,550
J37	ampm	standalone_selected_earlier_but_not_timely	201
J37	power7	no_earlier_selected_standalone_export	19,520
J37	power7	standalone_selected_but_late	4,888
J37	power7	standalone_selected_earlier_but_not_timely	157
J37	sandbox	no_earlier_selected_standalone_export	30,664
J37	sandbox	standalone_selected_but_late	4,756
J37	sandbox	standalone_selected_earlier_but_not_timely	176
J37	sms	no_earlier_selected_standalone_export	35,418
J37	sms	standalone_selected_but_late	7,275
J37	sms	standalone_selected_earlier_but_not_timely	212
J37	spp	no_earlier_selected_standalone_export	10,052
J37	spp	standalone_selected_but_late	1,527
J37	spp	standalone_selected_earlier_but_not_timely	61
J37	streamer	no_earlier_selected_standalone_export	18,354
J37	streamer	standalone_selected_but_late	3,998
J37	streamer	standalone_selected_earlier_but_not_timely	133
J37	stride	no_earlier_selected_standalone_export	6,890
J37	stride	standalone_selected_but_late	2,867
J37	stride	standalone_selected_earlier_but_not_timely	130
J35	ampm	no_earlier_selected_standalone_export	21,788
J35	ampm	standalone_selected_but_late	9,344
J35	ampm	standalone_selected_earlier_but_not_timely	162
J35	power7	no_earlier_selected_standalone_export	17,013
J35	power7	standalone_selected_but_late	5,961
J35	power7	standalone_selected_earlier_but_not_timely	119
J35	sandbox	no_earlier_selected_standalone_export	32,744
J35	sandbox	standalone_selected_but_late	6,071
J35	sandbox	standalone_selected_earlier_but_not_timely	122
J35	sms	no_earlier_selected_standalone_export	35,936
J35	sms	standalone_selected_but_late	9,076
J35	sms	standalone_selected_earlier_but_not_timely	161
J35	spp	no_earlier_selected_standalone_export	9,823
J35	spp	standalone_selected_but_late	1,854
J35	spp	standalone_selected_earlier_but_not_timely	46

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J35	streamer	no_earlier_selected_standalone_export	14,822
J35	streamer	standalone_selected_but_late	4,622
J35	streamer	standalone_selected_earlier_but_not_timely	94
J35	stride	no_earlier_selected_standalone_export	5,544
J35	stride	standalone_selected_but_late	3,062
J35	stride	standalone_selected_earlier_but_not_timely	94
J36	ampm	no_earlier_selected_standalone_export	25,526
J36	ampm	standalone_selected_but_late	9,351
J36	ampm	standalone_selected_earlier_but_not_timely	216
J36	power7	no_earlier_selected_standalone_export	19,734
J36	power7	standalone_selected_but_late	5,945
J36	power7	standalone_selected_earlier_but_not_timely	175
J36	sandbox	no_earlier_selected_standalone_export	36,556
J36	sandbox	standalone_selected_but_late	6,040
J36	sandbox	standalone_selected_earlier_but_not_timely	179
J36	sms	no_earlier_selected_standalone_export	40,108
J36	sms	standalone_selected_but_late	9,061
J36	sms	standalone_selected_earlier_but_not_timely	229
J36	spp	no_earlier_selected_standalone_export	10,681
J36	spp	standalone_selected_but_late	1,928
J36	spp	standalone_selected_earlier_but_not_timely	54
J36	streamer	no_earlier_selected_standalone_export	17,565
J36	streamer	standalone_selected_but_late	4,796
J36	streamer	standalone_selected_earlier_but_not_timely	137
J36	stride	no_earlier_selected_standalone_export	7,197
J36	stride	standalone_selected_but_late	3,087
J36	stride	standalone_selected_earlier_but_not_timely	145
J30	ampm	no_earlier_selected_standalone_export	19,732
J30	ampm	standalone_selected_but_late	9,056
J30	ampm	standalone_selected_earlier_but_not_timely	229
J30	power7	no_earlier_selected_standalone_export	15,499
J30	power7	standalone_selected_but_late	5,536
J30	power7	standalone_selected_earlier_but_not_timely	177
J30	sandbox	no_earlier_selected_standalone_export	31,103
J30	sandbox	standalone_selected_but_late	5,937
J30	sandbox	standalone_selected_earlier_but_not_timely	184
J30	sms	no_earlier_selected_standalone_export	33,707
J30	sms	standalone_selected_but_late	9,090
J30	sms	standalone_selected_earlier_but_not_timely	231
J30	spp	no_earlier_selected_standalone_export	9,712
J30	spp	standalone_selected_but_late	1,920
J30	spp	standalone_selected_earlier_but_not_timely	64
J30	streamer	no_earlier_selected_standalone_export	14,125
J30	streamer	standalone_selected_but_late	4,653
J30	streamer	standalone_selected_earlier_but_not_timely	140
J30	stride	no_earlier_selected_standalone_export	5,050
J30	stride	standalone_selected_but_late	2,612
J30	stride	standalone_selected_earlier_but_not_timely	129
J44	ampm	no_earlier_selected_standalone_export	19,110
J44	ampm	standalone_selected_but_late	2,687
J44	ampm	standalone_selected_earlier_but_not_timely	267
J44	power7	no_earlier_selected_standalone_export	15,277
J44	power7	standalone_selected_but_late	1,800
J44	power7	standalone_selected_earlier_but_not_timely	212
J44	sandbox	no_earlier_selected_standalone_export	25,965
J44	sandbox	standalone_selected_but_late	1,998
J44	sandbox	standalone_selected_earlier_but_not_timely	222
J44	sms	no_earlier_selected_standalone_export	28,846
J44	sms	standalone_selected_but_late	2,727
J44	sms	standalone_selected_earlier_but_not_timely	269
J44	spp	no_earlier_selected_standalone_export	8,530
J44	spp	standalone_selected_but_late	623
J44	spp	standalone_selected_earlier_but_not_timely	75
J44	streamer	no_earlier_selected_standalone_export	13,906
J44	streamer	standalone_selected_but_late	1,318
J44	streamer	standalone_selected_earlier_but_not_timely	179
J44	stride	no_earlier_selected_standalone_export	4,645
J44	stride	standalone_selected_but_late	826
J44	stride	standalone_selected_earlier_but_not_timely	176
J39	ampm	no_earlier_selected_standalone_export	16,164
J39	ampm	standalone_selected_but_late	2,765
J39	ampm	standalone_selected_earlier_but_not_timely	280
J39	power7	no_earlier_selected_standalone_export	12,707
J39	power7	standalone_selected_but_late	1,922
J39	power7	standalone_selected_earlier_but_not_timely	225
J39	sandbox	no_earlier_selected_standalone_export	21,013

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J39	sandbox	standalone_selected_but_late	2,049
J39	sandbox	standalone_selected_earlier_but_not_timely	237
J39	sms	no_earlier_selected_standalone_export	23,569
J39	sms	standalone_selected_but_late	2,937
J39	sms	standalone_selected_earlier_but_not_timely	273
J39	spp	no_earlier_selected_standalone_export	6,519
J39	spp	standalone_selected_but_late	561
J39	spp	standalone_selected_earlier_but_not_timely	67
J39	streamer	no_earlier_selected_standalone_export	11,336
J39	streamer	standalone_selected_but_late	1,459
J39	streamer	standalone_selected_earlier_but_not_timely	198
J39	stride	no_earlier_selected_standalone_export	4,832
J39	stride	standalone_selected_but_late	1,146
J39	stride	standalone_selected_earlier_but_not_timely	170
J32	ampm	no_earlier_selected_standalone_export	14,997
J32	ampm	standalone_selected_but_late	3,089
J32	ampm	standalone_selected_earlier_but_not_timely	255
J32	power7	no_earlier_selected_standalone_export	12,270
J32	power7	standalone_selected_but_late	1,843
J32	power7	standalone_selected_earlier_but_not_timely	196
J32	sandbox	no_earlier_selected_standalone_export	26,931
J32	sandbox	standalone_selected_but_late	2,399
J32	sandbox	standalone_selected_earlier_but_not_timely	218
J32	sms	no_earlier_selected_standalone_export	28,623
J32	sms	standalone_selected_but_late	3,476
J32	sms	standalone_selected_earlier_but_not_timely	251
J32	spp	no_earlier_selected_standalone_export	8,445
J32	spp	standalone_selected_but_late	757
J32	spp	standalone_selected_earlier_but_not_timely	70
J32	streamer	no_earlier_selected_standalone_export	11,079
J32	streamer	standalone_selected_but_late	1,717
J32	streamer	standalone_selected_earlier_but_not_timely	175
J32	stride	no_earlier_selected_standalone_export	4,291
J32	stride	standalone_selected_but_late	519
J32	stride	standalone_selected_earlier_but_not_timely	177
J45	ampm	no_earlier_selected_standalone_export	17,543
J45	ampm	standalone_selected_but_late	1,175
J45	ampm	standalone_selected_earlier_but_not_timely	293
J45	power7	no_earlier_selected_standalone_export	14,424
J45	power7	standalone_selected_but_late	870
J45	power7	standalone_selected_earlier_but_not_timely	250
J45	sandbox	no_earlier_selected_standalone_export	23,994
J45	sandbox	standalone_selected_but_late	994
J45	sandbox	standalone_selected_earlier_but_not_timely	256
J45	sms	no_earlier_selected_standalone_export	26,547
J45	sms	standalone_selected_but_late	1,338
J45	sms	standalone_selected_earlier_but_not_timely	283
J45	spp	no_earlier_selected_standalone_export	7,824
J45	spp	standalone_selected_but_late	279
J45	spp	standalone_selected_earlier_but_not_timely	81
J45	streamer	no_earlier_selected_standalone_export	13,246
J45	streamer	standalone_selected_but_late	778
J45	streamer	standalone_selected_earlier_but_not_timely	202
J45	stride	no_earlier_selected_standalone_export	4,415
J45	stride	standalone_selected_but_late	298
J45	stride	standalone_selected_earlier_but_not_timely	207
J40	ampm	no_earlier_selected_standalone_export	14,675
J40	ampm	standalone_selected_but_late	860
J40	ampm	standalone_selected_earlier_but_not_timely	317
J40	power7	no_earlier_selected_standalone_export	11,670
J40	power7	standalone_selected_but_late	643
J40	power7	standalone_selected_earlier_but_not_timely	244
J40	sandbox	no_earlier_selected_standalone_export	19,470
J40	sandbox	standalone_selected_but_late	768
J40	sandbox	standalone_selected_earlier_but_not_timely	262
J40	sms	no_earlier_selected_standalone_export	21,779
J40	sms	standalone_selected_but_late	1,098
J40	sms	standalone_selected_earlier_but_not_timely	303
J40	spp	no_earlier_selected_standalone_export	6,025
J40	spp	standalone_selected_but_late	221
J40	spp	standalone_selected_earlier_but_not_timely	81
J40	streamer	no_earlier_selected_standalone_export	10,443
J40	streamer	standalone_selected_but_late	525
J40	streamer	standalone_selected_earlier_but_not_timely	209
J40	stride	no_earlier_selected_standalone_export	4,357
J40	stride	standalone_selected_but_late	270

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J40	stride	standalone_selected_earlier_but_not_timely	187
J33	ampm	no_earlier_selected_standalone_export	14,196
J33	ampm	standalone_selected_but_late	1,196
J33	ampm	standalone_selected_earlier_but_not_timely	284
J33	power7	no_earlier_selected_standalone_export	11,724
J33	power7	standalone_selected_but_late	913
J33	power7	standalone_selected_earlier_but_not_timely	238
J33	sandbox	no_earlier_selected_standalone_export	26,175
J33	sandbox	standalone_selected_but_late	1,202
J33	sandbox	standalone_selected_earlier_but_not_timely	230
J33	sms	no_earlier_selected_standalone_export	27,739
J33	sms	standalone_selected_but_late	1,621
J33	sms	standalone_selected_earlier_but_not_timely	285
J33	spp	no_earlier_selected_standalone_export	8,263
J33	spp	standalone_selected_but_late	358
J33	spp	standalone_selected_earlier_but_not_timely	76
J33	streamer	no_earlier_selected_standalone_export	10,565
J33	streamer	standalone_selected_but_late	865
J33	streamer	standalone_selected_earlier_but_not_timely	195
J33	stride	no_earlier_selected_standalone_export	4,095
J33	stride	standalone_selected_but_late	195
J33	stride	standalone_selected_earlier_but_not_timely	196
J28	ampm	no_earlier_selected_standalone_export	9,630
J28	ampm	standalone_selected_but_late	14
J28	ampm	standalone_selected_earlier_but_not_timely	165
J28	ipcp	no_earlier_selected_standalone_export	970
J28	ipcp	standalone_selected_earlier_but_not_timely	84
J28	power7	no_earlier_selected_standalone_export	26,130
J28	power7	standalone_selected_but_late	55
J28	power7	standalone_selected_earlier_but_not_timely	343
J28	sandbox	no_earlier_selected_standalone_export	95,925
J28	sandbox	standalone_selected_but_late	76
J28	sandbox	standalone_selected_earlier_but_not_timely	888
J28	sms	no_earlier_selected_standalone_export	64,196
J28	sms	standalone_selected_but_late	47
J28	sms	standalone_selected_earlier_but_not_timely	636
J28	spp	no_earlier_selected_standalone_export	21
J28	streamer	no_earlier_selected_standalone_export	25,963
J28	streamer	standalone_selected_but_late	57
J28	streamer	standalone_selected_earlier_but_not_timely	347
J28	stride	no_earlier_selected_standalone_export	2,952
J28	stride	standalone_selected_earlier_but_not_timely	1
J27	ampm	no_earlier_selected_standalone_export	9,548
J27	ampm	standalone_selected_but_late	6
J27	ampm	standalone_selected_earlier_but_not_timely	299
J27	ipcp	no_earlier_selected_standalone_export	904
J27	ipcp	standalone_selected_earlier_but_not_timely	166
J27	power7	no_earlier_selected_standalone_export	23,907
J27	power7	standalone_selected_but_late	38
J27	power7	standalone_selected_earlier_but_not_timely	449
J27	sandbox	no_earlier_selected_standalone_export	91,903
J27	sandbox	standalone_selected_but_late	66
J27	sandbox	standalone_selected_earlier_but_not_timely	968
J27	sms	no_earlier_selected_standalone_export	60,546
J27	sms	standalone_selected_but_late	41
J27	sms	standalone_selected_earlier_but_not_timely	629
J27	spp	no_earlier_selected_standalone_export	20
J27	streamer	no_earlier_selected_standalone_export	23,629
J27	streamer	standalone_selected_but_late	39
J27	streamer	standalone_selected_earlier_but_not_timely	452
J27	stride	no_earlier_selected_standalone_export	2,894
J27	stride	standalone_selected_earlier_but_not_timely	13
J29	ampm	no_earlier_selected_standalone_export	12,564
J29	ampm	standalone_selected_but_late	7
J29	ampm	standalone_selected_earlier_but_not_timely	404
J29	ipcp	no_earlier_selected_standalone_export	1,220
J29	ipcp	standalone_selected_but_late	12
J29	ipcp	standalone_selected_earlier_but_not_timely	36
J29	power7	no_earlier_selected_standalone_export	31,747
J29	power7	standalone_selected_but_late	30
J29	power7	standalone_selected_earlier_but_not_timely	604
J29	sandbox	no_earlier_selected_standalone_export	104,443
J29	sandbox	standalone_selected_but_late	111
J29	sandbox	standalone_selected_earlier_but_not_timely	1,246
J29	sms	no_earlier_selected_standalone_export	70,300
J29	sms	standalone_selected_but_late	81

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J29	sms	standalone_selected_earlier_but_not_timely	700
J29	spp	no_earlier_selected_standalone_export	28
J29	streamer	no_earlier_selected_standalone_export	31,140
J29	streamer	standalone_selected_but_late	30
J29	streamer	standalone_selected_earlier_but_not_timely	610
J29	stride	no_earlier_selected_standalone_export	2,852
J29	stride	standalone_selected_earlier_but_not_timely	38
J26	ampm	no_earlier_selected_standalone_export	8,127
J26	ampm	standalone_selected_but_late	7
J26	ampm	standalone_selected_earlier_but_not_timely	538
J26	ipcp	no_earlier_selected_standalone_export	957
J26	ipcp	standalone_selected_earlier_but_not_timely	98
J26	power7	no_earlier_selected_standalone_export	23,786
J26	power7	standalone_selected_but_late	10
J26	power7	standalone_selected_earlier_but_not_timely	711
J26	sandbox	no_earlier_selected_standalone_export	92,013
J26	sandbox	standalone_selected_but_late	18
J26	sandbox	standalone_selected_earlier_but_not_timely	1,221
J26	sms	no_earlier_selected_standalone_export	61,592
J26	sms	standalone_selected_but_late	18
J26	sms	standalone_selected_earlier_but_not_timely	766
J26	spp	no_earlier_selected_standalone_export	14
J26	spp	standalone_selected_earlier_but_not_timely	1
J26	streamer	no_earlier_selected_standalone_export	23,295
J26	streamer	standalone_selected_but_late	9
J26	streamer	standalone_selected_earlier_but_not_timely	716
J26	stride	no_earlier_selected_standalone_export	1,826
J26	stride	standalone_selected_earlier_but_not_timely	3
J25	ampm	no_earlier_selected_standalone_export	14,209
J25	ampm	standalone_selected_earlier_but_not_timely	262
J25	ipcp	no_earlier_selected_standalone_export	2,071
J25	ipcp	standalone_selected_earlier_but_not_timely	2
J25	power7	no_earlier_selected_standalone_export	36,096
J25	power7	standalone_selected_but_late	19
J25	power7	standalone_selected_earlier_but_not_timely	429
J25	sandbox	no_earlier_selected_standalone_export	114,415
J25	sandbox	standalone_selected_but_late	20
J25	sandbox	standalone_selected_earlier_but_not_timely	526
J25	sms	no_earlier_selected_standalone_export	75,341
J25	sms	standalone_selected_but_late	15
J25	sms	standalone_selected_earlier_but_not_timely	250
J25	spp	no_earlier_selected_standalone_export	28
J25	spp	standalone_selected_earlier_but_not_timely	1
J25	streamer	no_earlier_selected_standalone_export	36,136
J25	streamer	standalone_selected_but_late	20
J25	streamer	standalone_selected_earlier_but_not_timely	431
J25	stride	no_earlier_selected_standalone_export	2,985
J24	ampm	no_earlier_selected_standalone_export	14,385
J24	ampm	standalone_selected_earlier_but_not_timely	180
J24	ipcp	no_earlier_selected_standalone_export	1,432
J24	ipcp	standalone_selected_earlier_but_not_timely	83
J24	power7	no_earlier_selected_standalone_export	35,479
J24	power7	standalone_selected_but_late	15
J24	power7	standalone_selected_earlier_but_not_timely	580
J24	sandbox	no_earlier_selected_standalone_export	113,527
J24	sandbox	standalone_selected_but_late	17
J24	sandbox	standalone_selected_earlier_but_not_timely	776
J24	sms	no_earlier_selected_standalone_export	75,204
J24	sms	standalone_selected_but_late	12
J24	sms	standalone_selected_earlier_but_not_timely	396
J24	spp	no_earlier_selected_standalone_export	25
J24	streamer	no_earlier_selected_standalone_export	35,257
J24	streamer	standalone_selected_but_late	15
J24	streamer	standalone_selected_earlier_but_not_timely	576
J24	stride	no_earlier_selected_standalone_export	2,985
J24	stride	standalone_selected_earlier_but_not_timely	1
J23	ampm	no_earlier_selected_standalone_export	8,127
J23	ampm	standalone_selected_but_late	7
J23	ampm	standalone_selected_earlier_but_not_timely	538
J23	ipcp	no_earlier_selected_standalone_export	957
J23	ipcp	standalone_selected_earlier_but_not_timely	98
J23	power7	no_earlier_selected_standalone_export	23,786
J23	power7	standalone_selected_but_late	10
J23	power7	standalone_selected_earlier_but_not_timely	711
J23	sandbox	no_earlier_selected_standalone_export	92,013
J23	sandbox	standalone_selected_but_late	18

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J23	sandbox	standalone_selected_earlier_but_not_timely	1,221
J23	sms	no_earlier_selected_standalone_export	61,592
J23	sms	standalone_selected_but_late	18
J23	sms	standalone_selected_earlier_but_not_timely	766
J23	spp	no_earlier_selected_standalone_export	14
J23	spp	standalone_selected_earlier_but_not_timely	1
J23	streamer	no_earlier_selected_standalone_export	23,295
J23	streamer	standalone_selected_but_late	9
J23	streamer	standalone_selected_earlier_but_not_timely	716
J23	stride	no_earlier_selected_standalone_export	1,826
J23	stride	standalone_selected_earlier_but_not_timely	3
J13	ampm	no_earlier_selected_standalone_export	6,832
J13	ampm	standalone_selected_but_late	46
J13	ampm	standalone_selected_earlier_but_not_timely	1,346
J13	ipcp	no_earlier_selected_standalone_export	1,471
J13	ipcp	standalone_selected_but_late	7
J13	ipcp	standalone_selected_earlier_but_not_timely	80
J13	power7	no_earlier_selected_standalone_export	7,042
J13	power7	standalone_selected_but_late	4
J13	power7	standalone_selected_earlier_but_not_timely	997
J13	sandbox	no_earlier_selected_standalone_export	70,368
J13	sandbox	standalone_selected_but_late	118
J13	sandbox	standalone_selected_earlier_but_not_timely	12,005
J13	sms	no_earlier_selected_standalone_export	27,235
J13	sms	standalone_selected_but_late	85
J13	sms	standalone_selected_earlier_but_not_timely	6,579
J13	spp	no_earlier_selected_standalone_export	620
J13	spp	standalone_selected_but_late	2
J13	spp	standalone_selected_earlier_but_not_timely	45
J13	streamer	no_earlier_selected_standalone_export	9,988
J13	streamer	standalone_selected_but_late	26
J13	streamer	standalone_selected_earlier_but_not_timely	1,604
J13	stride	no_earlier_selected_standalone_export	775
J13	stride	standalone_selected_earlier_but_not_timely	183
J12	ampm	no_earlier_selected_standalone_export	6,215
J12	ampm	standalone_selected_but_late	4
J12	ampm	standalone_selected_earlier_but_not_timely	1,202
J12	ipcp	no_earlier_selected_standalone_export	2,797
J12	ipcp	standalone_selected_earlier_but_not_timely	8
J12	power7	no_earlier_selected_standalone_export	7,930
J12	power7	standalone_selected_but_late	3
J12	power7	standalone_selected_earlier_but_not_timely	1,030
J12	sandbox	no_earlier_selected_standalone_export	88,909
J12	sandbox	standalone_selected_but_late	161
J12	sandbox	standalone_selected_earlier_but_not_timely	10,304
J12	sms	no_earlier_selected_standalone_export	41,152
J12	sms	standalone_selected_but_late	6
J12	sms	standalone_selected_earlier_but_not_timely	4,754
J12	spp	no_earlier_selected_standalone_export	696
J12	spp	standalone_selected_earlier_but_not_timely	31
J12	streamer	no_earlier_selected_standalone_export	11,066
J12	streamer	standalone_selected_but_late	6
J12	streamer	standalone_selected_earlier_but_not_timely	1,485
J12	stride	no_earlier_selected_standalone_export	742
J12	stride	standalone_selected_earlier_but_not_timely	201
J14	ampm	no_earlier_selected_standalone_export	4,340
J14	ampm	standalone_selected_but_late	121
J14	ampm	standalone_selected_earlier_but_not_timely	7,154
J14	ipcp	no_earlier_selected_standalone_export	2,663
J14	ipcp	standalone_selected_earlier_but_not_timely	48
J14	power7	no_earlier_selected_standalone_export	5,897
J14	power7	standalone_selected_but_late	27
J14	power7	standalone_selected_earlier_but_not_timely	5,121
J14	sandbox	no_earlier_selected_standalone_export	79,886
J14	sandbox	standalone_selected_but_late	200
J14	sandbox	standalone_selected_earlier_but_not_timely	35,119
J14	sms	no_earlier_selected_standalone_export	33,947
J14	sms	standalone_selected_but_late	134
J14	sms	standalone_selected_earlier_but_not_timely	15,110
J14	spp	no_earlier_selected_standalone_export	400
J14	spp	standalone_selected_but_late	4
J14	spp	standalone_selected_earlier_but_not_timely	311
J14	streamer	no_earlier_selected_standalone_export	7,894
J14	streamer	standalone_selected_but_late	115
J14	streamer	standalone_selected_earlier_but_not_timely	8,157
J14	stride	no_earlier_selected_standalone_export	418

Table 68: analysis/normal_only_timely_standalone_reason.csv — all 942 reason rows (cont.)

ID	normal_prefetcher	reason	normal_only_timely_miss_events
J14	stride	standalone_selected_but_late	6
J14	stride	standalone_selected_earlier_but_not_timely	816
J11	ampm	no_earlier_selected_standalone_export	2,011
J11	ampm	standalone_selected_but_late	83
J11	ampm	standalone_selected_earlier_but_not_timely	818
J11	ipcp	no_earlier_selected_standalone_export	420
J11	ipcp	standalone_selected_earlier_but_not_timely	21
J11	power7	no_earlier_selected_standalone_export	1,973
J11	power7	standalone_selected_but_late	17
J11	power7	standalone_selected_earlier_but_not_timely	768
J11	sandbox	no_earlier_selected_standalone_export	15,848
J11	sandbox	standalone_selected_but_late	60
J11	sandbox	standalone_selected_earlier_but_not_timely	2,002
J11	sms	no_earlier_selected_standalone_export	7,057
J11	sms	standalone_selected_but_late	168
J11	sms	standalone_selected_earlier_but_not_timely	1,126
J11	spp	no_earlier_selected_standalone_export	173
J11	spp	standalone_selected_but_late	3
J11	spp	standalone_selected_earlier_but_not_timely	25
J11	streamer	no_earlier_selected_standalone_export	2,926
J11	streamer	standalone_selected_but_late	40
J11	streamer	standalone_selected_earlier_but_not_timely	1,197
J11	stride	no_earlier_selected_standalone_export	384
J11	stride	standalone_selected_but_late	3
J11	stride	standalone_selected_earlier_but_not_timely	137
J10	ampm	no_earlier_selected_standalone_export	18,187
J10	ampm	standalone_selected_earlier_but_not_timely	108
J10	ipcp	no_earlier_selected_standalone_export	2,803
J10	ipcp	standalone_selected_earlier_but_not_timely	2
J10	power7	no_earlier_selected_standalone_export	16,162
J10	power7	standalone_selected_earlier_but_not_timely	95
J10	sandbox	no_earlier_selected_standalone_export	130,271
J10	sandbox	standalone_selected_but_late	14
J10	sandbox	standalone_selected_earlier_but_not_timely	7,792
J10	sms	no_earlier_selected_standalone_export	54,703
J10	sms	standalone_selected_but_late	1
J10	sms	standalone_selected_earlier_but_not_timely	3,950
J10	spp	no_earlier_selected_standalone_export	1,065
J10	spp	standalone_selected_earlier_but_not_timely	25
J10	streamer	no_earlier_selected_standalone_export	24,160
J10	streamer	standalone_selected_earlier_but_not_timely	132
J10	stride	no_earlier_selected_standalone_export	2,102
J10	stride	standalone_selected_earlier_but_not_timely	13
J09	ampm	no_earlier_selected_standalone_export	3,348
J09	ampm	standalone_selected_but_late	38
J09	ampm	standalone_selected_earlier_but_not_timely	1,411
J09	ipcp	no_earlier_selected_standalone_export	519
J09	ipcp	standalone_selected_earlier_but_not_timely	15
J09	power7	no_earlier_selected_standalone_export	3,041
J09	power7	standalone_selected_but_late	4
J09	power7	standalone_selected_earlier_but_not_timely	1,167
J09	sandbox	no_earlier_selected_standalone_export	20,169
J09	sandbox	standalone_selected_but_late	73
J09	sandbox	standalone_selected_earlier_but_not_timely	3,008
J09	sms	no_earlier_selected_standalone_export	9,243
J09	sms	standalone_selected_but_late	95
J09	sms	standalone_selected_earlier_but_not_timely	1,480
J09	spp	no_earlier_selected_standalone_export	258
J09	spp	standalone_selected_but_late	1
J09	spp	standalone_selected_earlier_but_not_timely	48
J09	streamer	no_earlier_selected_standalone_export	4,450
J09	streamer	standalone_selected_but_late	21
J09	streamer	standalone_selected_earlier_but_not_timely	1,847
J09	stride	no_earlier_selected_standalone_export	623
J09	stride	standalone_selected_earlier_but_not_timely	219

Table 69: resource summary: columns constant across all rows (shown once)

column	constant value
prefetch_reject_fraction	0

Table 70: replay_results/resource_summary.csv — PQ/MSHR occupancy and accept/reject/duplicate counters for all 92 runs
— part 1/3

variant	trace	family	demand_l2_loads	demand_mshr_occ_max	demand_mshr_occ_mean	demand_mshr_occ_p50	demand_mshr_occ_p95	demand_pq_occ_max	demand_pq_occ_mean	demand_pq_occ_p50
ampm	602.gcc.s-734B	normal	404,219	5	0.26717	0	2	10	3.73628	4
ipcp	602.gcc.s-734B	normal	404,219	3	0.1914	0	1	0	0	0
no_pref	602.gcc.s-734B	normal	404,219	3	0.1914	0	1	0	0	0
power7	602.gcc.s-734B	normal	404,219	12	0.2654	0	1	15	3.59287	4
sandbox	602.gcc.s-734B	normal	404,219	21	0.27104	0	1	11	5.01061	5
sms	602.gcc.s-734B	normal	404,219	19	0.19559	0	1	12	0.79478	0
spp	602.gcc.s-734B	normal	404,219	4	0.18144	0	1	19	5.04809	5
streamer	602.gcc.s-734B	normal	404,219	14	0.25571	0	1	10	4.72497	5
stride	602.gcc.s-734B	normal	404,219	3	0.18685	0	1	4	0.13384	0
J21	602.gcc.s-734B	standalone	404,219	9	0.28004	0	2	6	0.50154	0
J20	602.gcc.s-734B	standalone	404,219	9	0.27262	0	2	7	0.50113	0
J22	602.gcc.s-734B	standalone	404,219	9	0.26535	0	1	7	0.58743	0
J19	602.gcc.s-734B	standalone	404,219	9	0.25736	0	1	6	0.49113	0
J17	602.gcc.s-734B	standalone	404,219	9	0.28004	0	2	6	0.50154	0
J16	602.gcc.s-734B	standalone	404,219	9	0.27262	0	2	7	0.50113	0
J18	602.gcc.s-734B	standalone	404,219	4	0.17315	0	1	2	0.03709	0
J15	602.gcc.s-734B	standalone	404,219	9	0.25736	0	1	6	0.49113	0
ampm	605.mcf.s-994B	normal	985,273	15	0.86903	0	3	20	0.3256	0
ipcp	605.mcf.s-994B	normal	985,276	7	0.71986	0	3	1	3.045e-06	0
no_pref	605.mcf.s-994B	normal	985,276	7	0.71986	0	3	0	0	0
power7	605.mcf.s-994B	normal	985,411	32	1.63989	1	7	29	0.81822	0
sandbox	605.mcf.s-994B	normal	1,130,709	32	10.55553	6	32	32	7.01808	7
sms	605.mcf.s-994B	normal	985,312	32	1.58257	1	6	28	0.51107	0
spp	605.mcf.s-994B	normal	985,273	9	0.72666	0	3	8	0.23946	0
streamer	605.mcf.s-994B	normal	985,268	32	1.6891	1	7	27	0.87569	0
stride	605.mcf.s-994B	normal	985,274	8	0.7261	0	3	5	0.01277	0
J07	605.mcf.s-994B	standalone	985,275	14	1.09205	1	4	8	0.5202	0
J06	605.mcf.s-994B	standalone	985,272	15	1.16086	1	4	8	0.53261	0
J08	605.mcf.s-994B	standalone	985,275	14	1.09608	1	4	8	0.46938	0
J05	605.mcf.s-994B	standalone	985,277	14	1.0845	1	4	8	0.65539	0
J03	605.mcf.s-994B	standalone	985,275	14	0.92866	0	4	8	0.44894	0
J02	605.mcf.s-994B	standalone	985,274	15	0.91999	0	4	8	0.44274	0
J04	605.mcf.s-994B	standalone	985,275	14	0.83	0	3	8	0.26728	0
J01	605.mcf.s-994B	standalone	985,276	14	0.86766	0	3	8	0.58399	0
ampm	619.lbm.s-4268B	normal	295,027	32	15.48692	15	21	15	3.8295	4
ipcp	619.lbm.s-4268B	normal	295,039	15	10.82501	12	15	1	1.356e-05	0
no_pref	619.lbm.s-4268B	normal	295,037	15	10.81882	12	15	0	0	0
power7	619.lbm.s-4268B	normal	295,027	26	13.70647	14	19	15	1.8231	0
sandbox	619.lbm.s-4268B	normal	295,024	22	15.15489	15	19	15	6.02754	6
sms	619.lbm.s-4268B	normal	295,545	32	15.41309	14	27	24	1.16261	0
spp	619.lbm.s-4268B	normal	295,024	17	13.45221	14	16	12	4.01631	5
streamer	619.lbm.s-4268B	normal	295,032	21	13.85918	14	19	7	1.35722	0
stride	619.lbm.s-4268B	normal	295,026	18	11.39929	12	16	6	0.54674	0
J46	619.lbm.s-4268B	standalone	295,164	32	15.34938	15	20	16	3.43679	4
J41	619.lbm.s-4268B	standalone	295,187	32	15.50987	16	20	15	3.76665	4
J34	619.lbm.s-4268B	standalone	300,760	32	15.40981	15	22	17	3.43705	4
J43	619.lbm.s-4268B	standalone	295,040	28	14.70604	15	20	10	1.89578	2
J38	619.lbm.s-4268B	standalone	295,047	30	14.9605	15	20	11	2.09876	2
J47	619.lbm.s-4268B	standalone	295,033	28	16.51338	16	20	9	2.06306	2
J31	619.lbm.s-4268B	standalone	295,056	29	14.99424	15	20	10	1.8766	2
J42	619.lbm.s-4268B	standalone	295,040	28	14.70604	15	20	10	1.89578	2
J37	619.lbm.s-4268B	standalone	295,047	30	14.9605	15	20	11	2.09876	2
J35	619.lbm.s-4268B	standalone	295,055	29	14.85286	15	20	11	1.8302	2
J36	619.lbm.s-4268B	standalone	295,066	29	14.76321	15	20	11	1.82127	2
J30	619.lbm.s-4268B	standalone	295,056	29	14.99424	15	20	10	1.8766	2
J44	619.lbm.s-4268B	standalone	295,052	31	15.20464	15	20	14	2.80785	3
J39	619.lbm.s-4268B	standalone	295,075	32	15.49276	16	20	13	3.06108	3
J32	619.lbm.s-4268B	standalone	295,594	32	15.13262	15	21	15	2.74418	3
J45	619.lbm.s-4268B	standalone	295,164	32	15.34938	15	20	16	3.43679	4
J40	619.lbm.s-4268B	standalone	295,187	32	15.50987	16	20	15	3.76665	4
J33	619.lbm.s-4268B	standalone	300,760	32	15.40981	15	22	17	3.43705	4
ampm	620.omnetpp.s-874B	normal	451,359	15	0.52357	0	2	13	0.30955	0
ipcp	620.omnetpp.s-874B	normal	451,406	13	0.46162	0	2	6	0.02408	0
no_pref	620.omnetpp.s-874B	normal	451,398	13	0.446	0	2	0	0	0
power7	620.omnetpp.s-874B	normal	451,341	20	0.64706	0	3	21	0.855	0
sandbox	620.omnetpp.s-874B	normal	466,225	32	3.4948	1	17	32	7.57395	10

Table 70: replay_results/resource.summary.csv — PQ/MSHR occupancy and accept/reject/duplicate counters for all 92 runs
— part 1/3 (cont.)

variant	trace	family	demand_l2_loads	demand_mshr_occ_max	demand_mshr_occ_mean	demand_mshr_occ_p50	demand_mshr_occ_p95	demand_pq_occ_max	demand_pq_occ_mean	demand_pq_occ_p50
sms	620.omnetpp.s-874B	normal	451,284	20	0.70194	0	3	25	0.6199	0
spp	620.omnetpp.s-874B	normal	451,389	13	0.44337	0	2	7	0.14757	0
streamer	620.omnetpp.s-874B	normal	451,360	17	0.63092	0	3	15	0.80459	0
stride	620.omnetpp.s-874B	normal	451,396	13	0.44949	0	2	8	0.01758	0
J28	620.omnetpp.s-874B	standalone	451,413	13	0.47472	0	2	4	0.14823	0
J27	620.omnetpp.s-874B	standalone	451,441	13	0.46094	0	2	5	0.17573	0
J29	620.omnetpp.s-874B	standalone	451,386	13	0.51218	0	2	4	0.1886	0
J26	620.omnetpp.s-874B	standalone	451,464	13	0.43857	0	2	4	0.12326	0
J25	620.omnetpp.s-874B	standalone	451,433	13	0.44686	0	2	3	0.05915	0
J24	620.omnetpp.s-874B	standalone	451,410	13	0.45168	0	2	3	0.05589	0
J23	620.omnetpp.s-874B	standalone	451,464	13	0.43857	0	2	4	0.12326	0
ampm	623.xalancbmk.s-700B	normal	1,218,334	27	0.59512	0	2	16	0.49849	0
ipcp	623.xalancbmk.s-700B	normal	1,218,487	18	0.53065	0	2	6	0.07262	0
no_pref	623.xalancbmk.s-700B	normal	1,218,494	17	0.51516	0	2	0	0	0
power7	623.xalancbmk.s-700B	normal	1,218,439	20	0.73446	0	3	15	0.5946	0
sandbox	623.xalancbmk.s-700B	normal	1,218,513	32	1.84836	0	9	32	6.77927	7
sms	623.xalancbmk.s-700B	normal	1,218,827	28	0.87722	0	3	29	0.78068	0
spp	623.xalancbmk.s-700B	normal	1,218,483	17	0.51388	0	2	11	0.78041	1
streamer	623.xalancbmk.s-700B	normal	1,218,352	21	0.9003	0	4	15	1.05032	0
stride	623.xalancbmk.s-700B	normal	1,218,495	17	0.51539	0	2	9	0.16661	0
J13	623.xalancbmk.s-700B	standalone	1,219,257	19	0.91478	0	3	7	0.5239	0
J12	623.xalancbmk.s-700B	standalone	1,218,225	32	0.93106	0	3	5	0.45508	0
J14	623.xalancbmk.s-700B	standalone	1,245,962	32	2.68322	1	7	11	0.63431	0
J11	623.xalancbmk.s-700B	standalone	1,219,281	22	0.76146	0	2	8	0.69313	1
J10	623.xalancbmk.s-700B	standalone	1,218,369	17	0.5563	0	2	3	0.11083	0
J09	623.xalancbmk.s-700B	standalone	1,219,218	18	0.72896	0	2	7	0.6177	0

Table 71: replay_results/resource.summary.csv — PQ/MSHR occupancy and accept/reject/duplicate counters for all 92 runs
— part 2/3

variant	demand_pq_occ_p95	late_prefetch_demand	pf_mshr_occ_max	pf_mshr_occ_mean	pf_mshr_occ_p50	pf_mshr_occ_p95	pf_pq_occ_max	pf_pq_occ_mean	pf_pq_occ_p50	pf_pq_occ_p95
ampm	6	600	6	0.27776	0	2	10	2.68508	3	5
ipcp	0	1	1	1	1	1	1	1	1	1
no_pref	0	0	0	0	0	0	0	0	0	0
power7	7	126	13	0.26676	0	1	15	2.88119	3	6
sandbox	10	43	21	0.28903	0	1	11	3.29449	3	6
sms	4	173	20	0.79686	0	4	12	2.67915	3	5
spp	10	384	4	0.33618	0	1	19	4.15476	4	9
streamer	8	47	12	0.26167	0	1	10	3.27602	3	6
stride	2	164	3	0.79162	1	2	4	1.49195	1	2
J21	2	1	6	0.10874	0	1	8	1.4886	1	3
J20	2	1	6	0.0793	0	1	8	1.40299	1	3
J22	2	2	6	0.05296	0	0	7	1.31005	1	2
J19	2	0	6	0.06904	0	1	8	1.2762	1	3
J17	2	1	6	0.10874	0	1	8	1.4886	1	3
J16	2	1	6	0.0793	0	1	8	1.40299	1	3
J18	0	1	3	0.72583	1	1	2	1.00141	1	1
J15	2	0	6	0.06904	0	1	8	1.2762	1	3
ampm	4	1,267	15	0.73881	0	3	20	2.27804	2	4
ipcp	0	0	0	0	0	0	1	1	1	1
no_pref	0	0	0	0	0	0	0	0	0	0
power7	6	17,367	32	2.5532	1	9	30	3.70193	3	7
sandbox	21	133,738	32	7.9286	5	26	32	8.47732	7	21
sms	3	57,629	32	2.38491	1	7	28	2.11758	1	5
spp	1	1,105	9	1.29431	1	4	9	1.3498	1	3

Table 71: replay_results/resource_summary.csv — PQ/MSHR occupancy and accept/reject/duplicate counters for all 92 runs
— part 2/3 (cont.)

variant	<i>demand-pq-occ-p95</i>	<i>late-prefetch-demand</i>	<i>pf-mshr-occ-max</i>	<i>pf-mshr-occ-mean</i>	<i>pf-mshr-occ-p50</i>	<i>pf-mshr-occ-p95</i>	<i>pf-pq-occ-max</i>	<i>pf-pq-occ-mean</i>	<i>pf-pq-occ-p50</i>	<i>pf-pq-occ-p95</i>
streamer	5	17,792	32	2.41073	1	8	27	3.35979	3	6
stride	0	0	8	0.99284	1	3	5	1.49149	1	2
J07	2	24,576	15	1.78429	1	5	8	1.33771	1	3
J06	2	24,216	16	1.89029	1	5	8	1.30372	1	3
J08	2	25,342	15	1.92461	2	5	8	1.32106	1	3
J05	2	21,679	15	1.61215	1	5	8	1.34842	1	3
J03	2	2,387	15	1.64597	1	5	8	1.41112	1	3
J02	2	14,110	16	1.67288	1	5	8	1.40051	1	3
J04	2	2,596	15	1.92189	2	5	8	1.38865	1	3
J01	2	1,738	15	1.33184	1	4	8	1.43731	1	3
ampm	8	24,892	32	16.0699	16	22	15	3.61052	4	7
ipcp	0	0	16	12.81818	13	15	1	1	1	1
no-pref	0	0	0	0	0	0	0	0	0	0
power7	7	76,667	26	13.72642	14	19	15	3.74116	3	8
sandbox	10	76,994	23	15.23522	16	19	15	5.42301	6	10
sms	8	25,285	32	18.07701	17	30	24	4.61021	4	11
spp	8	63,324	17	14.02001	14	16	13	5.24506	6	8
streamer	5	112,350	20	11.97705	13	16	10	2.97056	3	5
stride	2	15,702	18	12.58396	14	17	6	1.86822	2	4
J46	7	1,656	32	15.64955	16	21	16	3.57798	3	7
J41	8	1,544	32	15.71585	16	20	15	3.57978	3	7
J34	8	2,451	32	15.71894	15	22	17	3.68169	3	8
J43	4	7,573	28	14.96789	15	20	10	2.11039	2	4
J38	4	8,820	30	15.12588	16	20	11	2.19665	2	5
J47	4	419	28	16.50927	16	20	9	1.95066	2	4
J31	4	10,921	29	15.36676	16	21	10	2.09958	2	4
J42	4	7,573	28	14.96789	15	20	10	2.11039	2	4
J37	4	8,820	30	15.12588	16	20	11	2.19665	2	5
J35	4	10,845	29	15.22145	16	20	11	2.07815	2	4
J36	4	10,729	30	15.14267	15	21	11	2.09841	2	4
J30	4	10,921	29	15.36676	16	21	10	2.09958	2	4
J44	6	3,348	32	15.49104	16	20	14	2.89863	3	6
J39	6	3,607	32	15.69014	16	20	13	2.92409	3	6
J32	6	4,516	32	15.63443	16	21	15	2.89517	3	6
J45	7	1,656	32	15.64955	16	21	16	3.57798	3	7
J40	8	1,544	32	15.71585	16	20	15	3.57978	3	7
J33	8	2,451	32	15.71894	15	22	17	3.68169	3	8
ampm	2	880	16	1.15924	1	3	13	2.32968	2	5
ipcp	0	292	13	0.9424	1	3	7	1.06309	1	2
no-pref	0	0	0	0	0	0	0	0	0	0
power7	6	2,005	21	1.29863	1	5	21	4.0599	4	8
sandbox	20	8,884	32	3.01794	1	12	32	8.25318	7	20
sms	4	1,934	20	1.50769	1	5	25	2.2273	2	6
spp	1	13	12	1.35339	1	3	7	1.23154	1	2
streamer	5	1,942	18	1.27961	1	4	15	3.66059	4	7
stride	0	0	7	0.69351	0	3	8	1.63793	1	3
J28	1	86	10	1.48986	1	4	5	1.46298	1	3
J27	1	123	9	1.30234	1	3	5	1.42291	1	3
J29	1	161	10	1.58011	1	4	5	1.47128	1	3
J26	1	53	8	1.31841	1	4	5	1.33951	1	3
J25	0	20	7	1.56088	1	4	4	1.32034	1	2
J24	0	18	7	1.62756	1	4	3	1.33867	1	3
J23	1	53	8	1.31841	1	4	5	1.33951	1	3
ampm	4	3,240	26	1.13944	0	7	16	3.07597	2	8
ipcp	1	80	18	1.44908	0	12	6	1.22618	1	2
no-pref	0	0	0	0	0	0	0	0	0	0
power7	4	7,360	20	1.33611	0	6	21	2.87548	3	6
sandbox	14	23,387	32	2.31573	1	10	32	7.01276	7	15
sms	3	10,936	28	1.53389	1	5	29	2.1978	2	5
spp	2	1,852	17	1.10931	1	3	11	1.59209	1	3
streamer	5	6,635	20	1.40642	0	6	15	3.6898	4	7
stride	2	7,486	17	1.28005	0	9	9	2.14523	2	4
J13	2	1,180	17	1.37163	1	3	7	1.34647	1	2
J12	2	684	32	1.53234	1	4	5	1.41709	1	3
J14	2	1,203	32	3.15371	3	7	11	1.53015	1	3
J11	2	805	22	0.85123	1	2	8	1.35163	1	2
J10	1	264	6	0.90256	1	3	3	1.142	1	2

Table 71: replay_results/resource_summary.csv — PQ/MSHR occupancy and accept/reject/duplicate counters for all 92 runs
— part 2/3 (cont.)

variant	<i>demand-pq-occ-p95</i>	<i>late-prefetch-demand</i>	<i>pf_mshr-occ-max</i>	<i>pf_mshr-occ-mean</i>	<i>pf_mshr-occ-p50</i>	<i>pf_mshr-occ-p95</i>	<i>pf_pq-occ-max</i>	<i>pf_pq-occ-mean</i>	<i>pf_pq-occ-p50</i>	<i>pf_pq-occ-p95</i>
J09	2	706	17	0.80113	1	2	7	1.30491	1	2

Table 72: replay_results/resource_summary.csv — PQ/MSHR occupancy and accept/reject/duplicate counters for all 92 runs
— part 3/3

variant	<i>prefetch-accepted</i>	<i>prefetch-accepted-per-l2-load</i>	<i>prefetch-attempts</i>	<i>prefetch-attempts-per-l2-load</i>	<i>prefetch-duplicate</i>	<i>prefetch-duplicate-fraction</i>	<i>timely-prefetch-demand</i>
ampm	1,424,170	3.52326	1,424,170	3.52326	1,015	7.127e-04	186,100
ipcp	2	4.948e-06	2	4.948e-06	0	0	0
no_pref	0	0	0	0	0	0	0
power7	1,419,281	3.51117	1,419,281	3.51117	38,580	0.02718	185,829
sandbox	1,926,225	4.7653	1,926,225	4.7653	3,542	0.00184	195,999
sms	343,446	0.84965	343,446	0.84965	0	0	161,972
spp	2,254,525	5.57748	2,254,525	5.57748	1,846	8.188e-04	132,253
streamer	1,808,879	4.475	1,808,879	4.475	847	4.682e-04	187,663
stride	122,156	0.3022	122,156	0.3022	0	0	34,854
J21	188,865	0.46723	188,865	0.46723	0	0	184,979
J20	189,926	0.46986	189,926	0.46986	0	0	185,459
J22	199,519	0.49359	199,519	0.49359	0	0	189,965
J19	189,461	0.46871	189,461	0.46871	0	0	187,013
J17	188,865	0.46723	188,865	0.46723	0	0	184,979
J16	189,926	0.46986	189,926	0.46986	0	0	185,459
J18	41,212	0.10195	41,212	0.10195	0	0	40,312
J15	189,461	0.46871	189,461	0.46871	0	0	187,013
ampm	298,378	0.30284	298,378	0.30284	601	0.00201	59,074
ipcp	3	3.045e-06	3	3.045e-06	0	0	0
no_pref	0	0	0	0	0	0	0
power7	975,958	0.99041	975,958	0.99041	53,320	0.05463	77,025
sandbox	9,261,364	8.19076	9,261,364	8.19076	960,875	0.10375	209,589
sms	765,510	0.77692	765,510	0.77692	0	0	99,548
spp	363,575	0.36901	363,575	0.36901	3,443	0.00947	5,221
streamer	1,061,254	1.07712	1,061,254	1.07712	44,035	0.04149	93,414
stride	14,393	0.01461	14,393	0.01461	4	2.779e-04	4,858
J07	699,538	0.70999	699,538	0.70999	9	1.287e-05	171,410
J06	788,620	0.80041	788,620	0.80041	9	1.141e-05	172,594
J08	708,864	0.71946	708,864	0.71946	5	7.054e-06	166,548
J05	819,983	0.83224	819,983	0.83224	9	1.098e-05	156,700
J03	493,807	0.50119	493,807	0.50119	8	1.620e-05	158,696
J02	497,031	0.50446	497,031	0.50446	6	1.207e-05	157,428
J04	323,579	0.32841	323,579	0.32841	2	6.181e-06	120,925
J01	573,234	0.5818	573,234	0.5818	10	1.744e-05	134,011
ampm	1,006,819	3.41263	1,006,819	3.41263	231,128	0.22956	235,949
ipcp	11	3.728e-05	11	3.728e-05	0	0	0
no_pref	0	0	0	0	0	0	0
power7	391,461	1.32686	391,461	1.32686	14,277	0.03647	160,540
sandbox	1,696,544	5.75053	1,696,544	5.75053	448,317	0.26425	209,323
sms	285,290	0.9653	285,290	0.9653	0	0	255,931
spp	1,906,368	6.46174	1,906,368	6.46174	632,336	0.3317	68,778
streamer	262,880	0.89102	262,880	0.89102	8	3.043e-05	142,314
stride	174,946	0.59299	174,946	0.59299	3	1.715e-05	67,410
J46	833,464	2.82373	833,464	2.82373	151,058	0.18124	252,101
J41	904,426	3.06391	904,426	3.06391	172,971	0.19125	260,265
J34	869,593	2.89132	869,593	2.89132	160,468	0.18453	249,511
J43	478,946	1.62333	478,946	1.62333	59,066	0.12332	228,439

Table 72: replay_results/resource_summary.csv — PQ/MSHR occupancy and accept/reject/duplicate counters for all 92 runs — part 3/3 (cont.)

variant	<i>prefetch-accepted</i>	<i>prefetch-accepted-per_l2-load</i>	<i>prefetch-attempts</i>	<i>prefetch-attempts-per_l2-load</i>	<i>prefetch-duplicate</i>	<i>prefetch-duplicate-fraction</i>	<i>timely-prefetch-demand</i>
J38	506,926	1.71812	506,926	1.71812	54,007	0.10654	236,578
J47	468,215	1.58699	468,215	1.58699	54,231	0.11582	291,124
J31	480,929	1.62996	480,929	1.62996	65,399	0.13598	234,227
J42	478,946	1.62333	478,946	1.62333	59,066	0.12332	228,439
J37	506,926	1.71812	506,926	1.71812	54,007	0.10654	236,578
J35	474,081	1.60675	474,081	1.60675	70,347	0.14839	230,274
J36	472,941	1.60283	472,941	1.60283	68,633	0.14512	226,333
J30	480,929	1.62996	480,929	1.62996	65,399	0.13598	234,227
J44	673,058	2.28115	673,058	2.28115	96,646	0.14359	247,243
J39	722,342	2.44799	722,342	2.44799	103,988	0.14396	256,078
J32	679,499	2.29876	679,499	2.29876	102,399	0.1507	246,430
J45	833,464	2.82373	833,464	2.82373	151,058	0.18124	252,101
J40	904,426	3.06391	904,426	3.06391	172,971	0.19125	260,265
J33	869,593	2.89132	869,593	2.89132	160,468	0.18453	249,511
ampm	158,198	0.35049	158,198	0.35049	10,386	0.06565	22,176
ipcp	14,185	0.03142	14,185	0.03142	0	0	2,150
no_pref	0	0	0	0	0	0	0
power7	404,526	0.89628	404,526	0.89628	55,098	0.1362	52,554
sandbox	4,271,882	9.1627	4,271,882	9.1627	530,053	0.12408	161,570
sms	412,799	0.91472	412,799	0.91472	0	0	90,340
spp	90,145	0.19971	90,145	0.19971	2,129	0.02362	50
streamer	378,952	0.83958	378,952	0.83958	48,338	0.12756	52,592
stride	9,896	0.02192	9,896	0.02192	1	1.011e-04	2,991
J28	70,159	0.15542	70,159	0.15542	0	0	40,825
J27	83,744	0.1855	83,744	0.1855	0	0	48,808
J29	102,407	0.22687	102,407	0.22687	1	9.765e-06	30,673
J26	57,133	0.12655	57,133	0.12655	0	0	45,870
J25	24,752	0.05483	24,752	0.05483	0	0	20,937
J24	24,815	0.05497	24,815	0.05497	0	0	20,939
J23	57,133	0.12655	57,133	0.12655	0	0	45,870
ampm	500,930	0.41116	500,930	0.41116	76,605	0.15293	22,845
ipcp	59,386	0.04874	59,386	0.04874	0	0	2,858
no_pref	0	0	0	0	0	0	0
power7	720,657	0.59146	720,657	0.59146	120,576	0.16731	27,781
sandbox	8,210,756	6.73834	8,210,756	6.73834	1,248,937	0.15211	368,129
sms	894,555	0.73395	894,555	0.73395	0	0	124,113
spp	804,305	0.66009	804,305	0.66009	42,036	0.05226	1,140
streamer	1,303,747	1.07009	1,303,747	1.07009	221,023	0.16953	49,555
stride	166,147	0.13635	166,147	0.13635	29,686	0.17867	2,119
J13	516,174	0.42335	516,174	0.42335	14	2.712e-05	264,876
J12	479,615	0.3937	479,615	0.3937	33	6.881e-05	197,550
J14	794,157	0.63738	794,157	0.63738	10	1.259e-05	183,090
J11	534,810	0.43863	534,810	0.43863	125	2.337e-04	416,605
J10	112,483	0.09232	112,483	0.09232	0	0	82,142
J09	481,918	0.39527	481,918	0.39527	98	2.034e-04	396,354

Table 73: analysis/top_residual_contexts.csv — all 4600 residual (PC, delta, page-offset) contexts, two records per line; tr codes: g=602 m=605 l=619 o=620 x=623

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	ampm	0x40f70a	1	1	372	l	J43	0x401cb0	3	3	1,504
g	ampm	0x40f70a	1	2	367	l	J43	0x401cb0	65	0	1,191
g	ampm	0x40f7b9	1	1	191	l	J43	0x401cb0	4	4	939
g	ampm	0x40f7b9	1	0	48	l	J43	0x401cb0	4	6	920
g	ampm	0x40f7b9	1	2	40	l	J43	0x401cb0	4	10	919
g	ampm	0x40f70a	1	0	13	l	J43	0x401cb0	4	7	917
g	ampm	0xa52221	1	2	13	l	J43	0x401cb0	4	5	916
g	ampm	0xa5222e	-1	34	10	l	J43	0x401cb0	4	9	915
g	ampm	0xa52221	1	1	10	l	J43	0x401cb0	4	8	914
g	ampm	0xa52221	1	0	9	l	J43	0x401cb0	68	2	898
g	ampm	0xa5222e	-1	62	9	l	J43	0x401cb0	67	1	896
g	ampm	0xa5222e	-1	53	8	l	J43	0x401cb0	3	4	869
g	ampm	0xa5222e	-1	16	8	l	J43	0x401cb0	4	11	854
g	ampm	0xa52221	1	52	8	l	J43	0x401cb0	66	1	723
g	ampm	0xa52221	1	61	8	l	J43	0x401cb0	4	51	656
g	ampm	0xa5222e	-1	32	7	l	J43	0x401e02	-2	2	591
g	ampm	0xa5222e	-1	41	7	l	J43	0x401e33	-1	1	580

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	ampm	0xa5222e	-1	43	7	1	J43	0x401cb0	4	42	568
g	ampm	0xa52221	1	25	7	1	J43	0x401e02	-2	1	554
g	ampm	0xa52221	1	26	7	1	J43	0x401e02	-2	4	542
g	ampm	0xa52221	1	3	7	1	J43	0x401cb0	2	2	531
g	ampm	0xa5222e	-1	0	7	1	J43	0x401e67	1	2	530
g	ampm	0xa52221	1	4	7	1	J43	0x401e67	1	1	520
g	ampm	0xa52221	1	57	7	1	J43	0x401e67	1	7	511
g	ampm	0xa52221	1	51	7	1	J43	0x401e33	-1	0	503
g	ampm	0xa52221	1	43	7	1	J43	0x401e02	-2	3	503
g	ampm	0xa5223e	1	1	7	1	J43	0x401cb0	3	6	495
g	ampm	0xa5222e	-1	37	6	1	J43	0x401e02	-2	0	494
g	ampm	0x40f70a	1	26	6	1	J43	0x401e33	-1	3	486
g	ampm	0xa5222e	-1	24	6	1	J43	0x401cb0	4	63	480
g	ampm	0xa52221	1	48	6	1	J43	0x401cb0	4	45	465
g	ampm	0xa52221	1	15	6	1	J43	0x401e67	1	8	464
g	ampm	0xa5222e	-1	21	6	1	J43	0x401cb0	4	43	458
g	ampm	0xa5222e	-1	39	6	1	J43	0x401e67	1	63	455
g	ampm	0xa5222e	-1	30	6	1	J43	0x401e33	-1	2	436
g	ampm	0xa52221	1	22	6	1	J43	0x401cb0	3	5	419
g	ampm	0xa52221	1	17	6	1	J43	0x401e67	-1	0	408
g	ampm	0xa5223e	1	30	6	1	J43	0x401cb0	4	16	402
g	ampm	0xa5222e	-1	36	6	1	J43	0x401e02	-2	5	399
g	ampm	0xa5222e	-1	58	5	1	J43	0x401cb0	4	62	385
g	ampm	0xa5222e	-1	10	5	1	J43	0x401cb0	4	54	372
g	ampm	0xa52221	1	23	5	1	J43	0x401cdb	-2	1	360
g	ampm	0xa5222e	-1	51	5	1	J43	0x401cdb	-2	4	356
g	ampm	0xa52221	1	12	5	1	J43	0x401d43	1	1	354
g	ampm	0xa52221	1	54	5	1	J43	0x401cdb	-2	3	350
g	ampm	0x40f70a	1	14	5	1	J43	0x401d14	-1	2	349
g	ampm	0xa5222e	-1	60	5	1	J43	0x401d14	-1	3	344
g	ampm	0xa5222e	-1	25	5	1	J43	0x401d43	1	2	341
g	ampm	0x40f70a	1	31	5	1	J43	0x401cb0	4	38	335
g	ampm	0xa52221	1	16	5	1	J43	0x401d14	-1	0	330
g	ipcp	0x40f70a	1	14	398	1	J38	0x401cb0	65	0	1,183
g	ipcp	0x40f70a	1	12	393	1	J38	0x401cb0	3	8	860
g	ipcp	0x40f70a	1	9	393	1	J38	0x401cb0	3	6	856
g	ipcp	0x40f70a	1	17	383	1	J38	0x401cb0	4	6	849
g	ipcp	0x40f70a	1	18	381	1	J38	0x401cb0	67	1	795
g	ipcp	0x40f70a	1	15	374	1	J38	0x401cb0	4	10	775
g	ipcp	0x40f70a	1	5	373	1	J38	0x401cb0	4	8	730
g	ipcp	0x40f70a	1	1	372	1	J38	0x401cb0	66	1	723
g	ipcp	0x40f70a	1	58	372	1	J38	0x401cb0	4	5	702
g	ipcp	0x40f70a	1	19	371	1	J38	0x401cb0	3	3	655
g	ipcp	0x40f70a	1	8	371	1	J38	0x401cb0	4	7	606
g	ipcp	0x40f70a	1	11	371	1	J38	0x401cb0	4	9	597
g	ipcp	0x40f70a	1	7	370	1	J38	0x401e67	1	62	543
g	ipcp	0x40f70a	1	16	370	1	J38	0x401e02	-2	3	531
g	ipcp	0x40f70a	1	21	368	1	J38	0x401e67	1	8	516
g	ipcp	0x40f70a	1	2	367	1	J38	0x401e02	-2	1	506
g	ipcp	0x40f70a	1	10	367	1	J38	0x401cb0	2	2	497
g	ipcp	0x40f70a	1	53	367	1	J38	0x401e67	1	6	497
g	ipcp	0x40f70a	1	13	367	1	J38	0x401cb0	3	18	484
g	ipcp	0x40f70a	1	61	365	1	J38	0x401cb0	4	63	479
g	ipcp	0x40f70a	1	63	365	1	J38	0x401e02	-2	0	476
g	ipcp	0x40f70a	1	59	364	1	J38	0x401cb0	3	63	451
g	ipcp	0x40f70a	1	54	363	1	J38	0x401e67	1	63	451
g	ipcp	0x40f70a	1	56	360	1	J38	0x401e67	1	5	428
g	ipcp	0x40f70a	1	20	360	1	J38	0x401e02	-2	2	407
g	ipcp	0x40f70a	1	6	358	1	J38	0x401e67	-1	0	402
g	ipcp	0x40f70a	1	57	358	1	J38	0x401cb0	3	7	393
g	ipcp	0x40f70a	1	4	357	1	J38	0x401cb0	4	13	382
g	ipcp	0x40f70a	1	46	356	1	J38	0x401d43	1	62	374
g	ipcp	0x40f70a	1	60	354	1	J38	0x401cdb	-2	3	362
g	ipcp	0x40f70a	1	62	354	1	J38	0x401d43	1	8	357
g	ipcp	0x40f70a	1	48	349	1	J38	0x401e02	-2	5	350
g	ipcp	0x40f70a	1	23	349	1	J38	0x401d43	1	6	336
g	ipcp	0x40f70a	1	24	347	1	J38	0x401cdb	-2	1	334
g	ipcp	0x40f70a	1	44	346	1	J38	0x401e67	1	4	323
g	ipcp	0x40f70a	1	50	345	1	J38	0x401cb0	68	2	316
g	ipcp	0x40f70a	1	22	345	1	J38	0x401cb0	67	2	316
g	ipcp	0x40f70a	1	52	344	1	J38	0x401cdb	-2	0	306
g	ipcp	0x40f70a	1	3	343	1	J38	0x401e33	-1	18	305
g	ipcp	0x40f70a	1	49	342	1	J38	0x401d43	1	4	279
g	ipcp	0x40f70a	1	45	342	1	J38	0x401cb0	3	17	278
g	ipcp	0x40f70a	1	51	341	1	J38	0x401cb0	3	5	276
g	ipcp	0x40f70a	1	42	341	1	J38	0x401cb0	67	0	274
g	ipcp	0x40f70a	1	47	340	1	J38	0x401cb0	4	4	272
g	ipcp	0x40f70a	1	43	340	1	J38	0x401d43	-1	0	270
g	ipcp	0x40f70a	1	31	340	1	J38	0x401e67	1	7	268
g	ipcp	0x40f70a	1	29	340	1	J38	0x401d43	1	63	264
g	ipcp	0x40f70a	1	41	339	1	J38	0x401cb0	4	14	264
g	ipcp	0x40f70a	1	55	339	1	J38	0x401cb0	3	4	262
g	ipcp	0x40f70a	1	25	337	1	J38	0x401e33	2	63	262
g	no.pref	0x40f70a	1	14	398	1	J47	0x401cb0	3	18	739
g	no.pref	0x40f70a	1	12	393	1	J47	0x401cb0	4	18	387
g	no.pref	0x40f70a	1	9	393	1	J47	0x401cb0	3	19	76
g	no.pref	0x40f70a	1	17	383	1	J47	0x401cb0	3	17	37
g	no.pref	0x40f70a	1	18	381	1	J47	0x401cb0	3	3	31
g	no.pref	0x40f70a	1	15	374	1	J47	0x401cb0	4	19	29
g	no.pref	0x40f70a	1	5	373	1	J47	0x401cb0	4	8	28
g	no.pref	0x40f70a	1	1	372	1	J47	0x401cb0	4	14	27
g	no.pref	0x40f70a	1	58	372	1	J47	0x401cb0	4	6	26
g	no.pref	0x40f70a	1	19	371	1	J47	0x401cb0	4	13	24
g	no.pref	0x40f70a	1	8	371	1	J47	0x401cb0	4	4	24
g	no.pref	0x40f70a	1	11	371	1	J47	0x401cdb	-2	17	24
g	no.pref	0x40f70a	1	7	370	1	J47	0x401cb0	4	17	24
g	no.pref	0x40f70a	1	16	370	1	J47	0x401e02	-2	17	22
g	no.pref	0x40f70a	1	21	368	1	J47	0x4023cc	1	43	21
g	no.pref	0x40f70a	1	2	367	1	J47	0x401cb0	4	9	21
g	no.pref	0x40f70a	1	10	367	1	J47	0x401cb0	4	7	20
g	no.pref	0x40f70a	1	53	367	1	J47	0x401cb0	4	12	20
g	no.pref	0x40f70a	1	13	367	1	J47	0x40208c	1	42	19
g	no.pref	0x40f70a	1	61	365	1	J47	0x401cb0	4	10	19
g	no.pref	0x40f70a	1	63	365	1	J47	0x401cb0	3	6	19
g	no.pref	0x40f70a	1	59	364	1	J47	0x401cb0	3	16	19
g	no.pref	0x40f70a	1	54	363	1	J47	0x401cb0	4	11	18
g	no.pref	0x40f70a	1	56	360	1	J47	0x401e02	-2	6	18
g	no.pref	0x40f70a	1	20	360	1	J47	0x401e02	-2	13	17
g	no.pref	0x40f70a	1	6	358	1	J47	0x401e02	-2	2	17
g	no.pref	0x40f70a	1	57	358	1	J47	0x401cb0	3	4	17
g	no.pref	0x40f70a	1	4	357	1	J47	0x401cb0	3	14	17
g	no.pref	0x40f70a	1	46	356	1	J47	0x401e02	-2	7	16
g	no.pref	0x40f70a	1	60	354	1	J47	0x401e02	-2	14	16
g	no.pref	0x40f70a	1	62	354	1	J47	0x401e67	1	13	15
g	no.pref	0x40f70a	1	48	349	1	J47	0x401cb0	4	5	15
g	no.pref	0x40f70a	1	23	349	1	J47	0x4023cc	2	43	15

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	no_pref	0x40f70a	1	24	347	1	J47	0x401e67	1	3	14
g	no_pref	0x40f70a	1	44	346	1	J47	0x401e02	-2	19	14
g	no_pref	0x40f70a	1	50	345	1	J47	0x401e02	-2	4	14
g	no_pref	0x40f70a	1	22	345	1	J47	0x401e33	-1	5	13
g	no_pref	0x40f70a	1	52	344	1	J47	0x401e33	-1	10	13
g	no_pref	0x40f70a	1	3	343	1	J47	0x401cb0	4	15	13
g	no_pref	0x40f70a	1	49	342	1	J47	0x401cb0	4	26	13
g	no_pref	0x40f70a	1	45	342	1	J47	0x401e02	-2	1	13
g	no_pref	0x40f70a	1	51	341	1	J47	0x401e02	-2	0	13
g	no_pref	0x40f70a	1	42	341	1	J47	0x401cb0	3	5	13
g	no_pref	0x40f70a	1	47	340	1	J47	0x401e02	-2	16	12
g	no_pref	0x40f70a	1	43	340	1	J47	0x401e33	-1	0	12
g	no_pref	0x40f70a	1	31	340	1	J47	0x401cdb	-2	4	12
g	no_pref	0x40f70a	1	29	340	1	J47	0x401e02	-2	5	11
g	no_pref	0x40f70a	1	41	339	1	J47	0x401e33	-1	12	11
g	no_pref	0x40f70a	1	55	339	1	J47	0x401cb0	4	21	10
g	no_pref	0x40f70a	1	25	337	1	J47	0x401e33	-1	1	10
g	power7	0x40f70a	1	1	372	1	J31	0x401cb0	3	3	1,386
g	power7	0x40f70a	1	2	309	1	J31	0x401cb0	65	0	1,191
g	power7	0x40f7b9	1	1	188	1	J31	0x401cb0	3	6	920
g	power7	0x40f7b9	1	0	48	1	J31	0x401cb0	3	5	907
g	power7	0x40f7b9	1	2	35	1	J31	0x401cb0	67	1	899
g	power7	0x40f70a	1	0	13	1	J31	0x401cb0	3	4	885
g	power7	0x40f70a	1	42	10	1	J31	0x401cb0	4	6	861
g	power7	0xa52221	1	0	9	1	J31	0x401cb0	4	8	847
g	power7	0x40f70a	1	25	9	1	J31	0x401cb0	4	7	816
g	power7	0xa5222e	-1	34	9	1	J31	0x401cb0	4	10	811
g	power7	0xa52221	1	3	9	1	J31	0x401cb0	3	8	777
g	power7	0xa52221	1	2	9	1	J31	0x401cb0	66	1	723
g	power7	0x40f70a	1	26	8	1	J31	0x401cb0	4	5	691
g	power7	0x40f70a	1	31	8	1	J31	0x401cb0	68	2	655
g	power7	0x40f70a	1	12	8	1	J31	0x401cb0	4	4	620
g	power7	0x40f70a	1	43	8	1	J31	0x401e67	1	3	571
g	power7	0x40f70a	1	23	8	1	J31	0x401e02	-2	2	554
g	power7	0xa52221	1	57	8	1	J31	0x401cb0	2	2	545
g	power7	0xa52221	1	1	7	1	J31	0x401cb0	4	9	524
g	power7	0xa52221	1	17	7	1	J31	0x401e67	1	6	519
g	power7	0x40f70a	1	36	7	1	J31	0x401e67	1	1	515
g	power7	0x40f70a	1	59	7	1	J31	0x401e02	-2	4	510
g	power7	0xa5222e	-1	62	7	1	J31	0x401e67	1	4	491
g	power7	0x40f70a	1	56	7	1	J31	0x401e67	1	5	478
g	power7	0x40f70a	1	35	7	1	J31	0x401e02	-2	3	450
g	power7	0xa5222e	-1	0	7	1	J31	0x401e67	1	2	440
g	power7	0x40f70a	1	18	7	1	J31	0x401e67	-1	0	403
g	power7	0x40f70a	1	13	7	1	J31	0x401d43	1	6	367
g	power7	0xa5222e	-1	16	6	1	J31	0x401e33	-1	5	362
g	power7	0xa52221	1	12	6	1	J31	0x401d43	1	4	359
g	power7	0x40f70a	1	41	6	1	J31	0x401d43	1	1	358
g	power7	0x40f70a	1	15	6	1	J31	0x401cdb	-2	4	354
g	power7	0x40f70a	1	46	6	1	J31	0x401d43	1	5	349
g	power7	0x40f70a	1	9	6	1	J31	0x401cdb	-2	3	346
g	power7	0x40f70a	1	52	6	1	J31	0x401cb0	67	2	337
g	power7	0x40f70a	1	21	6	1	J31	0x401d43	1	2	333
g	power7	0xa52221	1	52	6	1	J31	0x401e33	-1	3	332
g	power7	0x40f70a	1	53	6	1	J31	0x401d43	1	3	326
g	power7	0xa5222e	-1	21	6	1	J31	0x401e02	-2	7	319
g	power7	0xa5222e	-1	43	6	1	J31	0x401e33	-1	4	319
g	power7	0xa52221	1	22	6	1	J31	0x401cdb	-2	2	319
g	power7	0xa5222e	-1	36	6	1	J31	0x401d14	-1	3	316
g	power7	0x40f70a	1	49	6	1	J31	0x401d14	-1	2	315
g	power7	0xa5222e	-1	32	5	1	J31	0x401e02	-2	5	312
g	power7	0x40f70a	1	10	5	1	J31	0x401cdb	-2	5	295
g	power7	0x40f7b9	1	7	5	1	J31	0x401e33	-1	2	294
g	power7	0x40f7b9	1	10	5	1	J31	0x401d14	-1	5	293
g	power7	0xa5222e	-1	24	5	1	J31	0x401d14	-1	4	292
g	power7	0x40f70a	1	58	5	1	J31	0x401cb0	67	0	280
g	power7	0x40f70a	1	63	5	1	J31	0x401cb0	3	7	275
g	sandbox	0x40f7b9	1	0	48	1	J42	0x401cb0	3	3	1,504
g	sandbox	0x40f70a	1	1	16	1	J42	0x401cb0	65	0	1,191
g	sandbox	0x40f70a	1	0	13	1	J42	0x401cb0	4	4	939
g	sandbox	0xa52221	1	0	9	1	J42	0x401cb0	4	6	920
g	sandbox	0xa5222e	-1	0	7	1	J42	0x401cb0	4	10	919
g	sandbox	0x40f70a	1	26	5	1	J42	0x401cb0	4	7	917
g	sandbox	0xa5222e	-1	34	5	1	J42	0x401cb0	4	5	916
g	sandbox	0xa5222e	-1	43	5	1	J42	0x401cb0	4	9	915
g	sandbox	0xa522b6	59	12	5	1	J42	0x401cb0	4	8	914
g	sandbox	0x40f70a	1	23	5	1	J42	0x401cb0	68	2	898
g	sandbox	0xa5222e	-1	53	5	1	J42	0x401cb0	67	1	896
g	sandbox	0x40f70a	1	18	5	1	J42	0x401cb0	3	4	869
g	sandbox	0xa5222e	-1	18	5	1	J42	0x401cb0	4	11	854
g	sandbox	0xa5222e	-1	36	5	1	J42	0x401cb0	66	1	723
g	sandbox	0x40f7b9	1,624,154,079,928	0	5	1	J42	0x401cb0	4	51	656
g	sandbox	0x40f7b9	1,624,154,082,070	0	5	1	J42	0x401e02	-2	2	591
g	sandbox	0x40f7b9	1,624,154,084,776	0	5	1	J42	0x401e33	-1	1	580
g	sandbox	0x40f7b9	1,624,154,088,364	0	5	1	J42	0x401cb0	4	42	568
g	sandbox	0x40f7b9	1,624,154,091,214	0	5	1	J42	0x401e02	-2	1	554
g	sandbox	0x40f7b9	1,624,154,093,504	0	5	1	J42	0x401e02	-2	4	542
g	sandbox	0x40f7b9	1,624,154,094,946	0	5	1	J42	0x401cb0	2	2	531
g	sandbox	0x40f7b9	1,624,154,095,604	0	5	1	J42	0x401e67	1	2	530
g	sandbox	0x40f7b9	1,624,154,097,054	0	5	1	J42	0x401e67	1	1	520
g	sandbox	0x40f7b9	1,624,154,101,655	0	5	1	J42	0x401e67	1	7	511
g	sandbox	0x40f7b9	1,624,154,102,864	0	5	1	J42	0x401e33	-1	0	503
g	sandbox	0x40f7b9	1,624,154,103,796	0	5	1	J42	0x401e02	-2	3	503
g	sandbox	0x40f7b9	1,624,154,105,373	0	5	1	J42	0x401cb0	3	6	495
g	sandbox	0x40f7b9	1,624,154,110,039	0	5	1	J42	0x401e02	-2	0	494
g	sandbox	0x40f7b9	2,710,681,608,991	0	5	1	J42	0x401e33	-1	3	486
g	sandbox	0x40f7b9	2,802,730,789,712	0	5	1	J42	0x401cb0	4	63	480
g	sandbox	0x40f7b9	2,802,730,790,389	0	5	1	J42	0x401cb0	4	45	465
g	sandbox	0x40f7b9	2,802,730,790,920	0	5	1	J42	0x401e67	1	8	464
g	sandbox	0x40f7b9	1,332,083,806,374	0	5	1	J42	0x401cb0	4	43	458
g	sandbox	0x40f7b9	1,332,083,807,283	0	5	1	J42	0x401e67	1	63	455
g	sandbox	0x40f7b9	1,332,083,811,565	0	5	1	J42	0x401e33	-1	2	436
g	sandbox	0x40f7b9	329,063,421,022	0	5	1	J42	0x401cb0	3	5	419
g	sandbox	0x40f7b9	329,063,422,951	0	5	1	J42	0x401e67	-1	0	408
g	sandbox	0x40f7b9	822,036,543,466	0	5	1	J42	0x401cb0	4	16	402
g	sandbox	0x40f7b9	822,036,547,578	0	5	1	J42	0x401e02	-2	5	399
g	sandbox	0x40f7b9	822,036,554,801	0	5	1	J42	0x401cb0	4	62	385
g	sandbox	0x40f7b9	822,036,557,106	0	5	1	J42	0x401cb0	4	54	372
g	sandbox	0x40f7b9	822,036,558,640	0	5	1	J42	0x401cdb	-2	1	360
g	sandbox	0x40f7b9	822,036,561,947	0	5	1	J42	0x401e67	-1	4	356
g	sandbox	0x40f7b9	822,036,565,176	0	5	1	J42	0x401d43	1	1	354
g	sandbox	0x40f7b9	822,036,568,090	0	5	1	J42	0x401cdb	-2	3	350
g	sandbox	0x40f7b9	822,036,569,432	0	5	1	J42	0x401d14	-1	2	349
g	sandbox	0x40f7b9	822,036,572,405	0	5	1	J42	0x401d14	-1	3	344
g	sandbox	0x40f7b9	822,036,573,947	0	5	1	J42	0x401d43	1	2	341
g	sandbox	0x40f7b9	822,036,574,546	0	5	1	J42	0x401cb0	4	38	335

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	sandbox	0x40f7b9	799,077,457,239	0	5	1	J42	0x401d14	-1	0	330
g	sms	0x40f70a	1	32	331	1	J37	0x401cb0	65	0	1,183
g	sms	0x40f70a	1	31	114	1	J37	0x401cb0	3	8	860
g	sms	0x40f70a	1	63	100	1	J37	0x401cb0	3	6	856
g	sms	0x40f70a	1	29	98	1	J37	0x401cb0	4	6	849
g	sms	0x40f70a	1	61	97	1	J37	0x401cb0	67	1	795
g	sms	0x40f70a	1	60	88	1	J37	0x401cb0	4	10	775
g	sms	0x40f70a	1	58	87	1	J37	0x401cb0	4	8	730
g	sms	0x40f70a	1	28	87	1	J37	0x401cb0	66	1	723
g	sms	0x40f70a	1	26	83	1	J37	0x401cb0	4	5	702
g	sms	0x40f70a	1	59	81	1	J37	0x401cb0	3	3	655
g	sms	0x40f70a	1	21	80	1	J37	0x401cb0	4	7	606
g	sms	0x40f70a	1	56	78	1	J37	0x401cb0	4	9	597
g	sms	0x40f70a	1	27	78	1	J37	0x401e67	1	62	543
g	sms	0x40f70a	1	53	77	1	J37	0x401e02	-2	3	531
g	sms	0x40f70a	1	30	76	1	J37	0x401e67	1	8	516
g	sms	0x40f70a	1	23	72	1	J37	0x401e02	-2	1	506
g	sms	0x40f70a	1	24	72	1	J37	0x401cb0	2	2	497
g	sms	0x40f70a	1	55	70	1	J37	0x401e67	1	6	497
g	sms	0x40f70a	1	62	70	1	J37	0x401cb0	3	18	484
g	sms	0x40f70a	1	51	68	1	J37	0x401cb0	4	63	479
g	sms	0x40f70a	1	57	68	1	J37	0x401e02	-2	0	476
g	sms	0x40f70a	1	22	68	1	J37	0x401cb0	3	63	451
g	sms	0x40f70a	1	19	67	1	J37	0x401e67	1	63	451
g	sms	0x40f70a	1	54	65	1	J37	0x401e67	1	5	428
g	sms	0x40f70a	1	16	64	1	J37	0x401e02	-2	2	407
g	sms	0x40f70a	1	18	63	1	J37	0x401e67	-1	0	402
g	sms	0x40f70a	1	1	62	1	J37	0x401cb0	3	7	393
g	sms	0x40f70a	1	20	62	1	J37	0x401cb0	4	13	382
g	sms	0x40f70a	1	50	60	1	J37	0x401d43	1	62	374
g	sms	0x40f70a	1	14	56	1	J37	0x401cdb	-2	3	362
g	sms	0x40f70a	1	48	55	1	J37	0x401d43	1	8	357
g	sms	0x40f70a	1	46	54	1	J37	0x401e02	-2	5	350
g	sms	0x40f70a	1	17	52	1	J37	0x401d43	1	6	336
g	sms	0x40f70a	1	25	52	1	J37	0x401cdb	-2	1	334
g	sms	0x40f70a	1	49	51	1	J37	0x401e67	1	4	323
g	sms	0x40f7b9	1	0	48	1	J37	0x401cb0	68	2	316
g	sms	0x40f70a	1	52	47	1	J37	0x401cb0	67	2	316
g	sms	0x40f7b9	1	15	47	1	J37	0x401cdb	-2	0	306
g	sms	0x40f7b9	1	32	46	1	J37	0x401e33	-1	18	305
g	sms	0x40f70a	1	45	45	1	J37	0x401d43	1	4	279
g	sms	0x40f70a	1	13	43	1	J37	0x401cb0	3	17	278
g	sms	0x40f70a	1	12	41	1	J37	0x401cb0	3	5	276
g	sms	0x40f70a	1	11	40	1	J37	0x401cb0	67	0	274
g	sms	0x40f70a	1	47	39	1	J37	0x401cb0	4	4	272
g	sms	0x40f70a	1	9	37	1	J37	0x401d43	-1	0	270
g	sms	0x40f70a	1	43	33	1	J37	0x401e67	1	7	268
g	sms	0x40f70a	1	44	31	1	J37	0x401d43	1	63	264
g	sms	0x40f70a	1	33	30	1	J37	0x401cb0	4	14	264
g	sms	0x40f70a	1	2	30	1	J37	0x401cb0	3	4	262
g	sms	0x40f70a	1	41	30	1	J37	0x401e33	2	63	262
g	spp	0x40f70a	1	1	139	1	J35	0x401cb0	3	3	1,354
g	spp	0x40f70a	1	18	136	1	J35	0x401cb0	65	0	1,191
g	spp	0x40f70a	1	14	136	1	J35	0x401cb0	4	4	929
g	spp	0x40f70a	1	58	133	1	J35	0x401cb0	67	1	899
g	spp	0x40f70a	1	54	130	1	J35	0x401cb0	3	4	876
g	spp	0x40f70a	1	5	128	1	J35	0x401cb0	68	2	837
g	spp	0x40f70a	1	6	128	1	J35	0x401cb0	3	6	813
g	spp	0x40f70a	1	12	127	1	J35	0x401cb0	4	6	805
g	spp	0x40f70a	1	53	123	1	J35	0x401cb0	4	7	786
g	spp	0x40f70a	1	9	123	1	J35	0x401cb0	66	1	723
g	spp	0x40f70a	1	19	123	1	J35	0x401cb0	3	8	705
g	spp	0x40f70a	1	61	122	1	J35	0x401cb0	3	5	676
g	spp	0x40f70a	1	2	122	1	J35	0x401cb0	3	7	604
g	spp	0x40f70a	1	17	122	1	J35	0x401cb0	2	2	566
g	spp	0x40f70a	1	15	121	1	J35	0x401cb0	4	8	552
g	spp	0x40f70a	1	11	120	1	J35	0x401e02	-2	5	539
g	spp	0x40f70a	1	31	119	1	J35	0x401e02	-2	4	536
g	spp	0x40f70a	1	52	118	1	J35	0x401e33	-1	5	527
g	spp	0x40f70a	1	55	117	1	J35	0x401e02	-2	6	521
g	spp	0x40f70a	1	51	116	1	J35	0x401e02	-2	3	521
g	spp	0x40f70a	1	8	115	1	J35	0x401e67	1	1	516
g	spp	0x40f70a	1	16	115	1	J35	0x401e67	1	3	510
g	spp	0x40f70a	1	50	115	1	J35	0x401cb0	4	10	506
g	spp	0x40f70a	1	7	114	1	J35	0x401cb0	4	5	499
g	spp	0x40f70a	1	43	114	1	J35	0x401e02	-2	2	497
g	spp	0x40f70a	1	13	114	1	J35	0x401e67	1	4	488
g	spp	0x40f70a	1	62	114	1	J35	0x401e67	1	5	468
g	spp	0x40f70a	1	26	114	1	J35	0x401e33	-1	1	467
g	spp	0x40f70a	1	41	111	1	J35	0x401e67	1	2	462
g	spp	0x40f70a	1	47	111	1	J35	0x401e67	1	6	455
g	spp	0x40f70a	1	24	111	1	J35	0x401e67	-1	0	401
g	spp	0x40f70a	1	63	111	1	J35	0x401e33	-1	3	390
g	spp	0x40f70a	1	45	110	1	J35	0x401cb0	4	11	386
g	spp	0x40f70a	1	46	110	1	J35	0x401cb0	4	9	371
g	spp	0x40f70a	1	33	110	1	J35	0x401d43	1	4	366
g	spp	0x40f70a	1	21	110	1	J35	0x401e33	-2	5	365
g	spp	0x40f70a	1	3	109	1	J35	0x401e33	-1	0	360
g	spp	0x40f70a	1	25	108	1	J35	0x401d43	1	1	358
g	spp	0x40f70a	1	27	108	1	J35	0x401d43	1	6	358
g	spp	0x40f70a	1	37	107	1	J35	0x401cdb	-2	3	354
g	spp	0x40f70a	1	59	107	1	J35	0x401cdb	-2	4	352
g	spp	0x40f70a	1	44	106	1	J35	0x401d43	1	2	344
g	spp	0x40f70a	1	23	106	1	J35	0x401d43	1	5	342
g	spp	0x40f70a	1	39	105	1	J35	0x401e33	-1	2	339
g	spp	0x40f70a	1	32	104	1	J35	0x401cb0	67	2	336
g	spp	0x40f70a	1	60	104	1	J35	0x401d14	-1	5	329
g	spp	0x40f70a	1	10	103	1	J35	0x401d43	1	3	319
g	spp	0x40f70a	1	36	102	1	J35	0x401d14	-1	3	313
g	spp	0x40f70a	1	42	102	1	J35	0x401d14	-1	2	310
g	spp	0x40f70a	1	22	102	1	J35	0x401cdb	-2	6	301
g	streamer	0x40f70a	1	1	372	1	J36	0x401cb0	65	0	1,191
g	streamer	0x40f70a	1	2	367	1	J36	0x401cb0	3	3	1,132
g	streamer	0x40f7b9	1	1	191	1	J36	0x401cb0	3	5	903
g	streamer	0x40f7b9	1	0	48	1	J36	0x401cb0	67	1	899
g	streamer	0x40f7b9	1	2	40	1	J36	0x401cb0	3	4	876
g	streamer	0x40f70a	1	0	13	1	J36	0x401cb0	3	6	854
g	streamer	0xa52221	1	2	13	1	J36	0x401cb0	3	8	779
g	streamer	0xa52221	1	1	10	1	J36	0x401cb0	4	8	766
g	streamer	0xa52221	1	0	9	1	J36	0x401cb0	4	7	753
g	streamer	0xa5222e	-1	34	9	1	J36	0x401cb0	66	1	723
g	streamer	0xa52221	1	3	9	1	J36	0x401cb0	4	4	629
g	streamer	0xa52221	1	57	8	1	J36	0x401cb0	4	5	617
g	streamer	0xa52221	1	17	7	1	J36	0x401e02	-2	2	574
g	streamer	0xa5222e	-1	0	7	1	J36	0x401e67	1	3	568
g	streamer	0xa5223e	1	1	7	1	J36	0x401e33	-1	1	562

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	streamer	Oxa5222e	-1	16	6	1	J36	Ox401cb0	2	2	553
g	streamer	Oxa52221	1	12	6	1	J36	Ox401e02	-2	4	546
g	streamer	Oxa52221	1	52	6	1	J36	Ox401e02	-2	3	542
g	streamer	Oxa5222e	-1	21	6	1	J36	Ox401e67	1	6	541
g	streamer	Oxa5222e	-1	43	6	1	J36	Ox401e67	1	4	540
g	streamer	Oxa5222e	-1	62	6	1	J36	Ox401cb0	3	7	531
g	streamer	Oxa52221	1	22	6	1	J36	Ox401e67	1	2	527
g	streamer	Oxa5222e	-1	36	6	1	J36	Ox401e33	-1	6	519
g	streamer	Oxa5222e	-1	32	5	1	J36	Ox401e67	1	1	518
g	streamer	Ox40f70a	1	26	5	1	J36	Ox401e33	-1	4	510
g	streamer	Oxa5222e	-1	24	5	1	J36	Ox401cb0	4	10	508
g	streamer	Oxa52221	1	14	5	1	J36	Ox401e67	1	5	498
g	streamer	Oxa52221	1	15	5	1	J36	Ox401e33	-1	3	496
g	streamer	Oxa52221	1	4	5	1	J36	Ox401e02	-2	0	489
g	streamer	Oxa52221	1	61	5	1	J36	Ox401e02	-2	5	489
g	streamer	Oxa5222e	-1	37	5	1	J36	Ox401e02	-2	6	472
g	streamer	Ox40f70a	1	31	5	1	J36	Ox401e33	-1	5	462
g	streamer	Ox40f70a	1	12	5	1	J36	Ox401e02	2	63	441
g	streamer	Oxa522b6	59	12	5	1	J36	Ox401e33	-1	2	439
g	streamer	Oxa522f2	71	19	5	1	J36	Ox401e02	-2	8	409
g	streamer	Oxa5222e	-1	49	5	1	J36	Ox401e67	-1	0	405
g	streamer	Ox40f70a	1	23	5	1	J36	Ox401e67	1	63	392
g	streamer	Oxa5222e	-1	41	5	1	J36	Ox401d43	1	6	370
g	streamer	Oxa52221	1	51	5	1	J36	Ox401cdb	-2	3	367
g	streamer	Oxa5222e	-1	39	5	1	J36	Ox401d43	1	4	367
g	streamer	Oxa5222e	-1	53	5	1	J36	Ox401d14	-1	2	363
g	streamer	Oxa52221	1	43	5	1	J36	Ox401cdb	-2	4	362
g	streamer	Oxa5223e	1	2	5	1	J36	Ox401d43	1	2	359
g	streamer	Oxa5223e	1	30	5	1	J36	Ox401d43	1	1	358
g	streamer	Oxa52221	1	39	5	1	J36	Ox401cdb	-2	5	354
g	streamer	Oxa52221	1	48	5	1	J36	Ox401d14	-1	4	354
g	streamer	Ox40f70a	1	18	5	1	J36	Ox401d43	1	5	352
g	streamer	Oxa522b6	40	36	5	1	J36	Ox401d14	-1	3	349
g	streamer	Oxa5222e	-1	18	5	1	J36	Ox401d14	-1	6	339
g	streamer	Ox40f7b9	-138,185,628,694	2	5	1	J36	Ox401cb0	67	2	332
g	stride	Ox40f70a	1	1	368	1	J30	Ox401cb0	3	3	1,386
g	stride	Ox40f70a	1	14	235	1	J30	Ox401cb0	65	0	1,191
g	stride	Ox40f70a	1	17	227	1	J30	Ox401cb0	3	6	920
g	stride	Ox40f70a	1	18	222	1	J30	Ox401cb0	3	5	907
g	stride	Ox40f70a	1	21	217	1	J30	Ox401cb0	67	1	899
g	stride	Ox40f70a	1	5	214	1	J30	Ox401cb0	3	4	885
g	stride	Ox40f70a	1	12	212	1	J30	Ox401cb0	4	6	861
g	stride	Ox40f70a	1	16	211	1	J30	Ox401cb0	4	8	847
g	stride	Ox40f70a	1	2	210	1	J30	Ox401cb0	4	7	816
g	stride	Ox40f70a	1	7	209	1	J30	Ox401cb0	4	10	811
g	stride	Ox40f70a	1	11	207	1	J30	Ox401cb0	3	8	777
g	stride	Ox40f70a	1	15	207	1	J30	Ox401cb0	66	1	723
g	stride	Ox40f70a	1	19	206	1	J30	Ox401cb0	4	5	691
g	stride	Ox40f70a	1	3	206	1	J30	Ox401cb0	68	2	655
g	stride	Ox40f70a	1	9	206	1	J30	Ox401cb0	4	4	620
g	stride	Ox40f70a	1	13	205	1	J30	Ox401e67	1	3	571
g	stride	Ox40f70a	1	58	204	1	J30	Ox401e02	-2	2	554
g	stride	Ox40f70a	1	23	203	1	J30	Ox401cb0	2	2	545
g	stride	Ox40f70a	1	10	202	1	J30	Ox401cb0	4	9	524
g	stride	Ox40f70a	1	57	200	1	J30	Ox401e67	1	6	519
g	stride	Ox40f70a	1	63	200	1	J30	Ox401e67	1	1	515
g	stride	Ox40f70a	1	61	199	1	J30	Ox401e02	-2	4	510
g	stride	Ox40f70a	1	54	199	1	J30	Ox401e67	1	4	491
g	stride	Ox40f70a	1	8	197	1	J30	Ox401e67	1	5	478
g	stride	Ox40f70a	1	20	197	1	J30	Ox401e02	-2	3	450
g	stride	Ox40f70a	1	62	196	1	J30	Ox401e67	1	2	440
g	stride	Ox40f70a	1	46	195	1	J30	Ox401e67	-1	0	403
g	stride	Ox40f70a	1	4	195	1	J30	Ox401d43	1	6	367
g	stride	Ox40f70a	1	59	194	1	J30	Ox401e33	-1	5	362
g	stride	Ox40f70a	1	53	194	1	J30	Ox401d43	1	4	359
g	stride	Ox40f70a	1	24	193	1	J30	Ox401d43	1	1	358
g	stride	Ox40f70a	1	22	193	1	J30	Ox401cdb	-2	4	354
g	stride	Ox40f70a	1	52	192	1	J30	Ox401d43	1	5	349
g	stride	Ox40f70a	1	50	192	1	J30	Ox401cdb	-2	3	346
g	stride	Ox40f70a	1	6	192	1	J30	Ox401cb0	67	2	337
g	stride	Ox40f70a	1	60	190	1	J30	Ox401d43	1	2	333
g	stride	Ox40f70a	1	45	189	1	J30	Ox401e33	-1	3	332
g	stride	Ox40f70a	1	48	187	1	J30	Ox401d43	1	3	326
g	stride	Ox40f70a	1	29	187	1	J30	Ox401e02	-2	7	319
g	stride	Ox40f70a	1	56	186	1	J30	Ox401e33	-1	4	319
g	stride	Ox40f70a	1	44	186	1	J30	Ox401cdb	-2	2	319
g	stride	Ox40f70a	1	47	186	1	J30	Ox401d14	-1	3	316
g	stride	Ox40f70a	1	43	186	1	J30	Ox401d14	-1	2	315
g	stride	Ox40f70a	1	25	186	1	J30	Ox401e02	-2	5	312
g	stride	Ox40f70a	1	51	185	1	J30	Ox401cdb	-2	5	295
g	stride	Ox40f7b9	1	1	185	1	J30	Ox401e33	-1	2	294
g	stride	Ox40f70a	1	41	183	1	J30	Ox401d14	-1	5	293
g	stride	Ox40f70a	1	42	182	1	J30	Ox401d14	-1	4	292
g	stride	Ox40f70a	1	28	182	1	J30	Ox401cb0	67	0	280
g	stride	Ox40f70a	1	37	181	1	J30	Ox401cb0	3	7	275
g	J21	Ox40f70a	1	1	53	1	J44	Ox401cb0	65	0	1,191
g	J21	Ox40f70a	1	2	47	1	J44	Ox401cb0	4	4	924
g	J21	Ox40f7b9	1	1	32	1	J44	Ox401cb0	4	6	920
g	J21	Oxa52221	1	3	20	1	J44	Ox401cb0	4	10	919
g	J21	Oxa52221	1	17	19	1	J44	Ox401cb0	4	7	917
g	J21	Oxa52221	1	61	18	1	J44	Ox401cb0	4	5	916
g	J21	Oxa52221	1	44	18	1	J44	Ox401cb0	4	9	915
g	J21	Oxa52221	1	4	17	1	J44	Ox401cb0	4	8	914
g	J21	Oxa52221	1	43	17	1	J44	Ox401cb0	68	2	898
g	J21	Oxa52221	1	40	16	1	J44	Ox401cb0	3	3	758
g	J21	Oxa52221	1	62	16	1	J44	Ox401cb0	66	1	723
g	J21	Oxa52221	1	57	16	1	J44	Ox401cb0	4	11	716
g	J21	Oxa52221	1	26	16	1	J44	Ox401e33	-1	1	580
g	J21	Ox40f70a	1	12	15	1	J44	Ox401e02	-2	1	548
g	J21	Oxa52221	1	25	15	1	J44	Ox401e02	-2	2	510
g	J21	Oxa52221	1	41	15	1	J44	Ox401e33	-1	0	509
g	J21	Oxa52221	1	58	15	1	J44	Ox401e67	1	7	503
g	J21	Oxa52221	1	51	15	1	J44	Ox401e02	-2	0	499
g	J21	Oxa52221	1	5	15	1	J44	Ox401cb0	3	4	481
g	J21	Ox40f70a	1	5	14	1	J44	Ox401e67	1	1	478
g	J21	Oxa52221	1	48	14	1	J44	Ox401e67	1	8	457
g	J21	Oxa52221	1	35	14	1	J44	Ox401e67	1	63	444
g	J21	Oxa52221	1	28	14	1	J44	Ox401cb0	67	1	429
g	J21	Oxa52221	1	42	14	1	J44	Ox401e67	-1	0	398
g	J21	Ox40f70a	1	41	13	1	J44	Ox401cdb	-2	1	357
g	J21	Ox40f70a	1	14	13	1	J44	Ox401d14	-1	0	331
g	J21	Oxa52221	1	2	13	1	J44	Ox401d14	-1	1	327
g	J21	Oxa52221	1	54	13	1	J44	Ox401e02	-2	3	326
g	J21	Oxa52221	1	55	13	1	J44	Ox401cdb	-2	0	319
g	J21	Oxa52221	1	15	13	1	J44	Ox401d43	1	8	312
g	J21	Oxa52221	1	52	13	1	J44	Ox401cb0	4	42	308

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	J21	0x40f70a	1	50	13	1	J44	0x401e02	-2	4	305
g	J21	0xa52221	1	45	13	1	J44	0x401e67	1	2	301
g	J21	0xa52221	1	16	13	1	J44	0x401e02	-2	5	296
g	J21	0xa52221	1	27	13	1	J44	0x401cb0	2	2	296
g	J21	0xa52221	1	36	13	1	J44	0x401cb0	4	51	295
g	J21	0x40f70a	1	31	12	1	J44	0x401d43	1	7	290
g	J21	0x40f70a	1	9	12	1	J44	0x401e67	1	4	286
g	J21	0x40f70a	1	54	12	1	J44	0x401d43	1	1	283
g	J21	0xa52221	1	23	12	1	J44	0x401cb0	67	0	280
g	J21	0x40f70a	1	59	12	1	J44	0x401cb0	68	3	279
g	J21	0x40f70a	1	26	12	1	J44	0x401d43	-1	0	274
g	J21	0x40f70a	1	49	12	1	J44	0x401cdb	-2	2	268
g	J21	0xa52221	1	63	12	1	J44	0x401d43	1	63	263
g	J21	0xa52221	1	60	12	1	J44	0x401cb0	3	7	240
g	J21	0xa52221	1	56	12	1	J44	0x401cdb	-2	3	234
g	J21	0xa52221	1	46	12	1	J44	0x401e67	1	5	227
g	J21	0x40f70a	1	17	11	1	J44	0x401cb0	4	12	223
g	J21	0x40f70a	1	35	11	1	J44	0x401cb0	4	63	197
g	J21	0xa52221	1	18	11	1	J44	0x401cdb	-2	4	191
g	J20	0x40f70a	1	1	60	1	J39	0x401cb0	65	0	1,183
g	J20	0x40f70a	1	2	52	1	J39	0x401cb0	67	1	785
g	J20	0x40f7b9	1	1	35	1	J39	0x401cb0	4	6	764
g	J20	0xa52221	1	3	20	1	J39	0x401cb0	4	10	752
g	J20	0xa52221	1	17	19	1	J39	0x401cb0	66	1	722
g	J20	0xa52221	1	61	18	1	J39	0x401cb0	3	8	722
g	J20	0xa52221	1	44	18	1	J39	0x401cb0	4	5	622
g	J20	0xa52221	1	4	17	1	J39	0x401cb0	4	8	569
g	J20	0xa52221	1	43	17	1	J39	0x401cb0	4	9	553
g	J20	0xa52221	1	40	16	1	J39	0x401cb0	4	7	551
g	J20	0xa52221	1	62	16	1	J39	0x401e67	1	62	542
g	J20	0xa52221	1	57	16	1	J39	0x401cb0	3	6	486
g	J20	0xa52221	1	26	16	1	J39	0x401e67	1	63	450
g	J20	0xa52221	1	25	15	1	J39	0x401cb0	2	2	441
g	J20	0xa52221	1	41	15	1	J39	0x401e67	1	5	417
g	J20	0xa52221	1	58	15	1	J39	0x401cb0	3	3	411
g	J20	0xa52221	1	51	15	1	J39	0x401e67	-1	0	408
g	J20	0xa52221	1	5	15	1	J39	0x401e02	-2	0	407
g	J20	0x40f70a	1	59	14	1	J39	0x401cb0	4	63	385
g	J20	0xa52221	1	48	14	1	J39	0x401e67	1	6	374
g	J20	0xa52221	1	35	14	1	J39	0x401d43	1	62	373
g	J20	0xa52221	1	28	14	1	J39	0x401cb0	3	7	334
g	J20	0xa52221	1	42	14	1	J39	0x401cb0	68	2	314
g	J20	0xa52221	1	2	13	1	J39	0x401cdb	-2	0	285
g	J20	0xa52221	1	54	13	1	J39	0x401e02	-2	5	275
g	J20	0xa52221	1	55	13	1	J39	0x401cb0	67	0	274
g	J20	0xa52221	1	15	13	1	J39	0x401cb0	67	2	274
g	J20	0xa52221	1	52	13	1	J39	0x401d43	-1	0	271
g	J20	0xa52221	1	45	13	1	J39	0x401d43	1	63	264
g	J20	0xa52221	1	16	13	1	J39	0x401e02	-2	1	262
g	J20	0xa52221	1	27	13	1	J39	0x401e33	2	63	261
g	J20	0xa52221	1	36	13	1	J39	0x401e67	1	7	257
g	J20	0xa52221	1	23	12	1	J39	0x401e02	-2	3	256
g	J20	0x40f70a	1	7	12	1	J39	0x401cb0	4	4	247
g	J20	0xa52221	1	63	12	1	J39	0x401cb0	3	5	238
g	J20	0xa52221	1	60	12	1	J39	0x401e02	-2	2	230
g	J20	0x40f70a	1	50	12	1	J39	0x401d43	1	6	221
g	J20	0xa52221	1	56	12	1	J39	0x401e67	1	4	217
g	J20	0xa52221	1	46	12	1	J39	0x401cb0	3	4	201
g	J20	0x40f70a	1	31	11	1	J39	0x401cdb	-2	3	199
g	J20	0x40f70a	1	12	11	1	J39	0x401e02	-2	6	188
g	J20	0xa52221	1	18	11	1	J39	0x401cb0	66	0	185
g	J20	0xa52221	1	37	11	1	J39	0x401cdb	-2	5	180
g	J20	0x40f70a	1	58	11	1	J39	0x401cb0	68	1	179
g	J20	0xa52221	1	53	11	1	J39	0x401cdb	-2	1	174
g	J20	0xa52221	1	50	11	1	J39	0x401e02	-2	4	174
g	J20	0xa52221	1	49	11	1	J39	0x401d43	1	7	170
g	J20	0xa52221	1	34	11	1	J39	0x401cb0	68	0	157
g	J20	0xa52221	1	30	11	1	J39	0x401d43	1	5	156
g	J20	0xa52221	1	19	11	1	J39	0x401e67	1	8	152
g	J22	0x40f70a	1	1	44	1	J32	0x401cb0	3	3	1,386
g	J22	0x40f7b9	1	1	27	1	J32	0x401cb0	65	0	1,191
g	J22	0xa52221	1	3	20	1	J32	0x401cb0	3	6	919
g	J22	0xa52221	1	17	19	1	J32	0x401cb0	3	5	907
g	J22	0xa52221	1	61	18	1	J32	0x401cb0	67	1	899
g	J22	0xa52221	1	44	18	1	J32	0x401cb0	3	4	885
g	J22	0xa52221	1	4	17	1	J32	0x401cb0	4	6	861
g	J22	0xa52221	1	43	17	1	J32	0x401cb0	4	8	847
g	J22	0xa52221	1	40	16	1	J32	0x401cb0	4	7	816
g	J22	0xa52221	1	62	16	1	J32	0x401cb0	4	10	813
g	J22	0xa52221	1	57	16	1	J32	0x401cb0	3	8	744
g	J22	0xa52221	1	26	16	1	J32	0x401cb0	66	1	723
g	J22	0x40f70a	1	2	15	1	J32	0x401cb0	4	5	690
g	J22	0xa52221	1	25	15	1	J32	0x401cb0	68	2	655
g	J22	0xa52221	1	41	15	1	J32	0x401cb0	4	4	620
g	J22	0xa52221	1	58	15	1	J32	0x401e67	1	3	576
g	J22	0xa52221	1	51	15	1	J32	0x401e02	-2	2	545
g	J22	0xa52221	1	5	15	1	J32	0x401cb0	2	2	545
g	J22	0xa52221	1	48	14	1	J32	0x401cb0	4	9	524
g	J22	0xa52221	1	35	14	1	J32	0x401e67	1	1	518
g	J22	0xa52221	1	28	14	1	J32	0x401e67	1	6	496
g	J22	0xa52221	1	42	14	1	J32	0x401e67	1	4	491
g	J22	0xa52221	1	54	13	1	J32	0x401e02	-2	4	484
g	J22	0xa52221	1	55	13	1	J32	0x401e67	1	5	477
g	J22	0xa52221	1	15	13	1	J32	0x401e67	1	2	439
g	J22	0xa52221	1	52	13	1	J32	0x401e02	-2	3	438
g	J22	0xa52221	1	45	13	1	J32	0x401e67	-1	0	392
g	J22	0xa52221	1	16	13	1	J32	0x401d43	1	6	362
g	J22	0xa52221	1	27	13	1	J32	0x401e33	-1	5	362
g	J22	0xa52221	1	36	13	1	J32	0x401d43	1	4	359
g	J22	0xa52221	1	23	12	1	J32	0x401d43	1	1	356
g	J22	0xa52221	1	2	12	1	J32	0x401d43	1	5	349
g	J22	0xa52221	1	63	12	1	J32	0x401cdb	-2	3	344
g	J22	0xa52221	1	60	12	1	J32	0x401cdb	-2	4	342
g	J22	0xa52221	1	56	12	1	J32	0x401cb0	67	2	337
g	J22	0xa52221	1	46	12	1	J32	0x401d43	1	2	333
g	J22	0xa52221	1	18	11	1	J32	0x401e33	-1	3	332
g	J22	0xa52221	1	37	11	1	J32	0x401d43	1	3	326
g	J22	0xa52221	1	8	11	1	J32	0x401e33	-1	4	319
g	J22	0xa52221	1	53	11	1	J32	0x401d44	-1	3	316
g	J22	0xa52221	1	50	11	1	J32	0x401d44	-1	2	315
g	J22	0xa52221	1	49	11	1	J32	0x401cdb	-2	2	315
g	J22	0xa52221	1	34	11	1	J32	0x401e02	-2	5	311
g	J22	0xa52221	1	30	11	1	J32	0x401cdb	-2	5	295
g	J22	0xa52221	1	19	11	1	J32	0x401e33	-1	2	294
g	J22	0xa52221	1	29	11	1	J32	0x401d44	-1	5	293
g	J22	0xa52221	1	20	11	1	J32	0x401d44	-1	4	292

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	J22	Oxa5222e	-1	53	10	l	J32	Ox401cb0	67	0	280
g	J22	Oxa5222e	-1	41	10	l	J32	Ox401d43	-1	0	269
g	J22	Oxa5222e	-1	34	10	l	J32	Ox401cb0	3	7	252
g	J19	Ox40f70a	1	1	53	l	J45	Ox401cb0	65	0	1,164
g	J19	Ox40f70a	1	2	40	l	J45	Ox401cb0	4	10	919
g	J19	Ox40f7b9	1	1	31	l	J45	Ox401cb0	4	7	917
g	J19	Oxa52221	1	3	20	l	J45	Ox401cb0	4	9	916
g	J19	Oxa52221	1	17	19	l	J45	Ox401cb0	4	8	914
g	J19	Oxa52221	1	61	18	l	J45	Ox401cb0	4	5	904
g	J19	Oxa52221	1	44	18	l	J45	Ox401cb0	68	2	896
g	J19	Oxa52221	1	4	17	l	J45	Ox401cb0	4	4	865
g	J19	Oxa52221	1	43	17	l	J45	Ox401cb0	4	6	860
g	J19	Oxa52221	1	40	16	l	J45	Ox401cb0	66	1	723
g	J19	Oxa52221	1	62	16	l	J45	Ox401cb0	3	3	684
g	J19	Oxa52221	1	57	16	l	J45	Ox401e02	-2	1	548
g	J19	Oxa52221	1	26	16	l	J45	Ox401e67	1	7	503
g	J19	Oxa52221	1	25	15	l	J45	Ox401e02	-2	0	498
g	J19	Oxa52221	1	41	15	l	J45	Ox401e67	1	1	482
g	J19	Oxa52221	1	58	15	l	J45	Ox401cb0	4	11	469
g	J19	Oxa52221	1	51	15	l	J45	Ox401e33	-1	1	468
g	J19	Oxa52221	1	5	15	l	J45	Ox401e02	-2	2	457
g	J19	Oxa52221	1	48	14	l	J45	Ox401e33	-1	0	450
g	J19	Oxa52221	1	35	14	l	J45	Ox401e67	1	63	448
g	J19	Oxa52221	1	28	14	l	J45	Ox401e67	1	8	421
g	J19	Oxa52221	1	42	14	l	J45	Ox401e67	-1	0	404
g	J19	Oxa52221	1	2	13	l	J45	Ox401cb0	67	1	357
g	J19	Oxa52221	1	54	13	l	J45	Ox401cdb	-2	1	357
g	J19	Oxa52221	1	55	13	l	J45	Ox401cdb	-2	0	317
g	J19	Oxa52221	1	15	13	l	J45	Ox401e02	-2	3	316
g	J19	Oxa52221	1	52	13	l	J45	Ox401d14	-1	0	316
g	J19	Oxa52221	1	45	13	l	J45	Ox401d43	1	8	310
g	J19	Oxa52221	1	16	13	l	J45	Ox401e02	-2	4	290
g	J19	Oxa52221	1	27	13	l	J45	Ox401d43	1	7	290
g	J19	Oxa52221	1	36	13	l	J45	Ox401d43	1	1	286
g	J19	Oxa52221	1	23	12	l	J45	Ox401d14	-1	1	286
g	J19	Oxa52221	1	63	12	l	J45	Ox401e67	1	4	286
g	J19	Oxa52221	1	60	12	l	J45	Ox401e67	1	2	284
g	J19	Oxa52221	1	56	12	l	J45	Ox401d43	-1	0	271
g	J19	Oxa52221	1	46	12	l	J45	Ox401e02	-2	5	271
g	J19	Oxa52221	1	18	11	l	J45	Ox401cb0	67	0	265
g	J19	Oxa52221	1	37	11	l	J45	Ox401d43	1	63	264
g	J19	Ox40f70a	1	14	11	l	J45	Ox401cb0	68	3	259
g	J19	Oxa52221	1	53	11	l	J45	Ox401cb0	3	4	251
g	J19	Oxa52221	1	50	11	l	J45	Ox401cb0	3	7	240
g	J19	Oxa52221	1	49	11	l	J45	Ox401cdb	-2	3	233
g	J19	Oxa52221	1	34	11	l	J45	Ox401e67	1	5	227
g	J19	Oxa52221	1	30	11	l	J45	Ox401cdb	-2	2	227
g	J19	Oxa52221	1	19	11	l	J45	Ox401cb0	4	12	214
g	J19	Oxa52221	1	29	11	l	J45	Ox401cdb	-2	4	186
g	J19	Oxa52221	1	20	11	l	J45	Ox401cb0	66	0	185
g	J19	Oxa5222e	-1	53	10	l	J45	Ox401cdb	-2	5	181
g	J19	Oxa5222e	-1	41	10	l	J45	Ox401e02	-2	6	181
g	J19	Oxa5222e	-1	34	10	l	J45	Ox401cb0	68	1	179
g	J17	Ox40f70a	1	1	53	l	J40	Ox401cb0	65	0	1,183
g	J17	Ox40f70a	1	2	47	l	J40	Ox401cb0	67	1	769
g	J17	Ox40f7b9	1	1	32	l	J40	Ox401cb0	4	10	746
g	J17	Oxa52221	1	3	20	l	J40	Ox401cb0	66	1	721
g	J17	Oxa52221	1	17	19	l	J40	Ox401cb0	3	8	719
g	J17	Oxa52221	1	61	18	l	J40	Ox401cb0	4	6	678
g	J17	Oxa52221	1	44	18	l	J40	Ox401cb0	4	5	573
g	J17	Oxa52221	1	4	17	l	J40	Ox401cb0	4	8	542
g	J17	Oxa52221	1	43	17	l	J40	Ox401cb0	4	7	539
g	J17	Oxa52221	1	40	16	l	J40	Ox401cb0	4	9	467
g	J17	Oxa52221	1	62	16	l	J40	Ox401cb0	3	6	455
g	J17	Oxa52221	1	57	16	l	J40	Ox401e67	1	63	453
g	J17	Oxa52221	1	26	16	l	J40	Ox401cb0	2	2	435
g	J17	Ox40f70a	1	12	15	l	J40	Ox401e02	-2	0	404
g	J17	Oxa52221	1	25	15	l	J40	Ox401e67	1	5	402
g	J17	Oxa52221	1	41	15	l	J40	Ox401e67	-1	0	398
g	J17	Oxa52221	1	58	15	l	J40	Ox401cb0	3	3	397
g	J17	Oxa52221	1	51	15	l	J40	Ox401e67	1	6	374
g	J17	Oxa52221	1	5	15	l	J40	Ox401cb0	3	7	334
g	J17	Ox40f70a	1	5	14	l	J40	Ox401cb0	68	2	316
g	J17	Oxa52221	1	48	14	l	J40	Ox401cdb	-2	0	286
g	J17	Oxa52221	1	35	14	l	J40	Ox401cb0	67	0	274
g	J17	Oxa52221	1	28	14	l	J40	Ox401e02	-2	5	272
g	J17	Oxa52221	1	42	14	l	J40	Ox401d43	-1	0	269
g	J17	Ox40f70a	1	41	13	l	J40	Ox401cb0	67	2	269
g	J17	Ox40f70a	1	14	13	l	J40	Ox401d43	1	63	264
g	J17	Oxa52221	1	2	13	l	J40	Ox401e67	1	7	253
g	J17	Oxa52221	1	54	13	l	J40	Ox401e02	-2	3	247
g	J17	Oxa52221	1	55	13	l	J40	Ox401e02	-2	1	246
g	J17	Oxa52221	1	15	13	l	J40	Ox401cb0	3	5	233
g	J17	Oxa52221	1	52	13	l	J40	Ox401cb0	4	4	232
g	J17	Ox40f70a	1	50	13	l	J40	Ox401e33	2	63	224
g	J17	Oxa52221	1	45	13	l	J40	Ox401d43	1	6	220
g	J17	Oxa52221	1	16	13	l	J40	Ox401cb0	3	4	201
g	J17	Oxa52221	1	27	13	l	J40	Ox401cdb	-2	3	195
g	J17	Oxa52221	1	36	13	l	J40	Ox401e02	-2	6	186
g	J17	Ox40f70a	1	31	12	l	J40	Ox401cb0	66	0	185
g	J17	Ox40f70a	1	9	12	l	J40	Ox401e67	1	4	182
g	J17	Ox40f70a	1	54	12	l	J40	Ox401cdb	-2	5	180
g	J17	Oxa52221	1	23	12	l	J40	Ox401e02	-2	2	180
g	J17	Ox40f70a	1	59	12	l	J40	Ox401cb0	68	1	179
g	J17	Ox40f70a	1	26	12	l	J40	Ox401cdb	-2	1	169
g	J17	Ox40f70a	1	49	12	l	J40	Ox401d43	1	7	168
g	J17	Oxa52221	1	63	12	l	J40	Ox401e67	1	62	165
g	J17	Oxa52221	1	60	12	l	J40	Ox401cb0	68	0	157
g	J17	Oxa52221	1	56	12	l	J40	Ox401e02	-2	4	157
g	J17	Oxa52221	1	46	12	l	J40	Ox401d43	1	5	156
g	J17	Ox40f70a	1	17	11	l	J40	Ox401e33	-1	3	149
g	J17	Ox40f70a	1	35	11	l	J40	Ox401cdb	-2	4	137
g	J17	Oxa52221	1	18	11	l	J40	Ox401e33	-1	7	130
g	J16	Ox40f70a	1	1	60	l	J33	Ox401cb0	3	3	1,386
g	J16	Ox40f70a	1	2	52	l	J33	Ox401cb0	65	0	1,191
g	J16	Ox40f7b9	1	1	35	l	J33	Ox401cb0	3	6	919
g	J16	Oxa52221	1	3	20	l	J33	Ox401cb0	3	5	907
g	J16	Oxa52221	1	17	19	l	J33	Ox401cb0	67	1	899
g	J16	Oxa52221	1	61	18	l	J33	Ox401cb0	3	4	885
g	J16	Oxa52221	1	44	18	l	J33	Ox401cb0	4	6	861
g	J16	Oxa52221	1	4	17	l	J33	Ox401cb0	4	8	847
g	J16	Oxa52221	1	43	17	l	J33	Ox401cb0	4	7	817
g	J16	Oxa52221	1	40	16	l	J33	Ox401cb0	4	10	815
g	J16	Oxa52221	1	62	16	l	J33	Ox401cb0	3	8	732
g	J16	Oxa52221	1	57	16	l	J33	Ox401cb0	66	1	723
g	J16	Oxa52221	1	26	16	l	J33	Ox401cb0	4	5	690

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	J16	Oxa52221	1	25	15	l	J33	Ox401cb0	68	2	654
g	J16	Oxa52221	1	41	15	l	J33	Ox401cb0	4	4	620
g	J16	Oxa52221	1	58	15	l	J33	Ox401e67	1	3	569
g	J16	Oxa52221	1	51	15	l	J33	Ox401e02	-2	2	545
g	J16	Oxa52221	1	5	15	l	J33	Ox401cb0	2	2	545
g	J16	Ox40f70a	1	59	14	l	J33	Ox401cb0	4	9	524
g	J16	Oxa52221	1	48	14	l	J33	Ox401e67	1	1	513
g	J16	Oxa52221	1	35	14	l	J33	Ox401e67	1	6	497
g	J16	Oxa52221	1	28	14	l	J33	Ox401e67	1	4	488
g	J16	Oxa52221	1	42	14	l	J33	Ox401e02	-2	4	486
g	J16	Oxa52221	1	2	13	l	J33	Ox401e67	1	5	478
g	J16	Oxa52221	1	54	13	l	J33	Ox401e67	1	2	439
g	J16	Oxa52221	1	55	13	l	J33	Ox401e02	-2	3	438
g	J16	Oxa52221	1	15	13	l	J33	Ox401e67	-1	0	400
g	J16	Oxa52221	1	52	13	l	J33	Ox401e33	-1	5	362
g	J16	Oxa52221	1	45	13	l	J33	Ox401d43	1	6	360
g	J16	Oxa52221	1	16	13	l	J33	Ox401d43	1	1	359
g	J16	Oxa52221	1	27	13	l	J33	Ox401d43	1	4	358
g	J16	Oxa52221	1	36	13	l	J33	Ox401d43	1	5	349
g	J16	Oxa52221	1	23	12	l	J33	Ox401cdb	-2	3	344
g	J16	Ox40f70a	1	7	12	l	J33	Ox401cdb	-2	4	342
g	J16	Oxa52221	1	63	12	l	J33	Ox401cb0	67	2	337
g	J16	Oxa52221	1	60	12	l	J33	Ox401e33	-1	3	332
g	J16	Ox40f70a	1	50	12	l	J33	Ox401d43	1	2	331
g	J16	Oxa52221	1	56	12	l	J33	Ox401d43	1	3	326
g	J16	Oxa52221	1	46	12	l	J33	Ox401e33	-1	4	319
g	J16	Ox40f70a	1	31	11	l	J33	Ox401d14	-1	3	316
g	J16	Ox40f70a	1	12	11	l	J33	Ox401d14	-1	2	315
g	J16	Oxa52221	1	18	11	l	J33	Ox401cdb	-2	2	315
g	J16	Oxa52221	1	37	11	l	J33	Ox401e02	-2	5	310
g	J16	Ox40f70a	1	58	11	l	J33	Ox401cdb	-2	5	295
g	J16	Oxa52221	1	53	11	l	J33	Ox401e33	-1	2	293
g	J16	Oxa52221	1	50	11	l	J33	Ox401d14	-1	5	293
g	J16	Oxa52221	1	49	11	l	J33	Ox401d14	-1	4	292
g	J16	Oxa52221	1	34	11	l	J33	Ox401cb0	67	0	280
g	J16	Oxa52221	1	30	11	l	J33	Ox401d43	-1	0	270
g	J16	Oxa52221	1	19	11	l	J33	Ox401cb0	3	7	245
g	J18	Ox40f70a	1	14	325	o	ampm	Ox4ea019	1	4	178
g	J18	Ox40f70a	1	5	319	o	ampm	Ox4ea019	1	2	159
g	J18	Ox40f70a	1	9	319	o	ampm	Ox4ea019	1	8	154
g	J18	Ox40f70a	1	53	317	o	ampm	Ox54efc	18	61	141
g	J18	Ox40f70a	1	58	314	o	ampm	Ox4ea019	1	7	140
g	J18	Ox40f70a	1	59	311	o	ampm	Ox550ff8	-17	44	140
g	J18	Ox40f70a	1	2	311	o	ampm	Ox550ff8	1	63	138
g	J18	Ox40f70a	1	18	310	o	ampm	Ox4ea019	1	5	137
g	J18	Ox40f70a	1	8	309	o	ampm	Ox4ea019	1	6	133
g	J18	Ox40f70a	1	12	308	o	ampm	Ox4ea019	1	1	132
g	J18	Ox40f70a	1	15	307	o	ampm	Ox550ff8	1	53	131
g	J18	Ox40f70a	1	54	306	o	ampm	Ox550ff8	-17	2	130
g	J18	Ox40f70a	1	19	305	o	ampm	Ox550ff8	1	47	127
g	J18	Ox40f70a	1	48	303	o	ampm	Ox550ff8	1	62	126
g	J18	Ox40f70a	1	1	302	o	ampm	Ox4ea019	1	25	125
g	J18	Ox40f70a	1	63	299	o	ampm	Ox550ff8	-17	29	125
g	J18	Ox40f70a	1	13	297	o	ampm	Ox54efc	18	46	123
g	J18	Ox40f70a	1	44	296	o	ampm	Ox550ff8	-17	34	122
g	J18	Ox40f70a	1	17	295	o	ampm	Ox4ea019	1	40	121
g	J18	Ox40f70a	1	11	295	o	ampm	Ox54efc	18	36	121
g	J18	Ox40f70a	1	6	295	o	ampm	Ox550ff8	-17	19	121
g	J18	Ox40f70a	1	57	294	o	ampm	Ox550ff8	1	60	121
g	J18	Ox40f70a	1	61	293	o	ampm	Ox54efc	18	43	121
g	J18	Ox40f70a	1	4	293	o	ampm	Ox4ea019	1	11	120
g	J18	Ox40f70a	1	60	293	o	ampm	Ox550ff8	1	54	120
g	J18	Ox40f70a	1	16	293	o	ampm	Ox54efc	18	51	120
g	J18	Ox40f70a	1	20	293	o	ampm	Ox54efc	18	35	120
g	J18	Ox40f70a	1	10	291	o	ampm	Ox550ff8	-17	18	120
g	J18	Ox40f70a	1	23	289	o	ampm	Ox54efc	18	19	119
g	J18	Ox40f70a	1	46	288	o	ampm	Ox4ea019	1	3	119
g	J18	Ox40f70a	1	56	288	o	ampm	Ox54efc	18	25	118
g	J18	Ox40f70a	1	7	288	o	ampm	Ox550ff8	1	57	117
g	J18	Ox40f70a	1	43	286	o	ampm	Ox550ff8	1	52	117
g	J18	Ox40f70a	1	24	286	o	ampm	Ox54efc	18	52	117
g	J18	Ox40f70a	1	21	286	o	ampm	Ox4ea019	1	50	115
g	J18	Ox40f70a	1	62	285	o	ampm	Ox550ff8	-17	35	115
g	J18	Ox40f70a	1	47	285	o	ampm	Ox4ea019	1	43	114
g	J18	Ox40f70a	1	52	284	o	ampm	Ox54efc	18	62	114
g	J18	Ox40f70a	1	49	283	o	ampm	Ox550ff8	-17	26	114
g	J18	Ox40f70a	1	42	282	o	ampm	Ox4ea019	1	13	113
g	J18	Ox40f70a	1	39	282	o	ampm	Ox550ff8	-17	7	113
g	J18	Ox40f70a	1	29	281	o	ampm	Ox4ea019	1	61	112
g	J18	Ox40f70a	1	45	280	o	ampm	Ox550ff8	1	51	112
g	J18	Ox40f70a	1	50	280	o	ampm	Ox550ff8	-17	8	111
g	J18	Ox40f70a	1	51	279	o	ampm	Ox4ea019	1	30	111
g	J18	Ox40f70a	1	34	273	o	ampm	Ox550ff8	-17	45	111
g	J18	Ox40f70a	1	38	272	o	ampm	Ox4ea019	1	31	110
g	J18	Ox40f70a	1	28	271	o	ampm	Ox550ff8	1	48	110
g	J18	Ox40f70a	1	41	270	o	ampm	Ox54efc	18	24	110
g	J15	Ox40f70a	1	22	270	o	ampm	Ox550ff8	1	55	110
g	J15	Ox40f70a	1	1	53	o	ipcp	Ox4ea019	1	43	182
g	J15	Ox40f70a	1	2	40	o	ipcp	Ox4ea019	1	4	180
g	J15	Ox40f7b9	1	1	31	o	ipcp	Ox4ea019	1	25	176
g	J15	Oxa52221	1	3	20	o	ipcp	Ox4ea019	1	11	174
g	J15	Oxa52221	1	17	19	o	ipcp	Ox4ea019	1	22	169
g	J15	Oxa52221	1	61	18	o	ipcp	Ox4ea019	1	40	169
g	J15	Oxa52221	1	44	18	o	ipcp	Ox4ea019	1	8	166
g	J15	Oxa52221	1	4	17	o	ipcp	Ox4ea019	1	24	166
g	J15	Oxa52221	1	43	17	o	ipcp	Ox4ea019	1	61	163
g	J15	Oxa52221	1	40	16	o	ipcp	Ox4ea019	1	30	163
g	J15	Oxa52221	1	62	16	o	ipcp	Ox4ea019	1	36	162
g	J15	Oxa52221	1	57	16	o	ipcp	Ox4ea019	1	38	161
g	J15	Oxa52221	1	26	16	o	ipcp	Ox4ea019	1	45	161
g	J15	Oxa52221	1	25	15	o	ipcp	Ox4ea019	1	48	161
g	J15	Oxa52221	1	41	15	o	ipcp	Ox4ea019	1	13	159
g	J15	Oxa52221	1	58	15	o	ipcp	Ox4ea019	1	2	159
g	J15	Oxa52221	1	51	15	o	ipcp	Ox4ea019	1	50	156
g	J15	Oxa52221	1	5	15	o	ipcp	Ox4ea019	1	7	156
g	J15	Oxa52221	1	48	14	o	ipcp	Ox4ea019	1	16	156
g	J15	Oxa52221	1	35	14	o	ipcp	Ox4ea019	1	15	155
g	J15	Oxa52221	1	28	14	o	ipcp	Ox4ea019	1	26	155
g	J15	Oxa52221	1	42	14	o	ipcp	Ox4ea019	1	21	154
g	J15	Oxa52221	1	2	13	o	ipcp	Ox4ea019	1	53	153
g	J15	Oxa52221	1	54	13	o	ipcp	Ox4ea019	1	52	153
g	J15	Oxa52221	1	55	13	o	ipcp	Ox4ea019	1	51	152
g	J15	Oxa52221	1	15	13	o	ipcp	Ox4ea019	1	60	151
g	J15	Oxa52221	1	52	13	o	ipcp	Ox4ea019	1	39	151
g	J15	Oxa52221	1	45	13	o	ipcp	Ox4ea019	1	58	150
g	J15	Oxa52221	1	16	13	o	ipcp	Ox4ea019	1	34	150

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
g	J15	0xa52221	1	27	13	o	ipcp	0x4ea019	1	31	149
g	J15	0xa52221	1	36	13	o	ipcp	0x4ea019	1	57	147
g	J15	0xa52221	1	23	12	o	ipcp	0x4ea019	1	32	147
g	J15	0xa52221	1	63	12	o	ipcp	0x550ff8	-17	44	147
g	J15	0xa52221	1	60	12	o	ipcp	0x4ea019	1	35	147
g	J15	0xa52221	1	56	12	o	ipcp	0x4ea019	1	6	146
g	J15	0xa52221	1	46	12	o	ipcp	0x4ea019	1	19	146
g	J15	0xa52221	1	18	11	o	ipcp	0x4ea019	1	54	144
g	J15	0xa52221	1	37	11	o	ipcp	0x4ea019	1	33	144
g	J15	0x40f70a	1	14	11	o	ipcp	0x4ea019	1	42	144
g	J15	0xa52221	1	53	11	o	ipcp	0x4ea019	1	20	144
g	J15	0xa52221	1	50	11	o	ipcp	0x4ea019	1	47	143
g	J15	0xa52221	1	49	11	o	ipcp	0x4ea019	1	28	142
g	J15	0xa52221	1	34	11	o	ipcp	0x4ea019	1	14	142
g	J15	0xa52221	1	30	11	o	ipcp	0x4ea019	1	18	142
g	J15	0xa52221	1	19	11	o	ipcp	0x550ff8	1	63	142
g	J15	0xa52221	1	29	11	o	ipcp	0x4ea019	1	5	141
g	J15	0xa52221	1	20	11	o	ipcp	0x4ea019	1	29	141
g	J15	0xa5222e	-1	53	10	o	ipcp	0x54efc	18	61	141
g	J15	0xa5222e	-1	41	10	o	ipcp	0x4ea019	1	37	139
g	J15	0xa5222e	-1	34	10	o	ipcp	0x4ea019	1	10	139
m	ampm	0x40244f	1	5	1,670	o	no.pref	0x4ea019	1	43	182
m	ampm	0x40244f	1	6	1,638	o	no.pref	0x4ea019	1	4	180
m	ampm	0x40244f	1	13	1,632	o	no.pref	0x4ea019	1	25	176
m	ampm	0x40244f	1	51	1,631	o	no.pref	0x4ea019	1	11	174
m	ampm	0x40244f	1	44	1,628	o	no.pref	0x4ea019	1	22	169
m	ampm	0x40244f	1	41	1,627	o	no.pref	0x4ea019	1	40	169
m	ampm	0x40244f	1	11	1,626	o	no.pref	0x4ea019	1	8	166
m	ampm	0x40244f	1	9	1,615	o	no.pref	0x4ea019	1	24	166
m	ampm	0x40244f	1	58	1,615	o	no.pref	0x4ea019	1	61	163
m	ampm	0x40244f	1	35	1,611	o	no.pref	0x4ea019	1	30	163
m	ampm	0x40244f	1	56	1,604	o	no.pref	0x4ea019	1	36	162
m	ampm	0x40244f	1	61	1,604	o	no.pref	0x4ea019	1	38	161
m	ampm	0x40244f	1	15	1,602	o	no.pref	0x4ea019	1	45	161
m	ampm	0x40244f	1	25	1,593	o	no.pref	0x4ea019	1	48	161
m	ampm	0x40244f	1	34	1,592	o	no.pref	0x4ea019	1	13	159
m	ampm	0x40244f	1	37	1,584	o	no.pref	0x4ea019	1	2	159
m	ampm	0x40244f	1	46	1,583	o	no.pref	0x4ea019	1	50	156
m	ampm	0x40244f	1	55	1,582	o	no.pref	0x4ea019	1	7	156
m	ampm	0x40244f	1	10	1,581	o	no.pref	0x4ea019	1	16	156
m	ampm	0x40244f	1	14	1,581	o	no.pref	0x4ea019	1	15	155
m	ampm	0x40244f	1	2	1,575	o	no.pref	0x4ea019	1	26	155
m	ampm	0x40244f	1	48	1,572	o	no.pref	0x4ea019	1	21	154
m	ampm	0x40244f	1	26	1,568	o	no.pref	0x4ea019	1	53	153
m	ampm	0x40244f	1	8	1,566	o	no.pref	0x4ea019	1	52	153
m	ampm	0x40244f	1	57	1,561	o	no.pref	0x4ea019	1	51	152
m	ampm	0x40244f	1	3	1,558	o	no.pref	0x4ea019	1	60	151
m	ampm	0x40244f	1	16	1,551	o	no.pref	0x4ea019	1	39	151
m	ampm	0x40244f	1	31	1,549	o	no.pref	0x4ea019	1	58	150
m	ampm	0x40244f	1	53	1,548	o	no.pref	0x4ea019	1	34	150
m	ampm	0x40244f	1	43	1,548	o	no.pref	0x4ea019	1	31	149
m	ampm	0x40244f	1	4	1,542	o	no.pref	0x4ea019	1	57	147
m	ampm	0x40244f	1	63	1,540	o	no.pref	0x4ea019	1	32	147
m	ampm	0x40244f	1	60	1,533	o	no.pref	0x550ff8	-17	44	147
m	ampm	0x40244f	1	40	1,532	o	no.pref	0x4ea019	1	35	147
m	ampm	0x40244f	1	47	1,530	o	no.pref	0x4ea019	1	6	146
m	ampm	0x40244f	1	45	1,528	o	no.pref	0x4ea019	1	19	146
m	ampm	0x40244f	1	62	1,527	o	no.pref	0x4ea019	1	54	144
m	ampm	0x40244f	1	59	1,526	o	no.pref	0x4ea019	1	33	144
m	ampm	0x40244f	1	7	1,525	o	no.pref	0x4ea019	1	42	144
m	ampm	0x40244f	1	54	1,523	o	no.pref	0x4ea019	1	20	144
m	ampm	0x40244f	1	24	1,523	o	no.pref	0x4ea019	1	47	143
m	ampm	0x40244f	1	42	1,522	o	no.pref	0x4ea019	1	28	142
m	ampm	0x40244f	1	36	1,522	o	no.pref	0x4ea019	1	14	142
m	ampm	0x40244f	1	18	1,517	o	no.pref	0x4ea019	1	18	142
m	ampm	0x40244f	1	52	1,509	o	no.pref	0x550ff8	1	63	142
m	ampm	0x40244f	1	12	1,509	o	no.pref	0x4ea019	1	5	141
m	ampm	0x40244f	1	21	1,507	o	no.pref	0x4ea019	1	29	141
m	ampm	0x40244f	1	23	1,506	o	no.pref	0x54efc	18	61	141
m	ampm	0x40244f	1	33	1,504	o	no.pref	0x4ea019	1	37	139
m	ampm	0x40244f	1	49	1,501	o	no.pref	0x4ea019	1	10	139
m	ipcp	0x40244f	1	5	1,722	o	power7	0x4ea019	1	4	169
m	ipcp	0x40244f	1	51	1,697	o	power7	0x4ea019	1	2	159
m	ipcp	0x40244f	1	13	1,690	o	power7	0x54efc	18	61	141
m	ipcp	0x40244f	1	6	1,686	o	power7	0x4ea019	1	1	132
m	ipcp	0x40244f	1	11	1,678	o	power7	0x550ff8	-17	2	130
m	ipcp	0x40244f	1	44	1,668	o	power7	0x54efc	18	46	123
m	ipcp	0x40244f	1	41	1,665	o	power7	0x54efc	18	43	122
m	ipcp	0x40244f	1	58	1,658	o	power7	0x54efc	18	36	121
m	ipcp	0x40244f	1	15	1,656	o	power7	0x54efc	18	35	120
m	ipcp	0x40244f	1	9	1,656	o	power7	0x4ea019	1	3	119
m	ipcp	0x40244f	1	35	1,650	o	power7	0x54efc	18	19	118
m	ipcp	0x40244f	1	61	1,647	o	power7	0x54efc	18	25	117
m	ipcp	0x40244f	1	14	1,637	o	power7	0x54efc	18	51	117
m	ipcp	0x40244f	1	25	1,637	o	power7	0x54efc	18	52	117
m	ipcp	0x40244f	1	56	1,636	o	power7	0x54efc	18	62	114
m	ipcp	0x40244f	1	10	1,636	o	power7	0x54efc	18	24	110
m	ipcp	0x40244f	1	34	1,634	o	power7	0x550ff8	-17	3	108
m	ipcp	0x40244f	1	2	1,626	o	power7	0x54efc	18	29	106
m	ipcp	0x40244f	1	26	1,626	o	power7	0x54efc	18	45	105
m	ipcp	0x40244f	1	37	1,624	o	power7	0x54efc	18	34	105
m	ipcp	0x40244f	1	46	1,623	o	power7	0x54efc	18	54	104
m	ipcp	0x40244f	1	48	1,622	o	power7	0x54efc	18	58	103
m	ipcp	0x40244f	1	8	1,617	o	power7	0x54efc	18	57	103
m	ipcp	0x40244f	1	55	1,614	o	power7	0x550ff8	-17	0	103
m	ipcp	0x40244f	1	3	1,612	o	power7	0x54efc	18	48	101
m	ipcp	0x40244f	1	16	1,607	o	power7	0x54efc	18	53	101
m	ipcp	0x40244f	1	57	1,601	o	power7	0x4ea019	1	5	101
m	ipcp	0x40244f	1	31	1,598	o	power7	0x54efc	18	33	101
m	ipcp	0x40244f	1	53	1,596	o	power7	0x54efc	18	27	101
m	ipcp	0x40244f	1	63	1,588	o	power7	0x54efc	18	23	100
m	ipcp	0x40244f	1	43	1,584	o	power7	0x550ff8	-17	19	99
m	ipcp	0x40244f	1	7	1,581	o	power7	0x54efc	18	63	99
m	ipcp	0x40244f	1	4	1,581	o	power7	0x54efc	18	20	98
m	ipcp	0x40244f	1	59	1,580	o	power7	0x54efc	18	41	97
m	ipcp	0x40244f	1	18	1,580	o	power7	0x54efc	18	32	97
m	ipcp	0x40244f	1	45	1,576	o	power7	0x54efc	18	47	95
m	ipcp	0x40244f	1	36	1,576	o	power7	0x54efc	18	49	95
m	ipcp	0x40244f	1	54	1,570	o	power7	0x54efc	18	39	94
m	ipcp	0x40244f	1	40	1,570	o	power7	0x550ff8	-17	7	93
m	ipcp	0x40244f	1	24	1,569	o	power7	0x550ff8	-17	16	93
m	ipcp	0x40244f	1	47	1,569	o	power7	0x54efc	18	26	92
m	ipcp	0x40244f	1	33	1,567	o	power7	0x550ff8	-17	17	92
m	ipcp	0x40244f	1	42	1,566	o	power7	0x550ff8	-17	26	92
m	ipcp	0x40244f	1	62	1,566	o	power7	0x550ff8	-17	8	90
m	ipcp	0x40244f	1	60	1,565	o	power7	0x54efc	18	60	89

tr ID	PC	Δ	off	resid	tr ID	PC	Δ	off	resid
m ipcp	0x40244f	1	12	1,563	o power7	0x550ff8	-17	12	89
m ipcp	0x40244f	1	23	1,563	o power7	0x54efc	18	22	89
m ipcp	0x40244f	1	17	1,559	o power7	0x550ff8	-17	1	89
m ipcp	0x40244f	1	21	1,550	o power7	0x54efc	18	55	88
m ipcp	0x40244f	1	19	1,548	o power7	0x54efc	18	28	88
m no.pref	0x40244f	1	5	1,722	o sandbox	0x54efc	18	61	141
m no.pref	0x40244f	1	51	1,697	o sandbox	0x54efc	18	46	122
m no.pref	0x40244f	1	13	1,690	o sandbox	0x54efc	18	36	121
m no.pref	0x40244f	1	6	1,686	o sandbox	0x54efc	18	43	120
m no.pref	0x40244f	1	11	1,678	o sandbox	0x54efc	18	35	119
m no.pref	0x40244f	1	44	1,668	o sandbox	0x54efc	18	51	118
m no.pref	0x40244f	1	41	1,665	o sandbox	0x54efc	18	19	117
m no.pref	0x40244f	1	58	1,658	o sandbox	0x54efc	18	52	117
m no.pref	0x40244f	1	15	1,656	o sandbox	0x54efc	18	25	115
m no.pref	0x40244f	1	9	1,656	o sandbox	0x54efc	18	62	114
m no.pref	0x40244f	1	35	1,650	o sandbox	0x54efc	18	24	108
m no.pref	0x40244f	1	61	1,647	o sandbox	0x54efc	18	29	105
m no.pref	0x40244f	1	14	1,637	o sandbox	0x54efc	18	45	105
m no.pref	0x40244f	1	25	1,637	o sandbox	0x54efc	18	34	104
m no.pref	0x40244f	1	56	1,636	o sandbox	0x550ff8	-17	0	103
m no.pref	0x40244f	1	10	1,636	o sandbox	0x4ea019	1	1	102
m no.pref	0x40244f	1	34	1,634	o sandbox	0x54efc	18	57	102
m no.pref	0x40244f	1	2	1,626	o sandbox	0x54efc	18	54	101
m no.pref	0x40244f	1	26	1,626	o sandbox	0x54efc	18	58	101
m no.pref	0x40244f	1	37	1,624	o sandbox	0x54efc	18	27	101
m no.pref	0x40244f	1	46	1,623	o sandbox	0x54efc	18	48	100
m no.pref	0x40244f	1	48	1,622	o sandbox	0x54efc	18	53	100
m no.pref	0x40244f	1	8	1,617	o sandbox	0x54efc	18	33	100
m no.pref	0x40244f	1	55	1,614	o sandbox	0x54efc	18	63	99
m no.pref	0x40244f	1	3	1,612	o sandbox	0x54efc	18	20	98
m no.pref	0x40244f	1	16	1,607	o sandbox	0x54efc	18	41	97
m no.pref	0x40244f	1	57	1,601	o sandbox	0x54efc	18	32	97
m no.pref	0x40244f	1	31	1,598	o sandbox	0x54efc	18	23	96
m no.pref	0x40244f	1	53	1,596	o sandbox	0x54efc	18	47	95
m no.pref	0x40244f	1	63	1,588	o sandbox	0x54efc	18	39	93
m no.pref	0x40244f	1	43	1,584	o sandbox	0x54efc	18	26	92
m no.pref	0x40244f	1	7	1,581	o sandbox	0x54efc	18	49	91
m no.pref	0x40244f	1	4	1,581	o sandbox	0x4ea019	1	2	90
m no.pref	0x40244f	1	59	1,580	o sandbox	0x54efc	18	60	89
m no.pref	0x40244f	1	18	1,580	o sandbox	0x54efc	18	38	88
m no.pref	0x40244f	1	45	1,576	o sandbox	0x54efc	18	40	87
m no.pref	0x40244f	1	36	1,576	o sandbox	0x54efc	18	55	86
m no.pref	0x40244f	1	54	1,570	o sandbox	0x54efc	18	28	86
m no.pref	0x40244f	1	40	1,570	o sandbox	0x54efc	18	31	85
m no.pref	0x40244f	1	24	1,569	o sandbox	0x54efc	18	44	85
m no.pref	0x40244f	1	47	1,569	o sandbox	0x54efc	18	22	84
m no.pref	0x40244f	1	33	1,567	o sandbox	0x54efc	18	18	84
m no.pref	0x40244f	1	42	1,566	o sandbox	0x54efc	18	42	84
m no.pref	0x40244f	1	62	1,566	o sandbox	0x54efc	18	21	84
m no.pref	0x40244f	1	60	1,565	o sandbox	0x54efc	18	56	83
m no.pref	0x40244f	1	12	1,563	o sandbox	0x416e72	1	61	82
m no.pref	0x40244f	1	23	1,563	o sandbox	0x550ff8	-17	2	81
m no.pref	0x40244f	1	17	1,559	o sandbox	0x54efc	18	50	77
m no.pref	0x40244f	1	21	1,550	o sandbox	0x550ff8	-17	1	77
m no.pref	0x40244f	1	19	1,548	o sandbox	0x54efc	18	30	74
m power7	0x40244f	1	5	1,663	o sms	0x54efc	18	35	121
m power7	0x40244f	1	6	1,604	o sms	0x54efc	18	43	121
m power7	0x40244f	1	2	1,581	o sms	0x54efc	18	36	120
m power7	0x40244f	1	9	1,569	o sms	0x54efc	18	46	118
m power7	0x40244f	1	3	1,561	o sms	0x4ea019	1	43	116
m power7	0x40244f	1	11	1,550	o sms	0x4ea019	1	11	115
m power7	0x40244f	1	13	1,545	o sms	0x54efc	18	51	115
m power7	0x40244f	1	4	1,542	o sms	0x54efc	18	19	113
m power7	0x40244f	1	8	1,533	o sms	0x4ea019	1	4	112
m power7	0x40244f	1	10	1,532	o sms	0x550ff8	-17	34	109
m power7	0x40244f	1	15	1,514	o sms	0x4ea019	1	25	108
m power7	0x40244f	1	14	1,499	o sms	0x4ea019	1	40	107
m power7	0x40244f	1	7	1,492	o sms	0x4ea019	1	30	107
m power7	0x40244f	1	1	1,490	o sms	0x550ff8	-17	2	106
m power7	0x40244f	1	16	1,461	o sms	0x4ea019	1	7	106
m power7	0x40244f	1	41	1,455	o sms	0x4ea019	1	61	105
m power7	0x40244f	1	25	1,449	o sms	0x54efc	18	45	105
m power7	0x40244f	1	51	1,445	o sms	0x54efc	18	34	105
m power7	0x40244f	1	34	1,440	o sms	0x4ea019	1	2	103
m power7	0x40244f	1	44	1,440	o sms	0x4ea019	1	38	102
m power7	0x40244f	1	37	1,437	o sms	0x4ea019	1	45	102
m power7	0x40244f	1	12	1,435	o sms	0x54efc	18	48	101
m power7	0x40244f	1	26	1,432	o sms	0x54efc	18	33	99
m power7	0x40244f	1	35	1,429	o sms	0x54efc	18	52	99
m power7	0x40244f	1	58	1,427	o sms	0x4ea019	1	29	98
m power7	0x40244f	1	17	1,421	o sms	0x4ea019	1	32	98
m power7	0x40244f	1	56	1,415	o sms	0x4ea019	1	31	98
m power7	0x40244f	1	46	1,414	o sms	0x4ea019	1	24	98
m power7	0x40244f	1	24	1,410	o sms	0x4ea019	1	9	97
m power7	0x40244f	1	61	1,405	o sms	0x54efc	18	41	97
m power7	0x40244f	1	18	1,401	o sms	0x4ea019	1	36	97
m power7	0x40244f	1	55	1,394	o sms	0x54efc	18	32	97
m power7	0x40244f	1	21	1,394	o sms	0x4ea019	1	39	96
m power7	0x40244f	1	19	1,390	o sms	0x4ea019	1	48	96
m power7	0x40244f	1	23	1,390	o sms	0x550ff8	-17	32	96
m power7	0x40244f	1	31	1,379	o sms	0x54efc	18	47	95
m power7	0x40244f	1	47	1,377	o sms	0x4ea019	1	6	95
m power7	0x40244f	1	33	1,371	o sms	0x4ea019	1	34	95
m power7	0x40244f	1	43	1,371	o sms	0x4ea019	1	33	95
m power7	0x40244f	1	36	1,368	o sms	0x550ff8	-17	0	95
m power7	0x40244f	1	63	1,368	o sms	0x4ea019	1	35	95
m power7	0x40244f	1	48	1,368	o sms	0x54efc	18	49	95
m power7	0x40244f	1	42	1,366	o sms	0x4ea019	1	50	94
m power7	0x40244f	1	28	1,360	o sms	0x4ea019	1	8	94
m power7	0x40244f	1	40	1,360	o sms	0x4ea019	1	58	94
m power7	0x40244f	1	57	1,354	o sms	0x54efc	18	39	94
m power7	0x40244f	1	45	1,354	o sms	0x4ea019	1	57	93
m power7	0x40244f	1	22	1,353	o sms	0x4ea019	1	13	93
m power7	0x40244f	1	53	1,353	o sms	0x4ea019	1	10	92
m power7	0x40244f	1	29	1,349	o sms	0x4ea019	1	60	91
m sandbox	0x40244f	1	58	1,031	o spp	0x4ea019	1	43	182
m sandbox	0x40244f	1	5	1,014	o spp	0x4ea019	1	4	180
m sandbox	0x40244f	1	2	1,014	o spp	0x4ea019	1	25	176
m sandbox	0x40244f	1	61	995	o spp	0x4ea019	1	11	174
m sandbox	0x40244f	1	51	981	o spp	0x4ea019	1	22	169
m sandbox	0x40244f	1	63	980	o spp	0x4ea019	1	40	168
m sandbox	0x40244f	1	56	972	o spp	0x4ea019	1	8	166
m sandbox	0x40244f	1	6	967	o spp	0x4ea019	1	24	166
m sandbox	0x40244f	1	59	964	o spp	0x4ea019	1	61	163
m sandbox	0x40244f	1	62	960	o spp	0x4ea019	1	36	162
m sandbox	0x40244f	1	55	960	o spp	0x4ea019	1	30	162

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
m	sandbox	0x40244f	1	3	955	o	spp	0x4ea019	1	38	161
m	sandbox	0x40244f	1	10	954	o	spp	0x4ea019	1	45	161
m	sandbox	0x40244f	1	15	950	o	spp	0x4ea019	1	48	161
m	sandbox	0x40244f	1	8	950	o	spp	0x4ea019	1	13	159
m	sandbox	0x40244f	1	9	949	o	spp	0x4ea019	1	2	159
m	sandbox	0x40244f	1	14	939	o	spp	0x4ea019	1	50	156
m	sandbox	0x40244f	1	4	938	o	spp	0x4ea019	1	16	156
m	sandbox	0x40244f	1	57	932	o	spp	0x4ea019	1	15	155
m	sandbox	0x40244f	1	1	928	o	spp	0x4ea019	1	7	155
m	sandbox	0x40244f	1	13	926	o	spp	0x4ea019	1	21	154
m	sandbox	0x40244f	1	34	924	o	spp	0x4ea019	1	26	154
m	sandbox	0x40244f	1	46	919	o	spp	0x4ea019	1	53	153
m	sandbox	0x40244f	1	41	919	o	spp	0x4ea019	1	52	153
m	sandbox	0x40244f	1	44	916	o	spp	0x4ea019	1	51	152
m	sandbox	0x40244f	1	11	914	o	spp	0x4ea019	1	60	151
m	sandbox	0x40244f	1	35	911	o	spp	0x4ea019	1	39	151
m	sandbox	0x40244f	1	26	902	o	spp	0x4ea019	1	58	150
m	sandbox	0x40244f	1	7	901	o	spp	0x4ea019	1	34	150
m	sandbox	0x40244f	1	53	900	o	spp	0x4ea019	1	31	149
m	sandbox	0x40244f	1	16	899	o	spp	0x4ea019	1	57	147
m	sandbox	0x40244f	1	60	898	o	spp	0x550ff8	-17	44	147
m	sandbox	0x40244f	1	17	898	o	spp	0x4ea019	1	35	147
m	sandbox	0x40244f	1	48	894	o	spp	0x4ea019	1	6	146
m	sandbox	0x40244f	1	29	893	o	spp	0x4ea019	1	32	146
m	sandbox	0x40244f	1	54	892	o	spp	0x4ea019	1	19	146
m	sandbox	0x40244f	1	49	889	o	spp	0x4ea019	1	54	144
m	sandbox	0x40244f	1	36	888	o	spp	0x4ea019	1	33	144
m	sandbox	0x40244f	1	25	887	o	spp	0x4ea019	1	42	144
m	sandbox	0x40244f	1	40	886	o	spp	0x4ea019	1	47	143
m	sandbox	0x40244f	1	45	886	o	spp	0x4ea019	1	20	143
m	sandbox	0x40244f	1	31	880	o	spp	0x4ea019	1	28	142
m	sandbox	0x40244f	1	43	879	o	spp	0x4ea019	1	14	142
m	sandbox	0x40244f	1	12	877	o	spp	0x4ea019	1	18	142
m	sandbox	0x40244f	1	21	870	o	spp	0x550ff8	1	63	142
m	sandbox	0x40244f	1	47	867	o	spp	0x4ea019	1	29	141
m	sandbox	0x40244f	1	37	866	o	spp	0x54fec	18	61	141
m	sandbox	0x40244f	1	52	865	o	spp	0x4ea019	1	5	140
m	sandbox	0x40244f	1	23	865	o	spp	0x4ea019	1	37	139
m	sandbox	0x40244f	1	18	865	o	spp	0x4ea019	1	10	139
m	sms	0x40244f	1	32	1,405	o	streamer	0x4ea019	1	4	169
m	sms	0x40244f	1	5	1,273	o	streamer	0x4ea019	1	4	169
m	sms	0x40244f	1	58	1,260	o	streamer	0x54fec	18	2	159
m	sms	0x40244f	1	56	1,254	o	streamer	0x4ea019	1	61	141
m	sms	0x40244f	1	61	1,243	o	streamer	0x550ff8	-17	1	132
m	sms	0x40244f	1	51	1,241	o	streamer	0x54fec	18	2	130
m	sms	0x40244f	1	13	1,240	o	streamer	0x54fec	18	46	123
m	sms	0x40244f	1	15	1,237	o	streamer	0x54fec	18	43	122
m	sms	0x40244f	1	8	1,236	o	streamer	0x54fec	18	36	121
m	sms	0x40244f	1	59	1,230	o	streamer	0x54fec	18	51	120
m	sms	0x40244f	1	10	1,224	o	streamer	0x4ea019	1	35	120
m	sms	0x40244f	1	6	1,224	o	streamer	0x54fec	18	3	119
m	sms	0x40244f	1	14	1,216	o	streamer	0x54fec	18	19	118
m	sms	0x40244f	1	11	1,214	o	streamer	0x54fec	18	25	117
m	sms	0x40244f	1	44	1,213	o	streamer	0x54fec	18	52	117
m	sms	0x40244f	1	41	1,212	o	streamer	0x54fec	18	62	114
m	sms	0x40244f	1	9	1,212	o	streamer	0x550ff8	-17	24	110
m	sms	0x40244f	1	2	1,211	o	streamer	0x54fec	18	3	108
m	sms	0x40244f	1	55	1,209	o	streamer	0x54fec	18	29	106
m	sms	0x40244f	1	48	1,209	o	streamer	0x54fec	18	45	105
m	sms	0x40244f	1	25	1,201	o	streamer	0x54fec	18	34	105
m	sms	0x40244f	1	46	1,200	o	streamer	0x54fec	18	54	104
m	sms	0x40244f	1	53	1,199	o	streamer	0x54fec	18	58	103
m	sms	0x40244f	1	57	1,191	o	streamer	0x550ff8	-17	57	103
m	sms	0x40244f	1	3	1,190	o	streamer	0x54fec	18	0	103
m	sms	0x40244f	1	45	1,183	o	streamer	0x54fec	18	48	101
m	sms	0x40244f	1	63	1,181	o	streamer	0x4ea019	1	53	101
m	sms	0x40244f	1	26	1,179	o	streamer	0x54fec	18	5	101
m	sms	0x40244f	1	7	1,179	o	streamer	0x54fec	18	33	101
m	sms	0x40244f	1	62	1,178	o	streamer	0x54fec	18	27	101
m	sms	0x40244f	1	16	1,175	o	streamer	0x54fec	18	23	100
m	sms	0x40244f	1	17	1,175	o	streamer	0x54fec	18	63	99
m	sms	0x40244f	1	43	1,173	o	streamer	0x54fec	18	20	97
m	sms	0x40244f	1	4	1,172	o	streamer	0x54fec	18	41	97
m	sms	0x40244f	1	60	1,170	o	streamer	0x550ff8	-17	32	97
m	sms	0x40244f	1	18	1,165	o	streamer	0x54fec	18	19	97
m	sms	0x40244f	1	52	1,164	o	streamer	0x550ff8	-17	47	95
m	sms	0x40244f	1	19	1,158	o	streamer	0x54fec	18	16	95
m	sms	0x40244f	1	47	1,158	o	streamer	0x54fec	18	49	95
m	sms	0x40244f	1	12	1,157	o	streamer	0x550ff8	-17	39	94
m	sms	0x40244f	1	54	1,155	o	streamer	0x550ff8	-17	7	93
m	sms	0x40244f	1	21	1,155	o	streamer	0x550ff8	-17	26	93
m	sms	0x40244f	1	31	1,152	o	streamer	0x550ff8	-17	17	91
m	sms	0x40244f	1	49	1,150	o	streamer	0x54fec	18	8	90
m	sms	0x40244f	1	35	1,150	o	streamer	0x54fec	18	26	90
m	sms	0x40244f	1	34	1,149	o	streamer	0x550ff8	-17	60	89
m	sms	0x40244f	1	50	1,146	o	streamer	0x550ff8	-17	12	89
m	sms	0x40244f	1	23	1,146	o	streamer	0x54fec	18	1	89
m	sms	0x40244f	1	22	1,144	o	streamer	0x54fec	18	55	88
m	sms	0x40244f	1	36	1,142	o	streamer	0x54fec	18	28	88
m	spp	0x40244f	1	5	1,694	o	stride	0x4ea019	1	22	88
m	spp	0x40244f	1	13	1,655	o	stride	0x4ea019	1	43	182
m	spp	0x40244f	1	51	1,651	o	stride	0x4ea019	1	4	180
m	spp	0x40244f	1	44	1,647	o	stride	0x4ea019	1	25	176
m	spp	0x40244f	1	6	1,646	o	stride	0x4ea019	1	11	174
m	spp	0x40244f	1	11	1,640	o	stride	0x4ea019	1	22	169
m	spp	0x40244f	1	41	1,630	o	stride	0x4ea019	1	40	169
m	spp	0x40244f	1	9	1,627	o	stride	0x4ea019	1	8	166
m	spp	0x40244f	1	15	1,624	o	stride	0x4ea019	1	24	166
m	spp	0x40244f	1	58	1,620	o	stride	0x4ea019	1	61	163
m	spp	0x40244f	1	61	1,613	o	stride	0x4ea019	1	30	163
m	spp	0x40244f	1	35	1,612	o	stride	0x4ea019	1	36	162
m	spp	0x40244f	1	34	1,608	o	stride	0x4ea019	1	38	161
m	spp	0x40244f	1	10	1,600	o	stride	0x4ea019	1	45	161
m	spp	0x40244f	1	25	1,600	o	stride	0x4ea019	1	48	161
m	spp	0x40244f	1	14	1,598	o	stride	0x4ea019	1	13	159
m	spp	0x40244f	1	56	1,597	o	stride	0x4ea019	1	2	159
m	spp	0x40244f	1	55	1,597	o	stride	0x4ea019	1	50	156
m	spp	0x40244f	1	37	1,593	o	stride	0x4ea019	1	7	156
m	spp	0x40244f	1	8	1,586	o	stride	0x4ea019	1	16	156
m	spp	0x40244f	1	46	1,584	o	stride	0x4ea019	1	15	155
m	spp	0x40244f	1	48	1,583	o	stride	0x4ea019	1	26	155
m	spp	0x40244f	1	26	1,582	o	stride	0x4ea019	1	21	154
m	spp	0x40244f	1	16	1,574	o	stride	0x4ea019	1	53	153
m	spp	0x40244f	1	57	1,571	o	stride	0x4ea019	1	52	153
m	spp	0x40244f	1	43	1,562	o	stride	0x4ea019	1	51	152
m	spp	0x40244f	1	31	1,558	o	stride	0x4ea019	1	60	151
m	spp	0x40244f	1			o	stride	0x4ea019	1	39	151

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
m	spp	0x40244f	1	53	1,555	o	stride	0x4ea019	1	58	150
m	spp	0x40244f	1	7	1,554	o	stride	0x4ea019	1	34	150
m	spp	0x40244f	1	63	1,552	o	stride	0x4ea019	1	31	149
m	spp	0x40244f	1	4	1,552	o	stride	0x4ea019	1	57	147
m	spp	0x40244f	1	59	1,547	o	stride	0x4ea019	1	32	147
m	spp	0x40244f	1	2	1,546	o	stride	0x4ea019	1	35	147
m	spp	0x40244f	1	60	1,541	o	stride	0x4ea019	1	6	146
m	spp	0x40244f	1	24	1,541	o	stride	0x4ea019	1	19	146
m	spp	0x40244f	1	3	1,540	o	stride	0x550ff8	-17	44	146
m	spp	0x40244f	1	45	1,540	o	stride	0x4ea019	1	54	144
m	spp	0x40244f	1	47	1,539	o	stride	0x4ea019	1	33	144
m	spp	0x40244f	1	42	1,537	o	stride	0x4ea019	1	42	144
m	spp	0x40244f	1	36	1,537	o	stride	0x4ea019	1	20	144
m	spp	0x40244f	1	62	1,536	o	stride	0x4ea019	1	47	143
m	spp	0x40244f	1	54	1,535	o	stride	0x4ea019	1	28	142
m	spp	0x40244f	1	18	1,535	o	stride	0x4ea019	1	14	142
m	spp	0x40244f	1	40	1,533	o	stride	0x4ea019	1	18	142
m	spp	0x40244f	1	33	1,533	o	stride	0x550ff8	1	63	142
m	spp	0x40244f	1	12	1,531	o	stride	0x4ea019	1	5	141
m	spp	0x40244f	1	17	1,531	o	stride	0x4ea019	1	29	141
m	spp	0x40244f	1	23	1,521	o	stride	0x54efc	18	61	141
m	spp	0x40244f	1	52	1,515	o	stride	0x4ea019	1	37	139
m	spp	0x40244f	1	19	1,514	o	stride	0x4ea019	1	10	139
m	streamer	0x40244f	1	5	1,657	o	J28	0x54efc	18	61	139
m	streamer	0x40244f	1	6	1,599	o	J28	0x550ff8	-17	44	138
m	streamer	0x40244f	1	2	1,570	o	J28	0x550ff8	1	53	134
m	streamer	0x40244f	1	9	1,561	o	J28	0x550ff8	-17	2	127
m	streamer	0x40244f	1	3	1,557	o	J28	0x550ff8	-17	18	126
m	streamer	0x40244f	1	13	1,544	o	J28	0x550ff8	1	47	125
m	streamer	0x40244f	1	11	1,541	o	J28	0x550ff8	-17	29	124
m	streamer	0x40244f	1	4	1,535	o	J28	0x550ff8	-17	19	124
m	streamer	0x40244f	1	8	1,530	o	J28	0x550ff8	1	62	124
m	streamer	0x40244f	1	10	1,526	o	J28	0x550ff8	-17	34	122
m	streamer	0x40244f	1	15	1,507	o	J28	0x550ff8	-17	26	122
m	streamer	0x40244f	1	14	1,488	o	J28	0x54efc	18	36	121
m	streamer	0x40244f	1	7	1,488	o	J28	0x54efc	18	43	121
m	streamer	0x40244f	1	1	1,483	o	J28	0x550ff8	1	63	120
m	streamer	0x40244f	1	16	1,453	o	J28	0x54efc	18	35	120
m	streamer	0x40244f	1	25	1,443	o	J28	0x54efc	18	19	119
m	streamer	0x40244f	1	41	1,440	o	J28	0x550ff8	1	57	119
m	streamer	0x40244f	1	51	1,434	o	J28	0x54efc	18	46	119
m	streamer	0x40244f	1	34	1,428	o	J28	0x550ff8	1	54	119
m	streamer	0x40244f	1	12	1,425	o	J28	0x550ff8	-17	35	117
m	streamer	0x40244f	1	37	1,425	o	J28	0x550ff8	1	60	117
m	streamer	0x40244f	1	26	1,423	o	J28	0x550ff8	1	52	116
m	streamer	0x40244f	1	44	1,422	o	J28	0x54efc	18	51	116
m	streamer	0x40244f	1	58	1,421	o	J28	0x54efc	18	25	114
m	streamer	0x40244f	1	35	1,418	o	J28	0x54efc	18	62	114
m	streamer	0x40244f	1	17	1,412	o	J28	0x550ff8	-17	45	114
m	streamer	0x40244f	1	46	1,402	o	J28	0x550ff8	-17	7	113
m	streamer	0x40244f	1	56	1,399	o	J28	0x54efc	18	52	113
m	streamer	0x40244f	1	24	1,399	o	J28	0x550ff8	1	55	113
m	streamer	0x40244f	1	61	1,397	o	J28	0x54efc	18	24	110
m	streamer	0x40244f	1	18	1,395	o	J28	0x550ff8	-17	12	109
m	streamer	0x40244f	1	23	1,384	o	J28	0x550ff8	1	56	109
m	streamer	0x40244f	1	21	1,384	o	J28	0x550ff8	-17	3	108
m	streamer	0x40244f	1	19	1,381	o	J28	0x550ff8	1	58	107
m	streamer	0x40244f	1	55	1,377	o	J28	0x550ff8	-17	8	107
m	streamer	0x40244f	1	31	1,373	o	J28	0x550ff8	1	48	107
m	streamer	0x40244f	1	47	1,367	o	J28	0x54efc	18	29	106
m	streamer	0x40244f	1	48	1,364	o	J28	0x550ff8	-17	41	106
m	streamer	0x40244f	1	33	1,362	o	J28	0x550ff8	-17	28	106
m	streamer	0x40244f	1	36	1,358	o	J28	0x550ff8	-17	10	106
m	streamer	0x40244f	1	43	1,358	o	J28	0x550ff8	-17	17	106
m	streamer	0x40244f	1	42	1,357	o	J28	0x550ff8	-17	36	105
m	streamer	0x40244f	1	63	1,356	o	J28	0x550ff8	-17	16	105
m	streamer	0x40244f	1	22	1,349	o	J28	0x550ff8	1	61	105
m	streamer	0x40244f	1	57	1,349	o	J28	0x550ff8	1	49	105
m	streamer	0x40244f	1	28	1,346	o	J28	0x54efc	18	34	105
m	streamer	0x40244f	1	45	1,346	o	J28	0x550ff8	-17	31	104
m	streamer	0x40244f	1	40	1,343	o	J28	0x54efc	18	54	104
m	streamer	0x40244f	1	29	1,341	o	J28	0x550ff8	-17	22	104
m	streamer	0x40244f	1	53	1,341	o	J28	0x54efc	18	45	104
m	stride	0x40244f	1	5	1,722	o	J27	0x550ff8	-17	2	127
m	stride	0x40244f	1	51	1,694	o	J27	0x550ff8	-17	34	124
m	stride	0x40244f	1	13	1,690	o	J27	0x550ff8	-17	44	123
m	stride	0x40244f	1	6	1,682	o	J27	0x550ff8	-17	29	121
m	stride	0x40244f	1	11	1,678	o	J27	0x54efc	18	19	119
m	stride	0x40244f	1	44	1,668	o	J27	0x54efc	18	61	118
m	stride	0x40244f	1	41	1,665	o	J27	0x550ff8	-17	18	116
m	stride	0x40244f	1	58	1,658	o	J27	0x550ff8	-17	45	109
m	stride	0x40244f	1	9	1,656	o	J27	0x550ff8	-17	3	108
m	stride	0x40244f	1	15	1,652	o	J27	0x550ff8	-17	35	108
m	stride	0x40244f	1	35	1,650	o	J27	0x550ff8	-17	7	106
m	stride	0x40244f	1	61	1,645	o	J27	0x550ff8	-17	19	106
m	stride	0x40244f	1	25	1,637	o	J27	0x550ff8	-17	26	106
m	stride	0x40244f	1	56	1,636	o	J27	0x550ff8	-17	28	105
m	stride	0x40244f	1	10	1,635	o	J27	0x550ff8	1	53	103
m	stride	0x40244f	1	14	1,635	o	J27	0x550ff8	-17	8	103
m	stride	0x40244f	1	34	1,634	o	J27	0x550ff8	-17	0	103
m	stride	0x40244f	1	2	1,626	o	J27	0x550ff8	-17	16	102
m	stride	0x40244f	1	26	1,626	o	J27	0x550ff8	-17	36	101
m	stride	0x40244f	1	37	1,623	o	J27	0x54efc	18	20	99
m	stride	0x40244f	1	46	1,623	o	J27	0x416e72	1	61	99
m	stride	0x40244f	1	48	1,622	o	J27	0x54efc	18	24	97
m	stride	0x40244f	1	8	1,617	o	J27	0x550ff8	-17	10	97
m	stride	0x40244f	1	55	1,613	o	J27	0x550ff8	-17	17	97
m	stride	0x40244f	1	3	1,609	o	J27	0x550ff8	-17	40	95
m	stride	0x40244f	1	16	1,607	o	J27	0x550ff8	-17	12	93
m	stride	0x40244f	1	57	1,598	o	J27	0x54efc	18	53	93
m	stride	0x40244f	1	31	1,595	o	J27	0x550ff8	1	62	93
m	stride	0x40244f	1	53	1,594	o	J27	0x550ff8	-17	41	92
m	stride	0x40244f	1	63	1,588	o	J27	0x54efc	18	23	91
m	stride	0x40244f	1	43	1,582	o	J27	0x550ff8	-17	43	91
m	stride	0x40244f	1	7	1,581	o	J27	0x550ff8	-17	21	91
m	stride	0x40244f	1	4	1,581	o	J27	0x550ff8	-17	15	91
m	stride	0x40244f	1	18	1,579	o	J27	0x550ff8	1	63	91
m	stride	0x40244f	1	59	1,576	o	J27	0x550ff8	1	54	89
m	stride	0x40244f	1	45	1,575	o	J27	0x550ff8	-17	1	89
m	stride	0x40244f	1	36	1,573	o	J27	0x550ff8	-17	30	88
m	stride	0x40244f	1	40	1,570	o	J27	0x550ff8	-17	25	88
m	stride	0x40244f	1	54	1,569	o	J27	0x550ff8	-17	46	88
m	stride	0x40244f	1	24	1,569	o	J27	0x54efc	18	41	87
m	stride	0x40244f	1	47	1,568	o	J27	0x550ff8	-17	31	87
m	stride	0x40244f	1	33	1,567	o	J27	0x550ff8	-17	6	86
m	stride	0x40244f	1	42	1,566	o	J27	0x550ff8	1	47	86

tr ID	PC	Δ	off	resid	tr ID	PC	Δ	off	resid
m stride	0x40244f	1	62	1,564	o J27	0x54fecf	18	18	85
m stride	0x40244f	1	60	1,563	o J27	0x550ff8	1	61	85
m stride	0x40244f	1	12	1,563	o J27	0x416e72	1	39	85
m stride	0x40244f	1	23	1,563	o J27	0x550ff8	-17	38	84
m stride	0x40244f	1	17	1,559	o J27	0x550ff8	-17	11	84
m stride	0x40244f	1	21	1,549	o J27	0x550ff8	-17	22	84
m stride	0x40244f	1	19	1,548	o J27	0x550ff8	-17	32	84
m J07	0x40244f	1	9	1,113	o J29	0x54fecf	18	61	141
m J07	0x40244f	1	10	1,104	o J29	0x550ff8	-17	44	136
m J07	0x40244f	1	11	1,098	o J29	0x550ff8	1	53	130
m J07	0x40244f	1	34	1,098	o J29	0x550ff8	1	63	130
m J07	0x40244f	1	51	1,093	o J29	0x550ff8	-17	2	129
m J07	0x40244f	1	46	1,093	o J29	0x550ff8	1	62	127
m J07	0x40244f	1	5	1,089	o J29	0x4ea019	1	4	126
m J07	0x40244f	1	43	1,089	o J29	0x550ff8	1	47	124
m J07	0x40244f	1	13	1,088	o J29	0x54fecf	18	46	122
m J07	0x40244f	1	25	1,088	o J29	0x54fecf	18	43	122
m J07	0x40244f	1	16	1,087	o J29	0x54fecf	18	36	121
m J07	0x40244f	1	35	1,085	o J29	0x54fecf	18	35	121
m J07	0x40244f	1	6	1,084	o J29	0x550ff8	-17	18	121
m J07	0x40244f	1	3	1,078	o J29	0x550ff8	-17	29	120
m J07	0x40244f	1	41	1,074	o J29	0x54fecf	18	51	120
m J07	0x40244f	1	2	1,072	o J29	0x550ff8	-17	34	120
m J07	0x40244f	1	31	1,072	o J29	0x550ff8	-17	19	120
m J07	0x40244f	1	37	1,069	o J29	0x54fecf	18	19	119
m J07	0x40244f	1	55	1,067	o J29	0x550ff8	1	52	119
m J07	0x40244f	1	14	1,062	o J29	0x54fecf	18	25	118
m J07	0x40244f	1	26	1,062	o J29	0x550ff8	-17	26	118
m J07	0x40244f	1	62	1,061	o J29	0x54fecf	18	52	117
m J07	0x40244f	1	33	1,057	o J29	0x550ff8	1	60	117
m J07	0x40244f	1	42	1,055	o J29	0x550ff8	1	54	116
m J07	0x40244f	1	44	1,046	o J29	0x550ff8	-17	7	115
m J07	0x40244f	1	57	1,045	o J29	0x54fecf	18	62	114
m J07	0x40244f	1	54	1,040	o J29	0x550ff8	1	57	113
m J07	0x40244f	1	17	1,039	o J29	0x550ff8	1	51	113
m J07	0x40244f	1	53	1,038	o J29	0x550ff8	-17	8	111
m J07	0x40244f	1	15	1,035	o J29	0x550ff8	1	56	111
m J07	0x40244f	1	59	1,035	o J29	0x54fecf	18	24	110
m J07	0x40244f	1	49	1,035	o J29	0x550ff8	-17	35	110
m J07	0x40244f	1	38	1,034	o J29	0x550ff8	1	55	109
m J07	0x40244f	1	29	1,034	o J29	0x550ff8	-17	3	108
m J07	0x40244f	1	23	1,033	o J29	0x4ea019	1	36	108
m J07	0x40244f	1	58	1,030	o J29	0x4ea019	1	35	108
m J07	0x40244f	1	47	1,029	o J29	0x550ff8	1	48	108
m J07	0x40244f	1	21	1,029	o J29	0x550ff8	1	58	107
m J07	0x40244f	1	1	1,024	o J29	0x4ea019	1	52	107
m J07	0x40244f	1	4	1,018	o J29	0x4ea019	1	13	106
m J07	0x40244f	1	52	1,014	o J29	0x54fecf	18	29	106
m J07	0x40244f	1	61	1,013	o J29	0x550ff8	-17	45	106
m J07	0x40244f	1	18	1,013	o J29	0x550ff8	-17	16	105
m J07	0x40244f	1	63	1,013	o J29	0x54fecf	18	45	105
m J07	0x40244f	1	56	1,011	o J29	0x54fecf	18	34	105
m J07	0x40244f	1	48	1,010	o J29	0x54fecf	18	54	104
m J07	0x40244f	1	19	1,005	o J29	0x550ff8	-17	28	104
m J07	0x40244f	1	36	1,004	o J29	0x550ff8	-17	36	103
m J07	0x40244f	1	8	1,002	o J29	0x54fecf	18	58	103
m J07	0x40244f	1	7	999	o J29	0x54fecf	18	57	103
m J06	0x40244f	1	5	1,136	o J26	0x550ff8	-17	44	133
m J06	0x40244f	1	13	1,113	o J26	0x550ff8	1	53	132
m J06	0x40244f	1	35	1,112	o J26	0x550ff8	-17	29	131
m J06	0x40244f	1	11	1,105	o J26	0x550ff8	-17	34	126
m J06	0x40244f	1	51	1,103	o J26	0x550ff8	-17	2	125
m J06	0x40244f	1	6	1,102	o J26	0x550ff8	-17	18	125
m J06	0x40244f	1	58	1,095	o J26	0x54fecf	18	61	124
m J06	0x40244f	1	61	1,089	o J26	0x550ff8	-17	26	123
m J06	0x40244f	1	9	1,087	o J26	0x550ff8	1	47	120
m J06	0x40244f	1	3	1,086	o J26	0x550ff8	-17	35	119
m J06	0x40244f	1	2	1,085	o J26	0x54fecf	18	19	118
m J06	0x40244f	1	25	1,083	o J26	0x550ff8	1	63	118
m J06	0x40244f	1	47	1,082	o J26	0x550ff8	-17	8	117
m J06	0x40244f	1	55	1,079	o J26	0x550ff8	-17	19	116
m J06	0x40244f	1	10	1,074	o J26	0x550ff8	-17	45	115
m J06	0x40244f	1	53	1,072	o J26	0x550ff8	1	54	112
m J06	0x40244f	1	15	1,072	o J26	0x550ff8	1	62	111
m J06	0x40244f	1	63	1,070	o J26	0x550ff8	1	60	108
m J06	0x40244f	1	44	1,067	o J26	0x550ff8	-17	3	107
m J06	0x40244f	1	62	1,066	o J26	0x550ff8	1	52	107
m J06	0x40244f	1	41	1,065	o J26	0x550ff8	1	51	107
m J06	0x40244f	1	43	1,063	o J26	0x550ff8	-17	28	107
m J06	0x40244f	1	56	1,062	o J26	0x550ff8	-17	10	107
m J06	0x40244f	1	24	1,058	o J26	0x550ff8	-17	41	106
m J06	0x40244f	1	14	1,054	o J26	0x550ff8	1	56	106
m J06	0x40244f	1	37	1,053	o J26	0x550ff8	1	55	106
m J06	0x40244f	1	42	1,053	o J26	0x550ff8	-17	7	105
m J06	0x40244f	1	34	1,049	o J26	0x550ff8	-17	16	105
m J06	0x40244f	1	31	1,048	o J26	0x550ff8	-17	31	104
m J06	0x40244f	1	40	1,046	o J26	0x550ff8	-17	12	104
m J06	0x40244f	1	57	1,043	o J26	0x550ff8	-17	36	104
m J06	0x40244f	1	8	1,042	o J26	0x550ff8	1	48	104
m J06	0x40244f	1	16	1,042	o J26	0x550ff8	-17	17	104
m J06	0x40244f	1	52	1,040	o J26	0x550ff8	-17	22	103
m J06	0x40244f	1	26	1,039	o J26	0x550ff8	-17	40	103
m J06	0x40244f	1	4	1,039	o J26	0x550ff8	-17	32	102
m J06	0x40244f	1	60	1,034	o J26	0x550ff8	1	57	101
m J06	0x40244f	1	7	1,034	o J26	0x550ff8	-17	46	101
m J06	0x40244f	1	59	1,032	o J26	0x550ff8	1	49	100
m J06	0x40244f	1	33	1,031	o J26	0x416e72	1	61	100
m J06	0x40244f	1	17	1,030	o J26	0x550ff8	-17	30	99
m J06	0x40244f	1	45	1,027	o J26	0x54fecf	18	20	99
m J06	0x40244f	1	19	1,025	o J26	0x550ff8	1	58	99
m J06	0x40244f	1	46	1,024	o J26	0x550ff8	-17	0	99
m J06	0x40244f	1	23	1,024	o J26	0x550ff8	-17	43	98
m J06	0x40244f	1	48	1,023	o J26	0x54fecf	18	24	98
m J06	0x40244f	1	54	1,022	o J26	0x550ff8	-17	37	97
m J06	0x40244f	1	36	1,021	o J26	0x550ff8	1	61	97
m J06	0x40244f	1	49	1,021	o J26	0x550ff8	-17	9	96
m J06	0x40244f	1	21	1,021	o J26	0x550ff8	-17	6	93
m J08	0x40244f	1	35	1,155	o J25	0x550ff8	-17	44	147
m J08	0x40244f	1	15	1,146	o J25	0x550ff8	1	63	142
m J08	0x40244f	1	5	1,138	o J25	0x54fecf	18	61	141
m J08	0x40244f	1	42	1,120	o J25	0x550ff8	1	53	135
m J08	0x40244f	1	44	1,120	o J25	0x550ff8	1	47	133
m J08	0x40244f	1	61	1,111	o J25	0x550ff8	1	62	132
m J08	0x40244f	1	46	1,105	o J25	0x550ff8	-17	29	131
m J08	0x40244f	1	36	1,105	o J25	0x550ff8	-17	2	130
m J08	0x40244f	1	43	1,099	o J25	0x550ff8	-17	18	127

tr ID	PC	Δ	off	resid	tr ID	PC	Δ	off	resid
m J08	0x40244f	1	3	1,092	o J25	0x550ff8	-17	34	126
m J08	0x40244f	1	55	1,086	o J25	0x550ff8	-17	19	126
m J08	0x40244f	1	41	1,085	o J25	0x550ff8	1	52	125
m J08	0x40244f	1	25	1,084	o J25	0x550ff8	1	57	124
m J08	0x40244f	1	47	1,082	o J25	0x54efc	18	46	123
m J08	0x40244f	1	2	1,075	o J25	0x550ff8	1	60	123
m J08	0x40244f	1	21	1,075	o J25	0x550ff8	-17	26	123
m J08	0x40244f	1	23	1,071	o J25	0x550ff8	1	54	122
m J08	0x40244f	1	16	1,070	o J25	0x54efc	18	43	122
m J08	0x40244f	1	58	1,060	o J25	0x54efc	18	36	121
m J08	0x40244f	1	6	1,050	o J25	0x54efc	18	35	121
m J08	0x40244f	1	8	1,048	o J25	0x54efc	18	51	120
m J08	0x40244f	1	56	1,044	o J25	0x54efc	18	19	119
m J08	0x40244f	1	10	1,042	o J25	0x550ff8	-17	35	119
m J08	0x40244f	1	62	1,042	o J25	0x54efc	18	25	118
m J08	0x40244f	1	13	1,038	o J25	0x550ff8	-17	8	118
m J08	0x40244f	1	31	1,028	o J25	0x550ff8	1	51	118
m J08	0x40244f	1	37	1,023	o J25	0x54efc	18	52	117
m J08	0x40244f	1	11	1,019	o J25	0x550ff8	-17	45	116
m J08	0x40244f	1	24	1,016	o J25	0x550ff8	-17	7	116
m J08	0x40244f	1	9	1,016	o J25	0x550ff8	1	55	116
m J08	0x40244f	1	59	1,015	o J25	0x54efc	18	62	114
m J08	0x40244f	1	60	1,014	o J25	0x550ff8	1	58	113
m J08	0x40244f	1	27	1,014	o J25	0x550ff8	1	56	113
m J08	0x40244f	1	26	1,012	o J25	0x550ff8	1	48	112
m J08	0x40244f	1	17	1,006	o J25	0x54efc	18	24	110
m J08	0x40244f	1	51	998	o J25	0x550ff8	1	61	110
m J08	0x40244f	1	22	991	o J25	0x550ff8	-17	12	109
m J08	0x40244f	1	50	990	o J25	0x550ff8	-17	16	109
m J08	0x40244f	1	63	988	o J25	0x550ff8	-17	17	109
m J08	0x40244f	1	14	980	o J25	0x550ff8	-17	3	108
m J08	0x40244f	1	30	974	o J25	0x550ff8	-17	36	108
m J08	0x40244f	1	12	971	o J25	0x550ff8	-17	28	108
m J08	0x40244f	1	52	969	o J25	0x550ff8	1	49	108
m J08	0x40244f	1	7	966	o J25	0x550ff8	-17	10	107
m J08	0x40244f	1	29	961	o J25	0x550ff8	-17	31	106
m J08	0x40244f	1	49	960	o J25	0x54efc	18	29	106
m J08	0x40244f	1	54	958	o J25	0x550ff8	-17	41	106
m J08	0x40244f	1	19	958	o J25	0x54efc	18	45	105
m J08	0x40244f	1	39	957	o J25	0x54efc	18	34	105
m J08	0x40244f	1	20	955	o J25	0x54efc	18	54	104
m J05	0x40244f	1	51	1,214	o J24	0x550ff8	-17	44	147
m J05	0x40244f	1	5	1,170	o J24	0x550ff8	1	63	142
m J05	0x40244f	1	25	1,163	o J24	0x54efc	18	61	141
m J05	0x40244f	1	8	1,138	o J24	0x550ff8	1	53	135
m J05	0x40244f	1	60	1,136	o J24	0x550ff8	1	47	133
m J05	0x40244f	1	48	1,134	o J24	0x550ff8	1	62	132
m J05	0x40244f	1	2	1,133	o J24	0x550ff8	-17	29	131
m J05	0x40244f	1	44	1,131	o J24	0x550ff8	-17	2	130
m J05	0x40244f	1	15	1,113	o J24	0x550ff8	-17	18	127
m J05	0x40244f	1	6	1,113	o J24	0x550ff8	-17	34	126
m J05	0x40244f	1	17	1,113	o J24	0x550ff8	-17	19	126
m J05	0x40244f	1	13	1,112	o J24	0x550ff8	1	52	125
m J05	0x40244f	1	29	1,110	o J24	0x550ff8	1	57	124
m J05	0x40244f	1	59	1,109	o J24	0x54efc	18	46	123
m J05	0x40244f	1	14	1,105	o J24	0x550ff8	1	60	123
m J05	0x40244f	1	7	1,104	o J24	0x550ff8	-17	26	123
m J05	0x40244f	1	9	1,102	o J24	0x550ff8	1	54	122
m J05	0x40244f	1	12	1,101	o J24	0x54efc	18	43	122
m J05	0x40244f	1	3	1,097	o J24	0x54efc	18	36	121
m J05	0x40244f	1	10	1,096	o J24	0x54efc	18	35	121
m J05	0x40244f	1	11	1,095	o J24	0x54efc	18	51	120
m J05	0x40244f	1	46	1,095	o J24	0x54efc	18	19	119
m J05	0x40244f	1	58	1,089	o J24	0x550ff8	-17	35	119
m J05	0x40244f	1	61	1,086	o J24	0x54efc	18	25	118
m J05	0x40244f	1	16	1,085	o J24	0x550ff8	-17	8	118
m J05	0x40244f	1	26	1,084	o J24	0x550ff8	1	51	118
m J05	0x40244f	1	37	1,083	o J24	0x54efc	18	52	117
m J05	0x40244f	1	56	1,082	o J24	0x550ff8	-17	45	116
m J05	0x40244f	1	24	1,081	o J24	0x550ff8	-17	7	116
m J05	0x40244f	1	34	1,079	o J24	0x550ff8	1	55	116
m J05	0x40244f	1	47	1,075	o J24	0x54efc	18	62	114
m J05	0x40244f	1	63	1,075	o J24	0x550ff8	1	58	113
m J05	0x40244f	1	41	1,074	o J24	0x550ff8	1	56	113
m J05	0x40244f	1	62	1,073	o J24	0x550ff8	1	48	112
m J05	0x40244f	1	31	1,069	o J24	0x54efc	18	24	110
m J05	0x40244f	1	54	1,067	o J24	0x550ff8	1	61	110
m J05	0x40244f	1	22	1,065	o J24	0x4ea019	1	2	109
m J05	0x40244f	1	42	1,064	o J24	0x550ff8	-17	12	109
m J05	0x40244f	1	52	1,059	o J24	0x550ff8	-17	16	109
m J05	0x40244f	1	55	1,059	o J24	0x550ff8	-17	17	109
m J05	0x40244f	1	40	1,055	o J24	0x550ff8	-17	3	108
m J05	0x40244f	1	19	1,054	o J24	0x550ff8	-17	36	108
m J05	0x40244f	1	28	1,053	o J24	0x550ff8	-17	28	108
m J05	0x40244f	1	18	1,053	o J24	0x550ff8	1	49	108
m J05	0x40244f	1	53	1,052	o J24	0x4ea019	1	4	108
m J05	0x40244f	1	20	1,051	o J24	0x550ff8	-17	10	107
m J05	0x40244f	1	35	1,048	o J24	0x550ff8	-17	31	106
m J05	0x40244f	1	43	1,043	o J24	0x54efc	18	29	106
m J05	0x40244f	1	30	1,042	o J24	0x550ff8	-17	41	106
m J05	0x40244f	1	38	1,041	o J24	0x54efc	18	45	105
m J03	0x40244f	1	6	1,317	o J23	0x550ff8	-17	44	133
m J03	0x40244f	1	11	1,313	o J23	0x550ff8	1	53	132
m J03	0x40244f	1	37	1,313	o J23	0x550ff8	-17	29	131
m J03	0x40244f	1	35	1,300	o J23	0x550ff8	-17	34	126
m J03	0x40244f	1	9	1,296	o J23	0x550ff8	-17	2	125
m J03	0x40244f	1	25	1,290	o J23	0x550ff8	-17	18	125
m J03	0x40244f	1	34	1,284	o J23	0x54efc	18	61	124
m J03	0x40244f	1	13	1,278	o J23	0x550ff8	-17	26	123
m J03	0x40244f	1	51	1,277	o J23	0x550ff8	1	47	120
m J03	0x40244f	1	41	1,277	o J23	0x550ff8	-17	35	119
m J03	0x40244f	1	26	1,274	o J23	0x54efc	18	19	118
m J03	0x40244f	1	10	1,272	o J23	0x550ff8	1	63	118
m J03	0x40244f	1	43	1,272	o J23	0x550ff8	-17	8	117
m J03	0x40244f	1	46	1,272	o J23	0x550ff8	-17	19	116
m J03	0x40244f	1	5	1,270	o J23	0x550ff8	-17	45	115
m J03	0x40244f	1	16	1,267	o J23	0x550ff8	1	54	112
m J03	0x40244f	1	42	1,262	o J23	0x550ff8	1	62	111
m J03	0x40244f	1	44	1,258	o J23	0x550ff8	1	60	108
m J03	0x40244f	1	33	1,257	o J23	0x550ff8	-17	3	107
m J03	0x40244f	1	31	1,251	o J23	0x550ff8	1	52	107
m J03	0x40244f	1	14	1,248	o J23	0x550ff8	1	51	107
m J03	0x40244f	1	55	1,247	o J23	0x550ff8	-17	28	107
m J03	0x40244f	1	62	1,241	o J23	0x550ff8	-17	10	107
m J03	0x40244f	1	57	1,240	o J23	0x550ff8	-17	41	106
m J03	0x40244f	1	38	1,236	o J23	0x550ff8	1	56	106

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
m	J03	0x40244f	1	3	1,235	o	J23	0x550ff8	1	55	106
m	J03	0x40244f	1	15	1,234	o	J23	0x550ff8	-17	7	105
m	J03	0x40244f	1	54	1,233	o	J23	0x550ff8	-17	16	105
m	J03	0x40244f	1	53	1,227	o	J23	0x550ff8	-17	31	104
m	J03	0x40244f	1	2	1,226	o	J23	0x550ff8	-17	12	104
m	J03	0x40244f	1	17	1,226	o	J23	0x550ff8	-17	36	104
m	J03	0x40244f	1	58	1,223	o	J23	0x550ff8	1	48	104
m	J03	0x40244f	1	29	1,214	o	J23	0x550ff8	-17	17	104
m	J03	0x40244f	1	47	1,214	o	J23	0x550ff8	-17	22	103
m	J03	0x40244f	1	7	1,212	o	J23	0x550ff8	-17	40	103
m	J03	0x40244f	1	18	1,212	o	J23	0x550ff8	-17	32	102
m	J03	0x40244f	1	56	1,210	o	J23	0x550ff8	1	57	101
m	J03	0x40244f	1	59	1,210	o	J23	0x550ff8	-17	46	101
m	J03	0x40244f	1	63	1,209	o	J23	0x550ff8	1	49	100
m	J03	0x40244f	1	23	1,207	o	J23	0x416e72	1	61	100
m	J03	0x40244f	1	36	1,204	o	J23	0x550ff8	-17	30	99
m	J03	0x40244f	1	48	1,202	o	J23	0x54efcf	18	20	99
m	J03	0x40244f	1	61	1,202	o	J23	0x550ff8	1	58	99
m	J03	0x40244f	1	4	1,200	o	J23	0x550ff8	-17	0	99
m	J03	0x40244f	1	49	1,197	o	J23	0x550ff8	-17	43	98
m	J03	0x40244f	1	21	1,197	o	J23	0x54efcf	18	24	98
m	J03	0x40244f	1	40	1,191	o	J23	0x550ff8	-17	37	97
m	J03	0x40244f	1	50	1,189	o	J23	0x550ff8	1	61	97
m	J03	0x40244f	1	1	1,189	o	J23	0x550ff8	-17	9	96
m	J03	0x40244f	1	60	1,185	o	J23	0x550ff8	-17	6	93
m	J02	0x40244f	1	6	1,231	x	ampm	0x638ed1	1	17	1,747
m	J02	0x40244f	1	13	1,226	x	ampm	0x638ed1	1	62	1,611
m	J02	0x40244f	1	5	1,222	x	ampm	0x638eb0	1	19	1,474
m	J02	0x40244f	1	11	1,208	x	ampm	0x638ed1	1	18	1,381
m	J02	0x40244f	1	35	1,206	x	ampm	0x638eb0	1	52	1,358
m	J02	0x40244f	1	51	1,199	x	ampm	0x638ed1	1	1	1,265
m	J02	0x40244f	1	9	1,189	x	ampm	0x638ed1	1	41	1,255
m	J02	0x40244f	1	2	1,188	x	ampm	0x638ed1	1	12	1,186
m	J02	0x40244f	1	58	1,187	x	ampm	0x638ed1	1	45	1,147
m	J02	0x40244f	1	37	1,180	x	ampm	0x638ed1	1	9	1,134
m	J02	0x40244f	1	3	1,180	x	ampm	0x638ed1	1	6	1,117
m	J02	0x40244f	1	25	1,179	x	ampm	0x638ed1	1	47	1,099
m	J02	0x40244f	1	47	1,175	x	ampm	0x638ed1	1	4	1,081
m	J02	0x40244f	1	53	1,173	x	ampm	0x638eb0	1	47	1,052
m	J02	0x40244f	1	63	1,173	x	ampm	0x638ed1	1	34	1,022
m	J02	0x40244f	1	61	1,170	x	ampm	0x638ec0	1	53	1,008
m	J02	0x40244f	1	40	1,168	x	ampm	0x638eb0	1	46	987
m	J02	0x40244f	1	15	1,164	x	ampm	0x638eb0	1	21	975
m	J02	0x40244f	1	56	1,163	x	ampm	0x638eb0	1	17	963
m	J02	0x40244f	1	44	1,162	x	ampm	0x638ebd	1	19	962
m	J02	0x40244f	1	41	1,162	x	ampm	0x638eb0	1	42	945
m	J02	0x40244f	1	10	1,160	x	ampm	0x638ed1	1	52	925
m	J02	0x40244f	1	62	1,159	x	ampm	0x638ebd	1	25	920
m	J02	0x40244f	1	55	1,155	x	ampm	0x638eb0	1	5	907
m	J02	0x40244f	1	7	1,154	x	ampm	0x638ed1	1	38	896
m	J02	0x40244f	1	43	1,154	x	ampm	0x638ed1	1	23	890
m	J02	0x40244f	1	24	1,152	x	ampm	0x638ed1	1	59	865
m	J02	0x40244f	1	14	1,149	x	ampm	0x638eb0	1	18	863
m	J02	0x40244f	1	26	1,148	x	ampm	0x638eb0	1	10	863
m	J02	0x40244f	1	42	1,148	x	ampm	0x638ed1	1	57	846
m	J02	0x40244f	1	8	1,145	x	ampm	0x638ebd	1	59	841
m	J02	0x40244f	1	16	1,142	x	ampm	0x638eb0	1	20	838
m	J02	0x40244f	1	4	1,142	x	ampm	0x638ed1	1	7	777
m	J02	0x40244f	1	34	1,139	x	ampm	0x638eb0	1	14	776
m	J02	0x40244f	1	38	1,136	x	ampm	0x638ed1	1	25	773
m	J02	0x40244f	1	52	1,133	x	ampm	0x638ec0	1	34	771
m	J02	0x40244f	1	33	1,133	x	ampm	0x638ed1	1	3	769
m	J02	0x40244f	1	31	1,132	x	ampm	0x638ed1	1	40	766
m	J02	0x40244f	1	17	1,132	x	ampm	0x638ed1	1	58	759
m	J02	0x40244f	1	57	1,131	x	ampm	0x638eb0	1	11	743
m	J02	0x40244f	1	48	1,127	x	ampm	0x638eb0	1	58	738
m	J02	0x40244f	1	45	1,126	x	ampm	0x638eb0	1	1	730
m	J02	0x40244f	1	60	1,125	x	ampm	0x638ec0	1	17	725
m	J02	0x40244f	1	1	1,122	x	ampm	0x638eb0	1	6	676
m	J02	0x40244f	1	18	1,119	x	ampm	0x638eb0	1	33	653
m	J02	0x40244f	1	59	1,117	x	ampm	0x638ed1	1	49	643
m	J02	0x40244f	1	46	1,116	x	ampm	0x638ed1	1	21	630
m	J02	0x40244f	1	21	1,115	x	ampm	0x638ed1	1	33	626
m	J02	0x40244f	1	36	1,112	x	ampm	0x638ec0	1	39	624
m	J02	0x40244f	1	23	1,111	x	ampm	0x638ed1	1	50	602
m	J04	0x40244f	1	44	1,383	x	ipcp	0x638ed1	1	17	1,747
m	J04	0x40244f	1	5	1,383	x	ipcp	0x638ed1	1	62	1,611
m	J04	0x40244f	1	15	1,371	x	ipcp	0x638eb0	1	19	1,474
m	J04	0x40244f	1	35	1,370	x	ipcp	0x638ed1	1	18	1,381
m	J04	0x40244f	1	61	1,356	x	ipcp	0x638eb0	1	52	1,358
m	J04	0x40244f	1	42	1,355	x	ipcp	0x638ed1	1	34	1,279
m	J04	0x40244f	1	36	1,355	x	ipcp	0x638ed1	1	1	1,265
m	J04	0x40244f	1	46	1,347	x	ipcp	0x638ed1	1	41	1,258
m	J04	0x40244f	1	41	1,340	x	ipcp	0x638ed1	1	12	1,191
m	J04	0x40244f	1	6	1,337	x	ipcp	0x638ed1	1	4	1,184
m	J04	0x40244f	1	25	1,336	x	ipcp	0x638ed1	1	45	1,147
m	J04	0x40244f	1	55	1,329	x	ipcp	0x638ed1	1	6	1,139
m	J04	0x40244f	1	47	1,323	x	ipcp	0x638ed1	1	9	1,135
m	J04	0x40244f	1	58	1,319	x	ipcp	0x638ed1	1	47	1,099
m	J04	0x40244f	1	43	1,316	x	ipcp	0x638eb0	1	47	1,052
m	J04	0x40244f	1	37	1,306	x	ipcp	0x638ec0	1	53	1,034
m	J04	0x40244f	1	3	1,305	x	ipcp	0x638ed1	1	40	1,019
m	J04	0x40244f	1	2	1,298	x	ipcp	0x638eb0	1	46	987
m	J04	0x40244f	1	11	1,296	x	ipcp	0x638eb0	1	21	975
m	J04	0x40244f	1	23	1,292	x	ipcp	0x638eb0	1	17	963
m	J04	0x40244f	1	56	1,291	x	ipcp	0x638ebd	1	19	962
m	J04	0x40244f	1	13	1,289	x	ipcp	0x638eb0	1	42	945
m	J04	0x40244f	1	62	1,286	x	ipcp	0x638ed1	1	52	925
m	J04	0x40244f	1	16	1,285	x	ipcp	0x638ebd	1	25	920
m	J04	0x40244f	1	60	1,284	x	ipcp	0x638eb0	1	5	907
m	J04	0x40244f	1	51	1,284	x	ipcp	0x638ed1	1	38	896
m	J04	0x40244f	1	9	1,271	x	ipcp	0x638ed1	1	23	890
m	J04	0x40244f	1	63	1,270	x	ipcp	0x638ed1	1	57	879
m	J04	0x40244f	1	8	1,264	x	ipcp	0x638ed1	1	59	865
m	J04	0x40244f	1	26	1,262	x	ipcp	0x638eb0	1	18	863
m	J04	0x40244f	1	59	1,259	x	ipcp	0x638eb0	1	10	863
m	J04	0x40244f	1	10	1,258	x	ipcp	0x638ebd	1	59	841
m	J04	0x40244f	1	14	1,252	x	ipcp	0x638eb0	1	20	838
m	J04	0x40244f	1	31	1,251	x	ipcp	0x638eb0	1	14	781
m	J04	0x40244f	1	24	1,248	x	ipcp	0x638ed1	1	7	777
m	J04	0x40244f	1	50	1,246	x	ipcp	0x638ed1	1	25	773
m	J04	0x40244f	1	21	1,243	x	ipcp	0x638ec0	1	34	771
m	J04	0x40244f	1	27	1,238	x	ipcp	0x638ed1	1	3	769
m	J04	0x40244f	1	12	1,238	x	ipcp	0x638ed1	1	58	759
m	J04	0x40244f	1	17	1,234	x	ipcp	0x638ed1	1	33	754
m	J04	0x40244f	1	52	1,228	x	ipcp	0x638eb0	1	11	743

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
m	J04	0x40244f	1	39	1,223	x	ipcp	0x638eb0	1	58	738
m	J04	0x40244f	1	57	1,218	x	ipcp	0x638eb0	1	1	730
m	J04	0x40244f	1	22	1,215	x	ipcp	0x638ec0	1	17	725
m	J04	0x40244f	1	30	1,210	x	ipcp	0x638ed1	1	49	690
m	J04	0x40244f	1	54	1,209	x	ipcp	0x638eb0	1	6	677
m	J04	0x40244f	1	33	1,206	x	ipcp	0x638ec0	1	9	663
m	J04	0x40244f	1	29	1,205	x	ipcp	0x638eb0	1	33	655
m	J04	0x40244f	1	20	1,203	x	ipcp	0x64cc6c	1	40	649
m	J04	0x40244f	1	38	1,198	x	ipcp	0x638ed1	1	21	630
m	J01	0x40244f	1	51	1,386	x	no_pref	0x638ed1	1	17	1,747
m	J01	0x40244f	1	25	1,362	x	no_pref	0x638ed1	1	62	1,611
m	J01	0x40244f	1	5	1,331	x	no_pref	0x638eb0	1	19	1,474
m	J01	0x40244f	1	6	1,330	x	no_pref	0x638ed1	1	18	1,381
m	J01	0x40244f	1	11	1,311	x	no_pref	0x638eb0	1	52	1,358
m	J01	0x40244f	1	48	1,303	x	no_pref	0x638ed1	1	34	1,279
m	J01	0x40244f	1	37	1,296	x	no_pref	0x638ed1	1	1	1,265
m	J01	0x40244f	1	60	1,292	x	no_pref	0x638ed1	1	41	1,258
m	J01	0x40244f	1	44	1,292	x	no_pref	0x638ed1	1	12	1,191
m	J01	0x40244f	1	13	1,287	x	no_pref	0x638ed1	1	4	1,184
m	J01	0x40244f	1	17	1,286	x	no_pref	0x638ed1	1	45	1,147
m	J01	0x40244f	1	9	1,284	x	no_pref	0x638ed1	1	6	1,139
m	J01	0x40244f	1	26	1,282	x	no_pref	0x638ed1	1	9	1,135
m	J01	0x40244f	1	61	1,281	x	no_pref	0x638ed1	1	47	1,099
m	J01	0x40244f	1	8	1,279	x	no_pref	0x638eb0	1	47	1,052
m	J01	0x40244f	1	63	1,276	x	no_pref	0x638ec0	1	53	1,037
m	J01	0x40244f	1	41	1,275	x	no_pref	0x638ed1	1	40	1,019
m	J01	0x40244f	1	15	1,273	x	no_pref	0x638eb0	1	46	987
m	J01	0x40244f	1	14	1,273	x	no_pref	0x638eb0	1	21	975
m	J01	0x40244f	1	2	1,272	x	no_pref	0x638eb0	1	17	963
m	J01	0x40244f	1	29	1,270	x	no_pref	0x638ebd	1	19	962
m	J01	0x40244f	1	16	1,269	x	no_pref	0x638eb0	1	42	945
m	J01	0x40244f	1	58	1,269	x	no_pref	0x638ed1	1	52	925
m	J01	0x40244f	1	40	1,268	x	no_pref	0x638ebd	1	25	920
m	J01	0x40244f	1	7	1,268	x	no_pref	0x638eb0	1	5	907
m	J01	0x40244f	1	12	1,267	x	no_pref	0x638ed1	1	38	896
m	J01	0x40244f	1	34	1,264	x	no_pref	0x638ed1	1	23	890
m	J01	0x40244f	1	56	1,263	x	no_pref	0x638ed1	1	57	879
m	J01	0x40244f	1	24	1,263	x	no_pref	0x638ed1	1	59	865
m	J01	0x40244f	1	18	1,259	x	no_pref	0x638eb0	1	18	863
m	J01	0x40244f	1	42	1,258	x	no_pref	0x638eb0	1	10	863
m	J01	0x40244f	1	46	1,257	x	no_pref	0x638ebd	1	59	841
m	J01	0x40244f	1	54	1,253	x	no_pref	0x638eb0	1	20	838
m	J01	0x40244f	1	22	1,250	x	no_pref	0x638eb0	1	14	781
m	J01	0x40244f	1	62	1,246	x	no_pref	0x638ed1	1	7	777
m	J01	0x40244f	1	38	1,240	x	no_pref	0x638ed1	1	25	773
m	J01	0x40244f	1	28	1,240	x	no_pref	0x638ec0	1	34	771
m	J01	0x40244f	1	3	1,238	x	no_pref	0x638ed1	1	3	769
m	J01	0x40244f	1	47	1,236	x	no_pref	0x638ed1	1	58	759
m	J01	0x40244f	1	31	1,236	x	no_pref	0x638ed1	1	33	754
m	J01	0x40244f	1	59	1,229	x	no_pref	0x638eb0	1	11	743
m	J01	0x40244f	1	55	1,229	x	no_pref	0x638eb0	1	58	738
m	J01	0x40244f	1	50	1,228	x	no_pref	0x638eb0	1	1	730
m	J01	0x40244f	1	43	1,226	x	no_pref	0x638ec0	1	17	725
m	J01	0x40244f	1	35	1,224	x	no_pref	0x638ed1	1	49	690
m	J01	0x40244f	1	10	1,222	x	no_pref	0x638eb0	1	6	677
m	J01	0x40244f	1	53	1,220	x	no_pref	0x638ec0	1	9	663
m	J01	0x40244f	1	45	1,216	x	no_pref	0x638eb0	1	33	655
m	J01	0x40244f	1	52	1,214	x	no_pref	0x64cc6c	1	40	649
m	J01	0x40244f	1	33	1,212	x	no_pref	0x638ed1	1	21	630
l	ampm	0x401cb0	3	3	1,538	x	power7	0x638ed1	1	17	1,747
l	ampm	0x401cb0	65	0	1,191	x	power7	0x638eb0	1	19	1,474
l	ampm	0x401cb0	4	4	941	x	power7	0x638ed1	1	62	1,408
l	ampm	0x401cb0	3	6	920	x	power7	0x638ed1	1	18	1,381
l	ampm	0x401cb0	4	6	920	x	power7	0x638eb0	1	52	1,358
l	ampm	0x401cb0	3	8	920	x	power7	0x638ed1	1	1	1,265
l	ampm	0x401cb0	4	7	917	x	power7	0x638ed1	1	41	1,258
l	ampm	0x401cb0	3	5	917	x	power7	0x638ed1	1	12	1,186
l	ampm	0x401cb0	4	5	916	x	power7	0x638ed1	1	4	1,183
l	ampm	0x401cb0	3	7	916	x	power7	0x638ed1	1	34	1,173
l	ampm	0x401cb0	4	8	914	x	power7	0x638ed1	1	45	1,147
l	ampm	0x401cb0	68	2	899	x	power7	0x638ed1	1	6	1,135
l	ampm	0x401cb0	67	1	899	x	power7	0x638ed1	1	9	1,135
l	ampm	0x401cb0	3	4	886	x	power7	0x638ed1	1	47	1,098
l	ampm	0x401cb0	66	1	723	x	power7	0x638eb0	1	47	1,051
l	ampm	0x401e02	-2	2	590	x	power7	0x638ec0	1	53	1,035
l	ampm	0x401e33	-1	5	582	x	power7	0x638eb0	1	46	986
l	ampm	0x401e33	-1	1	581	x	power7	0x638eb0	1	21	975
l	ampm	0x401cb0	2	2	572	x	power7	0x638eb0	1	17	963
l	ampm	0x401e02	-2	1	554	x	power7	0x638ebd	1	19	962
l	ampm	0x401e02	-2	4	554	x	power7	0x638eb0	1	42	945
l	ampm	0x401e02	-2	5	550	x	power7	0x638ed1	1	52	925
l	ampm	0x401e33	-1	3	549	x	power7	0x638ebd	1	25	920
l	ampm	0x401e02	-2	3	547	x	power7	0x638eb0	1	5	907
l	ampm	0x401e33	-1	4	546	x	power7	0x638ed1	1	38	893
l	ampm	0x401e67	1	2	534	x	power7	0x638ed1	1	23	889
l	ampm	0x401e67	1	1	518	x	power7	0x638ed1	1	57	870
l	ampm	0x401e33	-1	0	509	x	power7	0x638ed1	1	59	865
l	ampm	0x401e33	-1	2	506	x	power7	0x638eb0	1	18	862
l	ampm	0x401e02	-2	0	500	x	power7	0x638eb0	1	10	850
l	ampm	0x401e67	-1	0	400	x	power7	0x638ebd	1	59	841
l	ampm	0x401d14	-1	2	383	x	power7	0x638eb0	1	20	838
l	ampm	0x401cdb	-2	5	370	x	power7	0x638ed1	1	40	810
l	ampm	0x401cdb	-2	3	368	x	power7	0x638eb0	1	14	778
l	ampm	0x401d14	-1	4	366	x	power7	0x638ed1	1	25	773
l	ampm	0x401cdb	-2	4	366	x	power7	0x638ec0	1	34	771
l	ampm	0x401d14	-1	3	365	x	power7	0x638ed1	1	3	769
l	ampm	0x401cdb	-2	1	360	x	power7	0x638ed1	1	58	759
l	ampm	0x401d43	1	2	360	x	power7	0x638eb0	1	58	738
l	ampm	0x401d43	1	1	359	x	power7	0x638eb0	1	1	730
l	ampm	0x401cb0	67	2	342	x	power7	0x638eb0	1	6	676
l	ampm	0x401d14	-1	5	337	x	power7	0x638ed1	1	49	655
l	ampm	0x401d14	-1	0	329	x	power7	0x638ed1	1	33	647
l	ampm	0x401d14	-1	1	327	x	power7	0x638ed1	1	21	630
l	ampm	0x401cdb	-2	2	326	x	power7	0x638ec0	1	9	629
l	ampm	0x401cdb	-2	0	318	x	power7	0x638ec0	1	39	623
l	ampm	0x401cb0	67	0	280	x	power7	0x638eb0	1	7	590
l	ampm	0x401cb0	68	3	279	x	power7	0x638ed1	1	8	574
l	ampm	0x401d43	-1	0	275	x	power7	0x638ed1	1	7	572
l	ampm	0x401d43	1	5	254	x	power7	0x638ec0	1	17	557
l	ipcp	0x401cb0	3	3	1,538	x	sandbox	0x638ed1	1	62	1,322
l	ipcp	0x401cb0	65	0	1,191	x	sandbox	0x638ed1	1	17	1,217
l	ipcp	0x401cb0	4	4	941	x	sandbox	0x638ed1	1	9	1,115
l	ipcp	0x401cb0	4	12	920	x	sandbox	0x638eb0	1	19	1,115
l	ipcp	0x401cb0	4	22	920	x	sandbox	0x638ed1	1	45	1,111
l	ipcp	0x401cb0	4	27	920	x	sandbox	0x638ed1	1	18	1,010
l	ipcp	0x401cb0	4	62	920	x	sandbox	0x638eb0	1	52	992

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
1	ipcp	0x401cb0	4	18	920	x	sandbox	0x638ed1	1	41	985
1	ipcp	0x401cb0	4	38	920	x	sandbox	0x638eb0	1	46	978
1	ipcp	0x401cb0	3	40	920	x	sandbox	0x638ed1	1	12	895
1	ipcp	0x401cb0	4	58	920	x	sandbox	0x638ed1	1	34	872
1	ipcp	0x401cb0	3	60	920	x	sandbox	0x638eb0	1	18	860
1	ipcp	0x401cb0	3	6	920	x	sandbox	0x638ed1	1	57	850
1	ipcp	0x401cb0	4	9	920	x	sandbox	0x638eb0	1	10	840
1	ipcp	0x401cb0	4	24	920	x	sandbox	0x638ebd	1	59	804
1	ipcp	0x401cb0	3	17	920	x	sandbox	0x638ebd	1	25	761
1	ipcp	0x401cb0	3	52	920	x	sandbox	0x638ed1	1	25	749
1	ipcp	0x401cb0	4	55	920	x	sandbox	0x638ed1	1	40	748
1	ipcp	0x401cb0	4	6	920	x	sandbox	0x638ed1	1	1	742
1	ipcp	0x401cb0	3	8	920	x	sandbox	0x638ed1	1	3	736
1	ipcp	0x401cb0	4	11	920	x	sandbox	0x638eb0	1	58	732
1	ipcp	0x401cb0	3	23	920	x	sandbox	0x638ed1	1	4	730
1	ipcp	0x401cb0	4	26	920	x	sandbox	0x638eb0	1	1	718
1	ipcp	0x401cb0	3	28	920	x	sandbox	0x638eb0	1	21	718
1	ipcp	0x401cb0	3	58	919	x	sandbox	0x638eb0	1	14	702
1	ipcp	0x401cb0	3	14	919	x	sandbox	0x638eb0	1	42	692
1	ipcp	0x401cb0	4	17	919	x	sandbox	0x638ed1	1	52	662
1	ipcp	0x401cb0	3	44	919	x	sandbox	0x638ed1	1	47	646
1	ipcp	0x401cb0	3	54	919	x	sandbox	0x638ed1	1	23	644
1	ipcp	0x401cb0	4	13	919	x	sandbox	0x638ed1	1	21	614
1	ipcp	0x401cb0	3	20	919	x	sandbox	0x638ed1	1	49	581
1	ipcp	0x401cb0	4	23	919	x	sandbox	0x638eb0	1	47	555
1	ipcp	0x401cb0	3	55	919	x	sandbox	0x638ed1	1	39	533
1	ipcp	0x401cb0	4	63	919	x	sandbox	0x638eb0	1	5	529
1	ipcp	0x401cb0	3	11	919	x	sandbox	0x638ed1	1	20	523
1	ipcp	0x401cb0	4	14	919	x	sandbox	0x638eb0	1	39	523
1	ipcp	0x401cb0	4	19	919	x	sandbox	0x638ed1	1	7	513
1	ipcp	0x401cb0	3	26	919	x	sandbox	0x638ed1	1	6	505
1	ipcp	0x401cb0	4	34	919	x	sandbox	0x638ebd	1	19	504
1	ipcp	0x401cb0	4	44	919	x	sandbox	0x638ebd	13	41	504
1	ipcp	0x401cb0	3	46	919	x	sandbox	0x638ed1	1	36	498
1	ipcp	0x401cb0	4	54	919	x	sandbox	0x638ebd	117	53	497
1	ipcp	0x401cb0	3	61	919	x	sandbox	0x638ed1	1	8	496
1	ipcp	0x401cb0	4	10	919	x	sandbox	0x638eb0	1	57	496
1	ipcp	0x401cb0	3	22	919	x	sandbox	0x638eb0	1	11	495
1	ipcp	0x401cb0	4	30	919	x	sandbox	0x638ec0	123	60	495
1	ipcp	0x401cb0	3	37	919	x	sandbox	0x638ebd	1	57	494
1	ipcp	0x401cb0	3	18	919	x	sandbox	0x638ebd	1	60	487
1	ipcp	0x401cb0	4	31	919	x	sandbox	0x638ebd	1	3	487
1	ipcp	0x401cb0	3	9	918	x	sandbox	0x638ebd	1	58	486
1	no_pref	0x401cb0	3	3	1,538	x	sms	0x638ed1	1	62	1,584
1	no_pref	0x401cb0	65	0	1,191	x	sms	0x638eb0	1	19	1,359
1	no_pref	0x401cb0	4	4	941	x	sms	0x638eb0	1	52	1,356
1	no_pref	0x401cb0	4	12	920	x	sms	0x638ed1	1	34	1,229
1	no_pref	0x401cb0	4	22	920	x	sms	0x638ed1	1	17	1,223
1	no_pref	0x401cb0	4	27	920	x	sms	0x638ed1	1	12	1,182
1	no_pref	0x401cb0	4	62	920	x	sms	0x638ed1	1	18	1,134
1	no_pref	0x401cb0	4	18	920	x	sms	0x638ed1	1	9	1,120
1	no_pref	0x401cb0	4	38	920	x	sms	0x638ed1	1	45	1,117
1	no_pref	0x401cb0	3	40	920	x	sms	0x638ed1	1	40	1,007
1	no_pref	0x401cb0	4	58	920	x	sms	0x638ed1	1	41	986
1	no_pref	0x401cb0	3	60	920	x	sms	0x638eb0	1	47	985
1	no_pref	0x401cb0	3	6	920	x	sms	0x638ed1	1	47	972
1	no_pref	0x401cb0	4	9	920	x	sms	0x638eb0	1	21	958
1	no_pref	0x401cb0	4	24	920	x	sms	0x638eb0	1	17	957
1	no_pref	0x401cb0	3	17	920	x	sms	0x638ebd	1	19	945
1	no_pref	0x401cb0	3	52	920	x	sms	0x638ed1	1	52	917
1	no_pref	0x401cb0	4	55	920	x	sms	0x638eb0	1	5	900
1	no_pref	0x401cb0	4	6	920	x	sms	0x638ed1	1	38	890
1	no_pref	0x401cb0	3	8	920	x	sms	0x638ebd	1	25	886
1	no_pref	0x401cb0	4	11	920	x	sms	0x638eb0	1	18	863
1	no_pref	0x401cb0	3	23	920	x	sms	0x638ed1	1	57	862
1	no_pref	0x401cb0	4	26	920	x	sms	0x638eb0	1	10	852
1	no_pref	0x401cb0	3	28	920	x	sms	0x638ebd	1	59	804
1	no_pref	0x401cb0	3	58	919	x	sms	0x638ed1	1	4	794
1	no_pref	0x401cb0	3	14	919	x	sms	0x638eb0	1	14	777
1	no_pref	0x401cb0	4	17	919	x	sms	0x638ed1	1	3	754
1	no_pref	0x401cb0	3	44	919	x	sms	0x638ed1	1	23	752
1	no_pref	0x401cb0	3	54	919	x	sms	0x638eb0	1	46	752
1	no_pref	0x401cb0	4	13	919	x	sms	0x638ed1	1	1	744
1	no_pref	0x401cb0	3	20	919	x	sms	0x638eb0	1	11	743
1	no_pref	0x401cb0	4	23	919	x	sms	0x638eb0	1	58	736
1	no_pref	0x401cb0	3	55	919	x	sms	0x638ed1	1	6	729
1	no_pref	0x401cb0	4	63	919	x	sms	0x638ed1	1	25	711
1	no_pref	0x401cb0	3	11	919	x	sms	0x638eb0	1	1	701
1	no_pref	0x401cb0	4	14	919	x	sms	0x638eb0	1	42	688
1	no_pref	0x401cb0	4	19	919	x	sms	0x638ec0	1	9	658
1	no_pref	0x401cb0	3	26	919	x	sms	0x638eb0	1	33	645
1	no_pref	0x401cb0	4	34	919	x	sms	0x638eb0	1	20	628
1	no_pref	0x401cb0	4	44	919	x	sms	0x638ed1	1	21	614
1	no_pref	0x401cb0	3	46	919	x	sms	0x638ec0	-2	39	614
1	no_pref	0x401cb0	4	54	919	x	sms	0x638ed1	1	59	606
1	no_pref	0x401cb0	3	61	919	x	sms	0x638ed1	1	19	588
1	no_pref	0x401cb0	4	10	919	x	sms	0x638ed1	1	49	585
1	no_pref	0x401cb0	3	22	919	x	sms	0x638ed1	1	8	551
1	no_pref	0x401cb0	4	30	919	x	sms	0x638ed1	1	39	539
1	no_pref	0x401cb0	3	37	919	x	sms	0x638eb0	1	41	535
1	no_pref	0x401cb0	3	18	919	x	sms	0x638ed1	1	55	531
1	no_pref	0x401cb0	4	31	919	x	sms	0x638ed1	1	20	528
1	no_pref	0x401cb0	3	9	918	x	sms	0x64cc6c	1	17	520
1	power7	0x401cb0	3	3	1,538	x	spp	0x638ed1	1	17	1,747
1	power7	0x401cb0	65	0	1,191	x	spp	0x638ed1	1	62	1,532
1	power7	0x401cb0	4	4	941	x	spp	0x638ed1	1	18	1,381
1	power7	0x401cb0	4	6	920	x	spp	0x638eb0	1	52	1,358
1	power7	0x401cb0	3	5	917	x	spp	0x638eb0	1	19	1,323
1	power7	0x401cb0	3	8	917	x	spp	0x638ed1	1	41	1,258
1	power7	0x401cb0	4	5	916	x	spp	0x638ed1	1	1	1,231
1	power7	0x401cb0	3	7	916	x	spp	0x638ed1	1	45	1,147
1	power7	0x401cb0	3	6	913	x	spp	0x638ed1	1	6	1,133
1	power7	0x401cb0	68	2	899	x	spp	0x638ed1	1	9	1,120
1	power7	0x401cb0	67	1	899	x	spp	0x638ed1	1	47	1,096
1	power7	0x401cb0	3	4	886	x	spp	0x638ed1	1	4	1,092
1	power7	0x401cb0	66	1	723	x	spp	0x638eb0	1	47	1,052
1	power7	0x401e02	-2	2	591	x	spp	0x638ed1	1	34	1,050
1	power7	0x401e33	-1	1	580	x	spp	0x638ed1	1	12	1,028
1	power7	0x401e67	1	3	578	x	spp	0x638ec0	1	53	987
1	power7	0x401e33	-1	5	574	x	spp	0x638eb0	1	21	967
1	power7	0x401cb0	2	2	572	x	spp	0x638ebd	1	19	961
1	power7	0x401e02	-2	1	554	x	spp	0x638eb0	1	17	957
1	power7	0x401e02	-2	4	554	x	spp	0x638eb0	1	42	945
1	power7	0x401e33	-1	6	549	x	spp	0x638ed1	1	40	941
1	power7	0x401e33	-1	3	548	x	spp	0x638ed1	1	52	925
1	power7	0x401e02	-2	3	547	x	spp	0x638eb0	1	46	917

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
1	power7	0x401e33	-1	4	546	x	spp	0x638ebd	1	25	917
1	power7	0x401e67	1	2	543	x	spp	0x638eb0	1	5	888
1	power7	0x401e67	1	4	542	x	spp	0x638ed1	1	59	865
1	power7	0x401e67	1	5	540	x	spp	0x638eb0	1	10	861
1	power7	0x401e33	-1	7	532	x	spp	0x638eb0	1	18	845
1	power7	0x401e67	1	1	525	x	spp	0x638ebd	1	59	841
1	power7	0x401e33	-1	0	511	x	spp	0x638eb0	1	20	838
1	power7	0x401e33	-1	2	506	x	spp	0x638ed1	1	57	836
1	power7	0x401e02	-2	0	503	x	spp	0x638ed1	1	23	822
1	power7	0x401e67	-1	0	413	x	spp	0x638ed1	1	7	777
1	power7	0x401d14	-1	2	383	x	spp	0x638ed1	1	25	773
1	power7	0x401d14	-1	7	380	x	spp	0x638ed1	1	3	769
1	power7	0x401cdb	-2	3	368	x	spp	0x638ec0	1	34	769
1	power7	0x401d43	1	4	368	x	spp	0x638ed1	1	58	758
1	power7	0x401d14	-1	4	366	x	spp	0x638eb0	1	11	743
1	power7	0x401cdb	-2	4	366	x	spp	0x638eb0	1	58	738
1	power7	0x401d43	1	5	366	x	spp	0x638ed1	1	38	709
1	power7	0x401d14	-1	3	365	x	spp	0x638ed1	1	33	699
1	power7	0x401d14	-1	6	362	x	spp	0x638ed1	1	49	687
1	power7	0x401cdb	-2	1	360	x	spp	0x638ab0	1	6	677
1	power7	0x401d43	1	2	360	x	spp	0x638ab0	1	33	655
1	power7	0x401d43	1	1	356	x	spp	0x638ab0	1	1	653
1	power7	0x401cb0	67	2	342	x	spp	0x64cc6c	1	40	631
1	power7	0x401d14	-1	5	337	x	spp	0x638ed1	1	21	630
1	power7	0x401d14	-1	0	330	x	spp	0x638ec0	1	39	619
1	power7	0x401d14	-1	1	327	x	spp	0x638ec0	1	9	614
1	power7	0x401cdb	-2	2	326	x	spp	0x638eb0	1	14	609
1	sandbox	0x401cb0	65	0	1,191	x	streamer	0x638ed1	1	17	1,747
1	sandbox	0x401cb0	68	2	899	x	streamer	0x638eb0	1	19	1,474
1	sandbox	0x401cb0	67	1	899	x	streamer	0x638ed1	1	62	1,384
1	sandbox	0x401cb0	66	1	723	x	streamer	0x638ed1	1	18	1,381
1	sandbox	0x401e02	-2	0	501	x	streamer	0x638eb0	1	52	1,358
1	sandbox	0x401cdb	-2	0	317	x	streamer	0x638ed1	1	1	1,264
1	sandbox	0x401e33	-1	0	281	x	streamer	0x638ed1	1	41	1,258
1	sandbox	0x401cb0	67	0	280	x	streamer	0x638ed1	1	12	1,186
1	sandbox	0x401cb0	3	3	250	x	streamer	0x638ed1	1	4	1,184
1	sandbox	0x401e67	-1	0	243	x	streamer	0x638ed1	1	34	1,150
1	sandbox	0x401d14	-1	0	230	x	streamer	0x638ed1	1	45	1,147
1	sandbox	0x401d43	-1	0	210	x	streamer	0x638ed1	1	9	1,134
1	sandbox	0x401cb0	66	0	185	x	streamer	0x638ed1	1	6	1,131
1	sandbox	0x401cb0	68	1	179	x	streamer	0x638ed1	1	47	1,098
1	sandbox	0x401d43	1	1	163	x	streamer	0x638ab0	1	47	1,051
1	sandbox	0x401e67	1	1	162	x	streamer	0x638ec0	1	53	1,035
1	sandbox	0x401cb0	68	0	157	x	streamer	0x638ab0	1	46	986
1	sandbox	0x401cb0	2	2	136	x	streamer	0x638eb0	1	21	975
1	sandbox	0x401cb0	4	4	133	x	streamer	0x638eb0	1	17	963
1	sandbox	0x401e02	-2	1	79	x	streamer	0x638ebd	1	19	962
1	sandbox	0x401e33	-1	1	53	x	streamer	0x638eb0	1	42	945
1	sandbox	0x401cb0	4	5	43	x	streamer	0x638ed1	1	52	925
1	sandbox	0x401e02	-2	2	41	x	streamer	0x638ebd	1	25	920
1	sandbox	0x401cb0	68	3	26	x	streamer	0x638eb0	1	5	906
1	sandbox	0x401d37	-2	0	26	x	streamer	0x638ed1	1	38	896
1	sandbox	0x401e67	1	2	24	x	streamer	0x638ed1	1	23	889
1	sandbox	0x401d43	65	0	21	x	streamer	0x638ed1	1	57	867
1	sandbox	0x401cb0	3	4	20	x	streamer	0x638ed1	1	59	865
1	sandbox	0x401e02	-1	0	18	x	streamer	0x638eb0	1	18	862
1	sandbox	0x401d4f	-1	0	16	x	streamer	0x638ebd	1	59	841
1	sandbox	0x401d37	-1	0	15	x	streamer	0x638eb0	1	20	838
1	sandbox	0x401ce6	-2	0	13	x	streamer	0x638eb0	1	14	778
1	sandbox	0x401cb0	67	2	13	x	streamer	0x638ed1	1	25	773
1	sandbox	0x401cb0	3	53	12	x	streamer	0x638ec0	1	34	771
1	sandbox	0x401cdb	-2	1	12	x	streamer	0x638ed1	1	3	769
1	sandbox	0x40209c	-1	42	12	x	streamer	0x638ed1	1	40	766
1	sandbox	0x401e74	-1	0	10	x	streamer	0x638ed1	1	58	759
1	sandbox	0x401cb0	4	16	9	x	streamer	0x638eb0	1	58	738
1	sandbox	0x401d60	-1	0	9	x	streamer	0x638ab0	1	1	730
1	sandbox	0x401cb0	3	54	9	x	streamer	0x638ab0	1	6	676
1	sandbox	0x401cb0	3	5	9	x	streamer	0x638ed1	1	49	643
1	sandbox	0x401cb0	4	45	9	x	streamer	0x638eb0	1	10	642
1	sandbox	0x401cb0	3	52	9	x	streamer	0x638ed1	1	21	630
1	sandbox	0x401cdb	-1	0	8	x	streamer	0x638ed1	1	33	626
1	sandbox	0x401cb0	3	19	8	x	streamer	0x638ec0	1	39	623
1	sandbox	0x401cb0	4	32	8	x	streamer	0x638ec0	1	9	615
1	sandbox	0x401cb0	4	53	8	x	streamer	0x638ed1	1	19	597
1	sandbox	0x401e33	-1	2	8	x	streamer	0x638eb0	1	7	591
1	sandbox	0x401e67	-3	0	8	x	streamer	0x638ed1	1	8	574
1	sandbox	0x401cb0	4	59	7	x	streamer	0x638ed1	1	39	550
1	sms	0x401cb0	65	0	1,191	x	stride	0x638ed1	1	17	1,747
1	sms	0x401cb0	4	34	919	x	stride	0x638ed1	1	62	1,611
1	sms	0x401cb0	4	33	917	x	stride	0x638eb0	1	19	1,474
1	sms	0x401cb0	3	32	915	x	stride	0x638ed1	1	18	1,381
1	sms	0x401cb0	3	33	913	x	stride	0x638eb0	1	52	1,358
1	sms	0x401cb0	4	32	912	x	stride	0x638ed1	1	34	1,279
1	sms	0x401cb0	68	2	899	x	stride	0x638ed1	1	1	1,265
1	sms	0x401cb0	67	1	899	x	stride	0x638ed1	1	41	1,258
1	sms	0x401cb0	66	1	723	x	stride	0x638ed1	1	12	1,191
1	sms	0x401cb0	3	3	311	x	stride	0x638ed1	1	4	1,184
1	sms	0x401cb0	67	0	280	x	stride	0x638ed1	1	45	1,147
1	sms	0x401e02	-2	32	228	x	stride	0x638ed1	1	6	1,139
1	sms	0x401e02	-2	0	205	x	stride	0x638ed1	1	9	1,135
1	sms	0x401e67	-1	0	200	x	stride	0x638ed1	1	47	1,099
1	sms	0x401e33	-1	0	194	x	stride	0x638eb0	1	47	1,052
1	sms	0x401cb0	66	0	185	x	stride	0x638ec0	1	53	1,037
1	sms	0x401e33	-1	32	184	x	stride	0x638ed1	1	40	1,019
1	sms	0x401cb0	68	1	179	x	stride	0x638eb0	1	46	987
1	sms	0x401cb0	2	2	177	x	stride	0x638eb0	1	21	975
1	sms	0x401d43	-1	0	175	x	stride	0x638eb0	1	17	963
1	sms	0x401cdb	-2	32	161	x	stride	0x638ebd	1	19	962
1	sms	0x401cb0	68	0	157	x	stride	0x638eb0	1	42	945
1	sms	0x401cdb	-2	0	152	x	stride	0x638ed1	1	52	925
1	sms	0x401d14	-1	32	152	x	stride	0x638ebd	1	25	920
1	sms	0x401cb0	4	4	151	x	stride	0x638eb0	1	5	907
1	sms	0x401d14	-1	0	148	x	stride	0x638ed1	1	38	896
1	sms	0x401cb0	4	36	147	x	stride	0x638ed1	1	23	890
1	sms	0x401e67	1	33	135	x	stride	0x638ed1	1	57	879
1	sms	0x401e67	1	1	124	x	stride	0x638ed1	1	59	865
1	sms	0x401cb0	4	35	111	x	stride	0x638eb0	1	18	863
1	sms	0x401e67	1	32	97	x	stride	0x638eb0	1	10	863
1	sms	0x401e02	-2	1	74	x	stride	0x638ebd	1	59	841
1	sms	0x401d43	1	1	72	x	stride	0x638eb0	1	20	838
1	sms	0x401cb0	4	5	67	x	stride	0x638eb0	1	14	781
1	sms	0x401d43	1	33	60	x	stride	0x638ed1	1	7	777
1	sms	0x401d43	1	32	59	x	stride	0x638ed1	1	25	773
1	sms	0x401cb0	3	36	56	x	stride	0x638ec0	1	34	771
1	sms	0x401cb0	3	4	55	x	stride	0x638ed1	1	3	769
1	sms	0x401e33	-1	1	50	x	stride	0x638ed1	1	58	759

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
1	sms	0x401e67	1	2	35	x	stride	0x638ed1	1	33	754
1	sms	0x401cb0	3	34	32	x	stride	0x638eb0	1	11	743
1	sms	0x401e02	-2	34	32	x	stride	0x638eb0	1	58	738
1	sms	0x401e02	-2	2	31	x	stride	0x638eb0	1	1	730
1	sms	0x401cb0	3	35	26	x	stride	0x638ec0	1	17	725
1	sms	0x401cb0	68	3	26	x	stride	0x638ed1	1	49	690
1	sms	0x401cb0	67	2	20	x	stride	0x638eb0	1	6	677
1	sms	0x401d43	65	0	19	x	stride	0x638ec0	1	9	663
1	sms	0x401cb0	3	5	19	x	stride	0x638eb0	1	33	655
1	sms	0x40208c	1	42	18	x	stride	0x638ed1	1	21	630
1	sms	0x4023cc	1	43	18	x	stride	0x638ec0	1	39	624
1	spp	0x401cb0	65	0	1,191	x	J13	0x638ed1	1	18	1,133
1	spp	0x401cb0	3	3	1,074	x	J13	0x638ed1	1	40	939
1	spp	0x401cb0	4	11	920	x	J13	0x638ed1	1	59	863
1	spp	0x401cb0	4	26	920	x	J13	0x638ed1	1	41	856
1	spp	0x401cb0	4	10	919	x	J13	0x638ed1	1	6	845
1	spp	0x401cb0	4	30	919	x	J13	0x638ebd	1	19	840
1	spp	0x401cb0	4	31	919	x	J13	0x638ed1	1	1	791
1	spp	0x401cb0	3	15	918	x	J13	0x638ed1	1	17	789
1	spp	0x401cb0	4	15	918	x	J13	0x638ab0	1	17	764
1	spp	0x401cb0	4	20	918	x	J13	0x638ed1	1	62	754
1	spp	0x401cb0	4	25	918	x	J13	0x638ed1	1	47	746
1	spp	0x401cb0	4	35	918	x	J13	0x638ec0	1	34	693
1	spp	0x401cb0	3	10	917	x	J13	0x638eb0	1	20	673
1	spp	0x401cb0	4	40	917	x	J13	0x638eb0	1	42	642
1	spp	0x401cb0	4	50	917	x	J13	0x638ed1	1	21	630
1	spp	0x401cb0	4	21	916	x	J13	0x638eb0	1	10	625
1	spp	0x401cb0	4	16	913	x	J13	0x638eb0	1	21	623
1	spp	0x401cb0	4	45	912	x	J13	0x638ebd	1	59	614
1	spp	0x401cb0	68	2	899	x	J13	0x638eb0	1	18	563
1	spp	0x401cb0	67	1	899	x	J13	0x638ed1	1	4	555
1	spp	0x401cb0	4	55	834	x	J13	0x638eb0	1	46	543
1	spp	0x401cb0	3	20	832	x	J13	0x638eb0	1	52	543
1	spp	0x401cb0	4	36	830	x	J13	0x638ed1	1	55	530
1	spp	0x401cb0	4	41	737	x	J13	0x638ed1	1	7	520
1	spp	0x401cb0	4	60	729	x	J13	0x638ebd	1	60	507
1	spp	0x401cb0	3	25	725	x	J13	0x638ebd	-1	10	505
1	spp	0x401cb0	66	1	723	x	J13	0x638ebd	1	47	502
1	spp	0x401cb0	4	46	638	x	J13	0x638ed1	1	33	497
1	spp	0x401cb0	3	30	614	x	J13	0x638ed1	1	36	494
1	spp	0x401e33	-1	1	581	x	J13	0x638ab0	1	6	493
1	spp	0x401cb0	4	51	576	x	J13	0x638ec0	1	35	490
1	spp	0x401cb0	2	2	572	x	J13	0x638ed1	1	12	489
1	spp	0x401e02	-2	13	570	x	J13	0x638ab0	1	19	485
1	spp	0x401e02	-2	8	557	x	J13	0x638ed1	1	52	485
1	spp	0x401e02	-2	4	554	x	J13	0x638eb0	1	47	479
1	spp	0x401e33	-1	9	551	x	J13	0x638eb0	1	61	479
1	spp	0x401e33	-1	14	542	x	J13	0x638eb0	1	57	462
1	spp	0x401e02	-2	9	542	x	J13	0x638ed1	1	61	460
1	spp	0x401e02	-2	14	532	x	J13	0x638ed1	1	45	434
1	spp	0x401cb0	3	35	528	x	J13	0x638ebd	1	55	431
1	spp	0x401e02	-2	18	526	x	J13	0x638ed1	1	34	411
1	spp	0x401e33	-1	19	492	x	J13	0x638ec0	-1	59	411
1	spp	0x401e02	2	62	492	x	J13	0x638ec0	1	53	402
1	spp	0x401e02	-2	23	452	x	J13	0x638eb0	1	14	398
1	spp	0x401e02	2	63	447	x	J13	0x638ebd	1	25	392
1	spp	0x401cb0	3	40	445	x	J13	0x638eb0	1	5	383
1	spp	0x401e33	-1	24	444	x	J13	0x638ed1	1	50	378
1	spp	0x401e02	-2	19	441	x	J13	0x638eb0	1	33	368
1	spp	0x401cb0	4	56	427	x	J13	0x638ed1	1	10	367
1	spp	0x401e67	-1	0	408	x	J13	0x638eb0	1	1	366
1	streamer	0x401cb0	3	3	1,538	x	J12	0x638ed1	1	18	1,279
1	streamer	0x401cb0	65	0	1,191	x	J12	0x638ed1	1	41	1,199
1	streamer	0x401cb0	4	4	941	x	J12	0x638ed1	1	17	1,170
1	streamer	0x401cb0	4	6	920	x	J12	0x638ed1	1	1	1,039
1	streamer	0x401cb0	3	5	917	x	J12	0x638ed1	1	62	1,020
1	streamer	0x401cb0	4	5	916	x	J12	0x638ed1	1	40	1,019
1	streamer	0x401cb0	3	7	916	x	J12	0x638eb0	1	21	947
1	streamer	0x401cb0	3	8	916	x	J12	0x638ed1	1	12	941
1	streamer	0x401cb0	3	6	911	x	J12	0x638eb0	1	19	939
1	streamer	0x401cb0	68	2	899	x	J12	0x638ed1	1	6	907
1	streamer	0x401cb0	67	1	899	x	J12	0x638eb0	1	52	888
1	streamer	0x401cb0	3	4	886	x	J12	0x638ed1	1	4	852
1	streamer	0x401cb0	66	1	723	x	J12	0x638eb0	1	10	813
1	streamer	0x401e02	-2	2	591	x	J12	0x638ed1	1	47	805
1	streamer	0x401e33	-1	1	580	x	J12	0x638eb0	1	20	782
1	streamer	0x401e67	1	3	579	x	J12	0x638ed1	1	45	771
1	streamer	0x401e33	-1	5	573	x	J12	0x638ebd	1	19	769
1	streamer	0x401cb0	2	2	572	x	J12	0x638ed1	1	59	747
1	streamer	0x401e02	-2	1	554	x	J12	0x638eb0	1	58	738
1	streamer	0x401e02	-2	4	554	x	J12	0x638eb0	1	42	686
1	streamer	0x401e33	-1	3	549	x	J12	0x638ed1	1	9	682
1	streamer	0x401e02	-2	3	547	x	J12	0x638eb0	1	5	671
1	streamer	0x401e33	-1	6	547	x	J12	0x638ebd	1	25	642
1	streamer	0x401e33	-1	4	546	x	J12	0x638ed1	1	25	562
1	streamer	0x401e67	1	2	544	x	J12	0x638eb0	1	47	555
1	streamer	0x401e67	1	4	542	x	J12	0x638ed1	1	50	537
1	streamer	0x401e67	1	5	542	x	J12	0x638ebd	1	59	528
1	streamer	0x401e33	-1	7	531	x	J12	0x638ed1	1	3	519
1	streamer	0x401e67	1	1	528	x	J12	0x638eb0	1	17	517
1	streamer	0x401e33	-1	0	511	x	J12	0x638ed1	1	34	502
1	streamer	0x401e33	-1	2	506	x	J12	0x638ed1	1	23	501
1	streamer	0x401e02	-2	0	503	x	J12	0x638ebd	1	57	500
1	streamer	0x401e67	-1	0	416	x	J12	0x638eb0	1	11	499
1	streamer	0x401d14	-1	2	383	x	J12	0x638eb0	1	46	497
1	streamer	0x401d14	-1	7	380	x	J12	0x638ed1	1	36	495
1	streamer	0x401cdb	-2	3	368	x	J12	0x638ec0	1	35	490
1	streamer	0x401d43	1	4	368	x	J12	0x638ec0	1	53	488
1	streamer	0x401d14	-1	4	366	x	J12	0x638ebd	-1	10	486
1	streamer	0x401cdb	-2	4	366	x	J12	0x638eb0	1	1	485
1	streamer	0x401d43	1	5	366	x	J12	0x638ec0	1	39	484
1	streamer	0x401d14	-1	3	365	x	J12	0x638ebd	1	38	473
1	streamer	0x401d14	-1	6	362	x	J12	0x638ed1	1	61	472
1	streamer	0x401cdb	-2	1	360	x	J12	0x638ed1	1	20	471
1	streamer	0x401d43	1	2	360	x	J12	0x638eb0	1	2	455
1	streamer	0x401d43	1	1	357	x	J12	0x638eb0	1	18	439
1	streamer	0x401cb0	67	2	342	x	J12	0x638ebd	1	61	439
1	streamer	0x401d14	-1	5	336	x	J12	0x638ed1	1	60	433
1	streamer	0x401d14	-1	0	331	x	J12	0x638ec0	1	9	430
1	streamer	0x401d14	-1	1	327	x	J12	0x638ed1	1	33	429
1	streamer	0x401cdb	-2	2	326	x	J12	0x638ed1	1	38	418
1	stride	0x401cb0	3	3	1,538	x	J14	0x638ed1	1	62	1,299
1	stride	0x401cb0	65	0	1,191	x	J14	0x638ed1	1	18	1,151
1	stride	0x401cb0	4	4	941	x	J14	0x638eb0	1	19	1,113
1	stride	0x401cb0	4	12	920	x	J14	0x638ed1	1	17	1,079
1	stride	0x401cb0	4	22	920	x	J14	0x638ed1	1	1	1,064

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
1	stride	0x401cb0	4	27	920	x	J14	0x638ed1	1	41	997
1	stride	0x401cb0	4	62	920	x	J14	0x638eb0	1	52	981
1	stride	0x401cb0	4	18	920	x	J14	0x638ed1	1	9	936
1	stride	0x401cb0	4	38	920	x	J14	0x638ebd	1	19	900
1	stride	0x401cb0	3	40	920	x	J14	0x638ed1	1	4	900
1	stride	0x401cb0	4	58	920	x	J14	0x638ed1	1	12	881
1	stride	0x401cb0	3	60	920	x	J14	0x638ed1	1	40	867
1	stride	0x401cb0	3	6	920	x	J14	0x638eb0	1	42	834
1	stride	0x401cb0	4	9	920	x	J14	0x638ed1	1	47	822
1	stride	0x401cb0	4	24	920	x	J14	0x638ed1	1	6	814
1	stride	0x401cb0	3	17	920	x	J14	0x638ed1	1	45	786
1	stride	0x401cb0	3	52	920	x	J14	0x638eb0	1	21	785
1	stride	0x401cb0	4	55	920	x	J14	0x638ebd	1	25	748
1	stride	0x401cb0	4	6	920	x	J14	0x638ed1	1	34	746
1	stride	0x401cb0	3	8	920	x	J14	0x638ed1	1	25	741
1	stride	0x401cb0	4	11	920	x	J14	0x638eb0	1	20	728
1	stride	0x401cb0	3	23	920	x	J14	0x638ec0	1	53	714
1	stride	0x401cb0	4	26	920	x	J14	0x638eb0	1	10	698
1	stride	0x401cb0	3	28	920	x	J14	0x638ebd	1	59	690
1	stride	0x401cb0	3	58	919	x	J14	0x638eb0	1	5	688
1	stride	0x401cb0	3	14	919	x	J14	0x638ed1	1	59	661
1	stride	0x401cb0	4	17	919	x	J14	0x638ed1	1	38	650
1	stride	0x401cb0	3	44	919	x	J14	0x638ed1	1	52	640
1	stride	0x401cb0	3	54	919	x	J14	0x638eb0	1	47	628
1	stride	0x401cb0	4	13	919	x	J14	0x638eb0	1	17	612
1	stride	0x401cb0	3	20	919	x	J14	0x638ed1	1	7	612
1	stride	0x401cb0	4	23	919	x	J14	0x638eb0	1	11	610
1	stride	0x401cb0	3	55	919	x	J14	0x638eb0	1	6	601
1	stride	0x401cb0	4	63	919	x	J14	0x638eb0	1	58	560
1	stride	0x401cb0	3	11	919	x	J14	0x638ec0	1	34	525
1	stride	0x401cb0	4	14	919	x	J14	0x638ec0	1	9	519
1	stride	0x401cb0	4	19	919	x	J14	0x638eb0	1	1	516
1	stride	0x401cb0	3	26	919	x	J14	0x638ed1	1	21	509
1	stride	0x401cb0	4	34	919	x	J14	0x638ed1	1	57	504
1	stride	0x401cb0	4	44	919	x	J14	0x638eb0	1	14	504
1	stride	0x401cb0	3	46	919	x	J14	0x638ed1	1	23	491
1	stride	0x401cb0	4	54	919	x	J14	0x638ed1	1	8	488
1	stride	0x401cb0	3	61	919	x	J14	0x638ed1	1	58	476
1	stride	0x401cb0	4	10	919	x	J14	0x638ebd	1	3	463
1	stride	0x401cb0	3	22	919	x	J14	0x638ec0	1	39	461
1	stride	0x401cb0	4	30	919	x	J14	0x638ebd	-1	19	450
1	stride	0x401cb0	3	37	919	x	J14	0x638ebd	1	58	448
1	stride	0x401cb0	3	18	919	x	J14	0x638ed1	1	33	444
1	stride	0x401cb0	4	31	919	x	J14	0x638ebd	1	60	444
1	stride	0x401cb0	3	9	918	x	J14	0x638eb0	1	18	443
1	J46	0x401cb0	65	0	1,164	x	J11	0x638ed1	1	18	384
1	J46	0x401cb0	4	10	919	x	J11	0x638ebd	-2,008	0	253
1	J46	0x401cb0	4	7	917	x	J11	0x638ec0	1,600,961,158,017	50	253
1	J46	0x401cb0	4	9	916	x	J11	0x638ebd	-24,255	18	239
1	J46	0x401cb0	4	8	914	x	J11	0x638ec0	1	21	236
1	J46	0x401cb0	4	5	904	x	J11	0x638ec0	2,028	20	236
1	J46	0x401cb0	68	2	896	x	J11	0x638ec0	-498,038,650,865	16	235
1	J46	0x401cb0	4	4	865	x	J11	0x638ec0	-9,346	19	224
1	J46	0x401cb0	4	6	860	x	J11	0x638ec0	94	51	221
1	J46	0x401cb0	66	1	723	x	J11	0x638ebd	99	59	220
1	J46	0x401cb0	3	3	684	x	J11	0x638ec0	-19,002	51	208
1	J46	0x401e02	-2	1	548	x	J11	0x638eb0	1	42	195
1	J46	0x401e67	1	7	503	x	J11	0x638ebd	522,111,659,342	14	186
1	J46	0x401e02	-2	0	498	x	J11	0x638ec0	-523,969,238,798	39	182
1	J46	0x401e67	1	1	482	x	J11	0x64cd40	-829	53	178
1	J46	0x401cb0	4	11	469	x	J11	0x638ec0	84	43	175
1	J46	0x401e33	-1	1	468	x	J11	0x64cc82	-4,189,474,377,000	50	169
1	J46	0x401e02	-2	2	457	x	J11	0x7d2230	596	8	169
1	J46	0x401e33	-1	0	450	x	J11	0x64cc6c	1	36	166
1	J46	0x401e67	1	63	448	x	J11	0x638ec0	90	60	159
1	J46	0x401e67	1	8	421	x	J11	0x638ebd	-523,969,233,149	17	156
1	J46	0x401e67	-1	0	404	x	J11	0x638ebd	92	63	156
1	J46	0x401cb0	67	1	357	x	J11	0x638ed1	1	48	145
1	J46	0x401c4b	-2	1	357	x	J11	0x638eb0	498,038,657,149	12	136
1	J46	0x401c4b	-2	0	317	x	J11	0x638ebd	-25,709	18	135
1	J46	0x401e02	-2	3	316	x	J11	0x638ec0	498,038,643,598	2	131
1	J46	0x401d14	-1	0	316	x	J11	0x638450	4,189,474,376,988	17	131
1	J46	0x401d43	1	8	310	x	J11	0x638ec0	74	18	130
1	J46	0x401e02	-2	4	290	x	J11	0x638ed1	-119	2	129
1	J46	0x401d43	1	7	290	x	J11	0x638ec0	-498,038,654,914	55	126
1	J46	0x401d43	1	1	286	x	J11	0x638ebd	1	31	126
1	J46	0x401d14	-1	1	286	x	J11	0x638ebd	1	25	125
1	J46	0x401e67	1	4	286	x	J11	0x638ebd	72	50	125
1	J46	0x401e67	1	2	284	x	J11	0x7d2253	98	35	124
1	J46	0x401d43	-1	0	271	x	J11	0x638eb0	1	20	123
1	J46	0x401e02	-2	5	271	x	J11	0x638ed1	1	4	123
1	J46	0x401cb0	67	0	265	x	J11	0x638ebd	523,969,216,714	4	123
1	J46	0x401d43	1	63	264	x	J11	0x638ec0	44	25	121
1	J46	0x401cb0	68	3	259	x	J11	0x638ec0	-498,038,638,224	17	121
1	J46	0x401cb0	3	4	251	x	J11	0x638ebd	523,969,232,260	18	121
1	J46	0x401cb0	3	7	240	x	J11	0x638ebd	49	21	120
1	J46	0x401c4b	-2	3	233	x	J11	0x638ebd	-1,319,306,398,996	29	119
1	J46	0x401e67	1	5	227	x	J11	0x638ec0	136	9	118
1	J46	0x401c4b	-2	2	227	x	J11	0x64cc6c	1	38	118
1	J46	0x401cb0	4	12	214	x	J11	0x638eb0	1	48	118
1	J46	0x401c4b	-2	4	186	x	J11	0x644140	1	52	117
1	J46	0x401cb0	66	0	185	x	J11	0x638ebd	1	51	116
1	J46	0x401c4b	-2	5	181	x	J11	0x638ec0	2	27	116
1	J46	0x401e02	-2	6	181	x	J11	0x638ec0	-200	48	116
1	J46	0x401cb0	68	1	179	x	J11	0x638ec0	-10,966	23	114
1	J41	0x401cb0	65	0	1,183	x	J10	0x638ed1	1	62	1,611
1	J41	0x401cb0	67	1	769	x	J10	0x638ed1	1	18	1,364
1	J41	0x401cb0	4	10	746	x	J10	0x638ed1	1	41	1,258
1	J41	0x401cb0	66	1	721	x	J10	0x638eb0	1	19	1,194
1	J41	0x401cb0	3	8	719	x	J10	0x638ed1	1	17	1,160
1	J41	0x401cb0	4	6	678	x	J10	0x638ed1	1	6	1,135
1	J41	0x401cb0	4	5	573	x	J10	0x638ed1	1	47	1,099
1	J41	0x401cb0	4	8	542	x	J10	0x638ed1	1	40	1,012
1	J41	0x401cb0	4	7	539	x	J10	0x638ed1	1	1	994
1	J41	0x401cb0	4	9	467	x	J10	0x638eb0	1	21	975
1	J41	0x401cb0	3	6	455	x	J10	0x638eb0	1	17	963
1	J41	0x401e67	1	63	453	x	J10	0x638ebd	1	19	962
1	J41	0x401cb0	2	2	435	x	J10	0x638eb0	1	42	945
1	J41	0x401e02	-2	0	404	x	J10	0x638ed1	1	23	880
1	J41	0x401e67	1	5	402	x	J10	0x638ed1	1	57	876
1	J41	0x401e67	-1	0	398	x	J10	0x638ed1	1	34	871
1	J41	0x401cb0	3	3	397	x	J10	0x638ec0	1	53	870
1	J41	0x401e67	1	6	374	x	J10	0x638ed1	1	9	866
1	J41	0x401cb0	3	7	334	x	J10	0x638ed1	1	12	866
1	J41	0x401cb0	68	2	316	x	J10	0x638ed1	1	59	865
1	J41	0x401c4b	-2	0	286	x	J10	0x638eb0	1	52	860

tr	ID	PC	Δ	off	resid	tr	ID	PC	Δ	off	resid
1	J41	0x401cb0	67	0	274	x	J10	0x638eb0	1	18	851
1	J41	0x401e02	-2	5	272	x	J10	0x638ed1	1	38	846
1	J41	0x401d43	-1	0	269	x	J10	0x638eb0	1	20	837
1	J41	0x401cb0	67	2	269	x	J10	0x638eb0	1	46	834
1	J41	0x401d43	1	63	264	x	J10	0x638ebd	1	59	823
1	J41	0x401e67	1	7	253	x	J10	0x638ed1	1	4	815
1	J41	0x401e02	-2	3	247	x	J10	0x638ed1	1	45	781
1	J41	0x401e02	-2	1	246	x	J10	0x638ec0	1	34	758
1	J41	0x401cb0	3	5	233	x	J10	0x638ebd	1	25	755
1	J41	0x401cb0	4	4	232	x	J10	0x638eb0	1	10	706
1	J41	0x401e33	2	63	224	x	J10	0x638eb0	1	6	677
1	J41	0x401d43	1	6	220	x	J10	0x638ed1	1	3	670
1	J41	0x401cb0	3	4	201	x	J10	0x64cc6c	1	40	649
1	J41	0x401cdb	-2	3	195	x	J10	0x638ed1	1	21	630
1	J41	0x401e02	-2	6	186	x	J10	0x638ec0	1	39	623
1	J41	0x401cb0	66	0	185	x	J10	0x638ed1	1	50	602
1	J41	0x401e67	1	4	182	x	J10	0x638eb0	1	14	599
1	J41	0x401cdb	-2	5	180	x	J10	0x638ed1	1	19	599
1	J41	0x401e02	-2	2	180	x	J10	0x64cc6c	1	17	574
1	J41	0x401cb0	68	1	179	x	J10	0x64cc6c	1	18	570
1	J41	0x401cdb	-2	1	169	x	J10	0x638ed1	1	52	569
1	J41	0x401d43	1	7	168	x	J10	0x638eb0	1	47	565
1	J41	0x401e67	1	62	165	x	J10	0x638eb0	1	5	563
1	J41	0x401cb0	68	0	157	x	J10	0x638ed1	1	55	537
1	J41	0x401e02	-2	4	157	x	J10	0x638eb0	1	41	535
1	J41	0x401d43	1	5	156	x	J10	0x638eb0	1	39	525
1	J41	0x401e33	-1	3	149	x	J10	0x638ed1	1	7	524
1	J41	0x401cdb	-2	4	137	x	J10	0x638ed1	1	20	517
1	J41	0x401e33	-1	7	130	x	J10	0x638ebd	1	57	516
1	J34	0x401cb0	3	3	1,386	x	J09	0x638ed1	1	18	385
1	J34	0x401cb0	65	0	1,191	x	J09	0x638ebd	-2,008	0	253
1	J34	0x401cb0	3	6	919	x	J09	0x638ec0	1,600,961,158,017	50	253
1	J34	0x401cb0	3	5	907	x	J09	0x638ec0	1	21	246
1	J34	0x401cb0	67	1	899	x	J09	0x64cc82	-4,189,474,377,000	50	244
1	J34	0x401cb0	3	4	885	x	J09	0x638ebd	-24,255	18	239
1	J34	0x401cb0	4	6	861	x	J09	0x638ec0	-498,038,650,865	16	238
1	J34	0x401cb0	4	8	847	x	J09	0x638ec0	2,028	20	237
1	J34	0x401cb0	4	7	817	x	J09	0x638ec0	-9,346	19	224
1	J34	0x401cb0	3	10	815	x	J09	0x638ec0	94	51	221
1	J34	0x401cb0	66	8	732	x	J09	0x638ebd	99	59	220
1	J34	0x401cb0	4	1	723	x	J09	0x64cc6c	1	23	219
1	J34	0x401cb0	68	5	690	x	J09	0x638ec0	-19,002	51	208
1	J34	0x401cb0	4	2	654	x	J09	0x638eb0	1	42	203
1	J34	0x401cb0	4	4	620	x	J09	0x64cc6c	1	36	192
1	J34	0x401e67	1	3	569	x	J09	0x64cc6c	1	53	190
1	J34	0x401e02	-2	2	545	x	J09	0x64cd40	-829	53	188
1	J34	0x401cb0	2	2	545	x	J09	0x638ebd	522,111,659,342	14	186
1	J34	0x401cb0	4	9	524	x	J09	0x638ec0	-523,969,238,798	39	182
1	J34	0x401e67	1	1	513	x	J09	0x638eb0	1	48	181
1	J34	0x401e67	1	6	497	x	J09	0x64cc6c	1	42	176
1	J34	0x401e67	1	4	488	x	J09	0x638ec0	84	43	175
1	J34	0x401e02	-2	4	486	x	J09	0x638ed1	1	4	173
1	J34	0x401e67	1	5	478	x	J09	0x7d2230	596	8	169
1	J34	0x401e67	1	2	439	x	J09	0x638ed1	1	50	169
1	J34	0x401e02	-2	3	438	x	J09	0x64cc6c	1	5	160
1	J34	0x401e67	-1	0	400	x	J09	0x638ec0	90	60	159
1	J34	0x401e33	-1	5	362	x	J09	0x64cc6c	1	38	157
1	J34	0x401d43	1	6	360	x	J09	0x638ebd	92	63	156
1	J34	0x401d43	1	1	359	x	J09	0x638eb0	1	33	154
1	J34	0x401d43	1	4	358	x	J09	0x638ed1	1	48	150
1	J34	0x401d43	1	5	349	x	J09	0x4983eb	107	63	138
1	J34	0x401cdb	-2	3	344	x	J09	0x638eb0	498,038,657,149	12	136
1	J34	0x401cdb	-2	4	342	x	J09	0x638ebd	-25,709	18	135
1	J34	0x401cb0	67	2	337	x	J09	0x638ed1	1	33	132
1	J34	0x401e33	-1	3	332	x	J09	0x638eb0	1	10	131
1	J34	0x401d43	1	2	331	x	J09	0x638ec0	498,038,643,598	2	131
1	J34	0x401d43	1	3	326	x	J09	0x638450	4,189,474,376,988	17	131
1	J34	0x401e33	-1	4	319	x	J09	0x64cc6c	1	30	131
1	J34	0x401d14	-1	3	316	x	J09	0x638ec0	74	18	130
1	J34	0x401d14	-1	2	315	x	J09	0x638ed1	-119	2	129
1	J34	0x401cdb	-2	2	315	x	J09	0x64cc6c	1	9	128
1	J34	0x401e02	-2	5	310	x	J09	0x638ec0	-498,038,654,914	55	126
1	J34	0x401cdb	-2	5	295	x	J09	0x638ebd	1	31	126
1	J34	0x401e33	-1	2	293	x	J09	0x638ebd	1	25	125
1	J34	0x401d14	-1	5	293	x	J09	0x638ebd	72	50	125
1	J34	0x401d14	-1	4	292	x	J09	0x7d2253	98	35	124
1	J34	0x401cb0	67	0	280	x	J09	0x638eb0	1	20	123
1	J34	0x401d43	-1	0	270	x	J09	0x638ebd	49	21	123
1	J34	0x401cb0	3	7	245	x	J09	0x638ebd	523,969,216,714	4	123

10.1 The 160M-row event table

analysis/normal_vs_standalone_event_rows.csv.gz holds one row per (NN job, normal prefetcher, no-prefetch demand-miss event): **159,914,809** rows, 2.15 GiB compressed, schema trace, standalone_variant, normal_prefetcher, demand_idx, base_event_id, pc, line, delta, page_offset, normal_state, standalone_state, standalone_selected_category, normal_only_reason. Every aggregate the archive derives from it — the 423-pair overlap/miss tables, the 942 reason rows, the 92-run unique-event outcomes and the 4,600 residual contexts — is printed in full above; per the archive’s own README (below) the row file adds per-event keys but no additional summary quantities.

11 Cross-run analysis (replay_results/v4_9_analysis/)

Table 74: v4.9_analysis/v4.8_all_candidate_replay_comparison.csv — columns beyond Table 7 (shared columns verified identical); 47 rows — part 1/2

ID	trace	best-normal-prefetcher	beats-best-normal-by-keyed-replay	keyed-replay-valid	fixed-no-pref-ipc	ipc-delta-vs-no-pref	event-coverage	event-timely	event-late	event-residual
J21	602.gcc.s-734B	sandbox	0	1	0.368	0.06035	0.91064	184,979	1	18,150
J20	602.gcc.s-734B	sandbox	0	1	0.368	0.06055	0.913	185,459	1	17,671
J22	602.gcc.s-734B	sandbox	0	1	0.368	0.06266	0.93517	189,962	2	13,083
J19	602.gcc.s-734B	sandbox	0	1	0.368	0.06124	0.92065	187,013	0	16,118
J15	602.gcc.s-734B	sandbox	0	1	0.368	0.06124	0.92065	187,013	0	16,118
J17	602.gcc.s-734B	sandbox	0	1	0.368	0.06035	0.91064	184,979	1	18,150
J16	602.gcc.s-734B	sandbox	0	1	0.368	0.06055	0.913	185,459	1	17,671
J18	602.gcc.s-734B	sandbox	0	1	0.368	0.0123	0.19845	40,312	1	162,773
J07	605.mcf.s-994B	ampm	1	1	0.18318	0.01041	0.21938	162,393	24,401	548,097
J06	605.mcf.s-994B	ampm	1	1	0.18318	0.01007	0.22095	163,549	24,016	547,031
J08	605.mcf.s-994B	ampm	1	1	0.18318	0.00941	0.21527	159,346	25,165	551,727
J05	605.mcf.s-994B	ampm	1	1	0.18318	0.00855	0.19904	147,332	21,477	566,639
J01	605.mcf.s-994B	ampm	1	1	0.18318	0.00617	0.16851	124,736	1,609	610,653
J03	605.mcf.s-994B	ampm	1	1	0.18318	0.0094	0.20307	150,314	2,272	583,788
J02	605.mcf.s-994B	ampm	1	1	0.18318	0.00916	0.20242	149,835	13,994	571,954
J04	605.mcf.s-994B	ampm	1	1	0.18318	0.00649	0.16107	119,231	2,575	615,671
J43	619.lbm.s-4268B	sms	0	1	0.322	0.04723	0.77396	228,038	7,540	58,387
J38	619.lbm.s-4268B	sms	0	1	0.322	0.04801	0.80147	236,144	8,770	49,023
J47	619.lbm.s-4268B	sms	1	1	0.322	0.06179	0.98504	290,231	416	2,972
J31	619.lbm.s-4268B	sms	0	1	0.322	0.05044	0.79333	233,746	10,890	49,264
J34	619.lbm.s-4268B	sms	0	1	0.322	0.05356	0.84473	248,889	2,439	42,442
J30	619.lbm.s-4268B	sms	0	1	0.322	0.05044	0.79333	233,746	10,890	49,264
J35	619.lbm.s-4268B	sms	0	1	0.322	0.05084	0.78029	229,904	10,801	53,252
J36	619.lbm.s-4268B	sms	0	1	0.322	0.04839	0.76678	225,923	10,703	57,285
J32	619.lbm.s-4268B	sms	0	1	0.322	0.0526	0.83437	245,839	4,505	43,446
J33	619.lbm.s-4268B	sms	0	1	0.322	0.05356	0.84473	248,889	2,439	42,442
J46	619.lbm.s-4268B	sms	0	1	0.322	0.05349	0.85367	251,524	1,645	40,622
J41	619.lbm.s-4268B	sms	0	1	0.322	0.05481	0.88103	259,585	1,538	32,602
J42	619.lbm.s-4268B	sms	0	1	0.322	0.04723	0.77396	228,038	7,540	58,387
J37	619.lbm.s-4268B	sms	0	1	0.322	0.04801	0.80147	236,144	8,770	49,023
J44	619.lbm.s-4268B	sms	0	1	0.322	0.05238	0.83735	246,715	3,331	43,758
J39	619.lbm.s-4268B	sms	0	1	0.322	0.05454	0.86703	255,462	3,578	34,731
J45	619.lbm.s-4268B	sms	0	1	0.322	0.05349	0.85367	251,524	1,645	40,622
J40	619.lbm.s-4268B	sms	0	1	0.322	0.05481	0.88103	259,585	1,538	32,602
J28	620.omnetpp.s-874B	sms	0	1	0.2375	0.00355	0.12835	40,630	85	275,413
J27	620.omnetpp.s-874B	sms	0	1	0.2375	0.00371	0.15317	48,487	120	267,360
J29	620.omnetpp.s-874B	sms	0	1	0.2375	0.00202	0.09619	30,449	161	285,460
J26	620.omnetpp.s-874B	sms	0	1	0.2375	0.00445	0.14463	45,782	53	270,343
J23	620.omnetpp.s-874B	sms	0	1	0.2375	0.00445	0.14463	45,782	53	270,343
J25	620.omnetpp.s-874B	sms	0	1	0.2375	0.00213	0.06614	20,937	20	295,524
J24	620.omnetpp.s-874B	sms	0	1	0.2375	0.00209	0.06615	20,939	18	295,440
J13	623.xalancbm.s-700B	spp	1	1	0.35321	0.00387	0.50588	227,823	1,028	218,108
J12	623.xalancbm.s-700B	spp	0	1	0.35321	-0.00194	0.37744	169,978	577	275,422
J14	623.xalancbm.s-700B	spp	0	1	0.35321	-0.01172	0.30195	135,984	613	310,237
J11	623.xalancbm.s-700B	spp	1	1	0.35321	0.02594	0.89075	401,150	749	42,845
J09	623.xalancbm.s-700B	spp	1	1	0.35321	0.02561	0.86354	388,893	652	55,607
J10	623.xalancbm.s-700B	spp	1	1	0.35321	0.00407	0.17943	80,808	263	367,905

Table 75: v4.9.analysis/v4.8.all_candidate_replay_comparison.csv — columns beyond Table 7 (shared columns verified identical); 47 rows — part 2/2

ID	event_timeliness	resource_attempts	resource_accepted	resource_duplicate	resource_duplicate_fraction	resource_rejected_fraction	resource_pq_p95	resource_mshr_p95	resource_prefetch_attempts_per_l2_load
J21	0.99999	188,865	188,865	0	0	0	3	1	0.46723
J20	0.99999	189,926	189,926	0	0	0	3	1	0.46986
J22	0.99999	199,519	199,519	0	0	0	2	0	0.49359
J19	1	189,461	189,461	0	0	0	3	1	0.46871
J15	1	189,461	189,461	0	0	0	3	1	0.46871
J17	0.99999	188,865	188,865	0	0	0	3	1	0.46723
J16	0.99999	189,926	189,926	0	0	0	3	1	0.46986
J18	0.99998	41,212	41,212	0	0	0	1	1	0.10195
J07	0.86937	699,538	699,538	9	1.287e-05	0	3	5	0.70999
J06	0.87196	788,620	788,620	9	1.141e-05	0	3	5	0.80041
J08	0.86361	708,864	708,864	5	7.054e-06	0	3	5	0.71946
J05	0.87277	819,983	819,983	9	1.098e-05	0	3	5	0.83224
J01	0.98727	573,234	573,234	10	1.744e-05	0	3	4	0.5818
J03	0.98511	493,807	493,807	8	1.620e-05	0	3	5	0.50119
J02	0.91458	497,031	497,031	6	1.207e-05	0	3	5	0.50446
J04	0.97886	323,579	323,579	2	6.181e-06	0	3	5	0.32841
J43	0.96799	478,946	478,946	59,066	0.12332	0	4	20	1.62333
J38	0.96419	506,926	506,926	54,007	0.10654	0	5	20	1.71812
J47	0.99857	468,215	468,215	54,231	0.11582	0	4	20	1.58699
J31	0.95548	480,929	480,929	65,399	0.13598	0	4	21	1.62996
J34	0.9903	869,593	869,593	160,468	0.18453	0	8	22	2.89132
J30	0.95548	480,929	480,929	65,399	0.13598	0	4	21	1.62996
J35	0.95513	474,081	474,081	70,347	0.14839	0	4	20	1.60675
J36	0.95477	472,941	472,941	68,633	0.14512	0	4	21	1.60283
J32	0.982	679,499	679,499	102,399	0.1507	0	6	21	2.29876
J33	0.9903	869,593	869,593	160,468	0.18453	0	8	22	2.89132
J46	0.9935	833,464	833,464	151,058	0.18124	0	7	21	2.82373
J41	0.99411	904,426	904,426	172,971	0.19125	0	7	20	3.06391
J42	0.96799	478,946	478,946	59,066	0.12332	0	4	20	1.62333
J37	0.96419	506,926	506,926	54,007	0.10654	0	5	20	1.71812
J44	0.98668	673,058	673,058	96,646	0.14359	0	6	20	2.28115
J39	0.98619	722,342	722,342	103,988	0.14396	0	6	20	2.44799
J45	0.9935	833,464	833,464	151,058	0.18124	0	7	21	2.82373
J40	0.99411	904,426	904,426	172,971	0.19125	0	7	20	3.06391
J28	0.99791	70,159	70,159	0	0	0	3	4	0.15542
J27	0.99753	83,744	83,744	0	0	0	3	3	0.1855
J29	0.99474	102,407	102,407	1	9.765e-06	0	3	4	0.22687
J26	0.99884	57,133	57,133	0	0	0	3	4	0.12655
J23	0.99884	57,133	57,133	0	0	0	3	4	0.12655
J25	0.99905	24,752	24,752	0	0	0	2	4	0.05483
J24	0.99914	24,815	24,815	0	0	0	3	4	0.05497
J13	0.99551	516,174	516,174	14	2.712e-05	0	2	3	0.42335
J12	0.99662	479,615	479,615	33	6.881e-05	0	3	4	0.3937
J14	0.99551	794,157	794,157	10	1.259e-05	0	3	7	0.63738
J11	0.99814	534,810	534,810	125	2.337e-04	0	2	2	0.43863
J09	0.99833	481,918	481,918	98	2.034e-04	0	2	2	0.39527
J10	0.99676	112,483	112,483	0	0	0	2	3	0.09232

Table 76: v4.9.analysis/v4.8.nn_vs_every_normal_ipc.csv — pair-level columns for all 423 (NN job, normal) pairs; per-job columns identical to Tables 7/74

ID	normal_prefetcher	normal_ipc	ipc	ipc_delta_nn_minus_normal	nn_beats_this_normal_by_keyed_replay
J21	no_pref	0.368	0.42835	0.06035	1
J21	stride	0.37006	0.42835	0.05829	1
J21	streamer	0.43161	0.42835	-0.00326	0
J21	ampm	0.43069	0.42835	-0.00234	0
J21	spp	0.42836	0.42835	-1.000e-05	0

Table 76: v4.9_analysis/v4.8_nn-vs.every_normal_ipc.csv — pair-level columns for all 423 (NN job, normal) pairs; per-job columns identical to Tables 7/74 (cont.)

ID	normal_prefetcher	normal_ipc	ipc	ipc_delta_nn_minus_normal	nn_beats_this_normal_by_keyed_replay
J21	ipcp	0.36796	0.42835	0.06039	1
J21	sms	0.42213	0.42835	0.00622	1
J21	sandbox	0.43628	0.42835	-0.00793	0
J21	power7	0.43075	0.42835	-0.0024	0
J20	no_pref	0.368	0.42855	0.06055	1
J20	stride	0.37006	0.42855	0.05849	1
J20	streamer	0.43161	0.42855	-0.00306	0
J20	ampm	0.43069	0.42855	-0.00214	0
J20	spp	0.42836	0.42855	1.900e-04	1
J20	ipcp	0.36796	0.42855	0.06059	1
J20	sms	0.42213	0.42855	0.00642	1
J20	sandbox	0.43628	0.42855	-0.00773	0
J20	power7	0.43075	0.42855	-0.0022	0
J22	no_pref	0.368	0.43066	0.06266	1
J22	stride	0.37006	0.43066	0.0606	1
J22	streamer	0.43161	0.43066	-9.500e-04	0
J22	ampm	0.43069	0.43066	-3.000e-05	0
J22	spp	0.42836	0.43066	0.0023	1
J22	ipcp	0.36796	0.43066	0.0627	1
J22	sms	0.42213	0.43066	0.00853	1
J22	sandbox	0.43628	0.43066	-0.00562	0
J22	power7	0.43075	0.43066	-9.000e-05	0
J19	no_pref	0.368	0.42924	0.06124	1
J19	stride	0.37006	0.42924	0.05918	1
J19	streamer	0.43161	0.42924	-0.00237	0
J19	ampm	0.43069	0.42924	-0.00145	0
J19	spp	0.42836	0.42924	8.800e-04	1
J19	ipcp	0.36796	0.42924	0.06128	1
J19	sms	0.42213	0.42924	0.00711	1
J19	sandbox	0.43628	0.42924	-0.00704	0
J19	power7	0.43075	0.42924	-0.00151	0
J15	no_pref	0.368	0.42924	0.06124	1
J15	stride	0.37006	0.42924	0.05918	1
J15	streamer	0.43161	0.42924	-0.00237	0
J15	ampm	0.43069	0.42924	-0.00145	0
J15	spp	0.42836	0.42924	8.800e-04	1
J15	ipcp	0.36796	0.42924	0.06128	1
J15	sms	0.42213	0.42924	0.00711	1
J15	sandbox	0.43628	0.42924	-0.00704	0
J15	power7	0.43075	0.42924	-0.00151	0
J17	no_pref	0.368	0.42835	0.06035	1
J17	stride	0.37006	0.42835	0.05829	1
J17	streamer	0.43161	0.42835	-0.00326	0
J17	ampm	0.43069	0.42835	-0.00234	0
J17	spp	0.42836	0.42835	-1.000e-05	0
J17	ipcp	0.36796	0.42835	0.06039	1
J17	sms	0.42213	0.42835	0.00622	1
J17	sandbox	0.43628	0.42835	-0.00793	0
J17	power7	0.43075	0.42835	-0.0024	0
J16	no_pref	0.368	0.42855	0.06055	1
J16	stride	0.37006	0.42855	0.05849	1
J16	streamer	0.43161	0.42855	-0.00306	0
J16	ampm	0.43069	0.42855	-0.00214	0
J16	spp	0.42836	0.42855	1.900e-04	1
J16	ipcp	0.36796	0.42855	0.06059	1
J16	sms	0.42213	0.42855	0.00642	1
J16	sandbox	0.43628	0.42855	-0.00773	0
J16	power7	0.43075	0.42855	-0.0022	0
J18	no_pref	0.368	0.3803	0.0123	1
J18	stride	0.37006	0.3803	0.01024	1
J18	streamer	0.43161	0.3803	-0.05131	0
J18	ampm	0.43069	0.3803	-0.05039	0
J18	spp	0.42836	0.3803	-0.04806	0
J18	ipcp	0.36796	0.3803	0.01234	1
J18	sms	0.42213	0.3803	-0.04183	0
J18	sandbox	0.43628	0.3803	-0.05598	0
J18	power7	0.43075	0.3803	-0.05045	0
J07	no_pref	0.18318	0.19359	0.01041	1
J07	stride	0.18359	0.19359	0.01	1
J07	streamer	0.1835	0.19359	0.01009	1
J07	ampm	0.18874	0.19359	0.00485	1
J07	spp	0.18874	0.19359	0.00485	1
J07	ipcp	0.18318	0.19359	0.01041	1
J07	sms	0.18777	0.19359	0.00582	1

Table 76: v4.9_analysis/v4.8_nn-vs.every_normal_ipc.csv — pair-level columns for all 423 (NN job, normal) pairs; per-job columns identical to Tables 7/74 (cont.)

ID	normal_prefetcher	normal_ipc	ipc	ipc_delta_nn_minus_normal	nn_beats_this_normal_by_keyed_replay
J07	sandbox	0.16905	0.19359	0.02454	1
J07	power7	0.18163	0.19359	0.01196	1
J06	no_pref	0.18318	0.19325	0.01007	1
J06	stride	0.18359	0.19325	0.00966	1
J06	streamer	0.1835	0.19325	0.00975	1
J06	ampm	0.18874	0.19325	0.00451	1
J06	spp	0.18874	0.19325	0.00451	1
J06	ipcp	0.18318	0.19325	0.01007	1
J06	sms	0.18777	0.19325	0.00548	1
J06	sandbox	0.16905	0.19325	0.0242	1
J06	power7	0.18163	0.19325	0.01162	1
J08	no_pref	0.18318	0.19259	0.00941	1
J08	stride	0.18359	0.19259	0.009	1
J08	streamer	0.1835	0.19259	0.00909	1
J08	ampm	0.18874	0.19259	0.00385	1
J08	spp	0.18874	0.19259	0.00385	1
J08	ipcp	0.18318	0.19259	0.00941	1
J08	sms	0.18777	0.19259	0.00482	1
J08	sandbox	0.16905	0.19259	0.02354	1
J08	power7	0.18163	0.19259	0.01096	1
J05	no_pref	0.18318	0.19173	0.00855	1
J05	stride	0.18359	0.19173	0.00814	1
J05	streamer	0.1835	0.19173	0.00823	1
J05	ampm	0.18874	0.19173	0.00299	1
J05	spp	0.18874	0.19173	0.00299	1
J05	ipcp	0.18318	0.19173	0.00855	1
J05	sms	0.18777	0.19173	0.00396	1
J05	sandbox	0.16905	0.19173	0.02268	1
J05	power7	0.18163	0.19173	0.0101	1
J01	no_pref	0.18318	0.18935	0.00617	1
J01	stride	0.18359	0.18935	0.00576	1
J01	streamer	0.1835	0.18935	0.00585	1
J01	ampm	0.18874	0.18935	6.100e-04	1
J01	spp	0.18874	0.18935	6.100e-04	1
J01	ipcp	0.18318	0.18935	0.00617	1
J01	sms	0.18777	0.18935	0.00158	1
J01	sandbox	0.16905	0.18935	0.0203	1
J01	power7	0.18163	0.18935	0.00772	1
J03	no_pref	0.18318	0.19258	0.0094	1
J03	stride	0.18359	0.19258	0.00899	1
J03	streamer	0.1835	0.19258	0.00908	1
J03	ampm	0.18874	0.19258	0.00384	1
J03	spp	0.18874	0.19258	0.00384	1
J03	ipcp	0.18318	0.19258	0.0094	1
J03	sms	0.18777	0.19258	0.00481	1
J03	sandbox	0.16905	0.19258	0.02353	1
J03	power7	0.18163	0.19258	0.01095	1
J02	no_pref	0.18318	0.19234	0.00916	1
J02	stride	0.18359	0.19234	0.00875	1
J02	streamer	0.1835	0.19234	0.00884	1
J02	ampm	0.18874	0.19234	0.0036	1
J02	spp	0.18874	0.19234	0.0036	1
J02	ipcp	0.18318	0.19234	0.00916	1
J02	sms	0.18777	0.19234	0.00457	1
J02	sandbox	0.16905	0.19234	0.02329	1
J02	power7	0.18163	0.19234	0.01071	1
J04	no_pref	0.18318	0.18967	0.00649	1
J04	stride	0.18359	0.18967	0.00608	1
J04	streamer	0.1835	0.18967	0.00617	1
J04	ampm	0.18874	0.18967	9.300e-04	1
J04	spp	0.18874	0.18967	9.300e-04	1
J04	ipcp	0.18318	0.18967	0.00649	1
J04	sms	0.18777	0.18967	0.0019	1
J04	sandbox	0.16905	0.18967	0.02062	1
J04	power7	0.18163	0.18967	0.00804	1
J43	no_pref	0.322	0.36923	0.04723	1
J43	stride	0.32465	0.36923	0.04458	1
J43	streamer	0.36368	0.36923	0.00555	1
J43	ampm	0.3748	0.36923	-0.00557	0
J43	spp	0.37933	0.36923	-0.0101	0
J43	ipcp	0.322	0.36923	0.04723	1
J43	sms	0.38105	0.36923	-0.01182	0
J43	sandbox	0.3744	0.36923	-0.00517	0
J43	power7	0.36128	0.36923	0.00795	1

Table 76: v4.9_analysis/v4.8_nn-vs.every_normal_ipc.csv — pair-level columns for all 423 (NN job, normal) pairs; per-job columns identical to Tables 7/74 (cont.)

ID	normal_prefetcher	normal_ipc	ipc	ipc_delta_nn_minus_normal	nn_beats_this_normal_by_keyed_replay
J38	no_pref	0.322	0.37001	0.04801	1
J38	stride	0.32465	0.37001	0.04536	1
J38	streamer	0.36368	0.37001	0.00633	1
J38	ampm	0.3748	0.37001	-0.00479	0
J38	spp	0.37933	0.37001	-0.00932	0
J38	ipcp	0.322	0.37001	0.04801	1
J38	sms	0.38105	0.37001	-0.01104	0
J38	sandbox	0.3744	0.37001	-0.00439	0
J38	power7	0.36128	0.37001	0.00873	1
J47	no_pref	0.322	0.38379	0.06179	1
J47	stride	0.32465	0.38379	0.05914	1
J47	streamer	0.36368	0.38379	0.02011	1
J47	ampm	0.3748	0.38379	0.00899	1
J47	spp	0.37933	0.38379	0.00446	1
J47	ipcp	0.322	0.38379	0.06179	1
J47	sms	0.38105	0.38379	0.00274	1
J47	sandbox	0.3744	0.38379	0.00939	1
J47	power7	0.36128	0.38379	0.02251	1
J31	no_pref	0.322	0.37244	0.05044	1
J31	stride	0.32465	0.37244	0.04779	1
J31	streamer	0.36368	0.37244	0.00876	1
J31	ampm	0.3748	0.37244	-0.00236	0
J31	spp	0.37933	0.37244	-0.00689	0
J31	ipcp	0.322	0.37244	0.05044	1
J31	sms	0.38105	0.37244	-0.00861	0
J31	sandbox	0.3744	0.37244	-0.00196	0
J31	power7	0.36128	0.37244	0.01116	1
J34	no_pref	0.322	0.37556	0.05356	1
J34	stride	0.32465	0.37556	0.05091	1
J34	streamer	0.36368	0.37556	0.01188	1
J34	ampm	0.3748	0.37556	7.600e-04	1
J34	spp	0.37933	0.37556	-0.00377	0
J34	ipcp	0.322	0.37556	0.05356	1
J34	sms	0.38105	0.37556	-0.00549	0
J34	sandbox	0.3744	0.37556	0.00116	1
J34	power7	0.36128	0.37556	0.01428	1
J30	no_pref	0.322	0.37244	0.05044	1
J30	stride	0.32465	0.37244	0.04779	1
J30	streamer	0.36368	0.37244	0.00876	1
J30	ampm	0.3748	0.37244	-0.00236	0
J30	spp	0.37933	0.37244	-0.00689	0
J30	ipcp	0.322	0.37244	0.05044	1
J30	sms	0.38105	0.37244	-0.00861	0
J30	sandbox	0.3744	0.37244	-0.00196	0
J30	power7	0.36128	0.37244	0.01116	1
J35	no_pref	0.322	0.37284	0.05084	1
J35	stride	0.32465	0.37284	0.04819	1
J35	streamer	0.36368	0.37284	0.00916	1
J35	ampm	0.3748	0.37284	-0.00196	0
J35	spp	0.37933	0.37284	-0.00649	0
J35	ipcp	0.322	0.37284	0.05084	1
J35	sms	0.38105	0.37284	-0.00821	0
J35	sandbox	0.3744	0.37284	-0.00156	0
J35	power7	0.36128	0.37284	0.01156	1
J36	no_pref	0.322	0.37039	0.04839	1
J36	stride	0.32465	0.37039	0.04574	1
J36	streamer	0.36368	0.37039	0.00671	1
J36	ampm	0.3748	0.37039	-0.00441	0
J36	spp	0.37933	0.37039	-0.00894	0
J36	ipcp	0.322	0.37039	0.04839	1
J36	sms	0.38105	0.37039	-0.01066	0
J36	sandbox	0.3744	0.37039	-0.00401	0
J36	power7	0.36128	0.37039	0.00911	1
J32	no_pref	0.322	0.3746	0.0526	1
J32	stride	0.32465	0.3746	0.04995	1
J32	streamer	0.36368	0.3746	0.01092	1
J32	ampm	0.3748	0.3746	-2.000e-04	0
J32	spp	0.37933	0.3746	-0.00473	0
J32	ipcp	0.322	0.3746	0.0526	1
J32	sms	0.38105	0.3746	-0.00645	0
J32	sandbox	0.3744	0.3746	2.000e-04	1
J32	power7	0.36128	0.3746	0.01332	1
J33	no_pref	0.322	0.37556	0.05356	1
J33	stride	0.32465	0.37556	0.05091	1

Table 76: v4.9_analysis/v4.8_nn-vs.every_normal_ipc.csv — pair-level columns for all 423 (NN job, normal) pairs; per-job columns identical to Tables 7/74 (cont.)

ID	normal_prefetcher	normal_ipc	ipc	ipc_delta_nn_minus_normal	nn_beats_this_normal_by_keyed_replay
J33	streamer	0.36368	0.37556	0.01188	1
J33	ampm	0.3748	0.37556	7.600e-04	1
J33	spp	0.37933	0.37556	-0.00377	0
J33	ipcp	0.322	0.37556	0.05356	1
J33	sms	0.38105	0.37556	-0.00549	0
J33	sandbox	0.3744	0.37556	0.00116	1
J33	power7	0.36128	0.37556	0.01428	1
J46	no_pref	0.322	0.37549	0.05349	1
J46	stride	0.32465	0.37549	0.05084	1
J46	streamer	0.36368	0.37549	0.01181	1
J46	ampm	0.3748	0.37549	6.900e-04	1
J46	spp	0.37933	0.37549	-0.00384	0
J46	ipcp	0.322	0.37549	0.05349	1
J46	sms	0.38105	0.37549	-0.00556	0
J46	sandbox	0.3744	0.37549	0.00109	1
J46	power7	0.36128	0.37549	0.01421	1
J41	no_pref	0.322	0.37681	0.05481	1
J41	stride	0.32465	0.37681	0.05216	1
J41	streamer	0.36368	0.37681	0.01313	1
J41	ampm	0.3748	0.37681	0.00201	1
J41	spp	0.37933	0.37681	-0.00252	0
J41	ipcp	0.322	0.37681	0.05481	1
J41	sms	0.38105	0.37681	-0.00424	0
J41	sandbox	0.3744	0.37681	0.00241	1
J41	power7	0.36128	0.37681	0.01553	1
J42	no_pref	0.322	0.36923	0.04723	1
J42	stride	0.32465	0.36923	0.04458	1
J42	streamer	0.36368	0.36923	0.00555	1
J42	ampm	0.3748	0.36923	-0.00557	0
J42	spp	0.37933	0.36923	-0.0101	0
J42	ipcp	0.322	0.36923	0.04723	1
J42	sms	0.38105	0.36923	-0.01182	0
J42	sandbox	0.3744	0.36923	-0.00517	0
J42	power7	0.36128	0.36923	0.00795	1
J37	no_pref	0.322	0.37001	0.04801	1
J37	stride	0.32465	0.37001	0.04536	1
J37	streamer	0.36368	0.37001	0.00633	1
J37	ampm	0.3748	0.37001	-0.00479	0
J37	spp	0.37933	0.37001	-0.00932	0
J37	ipcp	0.322	0.37001	0.04801	1
J37	sms	0.38105	0.37001	-0.01104	0
J37	sandbox	0.3744	0.37001	-0.00439	0
J37	power7	0.36128	0.37001	0.00873	1
J44	no_pref	0.322	0.37438	0.05238	1
J44	stride	0.32465	0.37438	0.04973	1
J44	streamer	0.36368	0.37438	0.0107	1
J44	ampm	0.3748	0.37438	-4.200e-04	0
J44	spp	0.37933	0.37438	-0.00495	0
J44	ipcp	0.322	0.37438	0.05238	1
J44	sms	0.38105	0.37438	-0.00667	0
J44	sandbox	0.3744	0.37438	-2.000e-05	0
J44	power7	0.36128	0.37438	0.0131	1
J39	no_pref	0.322	0.37654	0.05454	1
J39	stride	0.32465	0.37654	0.05189	1
J39	streamer	0.36368	0.37654	0.01286	1
J39	ampm	0.3748	0.37654	0.00174	1
J39	spp	0.37933	0.37654	-0.00279	0
J39	ipcp	0.322	0.37654	0.05454	1
J39	sms	0.38105	0.37654	-0.00451	0
J39	sandbox	0.3744	0.37654	0.00214	1
J39	power7	0.36128	0.37654	0.01526	1
J45	no_pref	0.322	0.37549	0.05349	1
J45	stride	0.32465	0.37549	0.05084	1
J45	streamer	0.36368	0.37549	0.01181	1
J45	ampm	0.3748	0.37549	6.900e-04	1
J45	spp	0.37933	0.37549	-0.00384	0
J45	ipcp	0.322	0.37549	0.05349	1
J45	sms	0.38105	0.37549	-0.00556	0
J45	sandbox	0.3744	0.37549	0.00109	1
J45	power7	0.36128	0.37549	0.01421	1
J40	no_pref	0.322	0.37681	0.05481	1
J40	stride	0.32465	0.37681	0.05216	1
J40	streamer	0.36368	0.37681	0.01313	1
J40	ampm	0.3748	0.37681	0.00201	1

Table 76: v4.9_analysis/v4.8_nn-vs.every_normal_ipc.csv — pair-level columns for all 423 (NN job, normal) pairs; per-job columns identical to Tables 7/74 (cont.)

ID	normal_prefetcher	normal_ipc	ipc	ipc_delta_nn_minus_normal	nn_beats_this_normal_by_keyed_replay
J40	spp	0.37933	0.37681	-0.00252	0
J40	ipcp	0.322	0.37681	0.05481	1
J40	sms	0.38105	0.37681	-0.00424	0
J40	sandbox	0.3744	0.37681	0.00241	1
J40	power7	0.36128	0.37681	0.01553	1
J28	no_pref	0.2375	0.24105	0.00355	1
J28	stride	0.23833	0.24105	0.00272	1
J28	streamer	0.24228	0.24105	-0.00123	0
J28	ampm	0.23976	0.24105	0.00129	1
J28	spp	0.23869	0.24105	0.00236	1
J28	ipcp	0.23771	0.24105	0.00334	1
J28	sms	0.24695	0.24105	-0.0059	0
J28	sandbox	0.24433	0.24105	-0.00328	0
J28	power7	0.24232	0.24105	-0.00127	0
J27	no_pref	0.2375	0.24121	0.00371	1
J27	stride	0.23833	0.24121	0.00288	1
J27	streamer	0.24228	0.24121	-0.00107	0
J27	ampm	0.23976	0.24121	0.00145	1
J27	spp	0.23869	0.24121	0.00252	1
J27	ipcp	0.23771	0.24121	0.0035	1
J27	sms	0.24695	0.24121	-0.00574	0
J27	sandbox	0.24433	0.24121	-0.00312	0
J27	power7	0.24232	0.24121	-0.00111	0
J29	no_pref	0.2375	0.23952	0.00202	1
J29	stride	0.23833	0.23952	0.00119	1
J29	streamer	0.24228	0.23952	-0.00276	0
J29	ampm	0.23976	0.23952	-2.400e-04	0
J29	spp	0.23869	0.23952	8.300e-04	1
J29	ipcp	0.23771	0.23952	0.00181	1
J29	sms	0.24695	0.23952	-0.00743	0
J29	sandbox	0.24433	0.23952	-0.00481	0
J29	power7	0.24232	0.23952	-0.0028	0
J26	no_pref	0.2375	0.24195	0.00445	1
J26	stride	0.23833	0.24195	0.00362	1
J26	streamer	0.24228	0.24195	-3.300e-04	0
J26	ampm	0.23976	0.24195	0.00219	1
J26	spp	0.23869	0.24195	0.00326	1
J26	ipcp	0.23771	0.24195	0.00424	1
J26	sms	0.24695	0.24195	-0.005	0
J26	sandbox	0.24433	0.24195	-0.00238	0
J26	power7	0.24232	0.24195	-3.700e-04	0
J23	no_pref	0.2375	0.24195	0.00445	1
J23	stride	0.23833	0.24195	0.00362	1
J23	streamer	0.24228	0.24195	-3.300e-04	0
J23	ampm	0.23976	0.24195	0.00219	1
J23	spp	0.23869	0.24195	0.00326	1
J23	ipcp	0.23771	0.24195	0.00424	1
J23	sms	0.24695	0.24195	-0.005	0
J23	sandbox	0.24433	0.24195	-0.00238	0
J23	power7	0.24232	0.24195	-3.700e-04	0
J25	no_pref	0.2375	0.23963	0.00213	1
J25	stride	0.23833	0.23963	0.0013	1
J25	streamer	0.24228	0.23963	-0.00265	0
J25	ampm	0.23976	0.23963	-1.300e-04	0
J25	spp	0.23869	0.23963	9.400e-04	1
J25	ipcp	0.23771	0.23963	0.00192	1
J25	sms	0.24695	0.23963	-0.00732	0
J25	sandbox	0.24433	0.23963	-0.0047	0
J25	power7	0.24232	0.23963	-0.00269	0
J24	no_pref	0.2375	0.23959	0.00209	1
J24	stride	0.23833	0.23959	0.00126	1
J24	streamer	0.24228	0.23959	-0.00269	0
J24	ampm	0.23976	0.23959	-1.700e-04	0
J24	spp	0.23869	0.23959	9.000e-04	1
J24	ipcp	0.23771	0.23959	0.00188	1
J24	sms	0.24695	0.23959	-0.00736	0
J24	sandbox	0.24433	0.23959	-0.00474	0
J24	power7	0.24232	0.23959	-0.00273	0
J13	no_pref	0.35321	0.35708	0.00387	1
J13	stride	0.3534	0.35708	0.00368	1
J13	streamer	0.33644	0.35708	0.02064	1
J13	ampm	0.34924	0.35708	0.00784	1
J13	spp	0.35391	0.35708	0.00317	1
J13	ipcp	0.35241	0.35708	0.00467	1

Table 76: v4.9_analysis/v4.8_nn-vs.every_normal_ipc.csv — pair-level columns for all 423 (NN job, normal) pairs; per-job columns identical to Tables 7/74 (cont.)

ID	normal_prefetcher	normal_ipc	ipc	ipc_delta_nn_minus_normal	nn_beats_this_normal_by_keyed_replay
J13	sms	0.34167	0.35708	0.01541	1
J13	sandbox	0.34524	0.35708	0.01184	1
J13	power7	0.3433	0.35708	0.01378	1
J12	no_pref	0.35321	0.35127	-0.00194	0
J12	stride	0.3534	0.35127	-0.00213	0
J12	streamer	0.33644	0.35127	0.01483	1
J12	ampm	0.34924	0.35127	0.00203	1
J12	spp	0.35391	0.35127	-0.00264	0
J12	ipcp	0.35241	0.35127	-0.00114	0
J12	sms	0.34167	0.35127	0.0096	1
J12	sandbox	0.34524	0.35127	0.00603	1
J12	power7	0.3433	0.35127	0.00797	1
J14	no_pref	0.35321	0.34149	-0.01172	0
J14	stride	0.3534	0.34149	-0.01191	0
J14	streamer	0.33644	0.34149	0.00505	1
J14	ampm	0.34924	0.34149	-0.00775	0
J14	spp	0.35391	0.34149	-0.01242	0
J14	ipcp	0.35241	0.34149	-0.01092	0
J14	sms	0.34167	0.34149	-1.800e-04	0
J14	sandbox	0.34524	0.34149	-0.00375	0
J14	power7	0.3433	0.34149	-0.00181	0
J11	no_pref	0.35321	0.37915	0.02594	1
J11	stride	0.3534	0.37915	0.02575	1
J11	streamer	0.33644	0.37915	0.04271	1
J11	ampm	0.34924	0.37915	0.02991	1
J11	spp	0.35391	0.37915	0.02524	1
J11	ipcp	0.35241	0.37915	0.02674	1
J11	sms	0.34167	0.37915	0.03748	1
J11	sandbox	0.34524	0.37915	0.03391	1
J11	power7	0.3433	0.37915	0.03585	1
J09	no_pref	0.35321	0.37882	0.02561	1
J09	stride	0.3534	0.37882	0.02542	1
J09	streamer	0.33644	0.37882	0.04238	1
J09	ampm	0.34924	0.37882	0.02958	1
J09	spp	0.35391	0.37882	0.02491	1
J09	ipcp	0.35241	0.37882	0.02641	1
J09	sms	0.34167	0.37882	0.03715	1
J09	sandbox	0.34524	0.37882	0.03358	1
J09	power7	0.3433	0.37882	0.03552	1
J10	no_pref	0.35321	0.35728	0.00407	1
J10	stride	0.3534	0.35728	0.00388	1
J10	streamer	0.33644	0.35728	0.02084	1
J10	ampm	0.34924	0.35728	0.00804	1
J10	spp	0.35391	0.35728	0.00337	1
J10	ipcp	0.35241	0.35728	0.00487	1
J10	sms	0.34167	0.35728	0.01561	1
J10	sandbox	0.34524	0.35728	0.01204	1
J10	power7	0.3433	0.35728	0.01398	1

Table 77: v4.9_analysis/v4.8_route_seed_variance.csv — replay IPC variance across seeds/routes (45 groups)

keyed_replay_count	model_size	policy_tag	replay_ipc_max	replay_ipc_mean	replay_ipc_min	replay_ipc_stdev	route_id	seeds
1	about_383k	assoc_control	0.42835	0.42835	0.42835	0	assoc_control	7 one_shot
1	about_383k	assoc_control	0.42835	0.42835	0.42835	0	assoc_control	7 one_shot
1	about_911k	assoc_control	0.42855	0.42855	0.42855	0	assoc_control	7 one_shot
1	about_911k	assoc_control	0.42855	0.42855	0.42855	0	assoc_control	7 one_shot
1	nearest_153k	assoc_control	0.43066	0.43066	0.43066	0	assoc_control	7 one_shot
1	nearest_153k	assoc_control	0.3803	0.3803	0.3803	0	assoc_control	7 one_shot
1	selected_full	assoc_control	0.42924	0.42924	0.42924	0	assoc_control	7 one_shot
1	selected_full	assoc_control	0.42924	0.42924	0.42924	0	assoc_control	7 one_shot
1	about_383k	support16_fixed_nextline	0.19359	0.19359	0.19359	0	support16_fixed_nextline	7 one_shot
1	about_383k	support16_fixed_nextline	0.19258	0.19258	0.19258	0	support16_fixed_nextline	7 one_shot
1	about_911k	support16_fixed_nextline	0.19325	0.19325	0.19325	0	support16_fixed_nextline	7 one_shot
1	about_911k	support16_fixed_nextline	0.19234	0.19234	0.19234	0	support16_fixed_nextline	7 one_shot
1	nearest_153k	support16_fixed_nextline	0.19259	0.19259	0.19259	0	support16_fixed_nextline	7 one_shot
1	nearest_153k	support16_fixed_nextline	0.18967	0.18967	0.18967	0	support16_fixed_nextline	7 one_shot
1	selected_full	support16_fixed_nextline	0.19173	0.19173	0.19173	0	support16_fixed_nextline	7 one_shot
1	selected_full	support16_fixed_nextline	0.18935	0.18935	0.18935	0	support16_fixed_nextline	7 one_shot
1	about_383k	adaptive_score_gap	0.37549	0.37549	0.37549	0	adaptive_score_gap	7 one_shot
1	about_911k	adaptive_score_gap	0.37681	0.37681	0.37681	0	adaptive_score_gap	7 one_shot
1	selected_full	adaptive_score_gap	0.37556	0.37556	0.37556	0	adaptive_score_gap	7 one_shot
1	about_383k	top2	0.36923	0.36923	0.36923	0	top2	7 one_shot
1	about_383k	top2	0.36923	0.36923	0.36923	0	top2	7 one_shot
1	about_911k	top2	0.37001	0.37001	0.37001	0	top2	7 one_shot
1	about_911k	top2	0.37001	0.37001	0.37001	0	top2	7 one_shot
1	nearest_153k	top2	0.38379	0.38379	0.38379	0	top2	7 one_shot
1	selected_full	top2	0.37244	0.37244	0.37244	0	top2	7 one_shot
3	selected_full	top2	0.37284	0.37189	0.37039	0.00107	top2	7;19;43
1	about_383k	top3	0.37438	0.37438	0.37438	0	top3	7 one_shot

Table 77: v4.9_analysis/v4.8_route_seed_variance.csv — replay IPC variance across seeds/routes (45 groups)

keyed_replay_count	model_size	policy_tag	replay_ipc_max	replay_ipc_mean	replay_ipc_min	replay_ipc_stdev	route_id	seeds	
1	about_911k	top3	0.37654	0.37654	0.37654	0	top3	7	one_sh
1	selected_full	top3	0.3746	0.3746	0.3746	0	top3	7	
1	about_383k	top4	0.37549	0.37549	0.37549	0	top4	7	one_sh
1	about_911k	top4	0.37681	0.37681	0.37681	0	top4	7	one_sh
1	selected_full	top4	0.37556	0.37556	0.37556	0	top4	7	
1	about_383k	region_control	0.24105	0.24105	0.24105	0	region_control	7	one.
1	about_383k	region_control	0.23963	0.23963	0.23963	0	region_control	7	
1	about_911k	region_control	0.24121	0.24121	0.24121	0	region_control	7	one.
1	about_911k	region_control	0.23959	0.23959	0.23959	0	region_control	7	
1	nearest_153k	region_control	0.23952	0.23952	0.23952	0	region_control	7	one.
1	selected_full	region_control	0.24195	0.24195	0.24195	0	region_control	7	one.
1	selected_full	region_control	0.24195	0.24195	0.24195	0	region_control	7	
1	about_383k	context_lru128	0.35708	0.35708	0.35708	0	context_lru128	7	one.
1	about_383k	context_lru128	0.35728	0.35728	0.35728	0	context_lru128	7	
1	about_911k	context_lru128	0.35127	0.35127	0.35127	0	context_lru128	7	one.
1	nearest_153k	context_lru128	0.34149	0.34149	0.34149	0	context_lru128	7	one.
1	selected_full	context_lru128	0.37915	0.37915	0.37915	0	context_lru128	7	one.
1	selected_full	context_lru128	0.37882	0.37882	0.37882	0	context_lru128	7	

Table 78: v4.9_analysis/v4.8_causal_route_decisions.csv — causal decision row per job —

ID	candidate_recall_before_policy	event_coverage	event_timeliness	ipc_delta_vs_best_normal		next_focus	rank_recall_top4	replay_ipc	resource_mshr_p95
J21	0.50851	0.91064	0.99999	-0.00793		ranking / calibration	0.48772	0.42835	1
J20	0.50851	0.913	0.99999	-0.00773		ranking / calibration	0.49048	0.42855	1
J22	0.50851	0.93517	0.99999	-0.00562		ranking / calibration	0.47544	0.43066	0
J19	0.50851	0.92065	1	-0.00704		ranking / calibration	0.49182	0.42924	1
J15	0.50851	0.92065	1	-0.00704		ranking / calibration	0.49182	0.42924	1
J17	0.50851	0.91064	0.99999	-0.00793		ranking / calibration	0.48772	0.42835	1
J16	0.50851	0.913	0.99999	-0.00773		ranking / calibration	0.49048	0.42855	1
J18	0.50851	0.19845	0.99998	-0.05598		ranking / calibration	0.47544	0.3803	1
J07	0.56117	0.21938	0.86937	0.00485		ranking / calibration	0.36318	0.19359	5
J06	0.56117	0.22095	0.87196	0.00451		ranking / calibration	0.36264	0.19325	5
J08	0.56117	0.21527	0.86361	0.00385		ranking / calibration	0.35841	0.19259	5
J05	0.56117	0.19904	0.87277	0.00299		ranking / calibration	0.34574	0.19173	5
J01	0.56117	0.16851	0.98727	6.100e-04		ranking / calibration	0.34574	0.18935	4
J03	0.56117	0.20307	0.98511	0.00384		ranking / calibration	0.36318	0.19258	5
J02	0.56117	0.20242	0.91458	0.0036		ranking / calibration	0.36264	0.19234	5
J04	0.56117	0.16107	0.97886	9.300e-04		ranking / calibration	0.35841	0.18967	5
J43	0.99678	0.77396	0.96799	-0.01182		cache pollution or PQ/MSHR resource behavior	0.99595	0.36923	20
J38	0.99678	0.80147	0.96419	-0.01104		cache pollution or PQ/MSHR resource behavior	0.99597	0.37001	20
J47	0.99678	0.98504	0.99857	0.00274	replay-confirmed route comparison; do not infer a cause from offline scores alone	cache pollution or PQ/MSHR resource behavior	0.99146	0.38379	20
J31	0.99678	0.79333	0.95548	-0.00861		cache pollution or PQ/MSHR resource behavior	0.99636	0.37244	21
J34	0.99678	0.84473	0.9903	-0.00549		cache pollution or PQ/MSHR resource behavior	0.99636	0.37556	22
J30	0.99678	0.79333	0.95548	-0.00861		cache pollution or PQ/MSHR resource behavior	0.99636	0.37244	21
J35	0.99678	0.78029	0.95513	-0.00821		cache pollution or PQ/MSHR resource behavior	0.99639	0.37284	20
J36	0.99678	0.76678	0.95477	-0.01066		cache pollution or PQ/MSHR resource behavior	0.99635	0.37039	21
J32	0.99678	0.83437	0.982	-0.00645		cache pollution or PQ/MSHR resource behavior	0.99636	0.3746	21
J33	0.99678	0.84473	0.9903	-0.00549		cache pollution or PQ/MSHR resource behavior	0.99636	0.37556	22
J46	0.99678	0.85367	0.9935	-0.00556		cache pollution or PQ/MSHR resource behavior	0.99595	0.37549	21
J41	0.99678	0.88103	0.99411	-0.00424		cache pollution or PQ/MSHR resource behavior	0.99597	0.37681	20
J42	0.99678	0.77396	0.96799	-0.01182		cache pollution or PQ/MSHR resource behavior	0.99595	0.36923	20
J37	0.99678	0.80147	0.96419	-0.01104		cache pollution or PQ/MSHR resource behavior	0.99597	0.37001	20
J44	0.99678	0.83735	0.98668	-0.00667		cache pollution or PQ/MSHR resource behavior	0.99595	0.37438	20
J39	0.99678	0.86703	0.98619	-0.00451		cache pollution or PQ/MSHR resource behavior	0.99597	0.37654	20
J45	0.99678	0.85367	0.9935	-0.00556		cache pollution or PQ/MSHR resource behavior	0.99595	0.37549	21
J40	0.99678	0.88103	0.99411	-0.00424		cache pollution or PQ/MSHR resource behavior	0.99597	0.37681	20
J28	0.42431	0.12835	0.99791	-0.0059		candidate-bank recall / representation	0.23775	0.24105	4
J27	0.42431	0.15317	0.99753	-0.00574		candidate-bank recall / representation	0.24312	0.24121	3
J29	0.42431	0.09619	0.99474	-0.00743		candidate-bank recall / representation	0.20403	0.23952	4
J26	0.42431	0.14463	0.99884	-0.005		candidate-bank recall / representation	0.26964	0.24195	4
J23	0.42431	0.14463	0.99884	-0.005		candidate-bank recall / representation	0.26964	0.24195	4
J25	0.42431	0.06614	0.99905	-0.00732		candidate-bank recall / representation	0.23775	0.23963	4
J24	0.42431	0.06615	0.99914	-0.00736		candidate-bank recall / representation	0.24312	0.23959	4
J13	0.92755	0.50588	0.99551	0.00317		ranking / calibration	0.69269	0.35708	3
J12	0.92755	0.37744	0.99662	-0.00264		ranking / calibration	0.70156	0.35127	4
J14	0.92755	0.30195	0.99551	-0.01242		ranking / calibration	0.63446	0.34149	7
J11	0.92755	0.89075	0.99814	0.02524	replay-confirmed route comparison; do not infer a cause from offline scores alone		0.8523	0.37915	2

Table 78: v4.9_analysis/v4.8_causal_route_decisions.csv — causal decision row per job — pa

ID	<i>candidate_recall_before_policy</i>	<i>event_coverage</i>	<i>event_timeliness</i>	<i>ipc_delta_vs_best_normal</i>		<i>next_focus</i>	<i>rank_recall_top4</i>	<i>replay_ipc</i>	<i>resource_mslr_p95</i>
J09	0.92755	0.86354	0.99833	0.02491	replay-confirmed route comparison; do not infer a cause from offline scores alone	0.8523	0.37882		2
J10	0.92755	0.17943	0.99676	0.00337	ranking / calibration	0.69269	0.35728		3

Table 79: v4.9_analysis/v4.8_causal_route_decisions.csv — causal decision row per job — part 2/2

ID	<i>trace</i>
J21	602.gcc.s-734B
J20	602.gcc.s-734B
J22	602.gcc.s-734B
J19	602.gcc.s-734B
J15	602.gcc.s-734B
J17	602.gcc.s-734B
J16	602.gcc.s-734B
J18	602.gcc.s-734B
J07	605.mcf.s-994B
J06	605.mcf.s-994B
J08	605.mcf.s-994B
J05	605.mcf.s-994B
J01	605.mcf.s-994B
J03	605.mcf.s-994B
J02	605.mcf.s-994B
J04	605.mcf.s-994B
J43	619.lbm.s-4268B
J38	619.lbm.s-4268B
J47	619.lbm.s-4268B
J31	619.lbm.s-4268B
J34	619.lbm.s-4268B
J30	619.lbm.s-4268B
J35	619.lbm.s-4268B
J36	619.lbm.s-4268B
J32	619.lbm.s-4268B
J33	619.lbm.s-4268B
J46	619.lbm.s-4268B
J41	619.lbm.s-4268B
J42	619.lbm.s-4268B
J37	619.lbm.s-4268B
J44	619.lbm.s-4268B
J39	619.lbm.s-4268B
J45	619.lbm.s-4268B
J40	619.lbm.s-4268B
J28	620.omnetpp.s-874B
J27	620.omnetpp.s-874B
J29	620.omnetpp.s-874B
J26	620.omnetpp.s-874B
J23	620.omnetpp.s-874B
J25	620.omnetpp.s-874B
J24	620.omnetpp.s-874B
J13	623.xalancbmk.s-700B
J12	623.xalancbmk.s-700B
J14	623.xalancbmk.s-700B
J11	623.xalancbmk.s-700B
J09	623.xalancbmk.s-700B
J10	623.xalancbmk.s-700B

Table 80: v4.9.analysis/v4.8_final.five.trace.comparison.csv — the five per-trace finalists (key columns; remaining columns identical to Tables 7/74) — part 1/2

ID	trace	best_normal_prefetcher	best_normal_ipc	ipc	ipc_delta_vs_best_normal	speedup_vs_best_normal	speedup_vs_no_pref	beats_best_normal_by_keyed_replay	model_size	parameters	accuracy
J22	602.gcc_s-734B	sandbox	0.43628	0.43066	-0.00562	0.98712	1.17027	0	nearest_153k	152,891	0.95211
J07	605.mcf_s-994B	ampm	0.18874	0.19359	0.00485	1.0257	1.05683	1	about_383k	382,549	0.24505
J47	619.lbm_s-4268B	sms	0.38105	0.38379	0.00274	1.00719	1.19189	1	nearest_153k	152,891	0.62177
J26	620.omnetpp_s-874B	sms	0.24695	0.24195	-0.005	0.97975	1.01874	0	selected_full	4,147,959	0.80988
J11	623.xalancbmk_s-700B	spp	0.35391	0.37915	0.02524	1.07132	1.07344	1	selected_full	4,147,959	0.77952

Table 81: v4.9.analysis/v4.8_final.five.trace.comparison.csv — the five per-trace finalists (key columns; remaining columns identical to Tables 7/74) — part 2/2

ID	selected_accuracy	timeliness	event_coverage	event_timeliness	pf_issued	pf_useful	pf_useless	pf_late	l2_load_miss	resource_pq-p95	resource_mshr-p95
J22	0.79469	0.99999	0.93517	0.99999	199,519	189,965	8,575	2	13,168	2	0
J07	0.15124	0.87461	0.21938	0.86937	699,538	171,424	336,523	24,576	623,232	3	5
J47	0.96866	0.99856	0.98504	0.99857	468,215	291,124	1,347	419	3,516	4	20
J26	0.59272	0.99886	0.14463	0.99884	57,133	46,271	7,465	53	271,764	3	4
J11	0.66306	0.99802	0.89075	0.99814	534,810	416,897	14,274	828	53,216	2	2

Empty files. v4.8_stage_b.acceptance.csv and v4.8_smallest_accepted_model_by_trace.csv are present in the archive but zero bytes: the analyzer wrote no Stage-B acceptance rows in this run (no strict-size rung passed all nine gates of Table 53 against its selected-full reference, so the smallest-accepted table is empty rather than fabricated).

A Artifact paths

Table 82: Artifact paths per NN job (lstm_summary path columns)

[illegible]

[illegible]

[illegible]

ID	paths
J10	<pre> history: /content/cache.arch/formal_NN.training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker_00.of_01/traces/623_xalancbmk.s-700b/exports/context_lru128/strict/about.383k/seed.7/training_history.c candidate_banq_metadata: /content/cache.arch/formal_NN.training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker_00.of_01/traces/623_xalancbmk.s-700b/exports/context_lru128/strict/about.383k/seed.7/can offline_diagnostics_path: /content/cache.arch/formal_NN.training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker_00.of_01/traces/623_xalancbmk.s-700b/exports/context_lru128/strict/about.383k/seed.7/can log: /home/qianruw/cache/formal_NN.training/results/prefetch.experiments/v4.9.5.one_shot.all_seed7.v4.9.5.1stm/v4.9.5-623_xalancbmk.s-700b_context_lru128_strict.about.383k_seed7.one_shot.strict.size/logs primary_export: /content/cache.arch/formal_NN.training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker_00.of_01/traces/623_xalancbmk.s-700b/exports/context_lru128/strict/about.383k/seed.7/prefetch.lis rich_list: /home/qianruw/cache/formal_NN.training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker_00.of_01/traces/623_xalancbmk.s-700b/exports/context_lru128/strict/about.383k/seed.7/prefetch.list.csv source_oracle: /content/cache.arch/formal_NN.training/results/standalone_nn.data/oracle/623_xalancbmk.s-700b.oracle.csv.gz; source_rel: formal_NN.training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/wo </pre>

Table 83: Normal-prefetcher log paths

trace	prefetcher	log
602.gcc.s-734B	no_pref	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.no.pref.log
602.gcc.s-734B	stride	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.stride.log
602.gcc.s-734B	streamer	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.streamer.log
602.gcc.s-734B	ampm	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.ampm.log
602.gcc.s-734B	spp	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.spp.log
602.gcc.s-734B	icpc	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.icpc.log
602.gcc.s-734B	sms	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.sms.log
602.gcc.s-734B	sandbox	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.sandbox.log
602.gcc.s-734B	power7	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/602.gcc.s-734B.power7.log
619.lbm.s-4268B	no_pref	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.no.pref.log
619.lbm.s-4268B	stride	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.stride.log
619.lbm.s-4268B	streamer	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.streamer.log
619.lbm.s-4268B	ampm	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.ampm.log
619.lbm.s-4268B	spp	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.spp.log
619.lbm.s-4268B	icpc	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.icpc.log
619.lbm.s-4268B	sms	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.sms.log
619.lbm.s-4268B	sandbox	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.sandbox.log
619.lbm.s-4268B	power7	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/619.lbm.s-4268B.power7.log
605.mcf.s-994B	no_pref	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.no.pref.log
605.mcf.s-994B	stride	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.stride.log
605.mcf.s-994B	streamer	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.streamer.log
605.mcf.s-994B	ampm	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.ampm.log
605.mcf.s-994B	spp	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.spp.log
605.mcf.s-994B	icpc	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.icpc.log
605.mcf.s-994B	sms	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.sms.log
605.mcf.s-994B	sandbox	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.sandbox.log
605.mcf.s-994B	power7	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/605.mcf.s-994B.power7.log
620.omnetpp.s-874B	no_pref	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.no.pref.log
620.omnetpp.s-874B	stride	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.stride.log
620.omnetpp.s-874B	streamer	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.streamer.log
620.omnetpp.s-874B	ampm	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.ampm.log
620.omnetpp.s-874B	spp	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.spp.log
620.omnetpp.s-874B	icpc	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.icpc.log
620.omnetpp.s-874B	sms	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.sms.log
620.omnetpp.s-874B	sandbox	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.sandbox.log
620.omnetpp.s-874B	power7	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/620.omnetpp.s-874B.power7.log
623.xalancbmk.s-700B	no_pref	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.no.pref.log
623.xalancbmk.s-700B	stride	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.stride.log
623.xalancbmk.s-700B	streamer	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.streamer.log
623.xalancbmk.s-700B	ampm	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.ampm.log
623.xalancbmk.s-700B	spp	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.spp.log
623.xalancbmk.s-700B	icpc	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.icpc.log
623.xalancbmk.s-700B	sms	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.sms.log
623.xalancbmk.s-700B	sandbox	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.sandbox.log
623.xalancbmk.s-700B	power7	/home/qianruw/cache/fornal_NN.training/results/prefetcher.baselines/logs/623.xalancbmk.s-700B.power7.log

Table 84: Event-file paths per resource-summary run

[illegible]

[illegible]

Table 85: Event/rich-list paths per unique-event-outcome run

[illegible]

Table 86: plan/v4.9_offline_diagnostic_inventory.csv — diagnostic file inventory

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ID	status	path
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/602.gcc.s-734B/exports/assoc.control/strict/selected.full/seed.7/offline.causal_diagn
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/602.gcc.s-734B/exports/assoc.control/strict/about.911k/seed.7/offline.causal_diagn
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/602.gcc.s-734B/exports/assoc.control/strict/about.383k/seed.7/offline.causal_diagn
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/602.gcc.s-734B/exports/assoc.control/strict/nearest.153k/seed.7/offline.causal_diagn
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/602.gcc.s-734B/exports/assoc.control/calibration.control/selected.full/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/602.gcc.s-734B/exports/assoc.control/calibration.control/about.383k/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/602.gcc.s-734B/exports/assoc.control/calibration.control/nearest.153k/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/620.omnetpp.s-874B/exports/region.control/strict/selected.full/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/620.omnetpp.s-874B/exports/region.control/strict/about.911k/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/620.omnetpp.s-874B/exports/region.control/strict/about.383k/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/620.omnetpp.s-874B/exports/region.control/calibration.control/selected.full/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/620.omnetpp.s-874B/exports/region.control/calibration.control/about.911k/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/620.omnetpp.s-874B/exports/region.control/calibration.control/about.383k/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/620.omnetpp.s-874B/exports/region.control/calibration.control/nearest.153k/seed.7/offline.causal
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/strict/selected.full/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/calibration.control/selected.full/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top3/strict/selected.full/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top4/strict/selected.full/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/adaptive.score.gap/strict/selected.full/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/strict/selected.full/seed.19/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/strict/selected.full/seed.43/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/strict/about.911k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/calibration.control/about.911k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top3/strict/about.911k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top4/strict/about.911k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/adaptive.score.gap/strict/about.911k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/strict/about.383k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top3/strict/about.383k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top4/strict/about.383k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/adaptive.score.gap/strict/about.383k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/calibration.control/nearest.153k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/623.xalanbmk.s-700B/exports/context.lru128/strict/about.911k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/620.omnetpp.s-874B/exports/region.control/strict/nearest.153k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top2/strict/nearest.153k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top3/strict/nearest.153k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/top4/strict/nearest.153k/seed.7/offline.causal_diagnostics
nan	present	/content/cache.arch/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/619.lbm.s-4268B/exports/adaptive.score.gap/strict/nearest.153k/seed.7/offline.causal_diagnostics

B Verbatim files

replay_results/analysis/README.json (verbatim)

```
{
  "scope": "L2C demand-event analysis",
  "event_rows_written": true,
  "replay_plan": "/home/qianruw/cache/formal_NN_training/artifacts/v4.9.5/runs/v4.9.5.one_shot.all_seed7/worker.00.of.01/traces/602.gcc.s-734B/exports/assoc.control/strict/selected.full/seed.7/offline.causal_diagnostics",
  "limit": "Standalone exports contain selections only. no_earlier_selected_standalone_export is not proof of candidate-bank absence.",
  "next_notebook_requirement": "Export a full decision ledger with every candidate, score/probabilities, candidate-source, and reject-reason to distinguish candidate absence from policy rejection."
}
```

plan/v4.9_stage_b_criteria.json (verbatim)

```
{
  "accuracy_ratio_min": 0.95,
  "coverage_ratio_min": 0.95,
  "duplicate_fraction_delta_max": 0.05,
  "ipc_ratio_min": 0.995,
  "issue_rate_ratio_max": 1.1,
  "issue_ratio_max": 1.1,
  "mshr_p95_ratio_max": 1.1,
  "one_shot_matrix_rule": "pre-replay matrix only; strict size rows are candidates, not accepted Stage-B results until keyed replay analysis",
  "pq_p95_ratio_max": 1.1,
  "reference_requirement": "one_shot_all emits selected_full and all smaller strict rungs before replay; acceptance is computed only after the resulting keyed replay, never from this offline plan",
  "rejected_fraction_delta_max": 0.05,
  "selected_accuracy_ratio_min": 0.95,
```

```

"strict_stage_b_only": true,
"timeliness_ratio_min": 0.95,
"zero_action_rule": "ABSTAIN_ZERO_ACTIONS; replay_eligible=0; never force a floor export"
}

```

evidence/trace_prefetch_evidence_report.md (verbatim, 185 lines)

Trace and prefetch evidence report

This report records measured associations from trace profiling, counter summaries, and L2C demand-event attribution. It does not claim a causal explanation beyond these measured counters and outcomes.

Inputs

```

- Trace profiles: 'formal_NN_training/results/trace_profiles'
- Normal baseline summary: '/home/qianruw/cache/formal_NN_training/results/prefetcher_baselines/summary.csv'
- Event attribution: '/home/qianruw/cache/formal_NN_training/results/prefetch_experiments/v4_9_5_one_shot_all_seed7_v4_9_5/analysis'
- Standalone replay summary: 'v4_9_5=/home/qianruw/cache/formal_NN_training/results/prefetch_experiments/v4_9_5_one_shot_all_seed7_v4_9_5/lstm_summary.'

```

602.gcc_s-734B

Trace window: instructions profiled; unique PCs=; branch fraction=; load-instruction fraction=; memory-instruction fraction= .

Best normal IPC: 'sandbox' at '0.43628'; L2 misses=7141; selected accuracy=0.10184300697703258; timeliness=0.999848780375281.

	standalone	IPC	delta vs best normal	selected accuracy	counter	timeliness	unique demand-event coverage	event timeliness	L2 misses
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__about_383k__seed7__one_shot_calibration_control	0.42835	-0.007930	0.95217748126969						
0.9999945940101632 0.9106389472803265 0.9999945940101632 18154									
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__about_911k__seed7__one_shot_calibration_control	0.42855	-0.007730	0.950880869391236						
0.9999946080017255 0.9130019544038084 0.9999946080017255 17673									
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__nearest_153k__seed7__one_shot_calibration_control	0.43066	-0.005620	0.7946912324139						
0.9999894718556381 0.9351699149809729 0.9999894716893727 13168									
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__selected_full__seed7__one_shot_calibration_control	0.42924	-0.007040	0.971096953990						
1.0 0.920652189966081 1.0 16118									
v4_9_5_602.gcc_s-734B__assoc_control__strict__selected_full__seed7__one_shot_full_reference	0.42924	-0.007040	0.9710969539905311						
1.0 0.920652189966081 1.0 16118									
v4_9_5_602.gcc_s-734B__assoc_control__strict__about_383k__seed7__one_shot_strict_size	0.42835	-0.007930	0.95217748126969	0.9999945940101632					
0.9106389472803265 0.9999945940101632 18154									
v4_9_5_602.gcc_s-734B__assoc_control__strict__about_911k__seed7__one_shot_strict_size	0.42855	-0.007730	0.9508808693912366	0.9999946080017255					
0.9130019544038084 0.9999946080017255 17673									
v4_9_5_602.gcc_s-734B__assoc_control__strict__nearest_153k__seed7__one_shot_strict_size	0.3803	-0.055980	0.9503057361933418	0.99997519410611					
0.1984532149204208 0.9999751941061196 162820									

Demand-event and cache-miss comparison against the best normal prefetcher:

	standalone	normal L2 misses	standalone L2 misses	NN - normal misses	both timely	normal-only timely	standalone-only timely
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__about_383k__seed7__one_shot_calibration_control	7141	18154	11013	182431			
13463 2548 NN has more L2 misses and lower measured timely-event coverage.							
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__about_911k__seed7__one_shot_calibration_control	7141	17673	10532	182922			
12972 2537 NN has more L2 misses and lower measured timely-event coverage.							
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__nearest_153k__seed7__one_shot_calibration_control	7141	13168	6027	187284			
8610 2678 NN has more L2 misses and lower measured timely-event coverage.							
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__selected_full__seed7__one_shot_calibration_control	7141	16118	8977	184466			
11428 2547 NN has more L2 misses and lower measured timely-event coverage.							
v4_9_5_602.gcc_s-734B__assoc_control__strict__selected_full__seed7__one_shot_full_reference	7141	16118	8977	184466	11428	2547	
NN has more L2 misses and lower measured timely-event coverage.							
v4_9_5_602.gcc_s-734B__assoc_control__strict__about_383k__seed7__one_shot_strict_size	7141	18154	11013	182431	13463	2548	
NN has more L2 misses and lower measured timely-event coverage.							
v4_9_5_602.gcc_s-734B__assoc_control__strict__about_911k__seed7__one_shot_strict_size	7141	17673	10532	182922	12972	2537	
NN has more L2 misses and lower measured timely-event coverage.							
v4_9_5_602.gcc_s-734B__assoc_control__strict__nearest_153k__seed7__one_shot_strict_size	7141	162820	155679	39822	156072	490	
NN has more L2 misses and lower measured timely-event coverage.							

Top residual contexts are preserved in 'top_residual_contexts.csv'; they are evidence for later candidate-representation analysis, not NN labels.

605.mcf_s-994B

Trace window: instructions profiled; unique PCs=; branch fraction=; load-instruction fraction=; memory-instruction fraction= .

Best normal IPC: 'ampm' at '0.18874'; L2 misses=687090; selected accuracy=0.1778740032117368; timeliness=0.9857843137254902.

	standalone	IPC	delta vs best normal	selected accuracy	counter	timeliness	unique demand-event coverage	event timeliness	L2 misses
v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__about_383k__seed7__one_shot_calibration_control	0.19359	0.004850							
0.15124273172146094 0.8746122448979592 0.21938388836877534 0.8693694658286669 623232									

```

| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__about_911k__seed7__one_shot_calibration_control | 0.19325 | 0.004510
| 0.13770486844559612 | 0.8769662236312645 | 0.22094557991308025 | 0.8719590541945459 | 619482 |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__nearest_153k__seed7__one_shot_calibration_control | 0.19259 | 0.003850
| 0.15485535188206848 | 0.867939571749428 | 0.21526756126194405 | 0.8636124675493602 | 620636 |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__selected_full__seed7__one_shot_calibration_control | 0.19173 | 0.002990
| 0.12049046466989728 | 0.878475492174537 | 0.19903731713280998 | 0.8727733710880344 | 634483 |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__selected_full__seed7__one_shot_full_reference | 0.18935 | 0.000610 | 0.15917112327302887
| 0.9871981850591476 | 0.16851138102977076 | 0.98726502829554 | 655741 |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__about_383k__seed7__one_shot_strict_size | 0.19258 | 0.003840 | 0.2025866324415046
| 0.9851828401522065 | 0.20306583286388022 | 0.9851100363073939 | 633604 |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__about_911k__seed7__one_shot_strict_size | 0.19234 | 0.003600 | 0.20823384487399463
| 0.917750886028726 | 0.20241873057173312 | 0.9145816674703502 | 629420 |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__nearest_153k__seed7__one_shot_strict_size | 0.18967 | 0.000930 | 0.2724193052566861
| 0.9789841814678691 | 0.16107443297492782 | 0.9788598262811355 | 644237 |

```

Demand-event and cache-miss comparison against the best normal prefetcher:

```

| standalone | normal L2 misses | standalone L2 misses | NN - normal misses | both timely | normal-only timely | standalone-only timely |
reading |
|---|---|---|---|---|---|---|
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__about_383k__seed7__one_shot_calibration_control | 687090 | 623232
| -63858 | 29654 | 28018 | 132739 | NN has fewer L2 misses; inspect overlap and resource counters before assigning cause. |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__about_911k__seed7__one_shot_calibration_control | 687090 | 619482
| -67608 | 29008 | 28664 | 134541 | NN has fewer L2 misses; inspect overlap and resource counters before assigning cause. |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__nearest_153k__seed7__one_shot_calibration_control | 687090 | 620636
| -66454 | 22702 | 34970 | 136644 | NN has fewer L2 misses; inspect overlap and resource counters before assigning cause. |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__selected_full__seed7__one_shot_calibration_control | 687090 | 634483
| -52607 | 23903 | 33769 | 123429 | NN has fewer L2 misses; inspect overlap and resource counters before assigning cause. |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__selected_full__seed7__one_shot_full_reference | 687090 | 655741 | -31349 | 9704
| 47968 | 115032 | NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__about_383k__seed7__one_shot_strict_size | 687090 | 633604 | -53486 | 25764 | 31908
| 124550 | NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__about_911k__seed7__one_shot_strict_size | 687090 | 629420 | -57670 | 24114 | 33558
| 125721 | NN has fewer L2 misses; inspect overlap and resource counters before assigning cause. |
| v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__nearest_153k__seed7__one_shot_strict_size | 687090 | 644237 | -42853 | 7183 |
50489 | 112048 | NN has fewer L2 misses; inspect overlap and resource counters before assigning cause. |

```

Top residual contexts are preserved in 'top_residual_contexts.csv'; they are evidence for later candidate-representation analysis, not NN labels.

619.lbm_s-4268B

Trace window: instructions profiled; unique PCs=; branch fraction=; load-instruction fraction=; memory-instruction fraction=.

Best normal IPC: 'sms' at '0.38105'; L2 misses=38705; selected accuracy=0.9421020849728089; timeliness=0.9534967009901484.

```

| standalone | IPC | delta vs best normal | selected accuracy | counter timeliness | unique demand-event coverage | event timeliness | L2
misses |
|---|---|---|---|---|---|---|
| v4_9_5_619.lbm_s-4268B__top2__calibration_control__about_383k__seed7__one_shot_619_calibration_control | 0.36923 | -0.011820 | 0.9985534011747883
| 0.9679126485093978 | 0.7739572833195877 | 0.9679936157026547 | 66193 |
| v4_9_5_619.lbm_s-4268B__top2__calibration_control__about_911k__seed7__one_shot_619_calibration_control | 0.37001 | -0.011040 | 0.9985441493123617
| 0.9640583867839184 | 0.8014689161991454 | 0.9641915121226227 | 58058 |
| v4_9_5_619.lbm_s-4268B__top2__calibration_control__nearest_153k__seed7__one_shot_619_calibration_control | 0.38379 | 0.002740 | 0.9686621688613148
| 0.9985628192067723 | 0.9850393193026042 | 0.998568710497614 | 3516 |
| v4_9_5_619.lbm_s-4268B__top2__calibration_control__selected_full__seed7__one_shot_619_calibration_control | 0.37244 | -0.008610 | 0.9996513018065888
| 0.9554514007864637 | 0.7933301429885385 | 0.9554848836638925 | 60410 |
| v4_9_5_619.lbm_s-4268B__adaptive_score_gap__strict__selected_full__seed7__one_shot_619_full_policy | 0.37556 | -0.005490 | 0.9997096476177907
| 0.9902723426548448 | 0.844725240039506 | 0.9902955500381971 | 45132 |
| v4_9_5_619.lbm_s-4268B__top2__strict__selected_full__seed7__one_shot_619_full_policy | 0.37244 | -0.008610 | 0.9996513018065888 | 0.9554514007864637
| 0.7933301429885385 | 0.9554848836638925 | 60410 |
| v4_9_5_619.lbm_s-4268B__top2__strict__selected_full__seed19__one_shot_619_full_policy | 0.37284 | -0.008210 | 0.9996587591471506 | 0.955022208950767
| 0.7802904571356812 | 0.9551276458735797 | 64362 |
| v4_9_5_619.lbm_s-4268B__top2__strict__selected_full__seed43__one_shot_619_full_policy | 0.37039 | -0.010660 | 0.999689633767846 | 0.9547417975044503
| 0.7667790075312501 | 0.9547682841276952 | 68307 |
| v4_9_5_619.lbm_s-4268B__top3__strict__selected_full__seed7__one_shot_619_full_policy | 0.3746 | -0.006450 | 0.9997106804443479 | 0.9820040964988483
| 0.834373589375473 | 0.9820047614482472 | 48208 |
| v4_9_5_619.lbm_s-4268B__top4__strict__selected_full__seed7__one_shot_619_full_policy | 0.37556 | -0.005490 | 0.9997096476177907 | 0.9902723426548448
| 0.844725240039506 | 0.9902955500381971 | 45132 |
| v4_9_5_619.lbm_s-4268B__adaptive_score_gap__strict__about_383k__seed7__one_shot_619_strict_size_policy | 0.37549 | -0.005560 | 0.9986674105860041
| 0.9934740716512254 | 0.8536683874164656 | 0.9935023640335112 | 42534 |
| v4_9_5_619.lbm_s-4268B__adaptive_score_gap__strict__about_911k__seed7__one_shot_619_strict_size_policy | 0.37681 | -0.004240 | 0.9987097813802359
| 0.9941025709582176 | 0.881027291023931 | 0.9941100554144982 | 34372 |
| v4_9_5_619.lbm_s-4268B__top2__strict__about_383k__seed7__one_shot_619_strict_size_policy | 0.36923 | -0.011820 | 0.9985534011747883 | 0.967912648509
| 0.7739572833195877 | 0.9679936157026547 | 66193 |
| v4_9_5_619.lbm_s-4268B__top2__strict__about_911k__seed7__one_shot_619_strict_size_policy | 0.37001 | -0.011040 | 0.9985441493123617 | 0.964058386783
| 0.8014689161991454 | 0.9641915121226227 | 58058 |
| v4_9_5_619.lbm_s-4268B__top3__strict__about_383k__seed7__one_shot_619_strict_size_policy | 0.37438 | -0.006670 | 0.9985290999485481 | 0.986639584023
| 0.8373467192055363 | 0.9866784511649856 | 47395 |
| v4_9_5_619.lbm_s-4268B__top3__strict__about_911k__seed7__one_shot_619_strict_size_policy | 0.37654 | -0.004510 | 0.998558060053594 | 0.986110094922
| 0.8670338957164531 | 0.9861874613959234 | 38561 |
| v4_9_5_619.lbm_s-4268B__top4__strict__about_383k__seed7__one_shot_619_strict_size_policy | 0.37549 | -0.005560 | 0.9986674105860041 | 0.993474071651
| 0.8536683874164656 | 0.9935023640335112 | 42534 |
| v4_9_5_619.lbm_s-4268B__top4__strict__about_911k__seed7__one_shot_619_strict_size_policy | 0.37681 | -0.004240 | 0.9987097813802359 | 0.994102570958
| 0.881027291023931 | 0.9941100554144982 | 34372 |

```

Demand-event and cache-miss comparison against the best normal prefetcher:

```
| standalone | normal L2 misses | standalone L2 misses | NN - normal misses | both timely | normal-only timely | standalone-only timely |
reading |
|---|---|---|---|---|---|---|
| v4_9_5_619.lbm_s-4268B_top2_calibration_control_about_383k_seed7_one_shot_619_calibration_control | 38705 | 66193 | 27488 | 208911
| 46359 | 19127 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top2_calibration_control_about_911k_seed7_one_shot_619_calibration_control | 38705 | 58058 | 19353 | 212365
| 42905 | 23779 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top2_calibration_control_nearest_153k_seed7_one_shot_619_calibration_control | 38705 | 3516 | -35189 | 252027
| 3243 | 38204 | NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
| v4_9_5_619.lbm_s-4268B_top2_calibration_control_selected_full_seed7_one_shot_619_calibration_control | 38705 | 60410 | 21705 | 212242
| 43028 | 21504 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_adaptive_score_gap_strict_selected_full_seed7_one_shot_619_full_policy | 38705 | 45132 | 6427 | 225625 | 29645
| 23264 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top2_strict_selected_full_seed7_one_shot_619_full_policy | 38705 | 60410 | 21705 | 212242 | 43028 | 21504 |
NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top2_strict_selected_full_seed19_one_shot_619_full_policy | 38705 | 64362 | 25657 | 210097 | 45173 | 19807
| NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top2_strict_selected_full_seed43_one_shot_619_full_policy | 38705 | 68307 | 29602 | 205872 | 49398 | 20051
| NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top3_strict_selected_full_seed7_one_shot_619_full_policy | 38705 | 48208 | 9503 | 222920 | 32350 | 22919 |
NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top4_strict_selected_full_seed7_one_shot_619_full_policy | 38705 | 45132 | 6427 | 225625 | 29645 | 23264 |
NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_adaptive_score_gap_strict_about_383k_seed7_one_shot_619_strict_size_policy | 38705 | 42534 | 3829 | 227102
| 28168 | 24422 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_adaptive_score_gap_strict_about_911k_seed7_one_shot_619_strict_size_policy | 38705 | 34372 | -4333 | 232090
| 23180 | 27495 | NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
| v4_9_5_619.lbm_s-4268B_top2_strict_about_383k_seed7_one_shot_619_strict_size_policy | 38705 | 66193 | 27488 | 208911 | 46359 | 19127
| NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top2_strict_about_911k_seed7_one_shot_619_strict_size_policy | 38705 | 58058 | 19353 | 212365 | 42905 | 23779
| NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top3_strict_about_383k_seed7_one_shot_619_strict_size_policy | 38705 | 47395 | 8690 | 223428 | 31842 | 23287
| NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top3_strict_about_911k_seed7_one_shot_619_strict_size_policy | 38705 | 38561 | -144 | 228491 | 26779 | 26971
| NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
| v4_9_5_619.lbm_s-4268B_top4_strict_about_383k_seed7_one_shot_619_strict_size_policy | 38705 | 42534 | 3829 | 227102 | 28168 | 24422
| NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_619.lbm_s-4268B_top4_strict_about_911k_seed7_one_shot_619_strict_size_policy | 38705 | 34372 | -4333 | 232090 | 23180 | 27495
| NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
```

Top residual contexts are preserved in 'top_residual_contexts.csv'; they are evidence for later candidate-representation analysis, not NN labels.

620.omnetpp.s-874B

Trace window: instructions profiled; unique PCs=; branch fraction=; load-instruction fraction=; memory-instruction fraction=.

Best normal IPC: 'sms' at '0.24695'; L2 misses=246242; selected accuracy=0.24469796531557284; timeliness=0.9818513148486534.

```
| standalone | IPC | delta vs best normal | selected accuracy | counter timeliness | unique demand-event coverage | event timeliness | L2
misses |
|---|---|---|---|---|---|---|
| v4_9_5_620.omnetpp.s-874B_region_control_calibration_control_about_383k_seed7_one_shot_calibration_control | 0.24105 | -0.005900 |
0.3381110462051377 | 0.9978852184058922 | 0.1283509290673945 | 0.9979123173277662 | 278633 |
| v4_9_5_620.omnetpp.s-874B_region_control_calibration_control_about_911k_seed7_one_shot_calibration_control | 0.24121 | -0.005740 |
0.32748126223325535 | 0.9975129910831632 | 0.153171338855298 | 0.9975312197831588 | 270804 |
| v4_9_5_620.omnetpp.s-874B_region_control_calibration_control_nearest_153k_seed7_one_shot_calibration_control | 0.23952 | -0.007430 |
0.14600128833278025 | 0.9948198198198198 | 0.09618895986150862 | 0.9947402809539366 | 292473 |
| v4_9_5_620.omnetpp.s-874B_region_control_calibration_control_selected_full_seed7_one_shot_calibration_control | 0.24195 | -0.005000 |
0.5927176651735723 | 0.9988558846386323 | 0.1446261933193073 | 0.9988436784116941 | 271764 |
| v4_9_5_620.omnetpp.s-874B_region_control_strict_selected_full_seed7_one_shot_full_reference | 0.24195 | -0.005000 | 0.5927176651735723
| 0.9988558846386323 | 0.1446261933193073 | 0.9988436784116941 | 271764 |
| v4_9_5_620.omnetpp.s-874B_region_control_strict_about_383k_seed7_one_shot_strict_size | 0.23963 | -0.007320 | 0.6202327084680026 |
0.9990015214910613 | 0.06614037415417275 | 0.999045664932958 | 296075 |
| v4_9_5_620.omnetpp.s-874B_region_control_strict_about_911k_seed7_one_shot_strict_size | 0.23959 | -0.007360 | 0.6338476413004068 |
0.9991484530229918 | 0.06614669219153763 | 0.9991410984396621 | 296018 |
```

Demand-event and cache-miss comparison against the best normal prefetcher:

```
| standalone | normal L2 misses | standalone L2 misses | NN - normal misses | both timely | normal-only timely | standalone-only timely |
reading |
|---|---|---|---|---|---|---|
| v4_9_5_620.omnetpp.s-874B_region_control_calibration_control_about_383k_seed7_one_shot_calibration_control | 246242 | 278633 | 32391
| 21128 | 64879 | 19502 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_620.omnetpp.s-874B_region_control_calibration_control_about_911k_seed7_one_shot_calibration_control | 246242 | 270804 | 24562
| 24791 | 61216 | 23696 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_620.omnetpp.s-874B_region_control_calibration_control_nearest_153k_seed7_one_shot_calibration_control | 246242 | 292473 | 46231
| 14926 | 71081 | 15523 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_620.omnetpp.s-874B_region_control_calibration_control_selected_full_seed7_one_shot_calibration_control | 246242 | 271764 |
25522 | 23631 | 62376 | 22151 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_620.omnetpp.s-874B_region_control_strict_selected_full_seed7_one_shot_full_reference | 246242 | 271764 | 25522 | 23631 | 62376
| 22151 | NN has more L2 misses and lower measured timely-event coverage. |
```

```

| v4_9_5_620.omnetpp_s-874B__region_control__strict__about_383k_seed7__one_shot_strict_size | 246242 | 296075 | 49833 | 10401 | 75606 |
10536 | NN has more L2 misses and lower measured timely-event coverage. |
| v4_9_5_620.omnetpp_s-874B__region_control__strict__about_911k_seed7__one_shot_strict_size | 246242 | 296018 | 49776 | 10395 | 75612 |
10544 | NN has more L2 misses and lower measured timely-event coverage. |

Top residual contexts are preserved in 'top_residual_contexts.csv'; they are evidence for later candidate-representation analysis, not NN
labels.

## 623.xalancbmk_s-700B

Trace window: instructions profiled; unique PCs=; branch fraction=; load-instruction fraction=; memory-instruction fraction= .

Best normal IPC: 'spp' at '0.35391'; L2 misses=439632; selected accuracy=0.004846407451661488; timeliness=0.806937848441049.

| standalone | IPC | delta vs best normal | selected accuracy | counter timeliness | unique demand-event coverage | event timeliness | L2
misses |
|---|---|---|---|---|---|---|---|
| v4_9_5_623.xalancbmk_s-700B__context_lru128__calibration_control__about_383k_seed7__one_shot_calibration_control | 0.35708 | 0.003170
| 0.38720320577410666 | 0.995569389739758 | 0.5058798712112801 | 0.9955079942844908 | 381025 |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__calibration_control__about_911k_seed7__one_shot_calibration_control | 0.35127 | -0.002640
| 0.31851952023800856 | 0.9965529579551582 | 0.37743532807816144 | 0.9966169270909677 | 481518 |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__calibration_control__nearest_153k_seed7__one_shot_calibration_control | 0.34149 | -0.012420
| 0.14130870077746713 | 0.9934365861583451 | 0.3019518152548018 | 0.9955123465376253 | 658974 |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__calibration_control__selected_full_seed7__one_shot_calibration_control | 0.37915 | 0.025240
| 0.6630559606525787 | 0.9980178346998624 | 0.8907516376151882 | 0.9981363476893448 | 53216 |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__strict__selected_full_seed7__one_shot_full_reference | 0.37882 | 0.024910 | 0.7365366639238762
| 0.998207664249195 | 0.8635350283113135 | 0.9983262524226982 | 62526 |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__strict__about_383k_seed7__one_shot_strict_size | 0.35728 | 0.003370 | 0.6546531062195512
| 0.9967965441506595 | 0.17943377373154212 | 0.9967559299872951 | 383674 |

Demand-event and cache-miss comparison against the best normal prefetcher:

| standalone | normal L2 misses | standalone L2 misses | NN - normal misses | both timely | normal-only timely | standalone-only timely |
reading |
|---|---|---|---|---|---|---|
| v4_9_5_623.xalancbmk_s-700B__context_lru128__calibration_control__about_383k_seed7__one_shot_calibration_control | 439632 | 381025 | -58607
| 465 | 667 | 227358 | NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__calibration_control__about_911k_seed7__one_shot_calibration_control | 439632 | 481518 | 41886
| 405 | 727 | 169573 | NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure. |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__calibration_control__nearest_153k_seed7__one_shot_calibration_control | 439632 | 658974 |
219342 | 417 | 715 | 135567 | NN has more L2 misses despite its selected-precision value; inspect target overlap, dedup, and resource pressure.
|
| v4_9_5_623.xalancbmk_s-700B__context_lru128__calibration_control__selected_full_seed7__one_shot_calibration_control | 439632 | 53216 |
-386416 | 931 | 201 | 400219 | NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__strict__selected_full_seed7__one_shot_full_reference | 439632 | 62526 | -377106 | 825 | 307
| 388068 | NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |
| v4_9_5_623.xalancbmk_s-700B__context_lru128__strict__about_383k_seed7__one_shot_strict_size | 439632 | 383674 | -55958 | 42 | 1090 | 80766
| NN has fewer L2 misses and no lower measured timely-event coverage/timeliness. |

Top residual contexts are preserved in 'top_residual_contexts.csv'; they are evidence for later candidate-representation analysis, not NN
labels.

```

logs/nohup_v4_9_5_one_shot_all_replay_final_20260708_201531.log (verbatim, 337 lines)

```

nohup: ignoring input
[start] Wed Jul 8 20:15:31 EDT 2026
[repo] /home/qianruw/cache
[RUN_DIR] /home/qianruw/cache/formal_NN_training/artifacts/v4_9_5/runs/v4_9_5_one_shot_all_seed7/worker_00_of_01
[OUT_ROOT] /home/qianruw/cache/formal_NN_training/results/prefetch_experiments/v4_9_5_one_shot_all_seed7/v4_9_5
[NORMAL_EVENT_ROOT] /home/qianruw/cache/formal_NN_training/results/prefetch_experiments/normal_full_evidence_20260705
[NORMAL_SUMMARY] /home/qianruw/cache/formal_NN_training/results/prefetcher_baselines/summary.csv
[MAX_JOBS] 1
[FORCE] 0
[CLEAN_OUT] 1
[normal event count]
45
[artifact counts]
13G /home/qianruw/cache/formal_NN_training/artifacts/v4_9_5/runs/v4_9_5_one_shot_all_seed7/worker_00_of_01
metadata.json: 54
prefetch lists: 54
[compile checks]
[tag uniqueness check]
/home/qianruw/cache/formal_NN_training/artifacts/v4_9_5/runs/v4_9_5_one_shot_all_seed7/worker_00_of_01/plan/v4_9_combined_replay_plan.csv
rows 47 duplicates 0 pipe_tags 0
/home/qianruw/cache/formal_NN_training/artifacts/v4_9_5/runs/v4_9_5_one_shot_all_seed7/worker_00_of_01/plan/v4_9_offline_export_acceptance.csv
rows 54 duplicates 0 pipe_tags 0
/home/qianruw/cache/formal_NN_training/artifacts/v4_9_5/runs/v4_9_5_one_shot_all_seed7/worker_00_of_01/v4_9_all_job_metadata.csv rows 54 duplicates
0 pipe_tags 0
[plan resolver test]
[plan] 47 entries
47 /home/qianruw/cache/formal_NN_training/artifacts/v4_9_5/runs/v4_9_5_one_shot_all_seed7/worker_00_of_01/plan/replay_plan_entries.preview.tsv
v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__about_383k_seed7__one_shot_calibration_control 602.gcc_s-734B /home/qianruw/cache/formal_N

```

```

v4_9_5__602.gcc_s-734B__assoc_control__calibration_control__about_911k__seed7__one_shot_calibration_control 602.gcc_s-734B /home/qianruw/cache/formal_N
v4_9_5__602.gcc_s-734B__assoc_control__calibration_control__nearest_153k__seed7__one_shot_calibration_control 602.gcc_s-734B /home/qianruw/cache/formal_N
[plan summary]

[file] /home/qianruw/cache/formal_NN_training/artifacts/v4_9_5/runs/v4_9_5_one_shot_all_seed7/worker_00_of_01/plan/v4_9_combined_replay_plan.csv
rows 47
eligible 47
by_trace {'602.gcc_s-734B': 8, '605.mcf_s-994B': 8, '619.lbm_s-4268B': 18, '620.omnetpp_s-874B': 7, '623.xalancbmk_s-700B': 6}
job_status {'ok': 47}

[file] /home/qianruw/cache/formal_NN_training/artifacts/v4_9_5/runs/v4_9_5_one_shot_all_seed7/worker_00_of_01/plan/v4_9_offline_export_acceptance.csv
rows 54
eligible 47
by_trace {'605.mcf_s-994B': 8, '623.xalancbmk_s-700B': 8, '602.gcc_s-734B': 8, '620.omnetpp_s-874B': 8, '619.lbm_s-4268B': 22}
job_status {'ok': 47, 'abstain': 7}
[clear previous OUT_ROOT]
[run replay] Wed Jul 8 20:15:31 EDT 2026
[plan] 47 entries
[skip] demand logger already present /home/qianruw/cache/external/ChampSim/src/cache.cc
11:// Enabled only when DEMAND_EVENT_LOG is set.
22: const char* path = std::getenv("DEMAND_EVENT_LOG");
30:static void demand_event_log_demand(const CACHE* cache, uint32_t cpu, uint64_t cycle, const char* op, uint32_t type,
705: demand_event_log_demand(this, read_cpu, current_core_cycle[read_cpu], "read", RQ.entry[index].type,
738: demand_event_log_demand(this, read_cpu, current_core_cycle[read_cpu], "read", RQ.entry[index].type,
Building single-core ChampSim...

rm -f -r obj
Generating dependencies for src/ini.cc...
Compiling src/ini.cc...
Generating dependencies for src/knobs.cc...
Compiling src/knobs.cc...
Generating dependencies for src/learning_engine_basic.cc...
Compiling src/learning_engine_basic.cc...
src/learning_engine_basic.cc: In member function 'void LearningEngineBasic::learn(uint32_t, uint32_t, int32_t, uint32_t, uint32_t)':
src/learning_engine_basic.cc:146:21: warning: variable 'Qsa1_old' set but not used [-Wunused-but-set-variable]
    float Qsa1, Qsa2, Qsa1_old;
           ~~~~~

Generating dependencies for src/learning_engine_featurewise.cc...
Compiling src/learning_engine_featurewise.cc...
Generating dependencies for src/main.cc...
Compiling src/main.cc...
Generating dependencies for src/ooo_cpu.cc...
Compiling src/ooo_cpu.cc...
Generating dependencies for src/util.cc...
Compiling src/util.cc...
Generating dependencies for src/cache.cc...
Compiling src/cache.cc...
Generating dependencies for src/bakshalipour_framework.cc...
Compiling src/bakshalipour_framework.cc...
Generating dependencies for src/bitmap.cc...
Compiling src/bitmap.cc...
Generating dependencies for src/block.cc...
Compiling src/block.cc...
Generating dependencies for src/dram_controller.cc...
Compiling src/dram_controller.cc...
Generating dependencies for src/feature_knowledge.cc...
Compiling src/feature_knowledge.cc...
src/feature_knowledge.cc: In member function 'void FeatureKnowledge::updateQ(State*, uint32_t, int32_t, State*, uint32_t)':
src/feature_knowledge.cc:127:20: warning: variable 'Qsa1_old' set but not used [-Wunused-but-set-variable]
    float Qsa1, Qsa2, Qsa1_old;
           ~~~~~

src/feature_knowledge.cc:129:8: warning: unused variable 'Qsa1_old_overall' [-Wunused-variable]
    float Qsa1_old_overall = retrieveQ(state1, action1);
           ~~~~~

src/feature_knowledge.cc:130:8: warning: unused variable 'Qsa2_old_overall' [-Wunused-variable]
    float Qsa2_old_overall = retrieveQ(state2, action2);
           ~~~~~

src/feature_knowledge.cc:145:8: warning: unused variable 'Qsa1_new_overall' [-Wunused-variable]
    float Qsa1_new_overall = retrieveQ(state1, action1);
           ~~~~~

Generating dependencies for src/learning_engine_base.cc...
Compiling src/learning_engine_base.cc...
Generating dependencies for src/uncore.cc...
Compiling src/uncore.cc...
Generating dependencies for branch/branch_predictor.cc...
Compiling branch/branch_predictor.cc...
Generating dependencies for replacement/base_replacement.cc...
Compiling replacement/base_replacement.cc...
Generating dependencies for replacement/llc_replacement.cc...
Compiling replacement/llc_replacement.cc...
Generating dependencies for prefetcher/ampm.cc...
Compiling prefetcher/ampm.cc...
Generating dependencies for prefetcher/bingo.cc...
Compiling prefetcher/bingo.cc...
Generating dependencies for prefetcher/bop.cc...

```

```

Compiling prefetcher/bop.cc...
Generating dependencies for prefetcher/dspatch.cc...
Compiling prefetcher/dspatch.cc...
prefetcher/dspatch.cc: In member function 'void DSPatch::add_to_spt(DSPatch_PBEntry*)':
prefetcher/dspatch.cc:340:11: warning: unused variable 'cov_bmp_acc' [-Wunused-variable]
    uint32_t cov_bmp_acc = 100 * (float)same_count_bmp_acc / pop_count_bmp_real;
    ~~~~~^
Generating dependencies for prefetcher/ipcp_L1.cc...
Compiling prefetcher/ipcp_L1.cc...
Generating dependencies for prefetcher/ipcp_L2.cc...
Compiling prefetcher/ipcp_L2.cc...
Generating dependencies for prefetcher/l1d_prefetcher.cc...
Compiling prefetcher/l1d_prefetcher.cc...
Generating dependencies for prefetcher/l2c_prefetcher.cc...
Compiling prefetcher/l2c_prefetcher.cc...
In file included from inc/ppf_dev.h:5,
                 from prefetcher/l2c_prefetcher.cc:14:
inc/ppf_dev_helper.h:48: warning: "PT_SET" redefined
#define PT_SET 2048

In file included from inc/spp_dev2.h:8,
                 from prefetcher/l2c_prefetcher.cc:13:
inc/spp_dev2_helper.h:30: note: this is the location of the previous definition
#define PT_SET 512

Generating dependencies for prefetcher/llc_prefetcher.cc...
Compiling prefetcher/llc_prefetcher.cc...
Generating dependencies for prefetcher/mlop.cc...
Compiling prefetcher/mlop.cc...
Generating dependencies for prefetcher/next_line.cc...
Compiling prefetcher/next_line.cc...
Generating dependencies for prefetcher/ppf_dev.cc...
Compiling prefetcher/ppf_dev.cc...
Generating dependencies for prefetcher/ppf_dev_helper.cc...
Compiling prefetcher/ppf_dev_helper.cc...
Generating dependencies for prefetcher/pref_power7.cc...
Compiling prefetcher/pref_power7.cc...
Generating dependencies for prefetcher/sandbox.cc...
Compiling prefetcher/sandbox.cc...
Generating dependencies for prefetcher/scooby.cc...
Compiling prefetcher/scooby.cc...
Generating dependencies for prefetcher/scooby_helper.cc...
Compiling prefetcher/scooby_helper.cc...
Generating dependencies for prefetcher/sms.cc...
Compiling prefetcher/sms.cc...
Generating dependencies for prefetcher/spp_dev2.cc...
Compiling prefetcher/spp_dev2.cc...
Generating dependencies for prefetcher/spp_dev2_helper.cc...
Compiling prefetcher/spp_dev2_helper.cc...
Generating dependencies for prefetcher/streamer.cc...
Compiling prefetcher/streamer.cc...
Generating dependencies for prefetcher/stride.cc...
Compiling prefetcher/stride.cc...
Generating dependencies for prefetcher/list_replayer.cc...
Compiling prefetcher/list_replayer.cc...
Linking bin/champsim...

ChampSim is successfully built
Branch Predictor: perceptron
L1D Prefetcher: no
L2C Prefetcher: standalone_nn_replayer
LLC Prefetcher: no
LLC Replacement: ship
Cores: 1
Binary: bin/perceptron-no-standalone_nn_replayer-no-ship-1core

[ok] built /home/qianruw/cache/external/ChampSim/bin/champsim.standalone_nn_replayer
[1stm] v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__about_383k__seed7__one_shot_calibration_control 602.gcc_s-734B
[1stm] v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__about_911k__seed7__one_shot_calibration_control 602.gcc_s-734B
[1stm] v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__nearest_153k__seed7__one_shot_calibration_control 602.gcc_s-734B
[1stm] v4_9_5_602.gcc_s-734B__assoc_control__calibration_control__selected_full__seed7__one_shot_calibration_control 602.gcc_s-734B
[1stm] v4_9_5_602.gcc_s-734B__assoc_control__strict__selected_full__seed7__one_shot_full_reference 602.gcc_s-734B
[1stm] v4_9_5_602.gcc_s-734B__assoc_control__strict__about_383k__seed7__one_shot_strict_size 602.gcc_s-734B
[1stm] v4_9_5_602.gcc_s-734B__assoc_control__strict__about_911k__seed7__one_shot_strict_size 602.gcc_s-734B
[1stm] v4_9_5_602.gcc_s-734B__assoc_control__strict__nearest_153k__seed7__one_shot_strict_size 602.gcc_s-734B
[1stm] v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__about_383k__seed7__one_shot_calibration_control 605.mcf_s-994B
[1stm] v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__about_911k__seed7__one_shot_calibration_control 605.mcf_s-994B
[1stm] v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__nearest_153k__seed7__one_shot_calibration_control 605.mcf_s-994B
[1stm] v4_9_5_605.mcf_s-994B__support16_fixed_nextline__calibration_control__selected_full__seed7__one_shot_calibration_control 605.mcf_s-994B
[1stm] v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__selected_full__seed7__one_shot_full_reference 605.mcf_s-994B
[1stm] v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__about_383k__seed7__one_shot_strict_size 605.mcf_s-994B
[1stm] v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__about_911k__seed7__one_shot_strict_size 605.mcf_s-994B
[1stm] v4_9_5_605.mcf_s-994B__support16_fixed_nextline__strict__nearest_153k__seed7__one_shot_strict_size 605.mcf_s-994B
[1stm] v4_9_5_619.lbm_s-4268B__top2__calibration_control__about_383k__seed7__one_shot_619_calibration_control 619.lbm_s-4268B
[1stm] v4_9_5_619.lbm_s-4268B__top2__calibration_control__about_911k__seed7__one_shot_619_calibration_control 619.lbm_s-4268B

```